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**Part 1: Conformance testing**
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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

The present document is part 1 of a multi-parts TS:

**3GPP TS 36.521-1: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception; Part 1: Conformance Testing.**

3GPP TS 36.521-2 [11]: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception; Part 2: Implementation Conformance Statement (ICS).

- 3GPP TS 36.521-3 [12]: Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification Radio transmission and reception; Part 3: Radio Resource Management (RRM) conformance testing.

# 1 Scope

The present document specifies the measurement procedures for the conformance test of the user equipment (UE) that contain transmitting characteristics, receiving characteristics and performance requirements as part of the 3G Long Term Evolution (3G LTE). Conformance test for the support of RRM (Radio Resource Management) are specified in TS 36.521-3 [12].

The requirements are listed in different clauses only if the corresponding parameters deviate. More generally, tests are only applicable to those mobiles that are intended to support the appropriate functionality. To indicate the circumstances in which tests apply, this is noted in the "*definition and applicability*" part of the test.

For example only Release 8 and later UE declared to support LTE shall be tested for this functionality. In the event that for some tests different conditions apply for different releases, this is indicated within the text of the test itself.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 36.101: "E-UTRA UE radio transmission and reception".

[3] ITU-R Recommendation SM.329-10, "Unwanted emissions in the spurious domain"

[4] 3GPP TS 36.133: "E-UTRA requirements for support of radio resource management".

[5] 3GPP TS 36.331: "E-UTRA Radio Resource Control (RRC): protocol specification".

[6] 3GPP TS 36.304: "E-UTRA UE procedures in idle mode".

[7] 3GPP TS 36.508: “Common test environments for User Equipment (UE)".

[8] 3GPP TS 36.211: "Physical Channels and Modulation".

[9] 3GPP TS 36.212: "E-UTRA Multiplexing and channel coding".

[10] 3GPP TS 36.213: "E-UTRA Physical layer procedures".

[11] 3GPP TS 36.521-2: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification; Radio transmission and reception; Part 2: Implementation Conformance Statement (ICS)".

[12] 3GPP TS 36.521-3: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification; Radio transmission and reception; Part 3: Radio Resource Management (RRM) conformance testing".

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

**Aggregated Channel Bandwidth:** The RF bandwidth in which a UE transmits and receives multiple contiguously aggregated carriers.

**Aggregated Transmission Bandwidth Configuration:** The number of resource block allocated within the aggregated channel bandwidth.

**Carrier aggregation:** Aggregation of two or more component carriers in order to support wider transmission bandwidths.

**Carrier aggregation band:** A set of one or more operating bands across which multiple carriers are aggregated with a specific set of technical requirements.

**Carrier aggregation bandwidth class:** A class defined by the aggregated transmission bandwidth configuration and maximum number of component carriers supported by a UE.

**Carrier aggregation configuration**: A combination of CA operating band(s) and CA bandwidth class(es) supported by a UE.

**Channel edge:** The lowest and highest frequency of the carrier, separated by the channel bandwidth.

**Channel bandwidth:** The RF bandwidth supporting a single E-UTRA RF carrier with the transmission bandwidth configured in the uplink or downlink of a cell. The channel bandwidth is measured in MHz and is used as a reference for transmitter and receiver RF requirements.

**Contiguous carriers:** A set of two or more carriers configured in a spectrum block where there are no RF requirements based on co-existence for un-coordinated operation within the spectrum block.

**Inter-band carrier aggregation:** Carrier aggregation of component carriers in different operating bands**.**

NOTE: Carriers aggregated in each band can be contiguous or non-contiguous.

**Intra-band contiguous carrier aggregation:** Contiguous carriers aggregated in the same operating band.

**Intra-band non-contiguous carrier aggregation:** Non-contiguous carriers aggregated in the same operating band.

**Maximum Output Power:** The mean power level per carrier of UE measured at the antenna connector in a specified reference condition.

**Mean power:** When applied to E-UTRA transmission this is the power measured in the operating system bandwidth of the carrier. The period of measurement shall be at least one subframe (1ms) unless otherwise stated.

**Occupied bandwidth:** The width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage $\beta/2$ of the total mean power of a given emission.

**Output power:** The mean power of one carrier of the UE, delivered to a load with resistance equal to the nominal load impedance of the transmitter.

**PMI delay:** The rate in basic time unit at which PMI is updated.

**Reference bandwidth**: The bandwidth in which an emission level is specified.

**Transmission bandwidth:** Bandwidth of an instantaneous transmission from a UE or BS, measured in Resource Block units.

**Transmission bandwidth configuration:** The highest transmission bandwidth allowed for uplink or downlink in a given channel bandwidth, measured in Resource Block units.

**Transmit Diversity:** Transmit diversity is based on space-frequency block coding techniques complemented with frequency-shift time diversity when four transmit antennas is used.

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

| | |
|---|---|
| $BW_{Channel}$ | Channel bandwidth |
| $BW_{Channel_CA}$ | Aggregated channel bandwidth, expressed in MHz. |
| $BW_{GB}$ | Virtual guard band to facilitate transmitter (receiver) filtering above / below edge CCs. |
| $E_{RS}$ | Transmitted energy per RE for reference symbols during the useful part of the symbol, i.e. excluding the cyclic prefix, (average power normalized to the subcarrier spacing) at the eNode B transmit antenna connector |
| $\hat{E}_s$ | The received energy per RE of the wanted signal during the useful part of the symbol, i.e. excluding the cyclic prefix, averaged across the allocated RB(s) (average power within the allocated RB(s), divided by the number of RE within this allocation, and normalized to the subcarrier spacing) at the UE antenna connector |
| F | Frequency |
| $F_{Interferer}$ (offset) | Frequency offset of the interferer |
| $F_{Interferer}$ | Frequency of the interferer |
| $F_C$ | Frequency of the carrier centre frequency |
| $F_{CA_low}$ | The centre frequency of the *lowest carrier*, expressed in MHz. |
| $F_{CA_high}$ | The centre frequency of the *highest carrier*, expressed in MHz. |
| $F_{DL_low}$ | The lowest frequency of the downlink operating band |
| $F_{DL_high}$ | The highest frequency of the downlink operating band |
| $F_{UL_low}$ | The lowest frequency of the uplink operating band |
| $F_{UL_high}$ | The highest frequency of the uplink operating band |
| $F_{edge_low}$ | The *lower edge* of aggregated channel bandwidth, expressed in MHz. |
| $F_{edge_high}$ | The *higher edge* of aggregated channel bandwidth, expressed in MHz. |
| $F_{offset}$ | Frequency offset from $F_{C_high}$ to the *higher edge* or $F_{C_low}$ to the *lower edge.* |
| $I_o$ | The power spectral density of the total input signal (power averaged over the useful part of the symbols within the transmission bandwidth configuration, divided by the total number of RE for this configuration and normalised to the subcarrier spacing) at the UE antenna connector, including the own-cell downlink signal |
| $I_{or}$ | The total transmitted power spectral density of the own-cell downlink signal (power averaged over the useful part of the symbols within the transmission bandwidth configuration, divided by the total number of RE for this configuration and normalised to the subcarrier spacing) at the eNode B transmit antenna connector |
| $\hat{I}_{or}$ | The total received power spectral density of the own-cell downlink signal (power averaged over the useful part of the symbols within the transmission bandwidth configuration, divided by the total number of RE for this configuration and normalised to the subcarrier spacing) at the UE antenna connector |
| $I_{ot}$ | The received power spectral density of the total noise and interference for a certain RE (average power obtained within the RE and normalized to the subcarrier spacing) as measured at the UE antenna connector |
| $L_{CRBs}$ | The length of a contiguous resource block allocation. |
| $N_{cp}$ | Cyclic prefix length |
| $N_{DL}$ | Downlink EARFCN |
| $N_{oc}$ | The power spectral density of a white noise source (average power per RE normalised to the subcarrier spacing), simulating interference from cells that are not defined in a test procedure, as measured at the UE antenna connector. |
| $N_{oc1}$ | The power spectral density of a white noise source (average power per RE normalized to the subcarrier spacing), simulating interference in non-CRS symbols in ABS subframe from cells that are not defined in a test procedure, as measured at the UE antenna connector. |

| | |
|---|---|
| $N_{oc2}$ | The power spectral density of a white noise source (average power per RE normalized to the subcarrier spacing), simulating interference in CRS symbols in ABS subframe from all cells that are not defined in a test procedure, as measured at the UE antenna connector. |
| $N_{oc3}$ | The power spectral density of a white noise source (average power per RE normalised to the subcarrier spacing), simulating interference in non-ABS subframe from cells that are not defined in a test procedure, as measured at the UE antenna connector. |
| $N_{Offs-DL}$ | Offset used for calculating downlink EARFCN |
| $N_{Offs-UL}$ | Offset used for calculating uplink EARFCN |
| $N_{otx}$ | The power spectral density of a white noise source (average power per RE normalised to the subcarrier spacing) simulating eNode B transmitter impairments as at the eNode B transmit antenna connector |
| $N_{RB}$ | Transmission bandwidth configuration, expressed in units of resource blocks |
| $N_{RB_agg}$ | Aggregated Transmission Bandwidth Configuration The number of the aggregated RBs within the fully allocated Aggregated Channel bandwidth. |
| $N_{RB_alloc}$ | Total number of simultaneously transmitted resource blocks in Aggregated Channel Bandwidth configuration. |
| $N_{UL}$ | Uplink EARFCN |
| $P$ | Number of cell-specific antenna ports |
| $p$ | Antenna port number |
| $P_{CMAX}$ | The measured configured maximum UE output power. |
| $P_{CMAX,c}$ | The configured maximum UE output power for serving cell *c*. |
| $P_{EMAX}$ | Maximum allowed UE output power signalled by higher layers. Same as IE *P-Max*, defined in [5]. |
| $P_{EMAX,c}$ | Maximum allowed UE output power signalled by higher layers for serving cell *c*. Same as IE *P-Max*, defined in [7]. |
| $P_{Interferer}$ | Modulated mean power of the interferer |
| $P_{PowerClass}$ | $P_{PowerClass}$ is the nominal UE power (i.e., no tolerance). |
| $P_{UMAX}$ | Maximum UE Power with possible power reduction due to modulation type, network signalling values and location near the edge of the band; it equals $P_{CMAX}$ when the IE *P-Max*, defined in [5], is not signalled. |
| Rav | Minimum average throughput per RB |
| RB # | Position of the RB in the channel bandwidth. |
| $RB_{start}$ | Indicates the lowest RB index of transmitted resource blocks. |
| $\Delta F_{OOB}$ | Δ Frequency of Out Of Band emission |
| $\Delta R_{IB,c}$ | Allowed reference sensitivity relaxation due to support for inter-band CA operation, for serving cell *c*. |
| $\Delta T_{IB,c}$ | Allowed maximum configured output power relaxation due to support for inter-band CA operation, for serving cell *c*. |
| $\Delta T_C$ | Allowed operating band edge transmission power relaxation. |
| $\Delta T_{C,c}$ | Allowed operating band edge transmission power relaxation for serving cell *c*.3.3 Abbreviations |

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| ABS | Almost Blank Subframe |
| ACLR | Adjacent Channel Leakage Ratio |
| ACS | Adjacent Channel Selectivity |
| A-MPR | Additional Maximum Power Reduction |
| AWGN | Additive White Gaussian Noise |
| BCCH | Broadcast Control Channel |
| BCH | Broadcast Channel |
| BS | Base Station |
| CA | Carrier Aggregation |
| CA_X | CA for band X where X is the applicable E-UTRA operating band |
| CA_X-Y | CA for band X and Band Y where X and Y are the applicable E-UTRA operating band |
| CC | Component CarriersCP Cyclic Prefix |

| | |
|---|---|
| CPE | Customer Premise Equipment |
| CPE_X | Customer Premise Equipment for E-UTRA operating band X |
| CQI | Channel Quality Indicator |
| CW | Continuous Wave |
| DCI | Downlink Control Information |
| eDL-MIMO | Down Link Multiple Antenna transmission |
| DL | Downlink |
| DTX | Discontinuous Transmission |
| DwPTS | Downlink Pilot Time-Slot |
| EARFCN | E-UTRA Absolute Radio Frequency Channel Number |
| EPRE | Energy Per Resource Element |
| E-UTRA | Evolved UMTS Terrestrial Radio Access |
| EUTRAN | Evolved UMTS Terrestrial Radio Access Network |
| EVM | Error Vector Magnitude |
| FDD | Frequency Division Duplex |
| FRC | Fixed Reference Channel |
| FSTD | Frequency-Shift Time Diversity |
| HARQ | Hybrid ARQ |
| HD-FDD | Half- Duplex FDD |
| MAC | Medium Access Control |
| MBMS | Multimedia Broadcast Multicast Service |
| MCS | Modulation and Coding Scheme |
| MOP | Maximum Output Power |
| MPR | Maximum Power Reduction |
| MSR | Maximum Sensitivity Reduction |
| OCNG | OFDMA Channel Noise Generator |
| OFDMA | Orthogonal Frequency Division Multiple Access |
| OOB | Out-of-band |
| P-MPR | Power Management Maximum Power Reduction |
| PA | Power Amplifier |
| PBCH | Physical Broadcast Channel |
| PCC | Primary Component Carrier |
| PCCH | Paging Control Channel |
| PCFICH | Physical Control Format Indicator Channel |
| PDCCH | Physical Downlink Control Channel |
| PDSCH | Physical Downlink Shared Channel |
| PDU | Packet Data Unit |
| PHICH | Physical Hybrid ARQ Indicator Channel |
| Pm-dsg | Probability of miss-detection of the Downlink Scheduling Grant |
| PMI | Precoding Matrix Indicator |
| PRACH | Physical Random Access Channel |
| PRB | Physical Resource Block |
| PSS | Primary Synchronization Signal |
| PSS_RA | PSS-to-EPRE ratio for the channel PSS |
| PUCCH | Physical Uplink Control Channel |
| PUSCH | Physical Uplink Shared ChannelRE Resource Element |
| REFSENS | Reference Sensitivity power level |
| RI | Rank Indicator |
| RLC | Radio Link Control |
| RMC | Reference Measurement Channel |
| r.m.s | Root Mean Square |
| RNTI | Radio Network Temporary Identifier |
| RRC | Radio Resource Control |
| RS | Reference Signal |
| RSRP | Reference Signal Received Power |
| SCC | Secondary Component Carrier |
| SCH | Synchronization Channel |
| SDU | Service Data Unit |
| SFBC | Space-Frequency Block Coding |
| SNR | Signal-to-Noise Ratio |
| SRS | Sounding Reference Signal |
| SSS | Secondary Synchronization Signal |

| | |
|---|---|
| SSS_RA | SSS-to-RS EPRE ratio for the channel SSS |
| TDD | Time Division Duplex |
| TPC | Transmit Power Control |
| TPMI | Transmitted Precoding Matrix Indicator |
| TTI | Transmission Time Interval |
| UE | User Equipment |
| UL | Uplink |
| UL-MIMO | Up Link Multiple Antenna transmission |
| UMTS | Universal Mobile Telecommunications System |
| UpPTS | Uplink Pilot Time-Slot |
| UTRA | UMTS Terrestrial Radio Access |
| UTRAN | UMTS Terrestrial Radio Access Network |
| xCH_RA | xCH-to-RS EPRE ratio for the channel xCH in all transmitted OFDM symbols not containing RS |
| xCH_RB | xCH-to-RS EPRE ratio for the channel xCH in all transmitted OFDM symbols containing RS |

# 4 General

Unless otherwise stated, the following reference conditions used by all test cases in this document are specified in TS 36.508 [7]:

- Connection Diagrams,
- Test Frequencies,
- Cell Settings,
- Reference Environments,
- Environmental Conditions,
- Generic Connection Setup Procedures,
- System Information (SI),
- Message Contents.

Where a test requires one of the above reference conditions that are different, this will be specified within the test itself.

The Minimum Requirements defined in each test make no allowance for Measurement Uncertainty. Therefore, Test Tolerances are used to relax the Minimum Requirements. If the Test Requirement differs from the Minimum Requirement then the Test Tolerance applied for that test is non-zero. For each test the Test Tolerances are individually calculated to create the Test Requirements. The Test Tolerance for each test and the explanation of how the Minimum Requirement has been relaxed by the Test Tolerance is given in Annex F.3.

Downlink and Uplink transmissions are organized into radio frames with $T_f = 307200 \times T_s = 10\,\text{ms}$ duration. Two radio frame structures are supported in this document:

- Type 1, applicable to FDD,
- Type 2, applicable to TDD.

In clauses 6 and 7 TX and RX test cases for FDD/TDD test cases are defined. FDD and TDD test scenarios/ requirements are included within the same test case. For test cases with any difference between the FDD and TDD branches the test description part of the test case has been separated in two sections to cover the two technologies. The applicability for the FDD and TDD branches are specified in TS 36.521-2.

In clause 8 the performance requirement test cases are defined. FDD and TDD performance requirement test cases are defined in different clauses accordingly to the requirements specified in TS 36.101.

Unless otherwise stated, each test case is tested for every operating band supported by the UE and repeated with the applicable test configurations (i.e. test environment, test frequencies, test channel bandwidths, channel bandwidth parameters) indicated in each test case. For test cases in clauses 6, 7, 8 the initial conditions of the downlink physical channels signal levels and downlink physical channels required are specified in Annex C.0, Annex C.1 and Annex C.2.

For test cases in clauses 6 and 7 that require measurements with maximum output power, the UE shall transmit at its maximum output power state with output power level of $P_{UMAX}$ level. This range of maximum output power shall not be modified for any further additional relaxation.

For test cases in clauses 6 and 7, the partial RB allocations refer to any RB allocation less than full RB allocation except 1 RB allocation.

## 4.1 Categorization of test requirements in CA, UL-MIMO, eDL-MIMO

The test requirements for Chapters 6 (Tx Characteristics) and 7 (Rx Characteristics), which are specific to CA, UL-MIMO, and eDL-MIMO are specified as suffix A, B, C, D where;

a) Suffix A additional requirements need to support CA

b) Suffix B additional requirements need to support UL-MIMO

c) Suffix C additional requirements need to support TBD

d) Suffix D additional requirements need to support eDL-MIMO

A terminal which supports the above features needs to meet both the general requirements and the additional requirement applicable to the additional sub-clause (suffix A, B, C and D).

A terminal which supports more than one feature (CA, UL-MIMO, and eDL-MIMO) shall meet all of the separate corresponding requirements.

NOTE 1: Test Case 6.5.2.1A, PUSCH-EVM with exclusion period, is a Release 8 non-CA test case and does not indicate a requirement to support CA.

NOTE 2: This categorization of test requirements reflects only the categorization of minimum requirements as done by RAN4 in 3GPP TS 36.101. For the categorization of the tests requirements done on behalf of RAN5 follow the information given in Annex I.

# 5 Frequency bands and channel arrangement

## 5.1 General

The channel arrangements presented in this clause are based on the frequency bands and channel bandwidths defined in the present release of specifications.

NOTE: Other operating bands and channel bandwidths may be considered in future releases.

## 5.2 Operating bands

E-UTRA is designed to operate in the operating bands defined in Table 5.2-1.

**Table 5.2-1: E-UTRA operating bands**

| E-UTRA Operating Band | Uplink (UL) eNode B receive UE transmit | | | Downlink (DL) eNode B transmit UE receive | | | Duplex Mode |
|---|---|---|---|---|---|---|---|
| | $F_{UL_low} - F_{UL_high}$ | | | $F_{DL_low} - F_{DL_high}$ | | | |
| 1 | 1920 MHz | – | 1980 MHz | 2110 MHz | – | 2170 MHz | FDD |
| 2 | 1850 MHz | – | 1910 MHz | 1930 MHz | – | 1990 MHz | FDD |
| 3 | 1710 MHz | – | 1785 MHz | 1805 MHz | – | 1880 MHz | FDD |
| 4 | 1710 MHz | – | 1755 MHz | 2110 MHz | – | 2155 MHz | FDD |
| 5 | 824 MHz | – | 849 MHz | 869 MHz | – | 894MHz | FDD |
| 6 | 830 MHz | – | 840 MHz | 875 MHz | – | 885 MHz | FDD |
| 7 | 2500 MHz | – | 2570 MHz | 2620 MHz | – | 2690 MHz | FDD |
| 8 | 880 MHz | – | 915 MHz | 925 MHz | – | 960 MHz | FDD |
| 9 | 1749.9 MHz | – | 1784.9 MHz | 1844.9 MHz | – | 1879.9 MHz | FDD |
| 10 | 1710 MHz | – | 1770 MHz | 2110 MHz | – | 2170 MHz | FDD |
| 11 | 1427.9 MHz | – | 1447.9 MHz | 1475.9 MHz | – | 1495.9 MHz | FDD |
| 12 | 699 MHz | – | 716 MHz | 729 MHz | – | 746 MHz | FDD |
| 13 | 777 MHz | – | 787 MHz | 746 MHz | – | 756 MHz | FDD |
| 14 | 788 MHz | – | 798 MHz | 758 MHz | – | 768 MHz | FDD |
| 15 | Reserved | | | Reserved | | | FDD |
| 16 | Reserved | | | Reserved | | | FDD |
| 17 | 704 MHz | – | 716 MHz | 734 MHz | – | 746 MHz | FDD |
| 18 | 815 MHz | – | 830 MHz | 860 MHz | – | 875 MHz | FDD |
| 19 | 830 MHz | – | 845 MHz | 875 MHz | – | 890 MHz | FDD |
| 20 | 832 MHz | – | 862 MHz | 791 MHz | – | 821 MHz | FDD |
| 21 | 1447.9 MHz | – | 1462.9 MHz | 1495.9 MHz | – | 1510.9 MHz | FDD |
| 22 | 3410 MHz | – | 3490 MHz | 3510 MHz | – | 3590 MHz | FDD |
| 23 | 2000 MHz | – | 2020 MHz | 2180 MHz | – | 2200 MHz | FDD |
| 24 | 1626.5 MHz | – | 1660.5 MHz | 1525 MHz | – | 1559 MHz | FDD |
| 25 | 1850 MHz | – | 1915 MHz | 1930 MHz | – | 1995 MHz | FDD |
| ... | | | | | | | |
| 33 | 1900 MHz | – | 1920 MHz | 1900 MHz | – | 1920 MHz | TDD |
| 34 | 2010 MHz | – | 2025 MHz | 2010 MHz | – | 2025 MHz | TDD |
| 35 | 1850 MHz | – | 1910 MHz | 1850 MHz | – | 1910 MHz | TDD |
| 36 | 1930 MHz | – | 1990 MHz | 1930 MHz | – | 1990 MHz | TDD |
| 37 | 1910 MHz | – | 1930 MHz | 1910 MHz | – | 1930 MHz | TDD |
| 38 | 2570 MHz | – | 2620 MHz | 2570 MHz | – | 2620 MHz | TDD |
| 39 | 1880 MHz | – | 1920 MHz | 1880 MHz | – | 1920 MHz | TDD |
| 40 | 2300 MHz | – | 2400 MHz | 2300 MHz | – | 2400 MHz | TDD |
| 41 | 2496 MHz | - | 2690 MHz | 2496 MHz | - | 2690 MHz | TDD |
| 42 | 3400 MHz | – | 3600 MHz | 3400 MHz | – | 3600 MHz | TDD |
| 43 | 3600 MHz | – | 3800 MHz | 3600 MHz | – | 3800 MHz | TDD |
| Note: Band 6 is not applicable. | | | | | | | |

# 5.2A Operating bands for CA

E-UTRA carrier aggregation is designed to operate in the operating bands defined in Tables 5.2A-1 and 5.2A-2.

**Table 5.2A-1: Intra band CA operating bands**

| E-UTRA CA Band | E-UTRA Band | Uplink (UL) operating band BS receive / UE transmit FUL_low – FUL_high | | | Downlink (DL) operating band BS transmit / UE receive FDL_low – FDL_high | | | Duplex Mode |
|---|---|---|---|---|---|---|---|---|
| CA_1 | 1 | 1920 MHz | – | 1980 MHz | 2110 MHz | – | 2170 MHz | FDD |
| CA_40 | 40 | 2300 MHz | – | 2400 MHz | 2300 MHz | – | 2400 MHz | TDD |

**Table 5.2A-2: Inter band CA operating bands**

| E-UTRA CA Band | E-UTRA Band | Uplink (UL) operating band BS receive / UE transmit FUL_low – FUL_high | | | Downlink (DL) operating band BS transmit / UE receive FDL_low – FDL_high | | | Duplex Mode |
|---|---|---|---|---|---|---|---|---|
| CA_1-5 | 1 | 1920 MHz | – | 1980 MHz | 2110 MHz | – | 2170 MHz | FDD |
| | 5 | 824 MHz | – | 849 MHz | 869 MHz | – | 894 MHz | |

# 5.2B Operating bands for UL-MIMO

E-UTRA UL-MIMO in Rel-10 is designed to operate in the operating bands defined in Table 5.2-1.

# 5.3 TX–RX frequency separation

a) The default EUTRA TX channel (carrier centre frequency) to RX channel (carrier centre frequency) separation is specified in Table 5.3-1 for the TX and RX channel bandwidths defined in Table 5.4.2.1-1

**Table 5.3-1: Default UE TX-RX frequency separation**

| E-UTRA Operating Band | TX - RX carrier centre frequency separation |
|---|---|
| 1 | 190 MHz |
| 2 | 80 MHz. |
| 3 | 95 MHz. |
| 4 | 400 MHz |
| 5 | 45 MHz |
| 6 | 45 MHz |
| 7 | 120 MHz |
| 8 | 45 MHz |
| 9 | 95 MHz |
| 10 | 400 MHz |
| 11 | 48 MHz |
| 12 | 30 MHz |
| 13 | -31 MHz |
| 14 | -30 MHz |
| 17 | 30 MHz |
| 18 | 45 MHz |
| 19 | 45 MHz |
| 20 | -41 MHz |
| 21 | 48 MHz |
| 22 | 100 MHz |
| 23 | 180 MHz |
| 24 | -101.5 MHz |
| 25 | 80 MHz |

b) The use of other TX channel to RX channel carrier centre frequency separation is not precluded and is intended to form part of a later release.

c) The range E-UTRA TX channel (carrier centre frequency) to RX channel (carrier centre frequency) separations for operating bands supporting variable duplex FDD is specified in Table 5.3.

**Table 5.3-2: TX-RX frequency separation for operating bands supporting variable duplex FDD**

| E-UTRA Operating Band | TX - RX carrier centre frequency separation | |
|---|---|---|
| | Allowed offset | Separation |
| 23 | -10 MHz | 170 MHz |
| | +10 MHz | 190 MHz |

# 5.3A TX–RX frequency separation for CA

For CA, the same TX-RX frequency separation as specified in Table 5.3-1 is applied to PCC and SCC, respectively.

# 5.4 Channel arrangement

## 5.4.1 Channel spacing

The spacing between carriers will depend on the deployment scenario, the size of the frequency block available and the channel bandwidths. The nominal channel spacing between two adjacent E-UTRA carriers is defined as following:

$$\text{Nominal Channel spacing} = (BW_{Channel(1)} + BW_{Channel(2)})/2$$

where $BW_{Channel(1)}$ and $BW_{Channel(2)}$ are the channel bandwidths of the two respective E-UTRA carriers. The channel spacing can be adjusted to optimize performance in a particular deployment scenario.

## 5.4.1A Channel spacing for intra-band contiguous CA

For CA Bandwidth Class C, the nominal channel spacing between two adjacent E-UTRA component carriers is defined as the following:

$$\text{Nominal channel spacing} = \left\lfloor \frac{BW_{Channel(1)} + BW_{Channel(2)} - 0.1\left|BW_{Channel(1)} - BW_{Channel(2)}\right|}{0.6} \right\rfloor 0.3 \quad [\text{MHz}]$$

where $BW_{Channel(1)}$ and $BW_{Channel(2)}$ are the channel bandwidths of the two respective E-UTRA component carriers according to Table 5.2-1 with values in MHz. The channel spacing for intra-band contiguous carrier aggregation can be adjusted to any multiple of 300 kHz less than the nominal channel spacing to optimize performance in a particular deployment scenario.

## 5.4.2 Channel bandwidth

Requirements in present document are specified for the channel bandwidths listed in Table 5.4.2-1

**Table 5.4.2-1 Transmission bandwidth configuration $N_{RB}$ in E-UTRA channel bandwidths**

| Channel bandwidth $BW_{Channel}$ [MHz] | 1.4 | 3 | 5 | 10 | 15 | 20 |
|---|---|---|---|---|---|---|
| Transmission bandwidth configuration $N_{RB}$ | 6 | 15 | 25 | 50 | 75 | 100 |

Figure 5.4.2-1 shows the relation between the Channel bandwidth ($BW_{Channel}$) and the Transmission bandwidth configuration ($N_{RB}$). The channel edges are defined as the lowest and highest frequencies of the carrier separated by the channel bandwidth, i.e. at $F_C$ +/- $BW_{Channel}$ /2.

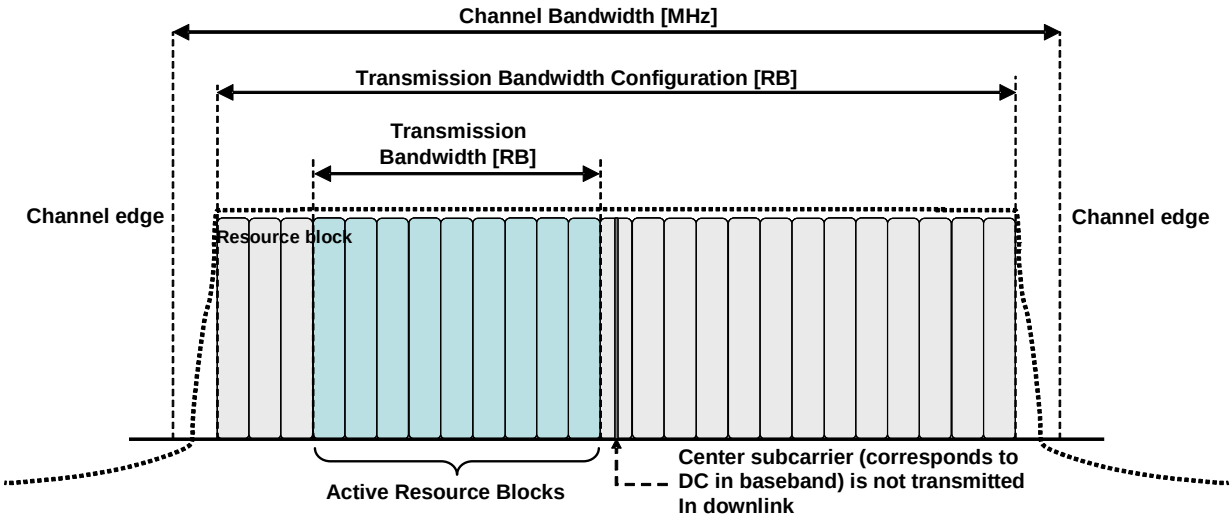



**Figure 5.4.2-1 Definition of Channel Bandwidth and Transmission Bandwidth Configuration for one E-UTRA carrier.**

### 5.4.2.1 Channel bandwidths per operating band

a) The requirements in this specification apply to the combination of channel bandwidths and operating bands shown in Table 5.4.2.1-1. The transmission bandwidth configuration in Table 5.4.2-1 shall be supported for each of the specified supported channel bandwidths. The same (symmetrical) channel bandwidth is specified for both the TX and RX path.

**Table 5.4.2.1-1: E-UTRA channel bandwidth**

| E-UTRA band / channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|
| **E-UTRA Band** | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| 1 | | | Yes | Yes | Yes | Yes |
| 2 | Yes | Yes | Yes | Yes | Yes[1] | Yes[1] |
| 3 | Yes | Yes | Yes | Yes | Yes[1] | Yes[1] |
| 4 | Yes | Yes | Yes | Yes | Yes | Yes |
| 5 | Yes | Yes | Yes | Yes[1] | | |
| 6 | | | Yes | Yes[1] | | |
| 7 | | | Yes | Yes | Yes[2] | Yes[1, 2] |
| 8 | Yes | Yes | Yes | Yes[1] | | |
| 9 | | | Yes | Yes | Yes[1] | Yes[1] |
| 10 | | | Yes | Yes | Yes | Yes |
| 11 | | | Yes | Yes[1] | | |
| 12 | Yes | Yes | Yes[1] | Yes[1] | | |
| 13 | | | Yes[1] | Yes[1] | | |
| 14 | | | Yes[1] | Yes[1] | | |
| ... | | | | | | |
| 17 | | | Yes[1] | Yes[1] | | |
| 18 | | | Yes | Yes[1] | Yes[1] | |
| 19 | | | Yes | Yes[1] | Yes[1] | |
| 20 | | | Yes | Yes[1] | Yes[1] | Yes[1] |
| 21 | | | Yes | Yes[1] | Yes[1] | |
| 22 | | | Yes | Yes | Yes[1] | Yes[1] |
| 23 | Yes | Yes | Yes | Yes | | |
| 24 | | | Yes | Yes | | |
| 25 | Yes | Yes | Yes | Yes | Yes[1] | Yes[1] |
| ... | | | | | | |
| 33 | | | Yes | Yes | Yes | Yes |
| 34 | | | Yes | Yes | Yes | |
| 35 | Yes | Yes | Yes | Yes | Yes | Yes |
| 36 | Yes | Yes | Yes | Yes | Yes | Yes |
| 37 | | | Yes | Yes | Yes | Yes |
| 38 | | | Yes | Yes | Yes[2] | Yes[2] |
| 39 | | | Yes | Yes | Yes | Yes |
| 40 | | | Yes | Yes | Yes | Yes |
| 41 | | | Yes | Yes | Yes | Yes |
| 42 | | | Yes | Yes | Yes | Yes |
| 43 | | | Yes | Yes | Yes | Yes |

Note 1: bandwidth for which a relaxation of the specified UE receiver sensitivity requirement (Clause 7.3) is allowed.

Note 2: bandwidth for which the uplink transmission bandwidth can be restricted by the network for some channel assignments in FDD/TDD co-existence scenarios in order to meet unwanted emissions requirements (Clause 6.6.3.2).

b) The use of different (asymmetrical)) channel bandwidth for the TX and RX is not precluded and is intended to form part of a later release.

## 5.4.2A Channel bandwidth for CA

For intra-band contiguously aggregated component carriers *Aggregated Channel Bandwidth*, *Aggregated Transmission Bandwidth Configuration* and *Guard Bands* are defined as follows, see Figure 5.4.2A-1.

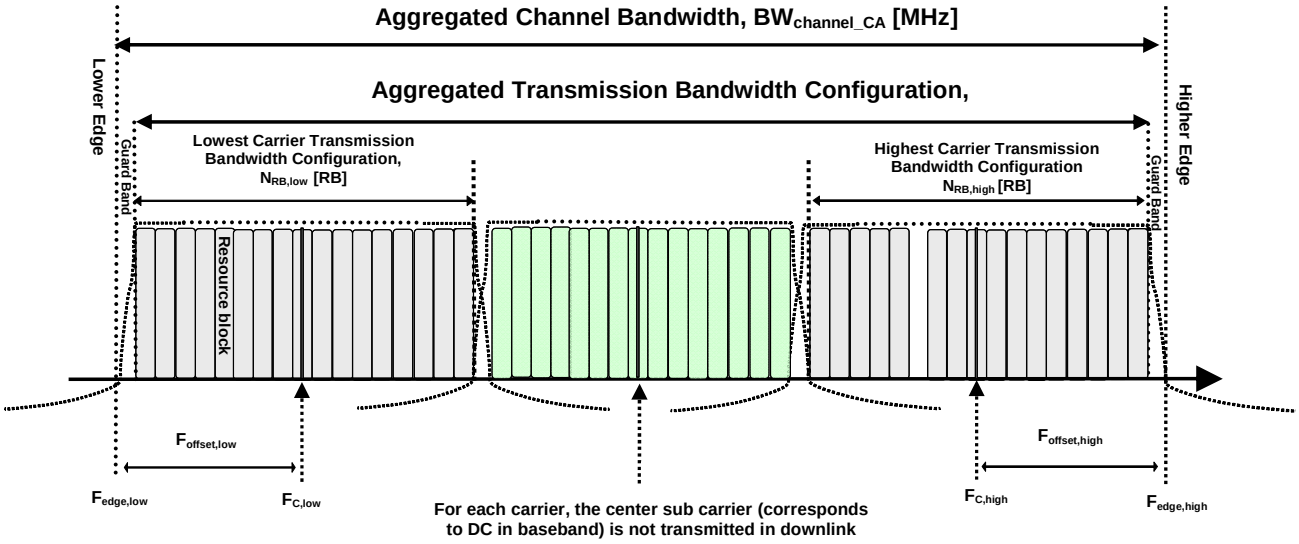



**Figure 5.4.2A-1. Definition of Aggregated Channel Bandwidth and Aggregated Channel Bandwidth Edges**

The *aggregated channel bandwidth*, $BW_{Channel_CA}$, is defined as

$$BW_{Channel_CA} = F_{edge,high} - F_{edge,low} \quad [MHz].$$

The lower bandwidth edge $F_{edge,low}$ and the upper bandwidth edge $F_{edge,high}$ of the aggregated channel bandwidth are used as frequency reference points for transmitter and receiver requirements and are defined by

$$F_{edge,low} = F_{C,low} - F_{offset,low}$$

$$F_{edge,high} = F_{C,high} + F_{offset,high}$$

The lower and upper frequency offsets depend on the transmission bandwidth configurations of the lowest and highest assigned edge component carrier and are defined as

$$F_{offset,low} = 0.18N_{RB,low}/2 + BW_{GB} \,[MHz]$$

$$F_{offset,high} = 0.18N_{RB,high}/2 + BW_{GB} \,[MHz]$$

where $N_{RB,low}$ and $N_{RB,high}$ are the transmission bandwidth configurations according to Table 5.2-1 for the lowest and highest assigned component carrier, respectively. $BW_{GB}$ denotes the *Nominal Guard Band* and is defined in Table 5.4.2A-1, and the factor 0.18 is the PRB bandwidth in MHz.

NOTE: The values of $BW_{Channel_CA}$ for UE and BS are the same if the lowest and the highest component carriers are identical.

Aggregated Transmission Bandwidth Configuration is the number of the aggregated RBs within the fully allocated Aggregated Channel bandwidth and is defined per CA Bandwidth Class (Table 5.4.2A-1).

**Table 5.4.2A-1: CA bandwidth classes and corresponding nominal guard bands**

| CA Bandwidth Class | Aggregated Transmission Bandwidth Configuration | Maximum number of CC | Nominal Guard Band $BW_{GB}$ |
|---|---|---|---|
| A | $N_{RB,agg} \leq 100$ | 1 | $0.05BW_{Channel(1)}$ |
| B | $N_{RB,agg} \leq 100$ | 2 | FFS |
| C | $100 < N_{RB,agg} \leq 200$ | 2 | $0.05\ max(BW_{Channel(1)}, BW_{Channel(2)})$ |
| D | $200 < N_{RB,agg} \leq [300]$ | FFS | FFS |
| E | $[300] < N_{RB,agg} \leq [400]$ | FFS | FFS |
| F | $[400] < N_{RB,agg} \leq [500]$ | FFS | FFS |
| Note 1: $BW_{Channel(1)}$ and $BW_{Channel(2)}$ are channel bandwidths of two E-UTRA component carriers according to Table 5.2-1. | | | |

The channel spacing between centre frequencies of contiguously aggregated component carriers is defined in clause 5.4.1A.

### 5.4.2A.1 Channel bandwidths per operating band for CA

The requirements in this specification apply to the combination of CA bandwidth class and CA operating bands shown in Table 5.4.2A.1-1.

Indexing letter in CA configuration acronym refers to supported CA bandwidth class. In case no CA bandwidth class is labelled acronym refers to all specified combinations of CA bandwidth class and CA operating band. CA configuration refers to a combination of CA operating band and CA bandwidth class supported by a UE.

DL component carrier combinations for a given CA operating band shall be symmetrical in relation to channel centre unless stated otherwise in table 5.4.2A.1-1 or 5.4.2A.1-2.

**Table 5.4.2A.1-1: Supported CC combinations per CA configuration for intra-band contiguous CA**

| CA Configuration / NRB_agg | | | | |
|---|---|---|---|---|
| E-UTRA CA Configuration | E-UTRA Band | 50RB+100RB (10 MHz + 20 MHz) | 75RB+75RB (15 MHz + 15 MHz) | 100RB+100RB (20 MHz + 20 MHz) |
| CA_1C | 1 | | Yes | Yes |
| CA_40C | 40 | Yes | Yes | Yes |

**Table 5.4.2A.1-2: Supported E-UTRA bandwidths per CA configuration for inter-band CA**

| CA operating / channel bandwidth | | | | | | | |
|---|---|---|---|---|---|---|---|
| E-UTRA CA Configuration | E-UTRA Bands | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| CA_1A-5A | 1 | | | | Yes | | |
| | 5 | | | | Yes | | |

## 5.4.2B Channel bandwidth for UL-MIMO

### 5.4.2B.1 Channel bandwidths per operating band for UL- MIMO

For UL-MIMO, the channel bandwidths specified in Table 5.4.2.1-1 in present document apply for the UL-MIMO operating bands.

## 5.4.3 Channel raster

The channel raster is 100 kHz for all bands, which means that the carrier centre frequency must be an integer multiple of 100 kHz.

## 5.4.3A Channel raster for CA

For LTE-A same channel raster as in E-UTRA Rel-9 is applied. Hence the channel raster is 100 kHz for all bands, which means that the carrier centre frequency must be an integer multiple of 100 kHz.

## 5.4.4 Carrier frequency and EARFCN

The carrier frequency in the uplink and downlink is designated by the E-UTRA Absolute Radio Frequency Channel Number (EARFCN) in the range 0 - 65535. The relation between EARFCN and the carrier frequency in MHz for the downlink is given by the following equation, where $F_{DL_low}$ and $N_{Offs-DL}$ are given in table 5.4.4-1 and $N_{DL}$ is the downlink EARFCN.

$$F_{DL} = F_{DL_low} + 0.1(N_{DL} - N_{Offs-DL})$$

The relation between EARFCN and the carrier frequency in MHz for the uplink is given by the following equation where $F_{UL_low}$ and $N_{Offs-UL}$ are given in table 5.4.4-1 and $N_{UL}$ is the uplink EARFCN.

$$F_{UL} = F_{UL_low} + 0.1(N_{UL} - N_{Offs-UL})$$

**Table 5.4.4-1 E-UTRA channel numbers**

| Band | Downlink | | | Uplink | | |
|---|---|---|---|---|---|---|
| | $F_{DL_low}$ (MHz) | $N_{Offs-DL}$ | Range of $N_{DL}$ | $F_{UL_low}$ (MHz) | $N_{Offs-UL}$ | Range of $N_{UL}$ |
| 1 | 2110 | 0 | 0 – 599 | 1920 | 18000 | 18000 – 18599 |
| 2 | 1930 | 600 | 600 – 1199 | 1850 | 18600 | 18600 – 19199 |
| 3 | 1805 | 1200 | 1200 – 1949 | 1710 | 19200 | 19200 – 19949 |
| 4 | 2110 | 1950 | 1950 – 2399 | 1710 | 19950 | 19950 – 20399 |
| 5 | 869 | 2400 | 2400 – 2649 | 824 | 20400 | 20400 – 20649 |
| 6 | 875 | 2650 | 2650 – 2749 | 830 | 20650 | 20650 – 20749 |
| 7 | 2620 | 2750 | 2750 – 3449 | 2500 | 20750 | 20750 – 20449 |
| 8 | 925 | 3450 | 3450 – 3799 | 880 | 21450 | 21450 – 21799 |
| 9 | 1844.9 | 3800 | 3800 – 4149 | 1749.9 | 21800 | 21800 – 22149 |
| 10 | 2110 | 4150 | 4150 – 4749 | 1710 | 22150 | 22150 – 22749 |
| 11 | 1475.9 | 4750 | 4750 – 4949 | 1427.9 | 22750 | 22750 – 22949 |
| 12 | 729 | 5010 | 5010 – 5179 | 699 | 23010 | 23010 – 23179 |
| 13 | 746 | 5180 | 5180 – 5279 | 777 | 23180 | 23180 – 23279 |
| 14 | 758 | 5280 | 5280 – 5379 | 788 | 23280 | 23280 – 23379 |
| … | | | | | | |
| 17 | 734 | 5730 | 5730 – 5849 | 704 | 23730 | 23730 – 23849 |
| 18 | 860 | 5850 | 5850 – 5999 | 815 | 23850 | 23850 – 23999 |
| 19 | 875 | 6000 | 6000 – 6149 | 830 | 24000 | 24000 – 24149 |
| 20 | 791 | 6150 | 6150 – 6449 | 832 | 24150 | 24150 – 24449 |
| 21 | 1495.9 | 6450 | 6450 – 6599 | 1447.9 | 24450 | 24450 – 24599 |
| 22 | 3510 | 6600 | 6600 – 7399 | 3410 | 24600 | 24600 – 25399 |
| 23 | 2180 | 7500 | 7500 – 7699 | 2000 | 25500 | 25500 – 25699 |
| 24 | 1525 | 7700 | 7700 - 8039 | 1626.5 | 25700 | 25700 – 26039 |
| 25 | 1930 | 8040 | 8040 - 8689 | 1850 | 26040 | 26040 - 26689 |
| ... | | | | | | |
| 33 | 1900 | 36000 | 36000 –36199 | 1900 | 36000 | 36000 – 36199 |
| 34 | 2010 | 36200 | 36200 –36349 | 2010 | 36200 | 36200 – 36349 |
| 35 | 1850 | 36350 | 36350 –36949 | 1850 | 36350 | 36350 – 36949 |
| 36 | 1930 | 36950 | 36950 –37549 | 1930 | 36950 | 36950 – 37549 |
| 37 | 1910 | 37550 | 37550 –37749 | 1910 | 37550 | 37550 – 37749 |
| 38 | 2570 | 37750 | 37750 –38249 | 2570 | 37750 | 37750 – 38249 |
| 39 | 1880 | 38250 | 38250 –38649 | 1880 | 38250 | 38250 – 38649 |
| 40 | 2300 | 38650 | 38650 –39649 | 2300 | 38650 | 38650 – 39649 |
| 41 | 2496 | 39650 | 39650 - 41589 | 2496 | 39650 | 39650 - 41589 |
| 42 | 3400 | 41590 | 41590 – 43589 | 3400 | 41590 | 41590 – 43589 |
| 43 | 3600 | 43590 | 43590 – 45589 | 3600 | 43590 | 43590 – 45589 |

Note: The channel numbers that designate carrier frequencies so close to the operating band edges that the carrier extends beyond the operating band edge shall not be used. This implies that the first 7, 15, 25, 50, 75 and 100 channel numbers at the lower operating band edge and the last 6, 14, 24, 49, 74 and 99 channel numbers at the upper operating band edge shall not be used for channel bandwidths of 1.4, 3, 5, 10, 15 and 20 MHz respectively.

# 6 Transmitter Characteristics

## 6.1 General

Unless otherwise stated, the transmitter characteristics are specified at the antenna connector of the UE with a single transmit antenna. For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed.

The transient periods due to power steps, OFF/ON and ON/OFF transitions could occur at slot or symbol boundary with transients, on one or both sides of the boundary. The measurement period and whether to exclude the transient periods are specified in the respective sections below.

Unless otherwise stated, the Test Equipment shall be synchronised to the Uplink signal for measurement of TDD transmitter characteristics.

## 6.2 Transmit power

### 6.2.1 Void

Editor's note: this "void" section was introduced because TS 36.101 v8.1.0 also contains a "void" sub-clause with in the transmit power clause 6.2, and there is a strong desire in RAN5 to keep the test cases clauses numbering matching their specific core requirements as much as possible.

### 6.2.2 UE Maximum Output Power

#### 6.2.2.1 Test purpose

To verify that the error of the UE maximum output power does not exceed the range prescribed by the specified nominal maximum output power and tolerance.

An excess maximum output power has the possibility to interfere to other channels or other systems. A small maximum output power decreases the coverage area.

#### 6.2.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

#### 6.2.2.3 Minimum conformance requirements

The following UE Power Classes defines the maximum output power for any transmission bandwidth within the channel bandwidth. The period of measurement shall be at least one sub frame (1ms).

**Table 6.2.2.3-1: UE Power Class**

| EUTRA band | Class 1 (dBm) | Tolerance (dB) | Class 2 (dBm) | Tolerance (dB) | Class 3 (dBm) | Tolerance (dB) | Class 4 (dBm) | Tolerance (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | 23 | ±2 | | |
| 2 | | | | | 23 | ±2[2] | | |
| 3 | | | | | 23 | ±2[2] | | |
| 4 | | | | | 23 | ±2 | | |
| 5 | | | | | 23 | ±2 | | |
| 6 | | | | | 23 | ±2 | | |
| 7 | | | | | 23 | ±2[2] | | |
| 8 | | | | | 23 | ±2[2] | | |
| 9 | | | | | 23 | ±2 | | |
| 10 | | | | | 23 | ±2 | | |
| 11 | | | | | 23 | ±2 | | |
| 12 | | | | | 23 | ±2[2] | | |
| 13 | | | | | 23 | ±2 | | |
| 14 | | | | | 23 | ±2 | | |
| … | | | | | | | | |
| 17 | | | | | 23 | ±2 | | |
| 18 | | | | | 23 | ±2 | | |
| 19 | | | | | 23 | ±2 | | |
| 20 | | | | | 23 | ±2[2] | | |
| 21 | | | | | 23 | ±2 | | |
| 22 | | | | | 23 | +2/-3.5[2] | | |
| 23 | | | | | 23 | ±2 | | |
| 24 | | | | | 23 | ±2 | | |
| 25 | | | | | 23 | ±2[2] | | |
| ... | | | | | | | | |
| 33 | | | | | 23 | ±2 | | |
| 34 | | | | | 23 | ±2 | | |
| 35 | | | | | 23 | ±2 | | |
| 36 | | | | | 23 | ±2 | | |
| 37 | | | | | 23 | ±2 | | |
| 38 | | | | | 23 | ±2 | | |
| 39 | | | | | 23 | ±2 | | |
| 40 | | | | | 23 | ±2 | | |
| 41 | | | | | 23 | ±2 | | |
| 42 | | | | | 23 | +2/-3 | | |
| 43 | | | | | 23 | +2/-3 | | |
| … | | | | | | | | |

Note 1: The above tolerances are applicable for UE(s) that support up to 4 E-UTRA operating bands. For UE(s) that support 5 or more E-UTRA bands the maximum output power is expected to decrease with each additional band and is FFS

Note 2: For transmission bandwidths (Figure 5.4.2-1) confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB

Note 3: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance

Note 4: For the UE which supports both Band 11 and Band 21 operating frequencies, the tolerance is FFS.

The normative reference for this requirement is TS 36.101 clause 6.2.2.

## 6.2.2.4 Test description

### 6.2.2.4.1 Initial condition

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.2.2.4.1-1. The details of the uplink reference

measurement channels (RMCs) are specified in Annexe A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.2.2.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Max UE output power testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 1 | 1 |
| 1.4MHz | | QPSK | 5 | 5 |
| 3MHz | | QPSK | 1 | 1 |
| 3MHz | | QPSK | 4 | 4 |
| 5MHz | | QPSK | 1 | 1 |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 1 | 1 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 1 | 1 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 1 | 1 |
| 20MHz | | QPSK | 18 | 18 |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: For E-UTRA bands not applied with Note 2 in Table 6.2.2.3-1:

- The 1 RB allocation shall be tested at RB#0 for low and mid range, RB #max for high range test frequency.
- The $RB_{start}$ of non-1RB allocation shall be RB #0 for low and mid range, RB# (max +1 - RB allocation) for high range test frequency.

Note 3: For E-UTRA bands applied with Note 2 in Table 6.2.2.3-1:

- If the test channel bandwidth is larger than 4MHz, then the 1 RB allocation shall be tested at both RB #0 and RB #max.
- If the test channel bandwidth is smaller or equal to 4MHz, then the 1 RB allocation shall be tested at RB #0.
- If the test channel bandwidth = ($F_{UL_high}$ - $F_{UL_low}$) specified by the operating band, then only one frequency range shall be tested and the 1 RB allocation shall be tested at RB #0, RB # $\left\lceil N_{RB}^{UL} / 2 \right\rceil$ and RB #max.
- For non-1RB allocation, test frequency is middle range, and the $RB_{start}$ shall be RB #0.

1. Connect the SS to the UE antenna connectors as shown in TS 36.508[7] Annex A Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channel is set according to Table 6.2.2.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RFaccording to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.2.2.4.3.

### 6.2.2.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.2.2.1.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in every uplink scheduling information to the UE; allow at least 200ms for the UE to reach $P_{UMAX}$ level.

3. Measure the mean power of the UE in the channel bandwidth of the radio access mode. The period of measurement shall be at least the continuous duration of one sub-frame (1ms). For TDD slots with transient periods are not under test.

#### 6.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.2.2.5 Test requirements

The maximum output power, derived in step 3 shall be within the range prescribed by the nominal maximum output power and tolerance in Table 6.2.2.5-1.

**Table 6.2.2.5-1: UE Power Class test requirements**

| EUTRA band | Class 1 (dBm) | Tolerance (dB) | Class 2 (dBm) | Tolerance (dB) | Class 3 (dBm) | Tolerance (dB) | Class 4 (dBm) | Tolerance (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | 23 | ±2.7 | | |
| 2 | | | | | 23 | ±2.7[2] | | |
| 3 | | | | | 23 | ±2.7[2] | | |
| 4 | | | | | 23 | ±2.7 | | |
| 5 | | | | | 23 | ±2.7 | | |
| 6 | | | | | 23 | ±2.7 | | |
| 7 | | | | | 23 | ±2.7[2] | | |
| 8 | | | | | 23 | ±2.7[2] | | |
| 9 | | | | | 23 | ±2.7 | | |
| 10 | | | | | 23 | ±2.7 | | |
| 11 | | | | | 23 | ±2.7 | | |
| 12 | | | | | 23 | ±2.7[2] | | |
| 13 | | | | | 23 | ±2.7 | | |
| 14 | | | | | 23 | ±2.7 | | |
| … | | | | | | | | |
| 17 | | | | | 23 | ±2.7 | | |
| 18 | | | | | 23 | ±2.7 | | |
| 19 | | | | | 23 | ±2.7 | | |
| 20 | | | | | 23 | ±2.7[2] | | |
| 21 | | | | | 23 | ±2.7 | | |
| 22 | | | | | 23 | +3.0/-4.5 | | |
| 23 | | | | | 23 | ±2.7 | | |
| 24 | | | | | 23 | ±2.7 | | |
| 25 | | | | | 23 | ±2.7[2] | | |
| ... | | | | | | | | |
| 33 | | | | | 23 | ±2.7 | | |
| 34 | | | | | 23 | ±2.7 | | |
| 35 | | | | | 23 | ±2.7 | | |
| 36 | | | | | 23 | ±2.7 | | |
| 37 | | | | | 23 | ±2.7 | | |
| 38 | | | | | 23 | ±2.7 | | |
| 39 | | | | | 23 | ±2.7 | | |
| 40 | | | | | 23 | ±2.7 | | |
| 41 | | | | | 23 | ±2.7 | | |
| 42 | | | | | 23 | +3.0/-4.0 | | |
| 43 | | | | | 23 | +3.0/-4.0 | | |
| … | | | | | | | | |

Note 1: The above tolerances are applicable for UE(s) that support up to 4 E-UTRA operating bands. For UE(s) that support 5 or more E-UTRA bands the maximum output power is expected to decrease with each additional band and is FFS

Note 2: For transmission bandwidths (Figure 5.4.2-1, Table 5.4.4-1) confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB

Note 3: For the UE which supports both Band 11 and Band 21 operating frequencies, the tolerance is FFS.

## 6.2.2A UE Maximum Output Power for intra-band contiguous CA

Editor's notes:

The following items are missing or incomplete:

- Initial conditions test setup for SCC is FFS

- References need to be updated

- Test State for CA RF testing is FFS

- Test procedure

- Message Contents

- Test tolerances

- Connection diagram is missing

- Test tolerances are incomplete

### 6.2.2A.1 Test purpose

To verify that the error of UE maximum output power in intra-band contiguous carrier aggregation does not exceed the range prescribed by the specified CA Power Class and tolerance.

An excess maximum output power has the possibility to interfere to other channels or other systems. A small maximum output power decreases the coverage area.

### 6.2.2A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

### 6.2.2A.3 Minimum conformance requirements

The following UE Power Classes define the maximum output power for any transmission bandwidth within the aggregated channel bandwidth.

- For CA Bandwidth Class A, the requirements in Clause 6.2.2 apply.
- For CA Bandwidth Class C, the maximum output power is specified in Table 6.2.2A.3-1.

The maximum output power is measured as the sum of the maximum output power at each UE antenna connector. The period of measurement shall be at least one sub frame (1ms).

**Table 6.2.2A.3-1: CA UE Power Class**

| EUTRA band | Class 1 (dBm) | Tolerance (dB) | Class 2 (dBm) | Tolerance (dB) | Class 3 (dBm) | Tolerance (dB) | Class 4 (dBm) | Tolerance (dB) |
|---|---|---|---|---|---|---|---|---|
| CA_1C | | | | | 23 | +2/-2 | | |
| CA_40C | | | | | 23 | +2/[-2] | | |

Note 1: The above tolerances are applicable for UE(s) that support up to 4 E-UTRA operating bands. For UE(s) that support 5 or more E-UTRA bands the maximum output power is expected to decrease with each additional band and is FFS

Note 2: For transmission bandwidths (Figure 5.4.2-1) confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB

Note 3: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance

Note 4: For intra-band contiguous carrier aggregation the maximum power requirement should apply to the total transmitted power over all component carriers (per UE).

The normative reference for this requirement is in TS 36.101 [2] clause 6.2.2A.

### 6.2.2A.4 Test description

#### 6.2.2A.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and CC combinations based on E-UTRA CA configurations specified in table 5.4.2A.1-1. All of these configurations shall be tested with applicable test parameters for each CA Configuration, and are shown in table 6.2.2A.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexe A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.2.2A.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | | | NC, TL/VL, TL/VH, TH/VL, TH/VH | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause [4.3.1] for different CA bandwidth classes. | | | C: Low and High range | | | | | |
| Test CC Combination setting ($N_{RB_agg}$) as specified in subclause 5.4.2A.1 for the CA Configuration | | | Lowest $N_{RB_agg}$<br>Highest $N_{RB_agg}$ | | | | | |
| Test Parameters for CA Configurations | | | | | | | | |
| **CA Configuration / $N_{RB_agg}$** | | **DL Allocation** | **CC MOD** | **UL Allocation** | | | | |
| **PCC $N_{RB}$** | **SCCs $N_{RB}$** | **PCC & SCC RB allocation** | | **$N_{RB_alloc}$** | **PCC & SCC RB allocations ($L_{CRB}$ @ $RB_{start}$)** | | | |
| 75 | 75 | N/A for this test | QPSK | 1 | P_1@0 | - | - | - |
| 75 | 75 | | QPSK | 16 | P_16@0 | - | - | - |
| 100 | 50 | | QPSK | 1 | P_1@0 | - | - | - |
| 100 | 50 | | QPSK | 12 | P_12@0 | - | - | - |
| 100 | 100 | | QPSK | 1 | P_1@0 | - | - | - |
| 100 | 100 | | QPSK | 18 | P_18@0 | - | - | - |
| Note 1: CA Configuration Test CC Combination settings are checked separately for each CA Configuration, which applicable aggregated channel bandwidths are specified in Table 5.4.2A.1-1 | | | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508[7] Annex A Figure [TBD].
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channel is set according to Table 6.2.2A.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in [State 3A-RF] according to TS 36.508 [7] clause [5.2A.2]. Message contents are defined in clause 6.2.2A.4.3.

### 6.2.2A.4.2 Test procedure

FFS

### 6.2.2A.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause [FFS].

## 6.2.2A.5 Test Requirements

The maximum output power for the CA configuration, derived in step [TBD] shall be within the range prescribed by the CA UE Power Class and tolerance in Table 6.2.2A.5-1.

**Table 6.2.2A.5-1: CA UE Power Class test requirements**

| CA Conf | Class 1 (dBm) | Tolerance (dB) | Class 2 (dBm) | Tolerance (dB) | Class 3 (dBm) | Tolerance (dB) | Class 4 (dBm) | Tolerance (dB) |
|---|---|---|---|---|---|---|---|---|
| CA_1C | | | | | 23 | [±2.7] | | |
| CA_40C | | | | | 23 | [±2.7] | | |

Note 1: The above tolerances are applicable for UE(s) that support up to 4 E-UTRA operating bands. For UE(s) that support 5 or more E-UTRA bands the maximum output power is expected to decrease with each additional band and is FFS

Note 2: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance

Note 3: For intra-band contiguous carrier aggregation the maximum power requirement should apply to the total transmitted power over all component carriers (per UE).

## 6.2.2B UE Maximum Output Power for UL-MIMO

Editor's notes: the following items are missing:

- Test Tolerance

- Connection Diagram

- Message Contents

### 6.2.2B.1 Test purpose

To verify that the error of UE maximum output power in UL MIMO does not exceed the range prescribed by the specified UL MIMO Power Class and tolerance.

An excess maximum output power has the possibility to interfere to other channels or other systems. A small maximum output power decreases the coverage area.

### 6.2.2B.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support UL MIMO.

### 6.2.2B.3 Minimum conformance requirements

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the maximum output power for any transmission bandwidth within the channel bandwidth is specified in Table 6.2.2B.3-1 with the UL-MIMO configurations specified in Table 6.2.2B.3-2. The maximum output power is measured as the sum of the maximum output power at each UE antenna connector. The period of measurement shall be at least one sub frame (1ms).

**Table 6.2.2B.3-1: UE Power Class for UL-MIMO in closed loop spatial multiplexing scheme**

| EUTRA band | Class 1 (dBm) | Tolerance (dB) | Class 2 (dBm) | Tolerance (dB) | Class 3 (dBm) | Tolerance (dB) | Class 4 (dBm) | Tolerance (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | 23 | +2/-3 | | |
| 2 | | | | | 23 | +2/-3[2] | | |
| 3 | | | | | 23 | +2/-3[2] | | |
| 4 | | | | | 23 | +2/-3 | | |
| 5 | | | | | 23 | +2/-3 | | |
| 6 | | | | | 23 | +2/-3 | | |
| 7 | | | | | 23 | +2/-3[2] | | |
| 8 | | | | | 23 | +2/-3[2] | | |
| 9 | | | | | 23 | +2/-3 | | |
| 10 | | | | | 23 | +2/-3 | | |
| 11 | | | | | 23 | +2/-3 | | |
| 12 | | | | | 23 | +2/-3[2] | | |
| 13 | | | | | 23 | +2/-3 | | |
| 14 | | | | | 23 | +2/-3 | | |
| | | | | | | | | |
| 17 | | | | | 23 | +2/-3 | | |
| 18 | | | | | 23 | +2/-3 | | |
| 19 | | | | | 23 | +2/-3 | | |
| 20 | | | | | 23 | +2/-3[2] | | |
| 21 | | | | | 23 | +2/-3 | | |
| … | | | | | | | | |
| 23 | | | | | 23 | +2/-3 | | |
| 24 | | | | | 23 | +2/-3 | | |
| 25 | | | | | 23 | +2/-3[2] | | |
| … | | | | | | | | |
| 33 | | | | | 23 | +2/-3 | | |
| 34 | | | | | 23 | +2/-3 | | |
| 35 | | | | | 23 | +2/-3 | | |
| 36 | | | | | 23 | +2/-3 | | |
| 37 | | | | | 23 | +2/-3 | | |
| 38 | | | | | 23 | +2/-3 | | |
| 39 | | | | | 23 | +2/-3 | | |
| 40 | | | | | 23 | +2/-3 | | |
| 41 | | | | | 23 | +2/-3 | | |
| 42 | | | | | 23 | +2/-3 | | |
| 43 | | | | | 23 | +2/-3 | | |

Note 1: The above tolerances are applicable for UE(s) that support up to 4 E-UTRA operating bands. For UE(s) that support 5 or more E-UTRA bands the maximum output power is expected to decrease with each additional band and is FFS

Note 2: For transmission bandwidths (Figure 5.6-1) confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB

Note 3: For the UE which supports both Band 11 and Band 21 operating frequencies, the tolerance is FFS.

Note 4: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance

**Table 6.2.2B.3-2: UL-MIMO configuration in closed-loop spatial multiplexing scheme**

| Transmission mode | DCI format | Codebook Index |
|---|---|---|
| Mode 2 | DCI format 4 | Codebook index 0 |

The normative reference for this requirement is TS 36.101 clause 6.2.2B.

## 6.2.2B.4 Test description

### 6.2.2B.4.1 Initial condition

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.2.2B.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexe A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.2.2B.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Max UE output power testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 1 | 1 |
| 1.4MHz | | QPSK | 5 | 5 |
| 3MHz | | QPSK | 1 | 1 |
| 3MHz | | QPSK | 4 | 4 |
| 5MHz | | QPSK | 1 | 1 |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 1 | 1 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 1 | 1 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 1 | 1 |
| 20MHz | | QPSK | 18 | 18 |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: For E-UTRA bands not applied with Note 2 in Table 6.2.2B.3-1:

- The 1 RB allocation shall be tested at RB#0 for low and mid range, RB #max for high range test frequency.
- The starting resource block of non-1RB allocation shall be RB #0 for low and mid range, RB# (max +1 - RB allocation) for high range test frequency.

Note 3: For E-UTRA bands applied with Note 2 in Table 6.2.2B.3-1:

- If the test channel bandwidth is larger than 4MHz, then the 1 RB allocation shall be tested at both RB #0 and RB #max.
- If the test channel bandwidth is smaller or equal to 4MHz, then the 1 RB allocation shall be tested at RB #0.
- If the test channel bandwidth = ($F_{UL_high}$ - $F_{UL_low}$) specified by the operating band, then only one frequency range shall be tested and the 1 RB allocation shall be tested at RB #0, RB #$\left\lceil N_{RB}^{UL} / 2 \right\rceil$ and RB #max.
- For non-1RB allocation, test frequency is middle range, and the starting resource block shall be RB #0.

1. Connect the SS to the UE antenna connectors as shown in TS 36.508[7] Annex A Figure [TBD].
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channel is set according to Table 6.2.2B.4.1-1.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RFaccording to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.2.2B.4.3.

### 6.2.2B.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 4 for C_RNTI to schedule the UL RMC according to Table 6.2.2B.1.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control “up” commands in every uplink scheduling information to the UE; allow at least 200ms for the UE to reach $P_{UMAX}$ level.

3. Measure the mean sum power of the UE in the channel bandwidth of the radio access mode. The period of measurement shall be at least the continuous duration of one sub-frame (1ms). For TDD slots with transient periods are not under test.

### 6.2.2B.4.3 Message contents

## 6.2.2B.5 Test requirements

The maximum output power, derived in step 3 shall be within the range prescribed by the nominal maximum output power and tolerance in Table 6.2.2B.5-1.

**Table 6.2.2B.5-1: UE Power Class for UL-MIMO in closed loop spatial multiplexing scheme**

| EUTRA band | Class 1 (dBm) | Tolerance (dB) | Class 2 (dBm) | Tolerance (dB) | Class 3 (dBm) | Tolerance (dB) | Class 4 (dBm) | Tolerance (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | 23 | (+2/-3)±TT | | |
| 2 | | | | | 23 | (+2/-3)±TT | | |
| 3 | | | | | 23 | (+2/-3)±TT | | |
| 4 | | | | | 23 | (+2/-3)±TT | | |
| 5 | | | | | 23 | (+2/-3)±TT | | |
| 6 | | | | | 23 | (+2/-3)±TT | | |
| 7 | | | | | 23 | (+2/-3)±TT | | |
| 8 | | | | | 23 | (+2/-3)±TT | | |
| 9 | | | | | 23 | (+2/-3)±TT | | |
| 10 | | | | | 23 | (+2/-3)±TT | | |
| 11 | | | | | 23 | (+2/-3)±TT | | |
| 12 | | | | | 23 | (+2/-3)±TT | | |
| 13 | | | | | 23 | (+2/-3)±TT | | |
| 14 | | | | | 23 | (+2/-3)±TT | | |
| | | | | | | | | |
| 17 | | | | | 23 | (+2/-3)±TT | | |
| 18 | | | | | 23 | (+2/-3)±TT | | |
| 19 | | | | | 23 | (+2/-3)±TT | | |
| 20 | | | | | 23 | (+2/-3)±TT | | |
| 21 | | | | | 23 | (+2/-3)±TT | | |
| … | | | | | | | | |
| 23 | | | | | 23 | (+2/-3)±TT | | |
| 24 | | | | | 23 | (+2/-3)±TT | | |
| 25 | | | | | 23 | (+2/-3)±TT | | |
| … | | | | | | | | |
| 33 | | | | | 23 | (+2/-3)±TT | | |
| 34 | | | | | 23 | (+2/-3)±TT | | |
| 35 | | | | | 23 | (+2/-3)±TT | | |
| 36 | | | | | 23 | (+2/-3)±TT | | |
| 37 | | | | | 23 | (+2/-3)±TT | | |
| 38 | | | | | 23 | (+2/-3)±TT | | |
| 39 | | | | | 23 | (+2/-3)±TT | | |
| 40 | | | | | 23 | (+2/-3)±TT | | |
| 41 | | | | | 23 | (+2/-3)±TT | | |
| 42 | | | | | 23 | (+2/-3)±TT | | |
| 43 | | | | | 23 | (+2/-3)±TT | | |

Note 1: The above tolerances are applicable for UE(s) that support up to 4 E-UTRA operating bands. For UE(s) that support 5 or more E-UTRA bands the maximum output power is expected to decrease with each additional band and is FFS

Note 2: For transmission bandwidths (Figure 5.6-1) confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB

Note 3: For the UE which supports both Band 11 and Band 21 operating frequencies, the tolerance is FFS.

Note 4: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance

## 6.2.3 Maximum Power Reduction (MPR)

### 6.2.3.1 Test purpose

The number of RB identified in Table 6.2.2.3-1 is based on meeting the requirements for adjacent channel leakage ratio and the maximum power reduction (MPR) due to Cubic Metric (CM).

Simple scaling can be used to derive the requirement for other bandwidth based on the previously agreed value for 5MHz channel bandwidth.

### 6.2.3.2 Test applicability

The requirements of this test apply in test cases 6.6.2.3 Adjacent Channel Leakage power Ratio to all types of E-UTRA UE release 8 and forward.

### 6.2.3.3 Minimum conformance requirements

For UE Power Class 3, the allowed Maximum Power Reduction (MPR) for the maximum output power in Table 6.2.2.3-1 due to higher order modulation and transmit bandwidth configuration (resource blocks) is specified in Table 6.2.3.3-1.

**Table 6.2.3.3-1: Maximum Power Reduction (MPR) for Power Class 3**

| Modulation | Channel bandwidth / Transmission bandwidth configuration [RB] | | | | | | MPR (dB) |
|---|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | |
| QPSK | > 5 | > 4 | > 8 | > 12 | > 16 | > 18 | ≤ 1 |
| 16 QAM | ≤ 5 | ≤ 4 | ≤ 8 | ≤ 12 | ≤ 16 | ≤ 18 | ≤ 1 |
| 16 QAM | > 5 | > 4 | > 8 | > 12 | > 16 | > 18 | ≤ 2 |

For the UE maximum output power modified by MPR, the power limits specified in subclause 6.2.5.3 apply.

The normative reference for this requirement is TS 36.101 clause 6.2.3.

### 6.2.3.4 Test description

#### 6.2.3.4.1 Initial condition

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.2.3.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.2.3.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, 10MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Maximum Power Reduction (MPR) test case | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 5 | 5 |
| 1.4MHz | | QPSK | 6 | 6 |
| 1.4MHz | | 16QAM | 5 | 5 |
| 1.4MHz | | 16QAM | 6 | 6 |
| 3.0MHz | | QPSK | 4 | 4 |
| 3.0MHz | | QPSK | 15 | 15 |
| 3.0MHz | | 16QAM | 4 | 4 |
| 3.0MHz | | 16QAM | 15 | 15 |
| 5MHz | | QPSK | 8 | 8 |
| 5MHz | | QPSK | 25 | 25 |
| 5MHz | | 16QAM | 8 | 8 |
| 5MHz | | 16QAM | 25 | 25 |
| 10MHz | | QPSK | 12 | 12 |
| 10MHz | | QPSK | 50 | 50 |
| 10MHz | | 16QAM | 12 | 12 |
| 10MHz | | 16QAM | 50 (Note 3) | 50 (Note 3) |
| 15MHz | | QPSK | 16 | 16 |
| 15MHz | | QPSK | 75 | 75 |
| 15MHz | | 16QAM | 16 | 16 |
| 15MHz | | 16QAM | 75 (Note 3) | 75 (Note 3) |
| 20MHz | | QPSK | 18 | 18 |
| 20MHz | | QPSK | 100 | 100 |
| 20MHz | | 16QAM | 18 | 18 |
| 20MHz | | 16QAM | 100 (Note 3) | 100 (Note 3) |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |
| Note 2: The $RB_{start}$ of partial RB allocation shall be RB#0 and RB# (max + 1 – RB allocation) of the channel bandwidth. | | | | |
| Note 3: Applies only for UE-Categories 2-5 | | | | |

1. Connect the SS and interfering sources to the UE antenna connectors as shown in TS 36.508[7] Annex A Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channels are set according to Table 6.2.3.4.1-1.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.2.2.4.3.

### 6.2.3.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.2.3.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE to ensure that the UE transmits at $P_{UMAX}$ level.

3. Measure the mean power of the UE in the channel bandwidth of the radio access mode. The period of measurement shall be at least the continuous duration one sub-frame (1ms). For TDD slots with transient periods are not under test.

#### 6.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.2.3.5 Test requirements

The maximum output power, derived in step 2 shall be within the range prescribed by the nominal maximum output power and tolerance in Table 6.2.3.5-1.

**Table 6.2.3.5-1: UE Power Class test requirements**

| E-UTRA Band | Class 1 (dBm) | Tol. (Db) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | QPSK partial RB allocation Tol. (dB)) | QPSK full RB allocation Tol. (dB)) | 16QAM partial RB allocation Tol. (dB) | 16QAM full RB allocation Tol. (dB) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 2 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| 3 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| 4 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 5 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 6 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 7 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| 8 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| 9 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 10 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 11 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 12 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| 13 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 14 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| … | | | | | | | | | |
| 17 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 18 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 19 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 20 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| 21 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 22 | | | | | 23 | +3.0/-4.5 | +3.0/-5.5 | +3.0/-5.5 | +3.0/-6.5 |
| 23 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 24 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 25 | | | | | 23 | ±2.7[1] | +2.7 / -3.7[1,2] | +2.7 / -3.7[1,2] | +2.7 / -4.7[1,2] |
| ... | | | | | | | | | |
| 33 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 34 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 35 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 36 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 37 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 38 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 39 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 40 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 41 | | | | | 23 | ±2.7 | +2.7 / -3.7 | +2.7 / -3.7 | +2.7 / -4.7 |
| 42 | | | | | 23 | +3.0 / -4.0 | +3.0 / -5.0 | +3.0 / -5.0 | +3.0 / -6.0 |
| 43 | | | | | 23 | +3.0 / -4.0 | +3.0 / -5.0 | +3.0 / -5.0 | +3.0 / -6.0 |
| Note 1: For transmission configurations (Figure 5.4.2-1) confined within FUL_low and FUL_low + 4 MHz or FUL_high – 4 MHz and FUL_high, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB.<br>Note 2: For the UE maximum output power modified by MPR, the power limits specified in Table 6.2.5.3-1 apply | | | | | | | | | |

## 6.2.3A Maximum Power Reduction (MPR) for CA

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The initial conditions for this test are undefined
- The Test Procedure for this test is undefined
- Message Contents for this test are undefined
- The Test Requirements for this test are undefined
- The Test Tolerances applicable to this test are undefined

### 6.2.3A.1 Test purpose

The number of RB identified in Table 6.2.3A.3-1 is based on meeting the requirements for Adjacent Channel Leakage power Ratio (ACLR) for CA and the maximum power reduction (MPR) for intra-band contiguous CA Bandwidth Class C due to Cubic Metric (CM).

### 6.2.3A.2 Test applicability

The requirements of this test apply in test case 6.6.2.3A Adjacent Channel Leakage power Ratio (ACLR) for CA to all types of E-UTRA UE release 10 and forward that support CA.

### 6.2.3A.3 Minimum conformance requirements

The following is specified for maximum power reduction (MPR) for CA.

- For CA Bandwidth Class A (Table 5.4.2A-1), the requirements in Clause 6.2.3 apply.

- For intra-band contiguous CA Bandwidth Class C, the allowed Maximum Power Reduction (MPR) for the maximum output power in Table 6.2.2A.3-1 due to higher order modulation and contiguously aggregated transmit bandwidth configuration (resource blocks) is specified in Table 6.2.3A.3-1. In case the modulation format is different on different component carriers then the MPR is determined by the rules applied to higher order of those modulations.

**Table 6.2.3A.3-1: Maximum Power Reduction (MPR) for Power Class 3**

| Modulation | CA bandwidth Class C | | | MPR (dB) |
|---|---|---|---|---|
| | 50 RB + 100 RB | 75 RB + 75 RB | 100 RB + 100 RB | |
| QPSK | > 12 and ≤ 50 | > 16 and ≤ 75 | > 18 and ≤ 100 | ≤ 1 |
| QPSK | > 50 | > 75 | > 100 | ≤ 2 |
| 16 QAM | ≤ 12 | ≤ 16 | ≤ 18 | ≤ 1 |
| 16 QAM | > 12 and ≤ 50 | > 16 and ≤ 75 | > 18 and ≤ 100 | ≤ 2 |
| 16 QAM | > 50 | > 75 | > 100 | ≤ 3 |

For intra-band contiguous CA Bandwidth Class C the allowed Maximum Power Reduction (MPR) for the maximum output power in Table 6.2.2A.3-1 due to multi cluster transmission is specified as follows:

$$MPR = CEIL \{M_{A,} 0.5\}$$

Where $M_A$ is defined as follows:

$M_A = 8.2$ ; $0 \leq A < 0.025$,

$9.3 - 42A$ ; $0.025 \leq A < 0.05$,

$8 - 16A$ ; $0.05 \leq A < 0.25$,

$4.83 - 3.33A$ ; $0.25 \leq A \leq 0.4$,

$3.83 - 0.83A$ ; $0.4 \leq A \leq 1$

Where

$A = N_{RB_alloc} / N_{RB_agg.}$

$CEIL\{M_{A,} 0.5\}$ means rounding upwards to closest 0.5dB, i.e. MPR$\in$[3.0, 3.5 4.0 4.5 5.0 5.5 6.0 6.5 7.0 7.5]

For the UE maximum output power modified by MPR, the power limits specified in sub-clause 6.2.5A apply.

The normative reference for this requirement is in TS 36.101 [2] clause 6.2.3A.

### 6.2.3A.4 Test description

#### 6.2.3A.4.1 Initial condition

#### 6.2.3A.4.2 Test procedure

#### 6.2.3A.4.3 Message contents

### 6.2.3A.5 Test Requirements

## 6.2.4 Additional Maximum Power Reduction (A-MPR)

### 6.2.4.1 Test purpose

Additional ACLR and spectrum emission requirements can be signalled by the network to indicate that the UE shall also meet additional requirements in a specific deployment scenario. To meet these additional requirements, Additional Maximum Power Reduction A-MPR is allowed for the output power as specified in Table 6.2.2.3-1. Unless stated otherwise, an A-MPR of 0 dB shall be used.

### 6.2.4.2 Test applicability

The requirements of this test apply in test case 6.6.2.2 Additional Spectrum Emission Mask for network signalled values NS_03, NS_04, NS_06 and NS_07 to all types of E-UTRA UE release 8 and forward.

The requirements of this test apply in test case 6.6.3.3 Additional Spurious Emissions for network signalled values NS_05, NS_07 and NS_08, NS_09 to all types of E-UTRA UE release 8 and forward.

### 6.2.4.3 Minimum conformance requirements

For UE Power Class 3 the specific requirements and identified sub-clauses are specified in Table 6.2.4.3-1 along with the allowed A-MPR values that may be used to meet these requirements. The allowed A-MPR values specified below in Table 6.2.4.3.-1 and 6.2.4.3-2 are in addition to the allowed MPR requirements specified in clause 6.2.3. For the UE maximum output power modified by A-MPR, the power limits specified in clause 6.2.5 apply.

**Table 6.2.4.3-1: Additional Maximum Power Reduction (A-MPR) / Spectrum Emission requirements**

| Network Signalling value | Requirements (sub-clause) | E-UTRA Band | Channel bandwidth (MHz) | Resources Blocks ($N_{RB}$) | A-MPR (dB) |
|---|---|---|---|---|---|
| NS_01 | 6.6.2.1.1 | Table 5.2-1 | 1.4,3,5,10,15,20 | Table 5.4.2-1 | NA |
| NS_03 | 6.6.2.2.3.1 | 2,4,10, 23, 25,35,36 | 3 | >5 | ≤ 1 |
| | | | 5 | >6 | ≤ 1 |
| | | | 10 | >6 | ≤ 1 |
| | | | 15 | >8 | ≤ 1 |
| | | | 20 | >10 | ≤ 1 |
| NS_04 | 6.6.2.2.3.2 | 41 | 5 | >6 | ≤1 |
| | | | 10, 15, 20 | See Table 6.2.4.3-4 | |
| NS_05 | 6.6.3.3.3.1 | 1 | 10,15,20 | ≥ 50 | ≤ 1 |
| NS_06 | 6.6.2.2.3.3 | 12, 13, 14, 17 | 1.4, 3, 5, 10 | Table 5.4.2-1 | n/a |
| NS_07 | 6.6.2.2.3.3 6.6.3.3.3.2 | 13 | 10 | Table 6.2.4.3-2 | Table 6.2.4.3-2 |
| NS_08 | 6.6.3.3.3.3 | 19 | 10, 15 | > 44 | ≤ 3 |
| NS_09 | 6.6.3.3.3.4 | 21 | 10, 15 | > 40 | ≤ 1 |
| | | | | > 55 | ≤ 2 |
| NS_10 | | 20 | 15, 20 | Table 6.2.4.3-3 | Table 6.2.4.3-3 |
| NS_11 | 6.6.2.2.1 | 23[1] | 1.4, 3, 5, 10 | Table 6.2.4.3-5 | Table 6.2.4.3-5 |
| .. | | | | | |
| NS_32 | - | - | - | - | - |
| Note 1: Applies to the lower block of Band 23, i.e. a carrier placed in the 2000-2010 MHz region. | | | | | |

**Table 6.2.4.3-2: A-MPR for "NS_07"**

| Parameters | Region A | | Region B | | | | Region C | |
|---|---|---|---|---|---|---|---|---|
| $RB_{start}$ [1] | 0 – 12 | | 13 –18 | | 19 – 42 | | 43 – 49 | |
| L_CRB[2] [RBs] | 6 – 8 | 1 to 5 and 9-50 | <8 | ≥8 | <18 | ≥18 | ≤2 | >2 |
| A-MPR [dB] | ≤8 | ≤12 | 0 | ≤12 | 0 | ≤6 | ≤3 | 0 |

Note 1: $RB_{start}$ indicates the lowest RB index of transmitted resource blocks
Note 2: L_CRB is the length of a contiguous resource block allocation
Note 3: For intra-subframe frequency hopping between two regions, notes 1 and 2 apply on a per slot basis.
Note 4: For intra-subframe frequency hopping between two regions, the larger A-MPR value of the two regions may be applied for both slots in the subframe.

**Table 6.2.4.3-3: A-MPR for "NS_10"**

| Channel BW | Parameters | Region A |
|---|---|---|

| | | |
|---|---|---|
| 15 | $RB_{start}$ [1] | 0 – 10 |
| | L_CRB [RBs] | 1 -20 |
| | A-MPR [dB] | ≤ 2 |
| 20 | $RB_{start}$ [1] | 0 – 15 |
| | L_CRB [RBs] | 1 -20 |
| | A-MPR [dB] | ≤ 5 |

Note 1: $RB_{start}$ indicates the lowest RB index of transmitted resource blocks.
Note 2: L_CRB is the length of a contiguous resource block allocation.
Note 3: For intra-subframe frequency hopping which intersects Region A, notes 1 and 2 apply on a per slot basis.
Note 4: For intra-subframe frequency hopping which intersect Region A, the larger A-MPR value may be applied for both slots in the subframe.

**Table 6.2.4.3-4: A-MPR for NS_04 for bandwidths > 5MHz**

| Channel BW | Parameters | Region A | Region B | | Region C |
|---|---|---|---|---|---|
| 10 | $RB_{start}$ [1] | 0 – 12 | 13 – 36 | | 37 – 49 |
| | $RB_{start}$ [1] + L_CRB [2] [RBs] | NA (Note 3) | 14 - 37 | >37 | NA (Note 3) |
| | A-MPR [dB] | ≤3dB | 0 | ≤2dB | ≤3dB |
| 15 | $RB_{start}$ [1] | 0 – 18 | 19 – 55 | | 56 – 74 |
| | $RB_{start}$ [1] + L_CRB [2] [RBs] | NA (Note 3) | 20 - 56 | >56 | NA (Note 3) |
| | A-MPR [dB] | ≤3dB | 0 | ≤2dB | ≤3dB |
| 20 | $RB_{start}$ [1] | 0 – 24 | 25 – 74 | | 75 – 99 |
| | $RB_{start}$ [1] + L_CRB [2] [RBs] | NA (Note 3) | 26 - 75 | >75 | NA (Note 3) |
| | A-MPR [dB] | ≤3dB | 0 | ≤2dB | ≤3dB |

Note 1: $RB_{start}$ indicates the lowest RB index of transmitted resource blocks
Note 2: L_CRB is the length of a contiguous resource block allocation
Note 3: Any RB allocation that starts in Region A or C is allowed the specified A-MPR
Note 4: For intra-subframe frequency hopping which intersects regions, notes 1 and 2 apply on a per slot basis
Note 5: For intra-subframe frequency hopping which intersects regions, the larger A-MPR value may be applied for both slots in the subframe

**Table 6.2.4.3-5: A-MPR for NS_11**

| Channel Bandwidth | Parameters | | | |
|---|---|---|---|---|
| 3 | Fc (MHz) | <2004 | ≥2004 | |
| | L_CRB (RBs) | 1-15 | >5 | |
| | A-MPR | ≤5 | ≤ 1 | |
| 5 | Fc (MHz) | <2004 | 2004 ≤ Fc <2007 | ≥2007 |
| | L_CRB (RBs) | 1-25 | 1-6 & 15-25 / 8-12 | >6 |
| | A-MPR | ≤7 | ≤ 4 / 0 | ≤ 1 |
| 10 | Fc (MHz) | 2005 | | |
| | $RB_{start}$ (RBs) | 0-49 | | |
| | L_CRB (RBs) | 1-50 | | |
| | A-MPR | ≤ 12 | | |

### 6.2.4.4 Test description

#### 6.2.4.4.1 Initial condition

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in tables 6.2.4.4.1-1 through table 6.2.4.4.1-6. The details of the uplink reference measurement channels (RMCs) are specified in Annex A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.2.4.4.1-1: Test Configuration Table (network signalled value "NS_03")**

| **Initial Conditions** | | | | | | | |
|---|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | | Lowest, 5MHz, 10MHz, Highest | | |
| **Test Parameters for NS_03 A-MPR** | | | | | | | |
| | | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Configuration ID | Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | | FDD | TDD | | FDD | TDD |
| 1 | 1.4MHz | N/A for A-MPR testing. | | | QPSK | 6 | 6 |
| 2 | 1.4MHz | | | | QPSK | 5 | 5 |
| 3 | 1.4MHz | | | | 16QAM | 5 | 5 |
| 4 | 3MHz | | | | QPSK | 15 | 15 |
| 5 | 3MHz | | | | QPSK | 4 | 4 |
| 6 | 3MHz | | | | 16QAM | 15 | 15 |
| 7 | 3MHz | | | | 16QAM | 4 | 4 |
| 8 | 5MHz | | | | QPSK | 25 | 25 |
| 9 | 5MHz | | | | QPSK | 8 | 8 |
| 10 | 5MHz | | | | QPSK | 6 | 6 |
| 11 | 5MHz | | | | 16QAM | 25 | 25 |
| 12 | 5MHz | | | | 16QAM | 8 | 8 |
| 13 | 10MHz | | | | QPSK | 50 | 50 |
| 14 | 10MHz | | | | QPSK | 12 | 12 |
| 15 | 10MHz | | | | QPSK | 6 | 6 |
| 16 | 10MHz | | | | 16QAM | 50 (Note 4) | 50 (Note 4) |
| 17 | 10MHz | | | | 16QAM | 12 | 12 |
| 18 | 15MHz | | | | QPSK | 75 | 75 |
| 19 | 15MHz | | | | QPSK | 16 | 16 |
| 20 | 15MHz | | | | QPSK | 8 | 8 |
| 21 | 15MHz | | | | 16QAM | 75 (Note 4) | 75 (Note 4) |
| 22 | 15MHz | | | | 16QAM | 16 | 16 |
| 23 | 20MHz | | | | QPSK | 100 | 100 |
| 24 | 20MHz | | | | QPSK | 18 | 18 |
| 25 | 20MHz | | | | QPSK | 10 | 10 |
| 26 | 20MHz | | | | 16QAM | 100 (Note 4) | 100 (Note 4) |
| 27 | 20MHz | | | | 16QAM | 18 | 18 |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE.

Note 3: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth.

Note 4: Applies only for UE-Categories 2-5

Note 5: For band 23, above table only applies to mid and high range test frequencies. Low range test frequencies will be covered by NS_11 test configuration table.

**Table 6.2.4.4.1-2: Test Configuration Table (network signalled value "NS_04")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | 5MHz, 10 MHz, 15 MHz, 20MHz | | |
| **Test Parameters for NS_04 A-MPR** | | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | | |
| Configuration ID | Ch BW | Mod'n | RB allocation TDD | Mod'n | RB allocation TDD | $RB_{start}$ TDD |
| 1 | 5MHz | N/A for A-MPR testing | | QPSK | 25 | Note 3 |
| 2 | 5MHz | | | QPSK | 8 | Note 3 |
| 3 | 5MHz | | | QPSK | 6 | Note 3 |
| 4 | 5MHz | | | 16QAM | 25 | Note 3 |
| 5 | 5MHz | | | 16QAM | 8 | Note 3 |
| 6 | 10MHz | | | QPSK | 1 | 0 |
| 7 | 10MHz | | | QPSK | 12 | 0 |
| 8 | 10MHz | | | QPSK | 50 | 0 |
| 9 | 10MHz | | | 16QAM | 50 (Note 4) | 0 |
| 10 | 10MHz | | | QPSK | 24 | 13 |
| 11 | 10MHz | | | 16QAM | 24 | 13 |
| 12 | 10MHz | | | QPSK | 36 | 13 |
| 13 | 10MHz | | | QPSK | 12 | 37 |
| 14 | 10MHz | | | QPSK | 1 | 49 |
| 15 | 15MHz | | | QPSK | 1 | 0 |
| 16 | 15MHz | | | QPSK | 16 | 0 |
| 17 | 15MHz | | | QPSK | 75 | 0 |
| 18 | 15MHz | | | 16QAM | 75 (Note 4) | 0 |
| 19 | 15MHz | | | QPSK | 36 | 19 |
| 20 | 15MHz | | | 16QAM | 36 (Note 4) | 19 |
| 21 | 15MHz | | | QPSK | 50 | 19 |
| 22 | 15MHz | | | QPSK | 18 | 56 |
| 23 | 15MHz | | | QPSK | 1 | 74 |
| 24 | 20MHz | | | QPSK | 1 | 0 |
| 25 | 20MHz | | | QPSK | 18 | 0 |
| 26 | 20MHz | | | QPSK | 100 | 0 |
| 27 | 20MHz | | | 16QAM | 100 (Note 4) | 0 |
| 28 | 20MHz | | | QPSK | 50 | 25 |
| 29 | 20MHz | | | 16QAM | 50 (Note 4) | 25 |
| 30 | 20MHz | | | QPSK | 75 | 25 |
| 31 | 20MHz | | | QPSK | 25 | 75 |
| 32 | 20MHz | | | QPSK | 1 | 99 |

Note 1: Test Channel Bandwidths are checked separately for E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: The configuration ID will be used to map the applicable Test Configuration to be corresponding Test Requirement in subclause 6.2.4 as not all combinations are necessarily required based on the applicability of the UE.

Note 3: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth.

Note 4: Applies only for UE-Categories 2-5

**Table 6.2.4.4.1-3: Test Configuration Table (network signalled value "NS_05")**

| **Initial Conditions** | | | | | |
|---|---|---|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | | | | Normal | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range<br><br>In case of Low range:<br>- For 5MHz channel bandwidth: UL 1927.2MHz (N_UL = 18072), DL 2117.2MHz (N_DL = 72) and UL 1931.1MHz (N_UL = 18111) DL 2121.1 MHz (N_DL = 111)<br>- For 10MHz: UL 1934.7MHz (N_UL = 18147), DL 2124.7MHz (N_DL = 147)<br>- For 20MHz channel bandwidth: Not available | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | | | | 5MHz, 10MHz, 15MHz, 20MHz | |
| **Test Parameters for NS_05 A-MPR** | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | |
| Configuration ID | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD |
| 1 | 5MHz | N/A for A-MPR testing | | QPSK | 1 |
| 2 | 5MHz | | | QPSK | 25 |
| 3 | 10MHz | | | QPSK | 1 |
| 4 | 10MHz | | | QPSK | 12 |
| 5 | 10MHz | | | QPSK | 48 |
| 6 | 10MHz | | | QPSK | 50 |
| 7 | 10MHz | | | 16QAM | 50<br>(Note 5) |
| 8 | 15MHz | | | QPSK | 1 |
| 9 | 15MHz | | | QPSK | 16 |
| 10 | 15MHz | | | QPSK | 48 |
| 11 | 15MHz | | | QPSK | 75 |
| 12 | 15MHz | | | 16QAM | 75<br>(Note 5) |
| 13 | 20MHz | | | QPSK | 1 |
| 14 | 20MHz | | | QPSK | 18 |
| 15 | 20MHz | | | QPSK | 48 |
| 16 | 20MHz | | | QPSK | 100 |
| 17 | 20MHz | | | 16QAM | 100<br>(Note 5) |

Note 1. The 1 RB allocation shall be tested at both RB #0 and RB #max.
Note 2. The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 - RB allocation) of the channel bandwidth.
Note 3: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE.
Note 4: Low range frequencies for 5MHz channel bandwidth in case of network signalled "NS_05" shall be reviewed after June 2012 because of PHS band operation change.
Note 5: Applies only for UE-Categories 2-5

**Table 6.2.4.4.1-4: Test Configuration Table (network signalled value "NS_06")**

| **Initial Conditions** | | | | | |
|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | Normal | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | Lowest, 5MHz, 10MHz, Highest | |
| **Test Parameters for NS_06 A-MPR** | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | |
| Configuration ID | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD |
| 1 | 1.4MHz | N/A for A-MPR testing | | QPSK | 6 |
| 2 | 1.4MHz | | | QPSK | 5 |
| 3 | 1.4MHz | | | 16QAM | 5 |
| 4 | 3MHz | | | QPSK | 15 |
| 5 | 3MHz | | | QPSK | 4 |
| 6 | 3MHz | | | 16QAM | 4 |
| 7 | 5MHz | | | QPSK | 25 |
| 8 | 5MHz | | | QPSK | 8 |
| 9 | 5MHz | | | 16QAM | 8 |
| 10 | 10MHz | | | QPSK | 50 |
| 11 | 10MHz | | | QPSK | 12 |
| 12 | 10MHz | | | 16QAM | 12 |
| 13 | 15MHz | | | QPSK | 75 |
| 14 | 15MHz | | | QPSK | 16 |
| 15 | 15MHz | | | 16QAM | 16 |
| 16 | 20MHz | | | QPSK | 100 |
| 17 | 20MHz | | | QPSK | 18 |
| 18 | 20MHz | | | 16QAM | 18 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | | |
| Note 2: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE. | | | | | |
| Note 3: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth. | | | | | |

**Table 6.2.4.4.1-5: Test Configuration Table (network signalled value "NS_07")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Mid range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | 10MHz | | |
| **Test Parameters for NS_07 A-MPR** | | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | | |
| Configuration ID | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD | $RB_{start}$ FDD |
| 1 | 10MHz | N/A for A-MPR testing | | QPSK | 1 | 0 |
| 2 | 10MHz | | | QPSK | 8 | 0 |
| 3 | 10MHz | | | QPSK | 6 | 13 |
| 4 | 10MHz | | | QPSK | 20 | 13 |
| 5 | 10MHz | | | QPSK | 12 | 13 |
| 6 | 10MHz | | | 16QAM | 36 (Note 2) | 13 |
| 7 | 10MHz | | | QPSK | 16 | 19 |
| 8 | 10MHz | | | QPSK | 12 | 19 |
| 9 | 10MHz | | | 16QAM | 16 | 19 |
| 10 | 10MHz | | | QPSK | 30 | 19 |
| 11 | 10MHz | | | 16QAM | 30 (Note 2) | 19 |
| 12 | 10MHz | | | QPSK | 6 | 43 |
| 13 | 10MHz | | | QPSK | 2 | 48 |
| 14 | 10MHz | | | QPSK | 50 | 0 |
| 15 | 10MHz | | | QPSK | 12 | 0 |
| 16 | 10MHz | | | 16QAM | 50 (Note 2) | 0 |
| Note 1: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE.<br>Note 2: Applies only for UE-Categories 2-5 | | | | | | |

**Table 6.2.4.4.1-6: Test Configuration Table (network signalled value "NS_08")**

| **Initial Conditions** | | | | | |
|---|---|---|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | | | | Normal | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | | | | High range | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | | | | 5MHz, 10MHz, 15MHz | |
| **Test Parameters for NS_08 A-MPR** | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | |
| Configuration ID | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD |
| 1 | 5MHz | N/A for A-MPR testing | | QPSK | 1 |
| 2 | 5MHz | | | QPSK | 8 |
| 3 | 5MHz | | | QPSK | 25 |
| 4 | 10MHz | | | QPSK | 1 |
| 5 | 10MHz | | | QPSK | 12 |
| 6 | 10MHz | | | QPSK | 40 |
| 7 | 10MHz | | | QPSK | 50 |
| 8 | 10MHz | | | 16QAM | 50<br>(Note 4) |
| 9 | 15MHz | | | QPSK | 1 |
| 10 | 15MHz | | | QPSK | 16 |
| 11 | 15MHz | | | QPSK | 40 |
| 12 | 15MHz | | | QPSK | 75 |
| 13 | 15MHz | | | 16QAM | 75<br>(Note 4) |

Note 1: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE.

Note 2. The 1 RB allocation shall be tested at both RB #0 and RB #max.

Note 3: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max + 1 - RB allocation) of the channel bandwidth

Note 4: Applies only for UE-Categories 2-5

**Table 6.2.4.4.1-7: Test Configuration Table (network signalled value "NS_09")**

| **Initial Conditions** | | | | | |
|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | Normal | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | High range | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | 5MHz, 10MHz, 15MHz | |
| **Test Parameters for Channel Bandwidths** | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | |
| Configuration ID | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD |
| 1 | 5MHz | N/A for A-MPR testing | | QPSK | 1 |
| 2 | 5MHz | | | QPSK | 8 |
| 3 | 5MHz | | | QPSK | 25 |
| 4 | 10MHz | | | QPSK | 1 |
| 5 | 10MHz | | | QPSK | 12 |
| 6 | 10MHz | | | QPSK | 40 |
| 7 | 10MHz | | | QPSK | 50 |
| 8 | 10MHz | | | 16QAM | 50 (Note 4) |
| 9 | 15MHz | | | QPSK | 1 |
| 10 | 15MHz | | | QPSK | 16 |
| 11 | 15MHz | | | QPSK | 40 |
| 12 | 15MHz | | | QPSK | 54 |
| 13 | 15MHz | | | QPSK | 75 |
| 14 | 15MHz | | | 16QAM | 75 (Note 4) |
| Note 1: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE. | | | | | |
| Note 2. The 1 RB allocation shall be tested at both RB #0 and RB #max. | | | | | |
| Note 3: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max + 1 - RB allocation) of the channel bandwidth | | | | | |
| Note 4: Applies only for UE-Categories 2-5 | | | | | |

**Table 6.2.4.4.1-8: Test Configuration Table (network signalled value "NS_10")**

FFS

1. Connect the SS to the UE antenna connectors as shown in Figure TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to the applicable table from Table 6.2.4.4.1-1 to Table 6.2.4.4.1-6.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.2.4.4.3.

**Table 6.2.4.4.1-9: Test Configuration Table (network signalled value "NS_11")**

| **Initial Conditions** | | | | | |
|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | Normal | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range<br>For 3 MHz Channel Banwidth:<br>a. UL 2001.5 MHz (N_UL=25515), DL 2181.5 MHz(N_DL=7515)<br>b.UL 2004.5 MHz (N_UL=25545), DL 2184.5 (N_DL=7545)<br>For 5 MHz Channel Bandwidth<br>a. UL 2002.5 MHz (N_UL=25525), DL 2182.5 MHz(N_DL=7525)<br>b. UL 2004.5 MHz (N_UL=25545), DL 2184.5 MHz(N_DL=7545)<br>c. UL 2007.5 MHz (N_UL=25575), DL 2187.5 MHz(N_DL=7575)<br>For 10 MHz Channel Bandwidth<br>UL 2005 MHz (N_UL=25550), DL 2185 MHz (N_DL=7550) | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | 1.4MHz, 3MHz, 5MHz, 10MHz | |
| **Test Parameters for NS_11 A-MPR** | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | |
| Configuration ID | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD |
| 1 | 3MHz | N/A for A-MPR testing | | QPSK | 6 |
| 2 | 3MHz | | | QPSK | 15 |
| 3 | 3MHz | | | 16QAM | 6 |
| 4 | 3MHz | | | 16QAM | 15 |
| 5 | 5MHz | | | QPSK | 1 |
| 6 | 5MHz | | | QPSK | 8 |
| 7 | 5MHz | | | QPSK | 25 |
| 8 | 5MHz | | | 16QAM | 8 |
| 9 | 5MHz | | | 16QAM | 25 |
| 10 | 10MHz | | | QPSK | 1 |
| 11 | 10MHz | | | QPSK | 12 |
| 12 | 10MHz | | | QPSK | 50 |
| 13 | 10MHz | | | 16QAM | 12 |
| 14 | 10MHz | | | 16QAM | 50<br>(Note 3) |

Note 1: The Configuration ID will be used to map the applicable Test Configuration to the corresponding Test Requirement in subclause 6.2.4.5 as not all combinations are necessarily required based on the applicability of the UE.
Note 2: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 - RB allocation) of the channel bandwidth.
Note 3: Applies only for UE-Categories 2-5

#### 6.2.4.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to the applicable table from Table 6.2.4.4.1-1 to Table 6.2.4.4.1-6. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.

3. Measure the mean power of the UE in the channel bandwidth of the radio access mode. The period of measurement shall be at least the continuous duration one sub-frame (1ms). For TDD slots with transient periods are not under test.

#### 6.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6, with the following exceptions for each network signalled value.

##### 6.2.4.4.3.1 Message contents exceptions (network signalled value "NS_03")

1. Information element additionalSpectrumEmission is set to NS_03. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.1-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_03"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 3 (NS_03) | | |

##### 6.2.4.4.3.2 Message contents exceptions (network signalled value "NS_04")

1. Information element additionalSpectrumEmission is set to NS_04. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.2-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_04"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 4 (NS_04) | | |

##### 6.2.4.4.3.3 Message contents exceptions (network signalled value "NS_05")

1. Information element additionalSpectrumEmission is set to NS_05. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.3-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_05"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 5 (NS_05) | | |

6.2.4.4.3.4 Message contents exceptions (network signalled value "NS_06")

1. Information element additionalSpectrumEmission is set to NS_06. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.4-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_06"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 6 (NS_06) | | |

6.2.4.4.3.5 Message contents exceptions (network signalled value "NS_07")

1. Information element additionalSpectrumEmission is set to NS_07. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.5-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_07"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 7 (NS_07) | | |

6.2.4.4.3.6 Message contents exceptions (network signalled value "NS_08")

1. Information element additionalSpectrumEmission is set to NS_08. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.6-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_08"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 8 (NS_08) | | |

6.2.4.4.3.7 Message contents exceptions (network signalled value "NS_09")

1. Information element additionalSpectrumEmission is set to NS_09. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.7-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_09"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 9 (NS_09) | | |

6.2.4.4.3.8 Message contents exceptions (network signalled value "NS_10")

1. Information element additionalSpectrumEmission is set to NS_10. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.8-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_10"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 10 (NS_10) | | |

6.2.4.4.3.9 Message contents exceptions (network signalled value "NS_11")

1. Information element additionalSpectrumEmission is set to NS_11. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.2.4.4.3.9-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_11"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 11 (NS_11) | | |

## 6.2.4.5 Test requirements

The maximum output power, derived in step 2 shall be within the range prescribed by the nominal maximum output power and tolerance in the applicable table from Table 6.2.4.5-1 to Table 6.2.4.5-8. The allowed A-MPR values specified in Table 6.2.4.3-1 are in addition to the allowed MPR requirements specified in clause 6.2.3. For the UE maximum output power modified by MPR and/or A-MPR, the power limits specified in Table 6.2.5.3-1 apply.

**Table 6.2.4.5-1: UE Power Class test requirements (network signalled value "NS_03") (for Bands 4, 10, 23, 35, and 36)**

| Configuration ID | EUTRA band | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -3.7 |
| 2 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -2.7 |
| 3 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -3.7 |
| 4 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -4.7 |
| 5 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -2.7 |
| 6 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -6.2 |
| 7 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -3.7 |
| 8 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -4.7 |
| 9 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -3.7 |
| 10 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -2.7 |
| 11 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -6.2 |
| 12 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -4.7 |
| 13 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -4.7 |
| 14 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -3.7 |
| 15 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -2.7 |
| 16 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -6.2 |
| 17 | 4,10, 23,35,36 | | | | | 23 | +2.7 / -4.7 |
| 18 | 4,10,35,36 | | | | | 23 | +2.7 / -4.7 |
| 19 | 4,10,35,36 | | | | | 23 | +2.7 / -3.7 |
| 20 | 4,10,35,36 | | | | | 23 | +2.7 / -2.7 |
| 21 | 4,10,35,36 | | | | | 23 | +2.7 / -6.2 |
| 22 | 4,10,35,36 | | | | | 23 | +2.7 / -4.7 |
| 23 | 4,10,35,36 | | | | | 23 | +2.7 / -4.7 |
| 24 | 4,10,35,36 | | | | | 23 | +2.7 / -3.7 |
| 25 | 4,10,35,36 | | | | | 23 | +2.7 / -2.7 |
| 26 | 4,10,35,36 | | | | | 23 | +2.7 / -6.2 |
| 27 | 4,10,35,36 | | | | | 23 | +2.7 / -4.7 |

**Table 6.2.4.5-2: UE Power Class test requirements (network signalled value "NS_03") (for Bands 2 and 25)**

| Configuration ID | EUTRA band | Test Freq. | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2, 25 | Mid | | | | | 23 | +2.7 / -3.7 |
| 1 | 2, 25 | Low, High | | | | | 23 | +2.7 / -5.7 |
| 2 | 2, 25 | Mid | | | | | 23 | +2.7 / -2.7 |
| 2 | 2, 25 | Low, High | | | | | 23 | +2.7 / -4.2 |
| 3 | 2, 25 | Mid | | | | | 23 | +2.7 / -3.7 |
| 3 | 2, 25 | Low, High | | | | | 23 | +2.7 / -5.7 |
| 4 | 2, 25 | Mid | | | | | 23 | +2.7 / -4.7 |
| 4 | 2, 25 | Low, High | | | | | 23 | +2.7 / -7.7 |
| 5 | 2, 25 | Mid | | | | | 23 | +2.7 / -2.7 |
| 5 | 2, 25 | Low, High | | | | | 23 | +2.7 / -4.2 |
| 6 | 2, 25 | Mid | | | | | 23 | +2.7 / -6.2 |
| 6 | 2, 25 | Low, High | | | | | 23 | +2.7 / -9.2 |
| 7 | 2, 25 | Mid | | | | | 23 | +2.7 / -3.7 |
| 7 | 2, 25 | Low, High | | | | | 23 | +2.7 / -5.7 |
| 8 | 2, 25 | All | | | | | 23 | +2.7 / -4.7 |
| 9 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -3.7 |
| 9 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -5.7 |
| 10 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -2.7 |
| 10 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -4.2 |
| 11 | 2, 25 | All | | | | | 23 | +2.7 / -6.2 |
| 12 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -4.7 |
| 12 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -7.7 |
| 13 | 2, 25 | All | | | | | 23 | +2.7 / -4.7 |
| 14 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -3.7 |
| 14 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB | | | | | 23 | +2.7 / -5.7 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | allocation) | | | | | | |
| 15 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -2.7 |
| 15 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -4.2 |
| 16 | 2, 25 | All | | | | | 23 | +2.7 / -6.2 |
| 17 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -4.7 |
| 17 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -7.7 |
| 18 | 2, 25 | All | | | | | 23 | +2.7 / -4.7 |
| 19 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -3.7 |
| 19 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -5.7 |
| 20 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -2.7 |
| 20 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -4.2 |
| 21 | 2, 25 | All | | | | | 23 | +2.7 / -6.2 |
| 22 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -4.7 |
| 22 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -7.7 |
| 23 | 2, 25 | All | | | | | 23 | +2.7 / -4.7 |
| 24 | 2, 25 | All | | | | | 23 | +2.7 / -3.7 |
| 25 | 2, 25 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -2.7 |
| 25 | 2, 25 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -4.2 |
| 26 | 2, 25 | All | | | | | 23 | +2.7 / -6.2 |
| 27 | 2, 25 | All | | | | | 23 | +2.7 / -4.7 |

**Table 6.2.4.5-3: UE Power Class test requirements (network signalled value "NS_04")**

| Configuration ID | EUTRA band | Bandwidth (MHz) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|
| 1 | 41 | 5 MHz | 23 | +2.7 / -4.7 |
| 2 | 41 | 5 MHz | 23 | +2.7 / -3.7 |
| 3 | 41 | 5 MHz | 23 | +2.7 / -2.7 |
| 4 | 41 | 5 MHz | 23 | +2.7 / -6.2 |
| 5 | 41 | 5 MHz | 23 | +2.7 / -4.7 |
| 6 | 41 | 10MHz | 23 | +2.7/ -6.2 |
| 7 | 41 | 10MHz | 23 | +2.7/ -6.2 |
| 8 | 41 | 10MHz | 23 | +2.7/ -8.2 |
| 9 | 41 | 10MHz | 23 | +2.7/ -9.7 |
| 10 | 41 | 10MHz | 23 | +2.7/ -3.7 |
| 11 | 41 | 10MHz | 23 | +2.7/ -4.7 |
| 12 | 41 | 10MHz | 23 | +2.7/ -6.2 |
| 13 | 41 | 10MHz | 23 | +2.7/ -6.2 |
| 14 | 41 | 10MHz | 23 | +2.7/ -6.2 |
| 15 | 41 | 15MHz | 23 | +2.7/ -6.2 |
| 16 | 41 | 15MHz | 23 | +2.7/ -6.2 |
| 17 | 41 | 15MHz | 23 | +2.7/ -8.2 |
| 18 | 41 | 15MHz | 23 | +2.7/ -9.7 |
| 19 | 41 | 15MHz | 23 | +2.7/ -3.7 |
| 20 | 41 | 15MHz | 23 | +2.7/ -4.7 |
| 21 | 41 | 15MHz | 23 | +2.7/ -6.2 |
| 22 | 41 | 15MHz | 23 | +2.7/ -8.2 |
| 23 | 41 | 15MHz | 23 | +2.7/ -6.2 |
| 24 | 41 | 20MHz | 23 | +2.7/ -6.2 |
| 25 | 41 | 20MHz | 23 | +2.7/ -6.2 |
| 26 | 41 | 20MHz | 23 | +2.7/ -8.2 |
| 27 | 41 | 20MHz | 23 | +2.7/ -9.7 |
| 28 | 41 | 20MHz | 23 | +2.7/ -3.7 |
| 29 | 41 | 20MHz | 23 | +2.7/ -4.7 |
| 30 | 41 | 20MHz | 23 | +2.7/ -6.2 |
| 31 | 41 | 20MHz | 23 | +2.7/ -8.2 |
| 32 | 41 | 20MHz | 23 | +2.7/ |

| | | | | |
|---|---|---|---|---|
| | | | | -6.2 |

**Table 6.2.4.5-4: UE Power Class test requirements (network signalled value "NS_05")**

| Configuration ID | EUTRA band | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 1 | | | | | 23 | +2.7 / -2.7 |
| 2 | 1 | | | | | 23 | +2.7 / -3.7 |
| 3 | 1 | | | | | 23 | +2.7 / -2.7 |
| 4 | 1 | | | | | 23 | +2.7 / -2.7 |
| 5 | 1 | | | | | 23 | +2.7 / -3.7 |
| 6 | 1 | | | | | 23 | +2.7 / -4.7 |
| 7 | 1 | | | | | 23 | +2.7 / -6.2 |
| 8 | 1 | | | | | 23 | +2.7 / -2.7 |
| 9 | 1 | | | | | 23 | +2.7 / -2.7 |
| 10 | 1 | | | | | 23 | +2.7 / -3.7 |
| 11 | 1 | | | | | 23 | +2.7 / -4.7 |
| 12 | 1 | | | | | 23 | +2.7 / -6.2 |
| 13 | 1 | | | | | 23 | +2.7 / -2.7 |
| 14 | 1 | | | | | 23 | +2.7 / -2.7 |
| 15 | 1 | | | | | 23 | +2.7 / -3.7 |
| 16 | 1 | | | | | 23 | +2.7 / -4.7 |
| 17 | 1 | | | | | 23 | +2.7 / -6.2 |

**Table 6.2.4.5-5: UE Power Class test requirements (network signalled value "NS_06") (for Bands 13, 14, and 17)**

| Configuration ID | EUTRA band | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 2 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 3 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 4 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 5 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 6 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 7 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 8 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 9 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 10 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 11 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 12 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 13 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 14 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 15 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 16 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |
| 17 | 13,14,17 | | | | | 23 | +2.7 / -2.7 |
| 18 | 13,14,17 | | | | | 23 | +2.7 / -3.7 |

**Table 6.2.4.5-6: UE Power Class test requirements (network signalled value "NS_06") (for Band 12)**

| Configuration ID | EUTRA band | Test Freq. | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|---|
| 1 | 12 | Mid | | | | | 23 | +2.7 / -3.7 |
| 1 | 12 | Low, High | | | | | 23 | +2.7 / -5.7 |
| 2 | 12 | Mid | | | | | 23 | +2.7 / -2.7 |
| 2 | 12 | Low, High | | | | | 23 | +2.7 / -4.2 |
| 3 | 12 | Mid | | | | | 23 | +2.7 / -2.7 |
| 3 | 12 | Low, High | | | | | 23 | +2.7 / -4.2 |
| 4 | 12 | Mid | | | | | 23 | +2.7 / -3.7 |
| 4 | 12 | Low, High | | | | | 23 | +2.7 / -5.7 |
| 5 | 12 | Mid | | | | | 23 | +2.7 / -2.7 |
| 5 | 12 | Low, High | | | | | 23 | +2.7 / -4.2 |
| 6 | 12 | Mid | | | | | 23 | +2.7 / -3.7 |
| 6 | 12 | Low, High | | | | | 23 | +2.7 / -5.7 |
| 7 | 12 | All | | | | | 23 | +2.7 / -3.7 |
| 8 | 12 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -2.7 |
| 8 | 12 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -4.2 |
| 9 | 12 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -3.7 |
| 9 | 12 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -5.7 |
| 10 | 12 | All | | | | | 23 | +2.7 / -3.7 |
| 11 | 12 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -2.7 |
| 11 | 12 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -4.2 |
| 12 | 12 | Low @ RB#(max+1-RB allocation), Mid, High @ RB#0 | | | | | 23 | +2.7 / -3.7 |
| 12 | 12 | Low @ RB#0, High @ RB#(max+1-RB allocation) | | | | | 23 | +2.7 / -5.7 |

**Table 6.2.4.5-7: UE Power Class test requirements (network signalled value "NS_07")**

| Configuration ID | EUTRA band | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 13 | | | | | 23 | +2.7 / -18.7 |
| 2 | 13 | | | | | 23 | +2.7 / -13.7 |
| 3 | 13 | | | | | 23 | +2.7 / -2.7 |
| 4 | 13 | | | | | 23 | +2.7 / -19.7 |
| 5 | 13 | | | | | 23 | +2.7 / -18.7 |
| 6 | 13 | | | | | 23 | +2.7 / -20.7 |
| 7 | 13 | | | | | 23 | +2.7 / -3.7 |
| 8 | 13 | | | | | 23 | +2.7 / -2.7 |
| 9 | 13 | | | | | 23 | +2.7 / -4.7 |
| 10 | 13 | | | | | 23 | +2.7 / -12.7 |
| 11 | 13 | | | | | 23 | +2.7 / -13.7 |
| 12 | 13 | | | | | 23 | +2.7 / -2.7 |
| 13 | 13 | | | | | 23 | +2.7 / -6.2 |
| 14 | 13 | | | | | 23 | +2.7 / -19.7 |
| 15 | 13 | | | | | 23 | +2.7 / -18.7 |
| 16 | 13 | | | | | 23 | +2.7 / -20.7 |

**Table 6.2.4.5-8: UE Power Class test requirements (network signalled value "NS_08")**

| Configuration ID | EUTRA band | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 19 | | | | | 23 | +2.7 / -2.7 |
| 2 | 19 | | | | | 23 | +2.7 / -2.7 |
| 3 | 19 | | | | | 23 | +2.7 / -3.7 |
| 4 | 19 | | | | | 23 | +2.7 / -2.7 |
| 5 | 19 | | | | | 23 | +2.7 / -2.7 |
| 6 | 19 | | | | | 23 | +2.7 / -3.7 |
| 7 | 19 | | | | | 23 | +2.7 / -8.2 |
| 8 | 19 | | | | | 23 | +2.7 / -9.7 |
| 9 | 19 | | | | | 23 | +2.7 / -2.7 |
| 10 | 19 | | | | | 23 | +2.7 / -2.7 |
| 11 | 19 | | | | | 23 | +2.7 / -3.7 |
| 12 | 19 | | | | | 23 | +2.7 / -8.2 |
| 13 | 19 | | | | | 23 | +2.7 / -9.7 |

**Table 6.2.4.5-9: UE Power Class test requirements (network signalled value "NS_09")**

| Configuration ID | EUTRA band | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|
| 1 | 21 | | | | | 23 | +2.7 / -2.7 |
| 2 | 21 | | | | | 23 | +2.7 / -2.7 |
| 3 | 21 | | | | | 23 | +2.7 / -3.7 |
| 4 | 21 | | | | | 23 | +2.7 / -2.7 |
| 5 | 21 | | | | | 23 | +2.7 / -2.7 |
| 6 | 21 | | | | | 23 | +2.7 / -3.7 |
| 7 | 21 | | | | | 23 | +2.7 / -4.7 |
| 8 | 21 | | | | | 23 | +2.7 / -6.2 |
| 9 | 21 | | | | | 23 | +2.7 / -2.7 |
| 19 | 21 | | | | | 23 | +2.7 / -2.7 |
| 11 | 21 | | | | | 23 | +2.7 / -3.7 |
| 12 | 21 | | | | | 23 | +2.7 / -4.7 |
| 13 | 21 | | | | | 23 | +2.7 / -6.2 |
| 14 | 21 | | | | | 23 | +2.7 / -8.2 |

**Table 6.2.4.5-10: UE Power Class test requirements (network signalled value "NS_10")**

FFS

**Table 6.2.4.5-11: UE Power Class test requirements (network signalled value "NS_11 for Band 23")**

| Configuration ID | EUTRA Band | Center Frequency | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|---|
| 1a | 23 | UL 2001.5 MHz<br>DL 2181.5 MHz | | | | | | + 2.7 /<br>-11.7 |
| 1b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-4.7 |
| 2a | 23 | UL 2001.5 MHz<br>DL 2181.5 MHz | | | | | | + 2.7 /<br>-11.7 |
| 2b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-4.7 |
| 3a | 23 | UL 2001.5 MHz<br>DL 2181.5 MHz | | | | | | + 2.7 /<br>-12.7 |
| 3b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-6.2 |
| 4a | 23 | UL 2001.5 MHz<br>DL 2181.5 MHz | | | | | | + 2.7 /<br>-12.7 |
| 4b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-6.2 |
| 5a | 23 | UL 2002.5 MHz<br>DL 2182.5 MHz | | | | | | + 2.7 /<br>-12.7 |
| 5b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-8.2 |
| 5c | 23 | UL 2007.5 MHz<br>DL 2187.5 MHz | | | | | | + 2.7 /<br>-2.7 |
| 6a | 23 | UL 2002.5 MHz<br>DL 2182.5 MHz | | | | | | + 2.7 /<br>-12.7 |
| 6b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-2.7 |
| 6c | 23 | UL 2007.5 MHz<br>DL 2187.5 MHz | | | | | | + 2.7 /<br>-3.7 |
| 7a | 23 | UL 2002.5 MHz<br>DL 2182.5 MHz | | | | | | + 2.7 /<br>-13.7 |
| 7b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-9.7 |
| 7c | 23 | UL 2007.5 MHz<br>DL 2187.5 MHz | | | | | | + 2.7 /<br>-4.7 |
| 8a | 23 | UL 2002.5 MHz<br>DL 2182.5 MHz | | | | | | + 2.7 /<br>-13.7 |
| 8b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-3.7 |
| 8c | 23 | UL 2007.5 MHz<br>DL 2187.5 MHz | | | | | | + 2.7 /<br>-4.7 |
| 9a | 23 | UL 2002.5 MHz<br>DL 2182.5 MHz | | | | | | + 2.7 /<br>-14.7 |

| Configuration ID | EUTRA Band | Center Frequency | Class 1 (dBm) | Tol. (dB) | Class 2 (dBm) | Tol. (dB) | Class 3 (dBm) | Tol. (dB) |
|---|---|---|---|---|---|---|---|---|
| 9b | 23 | UL 2004.5 MHz<br>DL 2184.5 MHz | | | | | | + 2.7 /<br>-11.7 |
| 9c | 23 | UL 2007.5 MHz<br>DL 2187.5 MHz | | | | | | + 2.7 /<br>-6.2 |
| 10 | 23 | UL 2005 MHz<br>DL 2185 MHz | | | | | | + 2.7 /<br>-18.7 |
| 11 | 23 | UL 2005 MHz<br>DL 2185 MHz | | | | | | + 2.7 /<br>-18.7 |
| 12 | 23 | UL 2005 MHz<br>DL 2185 MHz | | | | | | + 2.7 /<br>-19.7 |
| 13 | 23 | UL 2005 MHz<br>DL 2185 MHz | | | | | | + 2.7 /<br>-19.7 |
| 14 | 23 | UL 2005 MHz<br>DL 2185 MHz | | | | | | + 2.7 /<br>-20.7 |

## 6.2.4A Additional Maximum Power Reduction (A-MPR) for intra-band contiguous CA

<reserved for future use>

## 6.2.5 Configured UE transmitted Output Power

### 6.2.5.1 Test purpose

To verify the UE does not exceed the minimum between the $P_{EMAX}$ maximum allowed UL TX Power signalled by the E-UTRAN and the $P_{UMAX}$ maximum UE power for the UE power class.

### 6.2.5.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 6.2.5.3 Minimum conformance requirements

The UE is allowed to set its configured maximum output power $P_{CMAX}$. The configured maximum output power $P_{CMAX}$ is set within the following bounds:

$P_{CMAX_L} \leq PCMAX \leq P_{CMAX_H}$

Where

- $P_{CMAX_L} = MIN \{PEMAX - \Delta T_C, P_{PowerClass} - MPR - A\text{-}MPR - \Delta T_C\}$
- $P_{CMAX_H} = MIN \{P_{EMAX}, P_{PowerClass}\}$
- $P_{EMAX}$ is the value given to IE *P-Max*, defined in [5]
- $P_{PowerClass}$ is the maximum UE power specified in Table 6.2.2.3-1 without taking into account the tolerance specified in the Table 6.2.2.3-1
- MRP and A-MPR are specified in Section 6.2.3 and Section 6.2.4, respectively
- $\Delta T_C$ = 1.5 dB when Note 2 in Table 6.2.2.3-1 applies
- $\Delta T_C$ = 0 dB when Note 2 in Table 6.2.2.3-1 does not apply

The measured maximum output power $P_{CMAX}$ shall be within the following bounds:

$$P_{CMAX_L} - T(P_{CMAX_L}) \leq P_{CMAX} \leq P_{CMAX_H} + T(P_{CMAX_H})$$

Where

$T(P_{CMAX})$ is defined by the tolerance table below and applies to $P_{CMAX_L}$ and $P_{CMAX_H}$ separately.

**Table 6.2.5-1: $P_{CMAX}$ tolerance**

| $P_{CMAX}$ (dBm) | Tolerance $T(P_{CMAX})$ (dB) |
|---|---|
| $21 \leq P_{CMAX} \leq 23$ | 2.0 |
| $20 \leq P_{CMAX} < 21$ | 2.5 |
| $19 \leq P_{CMAX} < 20$ | 3.5 |
| $18 \leq P_{CMAX} < 19$ | 4.0 |
| $13 \leq P_{CMAX} < 18$ | 5.0 |
| $8 \leq P_{CMAX} < 13$ | 6.0 |
| $-40 \leq P_{CMAX} < 8$ | 7.0 |

The normative reference for this requirement is TS 36.101 [2] clause 6.2.5.

## 6.2.5.4 Test description

### 6.2.5.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.2.5.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annex A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.2.5.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Configured UE transmitted Output Power test case | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 5 | 5 |
| 3MHz | | QPSK | 4 | 4 |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 18 | 18 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |
| Note 2: For the uplink RB allocation the $RB_{start}$ shall be RB #0. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, in Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channel is set according to Table 6.2.5.4.1-1

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.2.5.4.3.

### 6.2.5.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.2.5.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.

2. Send transmit uplink power control "up" commands in every uplink scheduling information to the UE; allow at least 200ms for the UE to reach the Pumax level of the test point.

3. Measure the mean power of the UE in the channel bandwidth for each test point in table 6.2.5.5-1 according to the test configuration from Table 6.2.5.4.1-1. The period of measurement shall be at least continuous duration of one sub-frame (1ms). For TDD slots with transient periods are not under test.

### 6.2.5.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 6.2.5.4.3-1: SystemInformationBlockType1: Test point 1**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2, Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| p-Max | -10 | | |

**Table 6.2.5.4.3-2: SystemInformationBlockType1: Test point 2**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2, Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| p-Max | 10 | | |

**Table 6.2.5.4.3-3: SystemInformationBlockType1: Test point 3**

| Derivation Path: TS 36.508 [7] clause 4.4.3.2, Table 4.4.3.2-3 SystemInformationBlockType1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| p-Max | 15 | | |

## 6.2.5.5 Test requirement

The maximum output power measured shall not exceed the values specified in Table 6.2.5.5-1.

**Table 6.2.5.5-1: $P_{CMAX}$ configured UE output power**

| | Channel bandwidth / maximum output power | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Measured UE output power test point 1 | For carrier frequency f ≤ 3.0GHz: -10 dBm ± 7.7<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: -10 dBm ± 8.0 | | | | | |
| Measured UE output power test point 2 | For carrier frequency f ≤ 3.0GHz: 10 dBm ± 6.7<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: 10 dBm ± 7.0 | | | | | |
| Measured UE output power test point 3 | For carrier frequency f ≤ 3.0GHz: 15 dBm ± 5.7<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: 15 dBm ± 6.0 | | | | | |
| Note: In addition note 2 in Table 6.2.2.3-1 shall apply to the tolerances. | | | | | | |

# 6.3 Output Power Dynamics

## 6.3.1 Void

## 6.3.2 Minimum Output Power

### 6.3.2.1 Test purpose

To verify the UE's ability to transmit with a broadband output power below the value specified in the test requirement when the power is set to a minimum value.

### 6.3.2.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 6.3.2.3 Minimum conformance requirements

The minimum output power is defined as the mean power in one sub-frame (1ms). The minimum output power shall not exceed the values specified in Table 6.3.2.3-1.

**Table 6.3.2.3-1: Minimum output power**

| | Channel bandwidth / minimum output power / measurement bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Minimum output power | -40 dBm | | | | | |
| Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |

The normative reference for this requirement is TS 36.101 [2] clause 6.3.2.1.

Minimum output power test verifies the UE's ability to transmit with a broadband output power below the specified limit when the power is set to a minimum value. The broadband output power is defined as the power in the channel bandwidth, for all transmit bandwidth configurations (resource blocks).

An excess minimum output power potentially increases the Rise Over Thermal (RoT) and therefore reduces the cell coverage area for other UEs.

## 6.3.2.4 Test description

### 6.3.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.2.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.3.2.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for min output power test | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 3MHz | | QPSK | 15 | 15 |
| 5MHz | | QPSK | 25 | 25 |
| 10MHz | | QPSK | 50 | 50 |
| 15MHz | | QPSK | 75 | 75 |
| 20MHz | | QPSK | 100 | 100 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, in Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channel is set according to Table 6.3.2.4.1-1.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.2.4.3.

### 6.3.2.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.3.2.1.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC

2. Send continuous uplink power control "down" commands in the uplink scheduling information to the UE to ensure that the UE transmits at its minimum output power.

3. Measure the mean power of the UE in the associated measurement bandwidth specified in Table 6.3.2.5-1 for the specific channel bandwidth under test. The period of measurement shall be the continuous duration of one sub-frame (1ms). For TDD slots with transient periods are not under test.

#### 6.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.3.2.5 Test requirement

The minimum output power measured shall not exceed the values specified in Table 6.3.2.5-1.

**Table 6.3.2.5-1: Minimum output power**

| | **Channel bandwidth / minimum output power / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Minimum output power | For carrier frequency f ≤ 3.0GHz: ≤ -39 dBm<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: ≤ -38.7 dBm | | | | | |
| Measurement bandwidth (Note 1) | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| Note 1: Different implementations such as FFT or spectrum analyzer approach are allowed. For spectrum analyzer approach the measurement bandwidth is defined as an equivalent noise bandwidth. | | | | | | |

## 6.3.3 Transmit OFF power

### 6.3.3.1 Test purpose

To verify that the UE transmit OFF power is lower than the value specified in the test requirement.

### 6.3.3.2 Test applicability

The requirements of this test apply in test cases 6.3.4.1 General ON/OFF time mask and 6.3.4.2 PRACH and SRS time mask to all types of E-UTRA UE release 8 and forward.

### 6.3.3.3 Minimum conformance requirement

The transmit OFF power is defined as the mean power in a duration of at least one sub-frame (1ms) excluding any transient periods. The requirement for the transmit OFF power shall not exceed the values specified in Table 6.3.3.3-1.

**Table 6.3.3.3-1: Transmit OFF power**

| | **Channel bandwidth / Transmit OFF power / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Transmit OFF power | -50 dBm | | | | | |
| Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |

The normative reference for this requirement is TS 36.101 [2] clause 6.3.3.

Transmit OFF power is defined as the mean power when the transmitter is OFF. The transmitter is considered to be OFF when the UE is not allowed to transmit or during periods when the UE is not transmitting a sub-frame. During DTX and measurements gaps, the UE is not considered to be OFF.

An excess transmit OFF power power potentially increases the Rise Over Thermal (RoT) and therefore reduces the cell coverage area for other UEs

### 6.3.3.4 Test description

This test is covered by clause 6.3.4.1 General ON/OFF time mask and 6.3.4.2 PRACH and SRS time mask.

### 6.3.3.5 Test requirement

The requirement for the transmit OFF power shall not exceed the values specified in Table 6.3.3.5-1.

**Table 6.3.3.5-1: Transmit OFF power**

| | Channel bandwidth / Transmit OFF power / measurement bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Transmit OFF power | For carrier frequency f ≤ 3.0GHz: ≤ -48.5 dBm<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: ≤ -48.2 dBm | | | | | |
| Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |

## 6.3.3A UE Transmit OFF power for CA

Editor's notes:

The following items are missing or incomplete:

- Initial conditions (test frequencies, test configurations)

- Test procedure

- Message Contents

- Test requirements and test tolerances

- Connection diagram is missing

### 6.3.3A.1 Test purpose

For intra-band contiguous carrier aggregation, transmit OFF power is defined as the mean power per component carrier when the transmitter is OFF on both component carriers. The transmitter is considered to be OFF when the UE is not allowed to transmit or during periods when the UE is not transmitting a sub-frame. During measurements gaps, the UE is not considered to be OFF.

To verify that the UE transmit OFF power for CA is lower than the value specified in the test requirement.

### 6.3.3A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

### 6.3.3A.3 Minimum conformance requirements

The transmit OFF power is defined as the mean power in a duration of at least one sub-frame (1ms) excluding any transient periods. The transmit OFF power shall not exceed the values specified in Table 6.3.3A.3-1.

**Table 6.3.3A.3-1: Transmit OFF power for intra-band contiguous CA UE**

| | Channel bandwidth / Minimum output power / measurement bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Transmit OFF power | -50 dBm | | | | | |
| Measurement bandwidth | | | | 9.0 MHz | 13.5 MHz | 18 MHz |

The normative reference for this requirement is TS 36.101[2] clause 6.3.3A

### 6.3.3A.4 Test description

#### 6.3.3A.4.1 Initial condition

#### 6.3.3A.4.2 Test procedure

#### 6.2.3A.4.3 Message contents

### 6.3.3A.5 Test Requirements

## 6.3.4 ON/OFF time mask

### 6.3.4.1 General ON/OFF time mask

Editor's note: The measurement period in the minimum requirement is defined to be 1 subframe (14 symbols). Due to practical reasons the TDD measurement period for off power prior the PUSCH is 10 symbols. It is FFS, if this deviation is acceptable.

#### 6.3.4.1.1 Test purpose

To verify that the general ON/OFF time mask meets the requirements given in 6.3.4.1.5.

The time mask for transmit ON/OFF defines the ramping time allowed for the UE between transmit OFF power and transmit ON power.

Transmission of the wrong power increases interference to other channels, or increases transmission errors in the uplink channel.

#### 6.3.4.1.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

#### 6.3.4.1.3 Minimum conformance requirement

The General ON/OFF time mask defines the observation period between Transmit OFF and ON power and between Transmit ON and OFF power. ON/OFF scenarios include; the beginning or end of DTX, measurement gap, contiguous, and non contiguous transmission

The OFF power measurement period is defined in a duration of at least one sub-frame excluding any transient periods. The ON power is defined as the mean power over one sub-frame excluding any transient period.

There are no additional requirements on UE transmit power beyond that which is required in clause 6.2.2 and clause 6.6.2.3.

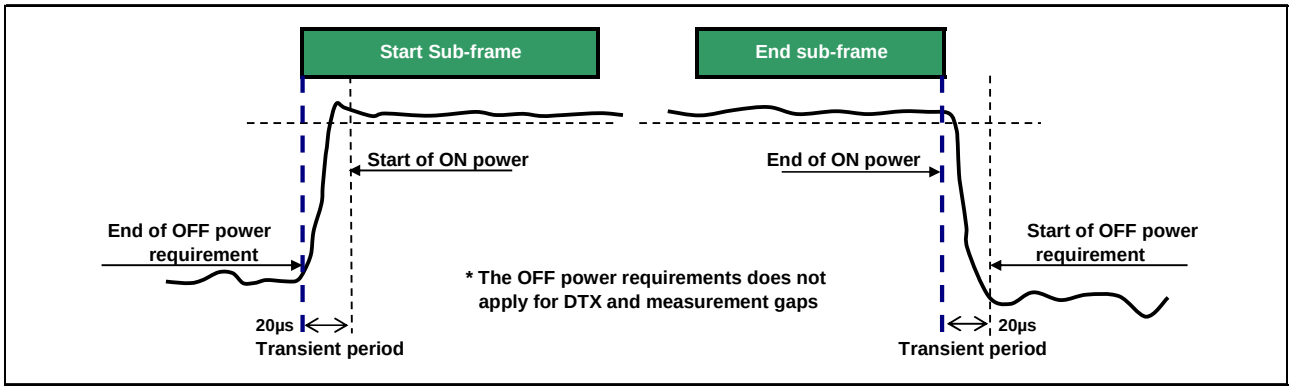



**Figure 6.3.4.1.3-1: General ON/OFF time mask**

The normative reference for this requirement is TS 36.101 [2] clause 6.3.4.1.

### 6.3.4.1.4 Test description

#### 6.3.4.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.4.1.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2

**Table 6.3.4.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for General On/Off Time Mask test case | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 3MHz | | QPSK | 15 | 15 |
| 5MHz | | QPSK | 25 | 25 |
| 10MHz | | QPSK | 50 | 50 |
| 15MHz | | QPSK | 75 | 75 |
| 20MHz | | QPSK | 100 | 100 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL and DL Reference Measurement channels are set according to Table 6.3.4.1.4.1-1.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.4.1.4.3. Note that PDCCH DCI format 0 sent after resetting uplink power with RRC Connection Reconfiguration, should have TPC command 0dB.

#### 6.3.4.1.4.2 Test procedure

1. SS sends uplink scheduling information via PDCCH DCI format 0 with TPC command 0dB for C_RNTI to schedule the UL RMC according to Table 6.3.4.1.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC. The UL assignment is such that the UE transmits on UL sub-frame 2 of every radio frame.

2. For FDD: Measure the UE transmission OFF power during the sub-frame prior to the PUSCH subframe. For TDD: Measure the UE transmission OFF power during the 10 SCFDMA symbols prior to the PUSCH subframe.

3. Measure the output power of the UE PUSCH transmission during one sub-frame, excluding a transient period of 20 µs at the beginning of the subframe.

4. Measure the UE transmission OFF power during one sub-frame following the PUSCH subframe, excluding a transient period of 20 µs at the beginning of the subframe.

#### 6.3.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 6.3.4.1.4.3-1: UplinkPowerControlCommon: Test point 1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::= SEQUENCE { p0-NominalPUSCH | -105 | Test point 1 to verify a UE relative low initial power transmission | |

**Table 6.3.4.1.4.3-2: PhysicalConfigDedicated**

| Derivation Path: TS 36.508 [7] clause 5.5.1.2, Table 5.5.1.2.1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| UplinkPowerControlDedicated | UplinkPowerControlDedicated-DEFAULT | See subclause 4.6.3 | SRB1 |
| | UplinkPowerControlDedicated-DEFAULT | See subclause 4.6.3 | RBC |

**Table 6.3.4.1.4.3-3: UplinkPowerControlDedicated**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-26 UplinkPowerControlDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UE-PUSCH | 1 | | SRB1 |
| | 0 | | RBC |
| } | | | |

**Table 6.3.4.1.3-4: TDD-Config-DEFAULT: On/OFF time mask measurement**

| Derivation Path: 36.508 clause 5.3.1 Table 5.3.1-1 (*SystemInformationBlockType1*) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | sa1 | | |
| specialSubframePatterns | ssp5 | To enable two symbol UpPTS, and to have 9 symbols GP. | |
| } | | | |

### 6.3.4.1.5 Test requirement

The requirement for the power measured in steps (2), (3) and (4) of the test procedure shall not exceed the values specified in Table 6.3.4.1.5-1.

**Table 6.3.4.1.5-1: General ON/OFF time mask**

| | **Channel bandwidth / minimum output power / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Transmit OFF power | For carrier frequency f ≤ 3.0GHz: ≤ -48.5 dBm<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: ≤ -48.2 dBm | | | | | |
| Transmission OFF Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| Expected Transmission ON Measured power | -14.8 dBm | -10.8 dBm | -8.6 dBm | -5.6 dBm | -3.9 dBm | -2.6 dBm |
| ON power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB |

## 6.3.4.2 PRACH and SRS time mask

### 6.3.4.2.1 PRACH time mask

#### 6.3.4.2.1.1 Test purpose

To verify that the PRACH time mask meets the requirements given in 6.3.4.2.1.5.

The time mask for PRACH time mask defines the ramping time allowed for the UE between transmit OFF power and transmit ON power when transmitting the PRACH.

Transmission of the wrong power increases interference to other channels, or increases transmission errors in the uplink channel.

#### 6.3.4.2.1.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

#### 6.3.4.2.1.3 Minimum conformance requirement

For the PRACH Power / Time mask defines the observation period for PRACH transmissions. The PRACH ON power is specified as the mean power over the PRACH measurement period excluding any transient periods. The measurement period for different PRACH preamble format is specified in Table 6.3.4.2.1.3-1.

There are no additional requirements on UE transmit power beyond that which is required in clause 6.2.2 and clause 6.6.2.3

**Table 6.3.4.2.1.3-1: PRACH ON power measurement period**

| PRACH preamble format | Measurement period (ms) |
|---|---|
| 0 | 0.9031 |
| 1 | 1.4844 |
| 2 | 1.8031 |
| 3 | 2.2844 |
| 4 | 0.1479 |

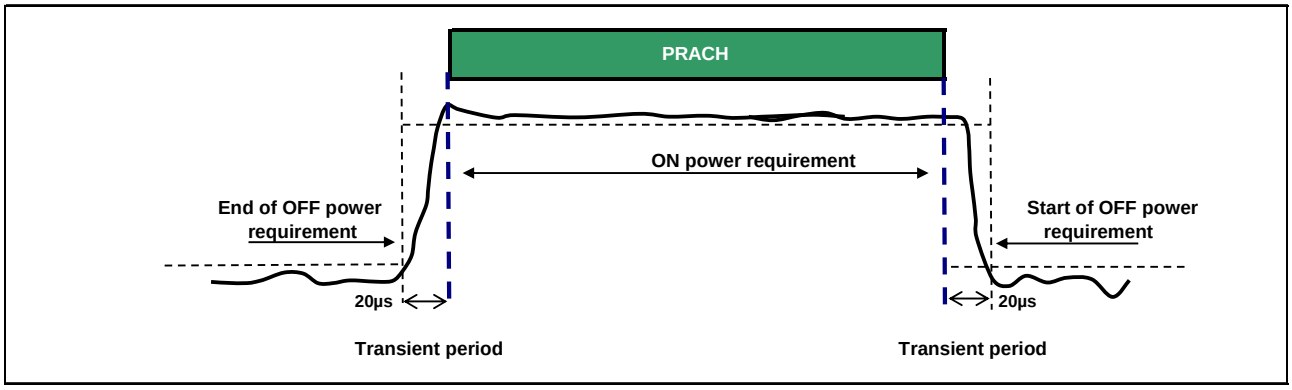



**Figure 6.3.4.2.1.3-1: PRACH ON/OFF time mask**

The normative reference for this requirement is TS 36.101 [2] clause 6.3.4.2.1.

### 6.3.4.2.1.4 Test description

#### 6.3.4.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.4.2.1.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3.

**Table 6.3.4.2.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | |
|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | Normal, TL/VL, TL/VH, TH/VL, TH/VH | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | Mid range | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | Lowest, 5MHz, Highest | |
| PRACH preamble format | | |
| | FDD | TDD |
| PRACH Configuration Index | 3 | 51 |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C.0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.4.2.1.4.3.

#### 6.3.4.2.1.4.2 Test procedure

1. The SS shall signal a Random Access Preamble ID via a PDCCH order to the UE and initiate a Non-contention based Random Access procedure.

2. The UE shall send the signalled preamble to the SS.

3. For FDD UE, the SS measure the UE transmission OFF power during the sub-frame preceding the PRACH preamble excluding a transient period of 20 µs according to Figure 6.3.4.2.1.3-1. For TDD UE, the SS measure the UE transmission OFF power starting (20us+the duration of 8 OFDM symbols) before the PRACH starts, and ending 20us before PRACH starts. Note, the nominal PRACH timing for TDD is not aligned with the sub frame and symbol raster.

4. Measure the output power of the transmitted PRACH preamble according to Figure 6.3.4.2.1.3-1.

5. Measure the UE transmission OFF power, starting 20 µs after the PRACH preamble ends for a measurement period of 980 µs..

#### 6.3.4.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 6.3.4.2.1.4.3-1: *RACH-ConfigCommon-DEFAULT:* PRACH measurement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| powerRampingParameters SEQUENCE { | | | |
| powerRampingStep | dB0 | | |
| preambleInitialReceivedTargetPower | dBm-104 | | PRACH Format 0 |
| | dBm-112 | | PRACH Format 4 |
| } | | | |
| ) | | | |

**Table 6.3.4.2.1.4.3-2: *PRACH-Config-DEFAULT:* PRACH measurement for TDD**

| Derivation Path: TS 36.508 [7] clause 5.3.1, Table 5.3.1-3: PRACH-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PRACH-Config-DEFAULT ::= SEQUENCE { | | | |
| prach-ConfigIndex | 51 | | TDD |
| } | | | |

**Table 6.3.4.2.1.4.3-3: *TDD-Config-DEFAULT:* PRACH measurement for TDD**

| Derivation Path: TS 36.508 [7] clause 5.3.1, Table 5.3.1-1: TDD-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | sa1 | | |
| specialSubframePatterns | ssp5 | To enable two symbol UpPTS, and to have 9 symbols GP. | |
| } | | | |

#### 6.3.4.2.1.5 Test requirement

The requirement for the power measured in steps (3), (4) and (5) of the test procedure shall not exceed the values specified in Table 6.3.4.2.1.5-1.

**Table 6.3.4.2.1.5-1: PRACH time mask**

| | **Channel bandwidth / Output Power [dBm] / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Transmit OFF power | For carrier frequency f ≤ 3.0GHz: ≤ -48.5 dBm<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: ≤ -48.2 dBm | | | | | |
| Transmission OFF Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| Expected PRACH Transmission ON Measured power | -1 dBm | -1 dBm | -1 dBm | -1 dBm | -1 dBm | -1 dBm |
| ON power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB |

### 6.3.4.2.2 SRS time mask

#### 6.3.4.2.2.1 Test purpose

To verify that the SRS time mask meets the requirements given in 6.3.4.2.2.5.

The time mask for SRS time mask defines the ramping time allowed for the UE between transmit OFF power and transmit ON power when transmitting the SRS.

Transmission of the wrong power increases interference to other channels, or increases transmission errors in the uplink channel.

#### 6.3.4.2.2.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

#### 6.3.4.2.2.3 Minimum conformance requirement

In the case a single SRS transmission, the ON power is defined as the mean power for each symbol duration excluding any transient period. Figure 6.3.4.2.2.3-1.

In the case a dual SRS transmission, the ON power is defined as the mean power for each symbol duration excluding any transient period. Figure 6.3.4.2.2.3-2.

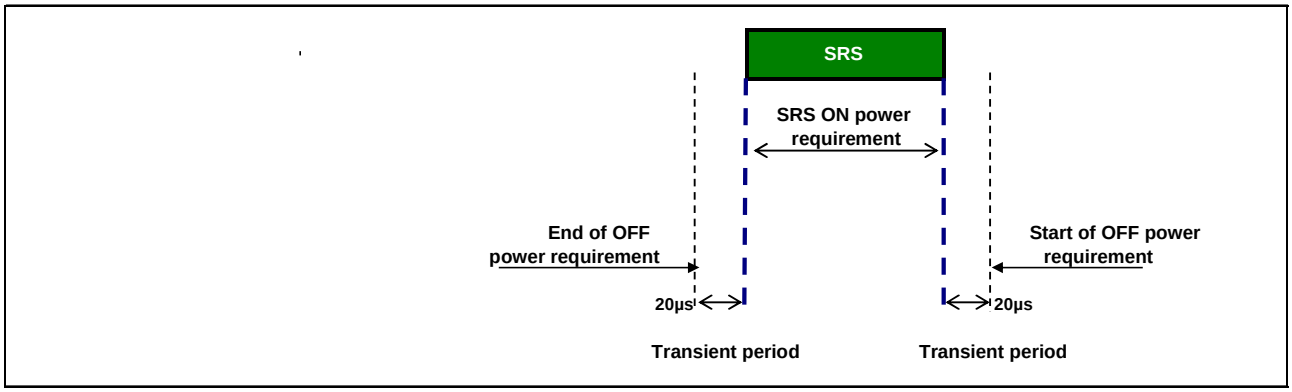



**Figure 6.3.4.2.2.3-1: Single SRS time mask**

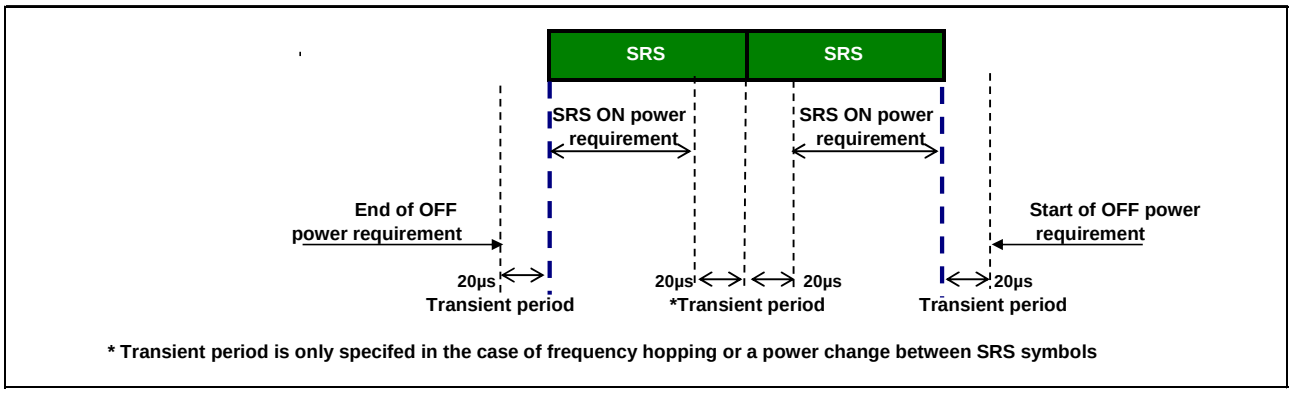



**Figure 6.3.4.2.2.3-2: Dual SRS time mask for the case of UpPTS transmissions**

### 6.3.4.2.2.4 Test description

#### 6.3.4.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.4.2.2.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3.

**Table 6.3.4.2.2.4.1-1: Test Configuration Table**

| **Initial Conditions** | | |
|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | Normal, TL/VL, TL/VH, TH/VL, TH/VH | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | Mid range | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | Lowest, 5MHz, Highest | |
| SRS configuration | | |
| | FDD | TDD |
| srs-BandwidthConfig | bw7 | bw7 (for BW 1.4 MHz)<br>bw5 (for BW 3 MHz)<br>bw2 (for BW 5 MHz)<br>bw0 (for BW 10, 15, 20 MHz) |
| srs-SubframeConfig | sc3 | sc0 |
| ackNackSRS-SimultaneousTransmission | FALSE | FALSE |
| srsMaxUpPts | N/A | N/A |
| srs-Bandwidth | bw3 | bw3 |
| srs-HoppingBandwidth | hbw3 | hbw0 |
| freqDomainPosition | 0 | 0 |
| Duration | TRUE | TRUE |
| srs-ConfigIndex | 7 | 0 |
| transmissionComb | 0 | 0 |
| cyclicShift | cs0 | cs0 |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.4.2.2.4.3. Note that PDCCH DCI format 0 sent after resetting uplink power with RRC Connection Reconfiguration, should have TPC command 0dB.

### 6.3.4.2.2.4.2 Test procedure

1. For FDD UE, the SS measure the UE transmission OFF power during the 13 symbols preceding the SRS symbol excluding a transient period of 20 µs according to Figure 6.3.4.2.2.3-1. For TDD UE, the SS measure the UE transmission OFF power during the 8 symbols preceding the two SRS symbols excluding a transient period of 20µs according to Figure 6.3.4.2.2.3-2.

2 Measure the output power of the transmitted SRS according to Figure 6.3.4.2.2.3-1 for FDD UE and according to Figure 6.3.4.2.2.3-2 for TDD UE, The transient periods are excluded from measurement accordingly.

3. Measure the UE transmission OFF power during the sub-frame following the SRS under test, excluding a transient period of 20 µs according to Figure 6.3.4.2.2.3-1 for FDD UE and according to Figure 6.3.4.2.2.3-2 for TDD UE.

### 6.3.4.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 6.3.4.2.2.4.3-1: *RadioResourceConfigCommonSIB-DEFAULT*: SRS measurement**

| Derivation Path: 36.508 clause 5.3.1 Table 5.3.1-2 RadioResourceConfigCommonSIB-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| rach-ConfigCommon | RACH-ConfigCommon-DEFAULT | | |
| bcch-Config | BCCH-Config-DEFAULT | | |
| pcch-Config | PCCH-Config-DEFAULT | | |
| prach-Config | PRACH-ConfigSIB-DEFAULT | | |
| pdsch-ConfigCommon | Not present | | |
| pusch-ConfigCommon | PUSCH-ConfigCommon-DEFAULT | | |
| pucch-ConfigCommon | PUCCH-ConfigCommon-DEFAULT | | |
| soundingRSUL-ConfigCommon | *SoundingRS-UL-ConfigCommon-DEFAULT* | | |
| uplinkPowerControlCommon | UplinkPowerControlCommon-DEFAULT | | |
| ul-CyclicPrefixLength | len1 | | |
| } | | | |

**Table 6.3.4.2.2.4.3-2: *SoundingRS-UL-ConfigCommon-DEFAULT*: SRS time mask measurement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-21 SoundingRS-UL-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| setup SEQUENCE { | | | |
| srs-BandwidthConfig | bw7 | | FDD |
| | bw7 | | TDD (BW 1.4 MHz) |
| | bw5 | | TDD (BW 3 MHz) |
| | bw2 | | TDD (BW 5 MHz) |
| | bw0 | | TDD (BW 10, 15, 20 MHz) |
| srs-SubframeConfig | sc3 | Periodicity of 5ms, with offset of 0. | FDD |
| | sc0 | Periodicity of 5ms, with offset of 1. | TDD |
| ackNackSRS-SimultaneousTransmission | FALSE | | |
| srsMaxUpPts | Not present | | |
| } | | | |
| } | | | |

**Table 6.3.4.2.2.4.3-3: PhysicalConfigDedicated-DEFAULT: SRS time mask measurement**

| Derivation Path: 36.508 clause 5.5.1 Table 5.5.1.2-1: PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | Not present | | |
| soundingRS-UL-ConfigDedicated | *SoundingRSUL-ConfigDedicated-DEFAULT* | | |
| } | | | |

**Table 6.3.4.2.2.4.3-4: *SoundingRSUL-ConfigDedicated-DEFAULT*: SRS time mask measurement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-22 SoundingRS-UL-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SoundingRS-UL-ConfigDedicated-DEFAULT ::= CHOICE { | | | |
| setup SEQUENCE { | | | |
| srs-Bandwidth | bw3 | bw3 used to ensure that the bandwidth is constantly 4 RBs irrespective of channel bandwidth. | |
| srs-HoppingBandwidth | hbw3 | This is selected so that hopping is disabled | FDD |
| | hbw0 | This is selected so that hopping is enabled | TDD |
| freqDomainPosition | 0 | | |
| Duration | TRUE | Indefinite duration | |
| srs-ConfigIndex | 7 | SRS periodicity of 10ms, Toffset=0. | FDD |
| | 0 | SRS periodicity of 2ms, Ksrs=0,1, this is two symbols UpPTS in first half subframe. | TDD |
| transmissionComb | 0 | | |
| cyclicShift | cs0 | No cyclic shift | |
| } | | | |
| } | | | |

**Table 6.3.4.2.2.4.3-5: TDD-Config-DEFAULT: SRS time mask measurement**

| Derivation Path: 36.508 331 clause 65.3.21 Table 5.3.1-1 (*SystemInformationBlockType1*) | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | sa1 | | |
| specialSubframePatterns | ssp5 | To enable two symbol UpPTS, and to have 9 symbols GP. | |
| } | | | |

**Table 6.3.4.2.2.4.3-6: PhysicalConfigDedicated**

| Derivation Path: TS 36.508 [7] clause 5.5.1.2, Table 5.5.1.2.1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| uplinkPowerControlDedicated | UplinkPowerControlDedicated-DEFAULT | See subclause 4.6.3 | SRB1 |
| | UplinkPowerControlDedicated-DEFAULT | See subclause 4.6.3 | RBC |

**Table 6.3.4.2.2.4.3-7: UplinkPowerControlDedicated**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-26 UplinkPowerControlDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UE-PUSCH | 1 | | SRB1 |
| | 0 | | RBC |
| } | | | |

| Condition | Explanation |
|---|---|
| FDD | FDD cell environment |
| TDD | TDD cell environment |

#### 6.3.4.2.2.5 Test requirement

The requirement for the power measured in steps (1), (2) and (3) of the test procedure shall not exceed the values specified in Table 6.3.4.2.2.5-1.

**Table 6.3.4.2.2.5-1: SRS time mask**

| | **Channel bandwidth / Output Power [dBm] / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Transmit OFF power | For carrier frequency f ≤ 3.0GHz: ≤ -48.5 dBm<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: ≤ -48.2 dBm | | | | | |
| Transmission OFF Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| Expected SRS Transmission ON Measured power | -2.6 dBm | -2.6 dBm | -2.6 dBm | -2.6 dBm | -2.6 dBm | -2.6 dBm |
| ON power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB | ± 7.5dB<br>± 7.8dB |

## 6.3.5 Power Control

Power control is used to limit the interference level and compensate the channel fading. The UE power is defined as the mean power in a subframe or ON power duration, whichever is available.

The UE transmission can be in two contiguity modes, i.e. contiguous transmission and non-contiguous transmission. The former has a transmission gap of 0 and the later has a transmission gap larger than 0. The transmission gap is the time interval between the end of the last UE transmission subframe and the beginning of the next UE transmission subframe or the UpPTS (for TDD).

### 6.3.5.1 Power Control Absolute power tolerance

#### 6.3.5.1.1 Test purpose

To verify the ability of the UE transmitter to set its initial output power to a specific value at the start of a contiguous transmission or non-contiguous transmission with a long transmission gap, i.e. transmission gap is larger than 20 ms.

#### 6.3.5.1.2 Minimum conformance requirement

Absolute power tolerance is the ability of the UE transmitter to set its initial output power to a specific value for the first sub-frame at the start of a contiguous transmission or non-contiguous transmission with a transmission gap larger than 20ms.

The minimum requirement on absolute power tolerance is given in Table 6.3.5.1.2-1 over the power range bounded by the Maximum output power as defined in sub-clause 6.2.2 and the Minimum output power as defined in sub clause 6.3.2.

For operating bands under Note 2 in Table 6.2.2.3-1, the absolute power tolerance as specified in Table 6.3.5.1.2-1 is relaxed by reducing the lower limit by 1.5 dB when the transmission bandwidth is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ − 4 MHz and $F_{UL_high}$.

**Table 6.3.5.1.2-1: Absolute power tolerance**

| Conditions | Tolerance |
|---|---|
| Normal conditions | ± 9.0 dB |
| Extreme conditions | ± 12.0 dB |

The normative reference for this requirement is TS 36.101 [2] clause 6.3.5.1.1.

#### 6.3.5.1.3 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

#### 6.3.5.1.4 Test description

##### 6.3.5.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.5.1.4.1-1. The details of the uplink reference measurement channel (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.3.5.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Power Control Absolute power tolerance test case | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 3MHz | | QPSK | 15 | 15 |
| 5MHz | | QPSK | 25 | 25 |
| 10MHz | | QPSK | 50 | 50 |
| 15MHz | | QPSK | 75 | 75 |
| 20MHz | | QPSK | 100 | 100 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C.0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channel is set according to Table 6.3.5.1.4.1-1.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.5.1.4.3. Note that PDCCH DCI format 0 sent after resetting uplink power with RRC Connection Reconfiguration, should have TPC command 0dB.

#### 6.3.5.1.4.2 Test procedure

1. SS sends uplink scheduling information via PDCCH DCI format 0 with TPC command 0dB for C_RNTI to schedule the UL RMC according to Table 6.3.5.1.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.

2. Measure the initial output power of the first subframe of UE PUSCH first transmission. The transient periods of 20us are excluded.

3. Repeat for the two test points as indicated in section 6.3.5.1.4.3. The timing of the execution between the two test points shall be larger than 20ms.

#### 6.3.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 6.3.5.1.4.3-1: UplinkPowerControlCommon: Test point 1**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::=<br>SEQUENCE {<br>p0-NominalPUSCH | -105 | Test point 1 to verify a UE relative low initial power transmission | |

**Table 6.3.5.1.4.3-2: UplinkPowerControlCommon: Test point 2**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-25 UplinkPowerControlCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlCommon-DEFAULT ::=<br>SEQUENCE {<br>p0-NominalPUSCH | -93 | Test point 2 to verify a UE relative high initial power transmission | |

**Table 6.3.5.1.4.3-3: PhysicalConfigDedicated**

| Derivation Path: TS 36.508 [7] clause 5.5.1.2, Table 5.5.1.2.1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| uplinkPowerControlDedicated | UplinkPowerControlDedicated-DEFAULT | See subclause 4.6.3 | SRB1 |
| | UplinkPowerControlDedicated-DEFAULT | See subclause 4.6.3 | RBC |

**Table 6.3.5.1.4.3-4: UplinkPowerControlDedicated**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-26 UplinkPowerControlDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UE-PUSCH | 1 | | SRB1 |
| | 0 | | RBC |
| } | | | |

## 6.3.5.1.5 Test requirement

The requirement for the power measured in step (2) of the test procedure is not to exceed the values specified in Table 6.3.5.1.5-1 and 6.3.5.1.5-2.

**Table 6.3.5.1.5-1: Absolute power tolerance: test point 1**

| | **Channel bandwidth / expected output power (dBm)** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Expected Measured power Normal conditions | -14.8 dBm | -10.8 dBm | -8.6 dBm | -5.6 dBm | -3.9 dBm | -2.6 dBm |
| Power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB |
| Expected Measured power Extreme conditions | -14.8 dBm | -10.8 dBm | -8.6 dBm | -5.6 dBm | -3.9 dBm | -2.6 dBm |
| Power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB |
| Note 1: The lower power limit shall not exceed the minimum output power requirements defined in sub-clause 6.3.2.3 | | | | | | |

**Table 6.3.5.1.5-2: Absolute power tolerance: test point 2**

| | **Channel bandwidth / expected output power (dBm)** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Expected Measured power Normal conditions | -2.8 dBm | 1.2 dBm | 3.4 dBm | 6.4 dBm | 8.2 dBm | 9.4 dBm |
| Power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB | ± 10.0dB<br>± 10.4dB |
| Expected Measured power Extreme conditions | -2.8 dBm | 1.2 dBm | 3.4 dBm | 6.4 dBm | 8.2 dBm | 9.4 dBm |
| Power tolerance<br>f ≤ 3.0GHz<br>3.0GHz < f ≤ 4.2GHz | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB | ± 13.0dB<br>± 13.4dB |
| Note 1: The lower power limit shall not exceed the minimum output power requirements defined in sub-clause 6.3.2.3 | | | | | | |

## 6.3.5.2 Power Control Relative power tolerance

### 6.3.5.2.1 Test purpose

To verify the ability of the UE transmitter to set its output power relatively to the power in a target sub-frame relatively to the power of the most recently transmitted reference sub-frame if the transmission gap between these sub-frames is ≤ 20 ms.

### 6.3.5.2.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 6.3.5.2.3 Minimum conformance requirement

The UE shall meet the requirements specified in Table 6.3.5.2.3-1.

To account for RF Power amplifier mode changes 2 exceptions are allowed for each of two test patterns. The test patterns are a monotonically increasing power sweep and a monotonically decreasing power sweep over a range bounded by the requirements of minimum power and maximum power specified in clauses 6.3.2.3 and 6.2.2.3. For these exceptions the power tolerance limit is a maximum of ±6.0 dB in Table 6.3.5.2.3-1.

**Table 6.3.5.2.3-1 Relative Power Tolerance for Transmission (normal conditions)**

| Power step ΔP (Up or down) [dB] | All combinations of PUSCH and PUCCH transitions [dB] | All combinations of PUSCH/PUCCH and SRS transitions between sub-frames [dB] | PRACH [dB] |
|---|---|---|---|
| ΔP < 2 | ±2.5 (Note 3) | ±3.0 | ±2.5 |
| 2 ≤ ΔP < 3 | ±3.0 | ±4.0 | ±3.0 |
| 3 ≤ ΔP < 4 | ±3.5 | ±5.0 | ±3.5 |
| 4 ≤ ΔP ≤ 10 | ±4.0 | ±6.0 | ±4.0 |
| 10 ≤ ΔP < 15 | ±5.0 | ±8.0 | ±5.0 |
| 15 ≤ ΔP | ±6.0 | ±9.0 | ±6.0 |

Note 1: For extreme conditions an additional ± 2.0 dB relaxation is allowed

Note 2: For operating bands under Note 2 in Table 6.2.2.3-1, the relative power tolerance is relaxed by increasing the upper limit by 1.5 dB if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges; if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges, then the tolerance is relaxed by reducing the lower limit by 1.5 dB.

Note 3: For PUSCH to PUSCH transitions with the allocated resource blocks fixed in frequency and no transmission gaps other than those generated by downlink subframes, DwPTS fields or Guard Periods for TDD: for a power step ΔP ≤ 1 dB, the relative power tolerance for transmission is ±1.0 dB.

The power step (ΔP) is defined as the difference in the calculated setting of the UE Transmit power between the target and reference sub-frames with the power setting according to Clause 5.1 of TS 36.213. The error is the difference between ΔP and the power change measured at the UE antenna port with the power of the cell-specific reference signals kept constant. The error shall be less than the relative power tolerance specified in Table 6.3.5.2.3-1.

The normative reference for this requirement is TS 36.101 clause 6.3.5.2.

### 6.3.5.2.4 Test description

#### 6.3.5.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.5.4.2.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.3.5.2.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal, TL/VL, TL/VH, TH/VL, TH/VH | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | High range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Power Control Relative power tolerance test case | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | See table 6.3.5.2.5-1 6.3.5.2.5-2 6.3.5.2.5-13 | See table 6.3.5.2.5-1 6.3.5.2.5-2 6.3.5.2.5-13 |
| 3MHz | | QPSK | See table 6.3.5.2.5-3 6.3.5.2.5-4 6.3.5.2.5-13 | See table 6.3.5.2.5-3 6.3.5.2.5-4 6.3.5.2.5-13 |
| 5MHz | | QPSK | See table 6.3.5.2.5-5 6.3.5.2.5-6 6.3.5.2.5-13 | See table 6.3.5.2.5-5 6.3.5.2.5-6 6.3.5.2.5-13 |
| 10MHz | | QPSK | See table 6.3.5.2.5-7 6.3.5.2.5-8 6.3.5.2.5-13 | See table 6.3.5.2.5-7 6.3.5.2.5-8 6.3.5.2.5-13 |
| 15MHz | | QPSK | See table 6.3.5.2.5-9 6.3.5.2.5-10 6.3.5.2.5-13 | See table 6.3.5.2.5-9 6.3.5.2.5-10 6.3.5.2.5-13 |
| 20MHz | | QPSK | See table 6.3.5.2.5-11 6.3.5.2.5-12 6.3.5.2.5-13 | See table 6.3.5.2.5-11 6.3.5.2.5-12 6.3.5.2.5-13 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1<br>Note 2: The starting resource block shall be RB# (max +1 - RB allocation). | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C.0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channel is set according to table 6.3.5.4.2.1-1

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.5.2.4.3.

### 6.3.5.2.4.2 Test procedure

The procedure is separated in various subtests to verify different aspects of relative power control. The power patterns of the subtests are described in figure 6.3.5.2.4.2-1.

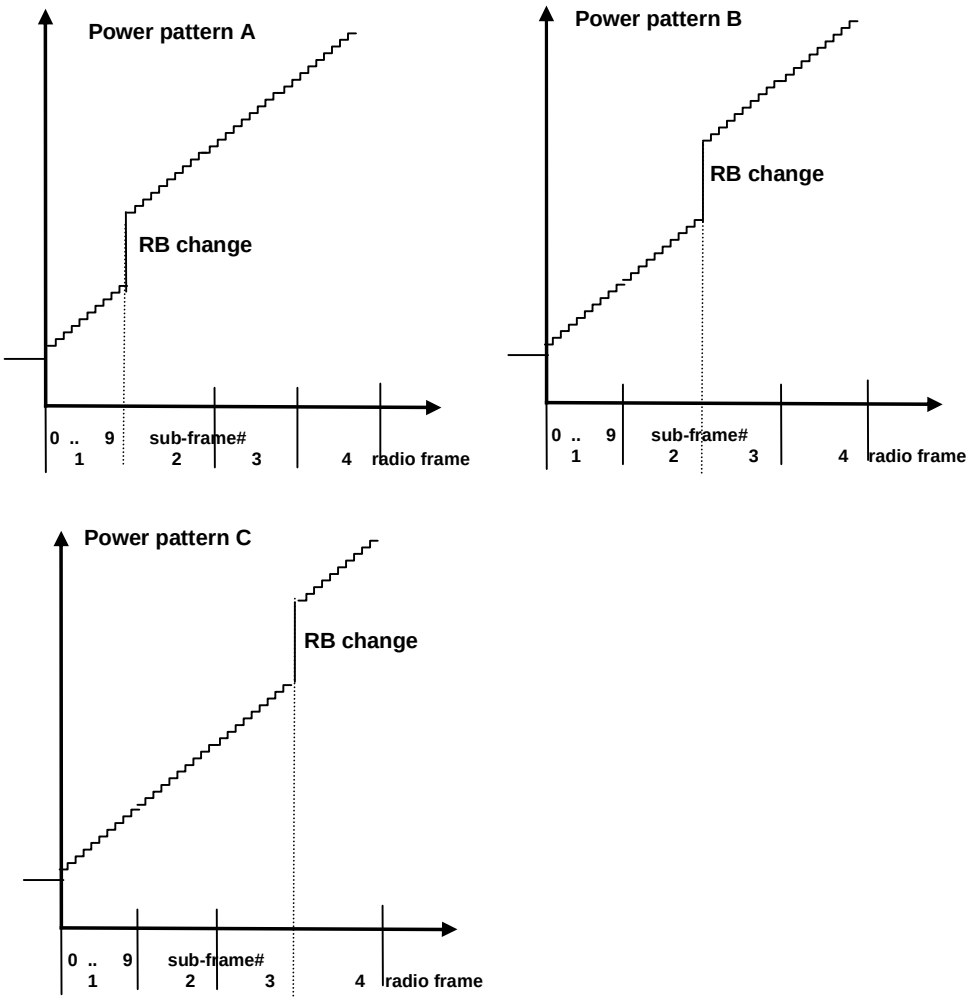



**Figure 6.3.5.2.4.2-1: FDD ramping up test power patterns**

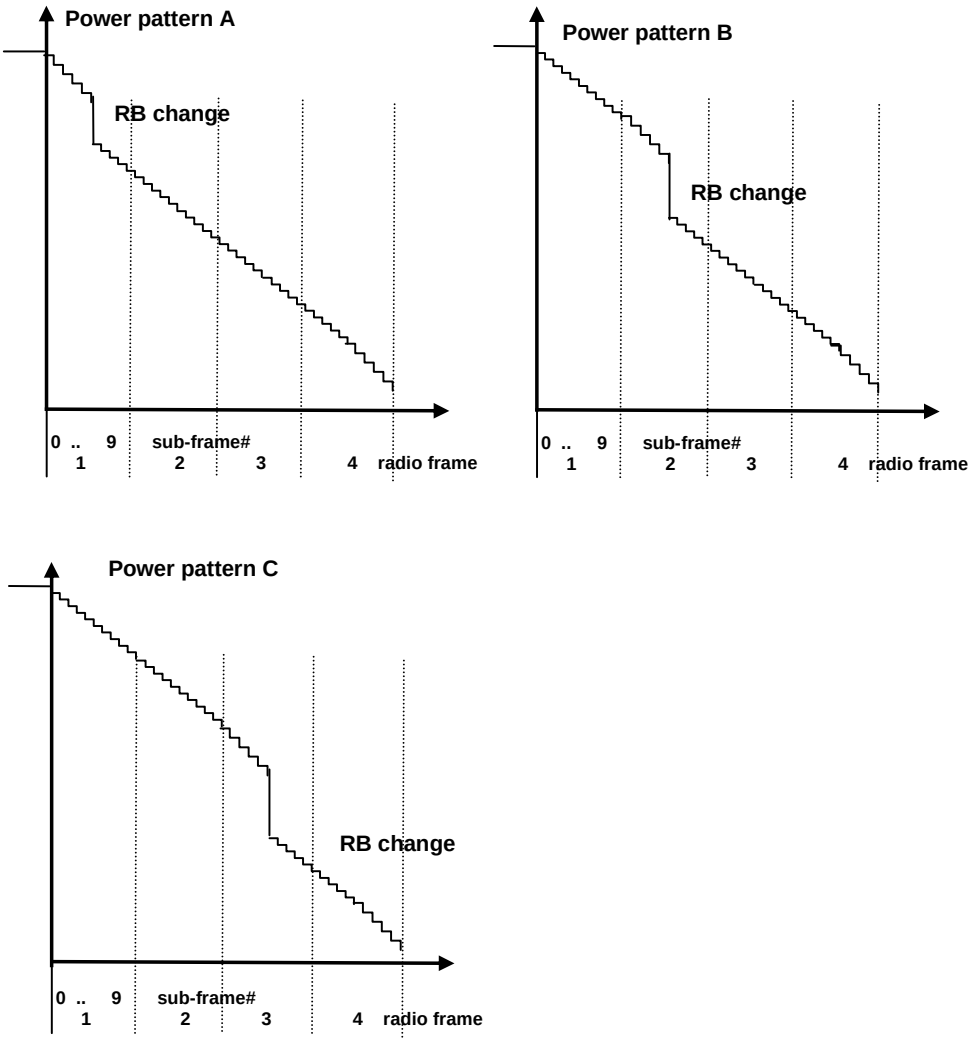



**Figure 6.3.5.2.4.2-2: FDD ramping down test power patterns**

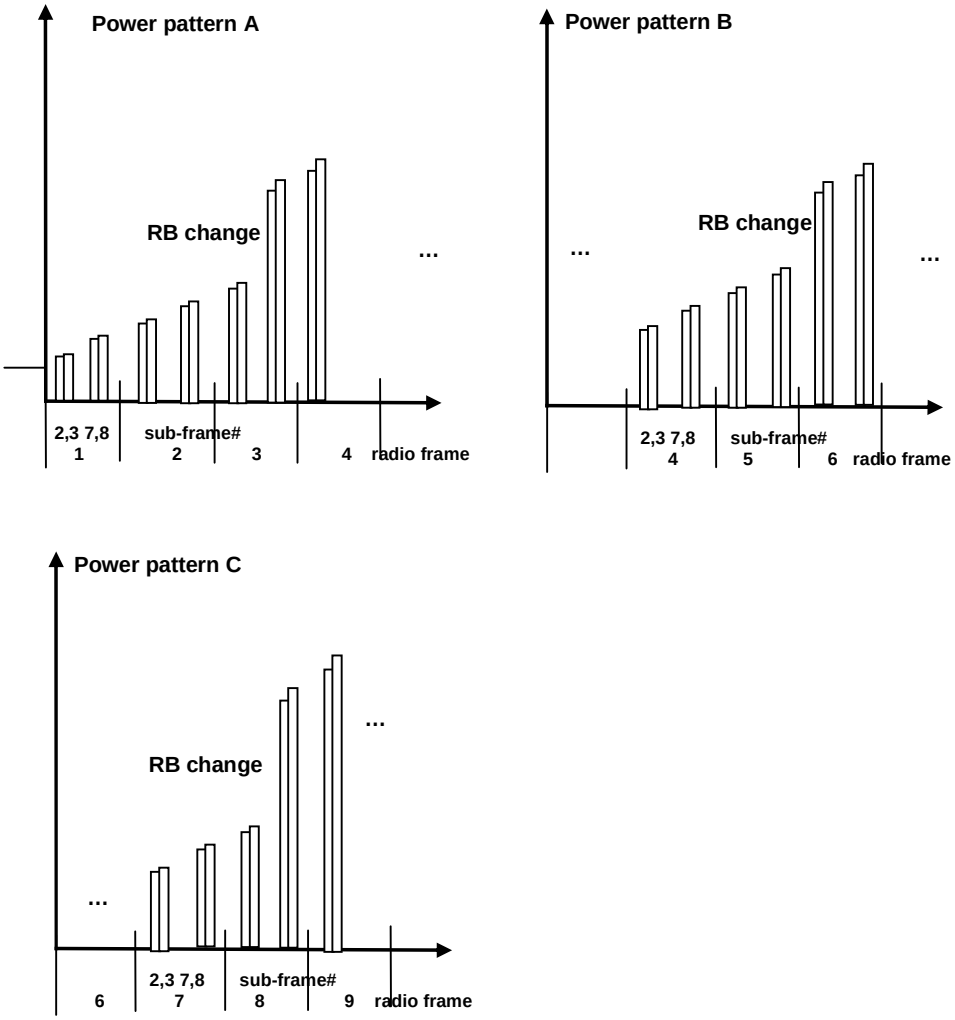



**Figure 6.3.5.2.4.2-3: TDD ramping up test power patterns**

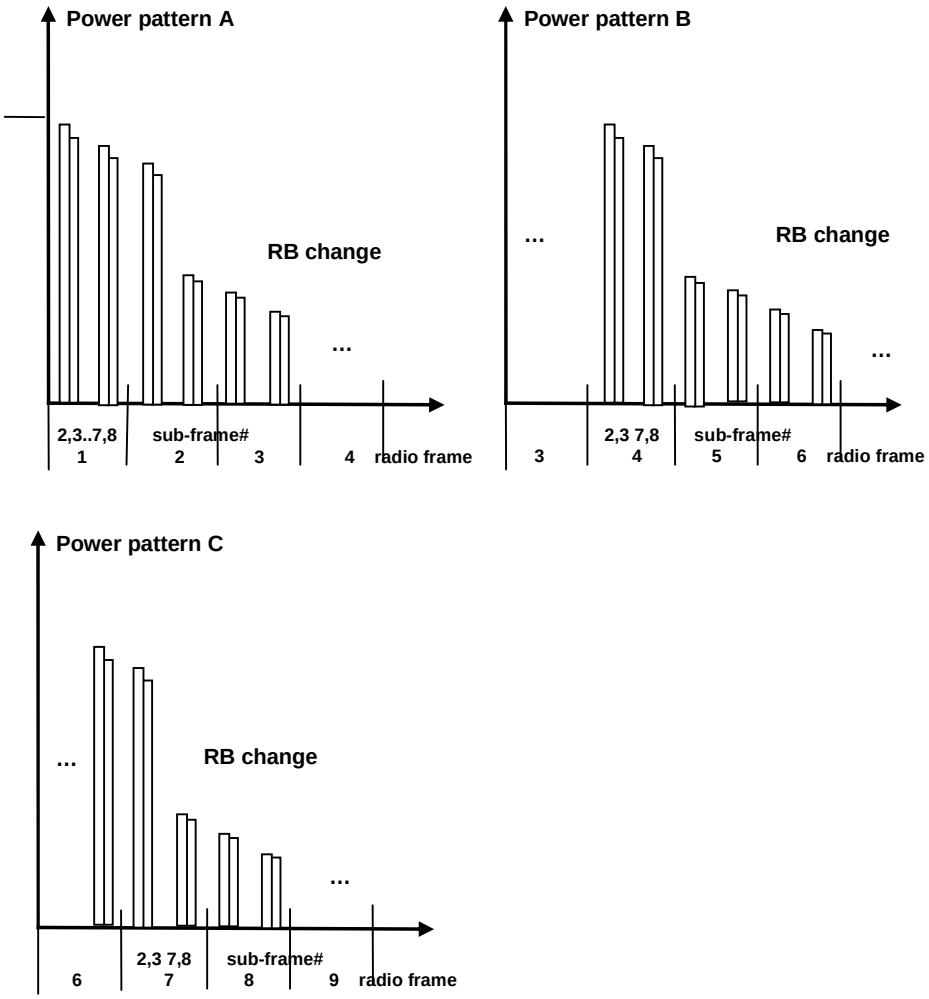



**Figure 6.3.5.2.4.2-4: TDD ramping down test power patterns**

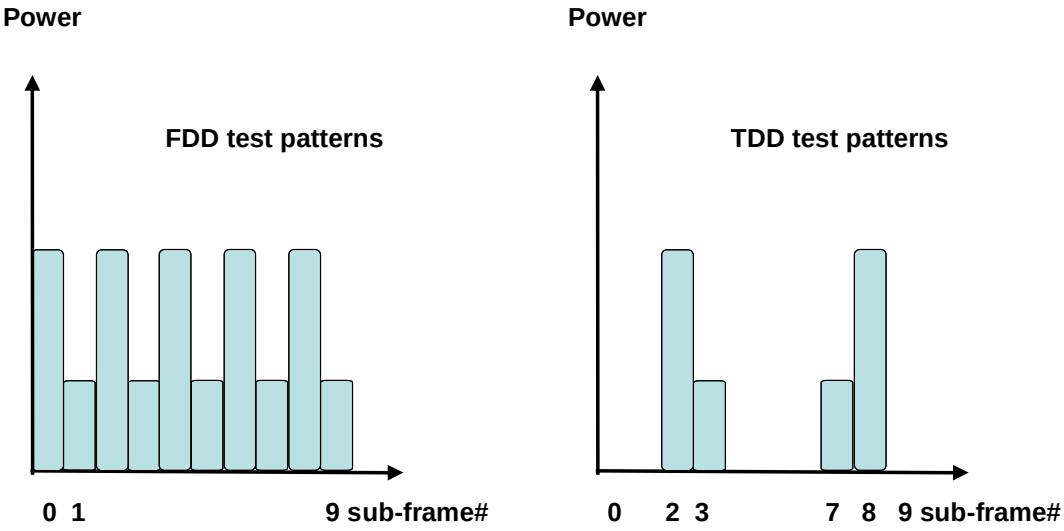



**Figure 6.3.5.2.4.2-5: Alternating Test Power patterns**

1. Sub test: ramping up pattern

1.1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the PUSCH. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC. Send the appropriate TPC commands for PUSCH to the UE to ensure that the UE transmits PUSCH at -36.8dBm +/- 3.2 dB for carrier frequency f $\leq$ 3.0GHz or at -36.5dBm +/- 3.5 dB for carrier frequency 3.0GHz < f $\leq$ 4.2GHz.

1.2. Schedule the UE's PUSCH data transmission as described in Figure 6.3.5.2.4.2-1 (FDD pattern A: sub-test is divided in 4 arbitrary radio frames with 10 active uplink sub-frames per radio frame) and Figure 6.3.5.2.4.2-3 (TDD pattern A: sub-test is divided in 10 arbitrary radio frames with 4 active uplink sub-frames per radio frame) with an uplink RB allocation as defined in tables 6.3.5.2.5-1 thru 6.3.5.2.5-12 depending on channel bandwidth. On the PDCCH format 0 for the scheduling of the PUSCH the SS will transmit a +1dB TPC command. Note that the measurement need not be done continuously, provided that interruptions are whole numbers of frames, and TPC commands of 0dB are sent during the interruption.

1.3. Measure the power of PUSCH transmissions to verify the UE relative power control meet test requirements 6.3.5.2.5. For power transients between subframes, transient periods of 40us between subframes are excluded. For ON/OFF or OFF/ON transients, transient periods of 20 us at the beginning of the subframe are excluded.

1.4. Repeat the subtest different pattern B, C to move the RB allocation change at different points in the pattern as described in Table 6.3.5.2.5-1 thru Table 6.3.5.2.5-12 to force bigger UE power steps at various points in the power range.

2. Sub test: ramping down pattern

2.1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the PUSCH. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC. Send the appropriate TPC commands for PUSCH to the UE to ensure that the UE transmits PUSCH at +18.0dBm +/- 3.2 dB for carrier frequency f $\leq$ 3.0GHz or at +17.7dBm +/- 3.5 dB for carrier frequency 3.0GHz < f $\leq$ 4.2GHz.

2.2. Schedule the UE's PUSCH data transmission as described in Figure 6.3.5.2.4.2-2 (FDD pattern A: sub-test is divided in 4 arbitrary radio frames with 10 active uplink sub-frames per radio frame) and Figure 6.3.5.2.4.2-4 (TDD pattern A: sub-test is divided in 10 arbitrary radio frames with 4 active uplink sub-frames per radio frame) with an uplink RB allocation as defined in tables 6.3.5.2.5-1 thru 6.3.5.2.5-12 depending on channel bandwidth. On the PDCCH format 0 for the scheduling of the PUSCH the SS will transmit a -1dB TPC command. Note that the measurement need not be done continuously, provided that interruptions are whole numbers of frames, and TPC commands of 0dB are sent during the interruption.

2.3. Measure the power of PUSCH transmissions to verify the UE relative power control meet test requirements 6.3.5.2.5. For power transients between subframes, transient periods of 40us between subframes are excluded. For ON/OFF or OFF/ON transients, transient periods of 20 us at the beginning of the subframe are excluded.

2.4. Repeat the subtest different pattern B, C to move the RB allocation change at different points in the pattern as described in Table 6.3.5.2.5-1 thru Table 6.3.5.2.5-12 to force bigger UE power steps at various points in the power range.

3. Sub test: alternating pattern

3.1 SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the PUSCH. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC. Send the appropriate TPC commands for PUSCH to the UE to ensure that the UE transmits PUSCH at -10dBm +/- 3.2 dB for carrier frequency f $\leq$ 3.0GHz or at -10dBm +/- 3.5 dB for carrier frequency 3.0GHz < f $\leq$ 4.2GHz. The initial uplink RB allocation is defined as the smaller uplink RB allocation value specified in tables 6.3.5.2.5-13. The power level and RB allocation are reset for each sub-test.

3.2. Schedule the UE's PUSCH data transmission as described in Figure 6.3.5.2.4.2-5for 10 sub-frames with an uplink RB allocation alternating pattern as defined in table 6.3.5.2.5-13 while transmitting 0dB TPC command for PUSCH via the PDCCH.

3.3. Measure the power of PUSCH transmissions to verify the UE relative power control meet test requirements specified in clause 6.3.5.2.5. For power transients between subframes, transient periods of 40us between subframes are excluded. For ON/OFF or OFF/ON transients, transient periods of 20 us at the beginning of the subframe are excluded.

#### 6.3.5.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.3.5.2.5 Test requirement

Each UE power step measured in the test procedure 6.3.5.2.4.2 should satisfy the test requirements specified in Table 6.3.5.2.5-1, thru 6.3.5.2.5-13 for normal conditions; for extreme conditions an additional ± 2.0 dB relaxation is allowed.

To account for RF Power amplifier mode changes 2 exceptions are allowed for each of ramping up and ramping down test patterns. For these exceptions the power tolerance limit is a maximum of ±6.7 dB. If there is an exception in the power step caused by the RB change for all test patterns (A, B, C) then fail the UE.

**Table 6.3.5.2.5-1: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 1.4MHz (ramping up)**

| Sub-test (ramp up) | Uplink RB allocation | TPC command | Expected power step size (Up) ΔP [dB] | Power step size range (Up) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 1 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 1 to 6 RBs | TPC=+1dB | 8.78 | 4 ≤ ΔP < 10 | 8.78 ± (4.7) Note 2<br>8.78 +6.2/-4.7 Note 3 |
| Subframes after RB change | Fixed = 6 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 10 active uplink subframes
Pattern B the position of RB uplink allocation change is after 20 active uplink subframes
Pattern C the position of RB uplink allocation change is after 30 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: N/A
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-2: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 1.4MHz (ramping down)**

| Sub-test (ramp down) | Uplink RB allocation | TPC command | Expected power step size (down) ΔP [dB] | Power step size range (down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 5 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 5 to 1 RBs | TPC=-1dB | 7.99 | 4 ≤ ΔP < 1 | 7.99 ± (4.7) Note 2<br>7.99 +4.7/-6.2 Note 4 |
| Subframes after RB change | Fixed = 1 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 6 active uplink subframes
Pattern B the position of RB uplink allocation change is after 16 active uplink subframes
Pattern C the position of RB uplink allocation change is after 26 active uplink subframes
Note 2: When Note 4 does not apply.
Note 3: N/A
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-3: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 3MHz (ramping up)**

| Sub-test (ramp up) | Uplink RB allocation | TPC command | Expected power step size (Up) ΔP [dB] | Power step size range (Up) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 1 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 1 to 4 RBs | TPC=+1dB | 7.02 | 4 ≤ ΔP < 10 | 7.02 ± (4.7) Note 2<br>7.02 +6.2/-4.7 Note 3 |
| Subframes after RB change | Fixed =4 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 10 active uplink subframes
Pattern B the position of RB uplink allocation change is after 20 active uplink subframes
Pattern C the position of RB uplink allocation change is after 30 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: N/A
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-4: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 3MHz (ramping down)**

| Sub-test (ramp down) | Uplink RB allocation | TPC command | Expected power step size (down) ΔP [dB] | Power step size range (down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 15 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 15 to 1 RBs | TPC=-1dB | 12.76 | 10 ≤ ΔP < 15 | 12.76 ± (5.7) Note 2<br>12.76 +5.7/-7.2 Note 4 |
| Subframes after RB change | Fixed =1 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 6 active uplink subframes
Pattern **A** the position of RB uplink allocation change is after 16 active uplink subframes
Pattern C the position of RB uplink allocation change is after 26 active uplink subframes
Note 2: When Note 4 does not apply.
Note 3: N/A
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-5: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 5MHz (ramping up)**

| Sub-test (ramp up) | Uplink RB allocation | TPC command | Expected power step size (Up) ΔP [dB] | Power step size range (Up) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 1 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 1 to 20 | TPC=+1dB | 14.01 | 10 ≤ ΔP < 15 | 14.01 ± (5.7) Note 2<br>14.01 +7.2/-5.7 Note 3 |
| Subframes after RB change | Fixed = 20 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 10 active uplink subframes
Pattern B the position of RB uplink allocation change is after 20 active uplink subframes
Pattern C the position of RB uplink allocation change is after 30 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: N/A
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-6: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 5MHz (ramping down)**

| Sub-test (ramp down) | Uplink RB allocation | TPC command | Expected power step size (down) ΔP [dB] | Power step size range (down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 25 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 25 to 1 | TPC=-1dB | 14.98 | 10 ≤ ΔP < 15 | 14.98 ± (5.7) Note 2<br>14.98 +5.7/-7.2 Note 4 |
| Subframes after RB change | Fixed = 1 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 6 active uplink subframes
Pattern B the position of RB uplink allocation change is after 16 active uplink subframes
Pattern C the position of RB uplink allocation change is after 26 active uplink subframes
Note 2: When Note 4 does not apply.
Note 3: N/A
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-7: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 10MHz (ramping up)**

| Sub-test (ramp up) | Uplink RB allocation | TPC command | Expected power step size (Up) ΔP [dB] | Power step size range (Up) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 1 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 1 to 25 | TPC=+1dB | 14.98 | 10 ≤ ΔP < 15 | 14.98 ± (5.7) Note 2<br>14.98 +7.2/-5.7 Note 3 |
| Subframes after RB change | Fixed = 25 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 10 active uplink subframes
Pattern B the position of RB uplink allocation change is after 20 active uplink subframes
Pattern C the position of RB uplink allocation change is after 30 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: N/A
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-8: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 10MHz (ramping down)**

| Sub-test (ramp down) | Uplink RB allocation | TPC command | Expected power step size (down) ΔP [dB] | Power step size range (down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 50 (UE Cat 2-5)<br><br>Fixed = 48 (UE Cat 1) | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 50 to 1 (UE cat 2-5)<br><br>Change from 48 to 1 (UE cat 1) | TPC=-1dB | 17.99<br><br>17.81 | 15 ≤ Δ | 17.99 ± (6.7) Note 2<br>17.99 +6.7/-8.2 Note 4<br><br>17.81 ± (6.7) Note 2<br>17.81 +6.7/-8.2 Note 4 |
| Subframes after RB change | Fixed = 1 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 6 active uplink subframes
Pattern B the position of RB uplink allocation change is after 16 active uplink subframes
Pattern C the position of RB uplink allocation change is after 26 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: N/A
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-9: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 15MHz (ramping up)**

| Sub-test (ramp up) | Uplink RB allocation | TPC command | Expected power step size (Up) ΔP [dB] | Power step size range (Up) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 1 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 1 to 50 | TPC=+1dB | 17.99 | 15 ≤ ΔP | 17.99±(6.7) Note 2<br>17.99 +8.2/-6.7 Note 3 |
| Subframes after RB change | Fixed = 50 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 10 active uplink subframes
Pattern B the position of RB uplink allocation change is after 20 active uplink subframes
Pattern C the position of RB uplink allocation change is after 30 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: N/A
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-10: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 15MHz (ramping down)**

| Sub-test (ramp down) | Uplink RB allocation | TPC command | Expected power step size (down) ΔP [dB] | Power step size range (down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 75 (UE Cat 2-5)<br><br>Fixed = 50 (UE Cat 1) | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 75 to 1 (UE Cat 2-5)<br><br>Change from 50 to 1 (UE Cat 1) | TPC=-1dB | 19.75<br><br>17.99 | 15 ≤ ΔP | 19.75 ± (6.7) Note 2<br>19.75 +6.7/-8.2 Note 4<br><br>17.99 ± (6.7) Note 2<br>17.99 +6.7/-8.2 Note 4 |
| Subframes after RB change | Fixed = 1 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 6 active uplink subframes
Pattern B the position of RB uplink allocation change is after 16 active uplink subframes
Pattern C the position of RB uplink allocation change is after 26 active uplink subframes
Note 2: When Note 4 does not apply.
Note 3: N/A
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-11: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 20MHz (ramping up)**

| Sub-test (ramp up) | Uplink RB allocation | TPC command | Expected power step size (Up) ΔP [dB] | Power step size range (Up) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 1 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 1 to 75 | TPC=+1dB | 19.75 | 15 ≤ ΔP | 19.75 ± (6.7) Note 2<br>19.75 +8.2/-6.7 Note 3 |
| Subframes after RB change | Fixed = 75 | TPC=+1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 10 active uplink subframes
Pattern B the position of RB uplink allocation change is after 20 active uplink subframes
Pattern C the position of RB uplink allocation change is after 30 active uplink subframes
Note 2: When Note 3 does not apply.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: N/A
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-12: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) channel bandwidth 20MHz (ramping down)**

| Sub-test (ramp down) | Uplink RB allocation | TPC command | Expected power step size (down) ΔP [dB] | Power step size range (down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| Subframes before RB change | Fixed = 100 (UE Cat 2-5)<br><br>Fixed = 75 (UE Cat 1) | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |
| RB change | Change from 100 to 1 (UE Cat 2-5)<br><br>Change from 75 to 1 (UE Cat 1) | TPC=-1dB | 21.0<br><br>19.75 | 15 ≤ ΔP | 21.0 ± (6.7) Note 2<br>21.0 +6.7/-8.2 Note 4<br><br>19.75 ± (6.7) Note 2<br>19.75 +6.7/-8.2 Note 4 |
| Subframes after RB change | Fixed = 1 | TPC=-1dB | 1 | ΔP < 2 | 1 ± (1.7) |

Note 1: Position of RB change:
Pattern A the position of RB uplink allocation change is after 6 active uplink subframes
Pattern B the position of RB uplink allocation change is after 16 active uplink subframes
Pattern C the position of RB uplink allocation change is after 26 active uplink subframes
Note 2: When Note 4 does not apply.
Note 3: N/A
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

**Table 6.3.5.2.5-13: Test Requirements Relative Power Tolerance for Transmission (normal conditions – Note 5) (Alternating pattern)**

| Sub-test | Uplink RB allocation | TPC command | Expected power step size (Up or down) ΔP [dB] | Power step size range (Up or down) ΔP [dB] | PUSCH [dB] |
|---|---|---|---|---|---|
| 1.4 MHz | **Alternating 1 and 6** | **TPC=0dB** | 7.78 | 4 ≤ ΔP < 10 | 7.78 ± (6.7) Note 1,2<br>7.78 +8.2/-6.7 Note 3<br>7.78 +6.7/-8.2 Note 4 |
| 3 MHz | **Alternating 1 and 15** | **TPC=0dB** | 11.76 | 10 ≤ ΔP < 15 | 11.76 ± (6.7) Note 1,2<br>11.76 +8.2/-6.7 Note 3<br>11.76 +6.7/-8.2 Note 4 |
| 5 MHZ | **Alternating 1 and 25** | **TPC=0dB** | 13.98 | 10 ≤ ΔP < 15 | 13.98 ± (6.7) Note 1<br>13.98 +8.2/-6.7 Note 2<br>13.98 +6.7/-8.2 Note 3 |
| 10 MHZ | **Alternating 1 and 50 (UE Cat 2-5)** | **TPC=0dB** | 16.99 | 15 ≤ ΔP | 16.99 ± (6.7) Note 1,2<br>16.99 +8.2/-6.7 Note 3<br>16.99 +6.7/-8.2 Note 4 |
| | **Alternating 1 and 48 (UE Cat 1)** | | 16,81 | | 16.81 ± (6.7) Note 1,2<br>16.81 +8.2/-6.7 Note 3<br>16.81 +6.7/-8.2 Note 4 |
| 15 MHZ | **Alternating 1 and 75 (UE Cat 2-5)** | **TPC=0dB** | 18.75 | 15 ≤ ΔP | 18.75 ± (6.7) Note 1,2<br>18.75 +8.2/-6.7 Note 3<br>18.75 +6.7/-8.2 Note 4 |
| | **Alternating 1 and 50 (UE Cat 1)** | | 16.99 | | 16.99 ± (6.7) Note 1,2<br>16.99 +8.2/-6.7 Note 3<br>16.99 +6.7/-8.2 Note 4 |
| 20 MHZ | **Alternating 1 and 100 (UE Cat 2-5)** | **TPC=0dB** | 20.00 | 15 ≤ ΔP | 20.00 ± (6.7) Note 1,2<br>20.00 +8.2/-6.7 Note 3<br>20.00 +6.7/-8.2 Note 4 |
| | **Alternating 1 and 75 (UE Cat 1)** | | 18.75 | | 18.75 ± (6.7) Note 1,2<br>18.75 +8.2/-6.7 Note 3<br>18.75 +6.7/-8.2 Note 4 |

Note 1: Test tolerance +/- 6.7 dB was selected to allow PA switch possible exceptions to occur.
Note 2: When neither Note 3 nor Note 4 applies.
Note 3: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the reference sub-frames is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the target sub-frame is not confined within any one of these frequency ranges.
Note 4: For operating bands under Note 2 in Table 6.2.2.3-1, if the transmission bandwidth of the target sub-frame is confined within $F_{UL_low}$ and $F_{UL_low}$ + 4 MHz or $F_{UL_high}$ – 4 MHz and $F_{UL_high}$ and the reference sub-frame is not confined within any one of these frequency ranges.
Note 5: For extreme conditions an additional ± 2.0 dB relaxation is allowed.
Note 6: The starting resource block shall be RB# (max +1 - RB allocation).

### 6.3.5.3 Aggregate power control tolerance

#### 6.3.5.3.1 Test purpose

To verify the ability of the UE to maintain its power level in non-contiguous transmission within 21 ms in response to 0 dB TPC commands with respect to the first UE transmission, when the power control parameters specified in TS 36.213 are constant.

#### 6.3.5.3.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

#### 6.3.5.3.3 Minimum conformance requirement

The UE shall meet the requirements specified in Table 6.3.5.3.3-1 for relative power control over the power range bounded by the minimum output power as defined in sub clause 6.3.2 and the maximum output power in sub-clause 6.2.2.

**Table 6.3.5.3.3-1: Power control tolerance**

| TPC command | UL channel | Aggregate power tolerance within 21 ms |
|---|---|---|
| 0 dB | PUCCH | ±2.5 dB |
| 0 dB | PUSCH | ±3.5 dB |
| Note: 1: The UE transmission gap is 4 ms. TPC command is transmitted via PDCCH 4 subframes preceding each PUCCH/PUSCH transmission. | | |

The normative reference for this requirement is TS 36.101 [2] clause 6.3.5.3.1.

#### 6.3.5.3.4 Test description

##### 6.3.5.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.3.5.3.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.3.5.3.4.1-1: Test Configuration Table: PUCCH sub-test**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** |
| Ch BW | Mod'n | RB allocation | | FDD: PUCCH format = Format 1a<br>TDD: PUCCH format = Format 1a/1b |
| | | FDD | TDD | |
| 1.4MHz | QPSK | 3 | 3 | |
| 3MHz | QPSK | 4 | 4 | |
| 5MHz | QPSK | 8 | 8 | |
| 10MHz | QPSK | 16 | 16 | |
| 15MHz | QPSK | 25 | 25 | |
| 20MHz | QPSK | 30 | 30 | |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

**Table 6.3.5.3.4.1-2: Test Configuration Table: PUSCH sub-test**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for PUSCH sub-test | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 1 | 1 |
| 3MHz | | QPSK | 4 | 4 |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 18 | 18 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL and DL Reference Measurement channels are set according to Table 6.3.5.3.4.1-1 (PUCCH sub-test) and Table 6.3.5.3.4.1-2 (PUSCH sub-test).

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.5.3.4.3.

### 6.3.5.3.4.2 Test procedure

The procedure is separated in two subtests to verify PUCCH and PUSCH aggregate power control tolerance respectively. The uplink transmission patterns are described in figure 6.3.5.3.4.2-1.

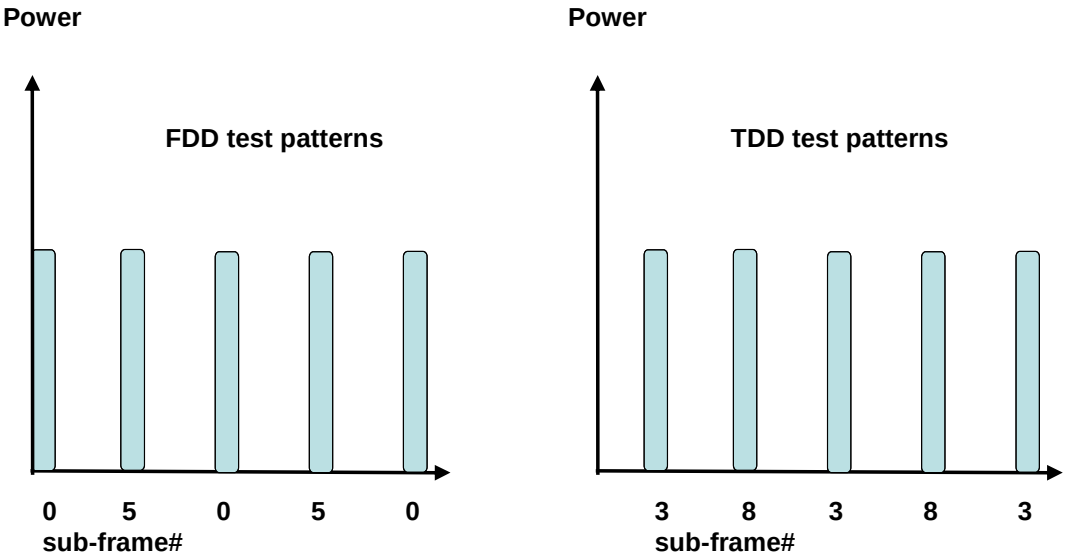



**Figure 6.3.5.3.4.2-1 Test uplink transmission**

1. PUCCH sub test:

1.1 The SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 6.3.5.3.4.1-1. The SS sends downlink MAC padding bits on the DL RMC. The transmission of PDSCH will make the UE send uplink ACK/NACK using PUCCH. Send the appropriate TPC commands for PUCCH to the UE to ensure that the UE transmits PUCCH at 0dBm +/- 3.2 dB for carrier frequency f $\leq$ 3.0GHz or at 0dBm +/- 3.5 dB for carrier frequency 3.0GHz < f $\leq$ 4.2GHz.

1.2. Every 5 subframes transmit to the UE downlink PDSCH MAC padding bits as well as 0 dB TPC command for PUCCH via the PDCCH to make the UE transmit ACK/NACK on the PUCCH with transmission gap of 4 subframes. The downlink transmission is scheduled in the appropriate sub-frames to make the UE transmit PUCCH as described in figure 6.3.5.3.4.2-1.

1.3. Measure the power of 5 consecutive PUCCH transmissions to verify the UE transmitted PUCCH power is maintained within 21 ms. The transient periods of 20us are excluded from the power measurement.

2. PUSCH sub test:

2.1. The SS sends uplink scheduling information via PDCCH DCI format 0 for C_RNTI to schedule the PUSCH. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC. Send the appropriate TPC commands for PUSCH to the UE to ensure that the UE transmits PUSCH at 0dBm +/- 3.2.dB for carrier frequency f $\leq$ 3.0GHz or at 0dBm +/- 3.5 dB for carrier frequency 3.0GHz < f $\leq$ 4.2GHz.

2.2. Every 5 subframes schedule the UE's PUSCH data transmission and transmit 0 dB TPC command for PUSCH via the PDCCH to make the UE transmit PUSCH with 4 subframes gap. The uplink transmission patterns are described in figure 6.3.5.3.4.2-1.

2.3. Measure the power of 5 consecutive PUSCH transmissions to verify the UE transmitted PUSCH power is maintained within 21 ms. The transient periods of 20us are excluded from the power measurement.

### 6.3.5.3.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

#### 6.3.5.3.5 Test requirement

The requirement for the power measurements made in step (1.3) and (2.3) of the test procedure shall not exceed the values specified in Table 6.3.5.3.5-1. The power measurement period shall be 1 sub-frame excluding transient periods.

**Table 6.3.5.3.5-1: Power control tolerance**

| TPC command | UL channel | Test requirement measured power |
|---|---|---|
| 0 dB | PUCCH | Given 5 power measurements in the pattern, the 2nd, 3rd, 4th, and 5th measurements shall be within ± 3.2 dB of the 1st measurement. |
| 0 dB | PUSCH | Given 5 power measurements in the pattern, the 2nd, 3rd, 4th, and 5th measurements shall be within ± 4.2 dB of the 1st measurement. |
| Note 1: The UE transmission gap is 4 ms. TPC command is transmitted via PDCCH 4 subframes preceding each PUCCH/PUSCH transmission. | | |

# 6.4 Void

# 6.5 Transmit signal quality

*Editor's note:*

*The test cases for transmit signal quality : frequency error, EVM, carrier leakage, IBE, EVM equalizer spectrum flatness are complete, except the following aspect is not determined:*

- *Reference signal EVM and PRACH EVM minimum requirements from the core specification are still in brackets*

In this clause a multitude of results are derived, all using one common algorithm returning these results: Global In-Channels TX-Test (Annex E). Each sub clause of this clause contains a procedure and test requirements described for a specific measurement. If all relevant test parameters in different sub clauses are the same, then the results, returned by the Global In-Channel TX-Test, may be used across the applicable sub clauses.

## 6.5.1 Frequency Error

### 6.5.1.1 Test purpose

This test verifies the ability of both, the receiver and the transmitter, to process frequency correctly.

Receiver: to extract the correct frequency from the stimulus signal, offered by the System simulator, under ideal propagation conditions and low level.

Transmitter: to derive the correct modulated carrier frequency from the results, gained by the receiver.

### 6.5.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.5.1.3 Minimum conformance requirements

The UE modulated carrier frequency shall be accurate to within ±0.1 PPM observed over a period of one time slot (0.5ms) compared to the carrier frequency received from the E-UTRA Node B.

The normative reference for this requirement is TS 36.101 clause 6.5.1

### 6.5.1.4 Test description

#### 6.5.1.4.1 Initial condition

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.5.1.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.5.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC, TL/VL, TL/VH, TH/VL, TH/VH | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 6 | 6 | QPSK | 6 | 6 |
| 3MHz | QPSK | 15 | 15 | QPSK | 15 | 15 |
| 5MHz | QPSK | 25 | 25 | QPSK | 25 | 25 |
| 5MHz | QPSK | 25 | N/A | QPSK | 20 | N/A |
| 5MHz | QPSK | 25 | N/A | QPSK | 15 | N/A |
| 10MHz | QPSK | 50 | 50 | QPSK | 50 | 50 |
| 10MHz | QPSK | 50 | N/A | QPSK | 25 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 20 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 15 | N/A |
| 15MHz | QPSK | 75 | 75 | QPSK | 75 | 75 |
| 15MHz | QPSK | 75 | N/A | QPSK | 50 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 25 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 20 | N/A |
| 20MHz | QPSK | 100 | 100 | QPSK | 100 | 100 |
| 20MHz | QPSK | 100 | N/A | QPSK | 75 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 50 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 25 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 20 | N/A |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.<br>Note 2. Depending on E-UTRA band, only the appropriate Uplink RB allocation value according to table 7.3.3-2 is tested per Test Channel Bandwidth. | | | | | | |

1. Connect the SS to the UE to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508[7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL and DL Reference Measurement channels are set according to Table 6.5.1.4.1-1.
5. Propagation conditions are set according to Annex B.0
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.5.1.4.3.

#### 6.5.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 6.5.1.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.5.1.4.1-1, since the UE has no payload data to send, the UE transmit uplink MAC padding bits on the UL RMC
3. Set the Downlink signal level to the appropriate REFSENS value defined in Table 7.3.5-1. Send continuously uplink power control "up" commands to the UE in every uplink scheduling information to the UE so that the UE transmits at $P_{UMAX}$ level for the duration of the test.
4. Measure the Frequency Error using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test.

#### 6.5.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the exceptions in subclause 7.3.4.3 and Table 7.3.3-3.

### 6.5.1.5 Test requirement

The 20 frequency error Δf results must fulfil the test requirement:

$|\Delta f| \leq (0.1\ \text{PPM} + 15\ \text{Hz})$

## 6.5.1A Frequency error for Intra-band CA

Editor's notes:

The following items are missing or incomplete:

- Initial conditions (test frequencies, test configurations)
- Test procedure
- Message Contents
- Test requirements and test tolerances
- Connection diagram is missing

### 6.5.1A.1 Test purpose

This test verifies the ability of both, receiver and the transmitter, to process frequency for intra-band CA correctly.

Receiver: to extract the correct frequency from the stimulus signal, offered by the System simulator, under ideal propagation conditions and low level.

Transmitter: to derive the correct modulated carrier frequency from the results, gained by the receiver.

### 6.5.1A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

### 6.5.1A.3 Minimum conformance requirements

The UE modulated carrier frequencies per band shall be accurate to within ±0.1 PPM observed over a period of one timeslot compared to the carrier frequency of primary component carrier received from the E-UTRA in the corresponding band.

The normative reference for this requirement is TS 36.101[2] clause 6.5.1A.

#### 6.5.1A.4 Test description

##### 6.5.1A.4.1 Initial condition

##### 6.5.1A.4.2 Test procedure

##### 6.5.1A.4.3 Message contents

#### 6.5.1A.5 Test Requirements

### 6.5.2 Transmit modulation

Transmit modulation defines the modulation quality for expected in-channel RF transmissions from the UE. This transmit modulation limit is specified in terms of:

- Error Vector Magnitude (EVM) for the allocated resources blocks (RB),
- EVM equalizer spectrum flatness derived from the equalizer coefficients generated by the EVM measurement process
- Carrier leakage (caused by IQ offset)

In-band emissions for the non-allocated RB

#### 6.5.2.1 Error Vector Magnitude (EVM)

##### 6.5.2.1.1 Test Purpose

The Error Vector Magnitude is a measure of the difference between the reference waveform and the measured waveform. This difference is called the error vector. Before calculating the EVM the measured waveform is corrected by the sample timing offset and RF frequency offset. Then the IQ origin offset shall be removed from the measured waveform before calculating the EVM.

The measured waveform is further modified by selecting the absolute phase and absolute amplitude of the Tx chain. The EVM result is defined after the front-end IDFT as the square root of the ratio of the mean error vector power to the mean reference power expressed as a %.

The basic EVM measurement interval in the time domain is one preamble sequence for the PRACH and is one slot for the PUCCH and PUSCH in the time domain. When the PUSCH or PUCCH transmission slot is shortened due to multiplexing with SRS, the EVM measurement interval is reduced by one symbol, accordingly. The PUSCH or PUCCH EVM measurement interval is also reduced when the mean power, modulation or allocation between slots is expected to change. In the case of PUSCH transmission, the measurement interval is reduced by a time interval equal to the sum of 5 μs and the applicable exclusion period defined in subclause 6.3.4, adjacent to the boundary where the power change is expected to occur. The PUSCH exclusion period is applied to the signal obtained after the front-end IDFT. In the case of PUCCH transmission, the PUCCH EVM measurement interval is reduced by one symbol adjacent to the slot boundary.

##### 6.5.2.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

##### 6.5.2.1.3 Minimum conformance requirements

EVM measurements are evaluated for 10 uplink sub-frames excluding any transient period for the average EVM case, and 60 subframes excluding any transient period for the reference signal EVM case, the different modulations schemes shall not exceed the values specified in Table 6.5.2.1.3-1 for the parameters defined in Table 6.5.2.1.3-2. For EVM evaluation purposes, [all PRACH preamble formats 0-4 and] all PUCCH formats 1, 1a, 1b, 2, 2a and 2b are considered to have the same EVM requirement as QPSK modulated.

**Table 6.5.2.1.3-1: Minimum requirements for Error Vector Magnitude**

| Parameter | Unit | Average EVM Level | Reference Signal EVM Level |
|---|---|---|---|
| QPSK or BPSK | % | 17.5 | [17.5] |
| 16QAM | % | 12.5 | [12.5] |

**Table 6.5.2.1.3-2: Parameters for Error Vector Magnitude**

| Parameter | Unit | Level |
|---|---|---|
| UE Output Power | dBm | ≥ -40 |
| Operating conditions | | Normal conditions |

The normative reference for this requirement is TS 36.101 [2] clause 6.5.2.1.1.

### 6.5.2.1.4 Test description

#### 6.5.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.5.2.1.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annex A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.5.2.1.4.1-1: Test Configuration Table for PUSCH**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | | **NC** | | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | | **See Table 6.5.1.4.1-1** | | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | | **See Table 6.5.1.4.1-1** | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for PUSCH EVM testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 1.4MHz | | QPSK | 1 | 1 |
| 1.4MHz | | 16QAM | 6 | 6 |
| 1.4MHz | | 16QAM | 1 | 1 |
| 3MHz | | QPSK | 15 | 15 |
| 3MHz | | QPSK | 4 | 4 |
| 3MHz | | 16QAM | 15 | 15 |
| 3MHz | | 16QAM | 4 | 4 |
| 5MHz | | QPSK | 25 | 25 |
| 5MHz | | QPSK | 8 | 8 |
| 5MHz | | 16QAM | 25 | 25 |
| 5MHz | | 16QAM | 8 | 8 |
| 10MHz | | QPSK | 50 | 50 |
| 10MHz | | QPSK | 12 | 12 |
| 10MHz | | 16QAM | 50<br>(Note 3) | 50<br>(Note 3) |
| 10MHz | | 16QAM | 12 | 12 |
| 15MHz | | QPSK | 75 | 75 |
| 15MHz | | QPSK | 16 | 16 |
| 15MHz | | 16QAM | 75<br>(Note 3) | 75<br>(Note 3) |
| 15MHz | | 16QAM | 16 | 16 |
| 20MHz | | QPSK | 100 | 100 |
| 20MHz | | QPSK | 18 | 18 |
| 20MHz | | 16QAM | 100<br>(Note 3) | 100<br>(Note 3) |
| 20MHz | | 16QAM | 18 | 18 |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: For partial RB allocation, the $RB_{start}$ shall be RB #0 and RB# (max+1 - RB allocation) of the channel bandwidth.

Note 3: Applies only for UE-Categories 2-5

**Table 6.5.2.1.4.1-2: Test Configuration Table for PUCCH**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | See Table 6.5.1.4.1-1 | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | See Table 6.5.1.4.1-1 | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** |
| Ch BW | Mod'n | RB allocation | | FDD: PUCCH format = Format 1a<br>TDD: PUCCH format = Format 1a / 1b |
| | | FDD | TDD | |
| 1.4MHz | QPSK | 3 | 3 | |
| 3MHz | QPSK | 4 | 4 | |
| 5MHz | QPSK | 8 | 8 | |
| 10MHz | QPSK | 16 | 16 | |
| 15MHz | QPSK | 25 | 25 | |
| 20MHz | QPSK | 30 | 30 | |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

**Table 6.5.2.1.4.1-3: Test Configuration for PRACH**

| **Initial Conditions** | | |
|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | NC | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | See Table 6.5.1.4.1-1 | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | See Table 6.5.1.4.1-1 | |
| PRACH preamble format | | |
| | FDD | TDD |
| PRACH Configuration Index | 4 | 53 |
| RS EPRE setting for test point 1 (dBm/15kHz) | -71 | -63 |
| RS EPRE setting for test point 2 (dBm/15kHz) | -86 | -78 |

1. Connect the SS to the UE to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508[7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to in Table 6.5.2.1.4.1-1.
5. Propagation conditions are set according to Annex B.0
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.5.2.1.4.3.

### 6.5.2.1.4.2 Test procedure

Test procedure for PUSCH:

1.1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.5.2.1.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

1.2 Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX\ level}$.

1.3 Measure the EVM and $\overline{EVM}_{DMRS}$ using Global In-Channel Tx-Test (Annex E).

1.4 Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is –36.8dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -36.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

1.5 Measure the EVM and $\overline{EVM}_{DMRS}$ using Global In-Channel Tx-Test (Annex E).

Test procedure for PUCCH:

2.1 PUCCH are set according to Table 6.5.2.1.4.1-2.

2.2 SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 6.5.2.1.4.1-2. The SS sends downlink MAC padding bits on the DL RMC. The transmission of PDSCH will make the UE send uplink ACK/NACK using PUCCH. There is no PUSCH transmission.

2.3 SS send appropriate TPC commands for PUCCH to the UE until the UE transmit PUCCH at $P_{UMAX}$ level.

2.4 Measure PUCCH EVM using Global In-Channel Tx-Test (Annex E).

2.5 Send the appropriate TPC commands for PUCCH to the UE until the UE transmits PUCCH at -36.8dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -36.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

2.6 Measure PUCCH EVM using Global In-Channel Tx-Test (Annex E).

Test procedure for PRACH:

3.1 The SS shall set RS EPRE according to Table 6.5.2.1.4.1-3.

3.2 PRACH is set according to Table 6.5.2.1.4.1-3.

3.3 The SS shall signal a Random Access Preamble ID via a PDCCH order to the UE and initiate a Non-contention based Random Access procedure

3.4 The UE shall send the signalled preamble to the SS.

3.5 In response to the preamble, the SS shall transmit a random access response not corresponding to the transmitted random access preamble, or send no response.

3.6 The UE shall consider the random access response reception not successful then re-transmit the preamble with the calculated PRACH transmission power.

3.7 Repeat step 5 and 6 until the SS collect enough PRACH preambles (2 preambles for format 0 and 10 preambles for format 4). Measure the EVM in PRACH channel using Global In-Channel Tx-Test (Annex E).

### 6.5.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 6.5.2.1.4.3-1: *PRACH-ConfCommonDEFAULT*: PRACH EVM measurement for FDD**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-7 PRACH-ConfCommonDEFAULT | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| PRACH-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 4 | | |

**Table 6.5.2.1.4.3-2: *PRACH-ConfCommonDEFAULT*: PRACH EVM measurement for TDD**

| Derivation Path: TS 36.508 [7] clause 5.3.1, Table 5.3.1-1 PRACH-ConfCommonDEFAULT | | | |
|---|---|---|---|
| Information Element | Value/remark | Comment | Condition |
| PRACH-ConfigInfo SEQUENCE { | | | |
| prach-ConfigIndex | 53 | | |

**Table 6.5.2.1.4.3-4: *RACH-ConfigCommon-DEFAULT:* PRACH EVM measurement**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-12 RACH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| preambleInfo SEQUENCE { | | | |
| numberOfRA-Preambles | n52 | | |
| preamblesGroupAConfig SEQUENCE {} | Not present | | |
| } | | | |
| powerRampingParameters SEQUENCE { | | | |
| powerRampingStep | dB0 | | |
| preambleInitialReceivedTargetPower | dBm-120 Test point 1<br>dBm-90 Test point 2 | | |
| } | | | |
| ra-SupervisionInfo SEQUENCE { | | | |
| preambleTransMax | n10 | | FDD |
| | n20 | | TDD |
| ra-ResponseWindowSize | Sf10 | | |
| mac-ContentionResolutionTimer | sf48 | | |
| } | | | |
| ra-SupervisionInfo SEQUENCE { | | | |

**Table 6.5.2.1.4.3-5: *TDD-Config-DEFAULT:* PRACH EVM measurement for TDD**

| Derivation Path: TS 36.508 [7] clause 5.3.1, Table 5.3.1-1: TDD-Config-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | sa1 | | |
| specialSubframePatterns | ssp5 | To enable two symbol UpPTS, and to have 9 symbols GP. | |
| } | | | |

### 6.5.2.1.5 Test requirement

The PUSCH EVM derived in E.4.2 shall not exceed 17,5 % for QPSK and BPSK, 12,5% for 16 QAM.

The PUSCH $\overline{EVM}_{DMRS}$ derived in E.4.6.2 shall not exceed [17,5 %] when embedded with data symbols of QPSK and BPSK, [12,5%] for 16 QAM.

The PUCCH EVM and derived in E.5.9.2 shall not exceed 17,5 %.

The PRACH EVM derived in FFS shall not exceed 17.5%.

## 6.5.2.1A PUSCH-EVM with exclusion period

### 6.5.2.1A.1 Test purpose

To verify the ability of the UE transmitter to keep the EVM minimum requirements, even in the presence of transients according to subclause 6.5.2.1.1 third paragraph:

…...In the case of PUSCH transmission, the measurement interval is reduced by a time interval equal to the sum of 5 μs and the applicable exclusion period defined in subclause 6.3.4, adjacent to the boundary where the power change is expected to occur. The PUSCH exclusion period is applied to the signal obtained after the front-end IDFT. …..

### 6.5.2.1A.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 6.5.2.1A.3 Minimum conformance requirement

EVM measurements are evaluated for 10 uplink sub-frames in a reduced time interval due to exclusion periods for the average EVM. The different modulations schemes shall not exceed the values specified in Table 6.5.2.1.3-1 for the parameters defined in Table 6.5.2.1.3-2.

### 6.5.2.1A.4 Test description

#### 6.5.2.1A.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.5.2.1A.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.5.2.1A.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal Conditions | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | 10 MHz | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 10MHz | | QPSK | Alternating 12 and 1 | Alternating 12 and 1 |
| 10MHz | | 16 QAM | Alternating 12 and 1 | Alternating 12 and 1 |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channel is set according to table 6.5.2.1A.4.1-1
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.3.5.2.4.3.

#### 6.5.2.1A.4.2 Test procedure

The test pattern is illustrated in figure 6.5.2.1A.4.2-1.

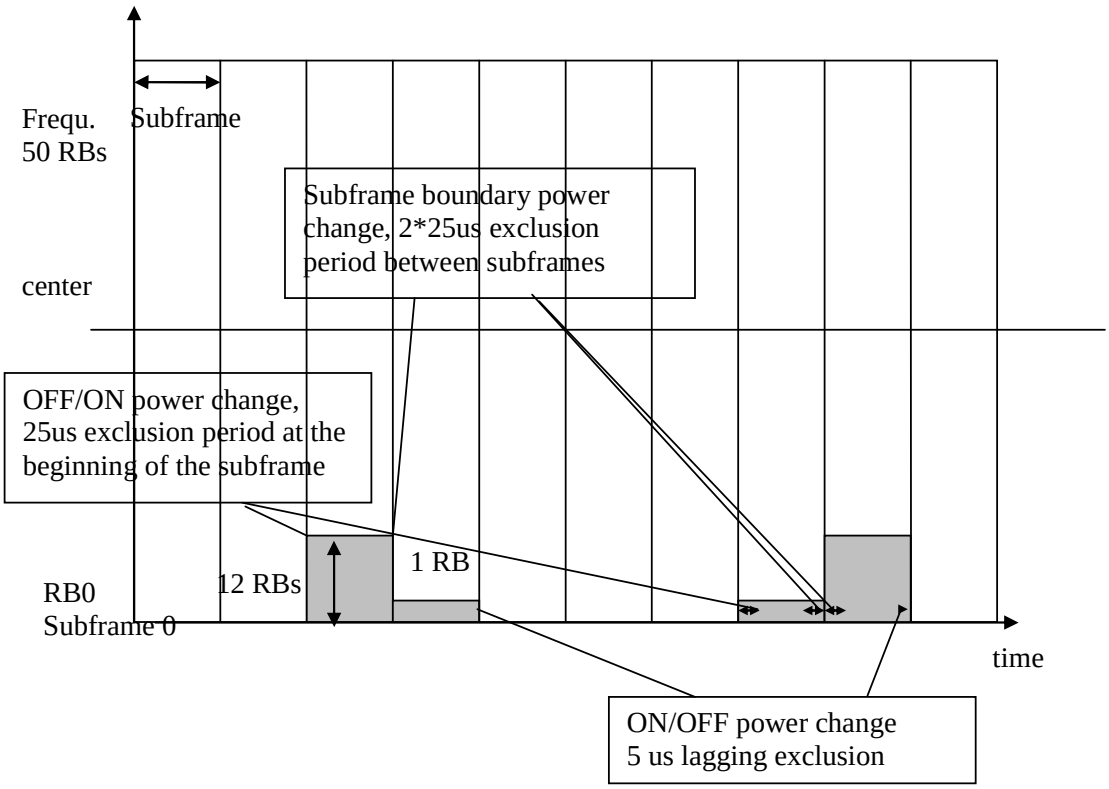



**Figure 6.5.2.1A.4.2-1: Test pattern**

NOTE 1: In TDD the free subframes are special subframes or DL, in FDD the free subframes are OFF.

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the PUSCH... Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC. The initial uplink RB allocation is 12 as specified in Table 6.3.5.2A.1.4.1. Send appropriate TPC commands for PUSCH to the UE to ensure the UE transmits PUSCH at 0dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or 0dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

2. Schedule the UE's PUSCH data transmission as described in Figure 6.5.2.1A.4.2-1 for 16[1)] active time slots with an uplink RB allocation alternating pattern as described in table 6.5.2.1A.4.1-1 while transmitting 0dB TPC command for PUSCH via the PDCCH.

3. Measure the EVM using Global In-Channel Tx-Test. The averaging across 16[1)] timeslots is done across mixed RB allocations, as illustrated in Figure 6.5.2.1A.4.2-1

NOTE 1: Averaging across 16 timeslots is used to represent each type of transition equally in the average.

### 6.5.2.1A.5 Test requirement

The PUSCH EVM derived in Annex E.4.2 taking into account Annex E.7 shall not exceed 17,5 % for QPSK and 12,5% for 16 QAM. The test requirements shall be fullfilled for early and late EVM window.

## 6.5.2.2 Carrier leakage

### 6.5.2.2.1 Test Purpose

Carrier leakage (the I/Q origin offset) is an interference caused by crosstalk or DC offset and expresses itself as unmodulated sine wave with the carrier frequency. It is an interference of approximately constant amplitude and independent of the amplitude of the wanted signal. I/Q origin offset interferes with the centre sub carriers of the UE under test (if allocated), especially, when their amplitude is small. The measurement interval is defined over one slot in the time domain.

The purpose of this test is to exercise the UE transmitter to verify its modulation quality in terms of carrier leakage.

### 6.5.2.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.5.2.2.3 Minimum conformance requirements

The relative carrier leakage power (IQ origin offset power) is a power ratio of the additive sinusoid waveform and the modulated waveform. The relative carrier leakage power shall not exceed the values specified in Table 6.5.2.2.3-1.

**Table 6.5.2.2.3-1: Minimum requirements for Relative Carrier Leakage Power**

| LO Leakage | Parameters | Relative Limit (dBc) |
|---|---|---|
| | Output power >0 dBm | -25 |
| | -30 dBm ≤ Output power ≤0 dBm | -20 |
| | -40 dBm ≤ Output power < -30 dBm | -10 |

The normative reference for this requirement is TS 36.101 clause 6.5.2.2.1

### 6.5.2.2.4 Test description

#### 6.5.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.5.2.2.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.5.2.2.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | See Table 6.5.1.4.1-1 | | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | See Table 6.5.1.4.1-1 | | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | See Table 6.5.1.4.1-1 | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for carrier leakage testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 1 | 1 |
| 3MHz | | QPSK | 4 | 4 |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 18 | 18 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |
| Note 2: For partial RB allocation, the $RB_{start}$ shall be RB #0 and RB# (max +1- RB allocation) of the channel bandwidth. | | | | |

1. Connect the SS to the UE to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508[7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to in Table 6.5.2.2.4.1-1.
5. Propagation conditions are set according to Annex B.0
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.5.2.2.4.3.

### 6.5.2.2.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.5.2.2.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC
2. Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is 3.2 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or 3.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.
3. Measure carrier leakage using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test.
4. Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is -26.8 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -26.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.
5. Measure carrier leakage using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test
6. Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is -36.8dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -36.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.
7. Measure carrier leakage using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test

#### 6.5.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.5.2.2.5 Test requirement

Each of the 20 IQ offset results, derived in Annex E.3.1, shall not exceed the values in table 6.5.2.2.5-1

**Table 6.5.2.2.5-1: Test requirements for Relative Carrier Leakage Power**

| LO Leakage | Parameters | Relative Limit (dBc) |
|---|---|---|
| | f ≤ 3.0GHz: 3.2 dBm ±3.2dB<br>3.0GHz < f ≤ 4.2GHz: 3.5 dBm ±3.5dB | -24.2 |
| | f ≤ 3.0GHz: -26.8 dBm ±3.2dB<br>3.0GHz < f ≤ 4.2GHz: -26.5 dBm ±3.5dB | -19.2 |
| | f ≤ 3.0GHz: -36.8dBm±3.2dB<br>3.0GHz < f ≤ 4.2GHz: -36.5 dBm ±3.5dB | -9.2 |

## 6.5.2.3 In-band emissions for non allocated RB

### 6.5.2.3.1 Test Purpose

The in-band emissions are a measure of the interference falling into the non-allocated resources blocks

The in-band emission is defined as the average across 12 sub-carrier and as a function of the RB offset from the edge of the allocated UL transmission bandwidth. The in-band emission is measured as the ratio of the UE output power in a non–allocated RB to the UE output power in an allocated RB. The basic in-band emissions measurement interval is defined over one slot in the time domain. When the PUSCH or PUCCH transmission slot is shortened due to multiplexing with SRS, the in-band emissions measurement interval is reduced by one SC-FDMA symbol, accordingly.

### 6.5.2.3.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.5.2.3.3 Minimum conformance requirements

The relative in-band emission shall not exceed the values specified in Table 6.5.2.3.3-1.

**Table 6.5.2.3.3-1: Minimum requirements for in-band emissions**

| Parameter Description | Unit | Limit (Note 1) | | Applicable Frequencies |
|---|---|---|---|---|
| General | dB | $\max\{ -25 - 10 \cdot \log_{10}(N_{RB} / L_{CRBs}),$ $20 \cdot \log_{10} EVM - 3 - 5 \cdot (\lvert\Delta_{RB}\rvert - 1) / L_{CRBs},$ $-57\ dBm\ /180\ kHz - P_{RB}\}$ | | Any non-allocated (Note 2) |
| IQ Image | dB | -25 | | Image frequencies (Notes 2, 3) |
| Carrier leakage | dBc | -25 | Output power > 0 dBm | LO frequency (Notes 4, 5) |
| | | -20 | -30 dBm ≤ Output power ≤ 0 dBm | |
| | | -10 | -40 dBm ≤ Output power < -30 dBm | |

Note 1: An in-band emissions combined limit is evaluated in each non-allocated RB. For each such RB, the minimum requirement is calculated as the higher of $P_{RB}$ - 30 dB and the power sum of all limit values (General, IQ Image or Carrier leakage) that apply. $P_{RB}$ is defined in Note 10.

Note 2: The measurement bandwidth is 1 RB and the limit is expressed as a ratio of measured power in one non-allocated RB to the measured average power per allocated RB, where the averaging is done across all allocated RBs.

Note 3: The applicable frequencies for this limit are those that are enclosed in the reflection of the allocated bandwidth, based on symmetry with respect to the centre carrier frequency, but excluding any allocated RBs.

Note 4: The measurement bandwidth is 1 RB and the limit is expressed as a ratio of measured power in one non-allocated RB to the measured total power in all allocated RBs.

Note 5: The applicable frequencies for this limit are those that are enclosed in the RBs containing the DC frequency if $N_{RB}$ is odd, or in the two RBs immediately adjacent to the DC frequency if $N_{RB}$ is even, but excluding any allocated RB.

Note 6: $L_{CRBs}$ is the Transmission Bandwidth (see Figure 5.4.2-1).

Note 7: $N_{RB}$ is the Transmission Bandwidth Configuration (see Figure 5.4.2-1).

Note 8: $EVM$ is the limit specified in Table 6.5.2.1.1-1 for the modulation format used in the allocated RBs.

Note 9: $\Delta_{RB}$ is the starting frequency offset between the allocated RB and the measured non-allocated RB (e.g. $\Delta_{RB} = 1$ or $\Delta_{RB} = -1$ for the first adjacent RB outside of the allocated bandwidth.

Note 10: $P_{RB}$ is the transmitted power per 180 kHz in allocated RBs, measured in dBm.

The normative reference for this requirement is TS 36.101 [2] clause 6.5.2.3.1.

The in-band emission is defined as the average across 12 sub-carrier and as a function of the RB offset from the edge of the allocated UL transmission bandwidth. The in-band emission is measured as the ratio of the UE output power in a non–allocated RB to the UE output power in an allocated RB. The basic in-band emissions measurement interval is defined over one slot in the time domain.

## 6.5.2.3.4 Test description

### 6.5.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.5.2.3.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.5.2.3.4.1-1: Test Configuration Table for PUSCH**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | See Table 6.5.1.4.1-1 | | | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | See Table 6.5.1.4.1-1 | | | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | See Table 6.5.1.4.1-1 | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for in-band emissions testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 1 | 1 |
| 3MHz | | QPSK | 4 | 4 |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 18 | 18 |
| Note 1. Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.<br>Note 2. For partial RB allocation, the starting resource block shall be RB #0 and RB# (max+1 - RB allocation) of the channel bandwidth. | | | | |

**Table 6.5.2.3.4.1-2: Test Configuration Table for PUCCH**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | See Table 6.5.1.4.1-1 | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | See Table 6.5.1.4.1-1 | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | See Table 6.5.1.4.1-1 | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** |
| Ch BW | Mod'n | RB allocation | | FDD: PUCCH format = Format 1a<br>TDD: PUCCH format = Format 1a / 1b |
| | | FDD | TDD | |
| 1.4MHz | QPSK | 3 | 3 | |
| 3MHz | QPSK | 4 | 4 | |
| 5MHz | QPSK | 8 | 8 | |
| 10MHz | QPSK | 16 | 16 | |
| 15MHz | QPSK | 25 | 25 | |
| 20MHz | QPSK | 30 | 30 | |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to in Table 6.5.2.3.4.1-1.
5. Propagation conditions are set according to Annex B.0
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.5.2.3.4.3.

### 6.5.2.3.4.2 Test procedure

Test procedure for PUSCH:

1.1 SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.5.2.3.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC

1.2 Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is 3.2 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or 3.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

1.3 Measure In-band emission using Global In-Channel Tx-Test (Annex E)

1.4 Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is -26.8 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -36.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

1.5 Measure In-band emission using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test

1.6 Send the appropriate TPC commands in the uplink scheduling information to the UE until UE output power is to -36.8 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -36.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

1,7 Measure In-band emission using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test

Test procedure for PUCCH:

2.1 PUCCH is set according to Table 6.5.2.3.4.1-2. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 6.5.2.3.4.1-2. The SS sends downlink MAC padding bits on the DL RMC. The transmission of PDSCH will make the UE send uplink ACK/NACK using PUCCH.

2.2 Send the appropriate TPC commands in the uplink scheduling information for PUCCH to the UE until UE output power is 3.2 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or 3.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

2.3 Measure In-band emission using Global In-Channel Tx-Test (Annex E)

2.4 Send the appropriate TPC commands for PUCCH in the uplink scheduling information to the UE until UE output power is -26.8 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -26.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

2.5 Measure In-band emission using Global In-Channel Tx-Test (Annex E)

2.6 Send the appropriate TPC commands for PUCCH in the uplink scheduling information to the UE until UE output power is to -36.8 dBm ±3.2dB for carrier frequency f ≤ 3.0GHz or -36.5dBm ±3.5 dB for carrier frequency 3.0GHz < f ≤ 4.2GHz.

2.7 Measure In-band emission using Global In-Channel Tx-Test (Annex E)

#### 6.5.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions:

**Table 4.6.3-8: *PUCCH-ConfigCommon*: PUCCH in-band emissions measurement**

| Derivation Path: 36.331 clause 6.3.2, Table 4.6.3-8: PUCCH-ConfigCommon-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PUCCH-ConfigCommon-DEFAULT ::= SEQUENCE { | | | |
| nRB-CQI | 0 | | |
| } | | | |

### 6.5.2.3.5 Test requirement

Each of the 20 In-band emissions results, derived in Annex E.4.3 shall not exceed the corresponding values in Table 6.5.2.3.5-1

**Table 6.5.2.3.5-1: Test requirements for in-band emissions**

| Parameter Description | Unit | Limit (Note 1) | | Applicable Frequencies |
|---|---|---|---|---|
| General | dB | $\max\{ -25 - 10 \cdot \log_{10}(N_{RB} / L_{CRBs}),$ $20 \cdot \log_{10} EVM - 3 - 5 \cdot (\lvert\Delta_{RB}\rvert - 1) / L_{CRBs},$ $-57\ dBm\ /180\ kHz - P_{RB}\}$ +0.8 | | Any non-allocated (Note 2) |
| IQ Image | dB | -24.2 | | Image frequencies (Notes 2, 3) |
| DC | dBc | -24.2 | Output power f ≤ 3.0GHz: 3.2dBm ±3.2dB 3.0GHz < f ≤ 4.2GHz: 3.5 dBm ±3.5dB | LO frequency (Notes 4, 5) |
| | | -19.2 | Output power f ≤ 3.0GHz: -26.8 dBm ±3.2dB 3.0GHz < f ≤ 4.2GHz: -26.5 dBm ±3.5dB | |
| | | -9.2 | Output power f ≤ 3.0GHz: -36.8 dBm ±3.2dB 3.0GHz < f ≤ 4.2GHz: -36.5 dBm ±3.5dB | |

Note 1: An in-band emissions combined limit is evaluated in each non-allocated RB. For each such RB, the test requirement is calculated as the higher of $P_{RB} - 29.2$ dB and the power sum of all limit values (General, IQ Image or Carrier leakage) that apply. $P_{RB}$ is defined in Note 10.

Note 2: The measurement bandwidth is 1 RB and the limit is expressed as a ratio of measured power in one non-allocated RB to the measured average power per allocated RB, where the averaging is done across all allocated RBs.

Note 3: The applicable frequencies for this limit are those that are enclosed in the reflection of the allocated bandwidth, based on symmetry with respect to the centre carrier frequency, but excluding any allocated RBs.

Note 4: The measurement bandwidth is 1 RB and the limit is expressed as a ratio of measured power in one non-allocated RB to the measured total power in all allocated RBs.

Note 5: The applicable frequencies for this limit are those that are enclosed in the RBs containing the DC frequency if $N_{RB}$ is odd, or in the two RBs immediately adjacent to the DC frequency if $N_{RB}$ is even, but excluding any allocated RB.

Note 6: $L_{CRBs}$ is the Transmission Bandwidth (see Figure 5.4.2-1).

Note 7: $N_{RB}$ is the Transmission Bandwidth Configuration (see Figure 5.4.2-1).

Note 8: $EVM$ is the limit specified in Table 6.5.2.1.1-1 for the modulation format used in the allocated RBs.

Note 9: $\Delta_{RB}$ is the starting frequency offset between the allocated RB and the measured non-allocated RB (e.g. $\Delta_{RB} = 1$ or $\Delta_{RB} = -1$ for the first adjacent RB outside of the allocated bandwidth.

Note 10: $P_{RB}$ is the transmitted power per 180 kHz in allocated RBs, measured in dBm.

## 6.5.2.4 EVM equalizer spectrum flatness

### 6.5.2.4.1 Test Purpose

The zero-forcing equalizer correction applied in the EVM measurement process (as described in Annex E) must meet a spectrum flatness requirement for the EVM measurement to be valid. The EVM equalizer spectrum flatness is defined in terms of the maximum peak-to-peak ripple of the equalizer coefficients (dB) across the allocated uplink block variation in dB of the equalizer coefficients generated by the EVM measurement process. The EVM equalizer spectrum flatness requirement does not limit the correction applied to the signal in the EVM measurement process but for the EVM result to be valid, the equalizer correction that was applied must meet the EVM equalizer spectrum flatness minimum requirements. The basic measurement interval is the same as for EVM.

### 6.5.2.4.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.5.2.4.3 Minimum conformance requirements

The peak-to-peak variation of the EVM equalizer coefficients contained within the frequency range of the uplink allocation shall not exceed the maximum ripple. The EVM equalizer spectrum flatness shall not exceed the values specified in Table 6.5.2.4.3-1 for normal conditions. For uplink allocations contained within both Range 1 and Range 2, the coefficients evaluated within each of these frequency ranges shall meet the corresponding ripple requirement and the following additional requirement: the relative difference between the maximum coefficient in Range 1 and the minimum coefficient in Range 2 must not be larger than 5 dB, and the relative difference between the maximum coefficient in Range 2 and the minimum coefficient in Range 1 must not be larger than 7 dB (see Figure 6.5.2.4.3-1).

The EVM equalizer spectrum flatness shall not exceed the values specified in Table 6.5.2.4.3-2 for extreme conditions. For uplink allocations contained within both Range 1 and Range 2, the coefficients evaluated within each of these frequency ranges shall meet the corresponding ripple requirement and the following additional requirement: the relative difference between the maximum coefficient in Range 1 and the minimum coefficient in Range 2 must not be larger than 6 dB, and the relative difference between the maximum coefficient in Range 2 and the minimum coefficient in Range 1 must not be larger than 10 dB (see Figure 6.5.2.4.3-1).

**Table 6.5.2.4.3-1: Minimum requirements for EVM equalizer spectrum flatness (normal conditions)**

| Frequency Range | Maximum Ripple [dB] |
|---|---|
| $F_{UL_Meas} - F_{UL_Low} \geq 3$ MHz and $F_{UL_High} - F_{UL_Meas} \geq 3$ MHz (Range 1) | 4 (p-p) |
| $F_{UL_Meas} - F_{UL_Low} < 3$ MHz or $F_{UL_High} - F_{UL_Meas} < 3$ MHz (Range 2) | 8 (p-p) |
| Note 1: $F_{UL_Meas}$ refers to the sub-carrier frequency for which the equalizer coefficient is evaluated<br>Note 2: $F_{UL_Low}$ and $F_{UL_High}$ refer to each E-UTRA frequency band specified in Table 5.2-1 | |

**Table 6.5.2.4.3-2: Minimum requirements for EVM equalizer spectrum flatness (extreme conditions)**

| | |
|---|---|
| $F_{UL_Meas} - F_{UL_Low} \geq 5$ MHz and $F_{UL_High} - F_{UL_Meas} \geq 5$ MHz (Range 1) | 4 (p-p) |
| $F_{UL_Meas} - F_{UL_Low} < 5$ MHz or $F_{UL_High} - F_{UL_Meas} < 5$ MHz (Range 2) | 12 (p-p) |
| Note 1: $F_{UL_Meas}$ refers to the sub-carrier frequency for which the equalizer coefficient is evaluated<br>Note 2: $F_{UL_Low}$ and $F_{UL_High}$ refer to each E-UTRA frequency band specified in Table 5.2-1 | |

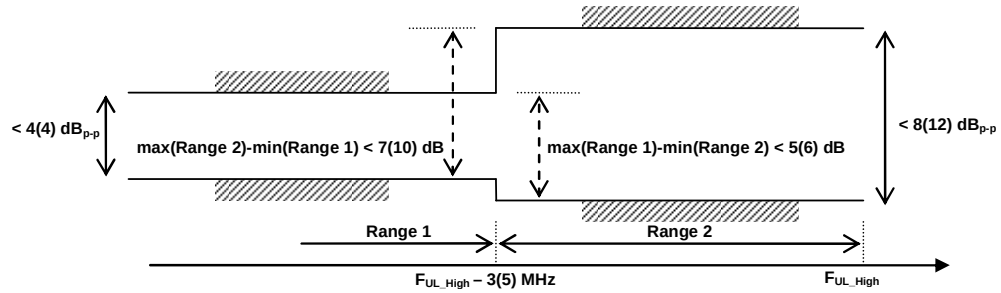



**Figure 6.5.2.4.3-1: The limits for EVM equalizer spectrum flatness with the maximum allowed variation of the coefficients indicated (the ETC minimum requirement within brackets).**

The normative reference for this requirement is TS 36.101 clause 6.5.2.4.1.

### 6.5.2.4.4 Test description

#### 6.5.2.4.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.4.2.4.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.5.2.4.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | See Table 6.5.1.4.1-1 | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | See Table 6.5.1.4.1-1 | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | See Table 6.5.1.4.1-1 | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for EVM equalizer spectrum flatness testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 3MHz | | QPSK | 15 | 15 |
| 5MHz | | QPSK | 25 | 25 |
| 10MHz | | QPSK | 50 | 50 |
| 15MHz | | QPSK | 75 | 75 |
| 20MHz | | QPSK | 100 | 100 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to in Table 6.5.2.4.4.1-1.
5. Propagation conditions are set according to Annex B.0
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.5.2.4.4.3.

#### 6.5.2.4.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.5.2.4.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC
2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.
3. Measure spectrum flatness using Global In-Channel Tx-Test (Annex E). For TDD slots with transient periods are not under test.

#### 6.5.2.4.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.5.2.4.5 Test requirement

Each of the 20 spectrum flatness functions, shall derive four ripple results in Annex E.4.4, The derived results shall not exceed the values in Figure 6.5.2.4.5-1:

For normal conditions, the maximum ripple in Range 1 and Range 2 shall not exceed the values specified in Table 6.5.2.4.5-1 and the following additional requirement: the relative difference between the maximum coefficient in Range 1 and the minimum coefficient in Range 2 must not be larger than 6.4 dB, and the relative difference between the maximum coefficient in Range 2 and the minimum coefficient in Range 1 must not be larger than 8.4 dB (see Figure 6.5.2.4.5-1).

For extreme conditions, the maximum ripple in Range 1 and Range 2 shall not exceed the values specified in Table 6.5.2.4.5-2 and the following additional requirement: the relative difference between the maximum coefficient in Range 1 and the minimum coefficient in Range 2 must not be larger than 7.4 dB, and the relative difference between the maximum coefficient in Range 2 and the minimum coefficient in Range 1 must not be larger than 11.4 dB (see Figure 6.5.2.4.5-1).

**Table 6.5.2.4.5-1: Test requirements for EVM equalizer spectrum flatness (normal conditions)**

| Frequency Range | Maximum Ripple [dB] |
|---|---|
| $F_{UL_Meas} - F_{UL_Low} \geq 3$ MHz and $F_{UL_High} - F_{UL_Meas} \geq 3$ MHz (Range 1) | 5.4 (p-p) |
| $F_{UL_Meas} - F_{UL_Low} < 3$ MHz or $F_{UL_High} - F_{UL_Meas} < 3$ MHz (Range 2) | 9.4 (p-p) |
| Note 1: $F_{UL_Meas}$ refers to the sub-carrier frequency for which the equalizer coefficient is evaluated<br>Note 2: $F_{UL_Low}$ and $F_{UL_High}$ refer to each E-UTRA frequency band specified in Table 5.2-1 | |

**Table 6.5.2.4.5-2: Test requirements for spectrum flatness (extreme conditions)**

| Frequency Range | Maximum Ripple [dB] |
|---|---|
| $F_{UL_Meas} - F_{UL_Low} \geq 5$ MHz and $F_{UL_High} - F_{UL_Meas} \geq 5$ MHz (Range 1) | 5.4 (p-p) |
| $F_{UL_Meas} - F_{UL_Low} < 5$ MHz or $F_{UL_High} - F_{UL_Meas} < 5$ MHz (Range 2) | 13.4 (p-p) |
| Note 1: $F_{UL_Meas}$ refers to the sub-carrier frequency for which the equalizer coefficient is evaluated<br>Note 2: $F_{UL_Low}$ and $F_{UL_High}$ refer to each E-UTRA frequency band specified in Table 5.2-1 | |

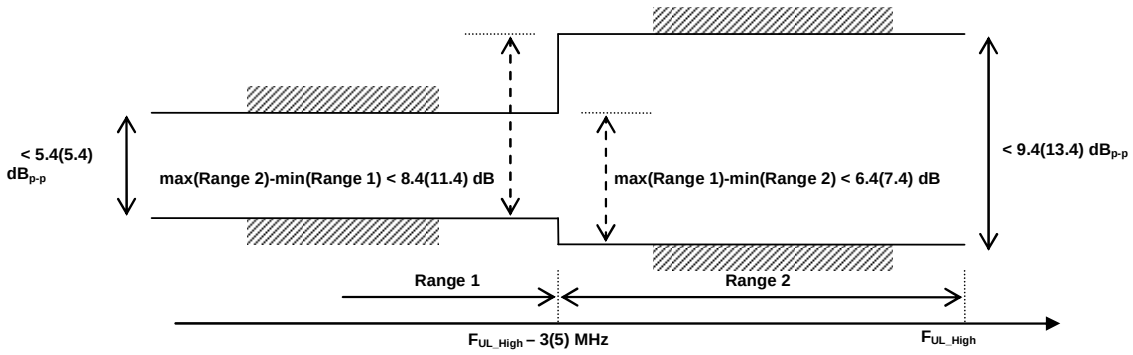



**Figure 6.5.2.4.5-1: The limits for EVM equalizer spectrum flatness with the maximum allowed variation of the coefficients indicated (the ETC minimum requirement within brackets).**

# 6.6 Output RF spectrum emissions

Unwanted emissions are divided into “Out-of-band emission” and “Spurious emissions” in 3GPP RF specifications. This notation is in line with ITU-R recommendations such as SM.329 [2] and the Radio Regulations [3].

ITU defines:

Out-of-band emission = Emission on a frequency or frequencies immediately outside the necessary bandwidth which results from the modulation process, but excluding spurious emissions.

Spurious emission = Emission on a frequency, or frequencies, which are outside the necessary bandwidth and the level of which may be reduced without affecting the corresponding transmission of information. Spurious emissions include harmonic emissions, parasitic emissions, intermodulation products and frequency conversion products but exclude out-of-band emissions.

Unwanted emissions = Consist of spurious emissions and out-of-band emissions.

The UE transmitter spectrum emission consists of the three components; the occupied bandwidth (channel bandwidth), the Out Of Band (OOB) emissions and the far out spurious emission domain.

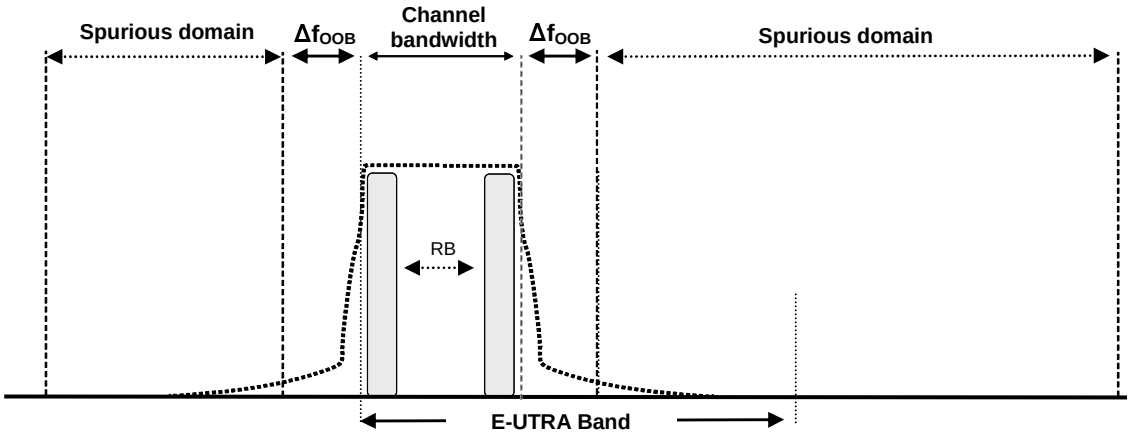



**Figure 6.6-1: Transmitter RF spectrum**

## 6.6.1 Occupied bandwidth

### 6.6.1.1 Test purpose

To verify that the UE occupied bandwidth for all transmission bandwidth configurations supported by the UE are less than their specific limits

### 6.6.1.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 6.6.1.2 Minimum conformance requirements

Occupied bandwidth is a measure of the bandwidth containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel. The occupied channel bandwidth for all transmission bandwidth configurations (Resources Blocks) should be less than the channel bandwidth specified in Table 6.6.1.2-1

**Table 6.6.1.2-1: Occupied channel bandwidth**

| | Occupied channel bandwidth / channel bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Channel bandwidth [MHz] | 1.4 | 3 | 5 | 10 | 15 | 20 |

The normative reference for this requirement is TS 36.101 [2] clause 6.6.1.

## 6.6.1.4 Test description

### 6.6.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.6. 1.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | All | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Occupied bandwidth | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 3MHz | | QPSK | 15 | 15 |
| 5MHz | | QPSK | 25 | 25 |
| 10MHz | | QPSK | 50 | 50 |
| 15MHz | | QPSK | 75 | 75 |
| 20MHz | | QPSK | 100 | 100 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channels are set according to Table 6.6.1.4.1-1.

5. Propagation conditions are set according to Annex B.0

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.1.4.3

#### 6.6.1.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.6.1.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.

2. Send continuously power control “up”commands to the UE until the UE transmits at $P_{UMAX}$ level.

3. Measure the power spectrum distribution within two times or more range over the requirement for Occupied Bandwidth specification centring on the current carrier frequency. The characteristic of the filter shall be approximately Gaussian (typical spectrum analyzer filter). Other methods to measure the power spectrum distribution are allowed. The measuring duration is one active uplink subframe. For TDD slots with transient periods are not under test.

4. Calculate the total power within the range of all frequencies measured in '3)' and save this value as "Total Power".

5. Sum up the power upward from the lower boundary of the measured frequency range in '3)' and seek the limit frequency point by which this sum becomes 0,5 % of "Total Power" and save this point as "Lower Frequency".

6. Sum up the power downward from the upper boundary of the measured frequency range in '3)' and seek the limit frequency point by which this sum becomes 0,5 % of "Total Power" and save this point as "Upper Frequency".

7. Calculate the difference ("Upper Frequency" – "Lower Frequency" = "Occupied Bandwidth") between two limit frequencies obtained in '5)' and '6)'.

#### 6.6.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6

### 6.6.1.5 Test requirement

The measured Occupied Bandwidth shall not exceed values in Table 6.6.1.5-1.

**Table 6.6.1.5-1: Occupied channel bandwidth**

| | Occupied channel bandwidth / channel bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Channel bandwidth [MHz] | 1.4 | 3 | 5 | 10 | 15 | 20 |

## 6.6.1A Occupied bandwidth for intra-band contiguous CA

Editor’s notes:

The following items are missing or incomplete:

- Initial conditions (test frequencies, test configurations)

- Test procedure

- Message Contents

- Test requirements and test tolerances

- Connection diagram is missing

### 6.6.1A.1 Test purpose

To verify that the UE occupied bandwidth for intra-band contiguous CA for all transmission bandwidth configurations supported by the UE are less than their specific limits

### 6.6.1A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

### 6.6.1A.3 Minimum conformance requirements

In the case carriers are contiguously aggregated in the uplink (intra-band), occupied bandwidth is a measure of the bandwidth containing 99 % of the total integrated power of the transmitted spectrum. The OBW for intra-band contiguously aggregated carriers shall be less than the aggregated channel bandwidth defined in section 5.4.2A.

The normative reference for this requirement is TS 36.101[2] clause 6.6.1A.

### 6.6.1A.4 Test description

#### 6.6.1A.4.1 Initial condition

#### 6.6.1A.4.2 Test procedure

#### 6.6.1A.4.3 Message contents

### 6.6.1A.5 Test Requirements

## 6.6.2 Out of band emission

Out of band emissions are unwanted emissions immediately outside the nominal channel resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out of band emission limit is specified in terms of a Spectrum Emission Mask and Adjacent Channel Leakage power Ratio.

### 6.6.2.1 Spectrum Emission Mask

#### 6.6.2.1.1 Test purpose

To verify that the power of any UE emission shall not exceed specified lever for the specified channel bandwidth.

#### 6.6.2.1.2 Test applicability

This test case applies to all types of E-UTRA FDD UE release 8 and forward.

#### 6.6.2.1.3 Minimum conformance requirements

The spectrum emission mask of the UE applies to frequencies ($\Delta f_{OOB}$) starting from the edge of the assigned E-UTRA channel bandwidth. For frequencies greater than ($\Delta f_{OOB}$) as specified in Table 6.6.2.1.3-1 the spurious requirements in clause 6.6.3 are applicable.

The power of any UE emission shall not exceed the levels specified in Table 6.6.2.1.3-1 for the specified channel bandwidth.

**Table 6.6.2.1.3-1: General E-UTRA spectrum emission mask**

| Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| ± 0-1 | -10 | -13 | -15 | -18 | -20 | -21 | 30 kHz |
| ± 1-2.5 | -10 | -10 | -10 | -10 | -10 | -10 | 1 MHz |
| ± 2.5-2.8 | -25 | -10 | -10 | -10 | -10 | -10 | 1 MHz |
| ± 2.8-5 | | -10 | -10 | -10 | -10 | -10 | 1 MHz |
| ± 5-6 | | -25 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 6-10 | | | -25 | -13 | -13 | -13 | 1 MHz |
| ± 10-15 | | | | -25 | -13 | -13 | 1 MHz |
| ± 15-20 | | | | | -25 | -13 | 1 MHz |
| ± 20-25 | | | | | | -25 | 1 MHz |

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

The normative reference for this requirement is TS 36.101 [2] clause 6.6.2.1.

### 6.6.2.1.4 Test description

#### 6.6.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.6.2.1.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2 respectively. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.2.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | Lowest, 5MHz, 10MHz, Highest | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for SEM testing | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 1.4MHz | | QPSK | 6 | 6 |
| 1.4MHz | | QPSK | 5 | 5 |
| 1.4MHz | | 16QAM | 5 | 5 |
| 1.4MHz | | 16QAM | 6 | 6 |
| 3MHz | | QPSK | 15 | 15 |
| 3MHz | | QPSK | 4 | 4 |
| 3MHz | | 16QAM | 4 | 4 |
| 3MHz | | 16QAM | 15 | 15 |
| 5MHz | | QPSK | 25 | 25 |
| 5MHz | | QPSK | 8 | 8 |
| 5MHz | | 16QAM | 8 | 8 |
| 5MHz | | 16QAM | 25 | 25 |
| 10MHz | | QPSK | 50 | 50 |
| 10MHz | | QPSK | 12 | 12 |
| 10MHz | | 16QAM | 12 | 12 |
| 10MHz | | 16QAM | 50 (Note 4) | 50 (Note 4) |
| 15MHz | | QPSK | 75 | 75 |
| 15MHz | | QPSK | 16 | 16 |
| 15MHz | | 16QAM | 16 | 16 |
| 15MHz | | 16QAM | 75 (Note 4) | 75 (Note 4) |
| 20MHz | | QPSK | 100 | 100 |
| 20MHz | | QPSK | 18 | 18 |
| 20MHz | | 16QAM | 18 | 18 |
| 20MHz | | 16QAM | 100 (Note 4) | 100 (Note 4) |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.
Note 2: The allowed MPR for maximum output power UE might apply is described in clause 6.2.3.3.
Note 3: The $RB_{start}$ of partial RB allocation shall be RB#0 and RB# (max+1 – RB allocation) of the channel bandwidth.
Note 4: Applies only for UE-Categories 2-5

1. Connect the SS to the UE antenna connectors as shown in Figure TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channels are set according to Table 6.6.2.1.4.1-1.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.2.1.4.3.

#### 6.6.2.1.4.2 Test procedure

1. SS sends uplink scheduling information via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.6.2.1.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.

3. Measure the mean power of the UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in Table 6.2.3.5-1. The period of the measurement shall be at least the continuous duration of one sub-frame (1ms). For TDD slots with transient periods are not under test.

4. Measure the power of the transmitted signal with a measurement filter of bandwidths according to table 6.6.2.1.5-1 or 6.6.2.1.5-2, as applicable. The centre frequency of the filter shall be stepped in continuous steps according to the same table. The measured power shall be recorded for each step. The measurement period shall capture the active TSs.

#### 6.6.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.6.2.1.5 Test requirements

The measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.3.5-1 as appropriate,

and

the power of any UE emission shall fullfil requirements in Table.6.6.2.1.5-1 or Table.6.6.2.1.5-2, as applicable.

**Table 6.6.2.1.5-1: General E-UTRA spectrum emission mask, E-UTRA bands ≤ 3GHz**

| | **Spectrum emission limit (dBm)/ Channel bandwidth** | | | | | | |
|---|---|---|---|---|---|---|---|
| **$\Delta f_{OOB}$ (MHz)** | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** | **Measurement bandwidth** |
| 0-1 | -8.5 | -11.5 | -13.5 | -16.5 | -18.5 | -19.5 | 30 kHz |
| 1-2.5 | -8.5 | -8.5 | -8.5 | -8.5 | -8.5 | -8.5 | 1 MHz |
| 2.5-2.8 | -23.5 | | | | | | 1 MHz |
| 2.8-5 | | | | | | | 1 MHz |
| 5-6 | | -23.5 | -11.5 | -11.5 | -11.5 | -11.5 | 1 MHz |
| 6-10 | | | -23.5 | | | | 1 MHz |
| 10-15 | | | | -23.5 | | | 1 MHz |
| 15-20 | | | | | -23.5 | | 1 MHz |
| 20-25 | | | | | | -23.5 | 1 MHz |
| Note 1: The first and last measurement position with a 30 kHz filter is at $\Delta f_{OOB}$ equals to 0.015 MHz and 0.985 MHz. | | | | | | | |
| Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively. | | | | | | | |
| Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel | | | | | | | |
| Note 4: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz. | | | | | | | |

**Table 6.6.2.1.5-2: General E-UTRA spectrum emission mask, 3GHz < E-UTRA bands ≤ 4.2GHz**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| 0-1 | -8.2 | -11.2 | -13.2 | -16.2 | -18.2 | -19.2 | 30 kHz |
| 1-2.5 | -8.2 | -8.2 | -8.2 | -8.2 | -8.2 | -8.2 | 1 MHz |
| 2.5-2.8 | -23.2 | | | | | | 1 MHz |
| 2.8-5 | | | | | | | 1 MHz |
| 5-6 | | -23.2 | -11.2 | -11.2 | -11.2 | -11.2 | 1 MHz |
| 6-10 | | | -23.2 | | | | 1 MHz |
| 10-15 | | | | -23.2 | | | 1 MHz |
| 15-20 | | | | | -23.2 | | 1 MHz |
| 20-25 | | | | | | -23.2 | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at $\Delta f_{OOB}$ equals to 0.015 MHz and 0.985 MHz.
Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.
Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel
Note 4: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

## 6.6.2.2 Additional Spectrum Emission Mask

### 6.6.2.2.1 Test purpose

To verify that the power of any UE emission shall not exceed specified level for the specified channel bandwidth under the deployment scenarios where additional requirements are specified.

### 6.6.2.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.6.2.2.3 Minimum conformance requirements

#### 6.6.2.2.3.1 Minimum requirement (network signalled value "NS_03")

When "NS_03" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.2.2.3.1-1.

**Table 6.6.2.2.3.1-1: Additional requirements (network signalled value "NS_03")**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| ± 0-1 | -10 | -13 | -15 | -18 | -20 | -21 | 30 kHz |
| ± 1-2.5 | -13 | -13 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 2.5-2.8 | -25 | -13 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 2.8-5 | | -13 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 5-6 | | -25 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 6-10 | | | -25 | -13 | -13 | -13 | 1 MHz |
| ± 10-15 | | | | -25 | -13 | -13 | 1 MHz |
| ± 15-20 | | | | | -25 | -13 | 1 MHz |
| ± 20-25 | | | | | | -25 | 1 MHz |

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

The normative reference for this requirement is TS 36.101 [2] clause 6.6.2.2.1.

### 6.6.2.2.3.2 Minimum requirement (network signalled value "NS_04")

When "NS_04" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.2.2.3.2-1.

**Table 6.6.2.2.3.2-1: Additional requirements (network signalled value "NS_04")**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| ± 0-1 | -10 | -13 | -15 | -18 | -20 | -21 | 30 kHz |
| ± 1-2.5 | -13 | -13 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 2.5-2.8 | -25 | -13 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 2.8-5 | | -13 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 5-6 | | -25 | -25 | -25 | -25 | -25 | 1 MHz |
| ± 6-10 | | | -25 | -25 | -25 | -25 | 1 MHz |
| ± 10-15 | | | | -25 | -25 | -25 | 1 MHz |
| ± 15-20 | | | | | -25 | -25 | 1 MHz |
| ± 20-25 | | | | | | -25 | 1 MHz |

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

The normative reference for this requirement is TS 36.101 [2] clause 6.6.2.2.2.

### 6.6.2.2.3.3 Minimum requirement (network signalled value "NS_06" or NS_07)

When "NS_06" or “NS_07” is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.2.2.3.3-1.

**Table 6.6.2.2.3.3-1: Additional requirements (network signalled value "NS_06" or "NS_07")**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | |
|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | Measurement bandwidth |
| ± 0-0.1 | -13 | -13 | -15 | -18 | 30 kHz |
| ± 0.1-1 | -13 | -13 | -13 | -13 | 100 kHz |
| ± 1-2.5 | -13 | -13 | -13 | -13 | 1 MHz |
| ± 2.5-2.8 | -25 | -13 | -13 | -13 | 1 MHz |
| ± 2.8-5 | | -13 | -13 | -13 | 1 MHz |
| ± 5-6 | | -25 | -13 | -13 | 1 MHz |
| ± 6-10 | | | -25 | -13 | 1 MHz |
| ± 10-15 | | | | -25 | 1 MHz |

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

The normative reference for this requirement is TS 36.101 [2] clause 6.6.2.2.3.

### 6.6.2.2.4 Test description

#### 6.6.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in tables 6.6.2.2.4.1-1, 6.6.2.2.4.1-2, and 6.6.2.2.4.1-3. The details of the uplink reference measurement channels (RMCs) are specified in Annex A.2 respectively. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.2.2.4.1-1: Test Configuration Table (network signalled value "NS_03")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | Lowest, 5MHz, 10MHz, Highest | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | N/A for Additional Spectrum Emission Mask testing. | | | QPSK | 6 | 6 |
| 1.4MHz | | | | QPSK | 5 | 5 |
| 1.4MHz | | | | 16QAM | 5 | 5 |
| 3MHz | | | | QPSK | 15 | 15 |
| 3MHz | | | | QPSK | 4 | 4 |
| 3MHz | | | | 16QAM | 15 | 15 |
| 3MHz | | | | 16QAM | 4 | 4 |
| 5MHz | | | | QPSK | 25 | 25 |
| 5MHz | | | | QPSK | 8 | 8 |
| 5MHz | | | | QPSK | 6 | 6 |
| 5MHz | | | | 16QAM | 25 | 25 |
| 5MHz | | | | 16QAM | 8 | 8 |
| 10MHz | | | | QPSK | 50 | 50 |
| 10MHz | | | | QPSK | 12 | 12 |
| 10MHz | | | | QPSK | 6 | 6 |
| 10MHz | | | | 16QAM | 50 (Note 3) | 50 (Note 3) |
| 10MHz | | | | 16QAM | 12 | 12 |
| 15MHz | | | | QPSK | 75 | 75 |
| 15MHz | | | | QPSK | 16 | 16 |
| 15MHz | | | | QPSK | 8 | 8 |
| 15MHz | | | | 16QAM | 75 (Note 3) | 75 (Note 3) |
| 15MHz | | | | 16QAM | 16 | 16 |
| 20MHz | | | | QPSK | 100 | 100 |
| 20MHz | | | | QPSK | 18 | 18 |
| 20MHz | | | | QPSK | 10 | 10 |
| 20MHz | | | | 16QAM | 100 (Note 3) | 100 (Note 3) |
| 20MHz | | | | 16QAM | 18 | 18 |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth.

Note 3: Applies only for UE-Categories 2-5

**Table 6.6.2.2.4.1-2: Test Configuration Table (network signalled value "NS_06")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | Lowest, 5MHz, 10MHz, Highest | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | N/A for Additional Spectrum Emission Mask testing. | | | QPSK | 6 | NA |
| 1.4MHz | | | | QPSK | 5 | |
| 1.4MHz | | | | 16QAM | 5 | |
| 3MHz | | | | QPSK | 15 | |
| 3MHz | | | | QPSK | 4 | |
| 3MHz | | | | 16QAM | 4 | |
| 5MHz | | | | QPSK | 25 | |
| 5MHz | | | | QPSK | 8 | |
| 5MHz | | | | 16QAM | 8 | |
| 10MHz | | | | QPSK | 50 | |
| 10MHz | | | | QPSK | 12 | |
| 10MHz | | | | 16QAM | 12 | |
| 15MHz | | | | QPSK | 75 | |
| 15MHz | | | | QPSK | 16 | |
| 15MHz | | | | 16QAM | 16 | |
| 20MHz | | | | QPSK | 100 | |
| 20MHz | | | | QPSK | 18 | |
| 20MHz | | | | 16QAM | 18 | |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1.

Note 2: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth.

**Table 6.6.2.2.4.1-3: Test Configuration Table (network signalled value "NS_07")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | | | | Mid range | | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | | | | 10MHz | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | | |
| Test Number | Ch BW | Mod'n | RB allocation FDD | Mod'n | RB allocation FDD | $RB_{start}$ |
| 1 | 10MHz | N/A for Additional Spectrum Emission Mask testing. | | QPSK | 1 | 0 |
| 2 | 10MHz | | | QPSK | 8 | 0 |
| 3 | 10MHz | | | QPSK | 6 | 13 |
| 4 | 10MHz | | | QPSK | 20 | 13 |
| 5 | 10MHz | | | QPSK | 12 | 13 |
| 6 | 10MHz | | | 16QAM | 36<br>(Note 1) | 13 |
| 7 | 10MHz | | | QPSK | 16 | 19 |
| 8 | 10MHz | | | QPSK | 12 | 19 |
| 9 | 10MHz | | | 16QAM | 16 | 19 |
| 10 | 10MHz | | | QPSK | 30 | 19 |
| 11 | 10MHz | | | 16QAM | 30<br>(Note 1) | 19 |
| 12 | 10MHz | | | QPSK | 6 | 43 |
| 13 | 10MHz | | | QPSK | 2 | 48 |
| 14 | 10MHz | | | QPSK | 50 | 0 |
| 15 | 10MHz | | | QPSK | 12 | 0 |
| 16 | 10MHz | | | 16QAM | 50<br>(Note 1) | 0 |
| Note 1: Applies only for UE-Categories 2-5 | | | | | | |

**Table 6.6.2.2.4.1-4: Test Configuration Table (network signalled value "NS_04")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | 5MHz, 10 MHz, 15 MHz, 20MHz | | |
| **Test Parameters for NS_04 A-MPR** | | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | | |
| Configuration ID | Ch BW | Mod'n | RB allocation TDD | Mod'n | RB allocation TDD | $RB_{start}$ TDD |
| 1 | 5MHz | N/A for Additional Spectrum Emission Mask testing. | | QPSK | 25 | Note 2 |
| 2 | 5MHz | | | QPSK | 8 | Note 2 |
| 3 | 5MHz | | | QPSK | 6 | Note 2 |
| 4 | 5MHz | | | 16QAM | 25 | Note 2 |
| 5 | 5MHz | | | 16QAM | 8 | Note 2 |
| 6 | 10MHz | | | QPSK | 1 | 0 |
| 7 | 10MHz | | | QPSK | 12 | 0 |
| 8 | 10MHz | | | QPSK | 50 | 0 |
| 9 | 10MHz | | | 16QAM | 50 (Note 3) | 0 |
| 10 | 10MHz | | | QPSK | 24 | 13 |
| 11 | 10MHz | | | 16QAM | 24 | 13 |
| 12 | 10MHz | | | QPSK | 36 | 13 |
| 13 | 10MHz | | | QPSK | 12 | 37 |
| 14 | 10MHz | | | QPSK | 1 | 49 |
| 15 | 15MHz | | | QPSK | 1 | 0 |
| 16 | 15MHz | | | QPSK | 16 | 0 |
| 17 | 15MHz | | | QPSK | 75 | 0 |
| 18 | 15MHz | | | 16QAM | 75 (Note 3) | 0 |
| 19 | 15MHz | | | QPSK | 36 | 19 |
| 20 | 15MHz | | | 16QAM | 36 (Note 3) | 19 |
| 21 | 15MHz | | | QPSK | 50 | 19 |
| 22 | 15MHz | | | QPSK | 18 | 56 |
| 23 | 15MHz | | | QPSK | 1 | 74 |
| 24 | 20MHz | | | QPSK | 1 | 0 |
| 25 | 20MHz | | | QPSK | 18 | 0 |
| 26 | 20MHz | | | QPSK | 100 | 0 |
| 27 | 20MHz | | | 16QAM | 100 (Note 3) | 0 |
| 28 | 20MHz | | | QPSK | 50 | 25 |
| 29 | 20MHz | | | 16QAM | 50 (Note 3) | 25 |
| 30 | 20MHz | | | QPSK | 75 | 25 |
| 31 | 20MHz | | | QPSK | 25 | 75 |
| 32 | 20MHz | | | QPSK | 1 | 99 |
| Note 1: Test Channel Bandwidths are checked separately for E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | | | |
| Note 2: The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth. | | | | | | |
| Note 3: Applies only for UE-Categories 2-5 | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The DL Reference Measurement channels are set according to Tables 6.6.2.2.4.1-1, 6.6.2.2.4.1-2, 6.6.2.2.4.1-3 and 6.6.2.2.4.1-4.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.2.2.4.3.

#### 6.6.2.2.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to the corresponding Table 6.6.2.2.4.1-1, 6.6.2.2.4.1-2, 6.6.2.2.4.1-3 or 6.6.2.2.4.1-4. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.

3. Measure the mean power of the UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in Table 6.2.4.5-1 to 6.2.4.5-9 as appropriate. The period of the measurement shall be at least one sub-frame (1ms).

4. Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table 6.6.2.2.5.1-1, 6.6.2.2.5.2-1, 6.6.2.2.5.3-1 or Table 6.6.2.2.5.1-2, 6.6.2.2.5.2-2, 6.6.2.2.5.3-2, as applicable. The centre frequency of the filter shall be stepped in continuous steps according to the same table. The measured power shall be recorded for each step. The measurement period shall capture the active TSs.

#### 6.6.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6, with the following exceptions for each network signalled value.

##### 6.6.2.2.4.3.1 Message contents exceptions (network signalled value "NS_03")

1. Information element additionalSpectrumEmission is set to NS_03. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.2.2.4.3.1-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 3 (NS_03) | | |

##### 6.6.2.2.4.3.2 Message contents exceptions (network signalled value "NS_04")

1. Information element additionalSpectrumEmission is set to NS_04. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.2.2.4.3.2-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 4 (NS_04) | | |

##### 6.6.2.2.4.3.3 Message contents exceptions (network signalled value "NS_06")

1. Information element additionalSpectrumEmission is set to NS_06. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.2.2.4.3.3-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 6 (NS_06) | | |

6.6.2.2.4.3.4 Message contents exceptions (network signalled value "NS_07")

a) 1. Information element additionalSpectrumEmission is set to NS_07. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.2.2.4.3.4-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 7 (NS_07) | | |

## 6.6.2.2.5 Test requirements

### 6.6.2.2.5.1 Test requirements (network signalled value "NS_03")

When "NS_03" is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-1 and 6.2.4.5-2 as appropriate,

and

- the power of any UE emission shall fulfil requirements in Table 6.6.2.2.5.1-1 or 6.6.2.2.5.1-2, as applicable.

**Table 6.6.2.2.5.1-1: Additional requirements (network signalled value "NS_03"), E-UTRA bands ≤ 3GHz**

| | **Spectrum emission limit (dBm)/ Channel bandwidth** | | | | | | |
|---|---|---|---|---|---|---|---|
| **$\Delta f_{OOB}$ (MHz)** | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** | **Measurement bandwidth** |
| 0-1 | -8.5 | -11.5 | -13.5 | -16.5 | -18.5 | -19.5 | 30 kHz |
| 1-2.5 | -11.5 | -11.5 | -11.5 | -11.5 | -11.5 | -11.5 | 1 MHz |
| 2.5-2.8 | -23.5 | | | | | | 1 MHz |
| 2.8-5 | | | | | | | 1 MHz |
| 5-6 | | -23.5 | | | | | 1 MHz |
| 6-10 | | | -23.5 | | | | 1 MHz |
| 10-15 | | | | -23.5 | | | 1 MHz |
| 15-20 | | | | | -23.5 | | 1 MHz |
| 20-25 | | | | | | -23.5 | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at ΔfOOB equals to 0.015 MHz and 0.985 MHz.
Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.
Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel
Note 4: Above SEM requirement applies to bands corresponding to network signalling value NS_03 as defined in TS 36.101 [2] subclause 6.2.4 Table 6.2.4-1.
Note 5: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

**Table 6.6.2.2.5.1-2: Additional requirements (network signalled value "NS_03"), 3GHz < E-UTRA bands ≤ 4.2GHz**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| 0-1 | -8.2 | -11.2 | -13.2 | -16.2 | -18.2 | -19.2 | 30 kHz |
| 1-2.5 | -11.2 | -11.2 | -11.2 | -11.2 | -11.2 | -11.2 | 1 MHz |
| 2.5-2.8 | -23.2 | | | | | | 1 MHz |
| 2.8-5 | | | | | | | 1 MHz |
| 5-6 | | -23.2 | | | | | 1 MHz |
| 6-10 | | | -23.2 | | | | 1 MHz |
| 10-15 | | | | -23.2 | | | 1 MHz |
| 15-20 | | | | | -23.2 | | 1 MHz |
| 20-25 | | | | | | -23.2 | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at ΔfOOB equals to 0.015 MHz and 0.985 MHz.
Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.
Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel
Note 4: Above SEM requirement applies to bands corresponding to network signalling value NS_03 as defined in TS 36.101 [2] subclause 6.2.4 Table 6.2.4-1.
Note 5: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

NOTE (only for emission measurement):

As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

#### 6.6.2.2.5.2 Test requirements (network signalled value "NS_04")

When "NS_04" is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-3 as appropriate,

and

- the power of any UE emission shall fulfil requirements in Table 6.6.2.2.5.2-1 or Table.6.6.2.2.5.2-2, as applicable.

**Table 6.6.2.2.5.2-1: Additional requirements (network signalled value "NS_04"), E-UTRA bands ≤ 3GHz**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| 0-1 | -8.5 | -11.5 | -13.5 | -16.5 | -18.5 | -19.5 | 30 kHz |
| 1-2.5 | -11.5 | -11.5 | -11.5 | -11.5 | -11.5 | -11.5 | 1 MHz |
| 2.5-2.8 | -23.5 | | | | | | 1 MHz |
| 2.8-5 | | | | | | | 1 MHz |
| 5-6 | | -23.5 | -23.5 | -23.5 | -23.5 | -23.5 | 1 MHz |
| 6-10 | | | | | | | 1 MHz |
| 10-15 | | | | | | | 1 MHz |
| 15-20 | | | | | | | 1 MHz |
| 20-25 | | | | | | | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at ΔfOOB equals to 0.015 MHz and 0.985 MHz.
Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.
Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel
Note 4: Above SEM requirement applies to bands corresponding to network signalling value NS_04 as defined in TS 36.101 [2] subclause 6.2.4 Table 6.2.4-1.
Note 5: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

**Table 6.6.2.2.5.2-2: Additional requirements (network signalled value "NS_04"), 3GHz < E-UTRA bands ≤ 4.2GHz**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | | | |
|---|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz | Measurement bandwidth |
| 0-1 | -8.2 | -11.2 | -13.2 | -16.2 | -18.2 | -19.2 | 30 kHz |
| 1-2.5 | -11.2 | -11.2 | -11.2 | -11.2 | -11.2 | -11.2 | 1 MHz |
| 2.5-2.8 | -23.2 | | | | | | 1 MHz |
| 2.8-5 | | | | | | | 1 MHz |
| 5-6 | | -23.2 | -23.2 | -23.2 | -23.2 | -23.2 | 1 MHz |
| 6-10 | | | | | | | 1 MHz |
| 10-15 | | | | | | | 1 MHz |
| 15-20 | | | | | | | 1 MHz |
| 20-25 | | | | | | | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at ΔfOOB equals to 0.015 MHz and 0.985 MHz.
Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.
Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel
Note 4: Above SEM requirement applies to bands corresponding to network signalling value NS_04 as defined in TS 36.101 [2] subclause 6.2.4 Table 6.2.4-1.
Note 5: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

NOTE (only for emission measurement):

As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

### 6.6.2.2.5.3 Test requirements (network signalled value "NS_06" or “NS_07”)

When "NS_06" or “NS_07” is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-5, 6.2.4.5-6 and 6.2.4.5-7 as appropriate l,

and

- the power of any UE emission shall fulfil requirements in Table 6.6.2.2.5.3-1 or Table.6.6.2.2.5.3-2, as applicable.

**Table 6.6.2.2.5.3-1: Additional requirements (network signalled value "NS_06" or "NS_07") , E-UTRA bands ≤ 3GHz**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | |
|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | Measurement bandwidth |
| 0-0.1 | -11.5 | -11.5 | -13.5 | -16.5 | 30 kHz |
| 0.1-1 | -11.5 | -11.5 | -11.5 | -11.5 | 100 kHz |
| 1-2.5 | -11.5 | -11.5 | -11.5 | -11.5 | 1 MHz |
| 2.5-2.8 | -23.5 | | | | 1 MHz |
| 2.8-5 | | | | | 1 MHz |
| 5-6 | | -23.5 | | | 1 MHz |
| 6-10 | | | -23.5 | | 1 MHz |
| 10-15 | | | | -23.5 | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at ΔfOOB equals to 0.015 MHz and 0.085 MHz.
The first and last measurement position with a 100 kHz filter is at ΔfOOB equals to 0.15 MHz and 0.95 MHz.

Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.

Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel

Note 4: Above SEM requirement applies to bands corresponding to network signalling value NS_06 and NS_07 as defined in TS 36.101 [2] subclause 6.2.4 Table 6.2.4-1.

Note 5: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

**Table 6.6.2.2.5.3-2: Additional requirements (network signalled value "NS_06" or "NS_07") , 3GHz < E-UTRA bands ≤ 4.2GHz**

| | Spectrum emission limit (dBm)/ Channel bandwidth | | | | |
|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | Measurement bandwidth |
| 0-0.1 | -11.2 | -11.2 | -13.2 | -16.2 | 30 kHz |
| 0.1-1 | -11.2 | -11.2 | -11.2 | -11.2 | 100 kHz |
| 1-2.5 | -11.2 | -11.2 | -11.2 | -11.2 | 1 MHz |
| 2.5-2.8 | -23.2 | | | | 1 MHz |
| 2.8-5 | | | | | 1 MHz |
| 5-6 | | -23.2 | | | 1 MHz |
| 6-10 | | | -23.2 | | 1 MHz |
| 10-15 | | | | -23.2 | 1 MHz |

Note 1: The first and last measurement position with a 30 kHz filter is at ΔfOOB equals to 0.015 MHz and 0.085 MHz. The first and last measurement position with a 100 kHz filter is at ΔfOOB equals to 0.15 MHz and 0.95 MHz.

Note 2: At the boundary of spectrum emission limit, the first and last measurement position with a 1 MHz filter is the inside of +0.5MHz and -0.5MHz, respectively.

Note 3: The measurements are to be performed above the upper edge of the channel and below the lower edge of the channel

Note 4: Above SEM requirement applies to bands corresponding to network signalling value NS_06 and NS_07 as defined in TS 36.101 [2] subclause 6.2.4 Table 6.2.4-1.

Note 5: For the 2.5-2.8 MHz offset range with 1.4 MHz channel bandwidth, the measurement position is at $\Delta f_{OOB}$ equals to 3 MHz.

NOTE (only for emission measurement):

As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

## 6.6.2.3 Adjacent Channel Leakage power Ratio

### 6.6.2.3.1 Test purpose

To verify that UE transmitter does not cause unacceptable interference to adjacent channels in terms of Adjacent Channel Leakage power Ratio (ACLR).

### 6.6.2.3.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.6.2.3.3 Minimum conformance requirements

ACLR requirements are specified for two scenarios for an adjacent E -$UTRA_{ACLR}$ and $UTRA_{ACLR1/2}$ as shown in Figure 6.6.2.3.3-1.

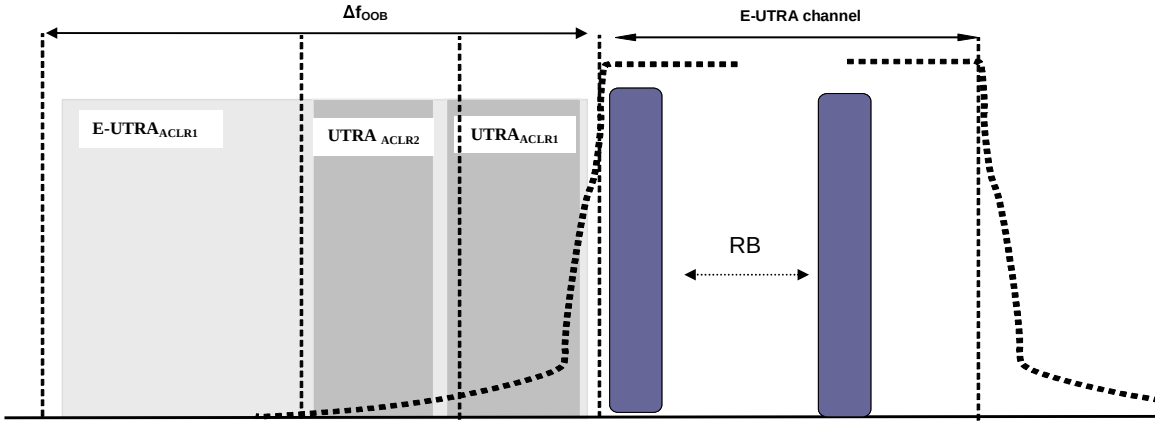



**Figure 6.6.2.3.3-1: Adjacent Channel Leakage Power Ratio requirements**

### 6.6.2.3.3.1 Minimum conformance requirements for E-UTRA

E-UTRA ACLR (E-UTRA$_{ACLR}$) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency at nominal channel spacing. The assigned E-UTRA channel power and adjacent E-UTRA channel power are measured with rectangular filters with measurement bandwidth specified in Table 6.6.2.3.3.1-1.

If the measured adjacent channel power is greater than –50dBm then the E-UTRA$_{ACLR}$ shall be higher than the valued specified in Table 6.6.2.3.3.1-1.

**Table 6.6.2.3.3.1-1: General requirements for E-UTRA$_{ACLR}$**

| | Channel bandwidth / E-UTRA$_{ACLR1}$ / measurement bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| E-UTRA$_{ACLR1}$ | 30 dB | 30 dB | 30 dB | 30 dB | 30 dB | 30 dB |
| E-UTRA channel Measurement bandwidth | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |

The normative reference for this requirement is TS 36.101 [2] subclause 6.6.2.3.1.

### 6.6.2.3.3.2 Minimum conformance requirements for UTRA

UTRA ACLR (UTRA$_{ACLR}$) is the ratio of the filtered mean power centred on the assigned E-UTRA channel frequency to the filtered mean power centred on an adjacent UTRA channel frequency.

UTRA ACLR is specified for both the first UTRA adjacent channel (UTRA$_{ACLR1}$) and the 2$^{nd}$ UTRA adjacent channel (UTRA$_{ACLR2}$). The UTRA channel power is measured with a RRC bandwidth filter with roll-off factor $\alpha = 0.22$. The assigned E-UTRA channel power is measured with a rectangular filter with measurement bandwidth specified in Table 6.6.2.3.3.2-1.

If the measured UTRA channel power is greater than –50dBm then the UTRA$_{ACLR1}$, and UTRA$_{ACLR2}$ shall be higher than the valued specified in Table 6.6.2.3.3.2-1.

**Table 6.6.2.3.3.2-1: General requirements for UTRA$_{ACLR1/2}$**

| | Channel bandwidth / UTRA$_{ACLR1/2}$ / measurement bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |

| | | | | | | |
|---|---|---|---|---|---|---|
| **$UTRA_{ACLR1}$** | 33 dB | 33 dB | 33 dB | 33 dB | 33 dB | 33 dB |
| **Adjacent channel centre frequency offset (in MHz)** | 0.7+$BW_{UTRA}$/2 / -0.7-$BW_{UTRA}$/2 | 1.5+$BW_{UTRA}$/2 / -1.5-$BW_{UTRA}$/2 | 2.5+$BW_{UTRA}$/2 / -2.5-$BW_{UTRA}$/2 | 5+$BW_{UTRA}$/2 / -5-$BW_{UTRA}$/2 | 7.5+$BW_{UTRA}$/2 / -7.5-$BW_{UTRA}$/2 | 10+$BW_{UTRA}$/2 / -10-$BW_{UTRA}$/2 |
| **$UTRA_{ACLR2}$** | - | - | 36 dB | 36 dB | 36 dB | 36 dB |
| **Adjacent channel centre frequency offset (in MHz)** | - | - | 2.5+3*$BW_{UTRA}$/2 / -2.5-3*$BW_{UTRA}$/2 | 5+3*$BW_{UTRA}$/2 / -5-3*$BW_{UTRA}$/2 | 7.5+3*$BW_{UTRA}$/2 / -7.5-3*$BW_{UTRA}$/2 | 10+3*$BW_{UTRA}$/2 / -10-3*$BW_{UTRA}$/2 |
| **E-UTRA channel Measurement bandwidth** | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| **UTRA 5MHz channel Measurement bandwidth[1]** | 3.84 MHz | 3.84 MHz | 3.84 MHz | 3.84 MHz | 3.84 MHz | 3.84 MHz |
| **UTRA 1.6MHz channel measurement bandwidth[2]** | 1.28 MHz | 1.28 MHz | 1.28 MHz | 1.28 MHz | 1.28 MHz | 1.28 MHz |
| Note 1: Applicable for E-UTRA FDD co-existence with UTRA FDD in paired spectrum.<br>Note 2: Applicable for E-UTRA TDD co-existence with UTRA TDD in unpaired spectrum. | | | | | | |

The normative reference for this requirement is TS 36.101 subclause 6.6.2.3.2.

### 6.6.2.3.4 Test description

#### 6.6.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.6.2.3.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in AnnexeA.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.2.3.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC, TL/VL, TL/VH, TH/VL, TH/VH | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | Lowest, 5MHz, 10MHz, Highest | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | N/A for ACLR testing | | | QPSK | 6 | 6 |
| 1.4MHz | | | | QPSK | 5 | 5 |
| 1.4MHz | | | | 16QAM | 6 | 6 |
| 1.4MHz | | | | 16QAM | 5 | 5 |
| 3MHz | | | | QPSK | 15 | 15 |
| 3MHz | | | | QPSK | 4 | 4 |
| 3MHz | | | | 16QAM | 15 | 15 |
| 3MHz | | | | 16QAM | 4 | 4 |
| 5MHz | | | | QPSK | 25 | 25 |
| 5MHz | | | | QPSK | 8 | 8 |
| 5MHz | | | | 16QAM | 25 | 25 |
| 5MHz | | | | 16QAM | 8 | 8 |
| 10MHz | | | | QPSK | 50 | 50 |
| 10MHz | | | | QPSK | 12 | 12 |
| 10MHz | | | | 16QAM | 50 (Note 3) | 50 (Note 3) |
| 10MHz | | | | 16QAM | 12 | 12 |
| 15MHz | | | | QPSK | 75 | 75 |
| 15MHz | | | | QPSK | 16 | 16 |
| 15MHz | | | | 16QAM | 75 (Note 3) | 75 (Note 3) |
| 15MHz | | | | 16QAM | 16 | 16 |
| 20MHz | | | | QPSK | 100 | 100 |
| 20MHz | | | | QPSK | 18 | 18 |
| 20MHz | | | | 16QAM | 100 (Note 3) | 100 (Note 3) |
| 20MHz | | | | 16QAM | 18 | 18 |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.
Note 2: The $RB_{start}$ of partial RB allocation shall be RB#0 and RB# (max + 1 – RB allocation) of the channel bandwidth.
Note 3: Applies only for UE-Categories 2-5

1. Connect the SS to the UE antenna connectors as shown in Figure TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to Table 6.6.2.3.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.2.3.4.3.

### 6.6.2.3.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.6.2.3.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.
3. Measure the mean power of the UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in Table 6.2.3.5-1. The period of the measurement shall be at least the continuous duration of one sub-frame (1ms). For TDD slots with transient periods are not under test.
4. Measure the rectangular filtered mean power for E-UTRA.
5. Measure the rectangular filtered mean power of the first E-UTRA adjacent channel.
6. Measure the RRC filtered mean power of the first and the second UTRA adjacent channel.
7. Calculate the ratio of the power between the values measured in step 4 over step 5 for $E\text{-}UTRA_{ACLR}$.
8. Calculated the ratio of the power between the values measured in step 4 over step 6 for $UTRA_{ACLR1}$, $UTRA_{ACLR2}$.

#### 6.6.2.3.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.6.2.3.5 Test requirement

#### 6.6.2.3.5.1 Test requirements E-UTRA

- The measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.3.5-1 as appropriate,

and

- if the measured adjacent channel power is greater than –50 dBm then the measured $E\text{-}UTRA_{ACLR}$, derived in step 7, shall be higher than the limits in table 6.6.2.3.5.1-1.

**Table 6.6.2.3.5.1-1: E-UTRA UE ACLR**

| | **Channel bandwidth / $E\text{-}UTRA_{ACLR1}$ / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| **$E\text{-}UTRA_{ACLR1}$** | 29.2 dB | 29.2 dB | 29.2 dB | 29.2 dB | 29.2 dB | 29.2 dB |
| **E-UTRA channel Measurement bandwidth** | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| **UE channel** | +1.4 MHz or -1.4 MHz | +3 MHz or -3 MHz | +5MHz or -5MHz | +10MHz or -10MHz | +15MHz or -15MHz | +20MHz or -20MHz |

#### 6.6.2.3.5.2 Test requirements UTRA

If the measured UTRA channel power is greater than –50dBm then the measured $UTRA_{ACLR1}$, $UTRA_{ACLR2}$, derived in step 8, shall be higher than the limits in table 6.6.2.3.5.2-1.

**Table 6.6.2.3.5.2-1: UTRA UE ACLR**

| | **Channel bandwidth / $UTRA_{ACLR1/2}$ / measurement bandwidth** | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3.0 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |

| | | | | | | |
|---|---|---|---|---|---|---|
| **$UTRA_{ACLR1}$** | 32.2 dB | 32.2 dB | 32.2 dB | 32.2 dB | 32.2 dB | 32.2 dB |
| **Adjacent channel centre frequency offset (in MHz)** | 0.7+$BW_{UTRA}$/2 / -0.7-$BW_{UTRA}$/2 | 1.5+$BW_{UTRA}$/2 / -1.5-$BW_{UTRA}$/2 | 2.5+$BW_{UTRA}$/2 / -2.5-$BW_{UTRA}$/2 | 5+$BW_{UTRA}$/2 / -5-$BW_{UTRA}$/2 | 7.5+$BW_{UTRA}$/2 / -7.5-$BW_{UTRA}$/2 | 10+$BW_{UTRA}$/2 / -10-$BW_{UTRA}$/2 |
| **$UTRA_{ACLR2}$** | - | - | 35.2 dB | 35.2 dB | 35.2 dB | 35.2 dB |
| **Adjacent channel centre frequency offset (in MHz)** | - | - | 2.5+3*$BW_{UTRA}$/2 / -2.5-3*$BW_{UTRA}$/2 | 5+3*$BW_{UTRA}$/2 / -5-3*$BW_{UTRA}$/2 | 7.5+3*$BW_{UTRA}$/2 / -7.5-3*$BW_{UTRA}$/2 | 10+3*$BW_{UTRA}$/2 / -10-3*$BW_{UTRA}$/2 |
| **E-UTRA channel Measurement bandwidth** | 1.08 MHz | 2.7 MHz | 4.5 MHz | 9.0 MHz | 13.5 MHz | 18 MHz |
| **UTRA 5MHz channel Measurement bandwidth[1]** | 3.84 MHz | 3.84 MHz | 3.84 MHz | 3.84 MHz | 3.84 MHz | 3.84 MHz |
| **UTRA 1.6MHz channel measurement bandwidth[2]** | 1.28 MHz | 1.28 MHz | 1.28 MHz | 1.28 MHz | 1.28 MHz | 1.28 MHz |
| Note 1: Applicable for E-UTRA FDD co-existence with UTRA FDD in paired spectrum.<br>Note 2: Applicable for E-UTRA TDD co-existence with UTRA TDD in unpaired spectrum.<br>Note 3: $BW_{UTRA}$ for UTRA FDD is 5MHz and for UTRA TDD is 1.6MHz. | | | | | | |

## 6.6.2.3A Adjacent Channel Leakage power Ratio for CA

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- The initial conditions for this test are undefined
- The Test Procedure for this test is undefined
- Message Contents for this test are undefined
- The Test Requirements for this test are undefined
- The Test Tolerances applicable to this test are undefined

### 6.6.2.3A.1 Test purpose

To verify that UE transmitter does not cause unacceptable interference to adjacent channels in terms of Adjacent Channel Leakage power Ratio (ACLR) for CA.

### 6.6.2.3A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

### 6.6.2.3A.3 Minimum conformance requirements

ACLR for CA requirements are specified for two scenarios for an adjacent E -$UTRA_{ACLR}$ and $UTRA_{ACLR1/2}$ as shown in Figure 6.6.2.3A.3-1.

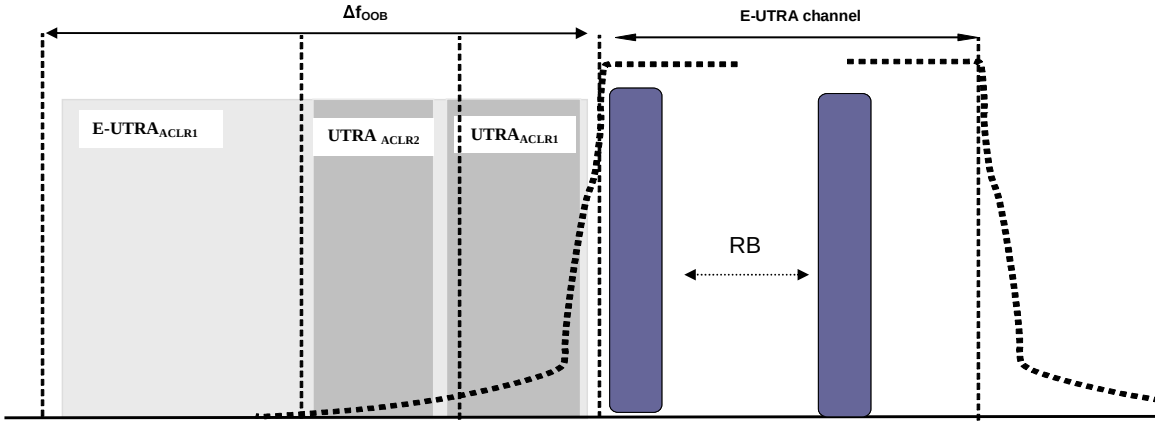



**Figure 6.6.2.3A.3-1: Adjacent Channel Leakage Power Ratio for CA requirements**

### 6.6.2.3A.3.1 Minimum conformance requirements of UTRA for CA

UTRA ACLR ($UTRA_{ACLR}$) for CA is the ratio of the filtered mean power centred on the assigned carrier aggregation E-UTRA channel frequency to the filtered mean power centred on an adjacent(s) UTRA channel frequency.

UTRA ACLR for CA is specified for both the first UTRA adjacent channel ($UTRA_{ACLR1}$) and the 2nd UTRA adjacent channel ($UTRA_{ACLR2}$). The UTRA channel power is measured with a RRC bandwidth filter with roll-off factor $\alpha$ = 0.22. The assigned carrier aggregated E-UTRA channel power is measured with a rectangular filter with measurement bandwidth specified in Table 6.6.2.3A.3.1-1.

If the measured UTRA channel power is greater than –50dBm then the $UTRA_{ACLR1}$ and $UTRA_{ACLR2}$ shall be higher than the valued specified in Table 6.6.2.3A.3.1-1.

**Table 6.6.2.3A.3.1-1: General requirements of $UTRA_{ACLR1/2}$ for CA**

| | **CA bandwidth class / $UTRA_{ACLR1/2}$ / measurement bandwidth** |
|---|---|
| | **CA bandwidth class C** |
| $UTRA_{ACLR1}$ | 33 dB |
| Adjacent channel centre frequency offset (in MHz) | + $BW_{Channel_CA}$ /2 + $BW_{UTRA}$/2<br>/<br>- $BW_{Channel_CA}$ / 2 - $BW_{UTRA}$/2 |
| $UTRA_{ACLR2}$ | 36 dB |
| Adjacent channel centre frequency offset (in MHz) | + $BW_{Channel_CA}$ /2 + 3*$BW_{UTRA}$/2<br>/<br>- $BW_{Channel_CA}$ /2 – 3*$BW_{UTRA}$/2 |
| CA E-UTRA channel Measurement bandwidth | $BW_{Channel_CA}$ - 2* $BW_{GB}$ |
| UTRA 5MHz channel Measurement bandwidth* | 3.84 MHz |
| UTRA 1.6MHz channel measurement bandwidth** | 1.28 MHz |
| * Note: Applicable for E-UTRA FDD co-existence with UTRA FDD in paired spectrum.<br>** Note: Applicable for E-UTRA TDD co-existence with UTRA TDD in unpaired spectrum. | |

The normative reference for this requirement is TS 36.101 [2] subclause 6.6.2.3.2A.

### 6.6.2.3A.3.2 Minimum conformance requirements of CA E-UTRA for CA

Carrier aggregated E-UTRA ACLR (CA $E\text{-}UTRA_{ACLR}$) is the ratio of the filtered mean power centred on the assigned aggregated E-UTRA channel frequency to the filtered mean power centred on an adjacent aggregated E-UTRA channel frequency at nominal channel spacing. The assigned aggregated E-UTRA channel power and adjacent aggregated E-UTRA channel power are measured with rectangular filters with measurement bandwidth specified in Table 6.6.2.3A.3.2-1.

If the measured adjacent channel power is greater than –50dBm then the E-UTRA$_{ACLR}$ shall be higher than the valued specified in Table 6.6.2.3A.3.2-1.

**Table 6.6.2.3A.3.2-1: General requirements of E-UTRA$_{ACLR}$ for CA**

| | **CA bandwidth class / CA E-UTRA$_{ACLR}$ / measurement bandwidth** |
|---|---|
| | **CA bandwidth class C** |
| CA E-UTRA$_{ACLR}$ | 30 dB |
| CA E-UTRA channel Measurement bandwidth | $BW_{Channel_CA}$ - 2* $BW_{GB}$ |
| Adjacent channel centre frequency offset (in MHz) | + $BW_{Channel_CA}$ / - $BW_{Channel_CA}$ |

The normative reference for this requirement is TS 36.101 subclause 6.6.2.3.3A.

### 6.6.2.3A.4 Test description

#### 6.6.2.3A.4.1 Initial condition

#### 6.6.2.3A.4.2 Test procedure

#### 6.6.2.3A.4.3 Message contents

### 6.6.2.3A.5 Test Requirements

### 6.6.2.4 Void

## 6.6.3 Spurious emissions

Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emissions, intermodulation products and frequency conversion products, but exclude out of band emissions. The spurious emission limits are specified in terms of general requirements inline with SM.329 [3] and E-UTRA operating band requirement to address UE co-existence.

Unless otherwise stated, the spurious emission limits apply for the frequency ranges that are more than $\Delta f_{OOB}$ (MHz) in Table6.6.3.1.3-1 from the edge of the channel bandwidth. To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

### 6.6.3.1 Transmitter Spurious emissions

#### 6.6.3.1.1 Test purpose

To verify that UE transmitter does not cause unacceptable interference to other channels or other systems in terms of transmitter spurious emissions.

#### 6.6.3.1.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

#### 6.6.3.1.3 Minimum conformance requirements

The spurious emission limits apply for the frequency ranges that are more than $\Delta f_{OOB}$ (MHz) from the edge of the channel bandwidth.

**Table 6.6.3.1.3-1: $\Delta f_{OOB}$ boundary between E-UTRA channel and spurious emission domain**

| Channel bandwidth | 1.4 MHz | 3.0 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
|---|---|---|---|---|---|---|
| $\Delta f_{OOB}$ (MHz) | 2.8 | 6 | 10 | 15 | 20 | 25 |

The spurious emission limits in Table 6.6.3.1.3-2 apply for all transmitter band configurations (RB) and channel bandwidths.

NOTE: In order that the measurement of spurious emissions falls within the frequency ranges that are more than ΔfOOB (MHz) from the edge of the channel bandwidth, the minimum offset of the measurement frequency from each edge of the channel should be ΔfOOB + MBW/2. MBW denotes the measurement bandwidth defined in Table 6.6.3.1.3-2.

**Table 6.6.3.1.3-2: Spurious emissions limits**

| Frequency Range | Maximum Level | Measurement Bandwidth | Notes |
|---|---|---|---|
| 9 kHz ≤ f < 150 kHz | -36 dBm | 1 kHz | |
| 150 kHz ≤ f < 30 MHz | -36 dBm | 10 kHz | |
| 30 MHz ≤ f < 1000 MHz | -36 dBm | 100 kHz | |
| 1 GHz ≤ f < 12.75 GHz | -30 dBm | 1 MHz | |
| 12.75 GHz ≤ f < 5th harmonic of the upper frequency edge of the UL operating band in GHz | -30 dBm | 1 MHz | Note 1 |
| Note 1: Applies for Band 22, Band 42 and Band 43 | | | |

The normative reference for this requirement is TS 36.101 [2] subclause 6.6.3.1.

### 6.6.3.1.4 Test description

#### 6.6.3.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in Table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in Table 6.6.3.1.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexe A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.3.1.4.1-1: Test Configuration Table**

| Initial Conditions | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | Lowest, 5MHz, Highest | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | N/A for Spurious Emissions testing | | | QPSK | 6 | 6 |
| 1.4MHz | | | | QPSK | 1 | 1 |
| 3MHz | | | | QPSK | 15 | 15 |
| 3MHz | | | | QPSK | 1 | 1 |
| 5MHz | | | | QPSK | 25 | 25 |
| 5MHz | | | | QPSK | 1 | 1 |
| 10MHz | | | | QPSK | 50 | 50 |
| 10MHz | | | | QPSK | 1 | 1 |
| 15MHz | | | | QPSK | 75 | 75 |
| 15MHz | | | | QPSK | 1 | 1 |
| 20MHz | | | | QPSK | 100 | 100 |
| 20MHz | | | | QPSK | 1 | 1 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | | | |
| Note 2: The 1 RB allocation shall be tested at both RB #0 and RB #max. | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in Figure TS 36.508 [7] Annex A, Figure A.7.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to Table 6.6.3.1.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.3.1.4.3.

### 6.6.3.1.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.6.3.1.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.

3. Measure the power of the transmitted signal with a measurement filter of bandwidths according to table 6.6.3.1.5-1. The centre frequency of the filter shall be stepped in contiguous steps according to table 6.6.3.1.5-1. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

### 6.6.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

## 6.6.3.1.5 Test requirement

The measured average power of spurious emission, derived in step 3, shall not exceed the described value in tables 6.6.3.1.5-1.

The spurious emission limits apply for the frequency ranges that are more than $\Delta f_{OOB}$ (MHz) from the edge of the channel bandwidth shown in Table 6.6.3.1.3-1.

**Table 6.6.3.1.5-1: General spurious emissions test requirements**

| Frequency Range | Maximum Level | Measurement Bandwidth | Notes |
|---|---|---|---|
| 9 kHz ≤ f < 150 kHz | -36 dBm | 1 kHz | |
| 150 kHz ≤ f < 30 MHz | -36 dBm | 10 kHz | |
| 30 MHz ≤ f < 1000 MHz | -36 dBm | 100 kHz | |
| 1 GHz ≤ f < 12.75 GHz | -30 dBm | 1 MHz | |
| 12.75 GHz ≤ f < 5th harmonic of the upper frequency edge of the UL operating band in GHz | -30 dBm | 1 MHz | Note 1 |
| Note 1: Applies for Band 22, Band 42 and Band 43 | | | |

NOTE: In order that the measurement of spurious emissions falls within the frequency ranges that are more than ΔfOOB (MHz) from the edge of the channel bandwidth, the minimum offset of the measurement frequency from each edge of the channel should be ΔfOOB + MBW/2. MBW denotes the measurement bandwidth defined in Table 6.6.3.1.3-2.

## 6.6.3.1A Transmitter Spurious emissions for CA

Editor's notes:

The following items are missing or incomplete:

- Initial conditions (test frequencies, test configurations)

- Test procedure

- Message Contents

- Test requirements and test tolerances

- Connection diagram is missing

### 6.6.3.1A.1 Test purpose

To verify that UE transmitter does not cause unacceptable interference to other channels or other systems in terms of transmitter spurious emissions.

### 6.6.3.1A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

### 6.6.3.1A.3 Minimum conformance requirements

The spurious emission limits apply for the frequency ranges that are more than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1A.3-1 from the ± edge of the aggregated channel bandwidth (Table 5.4.2A-1). For frequencies $\Delta f_{OOB}$ greater than $F_{OOB}$ as specified in Table 6.6.3.1A.3-1the spurious requirements in Table 6.6.3.1.5-1 are applicable.

**Table 6.6.3.1A.3-1: Boundary between E-UTRA $\Delta f_{OOB}$ and spurious emission domain for intra-band contiguous carrier aggregation**

| CA Bandwidth Class | OOB boundary $F_{OOB}$ [(MHz) |
|---|---|
| A | Table 6.6.3.1.3-1 |
| B | FFS |
| C | $BW_{Channel_CA}$ + 5 |

To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

NOTE: In order that the measurement of spurious emissions falls within the frequency ranges that are more than $\Delta f_{OOB}$ (MHz) from the edge of the channel bandwidth, the minimum offset of the measurement frequency from each edge of the channel should be $\Delta f_{OOB}$ + MBW/2. MBW denotes the measurement bandwidth defined in Table 6.6.3.1.3-2.

The normative reference for this requirement is TS 36.101[2] clause 6.6.3.1A

### 6.6.3.1A.4 Test description

#### 6.6.3.1A.4.1 Initial condition

#### 6.6.3.1A.4.2 Test procedure

#### 6.6.3.1A.4.3 Message contents

### 6.6.3.1A.5 Test Requirements

## 6.6.3.2 Spurious emission band UE co-existence

### 6.6.3.2.1 Test purpose

To verify that UE transmitter does not cause unacceptable interference to co-existing systems for the specified bands which has specific requirements in terms of transmitter spurious emissions.

### 6.6.3.2.2 Test applicability

This test case applies to all types of E-UTRA UE release 8.

### 6.6.3.2.3 Minimum conformance requirements

This clause specifies the requirements for the specified E-UTRA band for coexistence with protected bands as indicated in Table 6.6.3.2.3-1.

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

**Table 6.6.3.2.3-1: Spurious emission band UE co-existence limits**

| E-UTRA Band | Spurious emission | | | | | | |
|---|---|---|---|---|---|---|---|
| | Protected band | Frequency range (MHz) | | | Maximum Level (dBm) | MBW (MHz) | Comment |
| 1 | E-UTRA Band 1, 3, 7, 8, 9, 11, 20, 21, 22, 34, 38, 40, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 3, 9, 34 | FDL_low | - | FDL_high | -50 | 1 | Note 17 |
| | E-UTRA Band 33 | FDL_low | - | FDL_high | -50 | 1 | Note 3 |
| | E-UTRA Band 39 | FDL_low | - | FDL_high | -50 | 1 | Note 3 |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note 6, 7<br>Note17 |
| | | 1884.5 | - | 1915.7 | | | Note 6, 8<br>Note17 |
| 2 | E-UTRA Band 4, 5, 10, 12, 13, 14, 17, 22, 23,24, 41, 42 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 2, 25 | FDL_low | - | FDL_high | -50 | 1 | Note17 |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | Note2 |
| 3 | E-UTRA Band 1, 7, 8, 20, 33, 34, 38, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 3 | FDL_low | - | FDL_high | -50 | 1 | Note17 |
| | E-UTRA Band 11, 21 | FDL_low | - | FDL_high | -50 | 1 | Note13 |
| | Frequency range | 860 | - | 895 | -50 | 1 | Note13 |
| | E-UTRA Band 22, 42 | FDL_low | - | FDL_high | -50 | 1 | Note2 |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note13 |
| 4 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 22, 23, 24, 41 | FD, 43L_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 42 | FDL_low | - | FDL_high | -50 | 1 | Note2 |
| 5 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 22, 23, 24, 25, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 41 | FDL_low | - | FDL_high | -50 | 1 | Note2 |
| 6 | E-UTRA Band 1, 9, 11, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 875 | -37 | 1 | |
| | Frequency range | 875 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note7 |
| | | 1884.5 | - | 1915.7 | | | Note8 |
| 7 | E-UTRA Band 1, 3, 7, 8, 20, 22, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 2570 | - | 2575 | +1.6 | 5 | Note14<br>Note17 |
| | Frequency range | 2575 | - | 2595 | -15.5 | 5 | Note14<br>Note17 |
| 8 | E-UTRA Band 1, 20, 33, 34, 38, 39, 40 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 3 | FDL_low | - | FDL_high | -50 | 1 | Note 2 |
| | E-UTRA Band 7 | FDL_low | - | FDL_high | -50 | 1 | Note 2 |
| | E-UTRA Band 8 | FDL_low | - | FDL_high | -50 | 1 | Note17 |
| | E-UTRA Band 22, 42, 43 | | | | | | |
| 9 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note7 |
| | | 1884.5 | - | 1915.7 | | | Note8 |
| 10 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 25, 41, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 22, 42 | FDL_low | - | FDL_high | -50 | 1 | Note 2 |
| 11 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note7 |
| | | 1884.5 | - | 1915.7 | | | Note8 |
| 12 | E-UTRA Band 2, 5, 13, 14, 17, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 4, 10 | FDL_low | - | FDL_high | -50 | 1 | Note 2 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | E-UTRA Band 12 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| 13 | E-UTRA Band 2, 4, 5, 10, 12, 13, 17, 23, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 769 | - | 775 | -35 | 0.00625 | |
| | Frequency range | 799 | | 805 | -35 | 0.00625 | Note[11] |
| | E-UTRA Band 14 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| | E-UTRA Band 24 | FDL_low | - | FDL_high | -50 | 1 | Note[2] |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 14 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 769 | - | 775 | -35 | 0.00625 | Note[12] Note [17] |
| | Frequency range | 799 | | 805 | -35 | 0.00625 | Note[11] Note[12] Note [17] |
| 17 | E-UTRA Band 2, 5, 13, 14, 17, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 4, 10 | FDL_low | - | FDL_high | -50 | 1 | Note [2] |
| | E-UTRA Band 12 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| 18 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -40 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note[7] |
| | | 1884.5 | - | 1915.7 | | | Note[8] |
| 19 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -40 | 1 | Note[9] Note [17] |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note[7] |
| | | 1884.5 | - | 1915.7 | | | Note[8] |
| 20 | E-UTRA Band 1, 3, 7, 8, 22, 33, 34, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 20 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| | Frequency range 38, 42 | FDL_low | - | FDL_high | -50 | 1 | Note [2] |
| 21 | E-UTRA Band 11 | FDL_low | - | FDL_high | -35 | 1 | Note[10] Note [17] |
| | E-UTRA Band 1, 9, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 21 | FDL_low | - | FDL_high | -50 | 1 | Note [10] |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note[7] |
| | | 1884.5 | - | 1915.7 | | | Note[8] |
| 22 | E-UTRA Band 1, 3, 7, 8, 20, 33, 34, 38, 39, 40, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 3510 | - | 3525 | -40 | 1 | Note [17] |
| | Frequency range | 3525 | - | 3590 | -50 | 1 | |
| 23 | E-UTRA Band 4, 5, 10, 12, 13, 14, 17, 23, 24, 41 | FDL_low | - | FDL_high | -50 | 1 | Note[16] |
| | E-UTRA Band 2 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| | Frequency range | 1998 | - | 1999 | -21 | 1 | Note[16] |
| | Frequency range | 1997 | - | 1998 | -27 | 1 | Note[16] |
| | Frequency range | 1996 | - | 1997 | -32 | 1 | Note[16] |
| | Frequency range | 1995 | - | 1996 | -37 | 1 | Note[16] |
| 24 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 41, 25 | FDL_low | - | FDL_high | -50 | 1 | |
| 25 | E-UTRA Band 4, 5, 10,12, 13, 14, 17, 22, 24, 41, 42 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 2 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| | E-UTRA Band 25 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | Note [2] |
| ... | | | | | | | |
| 33 | E-UTRA Band 1, 7, 8, 20, 22, 34, 38, 39, 40, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | Note [5] |
| | E-UTR Band 3 | FDL_low | - | FDL_high | -50 | 1 | Note [17] |
| 34 | E-UTRA Band 1, 3, 7, 8, 9, 11, 20, 21, 33, 38,39, 40 | FDL_low | - | FDL_high | -50 | 1 | Note [5] |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | Note[7] |
| | | 1884.5 | - | 1915.7 | | | Note[8] |
| 35 | | | | | | | |
| 36 | | | | | | | |
| 37 | | | - | | | | |
| 38 | E-UTRA Band 1,3, 8, 33, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 2620 | - | 2645 | -15.5 | 5 | Note[15] Note[17] |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 39 | E-UTRA Band 34, 40, 42 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | Note 2 |
| 40 | E-UTRA Band 1, 3, 22, 33, 34, 39, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| 41 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 25 | FDL_low | - | FDL_high | -50 | 1 | |
| 42 | E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 10, 20, 25, 33, 34, 38,40 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | Note3 |
| 43 | E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 10, 20, 25, 33, 34, 38, 40 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 42 | FDL_low | - | FDL_high | -50 | 1 | Note3 |
| | E-UTRA Band 22 | FDL_low | - | FDL_high | [-50] | [1] | Note3 |

Note 1: FDL_low and FDL_high refer to each E-UTRA frequency band specified in Table 5.2-1.
Note 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.6.3.1.3-2 are permitted for each assigned E-UTRA carrier used in the measurement due to 2nd, 3rd or 4th harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.4.2-1) for which the 2nd, 3rd or 4th harmonic totally or partially overlaps the measurement bandwidth (MBW).
Note 3: To meet these requirements some restriction will be needed for either the operating band or protected band.
Note 4: N/A
Note 5: For non synchronised TDD operation to meet these requirements some restriction will be needed for either the operating band or protected band.
Note 6: Applicable when NS_05 in section 6.6.3.3.3.1 is signalled by the network.
Note 7: Applicable when co-existence with PHS system operating in 1884.5 -1919.6MHz.
Note 8: Applicable when co-existence with PHS system operating in 1884.5 -1915.7MHz.
Note 9: Applicable when NS_08 in section 6.6.3.3.3.3 is signalled by the network.
Note 10: Applicable when NS_09 in section 6.6.3.3.4 is signalled by the network.
Note 11: Whether the applicable frequency range should be 793-805MHz instead of 799-805MHz is TBD.
Note 12: The emissions measurement shall be sufficiently power averaged to ensure a standard deviation < 0.5 dB.
Note 13: Applicable when the assigned E-UTRA UL operating channel is ≥1749.9MHz and ≤ 1784.9MHz.
Note 14: This requirement is applicable for an uplink transmission bandwidth less than or equal to 54 RB for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2560.5 - 2562.5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2552 - 2560 MHz. This requirement is applicable without any other uplink transmission bandwidth restriction for channel bandwidths within the range 2500 - 2570 MHz.
Note 15: This requirement is applicable for an uplink transmission bandwidth less than or equal to 54 RB for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2605.5 - 2607.5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2597 – 2605 MHz. This requirement is applicable without any other uplink transmission bandwidth restriction for channel bandwidths within the range 2570 - 2615 MHz. For assigned carriers with bandwidths overlapping the frequency range 2615-2620 MHz the requirements apply with the maximum output power configured to +20 dBm in the IE *P-Max*.
Note 16: To meet this requirement NS_11 value shall be signalled when operating in 2000-2010 MHz.
Note 17: These requirements also apply for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

NOTE 1: Bands 1,6,9,11,34 in the tables shall be reviewed after June 2012 because of PHS band operation change.

NOTE 2: The restriction on the maximum uplink transmission to 54 RB in Notes 14 and 15 of Table 6.6.3.2.3-1 is intended for conformance testing and may be applied to network operation to facilitate coexistence when the aggressor and victim bands are deployed in the same geographical area. The applicable spurious emission requirement of -15.5 dBm/5MHz is a least restrictive technical condition for FDD/TDD coexistence and may have to be revised in the future.

The normative reference for this requirement is TS 36.101 [2] subclause 6.6.3.2.

## 6.6.3.2.4 Test description

### 6.6.3.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in Table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in Table 6.6.3.2.4.1-1. The details of the uplink reference measurement

channels (RMCs) are specified in Annexe A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.3.2.4.1-1: Test Configuration Table**

| Initial Conditions | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment (as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies (as specified in TS36.508 [7] subclause 4.3.1) | | | | Low range, Mid range, High range (Note 6) | | |
| Test Channel Bandwidths (as specified in TS 36.508 [7] subclause 4.3.1) | | | | Lowest, 5MHz, Highest | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | N/A for Spurious Emissions testing | | | QPSK | 6 | 6 |
| 1.4MHz | | | | QPSK | 1 | 1 |
| 3MHz | | | | QPSK | 15 | 15 |
| 3MHz | | | | QPSK | 1 | 1 |
| 5MHz | | | | QPSK | 25 | 25 |
| 5MHz | | | | QPSK | 1 | 1 |
| 10MHz | | | | QPSK | 50 | 50 |
| 10MHz | | | | QPSK | 1 | 1 |
| 15MHz | | | | QPSK | 75 | 75 |
| 15MHz | | | | QPSK | 54[3] | 54[4] |
| 15MHz | | | | QPSK | 1 | 1 |
| 20MHz | | | | QPSK | 100 | 100 |
| 20MHz | | | | QPSK | 54[3] | 54[4] |
| 20MHz | | | | QPSK | 1[5] | 1[5] |

Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.
Note 2: The 1 RB allocation shall be tested at both RB #0 and RB #max.
Note 3: To be used for requirements under note 14 in table 6.6.3.2.5-1 in high channel at RB#(full allocation – 54), instead of full allocation.
Note 4: To be used for requirements under note 15 in table 6.6.3.2.5-1 in high channel - 5MHz at RB#(full allocation – 54), instead of full allocation.
Note 5: For requirements under note 14 the allocation shall be in high channel at RB#max and for requirements under note 15 the allocation shall be in (high channel – 5MHz) at RB#max.
Note 6: Do not apply for requirements under Note 13 in Table 6.6.3.2.3-1 and 6.6.3.2.5-1. Test frequencies for these requirements are defined in Table 6.6.3.2.4.1-2.

**Table 6.6.3.2.4.1-2: Test frequencies for E-UTRA channel bandwidth for operating band 3 with Note 13 (in Table 6.6.3.2.3-1 and 6.6.3.2.5-1)**

| Bandwidth [MHz] | $N_{UL}$ | Frequency of Uplink [MHz] | $N_{DL}$ | Frequency of Downlink [MHz] |
|---|---|---|---|---|
| 1.4 | 19942 | 1784.2 | 1942 | 1879.2 |
| 3 | 19934 | 1783.4 | 1934 | 1878.4 |
| 5 | 19924 | 1782.4 | 1924 | 1877.4 |
| 10 | 19899 | 1779.9 | 1899 | 1874.9 |
| 15 [1] | 19874 | 1777.4 | 1874 | 1872.4 |
| 20 [1] | 19849 | 1774.9 | 1849 | 1869.9 |
| NOTE 1: Bandwidth for which a relaxation of the specified UE receiver sensitivity requirement (TS 36.101 [2] Clause 7.3) is allowed. | | | | |

1. Connect the SS to the UE to the UE antenna connectors as shown in Figure TS 36.508 [7] Annex A, Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.

4. The UL Reference Measurement channels are set according to Table 6.6.3.2.4.1-1.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.3.2.4.3.

#### 6.6.3.2.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.6.3.2.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.

3. 3. Measure the power of the transmitted signal with a measurement filter of bandwidths according to table 6.6.3.2.5- For band 14 measurements made in a bandwidth of 6.25kHz, measurement parameter settings defined in table 6.6.3.2.4.2-1 shall be used. 1. The centre frequency of the filter shall be stepped in contiguous steps according to table 6.6.3.2.5-1. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

**Table 6.6.3.2.4.2-1: Measurement setup for band 14**

| | **Option 1: Measurement with No RMS VBW available** | **Option2 Measurement with VBW Filtering on Power scale** |
|---|---|---|
| VBW | >=62.5 kHz<br><br>(10 times or more the RBW) | <=43Hz |
| RBW | <=6.25kHz | <=6.25kHz |
| Detector type | Averages signal envelope during each measurement point, such as "RMS detector" | Peak |
| Averaging mode (Trace averaging) | Power (RMS voltage) | Power (RMS voltage), as controlled by "Average Type" |
| Average Type (applies to detector) | Power (RMS voltage) (automatically occurs with "RMS detector") | Not applicable |
| Average Type (applies to VBW filter) | Not applicable | Power (RMS voltage) |
| Number of averages | 30, to reduce variance as required, or use an even longer sweep time | 1 or use an even narrower VBW filter, thus a longer sweep time |
| Sweep time | [Don't specify] | Sweep rate (span divided by sweep time) <= 0.8 * RBW*VBW |

#### 6.6.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

### 6.6.3.2.5 Test requirement

The measured average power of spurious emission, derived in step 3, shall not exceed the described value in tables 6.6.3.2.5-1.

**Table 6.6.3.2.5-1: Spurious emission band UE co-existence limits**

| E-UTRA Band | Spurious emission | | | | | | |
|---|---|---|---|---|---|---|---|
| | Protected band | Frequency range (MHz) | | | Maximum Level (dBm) | MBW (MHz) | Con |
| 1 | E-UTRA Band 1, 7, 8, 11, 20, 21, 22, 38, 40, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 3, 9, 34 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 33 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 39 | FDL_low | - | FDL_high | -50 | 1 | No |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | No<br>No |
| | | 1884.5 | - | 1915.7 | | | No<br>No |
| 2 | E-UTRA Band 4, 5, 10, 12, 13, 14, 17, 22, 23, 24, 41, 42 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 2, 25 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | No |
| 3 | E-UTRA Band 1, 7, 8, 20, 33, 34, 38, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 3 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 11, 21 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 22, 42 | FDL_low | - | FDL_high | -50 | 1 | No |
| | Frequency range | 860 | - | 895 | -50 | 1 | No |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | No |
| 4 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 22, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 42 | FDL_low | - | FDL_high | -50 | 1 | No |
| 5 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 22, 23, 24, 25 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 41 | FDL_low | - | FDL_high | -50 | 1 | No |
| 6 | E-UTRA Band 1, 9, 11, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 875 | -37 | 1 | |
| | Frequency range | 875 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | No |
| | | 1884.5 | - | 1915.7 | | | No |
| 7 | E-UTRA Band 1, 3, 7, 8, 20, 22, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 2570 | | 2575 | +1.6 | 5 | No<br>No |
| | Frequency range | 2575 | | 2595 | -15.5 | 5 | No<br>No |
| 8 | E-UTRA Band 1, 8, 20, 33, 34, 38, 39, 40 | FDL_low | | FDL_high | -50 | | |
| | E-UTRA Band 3 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 7 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 8 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 22, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | No |
| 9 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | No |
| | | 1884.5 | - | 1915.7 | | | No |
| 10 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 25, 41, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 22, 42 | FDL_low | - | FDL_high | -50 | 1 | No |
| 11 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | No |
| | | 1884.5 | - | 1915.7 | | | No |
| 12 | E-UTRA Band 2, 5, 13, 14, 17, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 4, 10 | FDL_low | - | FDL_high | -50 | 1 | No |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | E-UTRA Band 12 | FDL_low | - | FDL_high | -50 | 1 | No |
| 13 | E-UTRA Band 2, 4, 5, 10, 12, 13, 17, 23, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 769 | - | 775 | -35 | 0.00625 | No |
| | Frequency range | 799 | | 805 | -35 | 0.00625 | No<br>No |
| | E-UTRA Band 14 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 24 | FDL_low | - | FDL_high | -50 | 1 | N |
| 14 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23,24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 769 | - | 775 | -35 | 0.00625 | No<br>No |
| | Frequency range | 799 | | 805 | -35 | 0.00625 | No<br>No<br>No |
| 17 | E-UTRA Band 2, 5, 13, 14, 17, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 4, 10 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 12 | FDL_low | - | FDL_high | -50 | 1 | No |
| 18 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -40 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | N |
| | | 1884.5 | - | 1915.7 | | | N |
| 19 | E-UTRA Band 1, 9, 11, 21, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 860 | - | 895 | -40 | 1 | N<br>No |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | N |
| | | 1884.5 | - | 1915.7 | | | N |
| 20 | E-UTRA Band 1, 3, 7, 8, 22, 33, 34, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 20 | FDL_low | - | FDL_high | -50 | 1 | No |
| | Frequency range 38, 42 | FDL_low | - | FDL_high | -50 | 1 | No |
| 21 | E-UTRA Band 11 | FDL_low | - | FDL_high | -35 | 1 | No<br>No |
| | E-UTRA Band 1, 9, 34 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 21 | FDL_low | - | FDL_high | -50 | 1 | No |
| | Frequency range | 860 | - | 895 | -50 | 1 | |
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | N |
| | | 1884.5 | - | 1915.7 | | | N |
| 22 | E-UTRA Band 1, 3, 7, 8, 20, 33, 34, 38, 39, 40, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 3510 | - | 3525 | -40 | 1 | |
| | Frequency range | 3525 | - | 3590 | -50 | 1 | |
| 23 | E-UTRA Band 4, 5, 10, 12, 13, 14, 17, 23, 24, 41 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 2 | FDL_low | - | FDL_high | -50 | 1 | No |
| | Frequency range | 1998 | - | 1999 | -21 | 1 | No |
| | Frequency range | 1997 | - | 1998 | -27 | 1 | No |
| | Frequency range | 1996 | - | 1997 | -32 | 1 | No |
| | Frequency range | 1995 | - | 1996 | -37 | 1 | No |
| 24 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 25, 41 | FDL_low | - | FDL_high | -50 | 1 | |
| 25 | E-UTRA Band 4, 5, 10,12, 13, 14, 17, 22, 24, 41, 42 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 2 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 25 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | N |
| … | | | | | | | |
| 33 | E-UTRA Band 1, 7, 8, 20, ,22 34, 38, 39, 40, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | No |
| | E-UTRA Band 3 | FDL_low | - | FDL_high | -50 | 1 | No |
| 34 | E-UTRA Band 1, 3, 7, 8, 9, 11, 20, 21, 33, 38,39, 40, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | No |
| | Frequency range | 860 | - | 895 | -50 | 1 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Frequency range | 1884.5 | - | 1919.6 | -41 | 0.3 | No |
| | | 1884.5 | - | 1915.7 | | | No |
| 35 | | | | | | | |
| 36 | | | | | | | |
| 37 | | | - | | | | |
| 38 | E-UTRA Band 1, 3, 8, 20, 22, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | Frequency range | 2620 | - | 2645 | -15.5 | 5 | No<br>No |
| 39 | E-UTRA Band 34, 40, 42 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | N |
| 40 | E-UTRA Band 1, 3, 22, 33, 34, 39 , 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| 41 | E-UTRA Band 2, 4, 5, 10, 12, 13, 14, 17, 23, 24, 25 | FDL_low | - | FDL_high | -50 | 1 | |
| 42 | E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 10, 20, 25, 33, 34, 38,40 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 43 | FDL_low | - | FDL_high | -50 | 1 | N |
| 43 | E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 10, 20, 25, 33, 34, 38, 40 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 42 | FDL_low | - | FDL_high | -50 | 1 | N |
| | E-UTRA Band 22 | FDL_low | - | FDL_high | [-50] | [1] | N |

Note 1: FDL_low and FDL_high refer to each E-UTRA frequency band specified in Table 5.2-1.
Note 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.6.3.1.3-2 are permitted fo assigned E-UTRA carrier used in the measurement due to 2nd, 3rd or 4th harmonic spurious emissions. An exception is all there is at least one individual RB within the transmission bandwidth (see Figure 5.4.2-1) for which the 2nd, 3rd or 4th harm totally or partially overlaps the measurement bandwidth (MBW).Note 3: To meet these requirements some restriction will b needed for either the operating band or protected band.
Note 4: N/A
Note 5: For non synchronised TDD operation to meet these requirements some restriction will be needed for either the operating protected band.
Note 6: Applicable when NS_05 in section 6.6.3.3.3.1 is signalled by the network.
Note 7: Applicable when co-existence with PHS system operating in 1884.5 -1919.6MHz.
Note 8: Applicable when co-existence with PHS system operating in 1884.5 -1915.7MHz.
Note 9: Applicable when NS_08 in section 6.6.3.3.3.3 is signalled by the network.
Note 10: Applicable when NS_09 in section 6.6.3.3.4 is signalled by the network.
Note 11: Whether the applicable frequency range should be 793-805MHz instead of 799-805MHz is TBD
Note 12: The emissions measurement shall be sufficiently power averaged to ensure a standard deviation < 0.5 dB
Note 13: Applicable when the assigned E-UTRA UL operating channel is ≥1749.9MHz and ≤ 1784.9MHz.
Note 14 This requirement is applicable for an uplink transmission bandwidth less than or equal to 54 RB for carriers of 15 MHz ba when carrier centre frequency is within the range 2560.5 - 2562.5 MHz and for carriers of 20 MHz bandwidth when carri centre frequency is within the range 2552 - 2560 MHz. This requirement is applicable without any other uplink transmissio bandwidth restriction for channel bandwidths within the range 2500 - 2570 MHz.
Note 15 This requirement is applicable for an uplink transmission bandwidth less than or equal to 54 RB for carriers of 15 MHz ba when carrier centre frequency is within the range 2605.5 - 2607.5 MHz and for carriers of 20 MHz bandwidth when carri centre frequency is within the range 2597 – 2605 MHz. This requirement is applicable without any other uplink transmiss bandwidth restriction for channel bandwidths within the range 2500 – 2570 MHz. For assigned carriers with bandwidths overlapping the frequency range 2615-2620 MHz the requirements apply with the maximum output power configured to + dBm in the IE *P-Max*.
Note 16: To meet this requirement NS_11 value shall be signalled when operating in 2000-2010 MHz.
Note 17: These requirements also apply for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge channel bandwidth.

NOTE 1: Bands 1,6,9,11,34 in the tables shall be reviewed after June 2012 because of PHS band operation change

NOTE 2: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

NOTE 3: The frequency range applicable with network signalled values of NS_05, NS_08, and NS_09 are covered in 6.6.3.3 Additional Spurious Emissions.

NOTE 4: The restriction on the maximum uplink transmission to 54 RB in Notes 14 and 15 of Table 6.6.3.2.5-1 is intended for conformance testing and may be applied to network operation to facilitate coexistence when the aggressor and victim bands are deployed in the same geographical area. The applicable spurious emission requirement of -15.5 dBm/5MHz is a least restrictive technical condition for FDD/TDD coexistence and may have to be revised in the future.

## 6.6.3.2_1 Spurious emission band UE co-existence (Release 9 and forward)

*Editor's note: This test case is incomplete. The following aspects are either missing or not yet determined:*

- *The notes in the spurious emission band UE co-existence limits tables need further study*

### 6.6.3.2_1.1 Test purpose

Same test purpose as in 6.6.3.2.1.

### 6.6.3.2_1.2 Test applicability

This test case applies to all types of E-UTRA UE release 9 and forward.

### 6.6.3.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 6.6.3.2.3 for all E-UTRA bands with the following exceptions in table 6.6.3.2_1.3-1:

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

**Table 6.6.3.2_1.3-1: Spurious emission band UE co-existence limits**

| E-UTRA Band | Spurious emission | | | | | | |
|---|---|---|---|---|---|---|---|
| | Protected band | Frequency range (MHz) | | | Maximum Level (dBm) | MBW (MHz) | Comment |
| 7 | E-UTRA Band 1, 3, 7, 8, 20, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 38 | FDL_low | - | FDL_low | -50 | 1 | Note 3 |
| 38 | E-UTRA Band 1, 3, 8, 20, 22, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 7 | FDL_low | - | FDL_high | -50 | 1 | Note 3 |
| Note 1: FDL_low and FDL_high refer to each E-UTRA frequency band specified in Table 5.2-1 | | | | | | | |
| Note 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.6.3.1.3-2 are permitted for each assigned E-UTRA carrier used in the measurement due to 2nd, 3rd or 4th harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.4.2-1) for which the 2nd, 3rd or 4th harmonic totally or partially overlaps the measurement bandwidth (MBW). | | | | | | | |
| Note 3: To meet these requirements some restriction will be needed for either the operating band or protected band | | | | | | | |
| Note 4: N/A | | | | | | | |
| Note 5: For non synchronised TDD operation to meet these requirements some restriction will be needed for either the operating band or protected band | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] subclause 6.6.3.2.

### 6.6.3.2_1.4 Test description

Same test description as in clause 6.6.3.2.4 for all E-UTRA bands, with the following exceptions:

- Table 6.6.3.2_1.3-1
- Table 6.6.3.2_1.5-1

### 6.6.3.2_1.5 Test requirement

The measured average power of spurious emission, derived in step 3, shall not exceed the described value in table 6.6.3.2.5-1 with the following exceptions in E-UTRA bands in 6.6.3.2_1.5-1:

**Table 6.6.3.2_1.5-1: Spurious emission band UE co-existence limits**

| E-UTRA Band | Spurious emission | | | | | | |
|---|---|---|---|---|---|---|---|
| | Protected band | Frequency range (MHz) | | | Maximum Level (dBm) | MBW (MHz) | Comment |
| 7 | E-UTRA Band 1, 3, 7, 8, 20, 22, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 38 | FDL_low | - | FDL_low | -50 | 1 | Note 3 |
| 38 | E-UTRA Band 1, 3, 8, 20, 22, 33, 34, 42, 43 | FDL_low | - | FDL_high | -50 | 1 | |
| | E-UTRA Band 7 | FDL_low | - | FDL_high | -50 | 1 | Note3 |

Note 1: FDL_low and FDL_high refer to each E-UTRA frequency band specified in Table 5.2-1.
Note 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.6.3.1.3-2 are permitted for each assigned E-UTRA carrier used in the measurement due to 2nd, 3rd or 4th harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.4.2-1) for which the 2nd, 3rd or 4th harmonic totally or partially overlaps the measurement bandwidth (MBW).
Note 3: To meet these requirements some restriction will be needed for either the operating band or protected band.
Note 4: N/A
Note 5: For non synchronised TDD operation to meet these requirements some restriction will be needed for either the operating band or protected band.
Note 6: Applicable when NS_05 in section 6.6.3.3.3.1 is signalled by the network.
Note 7: Applicable when co-existence with PHS system operating in 1884.5 -1919.6MHz.
Note 8: Applicable when co-existence with PHS system operating in 1884.5 -1915.7MHz.
Note 9: Applicable when NS_08 in section 6.6.3.3.3.3 is signalled by the network.
Note 10: Applicable when NS_09 in section 6.6.3.3.4 is signalled by the network.
Note 11: Whether the applicable frequency range should be 793-805MHz instead of 799-805MHz is TBD
Note 12: The emissions measurement shall be sufficiently power averaged to ensure a standard deviation < 0.5 dB
Note 13: Applicable when the assigned E-UTRA UL operating channel is ≥1749.9MHz and ≤ 1784.9MHz.
Note 14: To meet this requirement NS_11 value shall be signalled when operating in 2000-2010 MHz.
Note 15: These requirements also apply for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

NOTE 1: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

NOTE 2: The frequency range applicable with network signalled values of NS_05, NS_08, and NS_09 are covered in 6.6.3.3 Additional Spurious Emissions.

## 6.6.3.3 Additional spurious emissions

### 6.6.3.3.1 Test purpose

To verify that UE transmitter does not cause unacceptable interference to other channels or other systems in terms of transmitter spurious emissions under the deployment scenarios where additional requirements are specified.

### 6.6.3.3.2 Test applicability

This test case applies to all types of E-UTRA UE release 8 and forward.

### 6.6.3.3.3 Minimum conformance requirements

#### 6.6.3.3.3.1 Minimum conformance requirements (network signalled value "NS_05")

When "NS_05" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.3.3.3.1-1. This requirement also applies for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.3.1-1: Additional requirements (PHS)**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | | | | Measurement bandwidth |
|---|---|---|---|---|---|
| | 5 MHz | 10 MHz | 15 MHz | 20 MHz | |
| $1884.5 \le f \le 1919.6$ [*1] | -41 | -41 | -41 | -41 | 300 KHz |
| $1884.5 \le f \le 1915.7$ [*2] | -41 | -41 | -41 | -41 | 300 KHz |
| Note 1: Applicable when the lower edge of the assigned E-UTRA UL channel bandwidth frequency is larger than or equal to the upper edge of PHS band (1919.6 MHz) + 4 MHz + the Channel BW assigned, where Channel BW is as defined in sub-clause 5.4.2. Operations below this point are for further study. | | | | | |
| Note 2: Applicable when the lower edge of the assigned E-UTRA UL channel bandwidth frequency is larger than or equal to the upper edge of PHS band (1915.7 MHz) + 4 MHz + the Channel BW assigned, where Channel BW is as defined in sub-clause 5.4.2. Operations below this point are for further study. | | | | | |

NOTE 1: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (300 kHz).

NOTE 2: Notes in the tables shall be reviewed after June 2012 because of PHS band operation change

The normative reference for this requirement is TS 36.101[2] subclause 6.6.3.3.1.

#### 6.6.3.3.3.2 Minimum conformance requirements (network signalled value “NS_07")

When “NS_07” is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.3.3.3.2-1. These requirements also apply for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.3.2-1: Additional requirements**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | Measurement bandwidth |
|---|---|---|
| | 10 MHz | |
| $769 \le f \le 775$ | -57 | 6.25 kHz |
| Note: The emissions measurement shall be sufficiently power averaged to ensure a standard deviation < 0.5 dB. | | |

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (6.25 kHz).

The normative reference for this requirement is TS 36.101[2] subclause 6.6.3.3.2.

#### 6.6.3.3.3.3 Minimum requirement (network signalled value "NS_08")

When "NS_08" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.3.3.3-1. This requirement also applies for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.3-1: Additional requirement**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | | | Measurement bandwidth |
|---|---|---|---|---|
| | 5MHz | 10MHz | 15MHz | |
| 860 ≤ f ≤ 895 | -40 | -40 | -40 | 1 MHz |

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (1 MHz).

#### 6.6.3.3.3.4 Minimum requirement (network signalled value "NS_09")

When "NS 09" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.6.3.3.3.4-1. This requirement also applies for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.3.4-1: Additional requirement**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | | | Measurement bandwidth |
|---|---|---|---|---|
| | 5MHz | 10MHz | 15MHz | |
| 1475.9 ≤ f ≤ 1510.9 | -35 | -35 | -35 | 1 MHz |

NOTE 1: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (1 MHz).

NOTE 2: To improve measurement accuracy, A-MPR values for NS_09 specified in Table 6.2.4.3-1 in sub-clause 6.2.4 are derived based on both the above NOTE 1 and 100 kHz RBW.

### 6.6.3.3.4 Test description

#### 6.6.3.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.6.3.3.4.1-1, 6.6.3.3.4.1-2, and 6.6.3.3.4.1-3. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.6.3.3.4.1-1: Test Configuration Table (network signalled value "NS_05")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range<br><br>In case of Low range:<br>- For 5MHz channel bandwidth: UL 1927.2MHz (N_UL = 18072), DL 2117.2MHz (N_DL = 72) and UL 1931.1MHz (N_UL = 18111) DL 2121.1 MHz (N_DL = 111)<br>- For 10MHz: UL 1934.7MHz (N_UL = 18147), DL 2124.7MHz (N_DL = 147)<br>- For 20MHz channel bandwidth: Not available | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | 5MHz, 10MHz, 20MHz | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 5MHz | N/A for Additional Spurious Emissions testing | | | QPSK | 1 | N/A |
| 5MHz | | | | QPSK | 25 | |
| 10MHz | | | | QPSK | 1 | |
| 10MHz | | | | QPSK | 12 | |
| 10MHz | | | | QPSK | 48 | |
| 10MHz | | | | QPSK | 50 | |
| 10MHz | | | | 16QAM | 50 (Note 4) | |
| 15MHz | | | | QPSK | 1 | |
| 15MHz | | | | QPSK | 16 | |
| 15MHz | | | | QPSK | 48 | |
| 15MHz | | | | QPSK | 75 | |
| 15MHz | | | | 16QAM | 75 (Note 4) | |
| 20MHz | | | | QPSK | 1 | |
| 20MHz | | | | QPSK | 18 | |
| 20MHz | | | | QPSK | 48 | |
| 20MHz | | | | QPSK | 100 | |
| 20MHz | | | | 16QAM | 100 (Note 4) | |

Note 1. The 1 RB allocation shall be tested at both RB #0 and RB #max.

Note 2. The $RB_{start}$ of partial RB allocation shall be RB# 0 and RB# (max +1 – RB allocation) of the channel bandwidth.

Note 3: Low range frequencies for 5MHz and 10MHz channel bandwidth in case of network signalled "NS_05" shall be reviewed after June 2012 because of PHS band operation change.

Note 4: Applies only for UE-Categories 2-5

**Table 6.6.3.3.4.1-2: Test Configuration Table (network signalled value "NS_07")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment<br>(as specified in TS 36.508 [7] subclause 4.1) | | | | NC | | |
| Test Frequencies<br>(as specified in TS36.508 [7] subclause 4.3.1) | | | | Mid range | | |
| Test Channel Bandwidths<br>(as specified in TS 36.508 [7] subclause 4.3.1) | | | | 10MHz | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | | **Downlink Configuration** | | **Uplink Configuration** | | |
| Test Number | Ch BW | Mod'n | RB allocation | Mod'n | RB allocation | $RB_{start}$ |
| 1 | 10MHz | N/A for Additional Spurious Emissions testing. | | QPSK | 1 | 0 |
| 2 | 10MHz | | | QPSK | 8 | 0 |
| 3 | 10MHz | | | QPSK | 6 | 13 |
| 4 | 10MHz | | | QPSK | 20 | 13 |
| 5 | 10MHz | | | QPSK | 12 | 13 |
| 6 | 10MHz | | | 16QAM | 36<br>(Note 1) | 13 |
| 7 | 10MHz | | | QPSK | 16 | 19 |
| 8 | 10MHz | | | QPSK | 12 | 19 |
| 9 | 10MHz | | | 16QAM | 16 | 19 |
| 10 | 10MHz | | | QPSK | 30 | 19 |
| 11 | 10MHz | | | 16QAM | 30<br>(Note 1) | 19 |
| 12 | 10MHz | | | QPSK | 6 | 43 |
| 13 | 10MHz | | | QPSK | 2 | 48 |
| 14 | 10MHz | | | QPSK | 50 | 0 |
| 15 | 10MHz | | | QPSK | 12 | 0 |
| 16 | 10MHz | | | 16QAM | 50<br>(Note 1) | 0 |
| Note 1: Applies only for UE-Categories 2-5 | | | | | | |

**Table 6.6.3.3.4.1-3: Test Configuration Table (network signalled value "NS_08")**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | High range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | 5MHz, 10MHz, 15MHz | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 5MHz | N/A for Additional Spurious Emissions testing | | | QPSK | 1 | N/A |
| 5MHz | | | | QPSK | 8 | |
| 5MHz | | | | QPSK | 25 | |
| 10MHz | | | | QPSK | 1 | |
| 10MHz | | | | QPSK | 12 | |
| 10MHz | | | | QPSK | 40 | |
| 10MHz | | | | QPSK | 50 | |
| 10MHz | | | | 16QAM | 50 (Note 3) | |
| 15MHz | | | | QPSK | 1 | |
| 15MHz | | | | QPSK | 16 | |
| 15MHz | | | | QPSK | 40 | |
| 15MHz | | | | QPSK | 75 | |
| 15MHz | | | | 16QAM | 75 (Note 3) | |
| Note 1: The 1 RB allocation shall be tested at both RB #0 and RB #max.<br>Note 2: The $RB_{start}$ of partial RB allocation shall be RB#0 and RB# (max + 1 - RB allocation) of the channel bandwidth<br>Note 3: Applies only for UE-Categories 2-5 | | | | | | |

**Table 6.6.3.3.4.1-4: Test Configuration Table (network signalled value "NS_09")**

| Initial Conditions | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | High range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | 5MHz, 10MHz, 15MHz | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 5MHz | N/A for Additional Spurious Emissions testing | | | QPSK | 1 | N/A |
| 5MHz | | | | QPSK | 8 | |
| 5MHz | | | | QPSK | 25 | |
| 10MHz | | | | QPSK | 1 | |
| 10MHz | | | | QPSK | 12 | |
| 10MHz | | | | QPSK | 40 | |
| 10MHz | | | | QPSK | 50 | |
| 10MHz | | | | 16QAM | 50 (Note 3) | |
| 15MHz | | | | QPSK | 1 | |
| 15MHz | | | | QPSK | 16 | |
| 15MHz | | | | QPSK | 40 | |
| 15MHz | | | | QPSK | 54 | |
| 15MHz | | | | QPSK | 75 | |
| 15MHz | | | | 16QAM | 75 (Note 3) | |
| Note 1: The 1 RB allocation shall be tested at both RB #0 and RB #max.<br>Note 2: The $RB_{start}$ of partial RB allocation shall be RB#0 and RB# (max + 1 - RB allocation) of the channel bandwidth<br>Note 3: Applies only for UE-Categories 2-5 | | | | | | |

1. Connect the SS to the UE to the UE antenna connectors as shown in Figure TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to Table 6.6.3.3.4.1-1, Table 6.6.3.3.4.1-2 or Table 6.6.3.3.4.1-3 depending on network signal value.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.6.3.3.4.3.

### 6.6.3.3.4.2 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.6.3.3.4.1-1 and Table 6.6.3.3.4.1-2. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.
2. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE until the UE transmits at $P_{UMAX}$ level.
3. Measure the mean power of the UE in the channel bandwidth of the radio access mode according to the test configuration, which shall meet the requirements described in Tables 6.2.4.5-1to 6.2.4.5-9 as appropriate. The period of the measurement shall be at least one sub-frame (1ms).
4. Measure the power of the transmitted signal with a measurement filter of bandwidths according to Table 6.6.3.3.5.1-1, 6.6.3.3.5.2-1, 6.6.3.3.5.3-1 and 6.6.3.3.5.4-1 as appropriate. The centre frequency of the filter shall be stepped in contiguous steps according to the same table. For NS_07 measurements made in a bandwidth of

6.25kHz, measurement parameter settings defined in table 6.6.3.3.4.2-1 shall be used. The measured power shall be verified for each step. The measurement period shall capture the active time slots.

**Table 6.6.3.3.4.2-1: Measurement setup for NS_07**

| | Option 1: Measurement with No RMS VBW available | Option2 Measurement with VBW Filtering on Power scale |
|---|---|---|
| VBW | >=62.5 kHz<br><br>(10 times or more the RBW) | <=43Hz |
| RBW | <=6.25kHz | <=6.25kHz |
| Detector type | Averages signal envelope during each measurement point, such as “RMS detector” | Peak |
| Averaging mode (Trace averaging) | Power (RMS voltage) | Power (RMS voltage), as controlled by “Average Type” |
| Average Type (applies to detector) | Power (RMS voltage) (automatically occurs with “RMS detector”) | Not applicable |
| Average Type (applies to VBW filter) | Not applicable | Power (RMS voltage) |
| Number of averages | 30, to reduce variance as required, or use an even longer sweep time | 1 or use an even narrower VBW filter, thus a longer sweep time |
| Sweep time | [Don’t specify] | Sweep rate (span divided by sweep time) <= 0.8 * RBW*VBW |

### 6.6.3.3.4.3 Message contents

#### 6.6.3.3.4.3.1 Message contents (network signalled value "NS_05")

Message contents are according to TS 36.508 [7] subclause 4.6, with the following exceptions:

1. Information element additionalSpectrumEmission is set to NS_05. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.3.3.4.3.1-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 5 (NS_05) | | |

#### 6.6.3.3.4.3.2 Message contents (network signalled value “NS_07")

Message contents are according to TS 36.508 [7] subclause 4.6, with the following exceptions:

1. Information element additionalSpectrumEmission is set to NS_07. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.3.3.4.3.2-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 7 (NS_07) | | |

#### 6.6.3.3.4.3.3 Message contents (network signalled value “NS_08")

Message contents are according to TS 36.508 [7] subclause 4.6, with the following exceptions:

1. Information element additionalSpectrumEmission is set to NS_08. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.3.3.4.3.3-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 8 (NS_08) | | |

#### 6.6.3.3.4.3.4 Message contents (network signalled value “NS_09")

Message contents are according to TS 36.508 [7] subclause 4.6, with the following exceptions:

1. Information element additionalSpectrumEmission is set to NS_09. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 6.6.3.3.4.3.4-1: *SystemInformationBlockType2* :Additional spurious emissions requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 9 (NS_09) | | |

### 6.6.3.3.5 Test requirement

#### 6.6.3.3.5.1 Test requirement (network signalled value "NS_05")

When "NS_05" is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-4 as appropriate,

and

the measured average power of spurious emission, derived in step 2, shall not exceed the described value in tables 6.6.3.3.5.1-1. This requirement also applies for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.5.1-1: Additional requirements (PHS) test requirements**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | | | | Measurement bandwidth |
|---|---|---|---|---|---|
| | 5 MHz | 10 MHz | 15 MHz | 20 MHz | |
| 1884.5 ≤ f ≤1919.6 *1 | -41 | -41 | -41 | -41 | 300 KHz |
| 1884.5 ≤ f ≤1915.7 *2 | -41 | -41 | -41 | -41 | 300 KHz |
| Note 1: Applicable when the lower edge of the assigned E-UTRA UL channel bandwidth frequency is larger than or equal to the upper edge of PHS band (1919.6 MHz) + 4 MHz + the Channel BW assigned, where Channel BW is as defined in sub-clause 5.4.2. Operations below this point are for further study.<br>Note 2: Applicable when the lower edge of the assigned E-UTRA UL channel bandwidth frequency is larger than or equal to the upper edge of PHS band (1915.7 MHz) + 4 MHz + the Channel BW assigned, where Channel BW is as defined in sub-clause 5.4.2. Operations below this point are for further study. | | | | | |

NOTE 1: Notes in the tables shall be reviewed after June 2012 because of PHS band operation change

NOTE 2 (only for testing requirements in Table 6.6.3.3.5.1-1): For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (300 kHz).

### 6.6.3.3.5.2 Test requirement (network signalled value “NS_07")

When "NS_07" is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-7 as appropriate,

and

- the measured average power of spurious emission, derived in step 4, shall not exceed the described value in tables 6.6.3.3.5.2-1. These requirements also apply for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.5.2-1: Additional requirements (network signalled value “NS_07”)**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | Measurement bandwidth |
|---|---|---|
| | 10 MHz | |
| 769 ≤ f ≤ 775 | -55.5 | 6.25 kHz |
| Note: The emissions measurement shall be sufficiently power averaged to ensure a standard deviation < 0.5 dB | | |

NOTE (only for testing requirements in Table 6.6.3.3.5.2-1): For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (6.25 kHz).

### 6.6.3.3.5.3 Test requirement (network signalled value “NS_08")

When "NS_08" is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-8 as appropriate,

and

the measured average power of spurious emission, derived in step 4, shall not exceed the described value in tables 6.6.3.3.5.3-1. This requirement also applies for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.5.3-1: Additional requirements (network signalled value "NS_08")**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | | | Measurement bandwidth |
|---|---|---|---|---|
| | 5MHz | 10MHz | 15MHz | |
| 860 ≤ f ≤ 895 | -40 | -40 | -40 | 1 MHz |

NOTE (only for testing requirements in Table 6.6.3.3.5.3-1): For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (1 MHz).

#### 6.6.3.3.5.4 Test requirement (network signalled value "NS_09")

When "NS_09" is indicated in the cell:

- the measured UE mean power in the channel bandwidth, derived in step 3, shall fulfil requirements in Table 6.2.4.5-9 as appropriate,

and

- the measured average power of spurious emission, derived in step 4, shall not exceed the described value in table 6.6.3.3.5.4-1. This requirement also applies for the frequency ranges that are less than $\Delta f_{OOB}$ (MHz) in Table 6.6.3.1.3-1 from the edge of the channel bandwidth.

**Table 6.6.3.3.5.4-1: Additional requirements (network signalled value "NS_09")**

| Frequency band (MHz) | Channel bandwidth / Spectrum emission limit (dBm) | | | Measurement bandwidth |
|---|---|---|---|---|
| | 5MHz | 10MHz | 15MHz | |
| 1475.9 ≤ f ≤ 1510.9 | -35 | -35 | -35 | 1 MHz |

NOTE 1 (only for testing requirements in Table 6.6.3.3.5.4-1): For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth (1 MHz).

NOTE 2: To improve measurement accuracy, A-MPR values for NS_09 specified in Table 6.2.4.3-1 in sub-clause 6.2.4 are derived based on both the above NOTE 1 and 100 kHz RBW.

# 6.7 Transmit intermodulation

## 6.7.1 Test purpose

To verify that the UE transmit intermodulation does not exceed the described value in the test requirement.

The transmit intermodulation performance is a measure of the capability of the transmitter to inhibit the generation of signals in its non linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter via the antenna.

## 6.7.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

## 6.7.3 Minimum conformance requirements

User Equipment(s) transmitting in close vicinity of each other can produce intermodulation products, which can fall into the UE, or eNode B receive band as an unwanted interfering signal. The UE intermodulation attenuation is defined by the ratio of the mean power of the wanted signal to the mean power of the intermodulation product when an interfering CW signal is added at a level below the wanted signal at each of the transmitter antenna port with the other antenna port(s) if any is terminated. Both the wanted signal power and the intermodulation product power are measured through E-UTRA rectangular filter with measurement bandwidth shown in Table 6.7.3-1.

The requirement of transmitting intermodulation is prescribed in Table 6.7.3-1.

**Table 6.7.3-1: Transmit Intermodulation**

| BWChannel (UL) | 5MHz | | 10MHz | | 15MHz | | 20MHz | |
|---|---|---|---|---|---|---|---|---|
| Interference Signal Frequency Offset | 5MHz | 10MHz | 10MHz | 20MHz | 15MHz | 30MHz | 20MHz | 40MHz |
| Interference CW Signal Level | -40dBc | | | | | | | |
| Intermodulation Product | -29dBc | -35dBc | -29dBc | -35dBc | -29dBc | -35dBc | -29dBc | -35dBc |
| Measurement bandwidth | 4.5MHz | 4.5MHz | 9.0MHz | 9.0MHz | 13.5MHz | 13.5MHz | 18MHz | 18MHz |

The normative reference for this requirement is TS 36.101 [2] clause 6.7.1.

## 6.7.4 Test description

### 6.7.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA operating bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 6.7.4.1-1. The details of the uplink reference measurement channels (RMCs) are specified in Annexes A.2. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 6.7.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | |
|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | Normal | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | 5MHz and Highest | | | |
| **Test Parameters for Channel Bandwidths** | | | | |
| | **Downlink Configuration** | **Uplink Configuration** | | |
| Ch BW | N/A for Transmit Intermodulation | Mod'n | RB allocation | |
| | | | FDD | TDD |
| 5MHz | | QPSK | 8 | 8 |
| 10MHz | | QPSK | 12 | 12 |
| 15MHz | | QPSK | 16 | 16 |
| 20MHz | | QPSK | 18 | 18 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, the applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.2.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C.0, C.1, and C.3.0, and uplink signals according to Annex H.1 and H.3.0.
4. The UL Reference Measurement channels are set according to Table 6.7.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 6.7.4.3.

## 6.7.4.3 Test procedure

1. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 6.7.4.1-1. Since the UE has no payload and no loopback data to send the UE sends uplink MAC padding bits on the UL RMC.
2. Send continuously uplink power control "up" commands to the UE until the UE transmits at its $P_{UMAX}$ level.
3. Measure the rectangular filtered mean power of the UE. For TDD slots with transient periods are not under test for the wanted signal and for the intermodulation product.
4. Set the interference signal frequency below the UL carrier frequency using the first offset in table 6.7.5-1.
5. Set the interference CW signal level according to table 6.7.5-1.
6. Search the intermodulation product signals below and above the UL carrier frequency, then measure the rectangular filtered mean power of transmitting intermodulation for both signals, and calculate the ratios with the power measured in step 3.
7. Set the interference signal frequency above the UL carrier frequency using the first offset in table 6.7.5-1.
8. Search the intermodulation product signals below and above the UL carrier frequency, then measure the rectangular filtered mean power of transmitting intermodulation for both signals, and calculate the ratios with the power measured in step 3.
9. Repeat the measurement using the second offset in table 6.7.5-1.

## 6.7.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6.

## 6.7.5 Test requirement

The ratio derived in step 5 and 7, shall not exceed the described value in table 6.7.5-1

**Table 6.7.5-1: Transmit Intermodulation**

| BWChannel (UL) | 5MHz | | 10MHz | | 15MHz | | 20MHz | |
|---|---|---|---|---|---|---|---|---|
| Interference Signal Frequency Offset | 5MHz | 10MHz | 10MHz | 20MHz | 15MHz | 30MHz | 20MHz | 40MHz (Note 1) |
| Interference CW Signal Level | -40dBc | | | | | | | |
| Intermodulation Product | -29dBc | -35dBc | -29dBc | -35dBc | -29dBc | -35dBc | -29dBc | -35dBc |
| Measurement bandwidth | 4.5MHz | 4.5MHz | 9.0MHz | 9.0MHz | 13.5MHz | 13.5MHz | 18MHz | 18MHz |
| Note 1: For Band 20, only applicable for interference signal frequency above the UL carrier frequency. | | | | | | | | |

# 7 Receiver Characteristics

## 7.1 General

Unless otherwise stated the receiver characteristics are specified at the antenna connector(s) of the UE. For UE(s) with an integral antenna only, a reference antenna(s) with a gain of 0 dBi is assumed for each antenna port(s). UE with an integral antenna(s) may be taken into account by converting these power levels into field strength requirements, assuming a 0 dBi gain antenna. For UEs with more than one receiver antenna connector, identical interfering signals shall be applied to each receiver antenna port if more than one of these is used (diversity).

Unless otherwise stated, the test signal levels are defined at each antenna port, and specified in the respective sections below. Any specific test conditions are defined in the paragraph for each test. Unless stated otherwise, power control of the Downlink is OFF.

In general, the UE is set into the correct state in the “Initial conditions” part of the test, using normal SS signalling procedures over the air interface under easy radio conditions to ensure reliable message exchange. In the “Test procedure” part of the test, specific radio conditions are applied according to the test requirement and the desired measurement is made or the desired response is tested.

The ACS, blocking, spurious emissions and intermodulation requirements in sections 7.5, 7.6, 7.7 and 7.8 are defined for full band width signals i.e. for signals where all resource blocks are allocated for a specific user.

With the exception of Clause 7.3, the requirements shall be verified with the network signalling value NS_01 configured (Table 6.2.4.3-1).

All the parameters in clause 7 are defined using the UL reference measurement channels specified in Annexes A.2.2 and A.2.3, the DL reference measurement channels specified in Annex A.3.2 and using the set-up specified in Annex C.3.1

## 7.2 Diversity characteristics

The requirements in Section 7 assume that the receiver is equipped with two Rx port as a baseline. Requirements for 4 ports are FFS. With the exception of clause 7.9, All requirements shall be verified by using both (all) antenna ports simultaneously.

## 7.3 Reference sensitivity level

Editor’s note: FDD/TDD aspects missing or not yet determined:

- The Maximum Sensitivity Degradation figures for large transmission configurations are not finalised in the core specification.

### 7.3.1 Test purpose

To verify the UE’s ability to receive data with a given average throughput for a specified reference measurement channel, under conditions of low signal level, ideal propagation and no added noise.

A UE unable to meet the throughput requirement under these conditions will decrease the effective coverage area of an e-NodeB.

### 7.3.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

## 7.3.3 Minimum conformance requirements

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.3.3-1, Table7.3.3-2 and Table 7.3.3-3.

**Table 7.3.3-1: Reference sensitivity QPSK $P_{REFSENS}$**

| Channel bandwidth | | | | | | | |
|---|---|---|---|---|---|---|---|
| **E-UTRA Band** | **1.4 MHz (dBm)** | **3 MHz (dBm)** | **5 MHz (dBm)** | **10 MHz (dBm)** | **15 MHz (dBm)** | **20 MHz (dBm)** | **Duplex Mode** |
| 1 | - | - | -100 | -97 | -95.2 | -94 | FDD |
| 2 | -102.7 | -99.7 | -98 | -95 | -93.2 | -92 | FDD |
| 3 | -101.7 | -98.7 | -97 | -94 | -92.2 | -91 | FDD |
| 4 | -104.7 | -101.7 | -100 | -97 | -95.2 | -94 | FDD |
| 5 | -103.2 | -100.2 | -98 | -95 | | | FDD |
| 6 | - | - | -100 | -97 | | | FDD |
| 7 | - | - | -98 | -95 | -93.2 | -92 | FDD |
| 8 | -102.2 | -99.2 | -97 | -94 | | | FDD |
| 9 | - | - | -99 | -96 | -94.2 | -93 | FDD |
| 10 | - | - | -100 | -97 | -95.2 | -94 | FDD |
| 11 | - | - | -100 | -97 | | | FDD |
| 12 | -101.7 | -98.7 | -97 | -94 | | | FDD |
| 13 | | | -97 | -94 | | | FDD |
| 14 | | - | -97 | -94 | | | FDD |
| ... | | | | | | | |
| 17 | - | - | -97 | -94 | | | FDD |
| 18 | - | - | -100 | -97 | -95.2 | - | FDD |
| 19 | - | - | -100 | -97 | -95.2 | - | FDD |
| 20 | | | -97 | -94 | -91.2 | -90 | FDD |
| 21 | | | -100 | -97 | -95.2 | | FDD |
| 22 | | | -97 | -94 | -92.2 | -91 | FDD |
| 23 | -104.7 | -101.7 | -100 | -97 | | | FDD |
| 24 | | | -100 | -97 | | | FDD |
| 25 | -101.2 | -98.2 | -96.5 | -93.5 | -91.7 | -90.5 | FDD |
| ... | | | | | | | |
| 33 | - | - | -100 | -97 | -95.2 | -94 | TDD |
| 34 | - | - | -100 | -97 | -95.2 | - | TDD |
| 35 | -106.2 | -102.2 | -100 | -97 | -95.2 | -94 | TDD |
| 36 | -106.2 | -102.2 | -100 | -97 | -95.2 | -94 | TDD |
| 37 | - | - | -100 | -97 | -95.2 | -94 | TDD |
| 38 | - | - | -100 | -97 | -95.2 | -94 | TDD |
| 39 | - | - | -100 | -97 | -95.2 | -94 | TDD |
| 40 | - | - | -100 | -97 | -95.2 | -94 | TDD |
| 41 | - | - | -99 | -96 | -94.2 | -93 | TDD |
| 42 | - | - | -99 | -96 | -94.2 | -93 | TDD |
| 43 | - | - | -99 | -96 | -94.2 | -93 | TDD |

Note 1: The transmitter shall be set to $P_{UMAX}$ as defined in clause 6.2.5
Note 2: The reference measurement channel is specified in A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1
Note 3: The signal power is specified per port
Note 4: For the UE which supports both Band 3 and Band 9 the reference sensitivity level is FFS.
Note 5: For the UE which supports both Band 11 and Band 21 the reference sensitivity level is FFS.

The reference receive sensitivity (REFSENS) requirement specified in Table 7.3.3-1 shall be met for an uplink transmission bandwidth less than or equal to that specified in Table 7.3.3-2.

NOTE: Table 7.3.3-2 does not necessarily reflect the operational conditions of the network, where the number of uplink and downlink allocated resource blocks will be practically constrained by other factors.

**Table 7.3.3-2: Uplink configuration for reference sensitivity**

| E-UTRA Band / Channel bandwidth / $N_{RB}$ / Duplex mode | | | | | | | |
|---|---|---|---|---|---|---|---|
| **E-UTRA Band** | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** | **Duplex Mode** |
| 1 | - | - | 25 | 50 | 75 | 100 | FDD |
| 2 | 6 | 15 | 25 | 50 | 50[1] | 50[1] | FDD |
| 3 | 6 | 15 | 25 | 50 | 50[1] | 50[1] | FDD |
| 4 | 6 | 15 | 25 | 50 | 75 | 100 | FDD |
| 5 | 6 | 15 | 25 | 25[1] | - | - | FDD |
| 6 | - | - | 25 | 25[1] | - | - | FDD |
| 7 | - | - | 25 | 50 | 75[1] | 75[1] | FDD |
| 8 | 6 | 15 | 25 | 25[1] | - | - | FDD |
| 9 | - | - | 25 | 50 | 50[1] | 50[1] | FDD |
| 10 | - | - | 25 | 50 | 75 | 100 | FDD |
| 11 | - | - | 25 | 25[1] | | | FDD |
| 12 | 6 | 15 | 20[1] | 20[1] | | | FDD |
| 13 | | | 20[1] | 20[1] | | | FDD |
| 14 | | - | 15[1] | 15[1] | | | FDD |
| ... | | | | | | | |
| 17 | - | - | 20[1] | 20[1] | | | FDD |
| 18 | - | - | 25 | 25[1] | 25[1] | - | FDD |
| 19 | - | - | 25 | 25[1] | 25[1] | - | FDD |
| 20 | | | 25 | 20[1] | 20[3] | 20[3] | FDD |
| 21 | | | 25 | 25[1] | 25[1] | | FDD |
| 22 | | | 25 | 50 | 50[1] | 50[1] | FDD |
| 23 | 6 | 15 | 25 | 50 | | | FDD |
| 24 | | | 25 | 50 | | | FDD |
| 25 | 6 | 15 | 25 | 50 | 50[1] | 50[1] | FDD |
| ... | | | | | | | |
| 33 | - | - | 25 | 50 | 75 | 100 | TDD |
| 34 | - | - | 25 | 50 | 75 | - | TDD |
| 35 | 6 | 15 | 25 | 50 | 75 | 100 | TDD |
| 36 | 6 | 15 | 25 | 50 | 75 | 100 | TDD |
| 37 | - | - | 25 | 50 | 75 | 100 | TDD |
| 38 | - | - | 25 | 50 | 75 | 100 | TDD |
| 39 | | | 25 | 50 | 75 | 100 | TDD |
| 40 | | | 25 | 50 | 75 | 100 | TDD |
| 41 | | | 25 | 50 | 75 | 100 | TDD |
| 42 | | | 25 | 50 | 75 | 100 | TDD |
| 43 | | | 25 | 50 | 75 | 100 | TDD |

Note 1: The UL resource blocks shall be located as close as possible to the downlink operating band but confined within the transmission bandwidth configuration for the channel bandwidth (Table 5.4.2-1).

Note 2: For the UE which supports both Band 11 and Band 21 the uplink configuration for reference sensitivity is FFS.

Note 3: For Band 20; in the case of 15MHz channel bandwidth, the UL resource blocks shall be located at $RB_{start}$=11 and in the case of 20MHz channel bandwidth, the UL resource blocks shall be located at $RB_{start}$=16

Unless given by Table 7.3.3-3, the minimum requirements specified in Tables 7.3.3-1 and 7.3.3-2 shall be verified with the network signalling value NS_01 (Table 6.2.4.3-1) configured.

**Table 7.3.3-3: Network Signalling Value for reference sensitivity**

| E-UTRA Band | Network Signalling value |
|---|---|
| 2 | NS_03 |
| 4 | NS_03 |
| 10 | NS_03 |
| 12 | NS_06 |
| 13 | NS_06 |
| 14 | NS_06 |
| 17 | NS_06 |
| 19 | NS_08 |
| 21 | NS_09 |
| 23 | NS_03 |

The normative reference for this requirement is TS 36.101 [2] clause 7.3.1.

## 7.3.4 Test description

### 7.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.3.4.1-1. The details of the downlink and uplink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.3.4.1-1: Test Configuration Table**

| Initial Conditions | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC, TL/VL, TL/VH, TH/VL, TH/VH | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 6 | 6 | QPSK | 6 | 6 |
| 3MHz | QPSK | 15 | 15 | QPSK | 15 | 15 |
| 5MHz | QPSK | 25 | 25 | QPSK | 25 | 25 |
| 5MHz | QPSK | 25 | N/A | QPSK | 20 | N/A |
| 5MHz | QPSK | 25 | N/A | QPSK | 15 | N/A |
| 10MHz | QPSK | 50 | 50 | QPSK | 50 | 50 |
| 10MHz | QPSK | 50 | N/A | QPSK | 25 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 20 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 15 | N/A |
| 15MHz | QPSK | 75 | 75 | QPSK | 75 | 75 |
| 15MHz | QPSK | 75 | N/A | QPSK | 50 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 25 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 20 | N/A |
| 20MHz | QPSK | 100 | 100 | QPSK | 100 | 100 |
| 20MHz | QPSK | 100 | N/A | QPSK | 75 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 50 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 25 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 20 | N/A |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | | | |
| Note 2. Depending on E-UTRA band, only the appropriate Uplink RB allocation value according to table 7.3.3-2 is tested per Test Channel Bandwidth. | | | | | | |
| Note 3: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used. | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.3.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1.
4. The UL and DL Reference Measurement channels are set according to Table 7.3.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.3.4.3.

## 7.3.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.3.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.3.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.
3. Set the Downlink signal level to the appropriate REFSENS value defined in Table 7.3.5-1. Send continuously uplink power control "up" commands in the uplink scheduling information to the UE to ensure the UE transmits $P_{UMAX}$ level for at least the duration of the Throughput measurement.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

### 7.3.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exceptions.

#### 7.3.4.3.1 Message contents exceptions (network signalled value "NS_01")

Message contents according to TS 36.508 [7] subclause 4.6 can be used without exceptions.

#### 7.3.4.3.2 Message contents exceptions (network signalled value "NS_03")

1. Information element additionalSpectrumEmission is set to NS_03. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 7.3.4.3.2-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_03"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 3 (NS_03) | | |

#### 7.3.4.3.3 Message contents exceptions (network signalled value "NS_06")

1. Information element additionalSpectrumEmission is set to NS_06. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 7.3.4.3.3-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_06"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | 6 (NS_06) | | |

#### 7.3.4.3.4 Message contents exceptions (network signalled value "NS_[09]")

1. Information element additionalSpectrumEmission is set to NS_[09]. This can be set in the *SystemInformationblockType2* as part of the cell broadcast message. This exception indicates that the UE shall meet the additional spurious emission requirement for a specific deployment scenario.

**Table 7.3.4.3.4-1: *SystemInformationBlockType2* :Additional spurious emissions test requirement for "NS_[09]"**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| additionalSpectrumEmission | TBD | | |

## 7.3.5 Test requirement

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Tables 7.3.5-1 and Table 7.3.5-2.

**Table 7.3.5-1: Reference sensitivity QPSK $P_{REFSENS}$**

| Channel bandwidth | | | | | | | |
|---|---|---|---|---|---|---|---|
| **E-UTRA Band** | **1.4 MHz (dBm)** | **3 MHz (dBm)** | **5 MHz (dBm)** | **10 MHz (dBm)** | **15 MHz (dBm)** | **20 MHz (dBm)** | **Duplex Mode** |
| 1 | - | - | -99.3 | -96.3 | -94.5 | -93.3 | FDD |
| 2 | -102.0 | -99.0 | -97.3 | -94.3 | -92.5 | -91.3 | FDD |
| 3 | -101.0 | -98.0 | -96.3 | -93.3 | -91.5 | -90.3 | FDD |
| 4 | -104.0 | -101 | -99.3 | -96.3 | -94.5 | -93.3 | FDD |
| 5 | -102.5 | -99.5 | -97.3 | -94.3 | | | FDD |
| 6 | - | - | -99.3 | -96.3 | | | FDD |
| 7 | - | - | -97.3 | -94.3 | -92.5 | -91.3 | FDD |
| 8 | -101.5 | -98.5 | -96.3 | -93.3 | | | FDD |
| 9 | - | - | -98.3 | -95.3 | -93.5 | -92.3 | FDD |
| 10 | - | - | -99.3 | -96.3 | -94.5 | -93.3 | FDD |
| 11 | - | - | -99.3 | -96.3 | | | FDD |
| 12 | -101.0 | -98.0 | -96.3 | -93.3 | | | FDD |
| 13 | | | -96.3 | -93.3 | | | FDD |
| 14 | | - | -96.3 | -93.3 | | | FDD |
| ... | | | | | | | |
| 17 | - | - | -96.3 | -93.3 | | | FDD |
| 18 | - | - | -99,3 | -96.3 | -94.5 | - | FDD |
| 19 | - | - | -99,3 | -96.3 | -94.5 | - | FDD |
| 20 | | | -96.3 | -93.3 | -90.5 | -89.3 | FDD |
| 21 | | | -99.3 | -96.3 | -94.5 | | FDD |
| 22 | | | -96.0 | -93.0 | -91.2 | -90.0 | FDD |
| 23 | -104.0 | -101 | -99.3 | -96.3 | | | FDD |
| 24 | | | -99.3 | -96.3 | | | FDD |
| 25 | -100.5 | -97.5 | -95.8 | -92.8 | -91.0 | -89.8 | FDD |
| ... | | | | | | | |
| 33 | - | - | -99,3 | -96.3 | -94.5 | -93.3 | TDD |
| 34 | - | - | -99.3 | -96.3 | -94.5 | - | TDD |
| 35 | -105.5 | -101.5 | -99.3 | -96.3 | -94.5 | -93.3 | TDD |
| 36 | -105.5 | -101.5 | -99.3 | -96.3 | -94.5 | -93.3 | TDD |
| 37 | - | - | -99.3 | -96.3 | -94.5 | -93.3 | TDD |
| 38 | - | - | -99.3 | -96.3 | -94.5 | -93.3 | TDD |
| 39 | - | - | -99.3 | -96.3 | -94.5 | -93.3 | TDD |
| 40 | - | - | -99.3 | -96.3 | -94.5 | -93.3 | TDD |
| 41 | - | - | -98.3 | -95.3 | -93.5 | -92.3 | TDD |
| 42 | - | - | -98.0 | -95.0 | -93.2 | -92.0 | TDD |
| 43 | - | - | -98.0 | -95.0 | -93.2 | -92.0 | TDD |

Note 1: The transmitter shall be set to maximum output power level (Table 7.3.5-2)
Note 2: The reference measurement channel is specified in A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1
Note 3: The signal power is specified per port
Note 4: For the UE which supports both Band 3 and Band 9 the reference sensitivity level is FFS.
Note 5: For the UE which supports both Band 11 and Band 21 the reference sensitivity level is FFS.

NOTE: The relation to the received PSD is $\left\langle \mathrm{REF}\,\hat{I}_{or} \right\rangle = P_{REFSENS}\left(N_{sc}^{RB} N_{RB} \Delta f\right)^{-1}$ with $N_{\mathrm{RB}}$ is the maximum transmission configuration according to Table 5.4.2-1.

The reference receive sensitivity (REFSENS) requirement specified in Table 7.3.5-1 shall be met for an uplink transmission bandwidth less than or equal to that specified in Table 7.3.5-2.

NOTE: Table 7.3.5-2 does not necessarily reflect the operational conditions of the network, where the number of uplink and downlink allocated resource blocks will be practically constrained by other factors.

**Table 7.3.5-2: Uplink configuration for reference sensitivity**

| E-UTRA Band / Channel bandwidth / $N_{RB}$ / Duplex mode | | | | | | | |
|---|---|---|---|---|---|---|---|
| **E-UTRA Band** | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** | **Duplex Mode** |
| 1 | - | - | 25 | 50 | 75 | 100 | FDD |
| 2 | 6 | 15 | 25 | 50 | 50[1] | 50[1] | FDD |
| 3 | 6 | 15 | 25 | 50 | 50[1] | 50[1] | FDD |
| 4 | 6 | 15 | 25 | 50 | 75 | 100 | FDD |
| 5 | 6 | 15 | 25 | 25[1] | - | - | FDD |
| 6 | - | - | 25 | 25[1] | - | - | FDD |
| 7 | - | - | 25 | 50 | 75[1] | 75[1] | FDD |
| 8 | 6 | 15 | 25 | 25[1] | - | - | FDD |
| 9 | - | - | 25 | 50 | 50[1] | 50[1] | FDD |
| 10 | - | - | 25 | 50 | 75 | 100 | FDD |
| 11 | - | - | 25 | 25[1] | | | FDD |
| 12 | 6 | 15 | 20[1] | 20[1] | | | FDD |
| 13 | | | 20[1] | 20[1] | | | FDD |
| 14 | | - | 15[1] | 15[1] | | | FDD |
| ... | | | | | | | |
| 17 | - | - | 20[1] | 20[1] | | | FDD |
| 18 | - | - | 25 | 25[1] | 25[1] | - | FDD |
| 19 | - | - | 25 | 25[1] | 25[1] | - | FDD |
| 20 | | | 25 | 20[1] | 20[3] | 20[3] | FDD |
| 21 | | | 25 | 25[1] | 25[1] | | FDD |
| 22 | | | 25 | 50 | 50[1] | 50[1] | FDD |
| 23 | 6 | 15 | 25 | 50 | | | FDD |
| 24 | | | 25 | 50 | | | FDD |
| 25 | 6 | 15 | 25 | 50 | 50[1] | 50[1] | FDD |
| … | | | | | | | |
| 33 | - | - | 25 | 50 | 75 | 100 | TDD |
| 34 | - | - | 25 | 50 | 75 | - | TDD |
| 35 | 6 | 15 | 25 | 50 | 75 | 100 | TDD |
| 36 | 6 | 15 | 25 | 50 | 75 | 100 | TDD |
| 37 | - | - | 25 | 50 | 75 | 100 | TDD |
| 38 | - | - | 25 | 50 | 75 | 100 | TDD |
| 39 | | | 25 | 50 | 75 | 100 | TDD |
| 40 | | | 25 | 50 | 75 | 100 | TDD |
| 41 | | | 25 | 50 | 75 | 100 | TDD |
| 42 | | | 25 | 50 | 75 | 100 | TDD |
| 43 | | | 25 | 50 | 75 | 100 | TDD |

Note 1: The UL resource blocks shall be located as close as possible to the downlink operating band but confined within the transmission bandwidth configuration for the channel bandwidth (Table 5.4.2-1).

Note 2: For the UE which supports both Band 11 and Band 21 the uplink configuration for reference sensitivity is FFS.

Note 3: For Band 20; in the case of 15MHz channel bandwidth, the UL resource blocks shall be located at $RB_{start}$=11 and in the case of 20MHz channel bandwidth, the UL resource blocks shall be located at $RB_{start}$=16

# 7.3A Reference sensitivity level for CA

Editor's notes:

The following items are missing or incomplete:

- Initial conditions (test frequencies, test configurations)

- Test procedure

- Message Contents

- Test requirements and test tolerances

- Connection diagram is missing

## 7.3A.1 Test purpose

To verify the ability of UE that support CA to receive data with a given average throughput for a specified reference measurement channel, under conditions of low signal level, ideal propagation and no added noise.

A UE unable to meet the throughput requirement under these conditions will decrease the effective coverage area of an e-NodeB.

## 7.3A.2 Test applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

## 7.3A.3 Minimum conformance requirements

For CA bandwidth class A the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.3.3-1 and table 7.3.3-2.

For the UE that supports inter band CA the reference sensitivity is defined to be met with both downlink component carriers active and either of the uplink component carriers active. The UE shall meet the requirements specified in clause 7.3.3.

For CA bandwidth class C the throughput of each component carrier shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.3.3-1 and table 7.3.A3-1. For UE(s) supporting one uplink, the uplink configuration of the Primary CC shall be in accordance with Table 7.3.3-2.

Table 7.3A.3-1 specifies the maximum number of allocated uplink resource blocks for which the intra-band contiguous CA reference receive sensitivity requirement must be met. The PCC allocation follows table 7.3.3-2. SCC and PCC transmission forms a contiguous allocation.

PCC and SCC TX–RX frequency separations are as defined in Table 5.3-1.

**Table 7.3A.3-1: Intra-band CA uplink configuration for reference sensitivity**

| CA Band / Aggregated channel bandwidth / NRB / Duplex mode | | | | | | | |
|---|---|---|---|---|---|---|---|
| **CA Band** | **100RB+50RB** | | **75RB+75RB** | | **100RB+100RB** | | **Duplex Mode** |
| CA_1C | n/a | n/a | PCC | SCC | PCC | SCC | FDD |
| | n/a | n/a | 75 | 55 | 100 | 30 | |
| CA_40C | PCC | SCC | PCC | SCC | PCC | SCC | TDD |
| | 100 | 50 | 75 | 75 | 100 | 100 | |
| NOTE 1: The carrier centre frequency of SCC in the UL operating band is configured closer to the DL operating band.<br>NOTE 2: The transmitted power over both PCC and SCC shall be set to $P_{UMAX}$ as defined in clause 6.2.5A.<br>NOTE 3: The UL resource blocks in both PCC and SCC shall be confined within the transmission bandwidth configuration for the channel bandwidth (Table 5.4.2-1). | | | | | | | |

The normative reference for this requirement is TS 36.101[2] clause 7.3.1A

## 7.3A.4 Test description

### 7.3A.4.1 Initial conditions

### 7.3A.4.2 Test procedure

### 7.3A.4.3 Message contents

## 7.3A.5 Test requirement

# 7.4 Maximum input level

## 7.4.1 Test purpose

Maximum input level tests the UE's ability to receive data with a given average throughput for a specified reference measurement channel, under conditions of high signal level, ideal propagation and no added noise.

A UE unable to meet the throughput requirement under these conditions will decrease the coverage area near to an e-NodeB.

## 7.4.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

## 7.4.3 Minimum conformance requirements

Maximum input level is defined as the maximum mean power received at the UE antenna port, at which the specified relative throughput shall meet or exceed the minimum requirements for the specified reference measurement channel.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.4.3-1.

**Table 7.4.3-1: Maximum input level**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | -25 | | | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5.<br>Note 2: Reference measurement channel is Annex A.3.2 64QAM R=3/4 variant with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.4.1.

## 7.4.4 Test description

### 7.4.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.4.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3 respectively. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.4.4.1-1: Test Configuration Table**

| Initial Conditions | | | | | | | |
|---|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | | UE Category |
| | | FDD | TDD | | FDD | TDD | |
| 1.4MHz | 64-QAM | 6 | 6 | QPSK | 5 | 5 | 1-5 |
| 3MHz | 64-QAM | 15 | 15 | QPSK | 4 | 4 | 1-5 |
| 5MHz | 64-QAM | 25 | 25 | QPSK | 8 | 8 | 2-5 |
| 5MHz | 64-QAM | 18 | 18 | QPSK | 8 | 8 | 1 |
| 10MHz | 64-QAM | 50 | 50 | QPSK | 12 | 12 | 2-5 |
| 10MHz | 64-QAM | 17 | 17 | QPSK | 12 | 12 | 1 |
| 15MHz | 64-QAM | 75 | 75 | QPSK | 16 | 16 | 2-5 |
| 15MHz | 64-QAM | 17 | 17 | QPSK | 16 | 16 | 1 |
| 20MHz | 64-QAM | 100 | 100 | QPSK | 18 | 18 | 3-5 |
| 20MHz | 64-QAM | 83 | 83 | QPSK | 18 | 18 | 2 |
| 20MHz | 64-QAM | 17 | 17 | QPSK | 18 | 18 | 1 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band. The applicable channel bandwidths are specified in Table 7.3.3-2.<br>Note 2: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used | | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Figure A.3.

2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1.

4. The UL and DL Reference Measurement channels are set according to Table 7.4.4.1-1.

5. Propagation conditions are set according to Annex B.0.

6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.4.4.3.

### 7.4.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.4.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.4.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

3. Set the Downlink signal level to the value defined in Table 7.4.5-1. Send Uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the target level in Table 7.4.5-1 for carrier frequency $f \leq 3.0$GHz or within +0, -4.0 dB of the target level for carrier frequency 3.0GHz < $f \leq 4.2$GHz, for at least the duration of the Throughput measurement.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

### 7.4.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception.

**Table 7.4.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

## 7.4.5 Test requirement

The throughput shall be $\geq$ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.4.5-1.

**Table 7.4.5-1: Maximum input level**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | For carrier frequency f ≤ 3.0GHz: -25.7<br>For carrier frequency 3.0GHz < f ≤ 4.2GHz: -26.0 | | | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5.<br>Note 2: Reference measurement channel is Annex A.3.2 64QAM R=3/4variant with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

# 7.4A Maximum input level for CA

Editor's notes:

The following items are missing or incomplete:

- References need to be updated

- Test procedure

- Message Contents

- Test requirements and test tolerances

## 7.4A.1 Test purpose

Maximum input level for CA test verifies the UE's ability to receive data with a given average throughput for a specified reference measurement channel, under conditions of high signal level, ideal propagation and no added noise.

A UE unable to meet the throughput requirement under these conditions will decrease the coverage area near to an e-NodeB.

## 7.4A.2 Test applicability

This test applies to all types of E-UTRA UE release 10 and forward that support CA.

## 7.4A.3 Minimum conformance requirements

UE maximum input level for CA is defined as the maximum mean power received at the UE antenna port over the aggregated channel bandwidth for intra-band contiguous carrier aggregation, at which the specified relative throughput shall meet or exceed the minimum requirements for the specified reference measurement channel over each component carrier.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels over each component carrier as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.4A.3-1. For UE(s) supporting one uplink, the uplink configuration of the Primary CC shall be in accordance with Table 7.3.3-2.

**Table 7.4A.3-1: Maximum input level for intra-band contiguous CA**

| Rx Parameter | Units | CA Bandwidth Class | | | | | |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | E | F |
| Power in Transmission Aggregated Bandwidth Configuration | dBm | | | -22 | | | |
| Note 1. The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3A.3-1 with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2 Reference measurement channel is Annex A.3.2: 64QAM, R=3/4 variant with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.4A.1.

## 7.4A.4 Test description

### 7.4A.4.1 Initial conditions

### 7.4A.4.2 Test procedure

### 7.4A.4.3 Message contents

## 7.4A.5 Test requirement

# 7.5 Adjacent Channel Selectivity (ACS)

## 7.5.1 Test purpose

Adjacent channel selectivity tests the UE's ability to receive data with a given average throughput for a specified reference measurement channel, in the presence of an adjacent channel signal at a given frequency offset from the centre frequency of the assigned channel, under conditions of ideal propagation and no added noise.

A UE unable to meet the throughput requirement under these conditions will decrease the coverage area when other e-NodeB transmitters exist in the adjacent channel.

## 7.5.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

## 7.5.3 Minimum conformance requirements

Adjacent Channel Selectivity (ACS) is a measure of a receiver's ability to receive an E-UTRA signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the centre frequency of the assigned channel. ACS is the ratio of the receive filter attenuation on the assigned channel frequency to the receive filter attenuation on the adjacent channel(s).

The UE shall fulfil the minimum requirement specified in Table 7.5.3-1 for all values of an adjacent channel interferer up to –25 dBm. However it is not possible to directly measure the ACS, instead the lower and upper range of test parameters are chosen in Table 7.5.3-2 and Table 7.5.3-3 where the throughput $R_{av}$ shall be $\geq$ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

**Table 7.5.3-1: Adjacent channel selectivity**

| | | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| **Rx Parameter** | **Units** | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| ACS | dB | 33.0 | 33.0 | 33.0 | 33.0 | 30 | 27 |

**Table 7.5.3-2: Test parameters for Adjacent channel selectivity, Case 1**

| **Rx Parameter** | **Units** | **Channel bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + 14 dB | | | | | |
| $P_{Interferer}$ | dBm | REFSENS +45.5dB | REFSENS +45.5dB | REFSENS +45.5dB* | REFSENS +45.5dB | REFSENS +42.5dB | REFSENS +39.5dB |
| $BW_{Interferer}$ | MHz | 1.4 | 3 | 5 | 5 | 5 | 5 |
| $F_{Interferer}$ (offset) | MHz | 1.4+0.0025 | 3+0.0075 | 5+0.0025 | 7.5+0.0075 | 10+0.0125 | 12.5+0.0025 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1. | | | | | | | |

**Table 7.5.3-3: Test parameters for Adjacent channel selectivity, Case 2**

| **Rx Parameter** | **Units** | **Channel bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Power in Transmission Bandwidth Configuration | dBm | -56.5 | -56.5 | -56.5 | -56.5 | -53.5 | -50.5 |
| $P_{Interferer}$ | dBm | -25 | | | | | |
| $BW_{Interferer}$ | MHz | 1.4 | 3 | 5 | 5 | 5 | 5 |
| $F_{Interferer}$ (offset) | MHz | 1.4+0.0025 | 3+0.0075 | 5+0.0025 | 7.5+0.0075 | 10+0.0125 | 12.5+0.0025 |
| Note 1: The transmitter shall be set to 24dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1. | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.5.1.

## 7.5.4 Test description

### 7.5.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.5.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3 respectively. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.5.4.1-1: Test Configuration Table**

| Initial Conditions | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | Full | Full | QPSK | 5 | 5 |
| 3MHz | QPSK | Full | Full | QPSK | 4 | 4 |
| 5MHz | QPSK | Full | Full | QPSK | 8 | 8 |
| 10MHz | QPSK | Full | Full | QPSK | 12 | 12 |
| 15MHz | QPSK | Full | Full | QPSK | 16 | 16 |
| 20MHz | QPSK | Full | Full | QPSK | 18 | 18 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band. The applicable channel bandwidths are specified in Table 7.3.3-2.<br>Note 2: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used. | | | | | | |

1. Connect the SS and interfering source to the UE antenna connectors as shown in TS 36.508 [7] Figure A.4.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1.
4. The UL and DL Reference Measurement channels are set according to Table 7.5.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.5.4.3.

### 7.5.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.5.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.5.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.
3. Set the Downlink signal level to the value as defined in Table 7.5.5-2 (Case 1). Send Uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the target level in Table 7.5.5-2 (Case 1) for carrier frequency $f \leq 3.0$GHz or within +0, - 4.0 dB of the target level for carrier frequency 3.0GHz $< f \leq$ 4.2GHz, for at least the duration of the Throughput measurement.
4. Set the Interferer signal level to the value as defined in Table 7.5.5-2 (Case 1), using a modulated interferer bandwidth as defined in Annex D of the present document.
5. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.
6. Set the Downlink signal level to the value as defined in Table 7.5.5-3 (Case 2). Send Uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the target level in Table 7.5.5-3 (Case 2) for carrier frequency $f \leq 3.0$GHz or within +0, - 4.0 dB of the target level for carrier frequency 3.0GHz $< f \leq$ 4.2GHz, for at least the duration of the Throughput measurement.

7. Set the Interferer signal level to the value as defined in Table 7.5.5-3 (Case 2), using a modulated interferer bandwidth as defined in Annex D of the present document.

8. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

9. Repeat for applicable channel bandwidths and operating band combinations in both Case 1 and Case 2.

### 7.5.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception

**Table 7.5.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

## 7.5.5 Test requirement

The throughput $R_{av}$ shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 under the conditions specified in table 7.5.5-2, and also under the conditions specified in table 7.5.5-3.

**Table 7.5.5-1: Adjacent channel selectivity**

| | | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| **Rx Parameter** | **Units** | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| ACS | dB | 33.0 | 33.0 | 33.0 | 33.0 | 30 | 27 |

**Table 7.5.5-2: Test parameters for Adjacent channel selectivity, Case 1**

| **Rx Parameter** | **Units** | **Channel bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + 14 dB | | | | | |
| $P_{Interferer}$ | dBm | REFSENS +45.5dB | REFSENS +45.5dB | REFSENS +45.5dB* | REFSENS +45.5dB | REFSENS +42.5dB | REFSENS +39.5dB |
| $BW_{Interferer}$ | MHz | 1.4 | 3 | 5 | 5 | 5 | 5 |
| $F_{Interferer}$ (offset) | MHz | 1.4+0.0025 | 3+0.0075 | 5+0.0025 | 7.5+0.0075 | 10+0.0125 | 12.5+0.0025 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5.<br>Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1. | | | | | | | |

**Table 7.5.5-3: Test parameters for Adjacent channel selectivity, Case 2**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | -56.5 | -56.5 | -56.5 | -56.5 | -53.5 | -50.5 |
| $P_{Interferer}$ | dBm | -25 | | | | | |
| $BW_{Interferer}$ | MHz | 1.4 | 3 | 5 | 5 | 5 | 5 |
| $F_{Interferer}$ (offset) | MHz | 1.4+0.0025 | 3+0.0075 | 5+0.0025 | 7.5+0.0075 | 10+0.0125 | 12.5+0.0025 |
| Note 1: The transmitter shall be set to 24dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1. | | | | | | | |

# 7.6 Blocking characteristics

The blocking characteristic is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency in the presence of an unwanted interferer on frequencies other than those of the spurious response or the adjacent channels, without this unwanted input signal causing a degradation of the performance of the receiver beyond a specified limit. The blocking performance shall apply at all frequencies except those at which a spurious response occur.

## 7.6.1 In-band blocking

### 7.6.1.1 Test Purpose

In-band blocking is defined for an unwanted interfering signal falling into the range from 15MHz below to 15MHz above the UE receive band, at which the relative throughput shall meet or exceed the requirement for the specified measurement channels.

The lack of in-band blocking ability will decrease the coverage area when other e-NodeB transmitters exist (except in the adjacent channels and spurious response).

### 7.6.1.2 Test Applicability

This test applies to all types of E-UTRA UE release 8 and forward..

### 7.6.1.3 Minimum Conformance Requirements

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.6.1.3-1 and 7.6.1.3-2.

**Table 7.6.1.3-1: In band blocking parameters**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 6 | 6 | 6 | 6 | 7 | 9 |
| $BW_{Interferer}$ | MHz | 1.4 | 3 | 5 | 5 | 5 | 5 |
| $F_{loffset, case\ 1}$ | MHz | 2.1+0.0125 | 4.5+0.0075 | 7.5+0.0125 | 7.5+0.0025 | 7.5+0.0075 | 7.5+0.0125 |
| $F_{loffset, case\ 2}$ | MHz | 3.5+0.0075 | 7.5+0.0075 | 12.5+0.0075 | 12.5+0.0125 | 12.5+0.0025 | 12.5+0.0075 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1. | | | | | | | |

**Table 7.6.1.3-2: In-band blocking**

| E-UTRA band | Parameter | Unit | Case 1 | Case 2 | Case 3 | Case 4 |
|---|---|---|---|---|---|---|
| | $P_{Interferer}$ | dBm | -56 | -44 | -30 | [-30] |
| | $F_{Interferer}$ (offset) | MHz | $=-BW/2 - F_{loffset,case\ 1}$ & $=+BW/2 + F_{loffset,case\ 1}$ | $\leq -BW/2 - F_{loffset,case\ 2}$ & $\geq +BW/2 + F_{loffset,case\ 2}$ | -BW/2 – 15 & -BW/2 – 9 | -BW/2 – 10 |
| 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 18, 19, 20, 21, 22, 23, 25,33, 34, 35, 36, 37, 38, 39, 40, 41 | $F_{Interferer}$ | MHz | (Note 2) | $F_{DL_low}$ – 15 to $F_{DL_high}$ + 15 | | |
| 12 | $F_{Interferer}$ | MHz | (Note 2) | $F_{DL_low}$ – 10 to $F_{DL_high}$ + 15 | | $F_{DL_low}$ – 10 |
| 17 | $F_{Interferer}$ | MHz | (Note 2) | $F_{DL_low}$ – 9 to $F_{DL_high}$ + 15 | $F_{DL_low}$ – 15 and $F_{DL_low}$ – 9 | |
| Note 1: For certain bands, the unwanted modulated interfering signal may not fall inside the UE receive band, but within the first 15 MHz below or above the UE receive band | | | | | | |
| Note 2: For each carrier frequency the requirement is valid for two frequencies: a. the carrier frequency -BW/2 - $F_{loffset, case\ 1}$ and b. the carrier frequency +BW/2 + $F_{loffset, case\ 1}$ | | | | | | |
| Note 3: $F_{Interferer}$ range values for unwanted modulated interfering signal are interferer centre frequencies | | | | | | |
| Note 4: Case 3 and Case 4 only apply to assigned UE channel bandwidth of 5 MHz | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.6.1.

## 7.6.1.4 Test Description

### 7.6.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.6.1.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3 respectively. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.6.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 6 | 6 | QPSK | 6 | 6 |
| 3MHz | QPSK | 15 | 15 | QPSK | 15 | 15 |
| 5MHz | QPSK | 25 | 25 | QPSK | 25 | 25 |
| 5MHz | QPSK | 25 | N/A | QPSK | 20 | N/A |
| 5MHz | QPSK | 25 | N/A | QPSK | 15 | N/A |
| 10MHz | QPSK | 50 | 50 | QPSK | 50 | 50 |
| 10MHz | QPSK | 50 | N/A | QPSK | 25 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 20 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 15 | N/A |
| 15MHz | QPSK | 75 | 75 | QPSK | 75 | 75 |
| 15MHz | QPSK | 75 | N/A | QPSK | 50 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 25 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 20 | N/A |
| 20MHz | QPSK | 100 | 100 | QPSK | 100 | 100 |
| 20MHz | QPSK | 100 | N/A | QPSK | 75 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 50 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 25 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 20 | N/A |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.<br>Note 2. Depending on E-UTRA band, only the appropriate Uplink RB allocation value according to table 7.3.3-2 is tested per Test Channel Bandwidth.<br>Note 3: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used. | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, in Figure A.4.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1.
4. The UL and DL Reference Measurement channels are set according to in Table 7.6.1.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.6.1.4.3.

### 7.6.1.4.2 Test Procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.6.1.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.6.1.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.
3. Set the parameters of the signal generator for an interfering signal below the wanted signal in Case 1 according to Tables 7.6.1.5-1 and 7.6.1.5-2.
4. Set the downlink signal level according to the table 7.6.1.5-1. Send uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the

target level in table 7.6.1.5-1 for carrier frequency f ≤ 3.0GHz or within +0, -4.0 dB of the target level for carrier frequency 3.0GHz < f ≤ 4.2GHz, for at least the duration of the throughput measurement.

5. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

6. Repeat steps from 3 to 5, using an interfering signal above the wanted signal in Case 1 at step 3.

7. Repeat steps from 3 to 6, using interfering signals in Case 2 at step 3and 6. The ranges of case 2 are covered in steps equal to the interferer bandwidth. The test frequencies are chosen in analogy to table 7.6.1.4.2-1.

8. Repeat steps from 3 to 5, using successively all interfering signals in Case 3 and Case 4 at step 3.

**Table 7.6.1.4.2-1: Example for interferer frequencies**

| | Lower frequency | Upper frequency |
|---|---|---|
| Band 1 DL | 2110 MHz | 2170 MHz |
| Band 1 Midrange | 2140 MHz | |
| Receive band wanted signal (BW 5MHz) | 2137.5 MHz | 2142.5 MHz |
| Interferer case 1 | 2129.9875 MHz | 2150.0125 MHz |
| Interferer case 2 (inner frequency) | 2124.9925 MHz | 2155.0075 MHz |
| Interferer case 2 (outer frequency) | 2099.9925 MHz | 2180.0075 MHz |
| Outer limit for inband blocking | 2095MHz | 2185MHz |
| Number of test frequencies case 2 | 6 | 6 |
| Number of test frequencies for Band 17(asymmetric!), BW 5MHz, case 2 | 0 | 2 |

### 7.6.1.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception

**Table 7.6.1.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

## 7.6.1.5 Test Requirement

The throughput measurement derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Tables 7.6.1.5-1 and 7.6.1.5-2.

**Table 7.6.1.5-1: In band blocking parameters**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 6 | 6 | 6 | 6 | 7 | 9 |
| $BW_{Interferer}$ | MHz | 1.4 | 3 | 5 | 5 | 5 | 5 |
| $F_{loffset, case\ 1}$ | MHz | 2.1+0.0125 | 4.5+0.0075 | 7.5+0.0125 | 7.5+0.0025 | 7.5+0.0075 | 7.5+0.0125 |
| $F_{loffset, case\ 2}$ | MHz | 3.5+0.0075 | 7.5+0.0075 | 12.5+0.0075 | 12.5+0.0125 | 12.5+0.0025 | 12.5+0.0075 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5.<br>Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1. | | | | | | | |

**Table 7.6.1.5-2: In-band blocking**

| E-UTRA band | Parameter | Unit | Case 1 | Case 2 | Case 3 | Case 4 |
|---|---|---|---|---|---|---|
| | $P_{Interferer}$ | dBm | -56 | -44 | -30 | [-30] |
| | $F_{Interferer}$ (offset) | MHz | $=-BW/2 - F_{loffset,case\ 1}$ & $=+BW/2 + F_{loffset,case\ 1}$ | $\leq -BW/2 - F_{loffset,case\ 2}$ & $\geq +BW/2 + F_{loffset,case\ 2}$ | -BW/2 – 15 & -BW/2 – 9 | -BW/2 – 10 |
| 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 18, 19, 20, 21, 22, 23, 25,33, 34, 35, 36, 37, 38,39, 40, 41 | $F_{Interferer}$ | MHz | (Note 2) | $F_{DL_low}$ – 15 to $F_{DL_high}$ + 15 | | |
| 12 | $F_{Interferer}$ | MHz | (Note 2) | $F_{DL_low}$ – 10 to $F_{DL_high}$ + 15 | | $F_{DL_low}$ – 10 |
| 17 | $F_{Interferer}$ | MHz | (Note 2) | $F_{DL_low}$ – 9 to $F_{DL_high}$ + 15 | $F_{DL_low}$ – 15 and $F_{DL_low}$ – 9 | |
| Note 1: For certain bands, the unwanted modulated interfering signal may not fall inside the UE receive band, but within the first 15 MHz below or above the UE receive band<br>Note 2: For each carrier frequency the requirement is valid for two frequencies:<br>a. the carrier frequency -BW/2 - $F_{loffset, case\ 1}$ and<br>b. the carrier frequency +BW/2 + $F_{loffset, case\ 1}$<br>Note 3: $F_{Interferer}$ range values for unwanted modulated interfering signal are interferer centre frequencies<br>Note 4: Case 3 and Case 4 only apply to assigned UE channel bandwidth of 5 MHz | | | | | | |

## 7.6.1A In-band blocking for CA

Editor's notes:

The following items are missing or incomplete:

- In minimum requirements, reference to UL configuration setting in Reference Sensitivity is missing

- References need to be updated

- Initial conditions test setup for SCC is FFS

- Test procedure

- Message Contents

- Test tolerances

- Connection diagram

- Inter-band test requirements handling and procedure is FFS

### 7.6.1A.1 Test Purpose

In-band blocking is defined for an unwanted interfering signal falling into the range from 15MHz below to 15MHz above the UE receive band, at which the relative throughput shall meet or exceed the requirement for the specified measurement channels.

The lack of in-band blocking ability will decrease the coverage area when other e-NodeB transmitters exist (except in the adjacent channels and spurious response).

### 7.6.1A.2 Test Applicability

This test applies to all types of E-UTRA UE release 10 and forward that support CA.

### 7.6.1A.3 Minimum Conformance Requirements

For intra-band contiguous CA (bandwidth Class C) the downlink Secondary CC shall be configured at nominal channel spacing to the Primary CC with the Primary CC configured closest to the uplink band. Downlink Primary CC and Secondary CC are both activated. The uplink output power shall be set as specified in Table 7.6.1.1A-1 with the uplink configuration set according to [Table 7.3.3A-1] for the applicable CA Band. For UE(s) supporting one uplink, the uplink configuration of the Primary CC shall be in accordance with Table 7.3.3-2.

The UE shall fulfil the minimum requirement in presence of an interfering signal specified in Tables 7.6.1A.3-1 and Tables 7.6.1A.3-2 being on either side of the aggregated signal. The throughput of each carrier shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.6.1A.3-1 and 7.6.1A.3-2.

**Table 7.6.1A.3-1: In band blocking parameters**

| Rx Parameter | Units | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated Transmission Bandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 12 | | | |
| $BW_{Interferer}$ | MHz | | 5 | | | |
| $F_{loffset, case\ 1}$ | MHz | | 7.5 | | | |
| $F_{loffset, case\ 2}$ | MHz | | 12.5 | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ as defined in clause 6.2.5A<br>Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1 | | | | | | |

**Table 7.6.1A.3-2: In-band blocking**

| CA operating band | Parameter | Unit | Case 1 | Case 2 |
|---|---|---|---|---|
| | $P_{Interferer}$ | dBm | -56 | -44 |
| | $F_{Interferer}$ | MHz | $=-F_{offset}-F_{Ioffset,case\ 1}$ & $=+F_{offset}+F_{Ioffset,case\ 1}$ | $\leq-F_{offset}-F_{Ioffset,case\ 2}$ & $\geq+F_{offset}+F_{Ioffset,case\ 2}$ |
| CA_1C, CA_40C | $F_{Interferer}$ (Range) | MHz | (Note 2) | $F_{DL_low}-15$ to $F_{DL_high}+15$ |

Note 1: For certain bands, the unwanted modulated interfering signal may not fall inside the UE receive band, but within the first 15 MHz below or above the UE receive band

Note 2: For each carrier frequency the requirement is valid for two frequencies:
a. the carrier frequency -BW/2 - $F_{Ioffset,\ case\ 1}$ and
b. the carrier frequency +BW/2 + $F_{Ioffset,\ case\ 1}$

Note 3: $F_{offset}$ is the frequency offset from the center frequency of the adjacent CC being tested to the edge of aggregated channel bandwidth.

Note 4: The $F_{interferer}$ (offset) is relative to the center frequency of the adjacent CC being tested and shall be further adjusted to $\lfloor F_{interferer}/0.015+0.5\rfloor 0.015+0.0075$ MHz to be offset from the sub-carrier raster.

For the UE that supports inter band CA with a single uplink in one band the in-band blocking requirements are defined with the single uplink active on the band other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.6.1 for each component carrier while both downlink carriers are active.

For the UE which supports inter band CA configuration in Table [7.3.3A-2], $P_{Interferer}$ power defined in table 7.6.1. 3-2 is increased by the amount given by $\Delta R_{IB}$ in Table [7.3.3A-2].

The normative reference for this requirement is TS 36.101 [2] clause 7.6.1.1A.

## 7.6.1A.4 Test Description

### 7.6.1A.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and CC combinations based on E-UTRA CA configurations specified in table 5.4.2A.1-1. All of these configurations shall be tested with applicable test parameters for each CA Configuration, and are shown in table 7.6.1A.4.1-1. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.6.1A.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | | |
|---|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause [4.3.1] for different CA bandwidth classes. | | C: Mid range | | | | | |
| Test CC Combination setting ($N_{RB_agg}$) as specified in subclause 5.4.2A.1 for the CA Configuration | | Lowest $N_{RB_agg}$<br>Highest $N_{RB_agg}$ | | | | | |
| **Test Parameters for CA Configurations** | | | | | | | |
| **CA Configuration / $N_{RB_agg}$** | | **DL Allocation** | | **UL Allocation** | | | |
| **PCC $N_{RB}$** | **SCC $N_{RB}$** | **CC MOD** | **PCC & SCC RB allocation** | **CC MOD** | **$N_{RB_alloc}$** | **PCC & SCC RB allocations ($L_{CRB}$ @ $RB_{start}$)** | |
| 75 | 75 | QPSK | 75+75 | QPSK | 130 | P_75@0 | S_55@0 |
| 75 | 75 | QPSK | 75+75 | QPSK | 150 | P_75@0 | S_75@0 |
| 100 | 50 | QPSK | 100+50 | QPSK | 150 | P_100@0 | S_50@0 |
| 100 | 100 | QPSK | 100+100 | QPSK | 130 | P_100@0 | S_30@0 |
| 100 | 100 | QPSK | 100+100 | QPSK | 200 | P_100@0 | S_100@0 |
| Note 1: CA Configuration Test CC Combination settings are checked separately for each CA Configuration, which applicable aggregated channel bandwidths are specified in Table 5.4.2A.1-1.<br>Note 2: Depending on CA Configuration only the appropriate Uplink RB allocation value according to table 7.3A.3-1 is tested per CA Configuration Test CC combination. | | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure [TBD].
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1
4. The UL and DL Reference Measurement channels are set according to Table 7.6.1A.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in [State 3A-RF] according to TS 36.508 [7] clause [5.2A.2]. Message contents are defined in clause 7.6.1A.4.3.

### 7.6.1A.4.2 Test Procedure

FFS

### 7.6.1A.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause [FFS].

## 7.6.1A.5 Test Requirement

The throughput measurement of each carrier derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.6.1A.5-1 and 7.6.1A.5-2.

**Table 7.6.1A.5-1: In band blocking parameter**

| Rx Parameter | Units | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated Transmission Bandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 12 | | | |
| $BW_{Interferer}$ | MHz | | 5 | | | |
| $F_{Ioffset, case\ 1}$ | MHz | | 7.5 | | | |
| $F_{Ioffset, case\ 2}$ | MHz | | 12.5 | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ as defined in clause 6.2.5A<br>Note 2: The interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1 | | | | | | |

**Table 7.6.1A.5-2: In-band blocking**

| CA operating band | Parameter | Unit | Case 1 | Case 2 |
|---|---|---|---|---|
| | $P_{Interferer}$ | dBm | -56 | -44 |
| | $F_{Interferer}$ | MHz | $=-F_{offset}-F_{Ioffset,case\ 1}$ & $=+F_{offset}+F_{Ioffset,case\ 1}$ | $\leq-F_{offset}-F_{Ioffset,case\ 2}$ & $\geq+F_{offset}+F_{Ioffset,case\ 2}$ |
| CA_1C, CA_40C | $F_{Interferer}$ (Range) | MHz | (Note 2) | $F_{DL_low}-15$ to $F_{DL_high}+15$ |
| Note 1: For certain bands, the unwanted modulated interfering signal may not fall inside the UE receive band, but within the first 15 MHz below or above the UE receive band<br>Note 2: For each carrier frequency the requirement is valid for two frequencies:<br>a. the carrier frequency -BW/2 - $F_{Ioffset, case\ 1}$ and<br>b. the carrier frequency +BW/2 + $F_{Ioffset, case\ 1}$<br>Note 3: $F_{offset}$ is the frequency offset from the center frequency of the adjacent CC being tested to the edge of aggregated channel bandwidth.<br>Note 4: The $F_{interferer}$ (offset) is relative to the center frequency of the adjacent CC being tested and shall be further adjusted to $\lfloor F_{interferer}/0.015+0.5\rfloor 0.015+0.0075$ MHz to be offset from the sub-carrier raster. | | | | |

## 7.6.2 Out-of-band blocking

### 7.6.2.1 Test Purpose

Out-of-band band blocking is defined for an unwanted CW interfering signal falling more than 15 MHz below or above the UE receive band, at which a given average throughput shall meet or exceed the requirement for the specified measurement channels.

For the first 15 MHz below or above the UE receive band the appropriate in-band blocking or adjacent channel selectivity in sub-clause 7.5.1 and sub-clause 7.6.1 shall be applied.

The lack of out-of-band blocking ability will decrease the coverage area when other e-NodeB transmitters exist (except in the adjacent channels and spurious response).

### 7.6.2.2 Test Applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 7.6.2.3 Minimum Conformance Requirements

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.6.2.3-1 and 7.6.2.3-2.

For Table 7.6.2.3-2 in frequency range 1, 2 and 3, up to $\max(24,\ 6\cdot\lceil N_{RB}/6\rceil)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size, where $N_{RB}$ is the number of resource blocks in the downlink transmission bandwidth configuration (see Figure 5.4.2-1). For these exceptions the requirements of clause 7.7 Spurious Response are applicable.

For Table 7.6.2.3-2 in frequency range 4, up to $\max(8,\ \lceil(N_{RB}+2\cdot L_{CRBs})/8\rceil)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size, where $N_{RB}$ is the number of resource blocks in the downlink transmission bandwidth configurations (see Figure 5.4.2-1) and $L_{CRBs}$ is the number of resource blocks allocated in the uplink. For these exceptions the requirements of clause 7.7 Spurious Response are applicable.

**Table 7.6.2.3-1: Out-of-band blocking parameters**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 6 | 6 | 6 | 6 | 7 | 9 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

**Table 7.6.2.3-2: Out of band blocking**

| E-UTRA band | Parameter | Units | Frequency | | | |
|---|---|---|---|---|---|---|
| | | | range 1 | range 2 | range 3 | range 4 |
| | $P_{Interferer}$ | dBm | -44 | -30 | -15 | -15 |
| 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 17, 18, 19, 20, 21, 22, 23, 24, 25, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43 | $F_{Interferer}$ (CW) | MHz | $F_{DL_low}$ -15 to $F_{DL_low}$ -60 | $F_{DL_low}$ -60 to $F_{DL_low}$ -85 | $F_{DL_low}$ -85 to 1 MHz | - |
| | | | $F_{DL_high}$ +15 to $F_{DL_high}$ + 60 | $F_{DL_high}$ +60 to $F_{DL_high}$ +85 | $F_{DL_high}$ +85 to +12750 MHz | - |
| 2, 5, 12, 17 | $F_{Interferer}$ | MHz | - | - | - | $F_{UL_low}$ - $F_{UL_high}$ |
| Note: For the UE which supports both Band 11 and Band 21 the out of blocking is FFS. | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.6.2.

## 7.6.2.4 Test Description

### 7.6.2.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.6.2.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3 respectively. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.6.2.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | One frequency chosen arbitrarily from low or high range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 6 | 6 | QPSK | 6 | 6 |
| 3MHz | QPSK | 15 | 15 | QPSK | 15 | 15 |
| 5MHz | QPSK | 25 | 25 | QPSK | 25 | 25 |
| 5MHz | QPSK | 25 | N/A | QPSK | 20 | N/A |
| 5MHz | QPSK | 25 | N/A | QPSK | 15 | N/A |
| 10MHz | QPSK | 50 | 50 | QPSK | 50 | 50 |
| 10MHz | QPSK | 50 | N/A | QPSK | 25 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 20 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 15 | N/A |
| 15MHz | QPSK | 75 | 75 | QPSK | 75 | 75 |
| 15MHz | QPSK | 75 | N/A | QPSK | 50 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 25 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 20 | N/A |
| 20MHz | QPSK | 100 | 100 | QPSK | 100 | 100 |
| 20MHz | QPSK | 100 | N/A | QPSK | 75 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 50 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 25 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 20 | N/A |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | | | |
| Note 2. Depending on E-UTRA band, only the appropriate Uplink RB allocation value according to table 7.3.3-2 is tested per Test Channel Bandwidth. | | | | | | |
| Note 3: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used. | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, in Figure A.5.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1.
4. The UL and DL Reference Measurement channels are set according to Table 7.6.2.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.6.2.4.3.

### 7.6.2.4.2 Test Procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.6.2.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.6.2.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.
3. Set the parameters of the CW signal generator for an interfering signal according to Table 7.6.2.5-2. The frequency step size is 1MHz.
4. Set the downlink signal level according to the table 7.6.2.5-1. Send uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the

target level in table 7.6.2.5-1 for carrier frequency f ≤ 3.0GHz or within +0, -4.0 dB of the target level for carrier frequency 3.0GHz < f ≤ 4.2GHz, for at least the duration of the throughput measurement.

5. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

6. Record the frequencies for which the throughput doesn't meet the requirements.

#### 7.6.2.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception.

**Table 7.6.2.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

### 7.6.2.5 Test Requirement

Except for the spurious response frequencies recorded at the final step of test procedure, the throughput measurement derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Tables 7.6.2.5-1 and 7.6.2.5-2.

For frequency range 1, 2, and 3, the number of spurious response frequencies recorded in the final step of test procedure shall not exceed $\max(24, \ 6 \cdot \lceil N_{RB} / 6 \rceil)$ in each assigned frequency channel when measured using a 1MHz step size. For these exceptions the requirements of clause 7.7 Spurious Response are applicable.

For frequency range 4, the number of spurious response frequencies recorded in the final step of test procedure shall not exceed $\max(8, \ \lceil (N_{RB} + 2 \cdot L_{CRBs}) / 8 \rceil)$ in each assigned frequency channel when measured using a 1MHz step size. For these exceptions the requirements of clause 7.7 Spurious Response are applicable.

**Table 7.6.2.5-1: Out-of-band blocking parameters**

| **Rx Parameter** | **Units** | **Channel bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 6 | 6 | 6 | 6 | 7 | 9 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

**Table 7.6.2.5-2: Out of band blocking**

| E-UTRA band | Parameter | Units | Frequency | | | |
|---|---|---|---|---|---|---|
| | | | range 1 | range 2 | range 3 | range 4 |
| | $P_{Interferer}$ | dBm | -44 | -30 | -15 | -15 |
| 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 17, 18, 19, 20, 21, 22, 23, 24, 25, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43 | $F_{Interferer}$ (CW) | MHz | $F_{DL_low}$ -15 to $F_{DL_low}$ -60 | $F_{DL_low}$ -60 to $F_{DL_low}$ -85 | $F_{DL_low}$ -85 to 1 MHz | - |
| | | | $F_{DL_high}$ +15 to $F_{DL_high}$ + 60 | $F_{DL_high}$ +60 to $F_{DL_high}$ +85 | $F_{DL_high}$ +85 to +12750 MHz | - |
| 2, 5, 12, 17 | $F_{Interferer}$ | MHz | - | - | - | $F_{UL_low}$ - $F_{UL_high}$ |
| Note 1: Range 3 shall be tested only with the highest channel bandwidth.<br>Note 2: For the UE which supports both Band 11 and Band 21 the out of blocking is FFS. | | | | | | |

## 7.6.2A Out-of-band blocking for CA

Editor's notes:

The following items are missing or incomplete:

- In minimum requirements, reference to UL configuration setting in Reference Sensitivity is missing

- References need to be updated

- Initial conditions test setup for SCC is FFS

- Test procedure

- Message Contents

- Test tolerances

- Connection diagram

- Inter-band CA requirements are FFS

### 7.6.2A.1 Test Purpose

Out-of-band band blocking for CA is defined for an unwanted CW interfering signal falling more than 15 MHz below or above the UE receive band, at which a given average throughput shall meet or exceed the requirement for the specified measurement channels in aggregated signals.

For the first 15 MHz below or above the UE receive band the appropriate in-band blocking or adjacent channel selectivity in sub-clause 7.5.1A and sub-clause 7.6.1A shall be applied.

The lack of out-of-band blocking ability will decrease the coverage area when other e-NodeB transmitters exist (except in the adjacent channels and spurious response).

### 7.6.2A.2 Test Applicability

This test applies to all types of E-UTRA UE release 10 and forward that support CA.

### 7.6.2A.3 Minimum Conformance Requirements

For intra-band contiguous CA (bandwidth Class C) the downlink Secondary CC shall be configured at nominal channel spacing to the Primary CC with the Primary CC configured closest to the uplink band. Downlink Primary CC and Secondary CC are both activated. The uplink output power shall be set as specified in Table 7.6.2A.3-1 with the uplink configuration set according to [Table 7.3.3A-1] for the applicable CA Band. For UE(s) supporting one uplink, the uplink configuration of the Primary CC shall be in accordance with Table 7.3.3-2.

The UE shall fulfil the minimum requirement in presence of an interfering signal specified in Tables 7.6.2A.3-1 and Tables 7.6.2A.3-2 being on either side of the aggregated signal. The throughput of each carrier shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.6.1A.3-1 and 7.6.1A.3-2.

For Table 7.6.2A.3-2 in frequency range 1, 2 and 3, up to $\max\left(24,\ 6\cdot\left\lceil N_{RB,agg}/6\right\rceil\right)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size, where $N_{RB,agg}$ is the number of aggregated resource blocks in the downlink transmission bandwidth configuration. For these exceptions the requirements of clause 7.7A Spurious response for CA are applicable.

For Table 7.6.2A.3-2 in frequency range 4, up to $\max\left(8,\ \left\lceil (N_{RB,agg}+2\cdot L_{CRBs})/8\right\rceil\right)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size, where $N_{RB,agg}$ is the number of aggregated resource blocks in the downlink transmission bandwidth configurations and $L_{CRBs}$ is the number of resource blocks allocated in the uplink. For these exceptions the requirements of clause 7.7A spurious response for CA are applicable.

**Table 7.6.2A.3-1: Out-of-band blocking parameters**

| Rx Parameter | Units | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated Transmission Bandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 9 | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in TS 36.101 [2] Table 7.3.1A-1, with $P_{CMAX_L}$ as defined in clause 6.2.5A.<br>Note 2: Reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2. | | | | | | |

**Table 7.6.2A.3-2: Out of band blocking**

| CA operating band | Parameter | Units | Frequency | | |
|---|---|---|---|---|---|
| | | | range 1 | range 2 | range 3 |
| CA_1C, CA_40C | $P_{Interferer}$ | dBm | -44 | -30 | -15 |
| | $F_{Interferer}$ (CW) | MHz | $F_{DL_low}$ -15 to $F_{DL_low}$ -60 | $F_{DL_low}$ -60 to $F_{DL_low}$ -85 | $F_{DL_low}$ -85 to 1 MHz |
| | | | $F_{DL_high}$ +15 to $F_{DL_high}$ + 60 | $F_{DL_high}$ +60 to $F_{DL_high}$ +85 | $F_{DL_high}$ +85 to +12750 MHz |

For the UE that supports inter-band CA, the out-of-bank blocking requirements are FFS.

The normative reference for this requirement is TS 36.101 [2] clause 7.6.2.1A.

## 7.6.2A.4 Test Description

### 7.6.2A.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and CC combinations based on E-UTRA CA configurations specified in table 5.4.2A.1-1. All of these configurations shall be tested with applicable test parameters for each CA Configuration, and are shown in table 7.6.2A.4.1-1. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.6.2A.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | | |
|---|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause [4.3.1] for different CA bandwidth classes. | | C: Mid range | | | | | |
| Test CC Combination setting ($N_{RB_agg}$) as specified in subclause 5.4.2A.1 for the CA Configuration | | Highest $N_{RB_agg}$ | | | | | |
| **Test Parameters for CA Configurations** | | | | | | | |
| **CA Configuration / $N_{RB_agg}$** | | **DL Allocation** | | **UL Allocation** | | | |
| **PCC $N_{RB}$** | **SCC $N_{RB}$** | **CC MOD** | **PCC & SCC RB allocation** | **CC MOD** | **$N_{RB_alloc}$** | **PCC & SCC RB allocations ($L_{CRB}$ @ $RB_{start}$)** | |
| 75 | 75 | QPSK | 75+75 | QPSK | 130 | P_75@0 | S_55@0 |
| 75 | 75 | QPSK | 75+75 | QPSK | 150 | P_75@0 | S_75@0 |
| 100 | 50 | QPSK | 100+50 | QPSK | 150 | P_100@0 | S_50@0 |
| 100 | 100 | QPSK | 100+100 | QPSK | 130 | P_100@0 | S_30@0 |
| 100 | 100 | QPSK | 100+100 | QPSK | 200 | P_100@0 | S_100@0 |
| Note 1: CA Configuration Test CC Combination settings are checked separately for each CA Configuration, which applicable aggregated channel bandwidths are specified in Table 5.4.2A.1-1. | | | | | | | |
| Note 2: Depending on CA Configuration only the appropriate Uplink RB allocation value according to table 7.3A.3-1 is tested per CA Configuration Test CC combination. | | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure [TBD].
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1
4. The UL and DL Reference Measurement channels are set according to Table 7.6.2A.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in [State 3A-RF] according to TS 36.508 [7] clause [5.2A.2]. Message contents are defined in clause 7.6.2A.4.3.

### 7.6.2A.4.2 Test Procedure

FFS

### 7.6.2A.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause [FFS].

## 7.6.2A.5 Test Requirement

Except for the spurious response frequencies recorded at the final step of test procedure, the throughput measurement of each carrier derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.6.2A.5-1 and 7.6.2A.5-2.

For Table 7.6.2A.5-2 in frequency range 1, 2 and 3, up to $\max\left(24,\ 6\cdot\left\lceil N_{RB,agg}/6\right\rceil\right)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size, where

$N_{RB,agg}$ is the number of aggregated resource blocks in the downlink transmission bandwidth configuration. For these exceptions the requirements of clause 7.7A Spurious response for CA are applicable.

For Table 7.6.2A.5-2 in frequency range 4, up to $\max\left(8, \left\lceil (N_{RB,agg} + 2 \cdot L_{CRBs})/8 \right\rceil\right)$ exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size, where $N_{RB,agg}$ is the number of aggregated resource blocks in the downlink transmission bandwidth configurations and $L_{CRBs}$ is the number of resource blocks allocated in the uplink. For these exceptions the requirements of clause 7.7A Spurious response for CA are applicable.

**Table 7.6.2A.5-1: Out-of-band blocking parameters**

| Rx Parameter | Units | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated Transmission Bandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 9 | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in TS 36.101 [2] Table 7.3.1A-1, with $P_{CMAX_L}$ as defined in clause 6.2.5A.<br>Note 2: Reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2. | | | | | | |

**Table 7.6.2A.5-2: Out of band blocking**

| CA operating band | Parameter | Units | Frequency | | |
|---|---|---|---|---|---|
| | | | range 1 | range 2 | range 3 |
| | $P_{Interferer}$ | dBm | -44 | -30 | -15 |
| CA_1C, CA_40C | $F_{Interferer}$ (CW) | MHz | $F_{DL_low}$ -15 to $F_{DL_low}$ -60 | $F_{DL_low}$ -60 to $F_{DL_low}$ -85 | $F_{DL_low}$ -85 to 1 MHz |
| | | | $F_{DL_high}$ +15 to $F_{DL_high}$ + 60 | $F_{DL_high}$ +60 to $F_{DL_high}$ +85 | $F_{DL_high}$ +85 to +12750 MHz |

For the UE that supports inter-band CA, the out-of-bank blocking requirements are FFS.

## 7.6.3 Narrow band blocking

### 7.6.3.1 Test Purpose

Verifies a receiver's ability to receive an E-UTRA signal at its assigned channel frequency in the presence of an unwanted narrow band CW interferer at a frequency, which is less than the nominal channel spacing.

The lack of narrow-band blocking ability will decrease the coverage area when other e-NodeB transmitters exist (except in the adjacent channels and spurious response).

### 7.6.3.2 Test Applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 7.6.3.3 Minimum Conformance Requirements

The relative throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.6.3.3-1.

**Table 7.6.3.3-1: Narrow-band blocking**

| Parameter | Unit | Channel Bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| $P_w$ | dBm | $P_{REFSENS}$ + channel-bandwidth specific value below | | | | | |
| | | 22 | 18 | 16 | 13 | 14 | 16 |
| $P_{uw}$ (CW) | dBm | -55 | -55 | -55 | -55 | -55 | -55 |
| $F_{uw}$ (offset for $\Delta f$ = 15 kHz) | MHz | 0.9075 | 1.7025 | 2.7075 | 5.2125 | 7.7025 | 10.2075 |
| $F_{uw}$ (offset for $\Delta f$ = 7.5 kHz) | MHz | | | | | | |
| Note 1: The transmitter shall be set a 4 dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.6.3.

## 7.6.3.4 Test Description

### 7.6.3.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.6.3.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3 respectively. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.6.3.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 6 | 6 | QPSK | 6 | 6 |
| 3MHz | QPSK | 15 | 15 | QPSK | 15 | 15 |
| 5MHz | QPSK | 25 | 25 | QPSK | 25 | 25 |
| 5MHz | QPSK | 25 | N/A | QPSK | 20 | N/A |
| 5MHz | QPSK | 25 | N/A | QPSK | 15 | N/A |
| 10MHz | QPSK | 50 | 50 | QPSK | 50 | 50 |
| 10MHz | QPSK | 50 | N/A | QPSK | 25 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 20 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 15 | N/A |
| 15MHz | QPSK | 75 | 75 | QPSK | 75 | 75 |
| 15MHz | QPSK | 75 | N/A | QPSK | 50 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 25 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 20 | N/A |
| 20MHz | QPSK | 100 | 100 | QPSK | 100 | 100 |
| 20MHz | QPSK | 100 | N/A | QPSK | 75 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 50 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 25 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 20 | N/A |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1.<br>Note 2. Depending on E-UTRA band, only the appropriate Uplink RB allocation value according to table 7.3.3-2 is tested per Test Channel Bandwidth.<br>Note 3: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used. | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.5.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1
4. The UL and DL Reference Measurement channels are set according to Table 7.6.3.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.6.3.4.3.

### 7.6.3.4.2 Test Procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.6.3.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.6.3.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.
3. Set the parameters of the CW signal generator for an interfering signal below the wanted signal according to Table 7.6.3.5-1.
4. Set the downlink signal level according to the table 7.6.3.5-1. Send uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the

target level in table 7.6.3.5-1 for carrier frequency f ≤ 3.0GHz or within +0, -4.0 dB of the target level for carrier frequency 3.0GHz < f ≤ 4.2GHz, for at least the duration of the throughput measurement.

5. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

6. Repeat steps from 3 to 5, using an interfering signal above the wanted signal at step 3.

#### 7.6.3.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception.

**Table 7.6.3.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

### 7.6.3.5 Test Requirement

The throughput measurement derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.6.3.5-1.

**Table 7.6.3.5-1: Narrow-band blocking**

| **Parameter** | **Unit** | **Channel Bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| $P_w$ | dBm | $P_{REFSENS}$ + channel-bandwidth specific value below | | | | | |
| | | 22 | 18 | 16 | 13 | 14 | 16 |
| $P_{uw}$ (CW) | dBm | -55 | -55 | -55 | -55 | -55 | -55 |
| $F_{uw}$ (offset for $\Delta f$ = 15 kHz) | MHz | 0.9075 | 1.7025 | 2.7075 | 5.2125 | 7.7025 | 10.2075 |
| $F_{uw}$ (offset for $\Delta f$ = 7.5 kHz) | MHz | | | | | | |
| Note 1: The transmitter shall be set a 4 dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5.<br>Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

## 7.6.3A Narrow band blocking for CA

Editor's notes:

The following items are missing or incomplete:

- In minimum requirements, reference to UL configuration setting in Reference Sensitivity is missing

- References need to be updated

- Initial conditions test setup for SCC is FFS

- Test procedure
- Message Contents
- Test tolerances
- Connection diagram
- Inter-band test requirements handling and procedure is FFS

## 7.6.3A.1 Test Purpose

Verifies a receiver's ability to receive an E-UTRA signal at its assigned CA channel frequencies in the presence of an unwanted narrow band CW interferer at a frequency, which is less than the nominal channel spacing.

The lack of narrow-band blocking ability will decrease the coverage area when other e-NodeB transmitters exist (except in the adjacent channels and spurious response).

## 7.6.3A.2 Test Applicability

This test case applies to all types of E-UTRA UE release 10 and forward that support CA.

## 7.6.3A.3 Minimum Conformance Requirements

For intra-band contiguous CA (bandwidth Class C) the downlink Secondary CC shall be configured at nominal channel spacing to the Primary CC with the Primary CC configured closest to the uplink band. Downlink Primary CC and Secondary CC are both activated. The uplink output power shall be set as specified in Table 7.6.3A.1-1 with the uplink configuration set according to Table [7.3.3A-1] for the applicable CA Band. For UE(s) supporting one uplink, the uplink configuration of the Primary CC shall be in accordance with Table 7.3.3-2.

The UE shall fulfil the minimum requirement in presence of an interfering signal specified in Table 7.6.3.1A-1 being on either side of the aggregated signal. The throughput of each carrier shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.6.1A.3-1 and 7.6.1A.3-2.

**Table 7.6.3A.3-1: Narrow-band blocking**

| Parameter | Unit | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated TransmissionBandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 16 | | | |
| $P_{uw}$ (CW) | dBm | | -55 | | | |
| $F_{uw}$ (offset for $\Delta f$ = 15 kHz) | MHz | | $-F_{offset} - 0.2$ / $+F_{offset} + 0.2$ | | | |
| $F_{uw}$ (offset for $\Delta f$ = 7.5 kHz) | MHz | | | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in TS 36.101 [2] Table 7.3.1A-1, with $P_{CMAX_L}$ as defined in clause 6.2.5A. | | | | | | |
| Note 2: Reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | |
| Note 3: The $F_{interferer}$ (offset) is relative to the center frequency of the adjacent CC being tested and shall be further adjusted to $\lfloor F_{interferer}/0.015 + 0.5 \rfloor 0.015 + 0.0075$ MHz to be offset from the sub-carrier raster. | | | | | | |

For the UE that supports inter-band CA with a single uplink in one band the narrow-band blocking requirements are defined with the single uplink active on the band other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.6.3.3 for each component carrier while both downlink carriers are active.

The normative reference for this requirement is TS 36.101 [2] clause 7.6.3.1A.

## 7.6.3A.4 Test Description

### 7.6.3A.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and CC combinations based on E-UTRA CA configurations specified in table 5.4.2A.1-1. All of these configurations shall be tested with applicable test parameters for each CA Configuration, and are shown in table 7.6.3A.4.1-1. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.6.3A.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | | |
|---|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | | | NC | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause [4.3.1] for different CA bandwidth classes. | | | C: Mid range | | | | |
| Test CC Combination setting ($N_{RB_agg}$) as specified in subclause 5.4.2A.1 for the CA Configuration | | | Lowest $N_{RB_agg}$<br>Highest $N_{RB_agg}$ | | | | |
| **Test Parameters for CA Configurations** | | | | | | | |
| **CA Configuration / $N_{RB_agg}$** | | **DL Allocation** | | **UL Allocation** | | | |
| **PCC $N_{RB}$** | **SCC $N_{RB}$** | **CC MOD** | **PCC & SCC RB allocation** | **CC MOD** | **$N_{RB_alloc}$** | **PCC & SCC RB allocations ($L_{CRB}$ @ $RB_{start}$)** | |
| 75 | 75 | QPSK | 75+75 | QPSK | 130 | P_75@0 | S_55@0 |
| 75 | 75 | QPSK | 75+75 | QPSK | 150 | P_75@0 | S_75@0 |
| 100 | 50 | QPSK | 100+50 | QPSK | 150 | P_100@0 | S_50@0 |
| 100 | 100 | QPSK | 100+100 | QPSK | 130 | P_100@0 | S_30@0 |
| 100 | 100 | QPSK | 100+100 | QPSK | 200 | P_100@0 | S_100@0 |
| Note 1: CA Configuration Test CC Combination settings are checked separately for each CA Configuration, which applicable aggregated channel bandwidths are specified in Table 5.4.2A.1-1.<br>Note 2: Depending on CA Configuration only the appropriate Uplink RB allocation value according to table 7.3A.3-1 is tested per CA Configuration Test CC combination. | | | | | | | |

1. Connect the SS to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure [TBD].
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1
4. The UL and DL Reference Measurement channels are set according to Table 7.6.3A.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in [State 3A-RF] according to TS 36.508 [7] clause [5.2A.2]. Message contents are defined in clause 7.6.3A.4.3.

### 7.6.3A.4.2 Test Procedure

FFS

#### 7.6.3A.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause [FFS].

### 7.6.3A.5 Test Requirement

The throughput measurement of each carrier derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.6.3A.5-1.

**Table 7.6.3A.5-1: Narrow-band blocking**

| Parameter | Unit | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated TransmissionBandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 16 | | | |
| $P_{uw}$ (CW) | dBm | | -55 | | | |
| $F_{uw}$ (offset for $\Delta f$ = 15 kHz) | MHz | | - $F_{offset}$ – 0.2 / + $F_{offset}$ + 0.2 | | | |
| $F_{uw}$ (offset for $\Delta f$ = 7.5 kHz) | MHz | | | | | |

Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in TS 36.101 [2] Table 7.3.1A-1, with $P_{CMAX_L}$ as defined in clause 6.2.5A.

Note 2: Reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1.

Note 3: The $F_{interferer}$ (offset) is relative to the center frequency of the adjacent CC being tested and shall be further adjusted to $\lfloor F_{interferer} / 0.015 + 0.5 \rfloor 0.015 + 0.0075$ MHz to be offset from the sub-carrier raster.

## 7.7 Spurious response

### 7.7.1 Test Purpose

Spurious response verifies the receiver's ability to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency at which a response is obtained i.e. for which the out of band blocking limit as specified in sub-clause 7.6.2 is not met.

The lack of the spurious response ability decreases the coverage area when other unwanted interfering signal exists at any other frequency.

### 7.7.2 Test Applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 7.7.3 Minimum Conformance Requirements

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.7.3-1 and 7.7.3-2.

**Table 7.7.3-1: Spurious response parameters**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 6 | 6 | 6 | 6 | 7 | 9 |
| Note 1:The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5.<br>Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

**Table 7.7.3-2: Spurious Response**

| Parameter | Unit | Level |
|---|---|---|
| $P_{Interferer}$ (CW) | dBm | -44 |
| $F_{Interferer}$ | MHz | Spurious response frequencies |

The normative reference for this requirement is TS 36.101 [2] clause 7.7.

## 7.7.4 Test Description

### 7.7.4.1 Initial Conditions

The initial conditions shall be the same as in clause 7.6.2.4.1 in order to test spurious responses obtained in clause 7.6.2 under the same conditions.

### 7.7.4.2 Test Procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.6.2.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.

2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.6.2.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.

3. Set the parameters of the CW signal generator for an interfering signal according to Table 7.7.5-2. The spurious frequencies are taken from records in the final step of test procedures in clause 7.6.2.4.2.

4. Set the downlink signal level according to the table 7.7.5-1. Send uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the target level in table 7.7.5-1 for carrier frequency $f \leq 3.0$GHz or within +0, -4.0 dB of the target level for carrier frequency 3.0GHz $< f \leq$ 4.2GHz, for at least the duration of the throughput measurement.

5. For the spurious frequency, measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

### 7.7.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception.

**Table 7.7.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

## 7.7.5 Test Requirement

The throughput measurement derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Tables 7.7.5-1 and 7.7.5-2.

**Table 7.7.5-1: Spurious response parameters**

| **Rx Parameter** | **Units** | **Channel bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 6 | 6 | 6 | 6 | 7 | 9 |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5. | | | | | | | |
| Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | | |

**Table 7.7.5-2: Spurious Response**

| **Parameter** | **Unit** | **Level** |
|---|---|---|
| $P_{Interferer}$ (CW) | dBm | -44 |
| $F_{Interferer}$ | MHz | Spurious response frequencies |

# 7.7A Spurious response for CA

Editor's notes:

The following items are missing or incomplete:

- References need to be updated

- Test procedure

- Message Contents

- Test tolerances

- Connection diagram- Inter-band test requirements handling and procedure is FFS

## 7.7A.1 Test Purpose

Spurious response for CA verifies the receiver's ability to receive a wanted aggregated signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency at which a response is obtained i.e. for which the out of band blocking limit as specified in sub-clause 7.6.2A is not met.

The lack of the spurious response ability decreases the coverage area when other unwanted interfering signal exists at any other frequency.

## 7.7A.2 Test Applicability

This test applies to all types of E-UTRA UE release 10 and forward that support CA.

## 7.7A.3 Minimum Conformance Requirements

For intra-band contiguous CA the throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.7A.3-1 and 7.7A.3-2. For UE(s) supporting one uplink, the uplink configuration of the Primary CC shall be in accordance with Table 7.3.1-2.

**Table 7.7A.3-1: Spurious response parameters**

| Rx Parameter | Units | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated Transmission Bandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 9 | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in TS 36.101 [2] Table 7.3.1A-1, with $P_{CMAX_L}$ as defined in clause 6.2.5A.<br>Note 2: Reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | |

**Table 7.7A.3-2: Spurious Response**

| Parameter | Unit | Level |
|---|---|---|
| $P_{Interferer}$ (CW) | dBm | -44 |
| $F_{Interferer}$ | MHz | Spurious response frequencies |

For the UE that supports inter-band CA with a single uplink in one band the spurious response requirements are defined with the single uplink active on the band other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.7 for each component carrier while both downlink carriers are active.

The normative reference for this requirement is TS 36.101 [2] clause 7.7.1A.

## 7.7A.4 Test Description

### 7.7A.4.1 Initial Conditions

The initial conditions shall be the same as in clause 7.6.2A.4.1 in order to test spurious responses obtained in clause 7.6.2A under the same conditions.

### 7.7A.4.2 Test Procedure

FFS

### 7.7A.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause [FFS].

## 7.7A.5 Test Requirement

The throughput measurement of each carrier derived in test procedure shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.7A.5-1 and 7.7A.5-2.

**Table 7.7A.5-1: Spurious response parameters**

| Rx Parameter | Units | CA Bandwidth Class | | | | |
|---|---|---|---|---|---|---|
| | | B | C | D | E | F |
| Power per CC in Aggregated Transmission Bandwidth Configuration | dBm | REFSENS + CA Bandwidth Class specific value below | | | | |
| | | | 9 | | | |
| Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in TS 36.101 [2] Table 7.3.1A-1, with $P_{CMAX_L}$ as defined in clause 6.2.5A.<br>Note 2: Reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1. | | | | | | |

**Table 7.7A.5-2: Spurious Response**

| Parameter | Unit | Level |
|---|---|---|
| $P_{Interferer}$ (CW) | dBm | -44 |
| $F_{Interferer}$ | MHz | Spurious response frequencies |

# 7.8 Intermodulation characteristics

## 7.8.1 Wide band Intermodulation

### 7.8.1.1 Test purpose

Intermodulation response tests the UE's ability to receive data with a given average throughput for a specified reference measurement channel, in the presence of two or more interfering signals which have a specific frequency relationship to the wanted signal, under conditions of ideal propagation and no added noise.

A UE unable to meet the throughput requirement under these conditions will decrease the coverage area when two or more interfering signals exist which have a specific frequency relationship to the wanted signal.

### 7.8.1.2 Test applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 7.8.1.3 Minimum conformance requirements

Intermodulation response rejection is a measure of the capability of the receiver to receiver a wanted signal on its assigned channel frequency in the presence of two or more interfering signals which have a specific frequency relationship to the wanted signal.

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.2.3 and A.3.2 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.8.1.3-1 for the specified wanted signal mean power in the presence of two interfering signals.

**Table 7.8.1.3-1: Wide band intermodulation**

| Rx Parameter | Units | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 12 | 8 | 6 | 6 | 7 | 9 |
| $P_{Interferer\ 1}$ (CW) | dBm | -46 | | | | | |
| $P_{Interferer\ 2}$ (Modulated) | dBm | -46 | | | | | |
| $BW_{Interferer\ 2}$ | | 1.4 | 3 | 5 | | | |
| $F_{Interferer\ 1}$ (Offset) | MHz | -BW/2 –2.1 / +BW/2+ 2.1 | -BW/2 –4.5 / +BW/2 + 4.5 | -BW/2 – 7.5 / +BW/2 + 7.5 | | | |
| $F_{Interferer\ 2}$ (Offset) | MHz | $2*F_{Interferer\ 1}$ | | | | | |

Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ at the minimum uplink configuration specified in Table 7.3.3-2 with $P_{CMAX_L}$ as defined in clause 6.2.5.

Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1.

Note 3: The modulated interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 with set-up according to Annex C.3.1.The interfering modulated signal is 5MHz E-UTRA signal as described in Annex D for channel bandwidth ≥5MHz

The normative reference for this requirement is TS 36.101 [2] clause 7.8.1 and TS 36.101 [2] Annexes A and D.

[FFS: Although it is not explicitly stated in TS 36.101 [2] whether the modulated interferer defined in 36.101 Annex D applies to wanted channel bandwidths of less than 5MHz, this test specification has assumed that the modulated interferer definition applies to all channel bandwidths. The content of TS 36.101 [2] Annex D.2 has been copied into Annex FFS of the present document]

## 7.8.1.4 Test description

### 7.8.1.4.1 Initial condition

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.8.4.1-1. The details of the uplink and downlink reference measurement channels (RMCs) are specified in Annexes A.2 and A.3 respectively. The details of the OCNG patterns used are specified in Annex A.5. Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

**Table 7.8.1.4.1-1: Test Configuration Table**

| **Initial Conditions** | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Mid range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Lowest, 5MHz, Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 6 | 6 | QPSK | 6 | 6 |
| 3MHz | QPSK | 15 | 15 | QPSK | 15 | 15 |
| 5MHz | QPSK | 25 | 25 | QPSK | 25 | 25 |
| 5MHz | QPSK | 25 | N/A | QPSK | 20 | N/A |
| 5MHz | QPSK | 25 | N/A | QPSK | 15 | N/A |
| 10MHz | QPSK | 50 | 50 | QPSK | 50 | 50 |
| 10MHz | QPSK | 50 | N/A | QPSK | 25 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 20 | N/A |
| 10MHz | QPSK | 50 | N/A | QPSK | 15 | N/A |
| 15MHz | QPSK | 75 | 75 | QPSK | 75 | 75 |
| 15MHz | QPSK | 75 | N/A | QPSK | 50 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 25 | N/A |
| 15MHz | QPSK | 75 | N/A | QPSK | 20 | N/A |
| 20MHz | QPSK | 100 | 100 | QPSK | 100 | 100 |
| 20MHz | QPSK | 100 | N/A | QPSK | 75 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 50 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 25 | N/A |
| 20MHz | QPSK | 100 | N/A | QPSK | 20 | N/A |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band, which applicable channel bandwidths are specified in Table 5.4.2.1-1. | | | | | | |
| Note 2. Depending on E-UTRA band, only the appropriate Uplink RB allocation value according to table 7.3.3-2 is tested per Test Channel Bandwidth. | | | | | | |
| Note 3: For the DL signal one sided dynamic OCNG Pattern OP.1 FDD/TDD is used. | | | | | | |

1. Connect the SS and interfering sources to the UE antenna connectors as shown in TS 36.508 [7] Figure A.6.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1, and uplink signals according to Annex H.1 and H.3.1.
4. The UL and DL Reference Measurement channels are set according to Table 7.8.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.8.1.4.3.

### 7.8.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Table 7.8.1.4.1-1. The SS sends downlink MAC padding bits on the DL RMC.
2. SS sends uplink scheduling information for each UL HARQ process via PDCCH DCI format 0 for C_RNTI to schedule the UL RMC according to Table 7.8.1.4.1-1. Since the UE has no payload data to send, the UE transmits uplink MAC padding bits on the UL RMC.
3. Set the Downlink signal level to the value as defined in Table 7.8.1.5-1. Send Uplink power control commands to the UE (less or equal to 1dB step size should be used), to ensure that the UE output power is within +0, - 3.4 dB of the target level in Table 7.8.1.5-1 for carrier frequency $f \leq 3.0$GHz or within +0, -4.0 dB of the target level for carrier frequency 3.0GHz $< f \leq$ 4.2GHz, for at least the duration of the Throughput measurement.

4. Set the Interfering signal levels to the values as defined in Table 7.8.1.5-1, using a modulated interferer bandwidth as defined in Annex D of the present document.

5. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.2.

#### 7.8.1.4.3 Message contents

Message contents are according to TS 36.508 [7] subclause 4.6 with the following exception.

**Table 7.8.1.4.3-1: UplinkPowerControlDedicated**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| UplinkPowerControlDedicated-DEFAULT ::= SEQUENCE { | | | |
| p0-UePUSCH | 0 | | |
| deltaMCS-Enabled | en0 | | |
| accumulationEnabled | TRUE | | |
| p0-uePUCCH | 0 | | |
| pSRS-Offset | 3 (-6 dB) | | |
| filterCoefficient | fc8 | larger filter length is used to reduce the RSRP measurement variation | |
| } | | | |

### 7.8.1.5 Test requirements

The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channels as specified in Annex A.3.2 with parameters specified in Table 7.8.1.5-1 for the specified wanted signal mean power in the presence of two interfering signals.

**Table 7.8.1.5-1: Test parameters for Wide band intermodulation**

| **Rx Parameter** | **Units** | **Channel bandwidth** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| Power in Transmission Bandwidth Configuration | dBm | REFSENS + channel bandwidth specific value below | | | | | |
| | | 12 | 8 | 6 | 6 | 7 | 9 |
| $P_{Interferer\ 1}$ (CW) | dBm | -46 | | | | | |
| $P_{Interferer\ 2}$ (Modulated) | dBm | -46 | | | | | |
| $BW_{Interferer\ 2}$ | | 1.4 | 3 | 5 | | | |
| $F_{Interferer\ 1}$ (Offset) | MHz | -BW/2 –2.1 / +BW/2+ 2.1 | -BW/2 –4.5 / +BW/2 + 4.5 | -BW/2 – 7.5 / +BW/2 + 7.5 | | | |
| $F_{Interferer\ 2}$ (Offset) | MHz | $2*F_{Interferer\ 1}$ | | | | | |

Note 1: The transmitter shall be set to 4dB below $P_{CMAX_L}$ with $P_{CMAX_L}$ as defined in clause 6.2.5.
Note 2: The reference measurement channel is specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1.
Note 3: The modulated interferer consists of the Reference measurement channel specified in Annex A.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1.The interfering modulated signal is 5MHz E-UTRA signal as described in Annex D for channel bandwidth ≥5MHz

### 7.8.2 Void

## 7.9 Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector.

### 7.9.1 Test Purpose

Test verifies the UE's spurious emissions meet the requirements described in clause 7.9.3.

Excess spurious emissions increase the interference to other systems.

### 7.9.2 Test Applicability

This test applies to all types of E-UTRA UE release 8 and forward.

### 7.9.3 Minimum Conformance Requirements

The power of any narrow band CW spurious emission shall not exceed the maximum level specified in Table 7.9.3-1

**Table 7.9.3-1: General receiver spurious emission requirements**

| Frequency Band | Measurement Bandwidth | Maximum level | Note |
|---|---|---|---|
| 30MHz ≤ f < 1GHz | 100 kHz | -57 dBm | |
| 1GHz ≤ f ≤ 12.75 GHz | 1 MHz | -47 dBm | |
| 12.75 GHz ≤ f ≤ 5$^{th}$ harmonic of the upper frequency edge of the DL operating band in GHz | 1 MHz | -47 dBm | Note 1 |
| Note 1: Applies only for Band 22, Band 42 and Band 43 | | | |

The normative reference for this requirement is TS 36.101 [2] clause 7.9.

### 7.9.4 Test Description

#### 7.9.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, and channel bandwidths based on E-UTRA bands specified in table 5.4.2.1-1. All of these configurations shall be tested with applicable test parameters for each channel bandwidth, and are shown in table 7.9.4.1-1. The details of the downlink and uplink reference measurement channels (RMCs) are specified in Annexes A.3 and A.2 respectively.

**Table 7.9.4.1-1: Test Configuration Table**

| Initial Conditions | | | | | | |
|---|---|---|---|---|---|---|
| Test Environment as specified in TS 36.508[7] subclause 4.1 | NC | | | | | |
| Test Frequencies as specified in TS36.508 [7] subclause 4.3.1 | Low range, Mid range, High range | | | | | |
| Test Channel Bandwidths as specified in TS 36.508 [7] subclause 4.3.1 | Highest | | | | | |
| **Test Parameters for Channel Bandwidths** | | | | | | |
| | **Downlink Configuration** | | | **Uplink Configuration** | | |
| Ch BW | Mod'n | RB allocation | | Mod'n | RB allocation | |
| | | FDD | TDD | | FDD | TDD |
| 1.4MHz | QPSK | 0 | 0 | QPSK | 0 | 0 |
| 3MHz | QPSK | 0 | 0 | QPSK | 0 | 0 |
| 5MHz | QPSK | 0 | 0 | QPSK | 0 | 0 |
| 10MHz | QPSK | 0 | 0 | QPSK | 0 | 0 |
| 15MHz | QPSK | 0 | 0 | QPSK | 0 | 0 |
| 20MHz | QPSK | 0 | 0 | QPSK | 0 | 0 |
| Note 1: Test Channel Bandwidths are checked separately for each E-UTRA band. The applicable channel bandwidths are specified in Table 7.3.3-2. | | | | | | |

1. Connect a spectrum analyzer (or other suitable test equipment) to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.8.
2. The parameter settings for the cell are set up according to TS 36.508 [7] subclause 4.4.3.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.1.
4. The DL Reference Measurement channels are set according to Table 7.9.4.1-1.
5. Propagation conditions are set according to Annex B.0.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 7.9.4.3.

### 7.9.4.2 Test Procedure

1. Sweep the spectrum analyzer (or equivalent equipment) over a frequency range and measure the average power of spurious emission.
2. Repeat step 1 for all E-UTRA Rx antennas of the UE.

### 7.9.4.3 Message Contents

Message contents are according to TS 36.508 [7] subclause 4.6.

## 7.9.5 Test Requirement

The measured spurious emissions derived in step 1), shall not exceed the maximum level specified in Table 7.9.5-1

**Table 7.9.5-1: General receiver spurious emission requirements**

| Frequency Band | Measurement Bandwidth | Maximum level | Note |
|---|---|---|---|
| 30MHz ≤ f < 1GHz | 100 kHz | -57 dBm | |
| 1GHz ≤ f ≤ 12.75 GHz | 1 MHz | -47 dBm | |
| 12.75 GHz ≤ f ≤ 5$^{th}$ harmonic of the upper frequency edge of the DL operating band in GHz | 1 MHz | -47 dBm | Note 1 |
| Note 1: Applies only for Band 22, Band 42 and Band 43 | | | |

# 7.10 Void

# 7.10A Receiver image for CA

TS 36.101 [2] clause 7.10.1A specifies minimum requirements for receiver image for CA but recommends that these requirements do not need to be tested.

# 8 Performance Requirement

## 8.1 General

The performance requirements for the physical channels specified in TS 36.211 [8] clause 6 (for downlink physical channels) shall be as defined in the respective sections below.

The requirements for the UE in this clause are specified for the downlink reference measurement channels specified in Annex A, the propagation conditions specified in Annex B and the downlink physical channels specified in Annex C.

Unelss otherwise stated the throughput measurements in clause 8 shall be performed according to the general rules for statistical testing in Annex G clause G.3.

The requirement for a UE that support E-UTRA in downlink shall be tested according to the declared UE PDSCH category and CA capabilities.

The fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated. The levels of the test signal applied to each of the antenna connectors shall be as defined in the respective test cases.

The UE performance in this section is considered to be operating band independent. Therefore, the required performance in the respective test cases can be verified in one of the operating bands supported by the UE under test. All the test points supported by the bands of the multiband UE (based on channel bandwidth, DL and UL configuration) need to be tested.

### 8.1.1 Dual-antenna receiver capability

The performance requirements are based on UE(s) that utilize a dual-antenna receiver.

For all test cases, the SNR is defined as:

$$SNR = \frac{\hat{E}_s^{(1)} + \hat{E}_s^{(2)}}{N_{oc}^{(1)} + N_{oc}^{(2)}},$$

where the superscript indicates the receiver antenna connector. The SNR requirement applies for the UE categories given for each test.

The applicability of the requirements with respect to CA capabilities is given as in Table 8.1.1-1. In case the CA capability is omitted, the requirement is applicable to a UE regardless of its CA capability.

**Table 8.1.1-1: Applicability of the requirement with respect to the CA capability**

| CA Capability | CA Capability Description |
|---|---|
| CL_X | The requirement is applicable to a UE that indicates a CA bandwidth class X on at least one E-UTRA band. |
| CL_X-Y | The requirement is applicable to a UE that indicates CA bandwidth classes X and Y on at least one E-UTRA band combination. |
| non-CA | The requirement is applicable to a UE that indicates the CA bandwidth class A (one component carrier) on all supported E-UTRA bands. |
| Note: The CA bandwidth classes are defined in Table 5.4.2A-1 | |

For test cases with more than one component carrier, "Fraction of Maximum Throughput" in the performance requirement refers to the ratio of the sum of throughput values of all component carriers to the sum of the nominal maximum throughput values of all component carriers. The normative reference for this requirement is TS 36.101 [2] clause 8.1.1.

#### 8.1.1.1 Simultaneous unicast and MBMS operations

#### 8.1.1.2 Dual-antenna receiver capability in idle mode

## 8.2 Demodulation of PDSCH (Cell-Specific Reference Symbols)

### 8.2.1 FDD (Fixed Reference Channel)

The parameters specified in Table 8.2.1-1 are valid for all FDD tests unless otherwise stated.

**Table 8.2.1-1: Common Test Parameters (FDD)**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| Inter-TTI Distance | | 1 | |
| Number of HARQ processes | Processes | 8 | For FDD, 8 HARQ processes in the DL, as specified in TS 36.213 [10] clause 7. All 8 HARQ processes are used. |
| Scheduling of retransmissions | | | 1. Retransmissions use the same Transport Block Size (TBS) as the initial transmission.<br>2. HARQ processes are scheduled consecutively, independent of the fact, whether retransmissions (for negatively acknowledged HARQ processes) or new transmissions (for positively acknowledged HARQ processes) occur. |
| Maximum number of HARQ transmission | | 4 | It is always 4 for FDD, as specified in TS 36.213 [10] clause 8 |
| Redundancy version coding sequence | | {0,1,2,3} for QPSK and 16QAM<br>{0,0,1,2} for 64QAM | |
| Number of OFDM symbols for PDCCH | OFDM symbols | 4 for 1.4 MHz bandwidth, 3 for 3 MHz and 5 MHz bandwidths, 2 for 10 MHz, 15 MHz and 20 MHz bandwidths | The PCFICH carries information about the number of OFDM symbols used for transmission of PDCCHs in a subframe, as specified in TS 36.211 [8] clause 6.7 |
| Cyclic Prefix | | Normal | CP consist of the following physical resource blocks (RBs) parameters: 12 consecutive subcarriers at a 15 kHz spacing and 7 OFDM symbols, as specified in TS 36.211 [8] clause 6.2.3 |
| Cell ID | | 0 | The Cell ID is uniquely defined by a number in the range of 0 to 503, representing the physical-layer cell identity, as specified in TS 36.211 [8] clause 6.11. |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.

#### 8.2.1.1 FDD PDSCH Single Antenna Port Performance (Cell-Specific Reference Symbols)

##### 8.2.1.1.1 FDD PDSCH Single Antenna Port Performance

###### 8.2.1.1.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on a single-antenna port with different channel models and MCS.

#### 8.2.1.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

#### 8.2.1.1.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.1, with the addition of the relevant parameters in Tables 8.2.1-1, 8.2.1.1.1.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.1.1.3-2 for the specified SNR. For QPSK and 64QAM performance the bandwidths specified in Table 5.4.2.1-1 are verified.

**Table 8.2.1.1.1.3-1: Test Parameters for Testing**

| Parameter | | Unit | Test 1- 5 | Test 6- 8 | Test 9- 15 | Test 16-18 |
|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | 0 | 0 | 0 |
| | $\rho_B$ | dB | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 | -98 | -98 |
| Symbols for unused PRBs | | | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) |
| Modulation | | | QPSK | 16QAM | 64QAM | 16QAM |

Note 1: $P_B = 0$

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

**Table 8.2.1.1.1.3-2: Minimum performance (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.2 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | -1.0 | 1-5 |
| 2 | 10 MHz | R.2 FDD | OP.1 FDD | ETU70 | 1x2 Low | 70 | -0.4 | 1-5 |
| 3 | 10 MHz | R.2 FDD | OP.1 FDD | ETU300 | 1x2 Low | 70 | 0.0 | 1-5 |
| 4 | 10 MHz | R.2 FDD | OP.1 FDD | HST | 1x2 Low | 70 | -2.4 | 1-5 |
| 5 | 1.4 MHz | R.4 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 0.0 | 1-5 |
| 6 | 10 MHz | R.3 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 6.7 | 2-5 |
| 7 | 10 MHz | R.3 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | 1.4 | 2-5 |
| 8 | 10 MHz | R.3 FDD | OP.1 FDD | ETU300 | 1x2 High | 70 | 9.4 | 2-5 |
| 9 | 3 MHz | R.5 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.6 | 1-5 |
| 10 | 5 MHz | R.6 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.4 | 2-5 |
| 11 | 10 MHz | R.7 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.7 | 2-5 |
| 12 | 10 MHz | R.7 FDD | OP.1 FDD | ETU70 | 1x2 Low | 70 | 19.0 | 2-5 |
| 13 | 10 MHz | R.7 FDD | OP.1 FDD | EVA5 | 1x2 High | 70 | 19.1 | 2-5 |
| 14 | 15 MHz | R.8 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.7 | 2-5 |
| 15 | 20 MHz | R.9 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.6 | 3-5 |
| 16 | 3 MHz | R.0 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | 1.9 | 1-5 |
| 17 | 10 MHz | R.1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | 1.9 | 1-5 |
| 18 | 20 MHz | R.1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | 1.9 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.1.

#### 8.2.1.1.1.4 Test description

##### 8.2.1.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested for full allocation: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Frequencies to be tested for 1PRB allocation: Low Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.1.1.3-2 as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Table 8.2.1-1and 8.2.1.1.1.3-1 as appropriate.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.1.1.4.3.

#### 8.2.1.1.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.1.1.3-1 and 8.2.1.1.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 8.2.1.1.1.5-1as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

4. Repeat steps from 1 to 3 for each subtest in Table 8.2.1.1.1.5-1 as appropriate.

#### 8.2.1.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

### 8.2.1.1.1.5 Test requirement

Table 8.2.1.1.1.3-1defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.1 for each throughput test shall meet or exceed the specified value in Table 8.2.1.1.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.1.1.5-1: Test requirement (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.2 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | -0.2 | 1-5 |
| 2 | 10 MHz | R.2 FDD | OP.1 FDD | ETU70 | 1x2 Low | 70 | +0.4 | 1-5 |
| 3 | 10 MHz | R.2 FDD | OP.1 FDD | ETU300 | 1x2 Low | 70 | +0.8 | 1-5 |
| 4 | 10 MHz | R.2 FDD | OP.1 FDD | HST | 1x2 Low | 70 | -1.8 | 1-5 |
| 5 | 1.4 MHz | R.4 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +0.8 | 1-5 |
| 6 | 10 MHz | R.3 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +7.5 | 2-5 |
| 7 | 10 MHz | R.3 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | +2.2 | 2-5 |
| 8 | 10 MHz | R.3 FDD | OP.1 FDD | ETU300 | 1x2 High | 70 | +10.2 | 2-5 |
| 9 | 3 MHz | R.5 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +18.41 | 1-5 |
| 10 | 5 MHz | R.6 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +18.2 | 2-5 |
| 11 | 10 MHz | R.7 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +18.5 | 2-5 |
| 12 | 10 MHz | R.7 FDD | OP.1 FDD | ETU70 | 1x2 Low | 70 | +19.8 | 2-5 |
| 13 | 10 MHz | R.7 FDD | OP.1 FDD | EVA5 | 1x2 High | 70 | +19.9 | 2-5 |
| 14 | 15 MHz | R.8 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +18.5 | 2-5 |
| 15 | 20 MHz | R.9 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +18.4 | 3-5 |
| 16 | 3 MHz | R.0 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | +2.7 | 1-5 |
| 17 | 10 MHz | R.1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | +2.7 | 1-5 |
| 18 | 20 MHz | R.1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | +2.7 | 1-5 |

## 8.2.1.1.1_1 FDD PDSCH Single Antenna Port Performance (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Test tolerance is undefined

### 8.2.1.1.1_1.1 Test purpose

Same test purpose as in clause 8.2.1.1.1.1.

### 8.2.1.1.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

### 8.2.1.1.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.1.1.1.3 with the following exceptions:

- Instead of Table 8.2.1.1.1.3-1 → use Table 8.2.1.1.1_1.3-1.

- Instead of Table 8.2.1.1.1.3-2 → use Table 8.2.1.1.1_1.3-2.

**Table 8.2.1.1.1_1.3-1: Test Parameters for Testing**

| Parameter | | Unit | Test 1- 5 | Test 6- 8 | Test 9- 15 | Test 16-18 |
|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | 0 | 0 | 0 |
| | $\rho_B$ | dB | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 | -98 | -98 |
| Symbols for unused PRBs | | | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) |
| Modulation | | | QPSK | 16QAM | 64QAM | 16QAM |
| Note 1: $P_B = 0$<br>Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. | | | | | | |

**Table 8.2.1.1.1_1.3-2: Minimum performance (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | N/A | | | | | | | |
| 2 | N/A | | | | | | | |
| 3 | N/A | | | | | | | |
| 4 | N/A | | | | | | | |
| 5 | N/A | | | | | | | |
| 6 | 5 MHz | R.3-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 6.7 | 1 |
| 7 | 5 MHz | R.3-1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | 1.4 | 1 |
| 8 | 5 MHz | R.3-1 FDD | OP.1 FDD | ETU300 | 1x2 High | 70 | 9.4 | 1 |
| 9 | N/A | | | | | | | |
| 10 | 5 MHz | R.6-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.5 | 1 |
| 11 | 10 MHz | R.7-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.5 | 1 |
| 12 | 10 MHz | R.7-1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 70 | 18.1 | 1 |
| 13 | 10 MHz | R.7-1 FDD | OP.1 FDD | EVA5 | 1x2 High | 70 | 17.8 | 1 |
| 14 | 15 MHz | R.8-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 16.8 | 1 |
| 15 | 20 MHz | R.9-2 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.3 | 2 |
| | 20 MHz | R.9-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 16.7 | 1 |
| 16 | N/A | | | | | | | |
| 17 | N/A | | | | | | | |
| 18 | N/A | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.1.

### 8.2.1.1.1_1.4 Test description

Same test description as in clause 8.2.1.1.1.4 with the following exceptions:

- Instead of Table 8.2.1.1.1.3-1 → use Table 8.2.1.1.1_1.3-1.

- Instead of Table 8.2.1.1.1.3-2 → use Table 8.2.1.1.1_1.3-2.
- Instead of Table 8.2.1.1.1.5-1 → use Table 8.2.1.1.1_1.5-1.

### 8.2.1.1.1_1.5 Test requirement

Same test requirements as in clause 8.2.1.1.1.5 with the following exceptions:

- Instead of Table 8.2.1.1.1.3-1 → use Table 8.2.1.1.1_1.3-1.
- Instead of Table 8.2.1.1.1.5-1 → use Table 8.2.1.1.1_1.5-1.

**Table 8.2.1.1.1_1.5-1: Test requirement (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | N/A | | | | | | | |
| 2 | N/A | | | | | | | |
| 3 | N/A | | | | | | | |
| 4 | N/A | | | | | | | |
| 5 | N/A | | | | | | | |
| 6 | 5 MHz | R.3-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | +7.5 | 1 |
| 7 | 5 MHz | R.3-1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 30 | +2.2 | 1 |
| 8 | 5 MHz | R.3-1 FDD | OP.1 FDD | ETU300 | 1x2 High | 70 | +10.2 | 1 |
| 9 | N/A | | | | | | | |
| 10 | 5 MHz | R.6-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.5+ TT | 1 |
| 11 | 10 MHz | R.7-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.5+ TT | 1 |
| 12 | 10 MHz | R.7-1 FDD | OP.1 FDD | ETU70 | 1x2 Low | 70 | 18.1+ TT | 1 |
| 13 | 10 MHz | R.7-1 FDD | OP.1 FDD | EVA5 | 1x2 High | 70 | 17.8+ TT | 1 |
| 14 | 15 MHz | R.8-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 16.8+ TT | 1 |
| 15 | 20 MHz | R.9-2 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 17.3+ TT | 2 |
| | 20 MHz | R.9-1 FDD | OP.1 FDD | EVA5 | 1x2 Low | 70 | 16.7+ TT | 1 |
| 16 | N/A | | | | | | | |
| 17 | N/A | | | | | | | |
| 18 | N/A | | | | | | | |

### 8.2.1.1.1_A FDD PDSCH Single Antenna Port Performance ( CA)

Editor's Note: This clause is incomplete. The following aspects are either missing or not yet determined

- Test 2 parameters are FFS.
- Connection diagram is FFS.
- Message Contents is FFS.
- Test tolerance are FFS
- Test frequencies are FFS
- State-3A RF condition in test description is to be confirmed

- Statistical significance related parameters are FFS

### 8.2.1.1.1_A.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on a single-antenna port with different channel models and MCS.

### 8.2.1.1.1_A.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward which support carrier aggregation.

### 8.2.1.1.1_A.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.1, with the addition of the relevant parameters in Tables 8.2.1-1, 8.2.1.1.1_A.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.1.1_A.3-2 for the specified SNR. For QPSK and 64QAM performance the bandwidths specified in Table 5.4.2.1-1 are verified.

**Table 8.2.1.1.1_A.3-1: Test Parameters for Testing**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | FFS |
| | $\rho_B$ | dB | 0 (Note 1) | FFS |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | FFS |
| Symbols for unused PRBs | | | OCNG (Note 2) | FFS |
| Modulation | | | QPSK | FFS |

Note 1: $P_B = 0$

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

**Table 8.2.1.1.1_A.3-2: Minimum performance (FRC)**

| Test num. | Band-width | Reference channel | OCNG pattern | Propa-gation condi-tion | Correlation matrix and antenna config. | Reference value | | UE cate-gory | CA capa-bility |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) | | |
| 1 | 2x10 MHz | R.2 FDD | FFS | EVA5 | 1x2 Low | 70 | -1.1 | 3-8 | CL_A-A |
| 2 | 2x20 MHz | R.42 FDD | OP.1 FDD (Note 1) | EVA5 | 1x2 Low | 70 | -1.3 | 5-8 | CL_A-A, CL_C |

Note 1: For CA capable UE, the OCNG pattern applies for each CC.

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.1.

### 8.2.1.1.1_A.4 Test description

#### 8.2.1.1.1_A.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested for full allocation: Mid Range, as defined in TS 36.508 [7][FFS].

Frequencies to be tested for 1PRB allocation: Low Range, as defined in TS 36.508 [7] [FFS]

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.1.1_A.3-2 as defined in TS 36.508 [7] clause [FFS].

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure FFS
2. The parameter settings for the cell are set up according to Table 8.2.1-1and 8.2.1.1.1_A.3-1 as appropriate.
3. Downlink signals for PCC are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2. Configuration of the Downlink signals for SCC are also defined in Annex C0, C1 and C3.2
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in[State 3A-RF] according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.1.1_A.4.3.

#### 8.2.1.1.1_A.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.1.1_A.3-1 and 8.2.1.1.1_A.3-2 on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR on each of the component carriers according to Tables 8.2.1.1.1_A.5-1as appropriate.
3. Measure the average throughput per component carrier for a duration sufficient to achieve statistical significance according to Annex G clause G.3 on both PCC and SCC. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest on both component carriers and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.
4. Repeat steps from 1 to 3 for each subtest in Table 8.2.1.1.1_A.5-1 depending on UE CA capability as defined in Table 8.1.1-1 and as appropriate.

#### 8.2.1.1.1_A.4.3 Message contents

Message contents are according to TS 36.508 [7] clause. [FFS]

### 8.2.1.1.1_A.5 Test requirement

Table 8.2.1.1.1_A.3-1defines the primary level settings.

The fraction of maximum throughput percentage measured per component carrier for the downlink reference measurement channels specified in Annex A clause A.3.3.1 for each throughput test shall meet or exceed the specified value in Table 8.2.1.1.1_A.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.1.1_A.5-1: Test requirement (FRC)**

| Test num. | Band-width | Reference channel | OCNG pattern | Propa-gation condi-tion | Correlation matrix and antenna config. | Reference value | | UE cate-gory | CA capa-bility |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) | | |
| 1 | 2x10 MHz | R.2 FDD | FFS | EVA5 | 1x2 Low | 70 | -1.1 + TT | 3-8 | CL_A-A |
| 2 | 2x20 MHz | R.42 FDD | OP.1 FDD (Note 1) | EVA5 | 1x2 Low | 70 | -1.3 + TT | 5-8 | CL_A-A, CL_C |
| Note 1: For CA capable UE, the OCNG pattern applies for each CC. | | | | | | | | | |

## 8.2.1.1.2 FDD PDSCH Single Antenna Port Performance with 1 PRB in presence of MBSFN

### 8.2.1.1.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on a single-antenna port with different channel models and MCS for 1 PRB allocation in presence of MBSFN.

### 8.2.1.1.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 8.2.1.1.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.1, with the addition of the relevant parameters in Tables 8.2.1-1, 8.2.1.1.2.3-1, and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.1.1.2.3-2, for the specified SNR.

**Table 8.2.1.1.2.3-1: Test Parameters for Testing 1 PRB allocation**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 |
| | $\rho_B$ | dB | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Symbols for MBSFN portion of MBSFN subframes (Note 2) | | | OCNG (Note 3) |
| Note 1: $P_B = 0$ | | | |
| Note 2: The MBSFN portion of an MBSFN subframe comprises the whole MBSFN subframe except the first two symbols in the first slot. | | | |
| Note 3: The MBSFN portion of the MBSFN subframes shall contain QPSK modulated data. Cell-specific reference signals are not inserted in the MBSFN portion of the MBSFN subframes, QPSK modulated MBSFN data is used instead. | | | |

**Table 8.2.1.1.2.3-2: Minimum performance 1 PRB allocation (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.29 FDD | OP.3 FDD | ETU70 | 1x2 Low | 30 | 2.0 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.1.

### 8.2.1.1.2.4 Test description

#### 8.2.1.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Low Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Tables 8.2.1.1.2.3-2as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Tables 8.2.1-1, 8.2.1.1.2.3-1as appropriate.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.1.2.4.3.

#### 8.2.1.1.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.1.2.3-1 and 8.2.1.1.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.2.1.1.2.5-1as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

#### 8.2.1.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.1.2.4.3-1: *SystemInformationBlockType2*: Additional FDD PDSCH Single Antenna Port Performance for 1 PRB allocation with MBSFN subframes test point 1 requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig ::= SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | 111111 | Subframe 1, 2, 3, 6, 7, 8 is used for MBSFN | FDD |
| } | | | |
| } | | | |

8.2.1.1.2.5 Test requirement

Table 8.2.1.1.2.3-1defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.1 for each throughput test shall meet or exceed the specified value in Tables 8.2.1.1.2.5-1for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.1.2.5-1: Test requirement 1 PRB allocation with MBSFN subframes (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.29 FDD | OP.3 FDD | ETU70 | 1x2 Low | 30 | +2.8 | 1-5 |

## 8.2.1.2 FDD PDSCH Transmit Diversity Performance (Cell-Specific Reference Symbols)

### 8.2.1.2.1 FDD PDSCH Transmit Diversity 2x2

#### 8.2.1.2.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using transmit diversity (SFBC).

#### 8.2.1.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

8.2.1.2.1.3Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1 and 8.2.1.2.1.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.3.2.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.2.1.3-2 for the specified SNR. For transmit diversity (SFBC) performance with 2 transmitter antennas as specified.

**Table 8.2.1.2.1.3-1: Test Parameters for Testing Transmit Diversity Performance**

| Parameter | | Unit | Test 1-2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.2.1.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11 FDD | OP.1 FDD | EVA5 | 2x2 Medium | 70 | 6.8 | 2-5 |
| 2 | 10 MHz | R.10 FDD | OP.1 FDD | HST | 2x2 Low | 70 | -2.3 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.2.

### 8.2.1.2.1.4 Test description

#### 8.2.1.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.2.1.3-2 as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.
2. The parameter settings for the cell are set up according to Tables 8.2.1-1 and 8.2.1.2.1.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.2.1.4.3.

#### 8.2.1.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.2.1.3-1 and 8.2.1.2.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.2.1.2.1.5-1 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

4. Repeat steps from 1 to 3 for each test interval in Table 8.2.1.2.1.5-1 as appropriate.

#### 8.2.1.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

### 8.2.1.2.1.5 Test requirement

Table 8.2.1.2.1.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Table 8.2.1.2.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.2.1.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11 FDD | OP.1 FDD | EVA5 | 2x2 Medium | 70 | 7.7 | 2-5 |
| 2 | 10 MHz | R.10 FDD | OP.1 FDD | HST | 2x2 Low | 70 | -1.7 | 1-5 |

## 8.2.1.2.1_1 FDD PDSCH Transmit Diversity 2x2 (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Test tolerance is undefined

### 8.2.1.2.1_1.1 Test purpose

Same test purpose as in clause 8.2.1.2.1.1.

### 8.2.1.2.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

### 8.2.1.2.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.1.2.1.3 with the following exceptions:

- Instead of Table 8.2.1.2.1.3-1 → use Table 8.2.1.2.1_1.3-1.
- Instead of Table 8.2.1.2.1.3-2 → use Table 8.2.1.2.1_1.3-2.

**Table 8.2.1.2.1_1.3-1: Test Parameters for Testing Transmit Diversity Performance**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.2.1_1.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 5 MHz | R.11.2 FDD | OP.1 FDD | EVA5 | 2x2 Medium | 70 | 5.9 | 1 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.2.

### 8.2.1.2.1_1.4 Test description

Same test description as in clause 8.2.1.2.1.4 with the following exceptions:

- Instead of Table 8.2.1.2.1.3-1 → use Table 8.2.1.2.1_1.3-1.
- Instead of Table 8.2.1.2.1.3-2 → use Table 8.2.1.2.1_1.3-2.
- Instead of Table 8.2.1.2.1.5-1 → use Table 8.2.1.2.1_1.5-1.

### 8.2.1.2.1_1.5 Test requirement

Same test requirements as in clause 8.2.1.2.1.5 with the following exceptions:

- Instead of Table 8.2.1.2.1.3-1 → use Table 8.2.1.2.1_1.3-1.
- Instead of Table 8.2.1.2.1.5-1 → use Table 8.2.1.2.1_1.5-1.

**Table 8.2.1.2.1_1.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 5 MHz | R.11.2 FDD | OP.1 FDD | EVA5 | 2x2 Medium | 70 | 5.9+ TT | 1 |

## 8.2.1.2.2 FDD PDSCH Transmit Diversity 4x2

### 8.2.1.2.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using transmit diversity (SFBC-FSTD).

### 8.2.1.2.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 8.2.1.2.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1 and 8.2.1.2.2.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.3.2.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.2.2.3-2 for the specified SNR. For transmit diversity (SFBC) performance with 4 transmitter antennas as specified.

**Table 8.2.1.2.2.3-1: Test Parameters for Testing Transmit Diversity Performance**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.2.2.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 1.4 MHz | R.12 FDD | OP.1 FDD | EPA5 | 4x2 Medium | 70 | 0.6 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.2.

### 8.2.1.2.2.4 Test description

#### 8.2.1.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.2.2.3-2 as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11 for antenna configuration 4x2.
2. The parameter settings for the cell are set up according to Tables 8.2.1-1 and 8.2.1.2.2.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.2.2.4.3.

#### 8.2.1.2.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.2.2.3-1 and 8.2.1.2.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.2.1.2.2.5-1 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

#### 8.2.1.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.2.2.4.3-1: *PDSCH-ConfigDedicated-DEFAULT*: Additional FDD PDSCH transmit diversity performance downlink power allocation test point 1 requirement**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-3 | | |
| } | | | |

#### 8.2.1.2.2.5 Test requirement

Table 8.2.1.2.2.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Table 8.2.1.2.2.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.2.2.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 1.4 MHz | R.12 FDD | OP.1 FDD | EPA5 | 4x2 Medium | 70 | 1.5 | 1-5 |

### 8.2.1.2.2_1 FDD PDSCH Transmit Diversity 4x2 (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

Test tolerance is undefined

#### 8.2.1.2.2_1.1 Test purpose

Same test purpose as in clause 8.2.1.2.2.1.

#### 8.2.1.2.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 8.2.1.2.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.1.2.2.3 with the following exceptions:

- Instead of Table 8.2.1.2.2.3-1 → use Table 8.2.1.2.2_1.3-1.
- Instead of Table 8.2.1.2.2.3-2 → use Table 8.2.1.2.2_1.3-2.

**Table 8.2.1.2.2_1.3-1: Test Parameters for Testing Transmit Diversity Performance**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.2.2_1.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 FDD | OP.1 FDD | ETU70 | 4x2 Low | 70 | -0.9 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.2.

### 8.2.1.2.2_1.4 Test description

Same test description as in clause 8.2.1.2.2.4 with the following exceptions:

- Instead of Table 8.2.1.2.2.3-1 → use Table 8.2.1.2.2_1.3-1.
- Instead of Table 8.2.1.2.2.3-2 → use Table 8.2.1.2.2_1.3-2.
- Instead of Table 8.2.1.2.2.5-1 → use Table 8.2.1.2.2_1.5-1.

### 8.2.1.2.2_1.5 Test requirement

Same test requirements as in clause 8.2.1.2.2.5 with the following exceptions:

- Instead of Table 8.2.1.2.2.3-1 → use Table 8.2.1.2.2_1.3-1.
- Instead of Table 8.2.1.2.2.5-1 → use Table 8.2.1.2.2_1.5-1.

**Table 8.2.1.2.2_1.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 FDD | OP.1 FDD | ETU70 | 4x2 Low | 70 | -0.9+ TT | 1-5 |

## 8.2.1.3 FDD PDSCH Open Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols)

### 8.2.1.3.1 FDD PDSCH Open Loop Spatial Multiplexing 2x2

#### 8.2.1.3.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using large delay CDD.

#### 8.2.1.3.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

#### 8.2.1.3.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1 and 8.2.1.3.1.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.3.1.3-1 for the specified SNR. For open-loop spatial multiplexing performance with large delay CDD is specified.

**Table 8.2.1.3.1.3-1: Test Parameters for Large Delay CDD (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.3.1.3-2: Minimum performance Large Delay CDD (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11 FDD | OP.1 FDD | EVA70 | 2x2 Low | 70 | 13.0 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.3.

#### 8.2.1.3.1.4 Test description

##### 8.2.1.3.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.3.1.3-2, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.
2. The parameter settings for the cell are set up according to Tables 8.2.1-1 and 8.2.1.3.1.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.3.1.4.3.

#### 8.2.1.3.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.3.1.3-1 and 8.2.1.3.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.2.1.3.1.5-1 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

#### 8.2.1.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.3.1.4.3-1: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH open loop spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm3 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm3 | 11 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.2.1.3.1.5 Test requirement

Table 8.2.1.3.1.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Tables 8.2.1.3.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.3.1.5-1: Test Requirement Large Delay CDD (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11 FDD | OP.1 FDD | EVA70 | 2x2 Low | 70 | 13.9 | 2-5 |

### 8.2.1.3.1_A FDD PDSCH Open Loop Spatial Multiplexing 2x2 (CA)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection diagram is FFS.
- Message contents for activating SCC is FFS.
- Test tolerance is undefined

- Test frequencies are FFS
- State 3A-RF condition in test description is to be confirmed
- Statistical significance related parameters are FFS

### 8.2.1.3.1_A.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using large delay CDD.

### 8.2.1.3.1_A.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 10 and forward which support carrier aggregation.

### 8.2.1.3.1_A.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1 and 8.2.1.3.1_A.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.3.1_A.3-1 for the specified SNR. For open-loop spatial multiplexing performance with large delay CDD is specified.

**Table 8.2.1.3.1_A.3-1: Test Parameters for Large Delay CDD (FRC)**

| Parameter | | Unit | Test 1 -2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.3.1_A.3-2: Minimum performance Large Delay CDD (FRC)**

| Test num. | Band-width | Reference channel | OCNG pattern | Propa-gation condi-tion | Correlation matrix and antenna config. | Reference value | | UE cate-gory | CA capa-bility |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) | | |
| 1 | 2x10 MHz | R.11 FDD | OP.1 FDD (Note 1) | EVA70 | 2x2 Low | 70 | 13.7 | 3-8 | CL_A-A |
| 2 | 2x20 MHz | R.30 FDD | OP.1 FDD (Note 1) | EVA70 | 2x2 Low | 70 | 13.2 | 5-8 | CL_A-A, CL_C |
| Note 1: For CA capable UE, the OCNG pattern applies for each CC. | | | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.3.

### 8.2.1.3.1_A.4 Test description

#### 8.2.1.3.1_A.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause [FFS].

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.3.1_A.3-1, as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, FFS.

2. The parameter settings for the cell are set up according to Tables 8.2.1-1 and 8.2.1.3.1_A.3-1 as appropriate.

3. Downlink signals for PCC are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2. Configuration of the Downlink signals for SCC are also defined in Annex C0, C1 and C3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in[State 3A-RF] according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.3.1_A.4.3.

### 8.2.1.3.1_A.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.3.1_A.3-1 and 8.2.1.3.1_A.3-2 on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the reference channel, the propagation condition, the correlation matrix and the SNR on each of the component carrier according to Table 8.2.1.3.1_A.5-1 as appropriate.

3. Measure the average throughput per component carrier for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval on both the component carrier and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

### 8.2.1.3.1_A.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions and FFS:

**Table 8.2.1.3.1_A.4.3-1: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH open loop spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm3 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm3 | 11 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.2.1.3.1_A.5 Test requirement

Table 8.2.1.3.1_A.3-1 defines the primary level settings.

The fraction of maximum throughput percentage measured per component carrier for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Tables 8.2.1.3.1_A.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.3.1_A.5-1: Test Requirement Large Delay CDD (FRC)**

| Test num. | Band-width | Reference channel | OCNG pattern | Propa-gation condi-tion | Correlation matrix and antenna config. | Reference value | | UE cate-gory | CA capa-bility |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) | | |
| 1 | 2x10 MHz | R.11 FDD | OP.1 FDD (Note 1) | EVA70 | 2x2 Low | 70 | 13.7 + TT | 3-8 | CL_A-A |
| 2 | 2x20 MHz | R.30 FDD | OP.1 FDD (Note 1) | EVA70 | 2x2 Low | 70 | 13.2 +TT | 5-8 | CL_A-A, CL_C |
| Note 1: For CA capable UE, the OCNG pattern applies for each CC. | | | | | | | | | |

## 8.2.1.3.2 FDD PDSCH Open Loop Spatial Multiplexing 4x2

### 8.2.1.3.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using large delay CDD.

### 8.2.1.3.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

### 8.2.1.3.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1 and 8.2.1.3.2.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.1.3.2.3-1 for the specified SNR. For open-loop spatial multiplexing performance with large delay CDD is specified.

**Table 8.2.1.3.2.3-1: Test Parameters for Large Delay CDD (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.1.3.2.3-2: Minimum performance Large Delay CDD (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.14 FDD | OP.1 FDD | EVA70 | 4x2 Low | 70 | 14.3 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.3.

#### 8.2.1.3.2.4 Test description

##### 8.2.1.3.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Table 8.2.1.3.2.3-2, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11 for antenna configuration 4x2.
2. The parameter settings for the cell are set up according to Tables 8.2.1-1 and 8.2.1.3.2.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.3.2.4.3.

##### 8.2.1.3.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2A for C_RNTI to transmit the DL RMC according to Tables 8.2.1.3.2.3-1 and 8.2.1.3.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.2.1.3.2.5-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

##### 8.2.1.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.3.2.4.3-1: *PDSCH-ConfigDedicated-DEFAULT*: Additional FDD PDSCH open loop spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-6 | | |
| } | | | |

**Table 8.2.1.3.2.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH open loop spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm3 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm3 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

8.2.1.3.2.5 Test requirement

Table 8.2.1.3.2.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Tables 8.2.1.3.2.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.3.2.5-1: Test Requirement Large Delay CDD (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.14 FDD | OP.1 FDD | EVA70 | 4x2 Low | 70 | 15.2 | 2-5 |

## 8.2.1.4 FDD PDSCH Closed Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols)

### 8.2.1.4.1 FDD PDSCH Closed Loop Single/Multi Layer Spatial Multiplexing 2x2

#### 8.2.1.4.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using closed-loop spatial multiplexing.

#### 8.2.1.4.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8.

#### 8.2.1.4.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1, 8.2.1.4.1.3-1 and 8.2.1.4.1.3-3 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.1.4.1.3-2 and 8.2.1.4.1.3-4 for the specified SNR. For single-layer spatial multiplexing closed loop rank-one performance with wideband and frequency selective precoding is specified. For multi-layer spatial multiplexing closed loop rank-two performance with wideband and frequency selective precoding is specified.

**Table 8.2.1.4.1.3-1: Test Parameters for Testing Single-Layer Spatial Multiplexing (FRC)**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 | -3 |
| | $\rho_B$ | dB | -3 (Note 1) | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 |
| Precoding granularity | | PRB | 6 | 50 |
| PMI delay (Note 2) | | ms | 8 | 8 |
| Reporting interval | | ms | 1 | 1 |
| Reporting mode | | | PUSCH 1-2 | PUSCH 3-1 |
| CodeBookSubsetRestriction bitmap | | | 001111 | 001111 |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

**Table 8.2.1.4.1.3-2: Minimum performance Single-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.10 FDD | OP.1 FDD | EVA5 | 2x2 Low | 70 | -2.5 | 1-5 |
| 2 | 10 MHz | R.10 FDD | OP.1 FDD | EPA5 | 2x2 High | 70 | -2.3 | 1-5 |

**Table 8.2.1.4.1.3-3: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 3 | Test 4 |
|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 | -3 |
| | $\rho_B$ | dB | -3 (Note 1) | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 |
| Precoding granularity | | PRB | 50 | 50 |
| PMI delay (Note 2) | | ms | 8 | 8 |
| Reporting interval | | ms | 1 | 1 |
| Reporting mode | | | PUSCH 3-1 | PUSCH 3-1 |
| CodeBookSubsetRestriction bitmap | | | 110000 | 110000 |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

**Table 8.2.1.4.1.3-4: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.11 FDD | OP.1 FDD | EVA5 | 2x2 Low | 70 | 12.9 | 2-5 |
| 4 | 10 MHz | R.11 FDD | OP.1 FDD | ETU70 | 2x2 Low | 70 | 14.3 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.4.

### 8.2.1.4.1.4 Test description

#### 8.2.1.4.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Tables 8.2.1.4.1.3-2 and 8.2.1.4.1.3-4 as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.

2. The parameter settings for the cell are set up according to Tables 8.2.1-1, 8.2.1.4.1.3-1 and 8.2.1.4.1.3.-3 as appropriate.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.4.1.4.3.

#### 8.2.1.4.1.4.2 Test procedure

1. For single-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.1.4.1.3-1 and 8.2.1.4.1.3-2. For multi-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.1.4.1.3-3 and 8.2.1.4.1.3-4. The SS sends downlink MAC padding bits on the DL RMC.

2. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.2.1.4.1.5-1 and 8.2.1.4.1.5-2 as appropriate.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

5. Repeat steps from 1 to 4 for each test interval in Tables 8.2.1.4.1.5-1 and 8.2.1.4.1.5-2 as appropriate.

#### 8.2.1.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.4.1.4.3-1: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH closed loop single-layer spatial multiplexing performance downlink power allocation for Test number 1, 2**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 001111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.1.4.1.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH closed loop multi-layer spatial multiplexing performance downlink power allocation for Test number 3, 4**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 110000 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.1.4.1.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional FDD PDSCH closed loop single -layer spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

**Table 8.2.1.4.1.4.3-4: *CQI-ReportConfig-DEFAULT:* Additional FDD PDSCH closed loop single/multi-layer spatial multiplexing performance downlink power allocation for Test number 2, 3, 4**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm31 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

#### 8.2.1.4.1.5 Test requirement

Tables 8.2.1.4.3-1 and 8.2.1.4.3-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Tables 8.2.1.4.1.5-1 and 8.2.1.4.1.5-2 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.4.1.5-1: Test requirement Single-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.10 FDD | OP.1 FDD | EVA5 | 2x2 Low | 70 | -1.6 | 1-5 |
| 2 | 10 MHz | R.10 FDD | OP.1 FDD | EPA5 | 2x2 High | 70 | -1.4 | 1-5 |

**Table 8.2.1.4.1.5-2: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.11 FDD | OP.1 FDD | EVA5 | 2x2 Low | 70 | 13.8 | 2-5 |
| 4 | 10 MHz | R.11 FDD | OP.1 FDD | ETU70 | 2x2 Low | 70 | 15.2 | 2-5 |

### 8.2.1.4.1_1 FDD PDSCH Closed Loop Multi Layer Spatial Multiplexing 2x2 (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

Test tolerance is undefined

#### 8.2.1.4.1_1.1 Test purpose

Same test purpose as in clause 8.2.1.4.1.1

#### 8.2.1.4.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 8.2.1.4.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.1.4.1.3 with the following exceptions:

- Instead of Table 8.2.1.4.1.3-3 → use Table 8.2.1.4.1_1.3-1.
- Instead of Table 8.2.1.4.1.3-4 → use Table 8.2.1.4.1_1.3-2.

**Table 8.2.1.4.1_1.3-1: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 3-4 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Precoding granularity | | PRB | 50 |
| PMI delay (Note 2) | | ms | 8 |
| Reporting interval | | ms | 1 |
| Reporting mode | | | PUSCH 3-1 |
| CodeBookSubsetRestriction bitmap | | | 110000 |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

**Table 8.2.1.4.1_1.3-2: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.35 FDD | OP.1 FDD | EPA5 | 2x2 Low | 70 | 18.9 | 2-5 |
| 4 | 10 MHz | R.11 FDD | OP.1 FDD | ETU70 | 2x2 Low | 70 | 14.3 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.4.

### 8.2.1.4.1_1.4 Test description

Same test description as in clause 8.2.1.4.1.4 with the following exceptions:

- Instead of Table 8.2.1.4.1.3-3 → use Table 8.2.1.4.1_1.3-1.
- Instead of Table 8.2.1.4.1.3-4 → use Table 8.2.1.4.1_1.3-2.
- Instead of Table 8.2.1.4.1.5-2 → use Table 8.2.1.4.1_1.5-1.

### 8.2.1.4.1_1.5 Test requirement

Same test requirement as in clause 8.3.2.1.1.5 with the following exceptions:

- Instead of Table 8.2.1.4.1.3-3 → use Table 8.2.1.4.1_1.3-1.
- Instead of Table 8.2.1.4.1.5-2 → use Table 8.2.1.4.1_1.5-1.

**Table 8.2.1.4.1_1.5-1: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.35 FDD | OP.1 FDD | EPA5 | 2x2 Low | 70 | 18.9 + TT | 2-5 |
| 4 | 10 MHz | R.11 FDD | OP.1 FDD | ETU70 | 2x2 Low | 70 | 14.3 + TT | 2-5 |

### 8.2.1.4.2 FDD PDSCH Closed Loop Single/Multi Layer Spatial Multiplexing 4x2

#### 8.2.1.4.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using closed-loop spatial multiplexing.

#### 8.2.1.4.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8.

#### 8.2.1.4.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.3.2, with the addition of the relevant parameters in Tables 8.2.1-1, 8.2.1.4.2.3-1 and 8.2.1.4.2.3-3 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.1.4.2.3-2 and 8.2.1.4.2.3-4 for the specified SNR. For single-layer spatial multiplexing closed loop rank-one performance with wideband and frequency selective precoding is specified. For multi-layer spatial multiplexing closed loop rank-two performance with wideband and frequency selective precoding is specified.

**Table 8.2.1.4.2.3-1: Test Parameters for Testing Single-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Precoding granularity | | PRB | 6 |
| PMI delay (Note 2) | | ms | 8 |
| Reporting interval | | ms | 1 |
| Reporting mode | | | PUSCH 1-2 |
| CodeBookSubsetRestriction bitmap | | | 00000000000000 00000000000000 00000000000000 00000011111111 11111111 |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

**Table 8.2.1.4.2.3-2: Minimum performance Single-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 FDD | OP.1 FDD | EVA5 | 4x2 Low | 70 | -3.2 | 1-5 |

**Table 8.2.1.4.2.3-3: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Precoding granularity | | PRB | 6 |
| PMI delay (Note 2) | | ms | 8 |
| Reporting interval | | ms | 1 |
| Reporting mode | | | PUSCH 1-2 |
| CodeBookSubsetRestriction bitmap | | | 00000000000000 00000000000000 00001111111111 11111100000000 00000000 |
| Note 1: $P_B = 1$ | | | |
| Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4) | | | |

**Table 8.2.1.4.2.3-4: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.14 FDD | OP.1 FDD | EVA5 | 4x2 Low | 70 | 10.5 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.4.

### 8.2.1.4.2.4 Test description

#### 8.2.1.4.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: As specified per test number in Tables 8.2.1.4.2.3-2 and 8.2.1.4.2.3-4 as defined in TS 36.508 [7] clause 4.3.1.1.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11 for antenna configuration 4x2.
2. The parameter settings for the cell are set up according to Tables 8.2.1-1, 8.2.1.4.2.3-1 and 8.2.1.4.2.3-3 as appropriate.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.1.4.2.4.3.

#### 8.2.1.4.2.4.2 Test procedure

1. For single-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.1.4.2.3-1 and 8.2.1.4.2.3-2. For multi-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.1.4.2.3-3 and 8.2.1.4.2.3-4. The SS sends downlink MAC padding bits on the DL RMC.

2. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.2.1.4.2.5-1 and 8.2.1.4.2.5-2 as appropriate.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

5. Repeat steps from 1 to 4 for each test interval in Tables 8.2.1.4.2.5-1 and 8.2.1.4.2.5-2 as appropriate.

#### 8.2.1.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.2.1.4.2.4.3-1: *PDSCH-ConfigDedicated-DEFAULT*: Additional FDD PDSCH closed loop spatial multiplexing performance downlink power allocation for Test numbers 1, 2**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-6 PDSCH-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-6 | | |
| } | | | |

**Table 8.2.1.4.2.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH closed loop single-layer spatial multiplexing performance downlink power for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 00000000000000000000 00000000000000000000 00000000111111111111 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.1.4.2.4.3-3: *PhysicalConfigDedicated-DEFAULT*: Additional FDD PDSCH closed loop multi-layer spatial multiplexing performance downlink power allocation Test number 2**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 00000000000000000000 00000000000011111111 11111111000000000000 0000 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.1.4.2.4.3-4: *CQI-ReportConfig-DEFAULT:* Additional FDD PDSCH closed loop single/multi-layer spatial multiplexing performance downlink power allocation Test number 1, 2**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

### 8.2.1.4.2.5 Test requirement

Tables 8.2.1.4.2.3-1 and 8.2.1.4.2.3-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.3.2 for each throughput test shall meet or exceed the specified value in Tables 8.2.1.4.25-1 and 8.2.1.4.2.5-2 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.1.4.2.5-1: Test requirement Single-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 FDD | OP.1 FDD | EVA5 | 4x2 Low | 70 | -2.3 | 1-5 |

**Table 8.2.1.4.2.5-2: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.14 FDD | OP.1 FDD | EVA5 | 4x2 Low | 70 | 11.4 | 2-5 |

### 8.2.1.4.2_1 FDD PDSCH Closed Loop Multi Layer Spatial Multiplexing 4x2 (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

Test tolerance is undefined

#### 8.2.1.4.2_1.1 Test purpose

Same test purpose as in clause 8.2.1.4.2.1

#### 8.2.1.4.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 8.2.1.4.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.1.4.2.3 with the following exceptions:

- Instead of Table 8.2.1.4.2.3-3 → use Table 8.2.1.4.2_1.3-1.
- Instead of Table 8.2.1.4.2.3-4 → use Table 8.2.1.4.2_1.3-2.

**Table 8.2.1.4.2_1.3-1: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Precoding granularity | | PRB | 6 |
| PMI delay (Note 2) | | ms | 8 |
| Reporting interval | | ms | 1 |
| Reporting mode | | | PUSCH 1-2 |
| CodeBookSubsetRestriction bitmap | | | 0000000000000 0000000000000 0000001111111 1111111110000 000000000000 |
| Note 1: $P_B = 1$<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4) | | | |

**Table 8.2.1.4.2_1.3-2: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.36 FDD | OP.1 FDD | EPA5 | 4x2 Low | 70 | 14.7 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.1.4.

#### 8.2.1.4.2_1.4 Test description

Same test description as in clause 8.2.1.4.2.4 with the following exceptions:

- Instead of Table 8.2.1.4.2.3-3 → use Table 8.2.1.4.2_1.3-1.

- Instead of Table 8.2.1.4.2.3-4 → use Table 8.2.1.4.2_1.3-2.
- Instead of Table 8.2.1.4.2.5-2 → use Table 8.2.1.4.2_1.5-1.

#### 8.2.1.4.2_1.5 Test requirement

Same test requirement as in clause 8.3.2.1.2.5 with the following exceptions:

- Instead of Table 8.2.1.4.2.3-3 → use Table 8.2.1.4.2_1.3-1.
- Instead of Table 8.2.1.4.2.5-2 → use Table 8.2.1.4.2_1.5-1.

**Table 8.2.1.4.2_1.5-1: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Band-width | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.36 FDD | OP.1 FDD | EPA5 | 4x2 Low | 70 | 14.7 +TT | 2-5 |

## 8.2.2 TDD (Fixed Reference Channel)

The parameters specified in Table 8.2.2-1 are valid for all TDD tests unless otherwise stated.

**Table 8.2.2-1: Common Test Parameters (TDD)**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | 1 | |
| Special subframe configuration (Note 2) | | 4 | |
| Inter-TTI Distance | | 1 | |
| Number of HARQ processes | Processes | 7 | For TDD, 7 HARQ processes in the DL, as specified in TS 36.213 [10] clause 7.<br>All 7 HARQ processes are used. |
| Scheduling of retransmissions | | | 1. Retransmissions use the same Transport Block Size (TBS) as the initial transmission.<br>2. HARQ processes are scheduled consecutively, independent of the fact, whether retransmissions (for negatively acknowledged HARQ processes) or new transmissions (for positively acknowledged HARQ processes) occur.<br>3. In case when the initial transmission and the retransmissions are scheduled in subframes with a different $N_{PRB}$ (in terms of TS 36.213 [10] subclause 7.1.7) $29 \le I_{\mathrm{MCS}} \le 31$ according to TS 36.213 [10] subclause 7.1.7.2 and the appropriate modulation is used. |
| Maximum number of HARQ transmission | | 4 | It is always 4 for TDD, as specified in TS 36.213 [10] clause 8 |
| Redundancy version coding sequence | | {0,1,2,3} for QPSK and 16QAM<br>{0,0,1,2} for 64QAM | |
| Number of OFDM symbols for PDCCH | OFDM symbols | 4 for 1.4 MHz bandwidth, 3 for 3 MHz and 5 MHz bandwidths,<br>2 for 10 MHz, 15 MHz and 20 MHz bandwidths | The PCFICH carries information about the number of OFDM symbols used for transmission of PDCCHs in a subframe, as specified in TS 36.211 [8] clause 6.7 |
| Cyclic Prefix | | Normal | CP consist of the following physical resource blocks (RBs) parameters: 12 consecutive subcarriers at a 15 kHz spacing and 7 OFDM symbols, as specified in TS 36.211 [8] clause 6.2.3 |
| Cell ID | | 0 | The Cell ID is uniquely defined by a number in the range of 0 to 503, representing the physical-layer cell identity, as specified in TS 36.211 [8] clause 6.11. |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8] | | | |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.

## 8.2.2.1 TDD PDSCH Single Antenna Port Performance (Cell-Specific Reference Symbols)

### 8.2.2.1.1 TDD PDSCH Single Antenna Port Performance

#### 8.2.2.1.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on a single-antenna port with different channel models and MCS and also for the transmission on a single-antenna port with full RB allocation.

#### 8.2.2.1.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.2.2.1.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.1, with the addition of the relevant parameters in Tables 8.2.2-1, 8.2.2.1.1.3-1 and the downlink physical channel setup according to table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.2.1.1.3-2 for the specified SNR.

**Table 8.2.2.1.1.3-1: Test Parameters**

| Parameter | | Unit | Test 1- 5 | Test 6- 8 | Test 9- 15 | Test 16- 18 |
|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | 0 | 0 | 0 |
| | $\rho_B$ | dB | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 | -98 | -98 |
| Symbols for unused PRBs | | | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) |
| Modulation | | | QPSK | 16QAM | 64QAM | 16QAM |
| ACK/NACK feedback mode | | | Multiplexing | Multiplexing | Multiplexing | Multiplexing |

Note 1: $P_B = 0$

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

**Table 8.2.2.1.1.3-2: Minimum performance (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.2 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | -1.2 | 1-5 |
| 2 | 10 MHz | R.2 TDD | OP.1 TDD | ETU70 | 1x2 Low | 70 | -0.6 | 1-5 |
| 3 | 10 MHz | R.2 TDD | OP.1 TDD | ETU300 | 1x2 Low | 70 | -0.2 | 1-5 |
| 4 | 10 MHz | R.2 TDD | OP.1 TDD | HST | 1x2 Low | 70 | -2.6 | 1-5 |
| 5 | 1.4 MHz | R.4 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 0.0 | 1-5 |
| 6 | 10 MHz | R.3 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 6.7 | 2-5 |
| 7 | 10 MHz | R.3 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 1.4 | 2-5 |
| 8 | 10 MHz | R.3 TDD | OP.1 TDD | ETU300 | 1x2 High | 70 | 9.3 | 2-5 |
| 9 | 3 MHz | R.5 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.6 | 1-5 |
| 10 | 5 MHz | R.6 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.6 | 2-5 |
| 11 | 10 MHz | R.7 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.6 | 2-5 |
| 12 | 10 MHz | R.7 TDD | OP.1 TDD | ETU70 | 1x2 Low | 70 | 19.1 | 2-5 |
| 13 | 10 MHz | R.7 TDD | OP.1 TDD | EVA5 | 1x2 High | 70 | 19.1 | 2-5 |
| 14 | 15 MHz | R.8 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.8 | 2-5 |
| 15 | 20 MHz | R.9 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.7 | 3-5 |
| 16 | 3 MHz | R.0 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.1 | 1-5 |
| 17 | 10 MHz | R.1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.0 | 1-5 |
| 18 | 20 MHz | R.1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.1 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.

#### 8.2.2.1.1.4 Test description

##### 8.2.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Frequencies to be tested for 1PRB allocation: Low Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Bandwidths to be tested: As specified per test number in Tables 8.2.2.1.1.3-2 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Tables 8.2.2-1, 8.2.2.1.1.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.1.1.4.3.

#### 8.2.2.1.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.2.1.1.3-1 and 8.2.2.1.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the reference channel, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 8.2.2.1.1.5-1as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.
4. Repeat steps from 1 to 3 for each subtest in Tables 8.2.2.1.1.5-1 as appropriate.

#### 8.2.2.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

### 8.2.2.1.1.5 Test requirement

Table 8.2.2.1.1.3-1 defines the primary level settings including test tolerances for all throughput tests.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Tables 8.2.2.1.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.1.1.5-1: Test Requirement (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.2 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | -0.4 | 1-5 |
| 2 | 10 MHz | R.2 TDD | OP.1 TDD | ETU70 | 1x2 Low | 70 | 0.2 | 1-5 |
| 3 | 10 MHz | R.2 TDD | OP.1 TDD | ETU300 | 1x2 Low | 70 | 0.6 | 1-5 |
| 4 | 10 MHz | R.2 TDD | OP.1 TDD | HST | 1x2 Low | 70 | -2.0 | 1-5 |
| 5 | 1.4 MHz | R.4 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 0.8 | 1-5 |
| 6 | 10 MHz | R.3 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 7.5 | 2-5 |
| 7 | 10 MHz | R.3 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.2 | 2-5 |
| 8 | 10 MHz | R.3 TDD | OP.1 TDD | ETU300 | 1x2 High | 70 | 10.1 | 2-5 |
| 9 | 3 MHz | R.5 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 18.4 | 1-5 |
| 10 | 5 MHz | R.6 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 18.4 | 2-5 |
| 11 | 10 MHz | R.7 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 18.4 | 2-5 |
| 12 | 10 MHz | R.7 TDD | OP.1 TDD | ETU70 | 1x2 Low | 70 | 19.9 | 2-5 |
| 13 | 10 MHz | R.7 TDD | OP.1 TDD | EVA5 | 1x2 High | 70 | 19.9 | 2-5 |
| 14 | 15 MHz | R.8 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 18.6 | 2-5 |
| 15 | 20 MHz | R.9 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 18.5 | 3-5 |
| 16 | 3 MHz | R.0 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.9 | 1-5 |
| 17 | 10 MHz | R.1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.8 | 1-5 |
| 18 | 20 MHz | R.1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.9 | 1-5 |

## 8.2.2.1.1_1 TDD PDSCH Single Antenna Port Performance (Release 9 and forward)

### 8.2.2.1.1_1.1 Test purpose

Same test purpose as in clause 8.2.2.1.1.1.

### 8.2.2.1.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.2.2.1.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.2.1.1.3 with the following exceptions:

- Instead of Table 8.2.2.1.1.3-1 → use Table 8.2.2.1.1_1.3-1.
- Instead of Table 8.2.2.1.1.3-2 → use Table 8.2.2.1.1_1.3-2.

**Table 8.2.2.1.1_1.3-1: Test Parameters**

| Parameter | | Unit | Test 1- 5 | Test 6- 8 | Test 9- 15 | Test 16- 18 |
|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | 0 | 0 | 0 |
| | $\rho_B$ | dB | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 | -98 | -98 |
| Symbols for unused PRBs | | | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) |
| Modulation | | | QPSK | 16QAM | 64QAM | 16QAM |
| ACK/NACK feedback mode | | | Multiplexing | Multiplexing | Multiplexing | Multiplexing |

Note 1: $P_B = 0$

Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.

**Table 8.2.2.1.1_1.3-2: Minimum performance (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | N/A | | | | | | | |
| 2 | N/A | | | | | | | |
| 3 | N/A | | | | | | | |
| 4 | N/A | | | | | | | |
| 5 | N/A | | | | | | | |
| 6 | 5 MHz | R.3-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 6.7 | 1 |
| 7 | 5 MHz | R.3-1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 1.4 | 1 |
| 8 | 5 MHz | R.3-1 TDD | OP.1 TDD | ETU300 | 1x2 High | 70 | 9.3 | 1 |
| 9 | N/A | | | | | | | |
| 10 | 5 MHz | R.6-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.5 | 1 |
| 11 | 10 MHz | R.7-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 16.7 | 1 |
| 12 | 10 MHz | R.7-1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 70 | 18.1 | 1 |
| 13 | 10 MHz | R.7-1 TDD | OP.1 TDD | EVA5 | 1x2 High | 70 | 17.8 | 1 |
| 14 | 15 MHz | R.8-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 16.8 | 1 |
| 15 | 20 MHz | R.9-2 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.3 | 2 |
| | 20 MHz | R.9-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 16.7 | 1 |
| 16 | N/A | | | | | | | |
| 17 | N/A | | | | | | | |
| 18 | N/A | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.

### 8.2.2.1.1_1.4 Test description

Same test description as in clause 8.2.2.1.1.4 with the following exceptions:

- Instead of Table 8.2.2.1.1.3-1 → use Table 8.2.2.1.1_1.3-1.
- Instead of Table 8.2.2.1.1.3-2 → use Table 8.2.2.1.1_1.3-2.
- Instead of Table 8.2.2.1.1.5-1 → use Table 8.2.2.1.1_1.5-1.

#### 8.2.2.1.1_1.5 Test requirement

Same test requirements as in clause 8.2.2.1.1.5 with the following exceptions:

- Instead of Table 8.2.2.1.1.3-1 → use Table 8.2.2.1.1_1.3-1.
- Instead of Table 8.2.2.1.1.5-1 → use Table 8.2.2.1.1_1.5-1.

**Table 8.2.2.1.1_1.5-1: Test Requirement (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | N/A | | | | | | | |
| 2 | N/A | | | | | | | |
| 3 | N/A | | | | | | | |
| 4 | N/A | | | | | | | |
| 5 | N/A | | | | | | | |
| 6 | 5 MHz | R.3-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 7.5 | 1 |
| 7 | 5 MHz | R.3-1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 30 | 2.2 | 1 |
| 8 | 5 MHz | R.3-1 TDD | OP.1 TDD | ETU300 | 1x2 High | 70 | 10.1 | 1 |
| 9 | N/A | | | | | | | |
| 10 | 5 MHz | R.6-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.5+ TT | 1 |
| 11 | 10 MHz | R.7-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 16.7+ TT | 1 |
| 12 | 10 MHz | R.7-1 TDD | OP.1 TDD | ETU70 | 1x2 Low | 70 | 18.1+ TT | 1 |
| 13 | 10 MHz | R.7-1 TDD | OP.1 TDD | EVA5 | 1x2 High | 70 | 17.8+ TT | 1 |
| 14 | 15 MHz | R.8-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 16.8+ TT | 1 |
| 15 | 20 MHz | R.9-2 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 17.3+ TT | 2 |
| | 20 MHz | R.9-1 TDD | OP.1 TDD | EVA5 | 1x2 Low | 70 | 16.7+ TT | 1 |
| 16 | N/A | | | | | | | |
| 17 | N/A | | | | | | | |
| 18 | N/A | | | | | | | |

### 8.2.2.1.2 TDD PDSCH Single Antenna Port Performance with 1 PRB in the presence of MBSFN

#### 8.2.2.1.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on a single-antenna port with different channel models and MCS and also for the transmission on a single-antenna port with single RB allocation in the presence of MBSFN.

#### 8.2.2.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.2.2.1.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.1, with the addition of the relevant parameters in Tables 8.2.2-1, 8.2.2.1.2.3-1 and the downlink physical channel setup according to table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.2.1.2.3-2 for the specified SNR.

**Table 8.2.2.1.2.3-1: Test Parameters for Testing 1 PRB allocation**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 |
| | $\rho_B$ | dB | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Symbols for MBSFN portion of MBSFN subframes (Note 2) | | | OCNG (Note 3) |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: $P_B = 0$<br>Note 2: The MBSFN portion of an MBSFN subframe comprises the whole MBSFN subframe except the first two symbols in the first slot.<br>Note 3: The MBSFN portion of the MBSFN subframes shall contain QPSK modulated data. Cell-specific reference signals are not inserted in the MBSFN portion of the MBSFN subframes, QPSK modulated MBSFN data is used instead. | | | |

**Table 8.2.2.1.2.3-2: Minimum performance 1 PRB allocation (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.29 TDD | OP.3 TDD | ETU70 | 1x2 Low | 30 | 2.0 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.

#### 8.2.2.1.2.4 Test description

##### 8.2.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Low Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Bandwidths to be tested: As specified per test number in Tables 8.2.2.1.2.3-2 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Tables 8.2.2-1, 8.2.2.1.2.3-1 as appropriate.

3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.1.2.4.3.

#### 8.2.2.1.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.2.1.2.3-1 and 8.2.2.1.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the reference channel, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 8.2.2.1.1.2.5-1as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

#### 8.2.2.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.2.2.1.2.4.3-1: SystemInformationBlockType2: Additional TDD PDSCH Single Antenna Port Performance for 1 PRB allocation with MBSFN subframes test point 1 requirement**

| Derivation Path: TS 36.508 [7] clause 4.4.3.3, Table 4.4.3.3-1 SystemInformationBlockType2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SystemInformationBlockType2 ::= SEQUENCE { | | | |
| mbsfn-SubframeConfig ::= SEQUENCE { | | | |
| radioframeAllocationPeriod | n1 | Every radio frame is with MBSFN subframe | |
| radioframeAllocationOffset | 0 | | |
| subframeAllocation CHOICE { | | | |
| oneFrame | 01001x | subframe 4 and 9 is used for MBSFN. | TDD |
| } | | | |
| } | | | |
| } | | | |

#### 8.2.2.1.2.5 Test requirement

Table 8.2.2.1.2.3-1 defines the primary level settings including test tolerances for all throughput tests.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Tables 8.2.2.1.2.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.1.2.5-1: Test Requirement 1PRB with MBSFN subframes (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.29 TDD | OP.3 TDD | ETU70 | 1x2 Low | 30 | 2.8 | 1-5 |

## 8.2.2.2 TDD PDSCH Transmit Diversity Performance (Cell-Specific Reference Symbols)

### 8.2.2.2.1 TDD PDSCH Transmit Diversity 2x2

#### 8.2.2.2.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using transmit diversity (SFBC).

#### 8.2.2.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.2.2.2.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.2, with the addition of the relevant parameters in Tables 8.2.2-1 and 8.2.2.2.1.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.3.2.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.2.2.1.3-2 for the specified SNR. For transmit diversity (SFBC) performance with 2 and 4 transmitter antennas as specified.

**Table 8.2.2.2.1.3-1: Test Parameters for Testing Transmit Diversity Performance (FRC)**

| Parameter | | Unit | Test 1-2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.2.2.1.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11 TDD | OP.1 TDD | EVA5 | 2x2 Medium | 70 | 6.8 | 2-5 |
| 2 | 10 MHz | R.10 TDD | OP.1 TDD | HST | 2x2 Low | 70 | -2.3 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.2.

#### 8.2.2.2.1.4 Test description

##### 8.2.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: As specified per test number in Table 8.2.2.2.1.3-2 as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.
2. The parameter settings for the cell are set up according to Tables 8.2.2-1 and 8.2.2.2.1.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.2.1.4.3.

#### 8.2.2.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.2.2.1.3-1 and 8.2.2.2.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Table 8.2.2.2.1.5-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.
4. Repeat steps from 1 to 3 for each test interval in Table 8.2.2.2.1.5-1 as appropriate.

#### 8.2.2.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

### 8.2.2.2.1.5 Test requirement

Table 8.2.2.2.1.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.2, for each throughput test shall meet or exceed the specified value in Table 8.2.2.2.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.2.1.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11 TDD | OP.1 TDD | EVA5 | 2x2 Medium | 70 | 7.7 | 2-5 |
| 2 | 10 MHz | R.10 TDD | OP.1 TDD | HST | 2x2 Low | 70 | -1.7 | 1-5 |

## 8.2.2.2.1_1 TDD PDSCH Transmit Diversity 2x2 (Release 9 and forward)

### 8.2.2.2.1_1.1 Test purpose

Same test purpose as in clause 8.2.2.2.1.1

### 8.2.2.2.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.2.2.2.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.2.2.1.3 with the following exceptions:

- Instead of Table 8.2.2.2.1.3-1 → use Table 8.2.2.2.1_1.3-1.
- Instead of Table 8.2.2.2.1.3-2 → use Table 8.2.2.2.1_1.3-2.

**Table 8.2.2.2.1_1.3-1: Test Parameters for Testing Transmit Diversity Performance (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.2.2.1_1.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 5 MHz | R.11.1 TDD | OP.1 TDD | EVA5 | 2x2 Medium | 70 | 6.8 | 1 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.2.

### 8.2.2.2.1_1.4 Test description

Same test description as in clause 8.2.2.2.1.4 with the following exceptions:

- Instead of Table 8.2.2.2.1.3-1 → use Table 8.2.2.2.1_1.3-1.
- Instead of Table 8.2.2.2.1.3-2 → use Table 8.2.2.2.1_1.3-2.
- Instead of Table 8.2.2.2.1.5-1 → use Table 8.2.2.2.1_1.5-1.

### 8.2.2.2.1_1.5 Test requirement

Same test requirements as in clause 8.2.1.2.1.5 with the following exceptions:

- Instead of Table 8.2.1.2.1.3-1 → use Table 8.2.1.2.1_1.3-1.
- Instead of Table 8.2.1.2.1.5-1 → use Table 8.2.1.2.1_1.5-1.

**Table 8.2.2.2.1_1.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 5 MHz | R.11.1 TDD | OP.1 TDD | EVA5 | 2x2 Medium | 70 | 6.8+TT | 1 |

### 8.2.2.2.2 TDD PDSCH Transmit Diversity 4x2

#### 8.2.2.2.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on four antenna ports using transmit diversity (SFBC-FSTD).

#### 8.2.2.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.2.2.2.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.2, with the addition of the relevant parameters in Tables 8.2.2-1 and 8.2.2.2.2.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.3.2.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Table 8.2.2.2.2.3-2 for the specified SNR. For transmit diversity (SFBC) performance with 2 and 4 transmitter antennas as specified.

**Table 8.2.2.2.2.3-1: Test Parameters for Testing Transmit Diversity Performance (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.2.2.2.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 1.4 MHz | R.12 TDD | OP.1 TDD | EPA5 | 4x2 Medium | 70 | 0.2 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.2.

#### 8.2.2.2.2.4 Test description

##### 8.2.2.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: As specified per test number in Table 8.2.2.2.2.3-2 as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11 for antenna configuration 4x2.
2. The parameter settings for the cell are set up according to Tables 8.2.2-1 and 8.2.2.2.2.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.2.2.4.3.

#### 8.2.2.2.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to Tables 8.2.2.2.2.3-1 and 8.2.2.2.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Table 8.2.2.2.2.5-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

#### 8.2.2.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.2.2.2.2.4.3-1: *PDSCH-ConfigDedicated-DEFAULT*: Additional TDD PDSCH transmit diversity performance downlink power allocation test point 1 requirement**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-3 | | |
| } | | | |

### 8.2.2.2.2.5 Test requirement

Table 8.2.2.2.2.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.2, for each throughput test shall meet or exceed the specified value in Table 8.2.2.2.2.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.2.2.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 1.4 MHz | R.12 TDD | OP.1 TDD | EPA5 | 4x2 Medium | 70 | 1.1 | 1-5 |

## 8.2.2.2.2_1 TDD PDSCH Transmit Diversity 4x2 (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Test tolerance is undefined

### 8.2.2.2.2_1.1 Test purpose

Same test purpose as in clause 8.2.2.2.2.1.

### 8.2.2.2.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.2.2.2.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.1.2.2.3 with the following exceptions:

- Instead of Table 8.2.2.2.2.3-1 → use Table 8.2.2.2.2_1.3-1.
- Instead of Table 8.2.2.2.2.3-2 → use Table 8.2.2.2.2_1.3-2.

**Table 8.2.2.2.2_1.3-1: Test Parameters for Testing Transmit Diversity Performance (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.2.2.2_1.3-2: Minimum performance Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 TDD | OP.1 TDD | ETU70 | 4x2 Low | 70 | -0.5 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.2.

### 8.2.2.2.2_1.4 Test description

Same test description as in clause 8.2.2.2.2.4 with the following exceptions:

- Instead of Table 8.2.2.2.2.3-1 → use Table 8.2.2.2.2_1.3-1.
- Instead of Table 8.2.2.2.2.3-2 → use Table 8.2.2.2.2_1.3-2.
- Instead of Table 8.2.2.2.2.5-1 → use Table 8.2.2.2.2_1.5-1.

### 8.2.2.2.2_1.5 Test requirement

Same test requirements as in clause 8.2.2.2.2.5 with the following exceptions:

- Instead of Table 8.2.2.2.2.3-1 → use Table 8.2.2.2.2_1.3-1.
- Instead of Table 8.2.2.2.2.5-1 → use Table 8.2.2.2.2_1.5-1.

**Table 8.2.2.2.2_1.5-1: Test requirement Transmit Diversity (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 TDD | OP.1 TDD | ETU70 | 4x2 Low | 70 | -0.5 +TT | 1-5 |

## 8.2.2.3 TDD PDSCH Open Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols)

### 8.2.2.3.1 TDD PDSCH Open Loop Spatial Multiplexing 2x2

#### 8.2.2.3.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using large delay CDD.

#### 8.2.2.3.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.2.2.3.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.2, with the addition of the relevant parameters in Tables 8.2.2-1 and 8.2.2.3.1.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.2.3.1.3-2 for the specified SNR. For open-loop spatial multiplexing performance with large delay CDD is specified.

**Table 8.2.2.3.1.3-1: Test Parameters for Large Delay CDD (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| ACK/NACK feedback mode | | | Bundling |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.2.3.1.3-2: Minimum performance Large Delay CDD (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11-1 TDD | OP.1 TDD | EVA70 | 2x2 Low | 70 | 13.1 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.3.

#### 8.2.2.3.1.4 Test description

##### 8.2.2.3.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: As specified per test number in Table 8.2.2.3.1.3-2 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.
2. The parameter settings for the cell are set up according to Tables 8.2.2-1 and 8.2.2.3.1.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.3.1.4.3.

##### 8.2.2.3.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2A for C_RNTI to transmit the DL RMC according to Tables 8.2.2.3.1.3-1 and 8.2.2.3.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Table 8.2.2.3.1.5-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

##### 8.2.2.3.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.2.2.3.1.4.3-1: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH open loop spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm3 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm3 | 11 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.2.2.3.1.5 Test requirement

Table 8.2.2.3.1.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.2, for each throughput test shall meet or exceed the specified value in Tables 8.2.2.3.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.3.5-1: Test requirement Large Delay CDD (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.11-1 TDD | OP.1 TDD | EVA70 | 2x2 Low | 70 | 14.0 | 2-5 |

### 8.2.2.3.2 TDD PDSCH Open Loop Spatial Multiplexing 4x2

#### 8.2.2.3.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on four antenna ports using large delay CDD.

#### 8.2.2.3.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.2.2.3.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.2, with the addition of the relevant parameters in Tables 8.2.2-1 and 8.2.2.3.2.3-1 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.2.3.2.3-2 for the specified SNR. For open-loop spatial multiplexing performance with large delay CDD is specified.

**Table 8.2.2.3.2.3-1: Test Parameters for Large Delay CDD (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| ACK/NACK feedback mode | | | Bundling |
| Note 1: $P_B = 1$ | | | |

**Table 8.2.2.3.2.3-2: Minimum performance Large Delay CDD (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.14 TDD | OP.1 TDD | EVA70 | 4x2 Low | 70 | 14.2 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.3.

#### 8.2.2.3.2.4 Test description

##### 8.2.2.3.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: As specified per test number in Table 8.2.2.3.2.3-2 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11 for antenna configuration 4x2.
2. The parameter settings for the cell are set up according to Tables 8.2.2-1 and 8.2.2.3.2.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.3.2.4.3.

##### 8.2.2.3.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2A for C_RNTI to transmit the DL RMC according to Tables 8.2.2.3.2.3-1 and 8.2.2.3.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Table 8.2.2.3.2.5-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

##### 8.2.2.3.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.2.2.3.2.4.3-1: *PDSCH-ConfigDedicated-DEFAULT*: Additional TDD PDSCH open loop spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-6 | | |
| } | | | |

**Table 8.2.2.3.2.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH open loop spatial multiplexing performance downlink power for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm3 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm3 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

8.2.2.3.2.5 Test requirement

Table 8.2.2.3.2.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.2, for each throughput test shall meet or exceed the specified value in Tables 8.2.2.3.2.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.3.2.5-1: Test requirement Large Delay CDD (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.14 TDD | OP.1 TDD | EVA70 | 4x2 Low | 70 | 15.1 | 2-5 |

## 8.2.2.4 TDD PDSCH Closed Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols)

### 8.2.2.4.1 TDD PDSCH Closed Loop Single/Multi Layer Spatial Multiplexing 2x2

#### 8.2.2.4.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on two antenna ports using closed loop spatial multiplexing with wideband and frequency selective precoding.

#### 8.2.2.4.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8.

#### 8.2.2.4.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.2, with the addition of the relevant parameters in Tables 8.2.2-1, 8.2.2.4.1.3-1 and 8.2.2.4.1.3-3 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.2.4.1.3-2 and 8.2.2.4.1.3-4 for the specified SNR. For single-layer spatial multiplexing closed loop rank-one performance with wideband and frequency selective precoding is specified. For multi-layer spatial multiplexing closed loop rank-two performance with wideband and frequency selective precoding is specified.

**Table 8.2.2.4.1.3-1: Test Parameters for Testing Single-Layer Spatial Multiplexing (FRC)**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 | -3 |
| | $\rho_B$ | dB | -3 (Note 1) | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 |
| Precoding granularity | | PRB | 6 | 50 |
| PMI delay (Note 2) | | ms | 10 or 11 | 10 or 11 |
| Reporting interval | | ms | 1 or 4 (Note 3) | 1or 4 (Note 3) |
| Reporting mode | | | PUSCH 1-2 | PUSCH 3-1 |
| CodeBookSubsetRestriction bitmap | | | 001111 | 001111 |
| ACK/NACK feedback mode | | | Multiplexing | Multiplexing |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

Note 3: For Uplink - downlink configuration 1 the reporting interval will alternate between 1ms and 4ms

**Table 8.2.2.4.1.3-2: Minimum performance Single-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.10 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | -3.1 | 1-5 |
| 2 | 10 MHz | R.10 TDD | OP.1 TDD | EPA5 | 2x2 High | 70 | -2.8 | 1-5 |

**Table 8.2.2.4.1.3-3: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 3 | Test 4 |
|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 | -3 |
| | $\rho_B$ | dB | -3 (Note 1) | -3 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 |
| Precoding granularity | | PRB | 50 | 50 |
| PMI delay (Note 2) | | ms | 10 or 11 | 10 or 11 |
| Reporting interval | | ms | 1 or 4 (Note 3) | 1 or 4 (Note 3) |
| Reporting mode | | | PUSCH 3-1 | PUSCH 3-1 |
| ACK/NACK feedback mode | | | Bundling | Bundling |
| CodeBookSubsetRestriction bitmap | | | 110000 | 110000 |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

Note 3: For Uplink - downlink configuration 1 the reporting interval will alternate between 1ms and 4ms

**Table 8.2.2.4.1.3-4: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.11-1 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | 12.8 | 2-5 |
| 4 | 10 MHz | R.11-1 TDD | OP.1 TDD | ETU70 | 2x2 Low | 70 | 13.9 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.4.

### 8.2.2.4.1.4 Test description

#### 8.2.2.4.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: As specified per test number in Tables 8.2.2.4.1.3-2 and 8.2.2.4.1.3-4 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.
2. The parameter settings for the cell are set up according to Tables 8.2.2-1, 8.2.2.4.1.3-1 and 8.2.2.4.1.3-3 as appropriate.
3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.4.1.4.3.

#### 8.2.2.4.1.4.2 Test procedure

1. For single-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.2.4.1.3-1 and 8.2.2.4.1.3-2. For multi-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.2.4.1.3-3 and 8.2.2.4.1.3-4. The SS sends downlink MAC padding bits on the DL RMC.
2. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4.
3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.2.2.4.1.5-1 and 8.2.2.4.1.5-2 as appropriate.
4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.
5. Repeat steps from 1 to 4 for each test interval in Tables 8.2.2.4.1.5-1 and 8.2.2.4.1.5-2 as appropriate.

#### 8.2.2.4.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.2.2.4.1.4.3-1: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH closed loop single-layer spatial multiplexing performance downlink power allocation for Test number 1,2**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 001111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.1.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH closed loop multi-layer spatial multiplexing performance downlink power allocation for Test number 3,4**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 110000 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.1.4.3-3: *CQI-ReportConfig-DEFAULT:* Additional TDD PDSCH closed loop single -layer spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

**Table 8.2.2.4.1.4.3-4: *CQI-ReportConfig-DEFAULT:* Additional TDD PDSCH closed loop single/multi-layer spatial multiplexing performance downlink power allocation for Test number 2, 3, 4**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm31 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

#### 8.2.2.4.1.5 Test requirement

Tables 8.2.2.4.1.3-1 and 8.2.2.4.1.3-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.2, for each throughput test shall meet or exceed the specified value in Tables 8.2.2.4.1.5-1 and 8.2.2.4.1.5-2 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.4.1.5-1: Test requirement Single-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.10 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | -2.2 | 1-5 |
| 2 | 10 MHz | R.10 TDD | OP.1 TDD | EPA5 | 2x2 High | 70 | -2.0 | 1-5 |

**Table 8.2.2.4.1.5-2: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.11-1 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | 13.7 | 2-5 |
| 4 | 10 MHz | R.11-1 TDD | OP.1 TDD | ETU70 | 2x2 Low | 70 | 14.8 | 2-5 |

### 8.2.2.4.1_1 TDD PDSCH Closed Loop Multi Layer Spatial Multiplexing 2x2 (Release 9 and forward)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Test tolerance is undefined

#### 8.2.2.4.1_1.1 Test purpose

Same test purpose as in clause 8.2.2.4.1.1.

#### 8.2.2.4.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 8.2.2.4.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.2.4.1.3 with the following exceptions:

- Instead of Table 8.2.2.4.1.3-3 → use Table 8.2.2.4.1_1.3-1.
- Instead of Table 8.2.2.4.1.3-4 → use Table 8.2.2.4.1_1.3-2.

**Table 8.2.2.4.1_1.3-1: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 3-4 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 (Note 1) |

| | | |
|---|---|---|
| $N_{oc}$ at antenna port | dBm/15kHz | -98 |
| Precoding granularity | PRB | 50 |
| PMI delay (Note 2) | ms | 10 or 11 |
| Reporting interval | ms | 1 or 4 (Note 3) |
| Reporting mode | | PUSCH 3-1 |
| ACK/NACK feedback mode | | Bundling |
| CodeBookSubsetRestriction bitmap | | 110000 |
| Note 1: $P_B = 1$<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)<br>Note 3: For Uplink - downlink configuration 1 the reporting interval will alternate between 1ms and 4ms | | |

**Table 8.2.2.4.1_1.3-2: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.35 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 19.5 | 2-5 |
| 4 | 10 MHz | R.11-1 TDD | OP.1 TDD | ETU70 | 2x2 Low | 70 | 13.9 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.4.

### 8.2.2.4.1_1.4 Test description

Same test description as in clause 8.2.2.4.1.4 with the following exceptions:

- Instead of Table 8.2.2.4.1.3-3 → use Table 8.2.2.4.1_1.3-1.
- Instead of Table 8.2.2.4.1.3-4 → use Table 8.2.2.4.1_1.3-2.
- Instead of Table 8.2.2.4.1.5-2 → use Table 8.2.2.4.1_1.5-1.

### 8.2.2.4.1_1.5 Test requirement

Same test requirements as in clause 8.2.2.4.1.5 with the following exceptions:

- Instead of Table 8.2.2.4.1.3-3 → use Table 8.2.2.4.1_1.3-1.
- Instead of Table 8.2.2.4.1.5-2 → use Table 8.2.2.4.1_1.5-1.

**Table 8.2.2.4.1_1.5-1: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 3 | 10 MHz | R.35 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 19.5 + TT | 2-5 |
| 4 | 10 MHz | R.11-1 TDD | OP.1 TDD | ETU70 | 2x2 Low | 70 | 13.9 + TT | 2-5 |

### 8.2.2.4.2 TDD PDSCH Closed Loop Single/Multi Layer Spatial Multiplexing 4x2

#### 8.2.2.4.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for transmission on four antenna ports using closed loop spatial multiplexing with wideband and frequency selective precoding.

#### 8.2.2.4.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8.

#### 8.2.2.4.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.2, with the addition of the relevant parameters in Tables 8.2.2-1, 8.2.2.4.2.3-1 and 8.2.2.4.2.3-3 and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.2.2.4.3-2 and 8.2.2.4.2.3-4 for the specified SNR. For single-layer spatial multiplexing closed loop rank-one performance with wideband and frequency selective precoding is specified. For multi-layer spatial multiplexing closed loop rank-two performance with wideband and frequency selective precoding is specified.

**Table 8.2.2.4.2.3-1: Test Parameters for Testing Single-Layer Spatial Multiplexing (FRC)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Precoding granularity | | PRB | 6 |
| PMI delay (Note 2) | | ms | 10 or 11 |
| Reporting interval | | ms | 1 or 4 (Note 3) |
| Reporting mode | | | PUSCH 1-2 |
| CodeBookSubsetRestriction bitmap | | | 000000000000 0000000000000 0000000000000 0000000001111 111111111111 |
| ACK/NACK feedback mode | | | Multiplexing |

Note 1: $P_B = 1$

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

Note 3: For Uplink - downlink configuration 1 the reporting interval will alternate between 1ms and 4ms

**Table 8.2.2.4.2.3-2: Minimum performance Single-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 TDD | OP.1 TDD | EVA5 | 4x2 Low | 70 | -3.5 | 1-5 |

**Table 8.2.2.4.2.3-3: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Precoding granularity | | PRB | 6 |
| PMI delay (Note 2) | | ms | 10 or 11 |
| Reporting interval | | ms | 1 or 4 (Note 3) |
| Reporting mode | | | PUSCH 1-2 |
| ACK/NACK feedback mode | | | Bundling |
| CodeBookSubsetRestriction bitmap | | | 0000000000000 0000000000000 0000001111111 1111111110000 000000000000 |
| Note 1: $P_B = 1$ | | | |
| Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4) | | | |
| Note 3: For Uplink - downlink configuration 1 the reporting interval will alternate between 1ms and 4ms | | | |

**Table 8.2.2.4.2.3-4: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.14 TDD | OP.1 TDD | EVA5 | 4x2 Low | 70 | 10.7 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.4.

### 8.2.2.4.2.4 Test description

#### 8.2.2.4.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: As specified per test number in Tables 8.2.2.4.2.3-2 and 8.2.2.4.2.3-4 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 Figure A.11 for antenna configuration 4x2.

2. The parameter settings for the cell are set up according to Tables 8.2.2-1, 8.2.2.4.2.3-1 and 8.2.2.4.2.3-3 as appropriate.

3. Downlink signals are initially set up according to Annex C.0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.2.2.4.2.4.3.

#### 8.2.2.4.2.4.2 Test procedure

1. For single-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.2.4.2.3-1 and 8.2.2.4.2.3-2. For multi-layer spatial multiplexing, SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Tables 8.2.2.4.2.3-3 and 8.2.2.4.2.3-4. The SS sends downlink MAC padding bits on the DL RMC.

2. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.2.2.4.2.5-1 and 8.2.2.4.2.5-2 as appropriate.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

5. Repeat steps from 1 to 4 for each test interval in Tables 8.2.2.4.2.5-1 and 8.2.2.4.2.5-2 as appropriate.

#### 8.2.2.4.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.2.2.4.2.4.3-1: *PDSCH-ConfigDedicated-DEFAULT*: Additional TDD PDSCH closed loop spatial multiplexing performance downlink power allocation for Test numbers 1,2**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-6 | | |
| } | | | |

**Table 8.2.2.4.2.4.3-2: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH closed loop single-layer spatial multiplexing performance downlink power allocation for Test number 1**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 00000000000000000000 00000000000000000000 00000000111111111111 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.2.4.3-3: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH closed loop multi-layer spatial multiplexing performance downlink power allocation for Test number 2**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 00000000000000000000 00000000000011111111 11111111000000000000 0000 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.4.2.4.3-4: *CQI-ReportConfig-DEFAULT:* Additional TDD PDSCH closed loop single/multi-layer spatial multiplexing performance downlink power allocation for Test number 1, 2**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

8.2.2.4.2.5 Test requirement

Tables 8.2.2.4.2.3-1 and 8.2.2.4.2.3-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.2, for each throughput test shall meet or exceed the specified value in Tables 8.2.2.4.2.5-1 and 8.2.2.4.2.5-2 for the specified SNR including test tolerances for all throughput tests.

**Table 8.2.2.4.2.5-1: Test requirement Single-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz | R.13 TDD | OP.1 TDD | EVA5 | 4x2 Low | 70 | -2.6 | 1-5 |

**Table 8.2.2.4.2.5-2: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.14 TDD | OP.1 TDD | EVA5 | 4x2 Low | 70 | 11.6 | 2-5 |

## 8.2.2.4.2_1 TDD PDSCH Closed Loop Multi Layer Spatial Multiplexing 4x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

### 8.2.2.4.2_1.1 Test purpose

Same test purpose as in clause 8.2.2.4.2.1.

### 8.2.2.4.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.2.2.4.2_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.2.2.4.2.3 with the following exceptions:

- Instead of Table 8.2.2.4.2.3-3 → use Table 8.2.2.4.2_1.3-1.
- Instead of Table 8.2.2.4.2.3-4 → use Table 8.2.2.4.2_1.3-2.

**Table 8.2.2.4.2_1.3-1: Test Parameters for Testing Multi-Layer Spatial Multiplexing**

| Parameter | | Unit | Test 2 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 (Note 1) |

| | | |
|---|---|---|
| $N_{oc}$ at antenna port | dBm/15kHz | -98 |
| Precoding granularity | PRB | 6 |
| PMI delay (Note 2) | ms | 10 or 11 |
| Reporting interval | ms | 1 or 4 (Note 3) |
| Reporting mode | | PUSCH 1-2 |
| ACK/NACK feedback mode | | Bundling |
| CodeBookSubsetRestriction bitmap | | 0000000000000<br>0000000000000<br>0000001111111<br>1111111110000<br>000000000000 |
| Note 1: $P_B = 1$<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)<br>Note 3: For Uplink - downlink configuration 1 the reporting interval will alternate between 1ms and 4ms | | |

**Table 8.2.2.4.2_1.3-2: Minimum performance Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.36 TDD | OP.1 TDD | EPA5 | 4x2 Low | 70 | 15.7 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.2.2.4.

### 8.2.2.4.2_1.4 Test description

Same test description as in clause 8.2.2.4.1.4 with the following exceptions:

- Instead of Table 8.2.2.4.2.3-3 → use Table 8.2.2.4.2_1.3-1.
- Instead of Table 8.2.2.4.2.3-4 → use Table 8.2.2.4.2_1.3-2.
- Instead of Table 8.2.2.4.2.5-2 → use Table 8.2.2.4.2_1.5-1.

### 8.2.2.4.2_1.5 Test requirement

Same test requirements as in clause 8.2.2.4.1.5 with the following exceptions:

- Instead of Table 8.2.2.4.2.3-3 → use Table 8.2.2.4.2_1.3-1.
- Instead of Table 8.2.2.4.2.5-2 → use Table 8.2.2.4.2_1.5-1.

**Table 8.2.2.4.2_1.5-1: Test requirement Multi-Layer Spatial Multiplexing (FRC)**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 2 | 10 MHz | R.36 TDD | OP.1 TDD | EPA5 | 4x2 Low | 70 | 15.7+ TT | 2-5 |

# 8.3 Demodulation of PDSCH (User-Specific Reference Symbols)

## 8.3.1 FDD

[FFS]

## 8.3.2 TDD

The parameters specified in Table 8.3.2-1 are valid for TDD unless otherwise stated.

**Table 8.3.2-1: Common Test Parameters for User-specific Reference Symbols**

| Parameter | Unit | Value |
|---|---|---|
| Uplink downlink configuration (Note 1) | | 1 |
| Special subframe configuration (Note 2) | | 4 |
| Cyclic prefix | | Normal |
| Cell ID | | 0 |
| Inter-TTI Distance | | 1 |
| Number of HARQ processes. All these HARQ processes are used. | Processes | 7 |
| Maximum number of HARQ transmission | | 4 |
| Redundancy version coding sequence | | {0,1,2,3} for QPSK and 16QAM<br>{0,0,1,2} for 64QAM |
| Number of OFDM symbols for PDCCH | OFDM symbols | 2 |
| Beamforming Model | | As specified in Section B.4 |
| Precoder update granularity | | Frequency domain: 1 PRB<br>Time domain: 1 ms |
| ACK/NACK feedback mode | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8] | | |

For all test cases, the SNR is defined as:

$$SNR = \frac{\hat{E}_s^{(1)} + \hat{E}_s^{(2)}}{N_{oc}^{(1)} + N_{oc}^{(2)}},$$

where the superscript indicates the receiver antenna connector. The SNR requirement applies for the UE categories given for each test.

The normative reference for this requirement is TS 36.101 [2] clause 8.3.2.

## 8.3.2.1 TDD PDSCH Single-layer Spatial Multiplexing Performance (UE-Specific Reference Symbols)

### 8.3.2.1.1 TDD PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 8 and forward)

#### 8.3.2.1.1.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for single-layer transmission on antenna port 5 using user-specific reference signals with full RB or single RB allocation.

#### 8.3.2.1.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.3.2.1.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.3.1, with the addition of the relevant parameters in Tables 8.3.2-1, 8.3.2.1. 1.3-1, and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.3.2.1. 1.3-2 for the specified SNR.

**Table 8.3.2.1.1.3-1: Test Parameters for Testing DRS**

| parameter | | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | 0 | 0 | 0 |
| | $\rho_B$ | dB | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) |
| Cell-specific reference signals | | Antenna port 0 | | | | |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 | -98 | -98 |
| Symbols for unused PRBs | | | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) | OCNG (Note 2) |
| Number of allocated resource blocks | | PRB | 50 | 50 | 50 | 1 (Note 2) |
| Note 1: $P_B = 0$ | | | | | | |
| Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. | | | | | | |

**Table 8.3.2.1.1.3-2: Minimum performance DRS (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz QPSK 1/3 | R.25 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | -0.8 | 1-5 |
| 2 | 10 MHz 16QAM 1/2 | R.26 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 7.0 | 2-5 |
| 3 | 10 MHz 64QAM 3/4 | R.27 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 17.0 | 2-5 |
| 4 | 10 MHz 16QAM 1/2 | R.28 TDD | OP.1 TDD | EPA5 | 2x2 Low | 30 | 1.7 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.3.2.

#### 8.3.2.1.1.4 Test description

##### 8.3.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10 for antenna configuration 2x2.

2. The parameter settings for the cell are set up according to Tables 8.3.2-1 and 8.3.2.1. 1.3-1 as appropriate.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.3.2.1.1.4.3.

##### 8.3.2.1.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1 for C_RNTI to transmit the DL RMC according to Tables 8.3.2.1.1.3-1, 8.3.2.1.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.3.2.1.1.5-1 as appropriate.

BCH/CRS/PDCCH/PCFICH are sent on antenna port 0 using one Tx antenna, while DRS/Dedicated data for the test UE are sent on antenna port 5 using two Tx antennas with beam-forming model as specified in Annex B.4.1 and precoder update granularity specified in Table 8.3.2-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

4. Repeat steps from 1 to 3 for each test interval in Tables 8.3.2.1.1.5-1 as appropriate.

##### 8.3.2.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.3.2.1.1.4.3-1: *PhysicalConfigDedicated-DEFAULT*: Additional TDD PDSCH DRS performance downlink power allocation test point 1 requirement for Test number 1 - 4**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm7 | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| Release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.3.2.1.1.5 Test requirement

Table 8.3.2.1.1.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.3.1 for each throughput test shall meet or exceed the specified value in Table 8.3.2.1.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.3.2.1.1.5-1: Test requirement DRS**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz QPSK 1/3 | R.25 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 0.1 | 1-5 |
| 2 | 10 MHz 16QAM 1/2 | R.26 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 7.9 | 2-5 |
| 3 | 10 MHz 64QAM 3/4 | R.27 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 17.9 | 2-5 |
| 4 | 10 MHz 16QAM 1/2 | R.28 TDD | OP.1 TDD | EPA5 | 2x2 Low | 30 | 2.6 | 1-5 |

## 8.3.2.1.1_1 TDD PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 9 and forward)

### 8.3.2.1.1_1.1 Test purpose

Same test purpose as in clause 8.3.2.1.1.1.

### 8.3.2.1.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.3.2.1.1_1.3 Minimum conformance requirements

Same minimum conformance requirements as in clause 8.3.2.1.1.3 with the following exceptions:

- Instead of Table 8.3.2.1.1.3-1 → use Table 8.3.2.1.1_1.3-1.
- Instead of Table 8.3.2.1.1.3-2 → use Table 8.3.2.1.1_1.3-2.

**Table 8.3.2.1.1_1.3-1: Test Parameters for Testing DRS (Antenna port 5)**

| Parameter | | Unit | Test 1 (Note 4) | Test 2 | Test 3 | Test 4 (Note 4) |
|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | N/A | 0 | 0 | NA |
| | $\rho_B$ | dB | | 0 (Note 1) | 0 (Note 1) | |
| Cell-specific reference signals | | Antenna port 0 | | | | |
| $N_{oc}$ at antenna port | | dBm/15kHz | N/A | -98 | -98 | NA |
| Symbols for unused PRBs | | | | OCNG (Note 2) | OCNG (Note 2) | |
| Number of allocated resource blocks | | PRB | | 50 | 50 | |
| Note 1: $P_B = 0$<br>Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.<br>Note 3: The modulation symbols of the signal under test are mapped onto antenna port 5.<br>Note 4: This tests are covered in subclause 8.3.2.1.1 | | | | | | |

**Table 8.3.2.1.1_1.3-2: Minimum performance DRS (FRC) (Antenna port 5)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 (Note 1) | N/A | | | | | | | |
| 2 | 5MHz 16QAM 1/2 | R.26-1 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 7.0 | 1 |
| 3 | 10 MHz 64QAM 3/4 | R.27-1 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 17.0 | 1 |
| 4 (Note 1) | N/A | | | | | | | |
| Note 1: These tests are covered in subclause 8.3.2.1.1 | | | | | | | | |

The normative reference for this requirement is TS 36.101 [2] clause 8.3.2.

### 8.3.2.1.1_1.4 Test description

Same test description as in clause 8.3.2.1.1.4 with the following exceptions:

- Instead of Table 8.3.2.1.1.3-1 → use Table 8.3.2.1.1_1.3-1.
- Instead of Table 8.3.2.1.1.3-2 → use Table 8.3.2.1.1_1.3-2.
- Instead of Table 8.3.2.1.1.5-1 → use Table 8.3.2.1.1_1.5-1.

### 8.3.2.1.1_1.5 Test requirement

Same test requirement as in clause 8.3.2.1.1.5 with the following exceptions:

- Instead of Table 8.3.2.1.1.3-1 → use Table 8.3.2.1.1_1.3-1.
- Instead of Table 8.3.2.1.1.5-1 → use Table 8.3.2.1.1_1.5-1.

**Table 8.3.2.1.1_1.5-1: Test requirement DRS (FRC) (Antenna port 5)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 (Note 1) | N/A | | | | | | | |
| 2 | 5MHz 16QAM 1/2 | R.26-1 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 7.9 | 1 |
| 3 | 10 MHz 64QAM 3/4 | R.27-1 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 17.9 | 1 |
| 4 (Note 1) | N/A | | | | | | | |
| Note 1: These tests are covered in subclause 8.3.2.1.1 | | | | | | | | |

### 8.3.2.1.2 TDD PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 without a simultaneous transmission

#### 8.3.2.1.2.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for single-layer transmission on antenna port 7 or 8 without a simultaneous transmission on the other antenna port using DM-RS with full RB allocation.

#### 8.3.2.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 8.3.2.1.2.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.3.1, with the addition of the relevant parameters in Tables 8.3.2-1, 8.3.2.1.2.3-1, and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.3.2.1.2.3-2 for the specified SNR.

**Table 8.3.2.1.2.3-1: Test Parameters for Testing CDM-multiplexed DM RS (single layer)**

| parameter | | Unit | Test 1 | Test 2 | Test 3 | Test 4 (Note 3) | Test 5 (Note 3) |
|---|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 | 0 | 0 | N/A | N/A |
| | $\rho_B$ | dB | 0 (Note 1) | 0 (Note 1) | 0 (Note 1) | N/A | N/A |
| Cell-specific reference signals | | | Antenna port 0 and antenna port 1 | | | | |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 | -98 | N/A | N/A |
| Symbols for unused PRBs | | | OCNG (Note 4,5) | OCNG (Note 4,5) | OCNG (Note 4,5) | N/A | N/A |
| Number of allocated resource blocks (Note 2) | | PRB | 50 | 50 | 50 | N/A | N/A |
| Simultaneous transmission | | | No | No | No | N/A | N/A |

Note 1: $P_B = 1$
Note 2: The modulation symbols of the signal under test are mapped onto antenna port 7 or 8.
Note 3: Test 4 and Test 5 are covered in subclause 8.3.2.1.3.
Note 4: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.
Note 5: OCNG_RA = $\rho_A$ – 3 dB, OCNG_RB = $\rho_B$ – 3 dB in order to have the same PDSCH and OCNG power pro subcarrier at the receiver.

**Table 8.3.2.1.2.3-2: Minimum performance for CDM-multiplexed DM RS without simultaneous transmission (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz QPSK 1/3 | R.31 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | -1.0 | 1-5 |
| 2 | 10 MHz 16QAM 1/2 | R.32 TDD | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 7.7 | 2-5 |
| | 5MHz 16QAM 1/2 | R.32-1 TDD | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 7.7 | 1 |
| 3 | 10 MHz 64QAM 3/4 | R.33 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 17.7 | 2-5 |
| | 10 MHz 64QAM 3/4 | R.33-1 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 17.7 | 1 |

The normative reference for this requirement is TS 36.101 [2] clause 8.3.2.

### 8.3.2.1.2.4 Test description

#### 8.3.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Tables 8.3.2-1 and 8.3.2.1.2.3-1 as appropriate.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.3.2.1.2.4.3.

#### 8.3.2.1.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2B for C_RNTI to transmit the DL RMC according to Tables 8.3.2.1.2.3-1, 8.3.2.1.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.3.2.1.2.5-1 as appropriate.
BCH/CRS/PDCCH/PCFICH are sent on antenna ports 0 and 1 using two Tx antennas, while DRS/Dedicated data for the test UE are sent on antenna port 7 (or 8) using two Tx antennas with beam-forming model as specified in Annex B.4.1 and precoder update granularity specified in Table 8.3.2-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

4. Repeat steps from 1 to 3 for each test interval in Tables 8.3.2.1.2.5-1 as appropriate.

#### 8.3.2.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.3.2.1.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm8 | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.3.2.1.2.5 Test requirement

Table 8.3.2.1.2.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.3.1 for each throughput test shall meet or exceed the specified value in Table 8.3.2.1.2.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.3.2.1.2.5-1: Test requirement for CDM-multiplexed DM RS without simultaneous transmission (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz QPSK 1/3 | R.31 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | -0.1 | 1-5 |
| 2 | 10 MHz 16QAM 1/2 | R.32 TDD | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 8.6 | 2-5 |
| | 5 MHz 16QAM 1/2 | R.32-1 TDD | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 8.6 | 1 |
| 3 | 10 MHz 64QAM 3/4 | R.33 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 18.6 | 2-5 |
| | 10 MHz 64QAM 3/4 | R.33-1 TDD | OP.1 TDD | EPA5 | 2x2 Low | 70 | 18.6 | 1 |

### 8.3.2.1.3 TDD PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 with a simultaneous transmission

#### 8.3.2.1.3.1 Test purpose

To verify the UE's ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for single-layer transmission on antenna port 7 or 8 with a simultaneous transmission on the other antenna port using DM-RS with full RB allocation.

#### 8.3.2.1.3.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 8.3.2.1.3.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.3.1, with the addition of the relevant parameters in Tables 8.3.2-1, 8.3.2.1.3.3-1, and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.3.2.1.3.3-2 for the specified SNR.

**Table 8.3.2.1.3.3-1: Test Parameters for Testing CDM-multiplexed DM RS (single layer)**

| parameter | | Unit | Test 1 (Note 6) | Test 2 (Note 6) | Test 3 (Note 6) | Test 4 | Test 5 |
|---|---|---|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | N/A | N/A | N/A | 0 | 0 |
| | $\rho_B$ | dB | N/A | N/A | N/A | 0 (Note 1) | 0 (Note 1) |
| Cell-specific reference signals | | | Antenna port 0 and antenna port 1 | | | | |
| $N_{oc}$ at antenna port | | dBm/15kHz | N/A | N/A | N/A | -98 | -98 |
| Symbols for unused PRBs | | | N/A | N/A | N/A | OCNG (Note 4,5) | OCNG (Note 4,5) |
| Number of allocated resource blocks (Note 2) | | PRB | N/A | N/A | N/A | 50 | 50 |
| Simultaneous transmission | | | N/A | N/A | N/A | Yes (Note 3) | Yes (Note 3) |

Note 1: $P_B = 1$
Note 2: The modulation symbols of the signal under test are mapped onto antenna port 7 or 8.
Note 3: Modulation symbols of an interference signal is mapped onto the antenna port (7 or 8) not used for the input signal under test.
Note 4: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated.
Note 5: OCNG_RA = $\rho_A$ – 3 dB, OCNG_RB = $\rho_B$ – 3 dB in order to have the same PDSCH and OCNG power pro subcarrier at the receiver.
Note 6: Test 1, Test 2 and Test 3 are covered in subclause 8.3.2.1.2.

**Table 8.3.2.1.3.3-2: Minimum performance for CDM-multiplexed DM RS with interfering simultaneous transmission (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 4 | 10 MHz 16QAM 1/2 | R.32 TDD (Note 1) | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 21.9 | 2-5 |
| 5 | 10 MHz 64QAM 1/2 | R.34 TDD (Note 1) | OP.1 TDD | EPA5 | 2x2 Low | 70 | 22.0 | 2-5 |

Note 1: The reference channel applies to both the input signal under test and the interfering signal.

The normative reference for this requirement is TS 36.101 [2] clause 8.3.2.

### 8.3.2.1.3.4 Test description

#### 8.3.2.1.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Tables 8.3.2-1 and 8.3.2.1.3.3-1 as appropriate.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.3.2.1.3.4.3.

#### 8.3.2.1.3.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2B for C_RNTI to transmit the DL RMC according to Tables 8.3.2.1.3.3-1, 8.3.2.1.3.3-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.3.2.1.3.5-1 as appropriate.
BCH/CRS/PDCCH/PCFICH are sent on antenna ports 0 and 1 using two Tx antennas, while DRS/Dedicated data for the test UE are sent on antenna port 7 (or 8) and another simultaneous transmission of DRS/Dedicated data not for the test UE is sent on antenna port 8 (or 7). The DRS/Dedicated data transmissions use two Tx antennas with different beam-forming model as specified in Annex B.4.1 and precoder update granularity specified in Table 8.3.2-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.

4. Repeat steps from 1 to 3 for each test interval in Tables 8.3.2.1.3.5-1 as appropriate.

#### 8.3.2.1.3.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions.

**Table 8.3.2.1.3.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::=<br>SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm8 | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.3.2.1.3.5 Test requirement

Table 8.3.2.1.3.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.3.1 for each throughput test shall meet or exceed the specified value in Table 8.3.2.1.3.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.3.2.1.3.5-1: Test requirement for CDM-multiplexed DM RS with interfering simultaneous transmission (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 4 | 10 MHz 16QAM 1/2 | R.32 TDD (Note 1) | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 22.8 | 2-5 |
| 5 | 10 MHz 64QAM 1/2 | R.34 TDD (Note 1) | OP.1 TDD | EPA5 | 2x2 Low | 70 | 22.9 | 2-5 |
| Note 1: The reference channel applies to both the input signal under test and the interfering signal. | | | | | | | | |

## 8.3.2.2 TDD PDSCH Dual-layer Spatial Multiplexing Performance (UE-Specific Reference Symbols)

### 8.3.2.2.1 TDD PDSCH Dual-layer Spatial Multiplexing

#### 8.3.2.2.1.1 Test purpose

To verify the UE's rank-2 performance and ability to receive a predefined test signal, representing a multi-path fading channel that is determined by the SNR with a percentage of the information bit throughput for a specified downlink Reference Measurement Channel (RMC) not falling below a specified value for dual-layer transmission on antenna ports 7 and 8 using DM-RS with full RB allocation.

#### 8.3.2.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 8.3.2.2.1.3 Minimum conformance requirements

The requirements are specified in terms of the percentage of information bit throughput for the downlink reference measurement channels specified in Annex A clause A.3.4.3.2, with the addition of the relevant parameters in Tables 8.3.2-1, 8.3.2.2.1.3-1, and the downlink physical channel setup according to Table C.3.2-1 in Annex C.

Using this configuration the fraction of maximum throughput percentage shall meet or exceed the minimum requirements specified in Tables 8.3.2.2.1.3-2 for the specified SNR.

**Table 8.3.2.2.1.3-1: Test Parameters for Testing CDM-multiplexed DM RS (dual layer)**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | -3 | -3 |
| | $\rho_B$ | dB | -3 (Note 1) | -3 (Note 1) |
| Cell-specific reference symbols | | | Antenna port 0 and antenna port 1 | |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 | -98 |
| Symbols for unused PRBs | | | OCNG (Note 2) | OCNG (Note 2) |
| Number of allocated resource blocks | | PRB | 50 | 50 |
| Note 1: $P_B = 1$<br>Note 2: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. | | | | |

**Table 8.3.2.2.1.3-2: Minimum performance for CDM-multiplexed DM RS (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz QPSK 1/3 | R.31 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | 4.5 | 2-5 |
| 2 | 10 MHz 16QAM 1/2 | R.32 TDD | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 21.7 | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.3.2.

### 8.3.2.2.1.4 Test description

#### 8.3.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Tables 8.3.2-1 and 8.3.2.2.1.3-1 as appropriate.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.3.2.2.1.4.3.

#### 8.3.2.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2B for C_RNTI to transmit the DL RMC according to Tables 8.3.2.2.1.3-1, 8.3.2.2.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 8.3.2.2.1.5-1 as appropriate.
   BCH/CRS/PDCCH/PCFICH are sent on antenna ports 0 and 1 using two Tx antennas, while DRS/Dedicated data for test UE are sent on antenna ports 7 and 8 using two Tx antennas with beam-forming model as specified in Annex B.4.2 and precoder update granularity specified in Table 8.3.2-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Count the number of NACKs, ACKs and statDTXs on the UL during each test interval and decide pass or fail according to Tables G.3.5 and G.3.6 in Annex G clause G.3.
4. Repeat steps from 1 to 3 for each test interval in Tables 8.3.2.2.1.5-1 as appropriate.

#### 8.3.2.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions .

**Table 8.3.2.2.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm8 | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.3.2.2.1.5 Test requirement

Table 8.3.2.2.1.3-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.4.3.2 for each throughput test shall meet or exceed the specified value in Table 8.3.2.2.1.5-1 for the specified SNR including test tolerances for all throughput tests.

**Table 8.3.2.2.1.5-1: Test requirement for CDM-multiplexed DM RS (FRC)**

| Test number | Bandwidth and MCS | Reference Channel | OCNG Pattern | Propagation Condition | Correlation Matrix and Antenna Configuration | Reference value | | UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of Maximum Throughput (%) | SNR (dB) | |
| 1 | 10 MHz QPSK 1/3 | R.31 TDD | OP.1 TDD | EVA5 | 2x2 Low | 70 | 5.4 | 2-5 |
| 2 | 10 MHz 16QAM 1/2 | R.32 TDD | OP.1 TDD | EPA5 | 2x2 Medium | 70 | 22.6 | 2-5 |

# 8.4 Demodulation of PCFICH/PDCCH

## 8.4.1 FDD

### 8.4.1.1 FDD PCFICH/PDCCH Single-antenna Port Performance

#### 8.4.1.1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for a single-antenna port with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.1 remains below a given reference value.

#### 8.4.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

#### 8.4.1.1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.1.1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Single antenna port |
|---|---|---|---|
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | 0 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.1.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.1.3-2.

**Table 8.4.1.1.3-2: Minimum performance PDCCH/PCFICH**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and Correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 8 CCE | R.15 FDD | OP.1 FDD | ETU70 | 1x2 Low | 1 | -1.7 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.1.

### 8.4.1.1.4 Test description

#### 8.4.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 8.4.1.1.3-1.
3. The downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.1.1.4.3.

#### 8.4.1.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1 for C_RNTI to transmit the DL RMC according to Table 8.4.1.1.3-2.The details of PDCCH and PDSCH are specified in Table A.3.5.1-1 and Table A.3.5.1-2 respectively. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.4.1.1.5-1 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.1.1.5-1, pass the UE. Otherwise fail the UE.
   If Pm-dsg is less than the value specified in table 8.4.1.1.5-1, pass the UE. Otherwise fail the UE.

#### 8.4.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

### 8.4.1.1.5 Test requirement

For the parameters specified in Table 8.4.1.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.1.5-1.

**Table 8.4.1.1.5-1: Test requirement PDCCH/PCFICH**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 8 CCE | R.15 FDD | OP.1 FDD | ETU70 | 1x2 Low | 1 | -0.9 |

## 8.4.1.2 FDD PCFICH/PDCCH Transmit Diversity Performance

### 8.4.1.2.1 FDD PCFICH/PDCCH Transmit Diversity 2x2

#### 8.4.1.2.1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.1 remains below a given reference value.

#### 8.4.1.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8.

#### 8.4.1.2.1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.1.2.1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.1.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.1.3-2.

**Table 8.4.1.2.1.3-2: Minimum performance PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 1.4 MHz | 2 CCE | R.16 FDD | OP.1 FDD | EPA5 | 2 x 2 Low | 1 | 4.3 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.1.

### 8.4.1.2.1.4 Test description

#### 8.4.1.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 1.4MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 8.4.1.2.1.3-1.
3. The downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clauses B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.1.2.1.4.3.

#### 8.4.1.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.1.2.1.3-2.The details of PDCCH and PDSCH are specified in Table A.3.5.1-1 and Table A.3.5.1-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.4.1.2.1.5-1.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX).
If Pm-dsg is less than the value specified in table 8.4.1.2.1.5-1, pass the UE. Otherwise fail the UE.

#### 8.4.1.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.1.2.1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.4.1.2.1.5 Test requirement

For the parameters specified in Table 8.4.1.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.1.5-1.

**Table 8.4.1.2.1.5-1: Test requirement PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 1.4 MHz | 2 CCE | R.16 FDD | OP.1 FDD | EPA5 | 2 x 2 Low | 1 | 5.3 |

### 8.4.1.2.1_1 FDD PCFICH/PDCCH Transmit Diversity 2x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

### 8.4.1.2.1_1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.1 remains below a given reference value.

### 8.4.1.2.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

### 8.4.1.2.1_1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.1.2.1_1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups (Ng) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.1.2.1_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.1_1.3-2.

**Table 8.4.1.2.1_1.3-2: Minimum performance PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.16_1 FDD | OP.1 FDD | EVA70 | 2 x 2 Low | 1 | -0.6 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.1

### 8.4.1.2.1_1.4 Test description

#### 8.4.1.2.1_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Table 8.4.1.2.1_1.3-1.

3. The downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clauses B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.1.2.1_1.4.3.

### 8.4.1.2.1_1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.1.2.1_1.3-2.The details of PDCCH and PDSCH are specified in Table A.3.5.1-1 and Table A.3.5.1-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.4.1.2.1_1.5-1.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX).
If Pm-dsg is less than the value specified in table 8.4.1.2.1_1.5-1, pass the UE. Otherwise fail the UE.

### 8.4.1.2.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.1.2.1_1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.4.1.2.1_1.5 Test requirement

For the parameters specified in Table 8.4.1.2.1_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.1_1.5-1.

**Table 8.4.1.2.1_1.5-1: Test requirement PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.16_1 FDD | OP.1 FDD | EVA70 | 2 x 2 Low | 1 | -0.6+TT |

## 8.4.1.2.2 FDD PCFICH/PDCCH Transmit Diversity 4x2

### 8.4.1.2.2.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.1 remains below a given reference value.

### 8.4.1.2.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8.

### 8.4.1.2.2.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.1.2.2.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.1.2.2.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.2.3-2.

**Table 8.4.1.2.2.3-2: Minimum performance PDCCH/PCFICH 4 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.17 FDD | OP.1 FDD | EVA5 | 4 x 2 Medium | 1 | 0.9 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.1.

### 8.4.1.2.2.4 Test description

#### 8.4.1.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.11.
2. The parameter settings for the cell are set up according to Table 8.4.1.2.2.3-1.
3. The downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clauses B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.1.2.2.4.3.

#### 8.4.1.2.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.1.2.2.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.1-1 and Table A.3.5.1-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.4.1.2.2.5-1.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.1.2.2.5-1, pass the UE. Otherwise fail the UE.

#### 8.4.1.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.1.2.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 11111111111111111111 11111111111111111111 11111111111111111111 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

8.4.1.2.2.5 Test requirement

For the parameters specified in Table 8.4.1.2.2.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.2.5-1.

**Table 8.4.1.2.2.5-1: Test requirement PDCCH/PCFICH 4 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.17 FDD | OP.1 FDD | EVA5 | 4 x 2 Medium | 1 | 1.9 |

## 8.4.1.2.2_1 FDD PCFICH/PDCCH Transmit Diversity 4x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

### 8.4.1.2.2_1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.1 remains below a given reference value.

### 8.4.1.2.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

### 8.4.1.2.2_1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.1.2.2_1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.1.2.2_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.2_1.3-2.

**Table 8.4.1.2.2_1.3-2: Minimum performance PDCCH/PCFICH 4 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 5 MHz | 2 CCE | R.17_1 FDD | OP.1 FDD | EPA5 | 4 x 2 Medium | 1 | 6.3 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.1.

### 8.4.1.2.2_1.4 Test description

#### 8.4.1.2.2_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 5MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.11.

2. The parameter settings for the cell are set up according to Table 8.4.1.2.2_1.3-1.

3. The downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clauses B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.1.2.2_1.4.3.

### 8.4.1.2.2_1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.1.2.2_1.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.1-1 and Table A.3.5.1-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.4.1.2.2_1.5-1.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.1.2.2_1.5-1, pass the UE. Otherwise fail the UE.

### 8.4.1.2.2_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 8.4.1.2.2_1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 1111111111111111111111111111111111111111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 8.4.1.2.2_1.5 Test requirement

For the parameters specified in Table 8.4.1.2.2_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.1.2.2_1.5-1.

**Table 8.4.1.2.2_1.5-1: Test requirement PDCCH/PCFICH 4 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 5 MHz | 2 CCE | R.17_1 FDD | OP.1 FDD | EPA5 | 4 x 2 Medium | 1 | 6.3+TT |

## 8.4.2 TDD

### 8.4.2.1 TDD PCFICH/PDCCH Single-antenna Port Performance

#### 8.4.2.1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for a single-antenna port with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.2 remains below a given reference value.

#### 8.4.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.4.2.1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.2.1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Single antenna port |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 0 |
| Special subframe configuration (Note 2) | | | 4 |
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | 0 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICCH_RB<br>OCNG_RB | dB | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.1.3-2.

**Table 8.4.2.1.3-2: Minimum performance PDCCH/PCFICH**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 8 CCE | R.15 TDD | OP.1 TDD | ETU70 | 1x2Low | 1 | -1.6 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.2.

### 8.4.2.1.4 Test description

#### 8.4.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to 8.4.2.1.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.2.1.4.3.

#### 8.4.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1 for C_RNTI to transmit the DL RMC according to Table 8.4.2.1.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.2-1 and Table A.3.5.2-2 respectively. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.4.2.1.5-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the radio (statDTX)/(NACK +ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.2.2.1.5-1, pass the UE. Otherwise fail the UE.

#### 8.4.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exception:.

**Table 8.4.2.1.4.3-1: TDD-Configuration-DEFAULT**

| Derivation Path: 36.508 clause 4.6.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Configuration-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | sa0 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

### 8.4.2.1.5 Test requirement

For the parameters specified in Table 8.4.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.1.5-1.

**Table 8.4.2.1.5-1: Test requirement PDCCH/PCFICH**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 8 CCE | R.15 TDD | OP.1 TDD | ETU70 | 1x2Low | 1 | -0.8 |

## 8.4.2.2 TDD PCFICH/PDCCH Transmit Diversity Performance

### 8.4.2.2.1 TDD PCFICH/PDCCH Transmit Diversity 2x2

#### 8.4.2.2.1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.2 remains below a given reference value.

#### 8.4.2.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8.

#### 8.4.2.2.1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.2.2.1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 0 |
| Special subframe configuration (Note 2) | | | 4 |
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group | | | |

For the parameters specified in Table 8.4.2.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.1.3-2.

**Table 8.4.2.2.1.3-2: Minimum performance PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 1.4 MHz | 2 CCE | R.16 TDD | OP.1 TDD | EPA5 | 2 x 2 Low | 1 | 4.2 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.2.

### 8.4.2.2.1.4 Test description

#### 8.4.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 1.4MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to 8.4.2.2.1.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.2.2.1.4.3.

#### 8.4.2.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.2.2.1.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.2-1 and Table A.3.5.2-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.

2. Set the parameters of the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.4.2.2.1.5-1 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the radio (statDTX)/(NACK +ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.2.2.1.5-1, pass the UE. Otherwise fail the UE

#### 8.4.2.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exception:.

**Table 8.4.2.2.1.4.3-1: TDD-Configuration-DEFAULT**

| Derivation Path: 36.508 clause 4.6.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Configuration-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa0 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

**Table 8.4.2.2.1.4.3-2: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.4.2.2.1.5 Test requirement

For the parameters specified in Table 8.4.2.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.1.5-1.

**Table 8.4.2.2.1.5-1: Test requirement PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 1.4 MHz | 2 CCE | [R.16 TDD] | OP.1 TDD | EPA5 | 2 x 2 Low | 1 | 5.2 |

## 8.4.2.2.1_1 TDD PCFICH/PDCCH Transmit Diversity 2x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

### 8.4.2.2.1_1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.2 remains below a given reference value.

### 8.4.2.2.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.4.2.2.1_1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.2.2.1_1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 0 |
| Special subframe configuration (Note 2) | | | 4 |
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group | | | |

For the parameters specified in Table 8.4.2.2.1_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.1_1.3-2.

**Table 8.4.2.2.1_1.3-2: Minimum performance PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.16_1 TDD | **OP.1 TDD** | EVA70 | 2 x 2 Low | 1 | 0.1 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.2.

### 8.4.2.2.1_1.4 Test description

#### 8.4.2.2.1_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to 8.4.2.2.1_1.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.2.2.1_1.4.3.

#### 8.4.2.2.1_1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.2.2.1_1.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.2-1 and Table A.3.5.2-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.
2. Set the parameters of the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.4.2.2.1_1.5-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the radio (statDTX)/(NACK +ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.2.2.1_1.5-1, pass the UE. Otherwise fail the UE

#### 8.4.2.2.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exception:.

**Table 8.4.2.2.1_1.4.3-1: TDD-Configuration-DEFAULT**

| Derivation Path: 36.508 clause 4.6.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Configuration-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa0 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

**Table 8.4.2.2.1_1.4.3-2: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

8.4.2.2.1_1.5 Test requirement

For the parameters specified in Table 8.4.2.2.1_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.1_1.5-1.

**Table 8.4.2.2.1_1.5-1: Test requirement PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.16_1 TDD | **OP.1 TDD** | EVA70 | 2 x 2 Low | 1 | 0.1+TT |

## 8.4.2.2.2 TDD PCFICH/PDCCH Transmit Diversity 4x2

### 8.4.2.2.2.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.2 remains below a given reference value.

### 8.4.2.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8.

### 8.4.2.2.2.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.2.2.2.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 0 |
| Special subframe configuration (Note 2) | | | 4 |
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.2.2.2.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.2.3-2.

**Table 8.4.2.2.2.3-2: Minimum performance PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.17 TDD | OP.1 TDD | EVA5 | 4 x 2 Medium | 1 | 1.2 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.2.

### 8.4.2.2.2.4 Test description

#### 8.4.2.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11.

2. The parameter settings for the cell are set up according to 8.4.2.2.2.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.2.2.2.4.3.

#### 8.4.2.2.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.2.2.2.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.2-1 and Table A.3.5.2-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.

2. Set the parameters of the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.4.2.2.2.5-1 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the radio (statDTX)/(NACK +ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.2.2.2.5-1, pass the UE. Otherwise fail the UE

#### 8.4.2.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exception:.

**Table 8.4.2.2.2.4.3-1: TDD-Configuration-DEFAULT**

| Derivation Path: 36.508 clause 4.6.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Configuration-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa0 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

**Table 8.4.2.2.2.4.3-2: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 11111111111111111111111111111111111111111111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.4.2.2.2.5 Test requirement

For the parameters specified in Table 8.4.2.2.2.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.2.5-1.

**Table 8.4.2.2.2.5-1: Test requirement PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 4 CCE | R.17 TDD | OP.1 TDD | EVA5 | 4 x 2 Medium | 1 | 2.2 |

### 8.4.2.2.2_1 TDD PCFICH/PDCCH Transmit Diversity 4x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

#### 8.4.2.2.2_1.1 Test purpose

This test verifies the demodulation performance of PCFICH/PDCCH for transmit diversity with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant, tested jointly on PDCCH and PCFICH of the specified reference measurement channels in A.3.5.2 remains below a given reference value.

#### 8.4.2.2.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 8.4.2.2.2_1.3 Minimum conformance requirements

The receiver characteristics of the PDCCH/PCFICH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg). PDCCH and PCFICH are tested jointly, i.e. a miss detection of PCFICH implies a miss detection of PDCCH.

**Table 8.4.2.2.2_1.3-1: Test Parameters for PDCCH/PCFICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 0 |
| Special subframe configuration (Note 2) | | | 4 |
| Number of PDCCH symbols | | symbols | 2 |
| Number of PHICH groups ($N_g$) | | | 1 |
| PHICH duration | | | Normal |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: PHICH power setting refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.4.2.2.2_1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.2_1.3-2.

**Table 8.4.2.2.2_1.3-2: Minimum performance PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 5 MHz | 2 CCE | R.17_1 TDD | **OP.1 TDD** | EPA5 | 4 x 2 Medium | 1 | 6.5 |

The normative reference for this requirement is TS 36.101 [2] clause 8.4.2.

### 8.4.2.2.2_1.4 Test description

#### 8.4.2.2.2_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 5 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 36.508 [7] Annex A, Figure A.11.

2. The parameter settings for the cell are set up according to 8.4.2.2.2_1.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.4.2.2.2_1.4.3.

### 8.4.2.2.2_1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to Table 8.4.2.2.2.3-2. The details of PDCCH and PDSCH are specified in Table A.3.5.2-1 and Table A.3.5.2-2 respectively. The SS sends downlink MAC padding bits on the DL RMC. Transmission scheme for the PDSCH shall be transmit diversity.

2. Set the parameters of the propagation condition, the correlation matrix, antenna configuration and the SNR according to Tables 8.4.2.2.2_1.5-1 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the radio (statDTX)/(NACK +ACK+statDTX). If Pm-dsg is less than the value specified in table 8.4.2.2.2_1.5-1, pass the UE. Otherwise fail the UE

### 8.4.2.2.2_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exception:.

**Table 8.4.2.2.2_1.4.3-1: TDD-Configuration-DEFAULT**

| Derivation Path: 36.508 clause 4.6.4 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Configuration-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa0 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

**Table 8.4.2.2.2_1.4.3-2: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| codebookSubsetRestriction CHOICE { | | | |
| n4TxAntenna-tm4 | 11111111111111111111111111111111111111111111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 8.4.2.2.2.5 Test requirement

For the parameters specified in Table 8.4.2.2.2.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 8.4.2.2.2_1.5-1.

**Table 8.4.2.2.2_1.5-1: Test requirement PDCCH/PCFICH 2 Tx Antenna Port**

| Test number | Bandwidth | Aggregation level | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 5 MHz | 2 CCE | R.17_1 TDD | **OP.1 TDD** | EPA5 | 4 x 2 Medium | 1 | 6.5+TT |

# 8.5 Demodulation of PHICH

## 8.5.1 FDD

### 8.5.1.1 FDD PHICH Single-antenna Port Performance

#### 8.5.1.1.1 Test purpose

This test verifies the demodulation performance of PHICH for a single antenna port with a given SNR for which the average probability of miss detection of Hybrid Indicator ("ACK to NACK") of the specified reference measurement channels remains below a specified value.

#### 8.5.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

#### 8.5.1.1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.1.1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Single antenna port |
|---|---|---|---|
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | 0 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | 0 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 1) | | | Ng = 1 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note 1: according to Clause 6.9 in TS 36.211 [8]<br>Note 2: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.1.1.3-1 the average probability of a miss-detecting an ACK for a NACK (Pm-an) shall be below the specified value in Table 8.5.1.1.3-2.

**Table 8.5.1.1.3-2: Minimum performance PHICH**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.18 | OP.1 FDD | ETU70 | 1 x 2 Low | 0.1 | 5.5 |
| 2 | 10 MHz | R.24 | OP.1 FDD | ETU70 | 1 x 2 Low | 0.1 | 0.6 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.

### 8.5.1.1.4 Test description

#### 8.5.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Table 8.5.1.1.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.1.1.4.3.

#### 8.5.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.1.1.5-1 Test 1 as appropriate.

2. SS shall schedule PUSCH transmissions according to Annex A.2.2.1.1 Table A.2.2.1.1-1 to happen during 8 consecutive uplink TTIs via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions will transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.1.1.4.2-1 indicates the transmissions for one cycle.

**Table 8.5.1.1.4.2-1: PHICH test pattern**

| TTI | 1-4 | 5-8 | 9-12 | 13-16 | 17-20 | 21-24 |
|---|---|---|---|---|---|---|
| PDCCH | S | S | - | - | S | S |
| PHICH | - | - | A | A | - | - |
| PUSCH | | T | T | R | R | T |
| UL HARQ Process | 1-4 | 5-8 | 1-4 | 5-8 | 1-4 | 5-8 |
| Note 1: This table gives an example test pattern for HARQ process for FDD PHICH test<br>Note 2: Following notation is used:<br>S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission<br>A: represents the ACK transmission on PHICH<br>T: represents a scheduled PUSCH transmission<br>R: represents a potential PUSCH re-transmission due to a missed ACK | | | | | | |

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. Such re-transmissions (if they occur) will potentially happen in TTI 13 to 20. DTXs on TTI 13 to 20 are counted as successful ACK receptions while any transmission on these TTIs is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).
If Pm-an is less than the value specified in table 8.5.1.1.5-1, pass the UE. Otherwise fail the UE.

5. Repeat the same procedure (steps 1 to 3 ) with test conditions according to the Table 8.5.1.1.5-1 for Test 2.

#### 8.5.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6, with the following exceptions

**Table 8.5.1.1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

#### 8.5.1.1.5 Test requirement

For the parameters specified in Table 8.5.1.1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-dsg) shall be below the specified value in Table 8.5.1.1.5-1.

**Table 8.5.1.1.5-1: Test requirement PHICH**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.18 | OP.1 FDD | ETU70 | 1 x 2 Low | 0.1 | 6.4 |
| 2 | 10 MHz | R.24 | OP.1 FDD | ETU70 | 1 x 2 Low | 0.1 | 1.5 |

### 8.5.1.2 FDD PHICH Transmit Diversity Performance

#### 8.5.1.2.1 FDD PHICH Transmit Diversity 2x2

##### 8.5.1.2.1.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which the average probability of miss detection of Hybrid Indicator (“ACK to NACK”) of the specified reference measurement channels remains below a specified value.

##### 8.5.1.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8.

##### 8.5.1.2.1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.1.2.1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 1) | | | Ng = 1 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note 1: according to Clause 6.9 in TS 36.211 [8]<br>Note 2: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.1.2.1.3-1 the average probability of a miss-detecting an ACK for a NACK (Pm-an) shall be below the specified value in Table 8.5.1.2.1.3-2

**Table 8.5.1.2.1.3-2: Minimum performance PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 1.4 MHz | R.19 | OP.1 FDD | EPA5 | 2 x 2 Low | 0.1 | 5.6 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.1.

### 8.5.1.2.1.4 Test description

#### 8.5.1.2.1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 1.4 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Table 8.5.1.2.1.3-1.

3. Downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clauses B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.1.2.1.4.3.

#### 8.5.1.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.1.2.1.5-1.

2. SS shall schedule PUSCH transmissions according to Annex A.2.2.1.1 Table A.2.2.1.1-1 to happen during 8 consecutive uplink TTIs via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions will transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.1.2.1.4.2-1 indicates the transmissions for one cycle.

**Table 8.5.1.2.1.4.2-1: PHICH test pattern**

| TTI | 1-4 | 5-8 | 9-12 | 13-16 | 17-20 | 21-24 |
|---|---|---|---|---|---|---|
| PDCCH | S | S | - | - | S | S |
| PHICH | - | - | A | A | - | - |
| PUSCH | | T | T | R | R | T |
| UL HARQ Process | 1-4 | 5-8 | 1-4 | 5-8 | 1-4 | 5-8 |

Note 1: This table gives an example test pattern for HARQ process for FDD PHICH test
Note 2: Following notation is used:
S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission
A: represents the ACK transmission on PHICH
T: represents a scheduled PUSCH transmission
R: represents a potential PUSCH re-transmission due to a missed ACK

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. Such re-transmissions (if they occur) will potentially happen in TTI 13 to 20. DTXs on TTI 13 to 20 are counted as successful ACK receptions while any transmission on these TTIs is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).
If Pm-an is less than the value specified in table 8.5.1.2.1.5-1, pass the UE. Otherwise fail the UE.

#### 8.5.1.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions

**Table 8.5.1.2.1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

#### 8.5.1.2.1.5 Test requirement

For the parameters specified in Table 8.5.1.2.1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-dsg) shall be below the specified value in Table 8.5.1.2.1.5-1.

**Table 8.5.1.2.1.5-1: Test requirement PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 1.4 MHz | R.19 | OP.1 FDD | EPA5 | 2 x 2 Low | 0.1 | 6.7 |

### 8.5.1.2.1_1 FDD PHICH Transmit Diversity 2x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

#### 8.5.1.2.1_1.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which the average probability of miss detection of Hybrid Indicator ("ACK to NACK") of the specified reference measurement channels remains below a specified value.

#### 8.5.1.2.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 8.5.1.2.1_1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.1.2.1_1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 1) | | | Ng = 1 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | OCNG | |
| Cell ID | | | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note 1: according to Clause 6.9 in TS 36.211 [8]<br>Note 2: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.1.2.1_1.3-1 the average probability of a miss-detecting an ACK for a NACK (Pm-an) shall be below the specified value in Table 8.5.1.2.1_1.3-2

**Table 8.5.1.2.1_1.3-2: Minimum performance PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.19_1 | OP.1 FDD | EVA70 | 2 x 2 Low | 0.1 | 4.4 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.1.

### 8.5.1.2.1_1.4 Test description

#### 8.5.1.2.1_1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Table 8.5.1.2.1_1.3-1.

3. Downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clauses B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.1.2.1_1.4.3.

### 8.5.1.2.1_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.1.2.1_1.5-1.

2. SS shall schedule PUSCH transmissions according to Annex A.2.2.1.1 Table A.2.2.1.1-1 to happen during 8 consecutive uplink TTIs via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions will transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.1.2.1_1.4.2-1 indicates the transmissions for one cycle.

**Table 8.5.1.2.1_1.4.2-1: PHICH test pattern**

| TTI | 1-4 | 5-8 | 9-12 | 13-16 | 17-20 | 21-24 |
|---|---|---|---|---|---|---|
| PDCCH | S | S | - | - | S | S |
| PHICH | - | - | A | A | - | - |
| PUSCH | | T | T | R | R | T |
| UL HARQ Process | 1-4 | 5-8 | 1-4 | 5-8 | 1-4 | 5-8 |
| Note 1: This table gives an example test pattern for HARQ process for FDD PHICH test<br>Note 2: Following notation is used:<br>S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission<br>A: represents the ACK transmission on PHICH<br>T: represents a scheduled PUSCH transmission<br>R: represents a potential PUSCH re-transmission due to a missed ACK | | | | | | |

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. Such re-transmissions (if they occur) will potentially happen in TTI 13 to 20. DTXs on TTI 13 to 20 are counted as successful ACK receptions while any transmission on these TTIs is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).
   If Pm-an is less than the value specified in table 8.5.1.2.1_1.5-1, pass the UE. Otherwise fail the UE.

### 8.5.1.2.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions

**Table 8.5.1.2.1_1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

### 8.5.1.2.1_1.5 Test requirement

For the parameters specified in Table 8.5.1.2.1_1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-dsg) shall be below the specified value in Table 8.5.1.2.1_1.5-1.

**Table 8.5.1.2.1_1.5-1: Test requirement PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.19_1 | OP.1 FDD | EVA70 | 2 x 2 Low | 0.1 | 4.4+TT |

## 8.5.1.2.2 FDD PHICH Transmit Diversity 4x2

### 8.5.1.2.2.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which the average probability of miss detection of Hybrid Indicator ("ACK to NACK") of the specified reference measurement channels remains below a specified value.

### 8.5.1.2.2.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8.

### 8.5.1.2.2.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.1.2.2.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 1) | | | Ng = 1 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | | OCNG |
| Cell ID | | | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note 1: according to Clause 6.9 in TS 36.211 [8]<br>Note 2: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.1.2.2.3-1 the average probability of a miss-detecting an ACK for a NACK (Pm-an) shall be below the specified value in Table 8.5.1.2.2.3-2.

**Table 8.5.1.2.2.3-2: Minimum performance PHICH 4 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.20 | OP.1 FDD | EVA5 | 4 x 2 Medium | 0.1 | 6.0 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.1.

### 8.5.1.2.2.4 Test description

#### 8.5.1.2.2.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.11.

2. The parameter settings for the cell are set up according to Table 8.5.1.2.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clauses B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.1.2.2.4.3.

#### 8.5.1.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.1.2.2.5-1.

2. SS shall schedule PUSCH transmissions according to Annex A.2.2.1.1 Table A.2.2.1.1-1 to happen during 8 consecutive uplink TTIs via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions will transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.1.2.2.4.2-1 indicates the transmissions for one cycle.

**Table 8.5.1.2.2.4.2-1: PHICH test pattern**

| TTI | 1-4 | 5-8 | 9-12 | 13-16 | 17-20 | 21-24 |
|---|---|---|---|---|---|---|
| PDCCH | S | S | - | - | S | S |
| PHICH | - | - | A | A | - | - |
| PUSCH | | T | T | R | R | T |
| UL HARQ Process | 1-4 | 5-8 | 1-4 | 5-8 | 1-4 | 5-8 |
| Note 1: This table gives an example test pattern for HARQ process for FDD PHICH test<br>Note 2: Following notation is used:<br>S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission<br>A: represents the ACK transmission on PHICH<br>T: represents a scheduled PUSCH transmission<br>R: represents a potential PUSCH re-transmission due to a missed ACK | | | | | | |

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. Such re-transmissions (if they occur) will potentially happen in TTI 13 to 20. DTXs on TTI 13 to 20 are counted as successful ACK receptions while any transmission on these TTIs is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK). If Pm-an is less than the value specified in table 8.5.1.2.2.5-1, pass the UE. Otherwise fail the UE.

#### 8.5.1.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions

**Table 8.5.1.2.2.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

#### 8.5.1.2.2.5 Test requirement

For the parameters specified in Table 8.5.1.2.2.3-1 the average probability of a miss-detecting ACK for NACK (Pm-dsg) shall be below the specified value in Table 8.5.1.2.2.5-1.

**Table 8.5.1.2.2.5-1: Test requirement PHICH 4 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.20 | OP.1 FDD | EVA5 | 4 x 2 Medium | 0.1 | 7.0 |

### 8.5.1.2.2_1 FDD PHICH Transmit Diversity 4x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

#### 8.5.1.2.2_1.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which the average probability of miss detection of Hybrid Indicator ("ACK to NACK") of the specified reference measurement channels remains below a specified value.

#### 8.5.1.2.2_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 8.5.1.2.2_1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.1.2.2_1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 1) | | | Ng = 1 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | OCNG | |
| Cell ID | | | 0 |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| Note 1: according to Clause 6.9 in TS 36.211 [8]<br>Note 2: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.1.2.2_1.3-1 the average probability of a miss-detecting an ACK for a NACK (Pm-an) shall be below the specified value in Table 8.5.1.2.2_1.3-2.

**Table 8.5.1.2.2_1.3-2: Minimum performance PHICH 4 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 5 MHz | R.20_1 | OP.1 FDD | EPA5 | 4 x 2 Medium | 0.1 | 6.1 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.1.

### 8.5.1.2.2_1.4 Test description

#### 8.5.1.2.2_1.4.1 Initial conditions

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 5 MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.11.
2. The parameter settings for the cell are set up according to Table 8.5.1.2.2_1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clauses B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.1.2.2_1.4.3.

8.5.1.2.2_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.1.2.2_1.5-1.

2. SS shall schedule PUSCH transmissions according to Annex A.2.2.1.1 Table A.2.2.1.1-1 to happen during 8 consecutive uplink TTIs via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions will transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.1.2.2_1.4.2-1 indicates the transmissions for one cycle.

**Table 8.5.1.2.2_1.4.2-1: PHICH test pattern**

| TTI | 1-4 | 5-8 | 9-12 | 13-16 | 17-20 | 21-24 |
|---|---|---|---|---|---|---|
| PDCCH | S | S | - | - | S | S |
| PHICH | - | - | A | A | - | - |
| PUSCH | | T | T | R | R | T |
| UL HARQ Process | 1-4 | 5-8 | 1-4 | 5-8 | 1-4 | 5-8 |
| Note 1: This table gives an example test pattern for HARQ process for FDD PHICH test<br>Note 2: Following notation is used:<br>S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission<br>A: represents the ACK transmission on PHICH<br>T: represents a scheduled PUSCH transmission<br>R: represents a potential PUSCH re-transmission due to a missed ACK | | | | | | |

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. Such re-transmissions (if they occur) will potentially happen in TTI 13 to 20. DTXs on TTI 13 to 20 are counted as successful ACK receptions while any transmission on these TTIs is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK). If Pm-an is less than the value specified in table 8.5.1.2.2_1.5-1, pass the UE. Otherwise fail the UE.

8.5.1.2.2_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions

**Table 8.5.1.2.2_1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

8.5.1.2.2_1.5 Test requirement

For the parameters specified in Table 8.5.1.2.2_1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-dsg) shall be below the specified value in Table 8.5.1.2.2_1.5-1.

**Table 8.5.1.2.2_1.5-1: Test requirement PHICH 4 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 5 MHz | R.20_1 | OP.1 FDD | EPA5 | 4 x 2 Medium | 0.1 | 6.1+TT |

## 8.5.2 TDD

### 8.5.2.1 TDD PHICH Single-antenna Port Performance

#### 8.5.2.1.1 Test purpose

This test verifies the demodulation performance of PHICH for a single antenna port with a given SNR for which a certain Hybrid Indicator detection error rate (i.e. missed detection of "NACK to ACK" and "ACK to NACK") of the specified reference measurement channels is achieved.

#### 8.5.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

#### 8.5.2.1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.2.1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Single antenna port |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 1 |
| Special subframe configuration (Note 2) | | | 4 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | 0 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | 0 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 3) | | | Ng = 1 |
| Cell ID | | | 0 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | | OCNG |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: according to Clause 6.9 in TS 36.211 [8]<br>Note 4: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.2.1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.1.3-2.

**Table 8.5.2.1.3-2: Minimum performance of PHICH**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.18 | OP.1 TDD | ETU70 | 1 x 2 Low | 0.1 | 5.8 |
| 2 | 10 MHz | R.24 | OP.1 TDD | ETU70 | 1 x 2 Low | 0.1 | 1.3 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.2.

### 8.5.2.1.4 Test description

#### 8.5.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.1

Bandwidths to be tested: As specified per test number in Tables 8.5.2.1.3-2 as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex a, Figure A.9.
2. The parameter settings for the cell are set up according to 8.5.2.1.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.2.1.4.3.

#### 8.5.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.2.1.5-1 Test 1 as appropriate.

2. In Each HARQ process (4 HARQ processes for UL/DL configuration 1); SS shall schedule PUSCH transmissions according to Annex A.2.3.1.1 Table A.2.3.1.1-1 via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions shall transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.2.1.4.2-1 indicates the transmissions for one cycle.

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. DTX from the UE side is counted as successful ACK reception, while any transmission on these subframes is counted as NACKs.

**Table 8.5.2.1.4.2-1: PHICH test pattern**

| Subframe Index | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PDCCH | | S | | | S | | S | | | S | | | | | | | | | | |
| PHICH | | | | | | | | | | | | A | | | A | | A | | | A |
| PUSCH | | | R? | R? | | | | T | T | | | | T | T | | | | R? | R? | |
| HARQ process | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 |

Note 1: This table gives an example test pattern for HARQ process for TDD PHICH test
Note 2: Following notation is used:
S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission
A: represents the ACK transmission on PHICH
T: represents a scheduled PUSCH transmission
R: represents a potential PUSCH re-transmission due to a missed ACK
Note 3:TDD UL/DL configuration 1 is used here, special subframe is denoted as blue

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).

   If Pm-an is less than the value specified in table 8.5.2.1.5-1, pass the UE. Otherwise fail the UE.

5. Repeat the same procedure (steps 1 to 4 ) with test conditions according to the Table 8.5.2.1.5-1 for Test 2.

### 8.5.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6, with the following exceptions

**Table 8.5.1.1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

## 8.5.2.1.5 Test requirement

For the parameters specified in Table 8.5.2.1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.1.5-1.

**Table 8.5.2.1.5-1: Test requirement of PHICH**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.18 | OP.1 TDD | ETU70 | 1 x 2 Low | 0.1 | 6.7 |
| 2 | 10 MHz | R.24 | OP.1 TDD | ETU70 | 1 x 2 Low | 0.1 | 2.2 |

## 8.5.2.2 TDD PHICH Transmit Diversity Performance

### 8.5.2.2.1 TDD PHICH Transmit Diversity 2x2

#### 8.5.2.2.1.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which a certain Hybrid Indicator detection error rate (i.e. missed detection of "NACK to ACK" and "ACK to NACK") of the specified reference measurement channels is achieved.

#### 8.5.2.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8.

#### 8.5.2.2.1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.2.2.1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 1 |
| Special subframe configuration (Note 2) | | | 4 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 3) | | | Ng = 1 |
| Cell ID | | | 0 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | | OCNG |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: according to Clause 6.9 in TS 36.211 [8]<br>Note 4: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.2.2.1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.1.3-2.

**Table 8.5.2.2.1.3-2: Minimum performance of PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 1.4 MHz | R.19 | OP.1 TDD | EPA5 | 2 x 2 Low | 0.1 | 5.3 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.2.

### 8.5.2.2.1.4 Test description

#### 8.5.2.2.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2

Bandwidths to be tested: 1.4 MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A Figure A.10.
2. The parameter settings for the cell are set up according to Table 8.5.2.2.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.2.2.1.4.3.

#### 8.5.2.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.2.2.1.5-1.
2. In Each HARQ process (4 HARQ processes for UL/DL configuration 1), SS shall schedule PUSCH transmissions according to Annex A.2.3.1.1 Table A.2.3.1.1-1 via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions shall transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.2.2.1.4.2-1 indicates the transmissions for one cycle.
3. SS will only monitor for uplink retransmissions due to ACK missed-detections. DTX from the UE side is counted as successful ACK reception, while any transmission on these subframes is counted as NACKs.
4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).

If Pm-an is less than the value specified in table 8.5.2.2.1.5-1, pass the UE. Otherwise fail the UE.

**Table 8.5.2.2.1.4.2-1: PHICH test pattern**

| Subframe Index | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PDCCH | | S | | | S | | S | | | S | | | | | | | | | | |
| PHICH | | | | | | | | | | | | A | | | A | | A | | | A |
| PUSCH | | | R? | R? | | | | T | T | | | | T | T | | | | R? | R? | |
| HARQ process | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 |

Note 1: This table gives an example test pattern for HARQ process for TDD PHICH test
Note 2: Following notation is used:
S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission
A: represents the ACK transmission on PHICH
T: represents a scheduled PUSCH transmission
R: represents a potential PUSCH re-transmission due to a missed ACK
Note 3: TDD UL/DL configuration 1 is used here, special subframe is denoted as blue

### 8.5.2.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6, with the following exceptions:

**Table 8.5.2.2.1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

### 8.5.2.2.1.5 Test requirement

For the parameters specified in Table 8.5.2.2.1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.1.5-1.

**Table 8.5.2.2.1.5-1: Test requirement of PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 1.4 MHz | R.19 | OP.1 TDD | EPA5 | 2 x 2 Low | 0.1 | 6.4 |

## 8.5.2.2.1_1 TDD PHICH Transmit Diversity 2x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

### 8.5.2.2.1_1.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which a certain Hybrid Indicator detection error rate (i.e. missed detection of "NACK to ACK" and "ACK to NACK") of the specified reference measurement channels is achieved.

8.5.2.2.1_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

8.5.2.2.1_1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.2.2.1_1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 1 |
| Special subframe configuration (Note 2) | | | 4 |
| Downlink power allocation | PCFICH_RA PDCCH_RA PHICH_RA OCNG_RA | dB | -3 |
| | PCFICH_RB PDCCH_RB PHICH_RB OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 3) | | | Ng = 1 |
| Cell ID | | | 0 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | OCNG | |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: according to Clause 6.9 in TS 36.211 [8]<br>Note 4: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.2.2.1_1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.1_1.3-2.

**Table 8.5.2.2.1_1.3-2: Minimum performance of PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.19_1 | OP.1 TDD | EVA70 | 2 x 2 Low | 0.1 | 4.2 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.2.

8.5.2.2.1_1.4 Test description

8.5.2.2.1_1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2

Bandwidths to be tested: 10 MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A Figure A.10.

2. The parameter settings for the cell are set up according to Table 8.5.2.2.1_1.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clause B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.2.2.1_1.4.3.

### 8.5.2.2.1_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.2.2.1_1.5-1.

2. In Each HARQ process (4 HARQ processes for UL/DL configuration 1), SS shall schedule PUSCH transmissions according to Annex A.2.3.1.1 Table A.2.3.1.1-1 via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions shall transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.2.2.1_1.4.2-1 indicates the transmissions for one cycle.

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. DTX from the UE side is counted as successful ACK reception, while any transmission on these subframes is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).

If Pm-an is less than the value specified in table 8.5.2.2.1_1.5-1, pass the UE. Otherwise fail the UE.

**Table 8.5.2.2.1_1.4.2-1: PHICH test pattern**

| **Subframe Index** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PDCCH | | S | | | S | | S | | | S | | | | | | | | | | |
| PHICH | | | | | | | | | | | | A | | | A | | A | | | A |
| PUSCH | | | R? | R? | | | | T | T | | | | T | T | | | | R? | R? | |
| HARQ process | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 |

Note 1: This table gives an example test pattern for HARQ process for TDD PHICH test
Note 2: Following notation is used:
S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission
A: represents the ACK transmission on PHICH
T: represents a scheduled PUSCH transmission
R: represents a potential PUSCH re-transmission due to a missed ACK
Note 3: TDD UL/DL configuration 1 is used here, special subframe is denoted as blue

### 8.5.2.2.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6, with the following exceptions:

**Table 8.5.2.2.1_1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

8.5.2.2.1_1.5 Test requirement

For the parameters specified in Table 8.5.2.2.1_1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.1_1.5-1.

**Table 8.5.2.2.1.5-1: Test requirement of PHICH 2 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.19_1 | OP.1 TDD | EVA70 | 2 x 2 Low | 0.1 | 4.2+TT |

## 8.5.2.2.2 TDD PHICH Transmit Diversity 4x2

### 8.5.2.2.2.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which a certain Hybrid Indicator detection error rate (i.e. missed detection of “NACK to ACK” and “ACK to NACK”) of the specified reference measurement channels is achieved.

### 8.5.2.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8.

### 8.5.2.2.2.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.2.2.2.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 1 |
| Special subframe configuration (Note 2) | | | 4 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 3) | | | Ng = 1 |
| Cell ID | | | 0 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6. | |
| Unused RE-s and PRB-s | | | OCNG |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: according to Clause 6.9 in TS 36.211 [8]<br>Note 4: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.2.2.2.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.2.3-2.

**Table 8.5.2.2.2.3-2: Minimum performance of PHICH 4 Tx Antenna port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.20 | OP.1 TDD | EVA5 | 4 x 2 Medium | 0.1 | 6.1 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.2.

### 8.5.2.2.2.4 Test description

#### 8.5.2.2.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2

Channel Bandwidths to be tested: 10 MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.11.

2. The parameter settings for the cell are set up according to Table 8.5.2.2.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clause B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.2.2.2.4.3.

#### 8.5.2.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.2.2.2.5-1.

2. In Each HARQ process (4 HARQ processes for UL/DL configuration 1), SS shall schedule PUSCH transmissions according to Annex A.2.3.1.1 Table A.2.3.1.1-1 via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions shall transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.2.2.2.4.2-1 indicates the transmissions for one cycle.

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. DTX from the UE side is counted as successful ACK reception, while any transmission on these subframes is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).

   If Pm-an is less than the value specified in table 8.5.2.2.2.5-1, pass the UE. Otherwise fail the UE.

**Table 8.5.2.2.2.4.2-1: PHICH test pattern**

| **Subframe Index** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PDCCH | | S | | | S | | S | | | S | | | | | | | | | | |
| PHICH | | | | | | | | | | | | A | | | A | | A | | | A |
| PUSCH | | | R? | R? | | | | T | T | | | | T | T | | | | R? | R? | |
| HARQ process | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 |

Note 1: This table gives an example test pattern for HARQ process for TDD PHICH test
Note 2: Following notation is used:
S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission
A: represents the ACK transmission on PHICH
T: represents a scheduled PUSCH transmission
R: represents a potential PUSCH re-transmission due to a missed ACK
Note 3: TDD UL/DL configuration 1 is used here, special subframe is denoted as blue

#### 8.5.2.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6, with the following exceptions:

**Table 8.5.2.2.2.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

#### 8.5.2.2.2.5 Test requirement

For the parameters specified in Table 8.5.2.2.2.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.2.5-1.

**Table 8.5.2.2.2.5-1: Test requirement of PHICH 4 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 10 MHz | R.20 | OP.1 TDD | EVA5 | 4 x 2 Medium | 0.1 | 7.1 |

### 8.5.2.2.2_1 TDD PHICH Transmit Diversity 4x2 (Release 9 and forward)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Test tolerance is undefined

#### 8.5.2.2.2_1.1 Test purpose

This test verifies the demodulation performance of PHICH for transmit diversity with a given SNR for which a certain Hybrid Indicator detection error rate (i.e. missed detection of "NACK to ACK" and "ACK to NACK") of the specified reference measurement channels is achieved.

#### 8.5.2.2.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 8.5.2.2.2_1.3 Minimum conformance requirements

The receiver characteristics of the PHICH are determined by the probability of miss-detecting an ACK for a NACK (Pm-an). It is assumed that there is no bias applied to the detection of ACK and NACK (zero-threshold detection).

**Table 8.5.2.2.2_1.3-1: Test Parameters for PHICH**

| Parameter | | Unit | Transmit diversity |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | | 1 |
| Special subframe configuration (Note 2) | | | 4 |
| Downlink power allocation | PCFICH_RA<br>PDCCH_RA<br>PHICH_RA<br>OCNG_RA | dB | -3 |
| | PCFICH_RB<br>PDCCH_RB<br>PHICH_RB<br>OCNG_RB | dB | -3 |
| PHICH duration | | | Normal |
| Number of PHICH groups (Note 3) | | | Ng = 1 |
| Cell ID | | | 0 |
| PDCCH Content | | UL Grant should be included with the proper information aligned with A.3.6, other PDCCH resource shall be occupied by non-zero data. | |
| Unused RE-s and PRB-s | | OCNG | |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Cyclic prefix | | | Normal |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [8]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [8]<br>Note 3: according to Clause 6.9 in TS 36.211 [8]<br>Note 4: PHICH power settings refer to PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | | |

For the parameters specified in Table 8.5.2.2.2_1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.2_1.3-2.

**Table 8.5.2.2.2_1.3-2: Minimum performance of PHICH 4 Tx Antenna port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 5 MHz | R.20_1 | OP.1 TDD | EPA5 | 4 x 2 Medium | 0.1 | 6.2 |

The normative reference for this requirement is TS 36.101 [2] clause 8.5.2.

### 8.5.2.2.2_1.4 Test description

#### 8.5.2.2.2_1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.2

Channel Bandwidths to be tested: 5 MHz, as defined in TS 36.508 [7] clause 4.3.1.2.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.11.

2. The parameter settings for the cell are set up according to Table 8.5.2.2.2_1.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clause B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 8.5.2.2.2_1.4.3.

### 8.5.2.2.2_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 8.5.2.2.2_1.5-1.

2. In Each HARQ process (4 HARQ processes for UL/DL configuration 1), SS shall schedule PUSCH transmissions according to Annex A.2.3.1.1 Table A.2.3.1.1-1 via PDCCH DCI format 0 with new data indicator set to true. Since the UE has no payload, the UE shall send uplink MAC padding bits in PUSCH. SS upon receiving the PUSCH transmissions shall transmit the associated ACKs. PHICH is set according to Annex 3.6 Table A.3.6-1. SS will only transmit PDCCH to schedule PUSCH transmission in the appropriate sub-frames. Table 8.5.2.2.2_1.4.2-1 indicates the transmissions for one cycle.

3. SS will only monitor for uplink retransmissions due to ACK missed-detections. DTX from the UE side is counted as successful ACK reception, while any transmission on these subframes is counted as NACKs.

4. Repeat steps 1 – 3 for a duration sufficient to achieve statistical significance according to Annex G clause G.4 and measure Pm-an. Pm-an is (NACK) / (ACK + NACK).

   If Pm-an is less than the value specified in table 8.5.2.2.2_1.5-1, pass the UE. Otherwise fail the UE.

**Table 8.5.2.2.2_1.4.2-1: PHICH test pattern**

| **Subframe Index** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PDCCH | | S | | | S | | S | | | S | | | | | | | | | | |
| PHICH | | | | | | | | | | | | A | | | A | | A | | | A |
| PUSCH | | | R? | R? | | | | T | T | | | | T | T | | | | R? | R? | |
| HARQ process | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 | | 1 | 3 | 4 | 2 | | 3 | 1 | 2 | 4 |

Note 1: This table gives an example test pattern for HARQ process for TDD PHICH test
Note 2: Following notation is used:
S: represents sending PDCCH DCI format 0 to schedule a future PUSCH transmission
A: represents the ACK transmission on PHICH
T: represents a scheduled PUSCH transmission
R: represents a potential PUSCH re-transmission due to a missed ACK
Note 3: TDD UL/DL configuration 1 is used here, special subframe is denoted as blue

### 8.5.2.2.2_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6, with the following exceptions:

**Table 8.5.2.2.2_1.4.3-1: MAC-MainConfig-RBC**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| MAC-MainConfig-RBC ::= SEQUENCE { | | | |
| dl-SCH-Config SEQUENCE {} | Not present | | |
| ul-SCH-Config SEQUENCE { | | | |
| maxHARQ-Tx | n2 | Only one retransmission per UL HARQ | |
| … | | | |

#### 8.5.2.2.2_1.5 Test requirement

For the parameters specified in Table 8.5.2.2.2_1.3-1 the average probability of a miss-detecting ACK for NACK (Pm-an) shall be below the specified value in Table 8.5.2.2.2_1.5-1.

**Table 8.5.2.2.2_1.5-1: Test requirement of PHICH 4 Tx Antenna Port**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Pm-an (%) | SNR (dB) |
| 1 | 5 MHz | R.20_1 | OP.1 TDD | EPA5 | 4 x 2 Medium | 0.1 | 6.2+TT |

## 8.6 Demodulation of PBCH

RAN4 will specify the PBCH performance requirements and has recommended that these requirements do not need to be tested.

## 8.7 Sustained downlink data rate provided by lower layers

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- The minimum requirements are not confirmed for UE category 5
- The number of frames to be tested are within [ ] and the relevant parts of Annex G on statistical testing are undefined
- The length of time the SS should wait to let any HARQ retransmissions and RLC retransmissions to finish is within [ ]
- The $TB_{size}$ per Codeword for 8.7.2, UE category 3, Test 3B is undefined
- The OCNG patterns are undefined

### 8.7.1 FDD

#### 8.7.1.1 FDD sustained data rate performance

##### 8.7.1.1.1 Test purpose

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum number of DL-SCH transport block bits received within a TTI for the UE category indicated. The sustained downlink data rate shall be verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the RF conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement. The size of the TB per TTI corresponds to the largest possible DL-SCH transport block for each UE category using the maximum number of layers for spatial multiplexing. Transmission modes 1 and 3 are used with radio conditions resembling a scenario where sustained maximum data rates are available.

##### 8.7.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE Release 9 and forward.

##### 8.7.1.1.3 Minimum requirements

The parameters specified in Table 8.7.1.1.3-1 are valid for all FDD tests unless otherwise stated.

**Table 8.7.1.1.3-1: Common Test Parameters (FDD)**

| Parameter | Unit | Value |
|---|---|---|
| Cyclic prefix | | Normal |
| Cell ID | | 0 |
| Inter-TTI Distance | | 1 |
| Number of HARQ processes | Processes | Downlink: 8<br>Uplink: 8 |
| Maximum number of HARQ transmission | | Downlink: 4<br>Uplink: 1 |
| Redundancy version coding sequence | | {0,0,1,2} for 64QAM |
| Number of OFDM symbols for PDCCH | OFDM symbols | 1 |

The requirements are specified in Table 8.7.1.1.3-3, with the addition of the parameters in Table 8.7.1.1.3-2 and the downlink physical channel setup according to Annex C.3.2. The PDCP SDU success rate shall be sustained during at least 300 frames.

**Table 8.7.1.1.3-2: Test Parameters for sustained downlink data rate (FDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3,4 | Test 3A |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | 10 | 20 | 10 |
| Transmission mode | | | 1 | 3 | 3 | 3 |
| Antenna configuration | | | 1 x 2 | 2 x 2 | 2 x 2 | 2 x 2 |
| Propagation condition | | | Static propagation condition (Note 1) | | | |
| CodeBookSubsetRestriction bitmap | | | n/a | 10 | 10 | 10 |
| Downlink power allocation | $\rho_A$ | dB | 0 | -3 | -3 | -3 |
| | $\rho_B$ | dB | 0 | -3 | -3 | -3 |
| $\hat{E}_s$ at antenna port | | dBm/15kHz | -85 | -85 | -85 | -85 |
| Symbols for unused PRBs | | | [OCNG] | [OCNG] | [OCNG] | [OCNG] |
| Note 1: No external noise sources are applied | | | | | | |

**Table 8.7.1.1.3-3: Minimum Requirement (FDD)**

| Test | UE Category | Number of bits of a DL-SCH transport block received within a TTI | Measurement channel | Reference value TB success rate [%] |
|---|---|---|---|---|
| 1 | Category 1 | 10296 | R31-1 FDD | 95 |
| 2 | Category 2 | 25456 | R31-2 FDD | 95 |
| 3 | Category 3 (Note 1) | 51024 | R31-3 FDD | 95 |
| 3A | Category 3 (Note 2) | 36696 (Note 4) | R31-3A FDD | 85 |
| 4 | Category 4 | 75376 (Note 5) | R31-4 FDD | 85 |
| 5 | Category 5 | FFS | FFS | FFS |

Note 1: If the operating band under test does not support 20 MHz channel bandwidth, then test is executed according to Test 3A.
Note 2: Applicable to operating bands supporting up to 10 MHz channel bandwidths.
Note 3: For 2 layer transmissions, 2 transport blocks are received within a TTI
Note 4: 35160 bits for sub-frame 5
Note 5: 71112 bits for sub-frame 5
Note 6: The TB success rate is defined as TB success rate = 100%*$N_{DL_correct_rx}$/ ($N_{DL_newtx}$ + $N_{DL_retx}$), where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

The normative reference for this requirement is TS 36.101[2] clause 8.7.1

### 8.7.1.1.4 Test description

#### 8.7.1.1.4.1 Initial conditions

**Table 8.7.1.1.4.1-1: Applicable test and Transport Block Size for different UE categories**

| UE Category | Applicable test | DL Measurement channel | UL Measurement channel | $TB_{size}$ per Codeword | Number of PDCP SDU per Codeword | PDCP SDU size [bits] Note 3 |
|---|---|---|---|---|---|---|
| 1 | Test 1 | R31-1 FDD | R.1-1 FDD | 10296 | 1 | 8*FLOOR(($TB_{size}$ - 96)/8) |
| 2 | Test 2 | R31-2 FDD | R.1-2 FDD | 25456 | 3 | 8*FLOOR(($TB_{size}$ – 152)/24)) |
| 3 | Test 3 (Note 1) | R31-3 FDD | R.1-3 FDD | 51024 | 5 | 8*FLOOR(($TB_{size}$ -– 208)/40)) |
| 3 | Test 3A (Note 2) | R31-3A FDD | R.1-3A FDD | 36696 (Note 5) | 4 | 8*FLOOR(($TB_{size}$ – 184)/32))" |
| 4 | Test 4 (Note 2) | R31-4 FDD | R.1-4 FDD | 75376 (Note 6) | 7 | 8*FLOOR(($TB_{size}$ – 264)/56)) |
| 5 | FFS | FFS | TBD | FFS | FFS | FFS |

Note 1: If the operating band under test does not support 20 MHz channel bandwidth, then test is executed according to Test 3A.

Note 2: Applicable to operating bands supporting up to 10 MHz channel bandwidths.

Note 3: Transport block size under test according to applicable Fixed Reference Channel for sustained data-rate test in annex A.3.9.

Note 4: Each PDCP SDU is limited to 1500 octets (to keep below maximum SDU size of ESM as specified in TS 24.301 clause 9.9.4.12).

The PDCP SDU size of each PDCP SDU is:

PDCP SDU size = (TBsize – N*PDCP header size - AMD PDU header size - MAC header size - Size of RLC STATUS PDU) / N,

where PDCP header size is 16 bits for the RLC AM and 12-bit SN case; AMD PDU header size is CEIL[(16+(N-1)*12)/8] bytes which includes 16 bit standard AM header and (N-1) Length indicators; and MAC header size = R/R/E/LCID/F/L MAC subheader (24 bits for MAC SDU for RLC STATUS PDU with 15 bit LI) + R/R/E/LCID MAC subheader (8 bits for MAC SDU for RLC Data PDU) = 32 bits. The size of RLC STATUS PDU including one ACK_SN field and one NACK_SN field is 32 bits (if no STATUS PDU is sent or if the size of the STATUS PDU is less than 32 bits then padding will be used to fill the 32 bits). This gives: PDCP SDU size = 8*FLOOR((TBsize – N*16- 8*CEIL((16+(N-1)*12)/8) – 64)/(8*N)) bits.

The calculation of PDCP SDU sizes does not consider timing advance MAC CE as timing advance is not transmitted by SS for RF test cases.

Note 5: 35160 bits for sub-frame 5

Note 6: 71112 bits for sub-frame 5

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: According to table 8.7.1.1.3-2.

1. Connect the SS to the UE antenna connector(s) as shown in TS 36.508 [7] Annex A, Figure A.3 for test 1 and Figure A.10 for tests 2-5 (without using faders and AWGN generators).
2. The parameter settings for the cell are set up according to Table 8.7.1.1.5-1 and Table 8.7.1.1.5-2.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 4 (Loopback activated) according to TS 36.508 [7] clause 4.5.4. Message contents are defined in clause 8.7.1.1.4.3.

#### 8.7.1.1.4.2 Test procedure

1. The SS looks up $TB_{size}$ in table 8.7.1.1.4.1-1 according to the UE category under test.
2. SS sets the counters $N_{DL_newtx}$ ,$N_{DL_retx}$, $N_{UL_PDCP}$, and $ND_{DL_PDCP}$ to 0.
3. If PHY requests a new DL HARQ transmission this TTI, the SS creates one or more PDCP SDUs to fill up the TB [Note 1], depending on $TB_{size}$, in accordance with Table 8.7.1.1.4.1-1. and cipher the PDCP SDUs. The SS then concatenates the resultant PDCP PDUs to form an RLC PDU and then a MAC PDU. The SS then transmits the MAC PDU. The SS then increments the transmitted DL subframe ($N_{DL_newtx}$) by one and $N_{DL_PDCP}$ by the number of PDCP SDUs included in the MAC PDU.
4. If PHY requests a DL HARQ retransmission in this TTI, the SS performs a HARQ retransmission and increments the retransmitted subframe ($N_{DL_retx}$) by one. [Note 2]
5. Steps 3 to 4 are repeated at every TTI for at least [300] frames or until statistical significance is fulfilled according to [Annex G FFS] and the SS waits for [300ms] to let any HARQ retransmissions and RLC retransmissions to finish.
6. For each PDCP SDU received at the SS, if the content of the data matches that of the truncated version of the original PDCP SDU generated at the SS, the SS increments $N_{UL_PDCP}$ by one
7. The SS calculates the TB success rate as $A = 100\%*N_{DL_newtx}/ (N_{DL_newtx} + N_{DL_retx})$
8. The SS calculates the PDCP SDU loss as $B = N_{DL_PDCP} - N_{UL_PDCP}$
9. The UE passes the test if $A \geq$ "corresponding TB success rate according to Table 8.7.1.1.3-3" and B = 0.

NOTE 1: if there is RLC PDU retransmission in this TTI, the SS forms as many number of new PDCP SDUs to fill the rest of the TB.

NOTE 2: the SS should prioritize the HARQ retransmissions over new HARQ transmissions. This is to minimize the RLC buffering/processing load at the UE in case of HARQ transmission error.

#### 8.7.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.7A, with the following exceptions:

**Table 8.7.1.1.4.3-1: CLOSE UE TEST LOOP (in the preamble)**

| Derivation Path: 36.509 clause 6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 1 1 1 1 | | |
| Skip indicator | 0 0 0 0 | | |
| Message type | 1 0 0 0 0 0 0 0 | | |
| UE test loop mode | 0 0 0 0 0 0 0 0 | UE test loop mode A | |
| UE test loop mode A LB setup | | | |
| Length of UE test loop mode A LB setup list in bytes | 0 0 0 0 0 0 1 1 | Length of one LB setup DRB (3 bytes) | |
| LB setup DRB | 0 0 0 0 0 0 0 0,<br>0 0 1 0 1 0 0 0,<br>0 0 0 Q4 Q3 Q2 Q1 Q0 | UL PDCP SDU size = 40 bits (5 bytes)<br>Q4..Q0 = Data Radio Bearer identity number for the default radio bearer. See 36.509 clause 6.1 | |
| UE test loop mode B LB setup | Not present | | |

**Table 8.7.1.1.4.3-2: *SecurityModeCommand* (in the preamble)**

| Derivation Path: TS36.508 clause 4.6.1 table 4.6.1-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SecurityModeCommand ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| securityModeCommand-r8 SEQUENCE { | | | |
| securityConfiguration SEQUENCE { | | | |
| cipheringAlgorithm | eea2 | | |
| nextHopChainingCount | Not present | | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.7.1.1.5 Test requirement

The requirements are specified in Table 8.7.1.1.5-1. The PDCP SDU success rate shall be sustained during at least [100] frames.

**Table 8.7.1.1.5-1: Test requirements for sustained downlink data rate (FDD)**

| Test | UE Category | Number of bits of a DL-SCH transport block received within a TTI per Codeword | Measurement channel | Number of PDCP SDU per Codeword | PDCP SDU size [Octets] | Reference value |
|---|---|---|---|---|---|---|
| | | | | | | TB success rate [%] |
| 1 | Category 1 | 10296 | R31-1 FDD | 1 | 1275 | 95 |
| 2 | Category 2 | 25456 | R31-2 FDD | 3 | 1054 | 95 |
| 3 | Category 3 (Note 1) | 51024 | R31-3 FDD | 5 | 1270 | 95 |
| 3A | Category 3 (Note 2) | 36696 (Note4) | R31-3A FDD | 4 | 1141 | 85 |
| 4 | Category 4 | 75376 (Note5) | R31-4 FDD | 7 | 1341 | 85 |

Note 1: If the operating band under test does not support 20 MHz channel bandwidth, then test is executed according to Test 3A.
Note 2: Applicable to operating bands supporting up to 10 MHz channel bandwidths
Note 3: For 2 layer transmissions, 2 transport blocks are received within a TTI
Note 4: 35160 bits for sub-frame 5
Note 5: 71112 bits for sub-frame 5
Note 6: The TB success rate is defined as TB success rate = $100\%*N_{DL_correct_rx}/ (N_{DL_newtx} + N_{DL_retx})$, where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

**Table 8.7.1.1.5-2: Test Parameters for sustained downlink data rate (FDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3,4 | Test 3A |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | 10 | 20 | 10 |
| Transmission mode | | | 1 | 3 | 3 | 3 |
| Antenna configuration | | | 1 x 2 | 2 x 2 | 2 x 2 | 2 x 2 |
| Propagation condition | | | Static propagation condition (Note 1) | | | |
| CodeBookSubsetRestriction bitmap | | | n/a | 10 | 10 | 10 |
| Downlink power allocation | $\rho_A$ | dB | 0 | -3 | -3 | -3 |
| | $\rho_B$ | dB | 0 | -3 | -3 | -3 |
| $\hat{E}_s$ at antenna port | | dBm/15kHz | -85 | -85 | -85 | -85 |
| Symbols for unused PRBs | | | [OCNG] | [OCNG] | [OCNG] | [OCNG] |
| Note 1: No external noise sources are applied | | | | | | |

# 8.7.2 TDD

## 8.7.2.1 TDD sustained data rate performance

### 8.7.2.1.1 Test purpose

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum number of DL-SCH transport block bits received within a TTI for the UE category indicated. The sustained downlink data rate shall be verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the RF conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement. The size of the TB per TTI corresponds to the largest possible DL-SCH transport block for each UE category using the maximum number of layers for spatial multiplexing. Transmission modes 1 and 3 are used with radio conditions resembling a scenario where sustained maximum data rates are available.

### 8.7.2.1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 8.7.2.1.3 Minimum requirements

The parameters specified in Table 8.7.1.1.3-1 are valid for all TDD tests unless otherwise stated.

**Table 8.7.2.1.3-1: Common Test Parameters (TDD)**

| Parameter | Unit | Value |
|---|---|---|
| Uplink downlink configuration (Note 1) | | 5 |
| Special subframe configuration (Note 2) | | 4 |
| Cyclic prefix | | Normal |
| Cell ID | | 0 |
| Inter-TTI Distance | | 1 |
| Number of HARQ processes | Processes | Downlink: 7<br>Uplink: 1 |
| Maximum number of HARQ transmission | | Downlink: 4<br>Uplink: 1 |
| Redundancy version coding sequence | | {0,0,1,2} for 64QAM |
| Number of OFDM symbols for PDCCH | OFDM symbols | 1 |
| Note 1: as specified in Table 4.2-2 in TS 36.211 [4]<br>Note 2: as specified in Table 4.2-1 in TS 36.211 [4] | | |

The requirements are specified in Table 8.7.2.1.3-3, with the addition of the parameters in Table 8.7.2.1.3-2 and the downlink physical channel setup according to Annex C.3.2. The PDCP SDU success rate shall be sustained during at least 300 frames.

**Table 8.7.2.1.3-2: test parameters for sustained downlink data rate (TDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3,4 | Test 3B |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | 10 | 20 | 15 |
| Transmission mode | | | 1 | 3 | 3 | 3 |
| Antenna configuration | | | 1 x 2 | 2 x 2 | 2 x 2 | 2 x 2 |
| Propagation condition | | | Static propagation condition (Note 1) | | | |
| CodeBookSubsetRestriction bitmap | | | n/a | 10 | 10 | 10 |
| Downlink power allocation | $\rho_A$ | dB | 0 | -3 | -3 | -3 |
| | $\rho_B$ | dB | 0 | -3 | -3 | -3 |
| $\hat{E}_s$ at antenna port | | dBm/15kHz | -85 | -85 | -85 | -85 |
| Symbols for unused PRBs | | | [OCNG] | [OCNG] | [OCNG] | [OCNG] |
| ACK/NACK feedback mode | | | Bundling | Bundling | Bundling | Bundling |
| Note 1: No external noise sources are applied | | | | | | |

**Table 8.7.2.1.3-3: Minimum requirement (TDD)**

| Test | UE Category | Number of bits of a DL-SCH transport block received within a TTI for normal/special sub-frame | Measurement channel | Reference value |
|---|---|---|---|---|
| | | | | TB success rate [%] |
| 1 | Category 1 | 10296/0 | R31-1 TDD | 95 |
| 2 | Category 2 | 25456/0 | R31-2 TDD | 95 |
| 3 | Category 3 (Note 1) | 51024/0 | R31-3 TDD | 95 |
| 3B | Category 3 (Note 2) | 51024/0 | R31-3B TDD | 85 |
| 4 | Category 4 | 75376/0 (Note 4) | R31-4 TDD | 90 |
| 5 | Category 5 | FFS | FFS | FFS |

Note 1: If the operating band under test does not support 20 MHz channel bandwidth, then test is executed according to Test 3B.
Note 2: Applicable to operating bands supporting up to 15 MHz channel bandwidths.
Note 3: For 2 layer transmissions, 2 transport blocks are received within a TTI
Note 4: 71112 bits for sub-frame 5
Note 5: The TB success rate is defined as TB success rate = 100%*$N_{DL_correct_rx}$/($N_{DL_newtx}$ + $N_{DL_retx}$), where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

The normative reference for this requirement is TS 36.101[2] clause 8.7.2

### 8.7.2.1.4 Test description

#### 8.7.2.1.4.1 Initial conditions

**Table 8.7.2.1.4.1-1: Applicable test and Transport Block Size for different UE categories**

| UE Category | Applicable test | DL Measurement channel | UL Measurement Channel | $TB_{size}$ per Codeword | Number of PDCP SDU per Codeword for normal/special sub-frame | PDCP SDU size [bits] Note 3 |
|---|---|---|---|---|---|---|
| 1 | Test 1 | R31-1 TDD | R.1-1 TDD | 10296 | 1/0 | 8*FLOOR(($TB_{size}$ - 96)/8) |
| 2 | Test 2 | R31-2 TDD | R.1-2 TDD | 25456 | 3/0 | 8*FLOOR(($TB_{size}$ – 152)/24)) |
| 3 | Test 3 (Note 1) | R31-3 TDD | R.1-3 TDD | 51024 | 5/0 | 8*FLOOR(($TB_{size}$ – 208)/40)) |
| 3 | Test 3B (Note 2) | R31-3B TDD | R.1-3B TDD | TBD | 4/0 | 8*FLOOR(($TB_{size}$ – 184)/32))" |
| 4 | Test 4 (Note 2) | R31-4 TDD | R.1-4 TDD | 75376 (Note 5) | 7/0 | 8*FLOOR(($TB_{size}$ – 264)/56)) |
| 5 | FFS | FFS | FFS | FFS | FFS | FFS |

Note 1: If the operating band under test does not support 20 MHz channel bandwidth, then test is executed according to Test 3B.

Note 2: Applicable to operating bands supporting up to 15 MHz channel bandwidths

Note 3: Transport block size under test according to applicable Fixed Reference Channel for sustained data-rate test in annex A.3.9.

Note 4: Each PDCP SDU is limited to 1500 octets (to keep below maximum SDU size of ESM as specified in TS 24.301 clause 9.9.4.12).

The PDCP SDU size of each PDCP SDU is:

PDCP SDU size = (TBsize – N*PDCP header size - AMD PDU header size - MAC header size – Size of Timing Advance - Size of RLC STATUS PDU) / N,

where PDCP header size is 16 bits for the RLC AM and 12-bit SN case; AMD PDU header size is CEIL[(16+(N-1)*12)/8] bytes which includes 16 bit standard AM header and (N-1) Length indicators; and MAC header size = R/R/E/LCID/F/L MAC subheader (24 bits for MAC SDU for RLC STATUS PDU with 15 bit LI) + R/R/E/LCID MAC subheader (8 bits for MAC SDU for RLC data PDU) = 32 bits. The size of RLC STATUS PDU including one ACK_SN field and one NACK_SN field is 32 bits (if no STATUS PDU is sent or if the size of the STATUS PDU is less than 32 bits then padding will be used to fill the 32 bits). This gives: PDCP SDU size = 8*FLOOR((TBsize – N*16- 8*CEIL((16+(N-1)*12)/8) – 64)/(8*N)) bits.

The calculation of PDCP SDU sizes does not consider timing advance MAC CE as timing advance is not transmitted by SS for RF test cases.

Note 5: 71112 bits for sub-frame 5

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: According to table 8.7.2.1.3-2.

1. Connect the SS, to the UE antenna connector(s) as shown in TS 36.508 [7] Annex A, Figure A.3 for test 1 and Figure A.10 for tests 2-5 (without using faders and AWGN generators).

2. The parameter settings for the cell are set up according to Table 8.7.2.1.5-1 and Table 8.7.2.1.5-2.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 4 (Loopback activated) according to TS 36.508 [7] clause 4.5.4. Message contents are defined in clause 8.7.2.1.4.3.

#### 8.7.2.1.4.2 Test procedure

1. The SS looks up $TB_{size}$ in table 8.7.2.1.4.1-1 according to the UE category under test.

2. SS sets the counters $N_{DL_newtx}$ ,$N_{DL_retx}$, $N_{UL_PDCP}$, and $ND_{DL_PDCP}$ to 0.

3. If PHY requests a new DL HARQ transmission this TTI, the SS creates one or more PDCP SDUs to fill up the TB [Note 1], depending on $TB_{size}$, in accordance with Table 8.7.2.1.4.1-1. and cipher the PDCP SDUs The SS then concatenates the resultant PDCP PDUs to form an RLC PDU and then a MAC PDU. The SS then transmits the MAC PDU. The SS then increments the transmitted DL subframe ($N_{DL_newtx}$) by one and $N_{DL_PDCP}$ by the number of PDCP SDUs included in the MAC PDU.

4. If PHY requests a DL HARQ retransmission in this TTI, the SS performs a HARQ retransmission and increments the retransmitted subframe ($N_{DL_retx}$) by one. [Note 2]

5. Steps 3 to 4 are repeated at every TTI for at least [300] frames or until statistical significance is fulfilled according to [Annex G FFS] and the SS waits for [300ms] to let any HARQ retransmissions and RLC retransmissions to finish.

6. For each PDCP SDU received at the SS, if the content of the data matches that of the truncated version of the original PDCP SDU generated at the SS, the SS increments $N_{UL_PDCP}$ by one.

7. The SS calculates the TB success rate as $A = 100\%*N_{DL_newtx}/ (N_{DL_newtx} + N_{DL_retx})$

8. The SS calculates the PDCP SDU loss as $B = N_{DL_PDCP} - N_{UL_PDCP}$

9. The UE passes the test if A ≥ "corresponding TB success rate according to Table 8.7.2.1.3-3" and B = 0

NOTE 1: if there is RLC PDU retransmission in this TTI, the SS forms as many number of new PDCP SDUs to fill the rest of the TB.

NOTE 2: the SS should prioritize the HARQ retransmissions over new HARQ transmissions. This is to minimize the RLC buffering/processing load at the UE in case of HARQ transmission error.

#### 8.7.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 and 4.7A, with the following exceptions:

**Table 8.7.2.1.4.3-1: CLOSE UE TEST LOOP (in the preamble)**

| Derivation Path: 36.509 clause 6.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 1 1 1 1 | | |
| Skip indicator | 0 0 0 0 | | |
| Message type | 1 0 0 0 0 0 0 0 | | |
| UE test loop mode | 0 0 0 0 0 0 0 0 | UE test loop mode A | |
| UE test loop mode A LB setup | | | |
| Length of UE test loop mode A LB setup list in bytes | 0 0 0 0 0 0 1 1 | Length of one LB setup DRB (3 bytes) | |
| LB setup DRB | 0 0 0 0 0 0 0 0,<br>0 0 101 0 0 0,<br>0 0 0 Q4 Q3 Q2 Q1 Q0 | UL PDCP SDU size = 40 bits (5 bytes)<br>Q4..Q0 = Data Radio Bearer identity number for the default radio bearer. See 36.509 clause 6.1. | |
| UE test loop mode B LB setup | Not present | | |

**Table 8.7.2.1.4.3-2: *SecurityModeCommand* (in the preamble)**

| Derivation Path: TS36.508 clause 4.6.1 table 4.6.1-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SecurityModeCommand ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE{ | | | |
| securityModeCommand-r8 SEQUENCE { | | | |
| securityConfiguration SEQUENCE { | | | |
| cipheringAlgorithm | eea2 | | |
| nextHopChainingCount | Not present | | |
| } | | | |
| nonCriticalExtension SEQUENCE {} | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

## 8.7.2.1.5 Test requirement

The requirements are specified in Table 8.7.2.1.5-1. The PDCP SDU success rate shall be sustained during at least [100] frames.

**Table 8.7.2.1.5-1: Test requirements for sustained downlink data rate (TDD)**

| Test | UE Category | Number of bits of a DL-SCH transport block received within a TTI per codeword for normal/special sub-frame | Measurement channel | Number of PDCP SDU per TTI per codeword for normal/special sub-frame | PDCP SDU size for normal/special sub-frame [Octets] | Reference value |
|---|---|---|---|---|---|---|
| | | | | | | TB success rate [%] |
| 1 | Category 1 | 10296/0 | R31-1 TDD | 1/0 | 1275/0 | 95 |
| 2 | Category 2 | 25456/0 | R31-2 TDD | 3/0 | 1054/0 | 95 |
| 3 | Category 3 (Note 1) | 51024/0 | R31-3 TDD | 5/0 | 1270/0 | 95 |
| 3B | Category 3 (Note 2) | 51024 | R31-3B TDD | 4/0 | 1588 | 85 |
| 4 | Category 4 | 75376/0 | R31-4 TDD | 7/0 | 1341/0 | 85 |
| 5 | Category 5 | FFS | FFS | FFS | FFS | FFS |

Note 1: If the operating band under test does not support 20 MHz channel bandwidth, then test is executed according to Test 3B.
Note 2: Applicable to operating bands supporting up to 15 MHz channel bandwidths.
Note 3: For 2 layer transmissions, 2 transport blocks are received within a TTI
Note 4: 71112 bits for sub-frame 5
Note 5: The TB success rate is defined as TB success rate = $100\%*N_{DL_correct_rx}/ (N_{DL_newtx} + N_{DL_retx})$, where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

**Table 8.7.2.1.5-2: test parameters for sustained downlink data rate (TDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3,4 | Test 3B |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | 10 | 20 | 15 |
| Transmission mode | | | 1 | 3 | 3 | 3 |
| Antenna configuration | | | 1 x 2 | 2 x 2 | 2 x 2 | 2 x 2 |
| Propagation condition | | | Static propagation condition (Note 1) | | | |
| CodeBookSubsetRestriction bitmap | | | n/a | 10 | 10 | 10 |
| Downlink power allocation | $\rho_A$ | dB | 0 | -3 | -3 | -3 |
| | $\rho_B$ | dB | 0 | -3 | -3 | -3 |
| $\hat{E}_s$ at antenna port | | dBm/15kHz | -85 | -85 | -85 | -85 |
| Symbols for unused PRBs | | | [OCNG] | [OCNG] | [OCNG] | [OCNG] |
| Note 1: No external noise sources are applied | | | | | | |

# 9 Reporting of Channel State Information

## 9.1 General

For the cases in this clause it is expected that the UE will not always detect the PDCCH, resulting in a statDTX for the uplink ACK/NACK transmission. The downlink configuration for evaluating CQI performance does not use retransmission. Therefore any BLER and Throughput calculations must exclude any packets where the UE may have attempted to combine data from more than one transmission due to missed new data indicators from lost PDCCH transmissions. Thus in all test cases in which there are not retransmission (Max number of HARQ transmissions = 1), if the number of consecutive discarded statDTX for any one process is an odd number including one, also discard the next response for that HARQ process regardless whether it is an ACK or NACK. In the calculation of Throughput the discarded ACK / NACK is considered as a statDTX.

This section includes requirements for the reporting of channel state information (CSI).

The fading of the signals and the AWGN signals applied to each receiver antenna connector shall be uncorrelated. The levels of the test signal applied to each of the antenna connectors shall be as defined in the respective test cases.

The UE performance in this section is considered to be operating band independent. Therefore, the required performance in the respective test cases can be verified in one of the operating bands supported by the UE under test. All the test points supported by the bands of the multiband UE (based on channel bandwidth, DL and UL configuration) need to be tested.

## 9.2 CQI Reporting under AWGN conditions

The reporting accuracy of the channel quality indicator (CQI) under frequency non-selective conditions is determined by the reporting variance and the BLER performance using the transport format indicated by the reported CQI median. The purpose is to verify that the reported CQI values are in accordance with the CQI definition given in TS 36.213[10] clause 7.2.To account for sensitivity of the input SNR the reporting definition is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

### 9.2.1 CQI Reporting under AWGN conditions - PUCCH 1-0

#### 9.2.1.1 FDD CQI Reporting under AWGN conditions – PUCCH 1-0

##### 9.2.1.1.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

##### 9.2.1.1.2 Test applicability

This test applies to E-UTRA FDD UE release 8 and forward of UE category 2-5.

##### 9.2.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.2.1.1.3-1, and using the downlink physical channels specified in tables C.3.2-1 and C.3.2-2, the reported CQI value according to Table A.4-1 shall be in the range of ±1 of the reported median more than 90% of the time. If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, the BLER using the transport format indicated by the (median CQI +1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, the BLER using transport format indicated by (median CQI –1) shall be less than or equal to 0.1

**Table 9.2.1.1.3-1: PUCCH 1-0 static test**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| PDSCH transmission mode | | | 1 | | | |
| Downlink power allocation | $\rho_A$ | dB | 0 | | | |
| | $\rho_B$ | dB | 0 | | | |
| Propagation condition and antenna configuration | | | AWGN (1 x 2) | | | |
| SNR (Note 2) | | dB | 0 | 1 | 6 | 7 |
| $\hat{I}_{or}^{(j)}$ | | dB[mW/15kHz] | -98 | -97 | -92 | -91 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 | | -98 | |
| Max number of HARQ transmissions | | | 1 | | | |
| Physical channel for CQI reporting | | | PUCCH Format 2 | | | |
| PUCCH Report Type | | | 4 | | | |
| Reporting periodicity | | ms | $N_P$ = 5 | | | |
| *cqi-pmi-ConfigurationIndex* | | | 6 | | | |

Note 1: Reference measurement channel according to Table A.4-1 with one sided dynamic OCNG Pattern OP.1 FDD as described in Annex A.5.1.1.

Note 2: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

## 9.2.1.1.4 Test description

### 9.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Table 9.2.1.1.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.2.1.1.4.3.

### 9.2.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 9.2.1.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband

CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and 1800 or more of the wideband CQI values are in the range (Median CQI - 1) ≤ Median CQI ≤ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

   For the filtered ACK and NACK responses if the ratio ( NACK / ACK + NACK) ≤ 0.1 then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK ) > 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK ) ≤ 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. If both tests have not been done, then repeat the same procedure (steps 1 to 8) with test conditions according to the table 9.2.1.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.2.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.1.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.2.1.1.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 6 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

### 9.2.1.1.5 Test requirement

The pass fail decision is as specified in the test procedure in clause 9.2.1.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 9.2.1.2 TDD CQI Reporting under AWGN conditions – PUCCH 1-0

### 9.2.1.2.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based on wideband median CQI-1 and wideband median CQI or the transport format based on wideband median CQI and wideband median CQI +1.

### 9.2.1.2.2 Test applicability

This test applies to E-UTRA TDD UE release 8 and forward of UE category 2-5.

### 9.2.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.2.1.2.3-1, and using the downlink physical channels specified in tables C.3.2-1 and C.3.2-2, the reported CQI value according to Table A.4-2 shall be in the range of ±1 of the reported median more than 90% of the time. If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, the BLER using the transport format indicated by the (median CQI +1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, the BLER using transport format indicated by (median CQI –1) shall be less than or equal to 0.1.

**Table 9.2.1.2.3-1: PUCCH 1-0 static test (TDD)**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| PDSCH transmission mode | | | 1 | | | |
| Uplink downlink configuration | | | 2 | | | |
| Special subframe configuration | | | 4 | | | |
| Downlink power allocation | $\rho_A$ | dB | 0 | | | |
| | $\rho_B$ | dB | 0 | | | |
| Propagation condition and antenna configuration | | | AWGN (1 x 2) | | | |
| SNR (Note 2) | | dB | 0 | 1 | 6 | 7 |
| $\hat{I}_{or}^{(j)}$ | | dB[mW/15kHz] | -98 | -97 | -92 | -91 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 | | -98 | |
| Maximum number of HARQ transmissions | | | 1 | | | |
| Physical channel for CQI reporting | | | PUSCH (Note 3) | | | |
| PUCCH Report Type | | | 4 | | | |
| Reporting periodicity | | ms | $N_P$ = 5 | | | |
| *cqi-pmi-ConfigurationIndex* | | | 3 | | | |
| ACK/NACK feedback mode | | | Multiplexing | | | |

Note 1: Reference measurement channel according to clause A.4-2 with one sided dynamic OCNG Pattern OP.1 TDD as described in Annex A.5.2.1

Note 2: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 3: To avoid collisions between CQI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#3 and #8 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#7 and #2.

## 9.2.1.2.4 Test description

### 9.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 9.2.1.2.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.2.1.2.4.3.

9.2.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 9.2.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH.Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and 1800 or more of the wideband CQI values are in the range (Median CQI - 1) ≤ Median CQI ≤ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. *In case statDTX can not be differentiated from NACK due to multiplexing effect, evaluate the feedback as a NACK* The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

   For the filtered ACK and NACK responses if the ratio ( NACK / ACK + NACK) ≤ 0.1 then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered. *In case statDTX can not be differentiated from NACK due to multiplexing effect, evaluate the feedback as a NACK*

   If the ratio (NACK /ACK + NACK ) > 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered. *In case statDTX can not be differentiated from NACK due to multiplexing effect, evaluate the feedback as a NACK*

   If the ratio (NACK /ACK + NACK ) ≤ 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. If both tests have not been done, then repeat the same procedure (steps 1 to 8 ) with test conditions according to the table 9.2.1.2.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.2.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.1.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.2.1.2.4.3-2: CQI-ReportConfig-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 3 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.1.2.4.3-3: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | sa2 | | |
| specialSubframePatterns | ssp4 | | |
| } | | | |

#### 9.2.1.2.5 Test requirement

The pass fail decision is as specified in the test procedure in clause 9.2.1.2.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 9.2.2 CQI Reporting under AWGN conditions - PUCCH 1-1

### 9.2.2.1 FDD CQI Reporting under AWGN conditions – PUCCH 1-1

#### 9.2.2.1.1 Test purpose

To verify the variance of the wideband spatial differential CQI between codeword #0 and codeword #1 are within the limits defined and for both codeword #0 and codeword #1, the PDSCH BLER using the transport format indicated by the respective median $CQI_0 - 1$ and median $CQI_1 - 1$ shall be less than or equal to 0.1 and the PDSCH BLER using the transport format indicated by the respective median $CQI_0 + 1$ and median $CQI_1 + 1$ shall be greater than or equal to 0.1.

### 9.2.2.1.2 Test applicability

This test applies to E-UTRA FDD UE release 8 and forward of UE category 2-5.

### 9.2.2.1.3 Minimum conformance requirements

The minimum requirements for dual codeword transmission are defined in terms of a reporting spread of the wideband CQI value for codeword #1, and their BLER performance using the transport format indicated by the reported CQI median of codeword #0 and codeword #1. The precoding used at the transmitter is a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. The propagation condition assumed for the minimum performance requirement is defined in subclause B.1.

For the parameters specified in table 9.2.2.1.3-1, and using the downlink physical channels specified in tables C.3.2-1 and C.3.2-2, the reported offset level of the wideband spatial differential CQI for codeword #1 (Table 7.2.2 in TS 36.213[10]) shall be used to determine the wideband CQI index for codeword #1 as

$$\text{wideband CQI}_1 = \text{wideband CQI}_0 - \text{Codeword 1 offset level}$$

The wideband $CQI_1$ shall be within the set {median $CQI_1$ -1, median $CQI_1$ +1} for more than 90% of the time, where the resulting wideband values $CQI_1$ shall be used to determine the median CQI values for codeword #1. For both codewords #0 and #1, the PDSCH BLER using the transport format indicated by the respective median $CQI_0 - 1$ and median $CQI_1 - 1$ shall be less than or equal to 0.1. Furthermore, for both codewords #0 and #1, the PDSCH BLER using the transport format indicated by the respective median $CQI_0 + 1$ and median $CQI_1 + 1$ shall be greater than or equal to 0.1.

**Table 9.2.2.1.3-1: PUCCH 1-1 static test (FDD)**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| PDSCH transmission mode | | | 4 | | | |
| Downlink power allocation | $\rho_A$ | dB | -3 | | | |
| | $\rho_B$ | dB | -3 | | | |
| Propagation condition and antenna configuration | | | Clause B.1 (2 x 2) | | | |
| CodeBookSubsetRestriction bitmap | | | 010000 | | | |
| SNR (Note 2) | | dB | 10 | 11 | 16 | 17 |
| $\hat{I}_{or}^{(j)}$ | | dB[mW/15kHz] | -88 | -87 | -82 | -81 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 | | -98 | |
| Max number of HARQ transmissions | | | 1 | | | |
| Physical channel for CQI/PMI reporting | | | PUCCH Format 2 | | | |
| PUCCH Report Type for CQI/PMI | | | 2 | | | |
| PUCCH Report Type for RI | | | 3 | | | |
| Reporting periodicity | | ms | $N_P = 5$ | | | |
| *cqi-pmi-ConfigurationIndex* | | | 6 | | | |
| *ri-ConfigIndex* | | | 1 (Note 3) | | | |

Note 1: Reference measurement channel according to Table A.4-1 with one sided dynamic OCNG Pattern OP.1 FDD as described in Annex A.5.1.1

Note 2: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 3: It is intended to have UL collisions between RI reports and HARQ-ACK, since the RI reports shall not be used by the eNB in this test.

#### 9.2.2.1.4 Test description

##### 9.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.2.2.1.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.2.2.1.4.3.

##### 9.2.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 9.2.2.1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC including two codewords with spatial multiplexing both using the transport format according to CQI value 8 of Annex A.4 Table A.4-3a and keep them regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. The SS shall use a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. Continue transmission of the PDSCH until 2000 wideband and wideband spatial differential CQI reports have been gathered. In this process the SS collects wideband and wideband spatial differential CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband and wideband spatial differential CQI reports respectively.
3. From each wideband CQI report, wideband $CQI_0$ is defined as Wideband CQI of codeword #0 and wideband $CQI_1$ is calculated according to clause 9.2.2.1.3. Codeword 1 offset level is selected from {0,1,2,3,-4,-3,-2,-1}. Set up a relative frequency distribution for the wideband CQI-values, reported. Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side) for each codewords. Wideband Median $CQI_0$ is based on the wideband $CQI_0$ and wideband median $CQI_1$ is based on the wideband $CQI_1$.
4. If Median CQI is not equal to 1 or 15 and 1800 or more of the wideband $CQI_1$ values are in the range (Median $CQI_1$ - 1) $\leq$ Median CQI $\leq$ ( Median $CQI_1$ + 1) then continue with step 5, otherwise go to step 7.
5. The SS shall transmit PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC including two codewords with spatial multiplexing where the transport format of codeword #0 is according to the wideband median- $CQI_0$– 1 and the transport format of codeword #1 is according to the wideband median $CQI_1$– 1. The SS sends downlink MAC padding bits on the DL RMC. The SS shall not react to the any wideband CQI reports from UE and shall use a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses for each codewords respectively. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses for each codewords reaches 1000.

   If the ratio (NACK /ACK + NACK ) $\leq$ 0.1 for both codeword #0 and codeword #1

   then and go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC including two codewords with spatial multiplexing where the transport format of codeword #0 is according to the wideband median-$CQI_0$ + 1 and the transport format of codeword #1 is according to the wideband median-$CQI_1$ + 1. The SS sends downlink MAC padding bits on the DL RMC. The SS shall not react to the any wideband CQI reports from UE and shall use a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses for each codewords respectively. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses for each codewords reaches 1000.

   If the ratio (NACK /ACK + NACK ) ≥ 0.1 for both codeword #0 and codeword #1

   then pass the UE for this test and go to step 8, otherwise go to step 7.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the table 9.2.2.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.2.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.2.1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| } | | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 010000 | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.2.1.4.3-2: *PDSCH-ConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-3 | | |
| } | | | |

**Table 9.2.2.1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 6 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 1 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

### 9.2.2.1.5 Test requirement

The pass fail decision is as specified in the test procedure in clause 9.2.2.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 9.2.2.2 TDD CQI Reporting under AWGN conditions – PUCCH 1-1

### 9.2.2.2.1 Test purpose

To verify the variance of the wideband spatial differential CQI between codeword #0 and codeword #1 are within the limits defined and for both codeword #0 and codeword #1, the PDSCH BLER using the transport format indicated by the respective wideband median $CQI_0 - 1$ and wideband median $CQI_1 - 1$ shall be less than or equal to 0.1 and the PDSCH BLER using the transport format indicated by the respective wideband median $CQI_0 + 1$ and wideband median $CQI_1 + 1$ shall be greater than or equal to 0.1.

### 9.2.2.2.2 Test applicability

This test applies to E-UTRA TDD UE release 8 and forward of UE category 2-5.

### 9.2.2.2.3 Minimum conformance requirements

For the parameters specified in table 9.2.2.2.3-1, and using the downlink physical channels specified in tables C.3.2-1 and C.3.2-2, the reported offset level of the wideband spatial differential CQI for codeword #1 (Table 7.2.2 in TS 36.213[10]) shall be used to determine the wideband CQI index for codeword #1 as

$$\text{wideband } CQI_1 = \text{wideband } CQI_0 - \text{Codeword 1 offset level}$$

The wideband $CQI_1$ shall be within the set {median $CQI_1$ -1, median $CQI_1$ +1} for more than 90% of the time, where the resulting wideband values $CQI_1$ shall be used to determine the median CQI values for codeword #1. For both codewords #0 and #1, the PDSCH BLER using the transport format indicated by the respective median $CQI_0 - 1$ and median $CQI_1 - 1$ shall be less than or equal to 0.1. Furthermore, for both codewords #0 and #1, the PDSCH BLER using the transport format indicated by the respective median $CQI_0 + 1$ and median $CQI_1 + 1$ shall be greater than or equal to 0.1.

**Table 9.2.2.2.3-1: PUCCH 1-1 static test (TDD)**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| PDSCH transmission mode | | | 4 | | | |
| Uplink downlink configuration | | | 2 | | | |
| Special subframe configuration | | | 4 | | | |
| Downlink power allocation | $\rho_A$ | dB | -3 | | | |
| | $\rho_B$ | dB | -3 | | | |
| Propagation condition and antenna configuration | | | Clause B.1 (2 x 2) | | | |
| CodeBookSubsetRestriction bitmap | | | 010000 | | | |
| SNR (Note 2) | | dB | 10 | 11 | 16 | 17 |
| $\hat{I}_{or}^{(j)}$ | | dB[mW/15kHz] | -88 | -87 | -82 | -81 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 | | -98 | |
| Maximum number of HARQ transmissions | | | 1 | | | |
| Physical channel for CQI/PMI reporting | | | PUSCH (Note 3) | | | |
| PUCCH Report Type | | | 2 | | | |
| Reporting periodicity | | ms | $N_P$ = 5 | | | |
| *cqi-pmi-ConfigurationIndex* | | | 3 | | | |
| *ri-ConfigIndex* | | | 805 (Note 4) | | | |
| ACK/NACK feedback mode | | | Multiplexing | | | |

Note 1: Reference measurement channel according to Table A.4-2 with one sided dynamic OCNG Pattern OP.1 TDD as described in Annex A.5.2.1

Note 2: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 3: To avoid collisions between CQI/PMI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#3 and #8 to allow periodic CQI/PMI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#7 and #2.

Note 4: RI reporting interval is set to the maximum allowable length of 160ms to minimise collisions between RI, CQI/PMI and HARQ-ACK reports. In the case when all three reports collide, it is expected that CQI/PMI reports will be dropped, while RI and HARQ-ACK will be multiplexed. At eNB, CQI report collection shall be skipped every 160ms during performance verification.

## 9.2.2.2.4 Test description

### 9.2.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Table 9.2.2.2.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2 and uplink signals according to Annex H1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.2.2.2.4.3.

#### 9.2.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 9.2.2.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC including two codewords with spatial multiplexing both using the transport format according to CQI value 8 of Annex A.4 Table A.4-3a and keep them regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. The SS shall use a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. Continue transmission of the PDSCH until 2000 wideband and wideband spatial differential CQI reports have been gathered. In this process the SS collects wideband and wideband spatial differential CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband and wideband spatial differential CQI reports respectively.

3. From each wideband CQI report, wideband $CQI_0$ is defined as Wideband CQI of codeword #0 and wideband $CQI_1$ is calculated according to clause 9.2.2.2.3. Codeword 1 offset level is selected from {0,1,2,3,-4,-3,-2,-1}. Set up a relative frequency distribution for the wideband CQI-values, reported. Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side) for each codewords. Wideband Median $CQI_0$ is based on the wideband $CQI_0$ and wideband median $CQI_1$ is based on the wideband $CQI_1$.

4. If Median CQI is not equal to 1 or 15 and 1800 or more of the wideband $CQI_1$ values are in the range (Median $CQI_1$ - 1) ≤ Median CQI ≤ ( Median $CQI_1$ + 1) then continue with step 5, otherwise go to step 7.

5. The SS shall transmit PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC including two codewords with spatial multiplexing where the transport format of codeword #0 is according to the wideband median- $CQI_0$– 1 and the transport format of codeword #1 is according to the wideband median $CQI_1$– 1. The SS sends downlink MAC padding bits on the DL RMC. The SS shall not react to the any wideband CQI reports from UE and shall use a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses for each codewords respectively. *In case statDTX can not be differentiated from NACK due to multiplexing effect, evaluate the feedback as a NACK* The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses for each codewords reaches 1000.

   If the ratio (NACK /ACK + NACK ) ≤ 0.1

   then,and go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC including two codewords with spatial multiplexing where the transport format of codeword #0 is according to the wideband median-$CQI_0$ + 1 and the transport format of codeword #1 is according to the wideband median-$CQI_1$ + 1. The SS sends downlink MAC padding bits on the DL RMC. The SS shall not react to the any wideband CQI reports from UE and shall use a fixed precoding matrix specified by the bitmap parameter *codebookSubsetRestriction*. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses for each codewords respectively. *In case statDTX can not be differentiated from NACK due to multiplexing effect, evaluate the feedback as a NACK* The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses for each codewords reaches 1000.

   If the ratio (NACK /ACK + NACK ) ≥ 0.1

   then pass the UE for this test and go to step 8, otherwise go to step 7.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the table 9.2.2.2.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.2.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.2.2.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| } | | | |
| codebookSubsetRestriction CHOICE { | | | |
| n2TxAntenna-tm4 | 010000 | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.2.2.4.3-2: *PDSCH-ConfigDedicated-DEFAULT***

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-6 PDSCH-ConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-3 | | |
| } | | | |

**Table 9.2.2.2.4.3-3: CQI-ReportConfig-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 3 | | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 805 | | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.2.2.2.4.3-4: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa2 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

#### 9.2.2.2.5 Test requirement

The pass fail decision is as specified in the test procedure in clause 9.2.1.2.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

# 9.3 CQI Reporting under fading conditions

## 9.3.1 Frequency-selective scheduling mode

The accuracy of sub-band channel quality indicator (CQI) reporting under frequency selective fading conditions is determined by a double-sided percentile of the reported differential CQI offset level 0 per sub-band, and the relative increase of the throughput obtained when transmitting on any one of the sub-bands with the highest reported differential CQI offset level the corresponding transport format compared to the case for which a fixed format is transmitted on any sub-band in set of TS 36.213[10]. To account for sensitivity of the input SNR the sub-band CQI reporting under frequency selective fading conditions is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

### 9.3.1.1 CQI Reporting under fading conditions – PUSCH 3-0

#### 9.3.1.1.1 FDD CQI Reporting under fading conditions – PUSCH 3-0

##### 9.3.1.1.1.1 Test purpose

To verify that preferred sub-bands can be used for frequently-selective scheduling.

##### 9.3.1.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

##### 9.3.1.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.3.1.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.1.1.1.3-2 and by the following

a) a sub-band differential CQI offset level of 0 shall be reported at least $\alpha$% of the time but less than $\beta$% for each sub-band;

b) the ratio of the throughput obtained when transmitting a randomly selected sub-band among the sub-bands with the highest differential CQI offset level the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected sub-band in set *S* shall be $\geq \gamma$;

c) when transmitting a randomly selected sub-band among the sub-bands with the highest differential CQI offset level the corresponding TBS, the average BLER for the indicated transport formats shall be greater or equal to 0.05.

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each TTI. The transport block size TBS(wideband CQI median) is that resulting from the

code rate which is closest to that indicated by the wideband CQI median and the $N_{PRB}$ entry in Table 7.1.7.2.1-1 of TS 36.213[10] that corresponds to the sub-band size.

**Table 9.3.1.1.1.3-1: Sub-band test for single antenna transmission (FDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| SNR (Note 3) | dB | 9 | 10 | 14 | 15 |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -89 | -88 | -84 | -83 |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | -98 | | -98 | |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s, $a$ = 1, $f_D = 5$ Hz | | | |
| Correlation | | Full | | | |
| Reporting interval | ms | 5 | | | |
| CQI delay | ms | 8 | | | |
| Reporting mode | | PUSCH 3-0 | | | |
| Max number of HARQ transmissions | | 1 | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-4 with one/two sided dynamic OCNG Pattern OP.1/2 FDD as described in Annex A.5.1.1/2Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

**Table 9.3.1.1.1.3-2: Minimum requirement (FDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.1 | 1.1 |
| UE Category | 1-5 | 1-5 |

### 9.3.1.1.1.4 Test description

#### 9.3.1.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Table 9.3.1.1.1.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.1.1.1.4.3.

#### 9.3.1.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 9.3.1.1.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI and full-size subband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. Check if "2000*$\alpha$%/100 ≤ number of CQI reports with index 0 for each full-size subband ≤ 2000*$\beta$%/100". (2000= No of full-size subband reports, 100 because of %) If yes, continue with step5, otherwise goto step 7.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI value in an each TTI randomly selected full-size subband regardless of UE wideband and full-size subband CQI report. Note that each full-size subband is selected in equal probability. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the highest UE reported full-size subband CQI value in one full-size subband selected among the subbands in which UE reports the highest full-size subband CQI. Subband differential CQI offset level is selected from {0, 1, 2, -1}. Note that the SS shall send PDSCH in the same full-size subband until next UE report is available. In case when same full-size subbands are reported subsequently as subbands with highest full-size subband CQI, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput and (NACK /(ACK + NACK)) according to Annex G.5.3. Declare the throughput as $t_{subband}$. If the ratio ($t_{subband}$ / $t_{median}$) ≥ $\gamma$ and (NACK /(ACK + NACK))≥ 0.05 , pass the UE for this test and go to step 8. Otherwise, go to step 7.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the table 9.3.1.1.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.3.1.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.1.1.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.1.1.1.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |
| } | | | |

9.3.1.1.1.5 Test requirement

**Table 9.3.1.1.1.5-1: Test requirement (FDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.09 | 1.09 |
| BLER | 0.05 | 0.05 |

To pass the test, $\alpha$ and $\beta$ and $\gamma$ and BLER must be fulfilled.

To pass the test, Test 1 and Test 2 must be passed.

To pass the test, one **or** the other SNR point within one test must be fulfilled.

## 9.3.1.1.2 TDD CQI Reporting under fading conditions – PUSCH 3-0

### 9.3.1.1.2.1 Test purpose

To verify that preferred sub-bands can be used for frequently-selective scheduling.

### 9.3.1.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 9.3.1.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.3.1.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.1.1.2.3-2 and by the following

a) a sub-band differential CQI offset level of 0 shall be reported at least $\alpha$% of the time but less than $\beta$% for each sub-band;

b) the ratio of the throughput obtained when transmitting on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected sub-band in set *S* shall be $\geq \gamma$;

c) when transmitting on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level the corresponding TBS, the average BLER for the indicated transport formats shall be greater or equal to 0.05.

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each available downlink transmission instance]. The transport block size TBS(wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\mathrm{PRB}}$ entry in Table 7.1.7.2.1-1 of TS 36.213[10] that corresponds to the sub-band size.

**Table 9.3.1.1.2.3-1: Sub-band test for single antenna transmission (TDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| Uplink downlink configuration | | 2 | | | |
| Special subframe configuration | | 4 | | | |
| SNR | dB | 9 | 10 | 14 | 15 |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | -98 | | -98 | |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -89 | -88 | -84 | -83 |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s $a$ = 1, $f_D = 5$ Hz | | | |
| Correlation | | Full | | | |
| Reporting interval | ms | 5 | | | |
| Minimum CQI delay | ms | 10 or 11 | | | |
| Reporting mode | | PUSCH 3-0 | | | |
| Max number of HARQ transmissions | | 1 | | | |
| ACK/NACK feedback mode | | Multiplexing | | | |
| Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)<br>Note 2: Reference measurement channel according to Table A.4-5 with one/two sided dynamic OCNG Pattern OP.1/2 TDD as described in Annex A.5.2.1/2.<br>Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level. | | | | | |

**Table 9.3.1.1.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.1 | 1.1 |
| UE Category | 1-5 | 1-5 |

### 9.3.1.1.2.4 Test description

#### 9.3.1.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 9.3.1.1.2.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.1.1.2.4.3.

#### 9.3.1.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 9.3.1.1.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and full-size subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI report for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI and full-size subband CQI reports.
3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.
4. For each subband, if subband CQI of index 0 is reported, at least $\alpha$% but less than $\beta$% of 2000 full-size subband CQI report, then continue to step 5, otherwise, go to step 7.
5. The SS shall send PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI value in an each available downlink transmission instance randomly selected full-size subband regardless of UE wideband and subband CQI report. Note that each full-size subband is selected in equal probability. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3 Declare the throughput as $t_{median}$.
6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the highest UE reported full-size subband CQI value in one full-size subband selected among the sub-bands in which UE report the highest full-size subband CQI. Subband differential CQI offset level is selected from {0, 1, 2, -1}. Note that the SS shall send PDSCH in the same full-size subband until next UE report is available. In case when same full-size subbands are reported subsequently as subbands with highest full-size subband CQI, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput and (NACK /(ACK + NACK)) according to Annex G.5.3. Declare the throughput as $t_{subband}$. If the ratio ($t_{subband}$ / $t_{median}$) $\geq \gamma$ and (NACK /(ACK + NACK))$\geq$ 0.05, pass the UE and go to step 8. Otherwise, go to step 7.
7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the table 9.3.1.1.2.3-1 for the other test as appropriate.

#### 9.3.1.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.1.1.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.1.1.2.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |
| } | | | |

**Table 9.3.1.1.2.4.3-3: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa2 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

#### 9.3.1.1.2.5 Test requirement

**Table 9.3.1.1.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.09 | 1.09 |
| BLER | 0.05 | 0.05 |

To pass the test, $\alpha$ and $\beta$ and $\gamma$ and BLER must be fulfilled.

To pass the test, Test 1 and Test 2 must be passed.

To pass the test, one **or** the other SNR point within one test must be fulfilled.

## 9.3.2 Frequency non-selective scheduling mode

The reporting accuracy of the channel quality indicator (CQI) under frequency non-selective fading conditions is determined by a double-sided percentile of the reported CQI, and the relative increase of the throughput obtained when the transport format transmitted is that indicated by the reported CQI compared to the case for which a fixed transport

format configured according to the reported median CQI is transmitted. In addition, the reporting accuracy is determined by a minimum BLER using the transport formats indicated by the reported CQI. To account for sensitivity of the input SNR the CQI reporting under frequency non-selective fading conditions is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

### 9.3.2.1 CQI Reporting under fading conditions – PUCCH 1-0

#### 9.3.2.1.1 FDD CQI Reporting under fading conditions – PUCCH 1-0

##### 9.3.2.1.1.1 Test purpose

To verify that the UE is tracking the channel variations and selecting the largest transport format possible according to the prevailing channel state for frequently non-selective scheduling

##### 9.3.2.1.1.2 Test applicability

This test applies to E-UTRA FDD UE release 8 of UE category 2-5.

##### 9.3.2.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.3.2.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.2.1.1.3-2 and by the following

a) CQI index not in the set {median CQI -1, median CQI +1} shall be reported at least $\alpha$% of the time;

b) the ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$;

c) when transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater or equal to 0.02

The transport block size TBS(wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\mathrm{PRB}}$ entry in Table 7.1.7.2.1-1 of 36.213 [10] that corresponds to the maximum transmission configuration (Table 5.6-1).

**Table 9.3.2.1.1.3-1 Fading test for single antenna (FDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| SNR (Note 3) | dB | 6 | 7 | 12 | 13 |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -92 | -91 | -86 | -85 |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | -98 | | -98 | |
| Propagation channel | | EPA5 | | | |
| Correlation | | High | | | |
| Reporting mode | | PUCCH 1-0 | | | |
| Reporting periodicity | ms | $N_P$ = 2 | | | |
| CQI delay | ms | 8 | | | |
| Physical channel for CQI reporting | | PUSCH (Note 4) | | | |
| PUCCH Report Type | | 4 | | | |
| *cqi-pmi-ConfigurationIndex* | | 1 | | | |
| Max number of HARQ transmissions | | 1 | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink SF not later than SF#(n-4), this reported wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-1 with one sided dynamic OCNG Pattern OP.1 FDD as described in Annex A.5.1.1

Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 4: To avoid collisions between CQI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#1, #3, #7 and #9 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#5, #7, #1 and #3.

**Table 9.3.2.1.1.3-2 Minimum requirement (FDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.05 | 1.05 |
| UE Category | 2-5 | 2-5 |

### 9.3.2.1.1.4 Test description

#### 9.3.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Table 9.3.2.1.1.3-1.

3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2, and uplink signals according to Annex H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.2.1.1.4.3.

#### 9.3.2.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.2.1.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #1,#3,#5 and subframe #7 according to Annex A.4-10. The UE will send ACK/NACK and periodic CQI report using PUSCH Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 2 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, reported. Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If less than (100-$\alpha$)/100*2000 of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 7.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI value regardless of UE wideband CQI report. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #1,#3,#5 and subframe #7 according to Annex A.4-10. The UE will send ACK/NACK and periodic CQI report using PUSCH. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband CQI value reported from UE. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #1,#3,#5 and subframe #7 according to Annex A.4-10. The UE will send ACK/NACK and periodic CQI report using PUSCH. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{wideband}$. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses.
   If the ratio ($t_{wideband}$ / $t_{median}$) $\geq \gamma$ and ratio (NACK /(ACK + NACK)) is greater or equal to 0.02, then pass the UE for this test and go to step 8. Otherwise go to step 7.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the table 9.3.2.1.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.3.2.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.2.1.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.2.1.1.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 1 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

9.3.2.1.1.5 Test requirement

**Table 9.3.2.1.1.5-1 Test requirement (FDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.04 | 1.04 |
| BLER | 0.02 | 0.02 |

To pass the test, $\alpha$ and $\gamma$ and BLER must be fulfilled.

To pass the test, Test 1 and Test 2 must be passed.

To pass the test, one **or** the other SNR point within one test must be fulfilled.

## 9.3.2.1.2 TDD CQI Reporting under fading conditions – PUCCH 1-0

### 9.3.2.1.2.1 Test purpose

To verify that the UE is tracking the channel variations and selecting the transport format according to the prevailing channel state for frequently non-selective scheduling

### 9.3.2.1.2.2 Test applicability

This test applies to E-UTRA TDD UE release 8 .of UE category 2-5.

#### 9.3.2.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.3.2.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.2.1.2.3-2 and by the following

a) a CQI index not in the set {median CQI -1, median CQI +1} shall be reported at least $\alpha$% of the time;

b) the ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$;

c) when transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater or equal to 0.02

The transport block size TBS(wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\mathrm{PRB}}$ entry in Table 7.1.7.2.1-1 of 36.213[10] that corresponds to the maximum transmission configuration (Table 5.6-1).

**Table 9.3.2.1.2.3-1: Fading test for single antenna (TDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| Uplink downlink configuration | | 2 | | | |
| Special subframe configuration | | 4 | | | |
| SNR | dB | 6 | 7 | 12 | 13 |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | -98 | | -98 | |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -92 | -91 | -86 | -85 |
| Propagation channel | | EPA5 | | | |
| Correlation | | High | | | |
| Reporting mode | | PUCCH 1-0 | | | |
| Reporting periodicity | ms | $N_P$ = 5 | | | |
| CQI delay | ms | 10 or 11 | | | |
| Physical channel for CQI reporting | | PUSCH (Note 4) | | | |
| PUCCH Report Type | | 4 | | | |
| *cqi-pmi-ConfigurationIndex* | | 3 | | | |
| Max number of HARQ transmissions | | 1 | | | |
| ACK/NACK feedback mode | | Multiplexing | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink SF not later than SF#(n-4), this reported wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-2 with one sided dynamic OCNG Pattern OP.1 TDD as described in Annex A.5.2.1.

Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 4: To avoid collisions between CQI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#3 and #8 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#7 and #2.

**Table 9.3.2.1.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.05 | 1.05 |
| UE Category | 2-5 | 2-5 |

### 9.3.2.1.2.4 Test description

#### 9.3.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 9.3.2.1.2.3-1.
3. Downlink signals are initially set up according to Annex C0, C.1 and Annex C.3.2, and uplink signals according to Annex H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.2.1.2.4.3.

#### 9.3.2.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.2.1.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI reports.
3. Set up a relative frequency distribution for the reported wideband CQI-values, reported. Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.
4. If less than (100-$\alpha$)/100*2000 of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 7.
5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband median-CQI value regardless of UE wideband CQI report. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH.Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Declare the throughput as $t_{median}$.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the wideband CQI value reported from UE. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.3. Declare the throughput as $t_{wideband}$ . For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses.
   If the ratio ( $t_{wideband}$ / $t_{median}$ ) $\geq \gamma$ and ratio (NACK /(ACK + NACK )) is greater or equal to 0.02, then pass the UE and go to step 8. Otherwise go to step 7.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, repeat the same procedure (steps 1 to 7) with test conditions according to the table 9.3.2.1.2.3-1 for the other test as appropriate.

#### 9.3.2.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.2.1.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.2.1.2.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 3 | (see Table 7.2.2-1C in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | NULL | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.2.1.2.4.3-3: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa2 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

9.3.2.1.2.5 Test requirement

**Table 9.3.2.1.1.5-1 Test requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.04 | 1.04 |
| BLER | 0.02 | 0.02 |

To pass the test, $\alpha$ and $\gamma$ and BLER must be fulfilled.

To pass the test, Test 1 and Test 2 must be passed.

To pass the test, one **or** the other SNR point within one test must be fulfilled.

## 9.3.3 Frequency-selective interference

The accuracy of sub-band channel quality indicator (CQI) reporting under frequency selective interference conditions is determined by a double-sided percentile of the reported differential CQI offset level +2 for a preferred sub-band, and the relative increase of the throughput obtained when transmitting on any one of the sub-bands with the highest reported differential CQI offset level the corresponding transport format compared to the case for which a fixed format is transmitted on any sub-band in set of TS 36.213[10]. The purpose is to verify that preferred sub-bands are used for frequently-selective scheduling under frequency-selective interference conditions.

### 9.3.3.1 CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0

#### 9.3.3.1.1 FDD CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0

##### 9.3.3.1.1.1 Test purpose

To verify that preferred sub-bands can be used for frequently-selective scheduling with frequency-selective interference situation.

##### 9.3.3.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

##### 9.3.3.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.3.3.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.3.1.1.3-2 and by the following:

a) a sub-band differential CQI offset level of +2 shall be reported at least $\alpha$% for at least one of the sub-bands of full size at the channel edges;

b) the ratio of the throughput obtained when transmitting on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected sub-band in set *S* shall be $\geq \gamma$;

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each TTI for FDD, each available downlink transmission instance for TDD. Sub-bands of a size smaller than full size are excluded from the test. The transport block size TBS(wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\text{PRB}}$ entry in Table 7.1.7.2.1-1 of TS 36.213 [10] that corresponds to the sub-band size.

**Table 9.3.3.1.1.3-1 Sub-band test for single antenna transmission (FDD)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| Bandwidth | MHz | 10 MHz | 10 MHz |
| Transmission mode | | 1 (port 0) | 1 (port 0) |
| $I_{ot}^{(j)}$ for RB 0…5 Note 3 | dB[mW/15kHz] | -102 | -93 |
| $I_{ot}^{(j)}$ for RB 6…41 Note 3 | dB[mW/15kHz] | -93 | -93 |
| $I_{ot}^{(j)}$ for RB 42…49 Note 3 | dB[mW/15kHz] | -93 | -102 |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -94 | -94 |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s, $a$ = 1, $f_D = 5$ Hz | |
| Correlation | | Full | |
| Reporting interval | ms | 5 | |
| Minimum CQI delay | ms | 8 | |
| Reporting mode | | PUSCH 3-0 | |
| Sub-band size | RB | 6 (full size) | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-4.with one/two sided dynamic OCNG Pattern OP.1/2 FDD as described in Annex A.5.1.1/2

Note 3: Iot shall be modelled as connecting Gaussian distributed uncorrelated interference source for each UE receive antenna port. The received power spectral density of the interfering signal as measured at the UE antenna connector is to be scaled accordingly for different RB groups.

**Table 9.3.3.1.1.3-2: Minimum requirement (FDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 60 | 60 |
| $\gamma$ | 1.6 | 1.6 |
| UE Category | 1-5 | 1-5 |

### 9.3.3.1.1.4 Test description

#### 9.3.3.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, interfering source and faders to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.21.

2. The parameter settings for the cell are set up according to Table 9.3.3.1.1.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.3.1.1.4.3.

#### 9.3.3.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.3.1.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI and full-size subband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If for at least one of the full-size subbands at the channel edges, a subband differential CQI offset level of +2 is reported in $\alpha$% or more of 2000 reports, then continue with step 5, otherwise fail the UE.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC on an each TTI randomly selected full-size subband using the transport format according to the wideband median-CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the highest UE reported full-size subband CQI value in one full-size subband selected among the sub-bands in which UE reports the highest full-size subband CQI. Subband differential CQI offset level is selected from {0, 1, 2, -1}. Note that the SS shall send PDSCH in the same full-size subband until next UE report is available. In case when same full-size subbands are reported subsequently as subbands with highest full-size subband CQI, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{subband}$. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval.
   If $t_{subband} / t_{median} \geq \gamma$, then pass the UE for this test and go to step 7. Otherwise fail the UE.

7. If both tests have not been done, then repeat the same procedure (steps 1 to 6) with test conditions according to the table 9.3.3.1.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

9.3.3.1.1.4.3 Message contents

**Table 9.3.3.1.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.3.1.1.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | Not present | | |
| } | | | |
| } | | | |

9.3.3.1.1.5 Test requirement

**Table 9.3.3.1.1.5-1: Test requirement (FDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 40 | 40 |
| $\gamma$ | 1.50 | 1.50 |

## 9.3.3.1.2 TDD CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0

### 9.3.3.1.2.1 Test purpose

To verify that preferred sub-bands can be used for frequently-selective scheduling with frequency-selective interference situation.

### 9.3.3.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 9.3.3.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.3.3.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.3.1.2.3-2 and by the following:

a) a sub-band differential CQI offset level of +2 shall be reported at least $\alpha$% for at least one of the sub-bands of full size at the channel edges;

b) the ratio of the throughput obtained when transmitting on any one of the sub-bands with the highest differential CQI offset level the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected sub-band in set *S* shall be $\geq \gamma$;

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each TTI for FDD, each available downlink transmission instance for TDD. Sub-bands of a size smaller than full size are excluded from the test. The transport block size TBS(wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\mathrm{PRB}}$ entry in Table 7.1.7.2.1-1 of TS 36.213 [10] that corresponds to the sub-band size.

**Table 9.3.3.1.2.3-1: Sub-band test for single antenna transmission (TDD)**

| Parameter | Unit | Test 1 | Test 2 |
|---|---|---|---|
| Bandwidth | MHz | 10 MHz | 10 MHz |
| Transmission mode | | 1 (port 0) | 1 (port 0) |
| Uplink downlink configuration | | 2 | |
| Special subframe configuration | | 4 | |
| $I_{ot}^{(j)}$ for RB 0…[5] Note 3 | dB[mW/15kHz] | -102 | -93 |
| $I_{ot}^{(j)}$ for RB 6…[41] Note 3 | dB[mW/15kHz] | -93 | -93 |
| $I_{ot}^{(j)}$ for RB [42]…49 Note 3 | dB[mW/15kHz] | -93 | -102 |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -94 | -94 |
| Max number of HARQ transmissions | | 1 | |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s, $a$ = 1, $f_D = 5$ Hz | |
| Correlation | | Full | |
| Reporting interval | ms | 5 | |
| CQI delay | ms | 10 or 11 | |
| Reporting mode | | PUSCH 3-0 | |
| Sub-band size | RB | 6 (full size) | |
| ACK/NACK feedback mode | | Multiplexing | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to table A.4-5 with one/two sided dynamic OCNG Pattern OP.1/2 TDD as described in Annex A.5.2.1/2.

Note 3: Iot shall be modelled as connecting Gaussian distributed uncorrelated interference source for each UE receive antenna port. The received power spectral density of the interfering signal as measured at the UE antenna connector is to be scaled accordingly for different RB groups.

**Table 9.3.3.1.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 60 | 60 |
| $\gamma$ | 1.6 | 1.6 |
| UE Category | 1-5 | 1-5 |

### 9.3.3.1.2.4 Test description

#### 9.3.3.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and interfering source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.21.

2. The parameter settings for the cell are set up according to Table 9.3.3.1.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.3.1.2.4.3.

#### 9.3.3.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.3.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI and full-size subband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If for at least one of the full-size subbands at the channel edges, a subband differential CQI offset level of +2 is reported in $\alpha$% or more of 2000 reports, then continue with step 5, otherwise fail the UE.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC in an each available downlink transmission instance randomly selected full-size subband using the transport format according to the wideband median-CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.

6. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to the highest UE reported full-size subband CQI value in one full-size subband selected among the sub-bands in which UE reports the highest full-size subband CQI. Subband differential CQI offset level is selected from {0, 1, 2, -1}. Note that the SS shall send PDSCH in the same full-size subband until next UE report is available. In case when same full-size subbands are reported subsequently as subbands with highest full-size subband CQI, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{subband}$. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval.
   If $t_{subband} / t_{median} \geq \gamma$, then pass the UE for this test and go to step 7. Otherwise fail the UE.

7. If both tests have not been done, then repeat the same procedure (steps 1 to 6) with test conditions according to the table 9.3.3.1.2.3-1 for the other Test as appropriate. Otherwise pass the UE.

9.3.3.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.3.1.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.3.1.2.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm30 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | Not present | | |
| } | | | |
| } | | | |

**Table 9.3.3.1.2.4.3-3: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa2 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

9.3.3.1.2.5 Test requirement

**Table 9.3.3.1.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 40 | 40 |
| $\gamma$ | 1.50 | 1.50 |

## 9.3.4 UE-selected subband CQI

The accuracy of UE-selected subband channel quality indicator (CQI) reporting under frequency-selective fading conditions is determined by the relative increase of the throughput obtained when transmitting on the UE-selected subbands with the corresponding transport format compared to the case for which a fixed format is transmitted on any subband in set *S* of TS 36.213 [10]. The purpose is to verify that correct subbands are accurately reported for frequency-selective scheduling. To account for sensitivity of the input SNR the subband CQI reporting under frequency-selective fading conditions is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

### 9.3.4.1 CQI Reporting under fading conditions – PUSCH 2-0

#### 9.3.4.1.1 FDD CQI Reporting under fading conditions – PUSCH 2-0

9.3.4.1.1.1 Test purpose

To verify that UE-selected sub-bands can be used for frequently-selective scheduling.

#### 9.3.4.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 9.3.4.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.3.4.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.4.1.1.3-2 and by the following:

a) the ratio of the throughput obtained when transmitting on a randomly selected subband among the best M subbands reported by the UE the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected subband in set *S* shall be $\geq \gamma$;

The requirements only apply for subbands of full size and the random scheduling across the subbands is done by selecting a new subband in each TTI for FDD. The transport block size TBS (wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\text{PRB}}$ entry in Table 7.1.7.2.1-1 of TS 36.213 [10] that corresponds to the subband size.

**Table 9.3.4.1.1.3-1: Subband test for single antenna transmission (FDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| SNR (Note 3) | dB | 9 | 10 | 14 | 15 |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -89 | -88 | -84 | -83 |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | -98 | | -98 | |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s, *a* = 1, $f_D = 5$ Hz | | | |
| Reporting interval | ms | 5 | | | |
| CQI delay | ms | 8 | | | |
| Reporting mode | | PUSCH 2-0 | | | |
| Max number of HARQ transmissions | | 1 | | | |
| Subband size (*k*) | RBs | 3 (full size) | | | |
| Number of preferred subbands (*M*) | | 5 | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-12 with one/two sided dynamic OCNG Pattern OP.1/2 FDD as described in Annex A.5.1.1/2.

Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

**Table 9.3.4.1.1.3-2: Minimum requirement (FDD)**

| | Test 1 | Test 2 |
|---|---|---|
| $\gamma$ | [1.2] | [1.2] |
| UE Category | 1-5 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.3.4.1.1.

#### 9.3.4.1.1.4 Test description

##### 9.3.4.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS and faders to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.21.
2. The parameter settings for the cell are set up according to Table 9.3.4.1.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.4.1.1.4.3.

##### 9.3.4.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.4.1.1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI and full-size subband CQI reports.
3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.
4. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC on a randomly selected full-size subband using the transport format according to the wideband median-CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4.
   Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.
5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC in one full-size subband selected among the M subbands reported by the UE and with the corresponding TBS. Note that the SS shall send PDSCH in the same full-size subband until next UE report is available. In case when same full-size subbands are reported subsequently, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. SS schedules the UL transmission every 5 ms to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{subband}$.
   Count the number of NACKs, ACKs and statDTXs on the UL during the test interval.
   If $t_{subband} / t_{median} \geq \gamma$, then pass the UE for this test and go to step 6. Otherwise fail the UE.

6. If both tests have not been done, then repeat the same procedure (steps 1 to 5) with test conditions according to the table 9.3.4.1.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.3.4.1.1.4.3 Message contents

**Table 9.3.4.1.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.4.1.1.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm20 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | Not present | | |
| } | | | |
| } | | | |

#### 9.3.4.1.1.5 Test requirement

**Table 9.3.4.1.1.5-1: Test requirement (FDD)**

| | **Test 1** | **Test 2** |
|---|---|---|
| $\gamma$ | [1.19] | [1.19] |
| UE Category | 1-5 | 1-5 |

### 9.3.4.1.2 TDD CQI Reporting under fading conditions – PUSCH 2-0

#### 9.3.4.1.2.1 Test purpose

To verify that UE-selected sub-bands can be used for frequently-selective scheduling.

#### 9.3.4.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 9.3.4.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.3.4.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.4.1.2.3-2 and by the following:

a) the ratio of the throughput obtained when transmitting on a randomly selected subband among the best M subbands reported by the UE the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected subband in set *S* shall be $\geq \gamma$;

The requirements only apply for subbands of full size and the random scheduling across the subbands is done by selecting a new subband in each available downlink transmission instance for TDD. The transport block size TBS (wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{PRB}$ entry in Table 7.1.7.2.1-1 of TS 36.213 [10] that corresponds to the subband size.

**Table 9.3.4.1.2.3-1: Sub-band test for single antenna transmission (TDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| Uplink downlink configuration | | 2 | | | |
| Special subframe configuration | | 4 | | | |
| SNR (Note 3) | dB | 9 | 10 | 14 | 15 |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | -89 | -88 | -84 | -83 |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | -98 | | -98 | |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s, $a$ = 1, $f_D = 5$ Hz | | | |
| Reporting interval | ms | 5 | | | |
| CQI delay | ms | 10 or 11 | | | |
| Reporting mode | | PUSCH 2-0 | | | |
| Max number of HARQ transmissions | | 1 | | | |
| Subband size ($k$) | RBs | 3 (full size) | | | |
| Number of preferred subbands ($M$) | | 5 | | | |
| ACK/NACK feedback mode | | Multiplexing | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-13 with one/two sided dynamic OCNG Pattern OP.1/2 TDD as described in Annex A.5.2.1/2.

Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

**Table 9.3.4.1.2.3-2: Minimum requirement (TDD)**

| | Test 1 | Test 2 |
|---|---|---|
| $\gamma$ | [1.2] | [1.2] |
| UE Category | 1-5 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.3.4.1.2.

### 9.3.4.1.2.4 Test description

#### 9.3.4.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.

2. The parameter settings for the cell are set up according to Table 9.3.4.1.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.4.1.2.4.3.

#### 9.3.4.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.4.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are counted as wideband CQI and full-size subband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC on a randomly selected full-size subband using the transport format according to the wideband median-CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.

5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC in one full-size subband selected among the M subbands reported by the UE and with the corresponding TBS. Note that the SS shall send PDSCH in the same full-size until next UE report is available. In case when same full-size subbands are reported subsequently, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #2 and subframe #7 to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{subband}$. Count the number of NACKs, ACKs and statDTXs on the UL during the test interval.
   If $t_{subband} / t_{median} \geq \gamma$, then pass the UE for this test and go to step 6. Otherwise fail the UE.

6. If both tests have not been done, then repeat the same procedure (steps 1 to 5) with test conditions according to the table 9.3.4.1.2.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.3.4.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.4.1.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.4.1.2.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm20 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | Not present | | |
| } | | | |
| } | | | |

**Table 9.3.4.1.2.4.3-3: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa2 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

9.3.4.1.2.5 Test requirement

**Table 9.3.4.1.2.5-1: Test requirement (TDD)**

| | **Test 1** | **Test 2** |
|---|---|---|
| $\gamma$ | [1.19] | [1.19] |
| UE Category | 1-5 | 1-5 |

## 9.3.4.2 CQI Reporting under fading conditions – PUCCH 2-0

### 9.3.4.2.1 FDD CQI Reporting under fading conditions – PUCCH 2-0

#### 9.3.4.2.1.1 Test purpose

To verify that UE-selected sub-bands can be used for frequently-selective scheduling.

#### 9.3.4.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 9.3.4.2.1.3 Minimum conformance requirements

For the parameters specified in Table 9.3.4.2.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.4.2.1.3-2 and by the following

a) the ratio of the throughput obtained when transmitting on subbands reported by the UE the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected subband in set *S* shall be $\geq \gamma$;

The requirements only apply for subbands of full size and the random scheduling across the subbands is done by selecting a new subband in each TTI for FDD. The transport block size TBS (wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{\mathrm{PRB}}$ entry in Table 7.1.7.2.1-1 of TS 36.213 [10] that corresponds to the subband size.

**Table 9.3.4.2.1.3-1: Subband test for single antenna transmission (FDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| SNR (Note 3) | dB | [8] | [9] | [13] | [14] |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | [-90] | [-89] | [-85] | [-84] |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | [-98] | | [-98] | |
| Propagation channel | | [Clause B.2.4 with $\tau_d = 0.45\,\mu$s, $a$ = 1, $f_D = 5$ Hz] | | | |
| Reporting periodicity | ms | $N_P$ = 2 | | | |
| CQI delay | ms | 8 | | | |
| Physical channel for CQI reporting | | PUSCH (Note 4) | | | |
| PUCCH Report Type for wideband CQI | | 4 | | | |
| PUCCH Report Type for subband CQI | | 1 | | | |
| Max number of HARQ transmissions | | 1 | | | |
| Subband size ($k$) | RBs | 6 (full size) | | | |
| Number of bandwidth parts ($J$) | | 3 | | | |
| K | | 1 | | | |
| *cqi-pmi-ConfigIndex* | | 1 | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-4 with one/two sided dynamic OCNG Pattern OP.1/2 FDD as described in Annex A.5.1.1/2.

Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 4: To avoid collisions between CQI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#1, #3, #7 and #9 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#5, #7, #1 and #3.

Note 5: CQI reports for the short subband (having 2RBs in the last bandwidth part) are to be disregarded and data scheduling according to the most recent subband CQI report for bandwidth part with j=1.

Note 6: In the case where wideband CQI is reported, data is to be scheduled according to the most recently used subband CQI report.

**Table 9.3.4.2.1.3-2: Minimum requirement (FDD)**

| | Test 1 | Test 2 |
|---|---|---|
| $\gamma$ | [1.15] | [1.15] |
| UE Category | 1-5 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.3.4.2.1.

#### 9.3.4.2.1.4 Test description

##### 9.3.4.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS and faders to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.21.
2. The parameter settings for the cell are set up according to Table 9.3.4.2.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.4.2.1.4.3.

##### 9.3.4.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.4.2.1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #1, #3, #5 and subframe #7 according to Annex A.4-10. The UE will send ACK/NACK and periodic CQI report using PUSCH. In this process the SS collects wideband CQI reports and also cases where UE transmits nothing in its wideband CQI timing are counted as wideband CQI reports.
3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.
4. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC on a randomly selected full-size subband using the transport format according to the wideband median-CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #1, #3, #5 and subframe #7 according to Annex A.4-10. The UE will send ACK/NACK and periodic CQI report using PUSCH. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.
5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC in the full-size subband reported by the UE and with the corresponding TBS. When the UE reports a non-full-size SB, the SS schedules the recent reported SB for bandwidth part with j=1 and with the corresponding TBS. Note that the SS shall send PDSCH in the same full-size subband until next subband UE report is available. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-6. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #1, #3, #5 and subframe #7 according to Annex A.4-10. The UE will send ACK/NACK and periodic CQI report using PUSCH. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{subband}$. Count the number of NACKs, ACKs and statDTXs on the UL during the test

interval.

If $t_{subband} / t_{median} \geq \gamma$, then pass the UE for this test and go to step 6. Otherwise fail the UE.

6. If both tests have not been done, then repeat the same procedure (steps 1 to 5) with test conditions according to the table 9.3.4.2.1.3-1 for the other Test as appropriate. Otherwise pass the UE.

9.3.4.1.1.4.3 Message contents

**Table 9.3.4.2.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.4.2.1.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 1 | (see Table 7.2.2-1A in TS 36.213 | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| subbandCQI | 1 | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213 | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

9.3.4.2.1.5 Test requirement

**Table 9.3.4.2.1.5-1: Test requirement (FDD)**

| | **Test 1** | **Test 2** |
|---|---|---|
| $\gamma$ | [1.14] | [1.14] |
| UE Category | 1-5 | 1-5 |

## 9.3.4.2.2 TDD CQI Reporting under fading conditions – PUCCH 2-0

### 9.3.4.2.2.1 Test purpose

To verify that UE-selected sub-bands can be used for frequently-selective scheduling.

### 9.3.4.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 9.3.4.2.2.3 Minimum conformance requirements

For the parameters specified in Table 9.3.4.2.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.3.4.2.2.3-2 and by the following:

a) the ratio of the throughput obtained when transmitting on subbands reported by the UE the corresponding TBS and that obtained when transmitting the TBS indicated by the reported wideband CQI median on a randomly selected subband in set *S* shall be $\geq \gamma$;

The requirements only apply for subbands of full size and the random scheduling across the subbands is done by selecting a new subband in each available downlink transmission instance for TDD. The transport block size TBS (wideband CQI median) is that resulting from the code rate which is closest to that indicated by the wideband CQI median and the $N_{PRB}$ entry in Table 7.1.7.2.1-1 of TS 36.213 [10] that corresponds to the subband size.

**Table 9.3.4.2.2.3-1: Sub-band test for single antenna transmission (TDD)**

| Parameter | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|
| Bandwidth | MHz | 10 MHz | | | |
| Transmission mode | | 1 (port 0) | | | |
| Uplink downlink configuration | | 2 | | | |
| Special subframe configuration | | 4 | | | |
| SNR (Note 3) | dB | [8] | [9] | [13] | [14] |
| $\hat{I}_{or}^{(j)}$ | dB[mW/15kHz] | [-90] | [-89] | [-85] | [-84] |
| $N_{oc}^{(j)}$ | dB[mW/15kHz] | [-98] | | [-98] | |
| Propagation channel | | Clause B.2.4 with $\tau_d = 0.45\,\mu$s, $a$ = 1, $f_D = 5$ Hz | | | |
| Reporting periodicity | ms | $N_P$ = 5 | | | |
| CQI delay | ms | 10 or 11 | | | |
| Physical channel for CQI reporting | | PUSCH (Note 4) | | | |
| PUCCH Report Type for wideband CQI | | 4 | | | |
| PUCCH Report Type for subband CQI | | 1 | | | |
| Max number of HARQ transmissions | | 1 | | | |
| Subband size ($k$) | RBs | 6 (full size) | | | |
| Number of bandwidth parts ($J$) | | 3 | | | |
| K | | 1 | | | |
| *cqi-pmi-ConfigIndex* | | 3 | | | |
| ACK/NACK feedback mode | | Multiplexing | | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on CQI estimation at a downlink subframe not later than SF#(n-4), this reported subband or wideband CQI cannot be applied at the eNB downlink before SF#(n+4)

Note 2: Reference measurement channel according to Table A.4-5 with one/two sided dynamic OCNG Pattern OP.1/2 TDD as described in Annex A.5.2.1/2.

Note 3: For each test, the minimum requirements shall be fulfilled for at least one of the two SNR(s) and the respective wanted signal input level.

Note 4: To avoid collisions between CQI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#3 and #8 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#7 and #2.

Note 5: CQI reports for the short subband (having 2RBs in the last bandwidth part) are to be disregarded and data scheduling according to the most recent subband CQI report for bandwidth part with j=1.

Note 6: In the case where wideband CQI is reported, data is to be scheduled according to the most recently used subband CQI report.

**Table 9.3.4.2.2.3-2: Minimum requirement (TDD)**

| | Test 1 | Test 2 |
|---|---|---|
| $\gamma$ | [1.15] | [1.15] |
| UE Category | 1-5 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.3.4.2.1.

#### 9.3.4.2.2.4 Test description

##### 9.3.4.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 9.3.4.2.2.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.3.4.2.2.4.3.

##### 9.3.4.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.3.4.2.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC according to CQI value 8 of Annex A.4 Table A.4-3 and keep it regardless of the wideband and subband CQI value sent by the UE. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. In this process the SS collects wideband CQI reports and also cases where UE transmits nothing in its wideband CQI timing are counted as wideband CQI reports.
3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.
4. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC on a randomly selected full-size subband using the transport format according to the wideband median-CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{median}$.
5. The SS shall transmit PDSCH via PDCCH DCI format 1A for C_RNTI to transmit the DL RMC in the full-size subband reported by the UE and with the corresponding TBS. When the UE reports a non-full-size SB, the SS schedules the recent reported SB for bandwidth part with j=1 and with the corresponding TBS. Note that the SS shall send PDSCH in the same full-size subband until next subband UE report is available. The SS sends downlink MAC padding bits on the DL RMC. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #2 and subframe #7 according to Annex A.4-11. The UE will send ACK/NACK and periodic CQI report using PUSCH. Measure the average throughput according to Annex G.5.3. Declare the throughput as $t_{subband}$. Count the number of NACKs, ACKs and

statDTXs on the UL during the test interval.

If $t_{subband} / t_{median} \geq \gamma$, then pass the UE for this test and go to step 6. Otherwise fail the UE.

6. If both tests have not been done, then repeat the same procedure (steps 1 to 5) with test conditions according to the table 9.3.4.2.2.3-1 for the other Test as appropriate. Otherwise pass the UE.

#### 9.3.4.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.3.4.2.2.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| } | | | |

**Table 9.3.4.2.2.4.3-2: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 3 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| subbandCQI | 1 | | |
| } | | | |
| ri-ConfigIndex | 484 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.3.4.2.2.4.3-3: TDD-Config-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TDD-Config-DEFAULT ::= SEQUENCE { | | | |
| subframeAssignment | Sa2 | | |
| specialSubframePatterns | Ssp4 | | |
| } | | | |

#### 9.3.4.2.2.5 Test requirement

**Table 9.3.4.2.2.5-1: Test requirement (TDD)**

| | **Test 1** | **Test 2** |
|---|---|---|
| $\gamma$ | [1.14] | [1.14] |
| UE Category | 1-5 | 1-5 |

# 9.4 Reporting of Precoding Matrix Indicator (PMI)

The minimum performance requirements of PMI reporting are defined based on the precoding gain, expressed as the relative increase in throughput when the transmitter is configured according to the UE reports compared to the case when the transmitter is using random precoding . Transmission mode 6 is used with a fixed transport format (FRC) configured. The requirements are specified in terms of the ratio $\gamma = \frac{t_{ue}}{t_{rnd}}$

In the definition of $\gamma$, for PUSCH 3-1 single PMI and PUSCH 1-2 multiple PMI requirements, $t_{rnd}$ is 60% of the maximum throughput obtained at $SNR_{rnd}$ using random precoding, and $t_{ue}$ the throughput measured at $SNR_{rnd}$ with precoders configured according to the UE reports.

For the PUCCH 2-1 single PMI requirement, $t_{rnd}$ is [60]% of the maximum throughput obtained at $SNR_{rnd}$ using random precoding on a randomly selected full-size subband in set S subbands, and $t_{ue}$ the throughput measured at $SNR_{rnd}$ with both the precoder and the preferred full-size subband applied according to the UE reports;

For PUSCH 2-2 multiple PMI requirements, $t_{rnd}$ is [60]% of the maximum throughput obtained at $SNR_{rnd}$ using random precoding on a randomly selected full-size subband in set S subbands, and $t_{ue}$ the throughput measured at $SNR_{rnd}$ with both the subband precoder and a randomly selected full-size subband (within the preferred subbands) applied according to the UE reports.

## 9.4.1 Single PMI

### 9.4.1.1 PMI Reporting – PUSCH 3-1 (Single PMI)

#### 9.4.1.1.1 FDD PMI Reporting – PUSCH 3-1 (Single PMI)

##### 9.4.1.1.1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

##### 9.4.1.1.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 8 and forward.

##### 9.4.1.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.4.1.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.1.1.1.3-2.

**Table 9.4.1.1.1.3-1: PMI test for single-layer (FDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Transmission mode | | | 6 |
| Propagation channel | | | EVA5 |
| Precoding granularity | | PRB | 50 |
| Correlation and antenna configuration | | | Low 2 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| Reporting mode | | | PUSCH 3-1 |
| Reporting interval | | ms | 1 |
| PMI delay (Note 2) | | ms | 8 |
| Measurement channel | | | R.10 FDD |
| OCNG Pattern | | | OP.1 FDD |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| Note 1: For random precoder selection, the precoder shall be updated in each TTI (1 ms granularity)<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4). | | | |

**Table 9.4.1.1.1.3-2: Minimum requirement (FDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.1 |
| UE Category | 1-5 |

### 9.4.1.1.1.4 Test description

#### 9.4.1.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.1.1.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.1.1.1.4.3.

#### 9.4.1.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.4.1.1.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC.SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2

3. Set SNR to $SNR_{rnd}$ . The SS shall transmit PDCCH with DCI format 0 in which CQI request bit is set to true [every subframe]. Then the SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3

4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$ If the ratio ( throughput / $t_{rnd}$ ) ≥ γ which is specified in table 9.4.1.1.1.5-1, then the test is pass. Otherwise, the test is fail.

#### 9.4.1.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.1.1.1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.1.1.1.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n2TxAntenna-tm6 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |

**Table 9.4.1.1.1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm31 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.1.1.1.5 Test requirement

**Table 9.4.1.1.1.5-1: Test requirement (FDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.09 |

The ratio of throughput using precoding matrix of PMI reports from the UE to using random precoding matrix shall equal or exceed the value specified in table 9.4.1.1.1.5.

## 9.4.1.1.2 TDD PMI Reporting – PUSCH 3-1 (Single PMI)

### 9.4.1.1.2.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 9.4.1.1.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 8 and forward.

### 9.4.1.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.4.1.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in 9.4.1.1.2.3-2.

**Table 9.4.1.1.2.3-1: PMI test for single-layer (TDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Transmission mode | | | 6 |
| Uplink downlink configuration | | | 1 |
| Special subframe configuration | | | 4 |
| Propagation channel | | | EVA5 |
| Precoding granularity | | PRB | 50 |
| Correlation and antenna configuration | | | Low 2 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| Reporting mode | | | PUSCH 3-1 |
| Reporting interval | | ms | 1 |
| Minimum PMIdelay (Node-2) | | ms | 10 or 11 |
| Measurement channel | | | R.10 TDD |
| OCNG Pattern | | | OP.1 TDD |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| ACK/NACK feedback mode | | | Multiplexing |

Note 1: For random precoder selection, the precoder shall be updated in each available downlink transmission instance

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

**Table 9.4.1.1.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.1 |
| UE Category | 1-5 |

### 9.4.1.1.2.4 Test description

#### 9.4.1.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.1.1.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clause B.0

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2.Message contents are defined in clause 9.4.1.1.2.4.3.

#### 9.4.1.1.2.4.2 Test procedure

1. Set the bandwidth, propagation condition, antenna configuration and measurement channel parameters according to Table 9.4.1.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. .SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2.

3. Set SNR to $SNR_{rnd}$ . The SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. . SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3

4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

#### 9.4.1.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.1.1.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.1.1.2.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n2TxAntenna-tm6 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |

**Table 9.4.1.1.2.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm31 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.1.1.2.5 Test requirement

**Table 9.4.1.1.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.09 |

## 9.4.1.2 PMI Reporting – PUCCH 2-1 (Single PMI)

### 9.4.1.2.1 FDD PMI Reporting – PUCCH 2-1 (Single PMI)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Throughput ratio is still conducted in square brackets by RAN4

#### 9.4.1.2.1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

#### 9.4.1.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 9.4.1.2.1.3 Minimum conformance requirements

For the parameters specified in Table 9.4.1.2.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.1.2.1.3-2.

**Table 9.4.1.2.1.3-1: PMI test for single-layer (FDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Transmission mode | | | 6 |
| Propagation channel | | | [EVA5] |
| Correlation and antenna configuration | | | Low 4 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| PMI delay | | ms | [8 or 9] |
| Reporting mode | | | PUCCH 2-1 (Note 6) |
| Reporting periodicity | | ms | $N_P$ = 2 |
| Physical channel for CQI reporting | | | PUSCH (Note 3) |
| PUCCH Report Type for wideband CQI/PMI | | | 2 |
| PUCCH Report Type for subband CQI | | | 1 |
| Measurement channel | | | [R.14-1 FDD] |
| OCNG Pattern | | | [OP.1/2 FDD] |
| Precoding granularity | | PRB | 6 (full size) |
| Number of bandwidth parts (*J*) | | | 3 |
| K | | | 1 |
| *cqi-pmi-ConfigIndex* | | | 1 |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |

Note 1: For random precoder selection, the precoder shall be updated every two TTI (2 ms granularity)

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4).

Note 3: To avoid collisions between HARQ-ACK and wideband CQI/PMI or subband CQI, it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#1, #3, #7 and #9 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#5, #7, #1 and #3.

Note 4: Reports for the short subband (having 2RBs in the last bandwidth part) are to be disregarded and instead data is to be transmitted on the most recently used subband for bandwidth part with j=1.

Note 5: In the case where wideband PMI is reported, data is to be transmitted on the most recently used subband.

Note 6: The bit field for PMI confirmation in DCI format 1B shall be mapped to “0” and TPMI information shall indicate the codebook index used in Table 6.3.4.2.3-2 of TS36.211 [4] according to the latest PMI report on PUCCH.

**Table 9.4.1.2.1.3-2: Minimum requirement (FDD)**

| | Test 1 |
|---|---|
| $\gamma$ | [1.2] |
| UE Category | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.1.2.1.

#### 9.4.1.2.1.4 Test description

##### 9.4.1.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.1.2.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.1.2.1.4.3.

##### 9.4.1.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.4.1.2.1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #1, #3, #5 and #7 to carry the both ACK and CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2
3. Set SNR to $SNR_{rnd}$. The SS shall transmit PDCCH with DCI format 0 in which CQI request bit is set to true [every subframe]. Then the SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #1, #3, #5 and #7 to carry the both ACK and CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3
4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$ If the ratio ( throughput / $t_{rnd}$ ) $\geq \gamma$ which is specified in table 9.4.1.2.2.5-1, then the test is pass. Otherwise, the test is fail.

##### 9.4.1.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.1.2.1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.1.2.1.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n4TxAntenna-tm6 | 1111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.1.2.1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 1 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| subbandCQI | 1 | | |
| } | | | |
| ri-ConfigIndex | 483 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

9.4.1.2.1.5 Test requirement

**Table 9.4.1.2.1.5-1: Test requirement (FDD)**

| | **Test 1** |
|---|---|
| $\gamma$ | [1.19] |
| UE Category | 1-5 |

The ratio of throughput using precoding matrix of PMI reports from the UE to using random precoding matrix shall equal or exceed the value specified in table 9.4.1.2.1.5.

### 9.4.1.2.2 TDD PMI Reporting – PUCCH 2-1 (Single PMI)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Throughput ratio is still conducted in square brackets by RAN4

#### 9.4.1.2.2.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

#### 9.4.1.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

#### 9.4.1.2.2.3 Minimum conformance requirements

For the parameters specified in Table 9.4.1.2.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in 9.4.1.2.2.3-2.

**Table 9.4.1.2.2.3-1: PMI test for single-layer (TDD)**

| Parameter | Unit | Test 1 |
|---|---|---|
| Bandwidth | MHz | 10 |
| Transmission mode | | 6 |
| Uplink downlink configuration | | 1 |
| Special subframe configuration | | 4 |

| | | | |
|---|---|---|---|
| Propagation channel | | | [EVA5] |
| Correlation and antenna configuration | | | Low 4 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| PMI delay | | ms | [10] |
| Reporting mode | | | PUCCH 2-1 (Note 6) |
| Reporting periodicity | | ms | $N_P$ = 5 |
| Physical channel for CQI reporting | | | PUSCH (Note 3) |
| PUCCH Report Type for wideband CQI/PMI | | | 2 |
| PUCCH Report Type for subband CQI | | | 1 |
| Measurement channel | | | [R.14-1 TDD] |
| OCNG Pattern | | | [OP.1/2 TDD] |
| Precoding granularity | | PRB | 6 (full size) |
| Number of bandwidth parts (*J*) | | | 3 |
| K | | | 1 |
| *cqi-pmi-ConfigIndex* | | | 4 |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| ACK/NACK feedback mode | | | Multiplexing |

Note 1: For random precoder selection, the precoder shall be updated in each available downlink transmission instance

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

Note 3: To avoid collisions between HARQ-ACK and wideband CQI/PMI or subband CQI it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#4 and #9 to allow periodic CQI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#8 and #3.

Note 4: Reports for the short subband (having 2RBs in the last bandwidth part) are to be disregarded and instead data is to be transmitted on the most recently used subband for bandwidth part with j=1.

Note 5: In the case where wideband PMI is reported, data is to be transmitted on the most recently used subband.

Note 6: The bit field for PMI confirmation in DCI format 1B shall be mapped to “0” and TPMI information shall indicate the codebook index used in Table 6.3.4.2.3-2 of TS36.211 [4] according to the latest PMI report on PUCCH.

**Table 9.4.1.2.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | [1.2] |
| UE Category | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.1.2.2.

#### 9.4.1.2.2.4 Test description

##### 9.4.1.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.1.2.2.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2.Message contents are defined in clause 9.4.1.2.2.4.3.

##### 9.4.1.2.2.4.2 Test procedure

1. Set the bandwidth, propagation condition, antenna configuration and measurement channel parameters according to Table 9.4.1.2.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #3 and #8 to carry the both ACK and CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2
3. Set SNR to $SNR_{rnd}$ . The SS shall transmit PDCCH with DCI format 0 in which CQI request bit is set to true [every subframe]. Then the SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission in subframe #3 and #8 to carry the both ACK and CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3
4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$ If the ratio ( throughput / $t_{rnd}$ ) $\geq \gamma$ which is specified in table 9.4.1.2.2.5-1, then the test is pass. Otherwise, the test is fail.

##### 9.4.1.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.1.2.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.1.2.2.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n4TxAntenna-tm6 | 1111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.1.2.2.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 4 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| subbandCQI | 1 | | |
| } | | | |
| ri-ConfigIndex | 484 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

9.4.1.2.2.5 Test requirement

**Table 9.4.1.2.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | [1.19] |
| UE Category | 1-5 |

## 9.4.2 Multiple PMI

### 9.4.2.1 PMI Reporting – PUSCH 1-2 (Multiple PMI)

#### 9.4.2.1.1 FDD PMI Reporting – PUSCH 1-2 (Multiple PMI)

##### 9.4.2.1.1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

##### 9.4.2.1.1.2 Test applicability

This test applies to E-UTRA FDD UE release 8 of UE category 2-5.

##### 9.4.2.1.1.3 Minimum conformance requirements

For the parameters specified in Table 9.4.2.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.2.1.1.3-2.

**Table 9.4.2.1.1.3-1 PMI test for single-layer (FDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 20 |
| Transmission mode | | | 6 |
| Propagation channel | | | EPA5 |
| Precoding granularity (only for reporting and following PMI) | | PRB | 8 |
| Correlation and antenna configuration | | | Low 2 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| Reporting mode | | | PUSCH 1-2 |
| Reporting interval | | ms | 1 |
| PMI delay | | ms | 8 |
| Measurement channel | | | R.30 FDD |
| OCNG Pattern | | | OP.1 FDD |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| Note 1: For random precoder selection, the precoders shall be updated in each TTI (1 ms granularity)<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4) | | | |

**Table 9.4.2.1.1.3-2: Minimum requirement (FDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.2 |
| UE Category | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.2.1.1.

#### 9.4.2.1.1.4 Test description

##### 9.4.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 20MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.2.1.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.2.1.1.4.3.

##### 9.4.2.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.4.2.1.1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2
3. Set SNR to $SNR_{rnd}$ . The SS shall transmit PDCCH with DCI format 0 in which CQI request bit is set to true [every subframe]. Then the SS shall transmit PDSCH with precoding matrix according to PMI report from the UE.The SS sends downlink MAC padding bits on the DL RMC SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. . Measure the average throughput.
   Measure $t_{ue}$ according to Annex G.5.3
4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

##### 9.4.2.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.2.1.1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.1.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n2TxAntenna-tm6 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.2.1.1.5 Test requirement

**Table 9.4.2.1.1.5-1: Test requirement (FDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.19 |
| UE Category | 2-5 |

## 9.4.2.1.1_1 FDD PMI Reporting – PUSCH 1-2 (Multiple PMI) (Release 9 and forward)

### 9.4.2.1.1_1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 9.4.2.1.1_1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

### 9.4.2.1.1_1.3 Minimum conformance requirements

For the parameters specified in Table 9.4.2.1.1_1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.2.1.1_1.3-2.

**Table 9.4.2.1.1_1.3-1: PMI test for single-layer (FDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Transmission mode | | | 6 |
| Propagation channel | | | EPA5 |
| Precoding granularity (only for reporting and following PMI) | | PRB | 6 |
| Correlation and antenna configuration | | | Low 2 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| Reporting mode | | | PUSCH 1-2 |
| Reporting interval | | ms | 1 |
| PMI delay | | ms | 8 |
| Measurement channel | | | R.11-3 FDD for UE Category 1, R.11 FDD for UE Category 2-5 |
| OCNG Pattern | | | OP.1 FDD |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |

Note 1: For random precoder selection, the precoders shall be updated in each TTI (1 ms granularity)

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

Note 3: One/two sided dynamic OCNG Pattern OP.1/2 FDD as described in Annex A.5.1.1/2 shall be used.

**Table 9.4.2.1.1_1.3-2: Minimum requirement (FDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.2 |
| UE Category | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.2.1.1.

### 9.4.2.1.1_1.4 Test description

#### 9.4.2.1.1_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.2.1.1_1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.2.1.1_1.4.3.

### 9.4.2.1.1_1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.4.2.1.1_1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2
3. Set SNR to $SNR_{rnd}$ . The SS shall transmit PDCCH with DCI format 0 in which CQI request bit is set to true [every subframe]. Then the SS shall transmit PDSCH with precoding matrix according to PMI report from the UE.The SS sends downlink MAC padding bits on the DL RMC SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. . Measure the average throughput.
   Measure $t_{ue}$ according to Annex G.5.3
4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

### 9.4.2.1.1_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.2.1.1_1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.1_1.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | Tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n2TxAntenna-tm6 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.1_1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.2.1.1_1.5 Test requirement

**Table 9.4.2.1.1_1.5-1: Test requirement (FDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.19 |
| UE Category | 1-5 |

## 9.4.2.1.2 TDD PMI Reporting – PUSCH 1-2 (Multiple PMI)

### 9.4.2.1.2.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 9.4.2.1.2.2 Test applicability

This test applies to E-UTRA TDD UE release 8 of UE category 2-5.

### 9.4.2.1.2.3 Minimum conformance requirements

For the parameters specified in Table 9.4.2.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.2.1.2.3-2.

**Table 9.4.2.1.2.3-1: PMI test for single-layer (TDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 20 |
| Transmission mode | | | 6 |
| Uplink downlink configuration | | | 1 |
| Special subframe configuration | | | 4 |
| Propagation channel | | | EPA5 |
| Precoding granularity (only for reporting and following PMI) | | PRB | 8 |
| Correlation and antenna configuration | | | Low 2 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| Reporting mode | | | PUSCH 1-2 |
| Reporting interval | | ms | 1 |
| Minimum PMI delay | | ms | 10 or 11 |
| Measurement channel | | | R.30 TDD |
| OCNG Pattern | | | OP.1 TDD |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: For random precoder selection, the precoders shall be updated in each available downlink transmission instance<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4) | | | |

**Table 9.4.2.1.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.2 |
| UE Category | 2-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.2.1.2.

### 9.4.2.1.2.4 Test description

#### 9.4.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 20MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Table 9.4.2.1.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B clause B.0

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.2.1.2.4.3.

#### 9.4.2.1.2.4.2 Test procedure

1. Set the bandwidth, propagation condition, antenna configuration and measurement channel parameters according to Table 9.4.2.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2

3. Set SNR to $SNR_{rnd}$. The SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3

4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

#### 9.4.2.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.2.1.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.2.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n2TxAntenna-tm6 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |

**Table 9.4.2.1.2.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.2.1.2.5 Test requirement

**Table 9.4.2.1.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.19 |
| UE Category | 2-5 |

## 9.4.2.1.2_1 TDD PMI Reporting – PUSCH 1-2 (Multiple PMI) (Release 9 and forward)

### 9.4.2.1.2_1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 9.4.2.1.2_1.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 9.4.2.1.2_1.3 Minimum conformance requirements

For the parameters specified in Table 9.4.2.1.2_1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.2.1.2_1.3-2.

**Table 9.4.2.1.2_1.3-1: PMI test for single-layer (TDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Transmission mode | | | 6 |
| Uplink downlink configuration | | | 1 |
| Special subframe configuration | | | 4 |
| Propagation channel | | | EPA5 |
| Precoding granularity (only for reporting and following PMI) | | PRB | 6 |
| Correlation and antenna configuration | | | Low 2 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -3 |
| | $\rho_B$ | dB | -3 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| Reporting mode | | | PUSCH 1-2 |
| Reporting interval | | ms | 1 |
| PMI delay | | ms | 10 or 11 |
| Measurement channel | | | R.11-3 TDD for UE Category 1 R.11 TDD for UE Category 2-5 |
| OCNG Pattern | | | OP.1 TDD |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| ACK/NACK feedback mode | | | Multiplexing |

Note 1: For random precoder selection, the precoders shall be updated in each available downlink transmission instance

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

Note 3: One/two sided dynamic OCNG Pattern OP.1/2 TDD as described in Annex A.5.2.1/2 shall be used.

**Table 9.4.2.1.2_1.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.2 |
| UE Category | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.2.1.2.

### 9.4.2.1.2_1.4 Test description

#### 9.4.2.1.2_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.2.1.2_1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.2.1.2_1.4.3.

### 9.4.2.1.2_1.4.2 Test procedure

1. Set the bandwidth, propagation condition, antenna configuration and measurement channel parameters according to Table 9.4.2.1.2_1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2
3. Set SNR to $SNR_{rnd}$. The SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3
4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

### 9.4.2.1.2_1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.2.1.2_1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.2_1.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n2TxAntenna-tm6 | 1111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.2.1.2_1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm12 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.2.1.2_1.5 Test requirement

**Table 9.4.2.1.2_1.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.19 |
| UE Category | 1-5 |

## 9.4.2.2 PMI Reporting – PUSCH 2-2 (Multiple PMI)

### 9.4.2.2.1 FDD PMI Reporting – PUSCH 2-2 (Multiple PMI)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Throughput ratio is still conducted in square brackets by RAN4

#### 9.4.2.2.1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

#### 9.4.2.2.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE release 9 and forward.

#### 9.4.2.2.1.3 Minimum conformance requirements

For the parameters specified in Table 9.4.2.2.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.2.2.1.3-2.

**Table 9.4.2.2.1.3-1: PMI test for single-layer (FDD)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Transmission mode | | | 6 |
| Propagation channel | | | [EVA5] |
| Correlation and antenna configuration | | | Low 4 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| PMI delay | | ms | 8 |
| Reporting mode | | | PUSCH 2-2 |
| Reporting interval | | ms | 1 |
| Measurement channel | | | [R.14-2 FDD] |
| OCNG Pattern | | | [OP.1/2 FDD] |
| Subband size (*k*) | | RBs | 3 (full size) |
| Number of preferred subbands (*M*) | | | 5 |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |

Note 1: For random precoder selection, the precoder shall be updated in each TTI (1 ms granularity)

Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4)

**Table 9.4.2.2.1.3-2: Minimum requirement (FDD)**

| | Test 1 |
|---|---|
| $\gamma$ | [1.2] |
| UE Category | [1-5] |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.2.2.1.

### 9.4.2.2.1.4 Test description

#### 9.4.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.2.2.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.2.2.1.4.3.

#### 9.4.2.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 9.4.2.2.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2

3. Set SNR to $SNR_{rnd}$ . The SS shall PDCCH with DCI format 0 in which CQI request bit is set to true [every subframe]. Then the SS shall transmit PDSCH with precoding matrix according to PMI report from the UE.The SS sends downlink MAC padding bits on the DL RMC SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. . Measure the average throughput.
   Measure $t_{ue}$ according to Annex G.5.3

4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

#### 9.4.2.2.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.2.2.1.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.2.2.1.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n4TxAntenna-tm6 | 1111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.2.2.1.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm22 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.2.2.1.5 Test requirement

**Table 9.4.2.2.1.5-1: Test requirement (FDD)**

| | **Test 1** |
|---|---|
| $\gamma$ | [1.19] |
| UE Category | [1-5] |

## 9.4.2.2.2 TDD PMI Reporting – PUSCH 2-2 (Multiple PMI)

*Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:*

- Throughput ratio is still conducted in square brackets by RAN4

### 9.4.2.2.2.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 9.4.2.2.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE release 9 and forward.

### 9.4.2.2.2.3 Minimum conformance requirements

For the parameters specified in Table 9.4.2.2.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.4.2.2.2.3-2.

**Table 9.4.2.2.2.3-1: PMI test for single-layer (TDD)**

| **Parameter** | **Unit** | **Test 1** |
|---|---|---|
| Bandwidth | MHz | 10 |
| Transmission mode | | 6 |
| Uplink downlink configuration | | 1 |
| Special subframe configuration | | 4 |

| | | | |
|---|---|---|---|
| Propagation channel | | | [EVA5] |
| Correlation and antenna configuration | | | Low 4 x 2 |
| Downlink power allocation | $\rho_A$ | dB | -6 |
| | $\rho_B$ | dB | -6 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 |
| PMI delay | | ms | [10] |
| Reporting mode | | | PUSCH 2-2 |
| Reporting interval | | ms | 1 |
| Measurement channel | | | [R.14-2 TDD] |
| OCNG Pattern | | | [OP.1/2 FDD] |
| Subband size (*k*) | | RBs | 3 (full size) |
| Number of preferred subbands (*M*) | | | 5 |
| Max number of HARQ transmissions | | | 4 |
| Redundancy version coding sequence | | | {0,1,2,3} |
| ACK/NACK feedback mode | | | Multiplexing |
| Note 1: For random precoder selection, the precoders shall be updated in each available downlink transmission instance<br>Note 2: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI estimation at a downlink SF not later than SF#(n-4), this reported PMI cannot be applied at the eNB downlink before SF#(n+4) | | | |

**Table 9.4.2.2.2.3-2: Minimum requirement (TDD)**

| | **Test 1** |
|---|---|
| $\gamma$ | [1.15] |
| UE Category | [1-5] |

The normative reference for this requirement is TS 36.101 [2] clause 9.4.2.2.2.

### 9.4.2.2.2.4 Test description

#### 9.4.2.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.4.2.2.2.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.4.2.2.2.4.3.

9.4.2.2.2.4.2 Test procedure

1. Set the bandwidth, propagation condition, antenna configuration and measurement channel parameters according to Table 9.4.2.2.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1B for C_RNTI to transmit the DL RMC with randomly selected precoding matrix from codebook (Table 6.3.4.2.3-1 in TS 36.211 [8]) every subframe regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Establish $t_{rnd}$ and $SNR_{rnd}$ according to annex G.5.2

3. Set SNR to $SNR_{rnd}$. The SS shall transmit PDSCH with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{ue}$ according to Annex G.5.3

4. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$

9.4.2.2.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.4.2.2.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| explicitValue | AntennaInfoDedicated | | |
| } | | | |
| } | | | |

**Table 9.4.2.2.2.4.3-2: *AntennaInfoDedicated***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| AntennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm6 | | |
| codebookSubsetRestriction CHOICE{ | | | |
| n4TxAntenna-tm6 | 1111111111111111 | | |
| } | | | |
| ue-TransmitAntennaSelection CHOICE{ | | | |
| release | NULL | | |
| } | | | |
| } | | | |

**Table 9.4.2.2.2.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm22 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic | Not present | | |
| } | | | |

9.4.2.2.2.5 Test requirement

**Table 9.4.2.2.2.5-1: Test requirement (TDD)**

| | Test 1 |
|---|---|
| $\gamma$ | [1.14] |
| UE Category | [1-5] |

# 9.5 Reporting of Rank Indicator (RI)

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI (CQI) reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission. Transmission mode 4 is used with the specified CodebookSubSetRestriction.

For fixed rank 1 transmission, the RI and PMI reporting is restricted to two single-layer precoders, For fixed rank 2 transmission, the RI and PMI reporting is restricted to one two-layer precoder, For follow RI transmission, the RI and PMI reporting is restricted to select the union of these precoders. Channels with low and high correlation are used to ensure that RI reporting reflects the channel condition.

## 9.5.1 RI Reporting

### 9.5.1.1 FDD RI Reporting– PUCCH 1-1

#### 9.5.1.1.1 Test purpose

To verify that the reported rank indicator accurately represents the channel rank.

#### 9.5.1.1.2 Test applicability

This test applies to E-UTRA FDD UE release 8 and forward of UE category 2-5.

#### 9.5.1.1.3 Minimum conformance requirements

The minimum performance requirement in Table 9.5.1.1.3-2 is defined as

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

TBS selection is based on the UE wideband CQI feedback. The transport block size TBS for wideband CQI is selected according to Table A.4-3a.

For the parameters specified in Table 9.5.1.1.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.5.1.1.3-2.

**Table 9.5.1.1.3-1: RI Test (FDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | |
| PDSCH transmission mode | | | 4 | | |
| Downlink power allocation | $\rho_A$ | dB | -3 | | |
| | $\rho_B$ | dB | -3 | | |
| CodeBookSubsetRestriction bitmap | | | 000011 for fixed RI = 1<br>010000 for fixed RI = 2<br>010011 for UE reported RI | | |
| Propagation condition and antenna configuration | | | 2 x 2 EPA5 | | |
| Antenna correlation | | | Low | Low | High |
| RI configuration | | | Fixed RI=2 and follow RI | Fixed RI=1 and follow RI | Fixed RI=2 and follow RI |
| SNR | | dB | 0 | 20 | 20 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 | -98 | -98 |
| $\hat{I}_{or}^{(j)}$ | | dB[mW/15kHz] | -98 | -78 | -78 |
| Maximum number of HARQ transmissions | | | 1 | | |
| Reporting mode | | | PUCCH 1-1 (Note 4) | | |
| Physical channel for CQI/PMI reporting | | | PUCCH Format 2 | | |
| PUCCH Report Type for CQI/PMI | | | 2 | | |
| Physical channel for RI reporting | | | PUSCH (Note 3) | | |
| PUCCH Report Type for RI | | | 3 | | |
| Reporting periodicity | | ms | $N_P$ = 5 | | |
| PMI and CQI delay | | ms | 8 | | |
| *cqi-pmi-ConfigurationIndex* | | | 6 | | |
| *ri-ConfigurationInd* | | | 1 (Note 5) | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI and CQI estimation at a downlink subframe not later than SF#(n-4), this reported PMI and wideband CQI cannot be applied at the eNB downlink before SF#(n+4).

Note 2: Reference measurement channel according to Table A.4-1 with one sided dynamic OCNG Pattern OP.1 FDD as described in Annex A.5.1.1.

Note 3: To avoid collisions between RI reports and HARQ-ACK it is necessary to report both on PUSCH instead of PUCCH. PDCCH DCI format 0 shall be transmitted in downlink SF#4 and #9 to allow periodic RI to multiplex with the HARQ-ACK on PUSCH in uplink subframe SF#8 and #3.

Note 4: The bit field for precoding information in DCI format 2 shall be mapped as:
- For reported RI = 1 and PMI = 0 >> precoding information bit field index = 1
- For reported RI = 1 and PMI = 1 >> precoding information bit field index = 2
- For reported RI = 2 and PMI = 0 >> precoding information bit field index = 0

Note 5: To avoid the ambiguity of SS behaviour when applying CQI and PMI during rank switching, RI reports are to be applied at the SS with one subframe delay in addition to Note 1 to align with CQI and PMI reports.

**Table 9.5.1.1.3-2: Minimum requirement (FDD)**

| Parameter | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | N/A |
| $\gamma_2$ | 1 | N/A | 1.1 |
| UE Category | 2-5 | 2-5 | 2-5 |

#### 9.5.1.1.4 Test description

##### 9.5.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.1.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.1

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.
2. The parameter settings for the cell are set up according to Table 9.5.1.1.3-1.
3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.5.1.1.4.3.

##### 9.5.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, antenna correlation, CodeBookSubsetRestriction for fixed Rank and the SNR according to Table 9.5.1.1.3-1 as appropriate.
2. The SS shall send PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to the UE reported CQI, PMI and RI. In case of CQI reports for two codewords (codeword #0 and #1), the CQI offset level for codeword #1 is selected from {0,1,2,3,-4,-3,-2,-1}. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-3a. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #3 and #8 according to Annex A.4-10. Measure the $t_{fix}$ according to annex G.5.3.
3. Propagation conditions are set according to Annex B.1.
4. The SS sends uplink scheduling information via PDCCH DCI format 0 to schedule UL RMC in every subframe according to Annex A.4-10.
5. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 9.5.1.1.3-1.
6. The UE shall transmit RRC Connection Reconfiguration Complete message.
7. Propagation conditions are set according to Table 9.5.1.1.3-1.
8. The SS shall send PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to the UE reported CQI, PMI and RI. In case of CQI reports for two codewords (codeword #0 and #1), the CQI offset level for codeword #1 is selected from {0,1,2,3,-4,-3,-2,-1}. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-3a. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit set to 0 to schedule UL RMC in subframe #3 and #8 according to Annex A.4-10. Measure $t_{reported}$ according to Annex G.5.3

   If the ratio ($t_{reported}$ / $t_{fix}$) satisfies the requirement in Table 9.5.1. 1.5-1, then pass the UE for this test and go to step 9. Otherwise, fail the UE.
9. If all tests have not been done, then repeat the same procedure (steps 1 to 8) with test conditions according to the Table 9.5.1.1.3-2 for the other Tests as appropriate. Otherwise pass the UE.

9.5.1.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.5.1.1.4.3-1: PhysicalConfigDedicated-DEFAULT**

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-SRB2-DRB(n, m) ::= SEQUENCE { | | | |
| physicalConfigDedicated | PhysicalConfigDedicated - DEFAULT using condition RBC | | |
| } | | | |

**Table 9.5.1.1.4.3-2: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | RBC |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| } | | | |
| codebookSubsetRestriction CHOICE { | | | |
| N2TxAntenna-tm4 | According to each test | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.1.4.3-3: *PDSCH-ConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-3 | | |
| } | | | |

**Table 9.5.1.1.4.3-4: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | Not present | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | | | |
| setup SEQUENCE { | | | |
| cqi-PUCCH-ResourceIndex | 0 | | |
| cqi-pmi-ConfigIndex | 6 | (see Table 7.2.2-1A in TS 36.213) | |
| cqi-FormatIndicatorPeriodic CHOICE { | | | |
| widebandCQI | NULL | | |
| } | | | |
| ri-ConfigIndex | 1 | (see Table 7.2.2-1B in TS 36.213) | |
| simultaneousAckNackAndCQI | FALSE | | |
| } | | | |
| } | | | |
| } | | | |

### 9.5.1.1.5 Test requirement

**Table 9.5.1.1.5-1: Test requirement (FDD)**

| Parameter | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | N/A |
| $\gamma_2$ | 0.99 | N/A | 1.09 |
| UE Category | 2-5 | 2-5 | 2-5 |

## 9.5.1.2 TDD RI Reporting – PUSCH 3-1

### 9.5.1.2.1 Test purpose

To verify that the reported rank indicator accurately represents the channel rank.

### 9.5.1.2.2 Test applicability

This test applies to E-UTRA TDD UE release 8 and forward, which support UE Category 2-5.

### 9.5.1.2.3 Minimum conformance requirements

The minimum performance requirement in Table 9.5.1.2.3-2 is defined as

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

TBS selection is based on the UE wideband CQI feedback. The transport block size TBS is that resulting from the code rate which is closest to that indicated by M = wideband CQI and the $N_{PRB}$ entry in Table 7.1.7.2.1-1 of TS36.213 [10] that corresponds to the transmission bandwidth configuration in Table 5.6-1 of TS36.101 [2].

For the parameters specified in Table 9.5.1.2.3-1, and using the downlink physical channels specified in Annex C, the minimum requirements are specified in Table 9.5.1.2.3-2.

**Table 9.5.1.2.3-1: RI Test (TDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | |
| PDSCH transmission mode | | | 4 | | |
| Downlink power allocation | $\rho_A$ | dB | -3 | | |
| | $\rho_B$ | dB | -3 | | |
| Uplink downlink configuration | | | 2 | | |
| Special subframe configuration | | | 4 | | |
| Propagation condition and antenna configuration | | | 2 x 2 EPA5 | | |
| CodeBookSubsetRestriction bitmap | | | 000011 for fixed RI = 1<br>010000 for fixed RI = 2<br>010011 for UE reported RI | | |
| Antenna correlation | | | Low | Low | High |
| RI configuration | | | Fixed RI=2 and follow RI | Fixed RI=1 and follow RI | Fixed RI=2 and follow RI |
| SNR | | dB | 0 | 20 | 20 |
| $N_{oc}^{(j)}$ | | dB[mW/15kHz] | -98 | -98 | -98 |
| $\hat{I}_{or}^{(j)}$ | | dB[mW/15kHz] | -98 | -78 | -78 |
| Maximum number of HARQ transmissions | | | 1 | | |
| Reporting mode | | | PUSCH 3-1 (Note 3) | | |
| Reporting interval | | ms | 5 | | |
| PMI and CQI delay | | ms | 10 or 11 | | |
| ACK/NACK feedback mode | | | Bundling | | |

Note 1: If the UE reports in an available uplink reporting instance at subframe SF#n based on PMI and CQI estimation at a downlink subframe not later than SF#(n-4), this reported PMI and wideband CQI cannot be applied at the eNB downlink before SF#(n+4).
Note 2: Reference measurement channel according to Table A.4-2 with one sided dynamic OCNG Pattern OP.1 TDD as described in Annex A.5.2.1.
Note 3: Reported wideband CQI and PMI are used and sub-band CQI is discarded.

**Table 9.5.1.2.3-2: Minimum requirement (TDD)**

| Parameter | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | N/A |
| $\gamma_2$ | 1 | N/A | 1.1 |
| UE Category | 2-5 | 2-5 | 2-5 |

## 9.5.1.2.4 Test description

### 9.5.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.2.

Test Environment: Normal as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 36.508 [7] clause 4.3.1.2.

Channel Bandwidths to be tested: 10MHz, as defined in TS 36.508 [7] clause 4.3.1.2

1. Connect the SS, faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.10.

2. The parameter settings for the cell are set up according to Table 9.5.1.2.3-1.

3. Downlink signals are initially set up according to Annex C.1 and Annex C.3.2, and uplink signals according to Annex H.1 and H.3.2.

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 9.5.1.2.4.3.

#### 9.5.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration, antenna correlation, CodeBookSubsetRestriction for fixed Rank and the SNR according to Table 9.5.1.2.3-1 as appropriate.

2. The SS shall send PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-3a. The SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure the $t_{fix}$ according to annex G.5.3

3. Propagation conditions are set according to Annex B.1.

4. The SS sends uplink scheduling information via PDCCH DCI format 0 with CQI request bit to 0 to schedule UL RMC in subframe#2 and #7 according to Annex A.4-11.

5. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 9.5.1.2.3-1.

6. The UE shall transmit RRC Connection Reconfiguration Complete message.

7. Propagation conditions are set according to Table 9.5.1.2.3-1.

8. The SS shall send PDSCH via PDCCH DCI format 2 for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC. The transport format to be used is defined in Annex A.4 Table A.4-3a. The SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0 with CQI request bit set to 1 and I_MCS=29 and N_PRB allocated to be less or equal to 4. Measure $t_{reported}$ according to Annex G.5.3

   If the ratio ($t_{reported}$ / $t_{fix}$) satisfies the requirement in Table 9.5.1.2.5-1, then pass the UE for this test and go to step 9. Otherwise, fail the UE.

9. If all tests have not been done, then repeat the same procedure (steps 1 to 8) with test conditions according to the Table 9.5.1.2.3-2 for the other Tests as appropriate. Otherwise pass the UE.

#### 9.5.1.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6 with the following exceptions:

**Table 9.5.1.2.4.3-1: *PhysicalConfigDedicated-DEFAULT***

| Derivation Path: 36.508 clause 5.5.1.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportConfig | *CQI-ReportConfig-DEFAULT* | | |
| antennaInfo CHOICE { | | | |
| antennaInfoDedicated ::= SEQUENCE { | | | |
| transmissionMode | tm4 | | |
| } | | | |
| codebookSubsetRestriction CHOICE { | | | |
| N2TxAntenna-tm4 | According to each test | | |
| ue-TransmitAntennaSelection CHOICE { | | | |
| release | NULL | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.5.1.2.4.3-2: *PDSCH-ConfigDedicated-DEFAULT***

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| p-a | dB-3 | | |
| } | | | |

**Table 9.5.1.2.4.3-3: *CQI-ReportConfig-DEFAULT***

| Derivation Path: 36.508 clause 4.6.3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | rm31 | | |
| nomPDSCH-RS-EPRE-Offset | 0 | | |
| cqi-ReportPeriodic CHOICE { | Not Present | | |
| } | | | |
| } | | | |

**Table 9.5.1.2.4.3-4: PUCCH-ConfigDedicated-DEFAULT**

| Derivation Path: 36.331 clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PUCCH-ConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| ackNackRepetition CHOICE { | | | |
| release | NULL | | |
| } | | | |
| tddAckNackFeedbackMode | Bundling | Multiplexing is selected as default to align with RAN4's assumptions in RF tests. | TDD |
| } | | | |

#### 9.5.1.2.5 Test requirement

**Table 9.5.1.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | N/A |
| $\gamma_2$ | 0.99 | N/A | 1.09 |
| UE Category | 2-5 | 2-5 | 2-5 |

# 10 MBMS Performance

## 10.1 FDD MBMS performance (Fixed Reference Channel)

### 10.1.1 Test purpose

This test verifies the performance of FDD MBMS with a given SNR for which the average BLER remains below a given reference value.

### 10.1.2 Test applicability

This test applies to all types of E-UTRA FDD UE supporting MBMS release 9 and forward.

### 10.1.3 Minimum conformance requirements

The parameters specified in Table 10.1.3-1 are valid for all FDD tests unless otherwise stated. For the requirements defined in this section, the difference between CRS EPRE and the MBSFN RS EPRE should be set to 0 dB as the UE demodulation performance might be different when this condition is not met (e.g. in scenarios where power offsets are present, such as scenarios when reserved cells are present).

**Table 10.1.3-1: Common Test Parameters (FDD)**

| Parameter | Unit | Value |
|---|---|---|
| Number of HARQ processes | Processes | None |
| Subcarrier spacing | kHz | 15 kHz |
| Allocated subframes per Radio Frame (Note 1) | | 6 subframes |
| Number of OFDM symbols for PDCCH (Note 2) | | 2 symbols in the case of 3 PHICH symbols or 4 RS Ports;<br>1 or 2 symbols for other scenarios. |
| Cyclic Prefix | | Extended |
| Note1: For FDD mode, up to 6 subframes (#1/2/3/6/7/8) are available for MBMS, in line with TS 36.331.<br>Note2: 2 OFDM symbols are reserved for PDCCH in this subclause. | | |

The receive characteristic of MBMS is determined by the BLER. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

For the parameters specified in Table 10.1.3-1 and Table 10.1.3-2 and Annex A.3.8.1, the average downlink SNR shall be below the specified value for the BLER shown in Table 10.1.3-3.

**Table 10.1.3-2: Test Parameters for Testing**

| Parameter | | Unit | Test 1-4 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 |
| | $\rho_B$ | dB | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 0$ | | | |

**Table 10.1.3-3: Minimum performance**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation condition | Correlation Matrix and antenna | Reference value | | MBMS UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR(dB) | |
| 1 | 10 MHz | R.37 FDD | OP.4 FDD | MBSFN channel model (Table B.2.6-1) | 1x2 low | 1 | 4.1 | 1-5 |
| 2 | 10 MHz | R.38 FDD | OP.4 FDD | | | | 11.0 | 1-5 |
| 3 | 10 MHz | R.39 FDD | OP.4 FDD | | | | 20.1 | 2-5 |
| | 5.0MHz | R.39-1 FDD | OP.4 FDD | | | | 20.5 | 1 |
| 4 | 1.4 MHz | R.40 FDD | OP.4 FDD | | | | 6.6 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 10.1.

## 10.1.4 Test description

### 10.1.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.

Channel Bandwidths to be tested: As specified per test number in Table 10.1.3-3 as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 10.1.3-1.
3. The downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0.
5. SS configures 10 MBMS packets are included in one TB.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 10.1.4.3.

### 10.1.4.2 Test procedure

1. Initiate the variables $M_{tot}$ to 0. SNR is set according to table 10.1.5-1.Propagation conditions are set according to Annex B clause B.2.6
2. SS shall send MBMS Packets on the MTCH radio bearer for the test time specified in Table G.6.4-1) . SS stores the count of transmitted MBMS Packets on the MTCH in the variable $M_{tot}$.

3. SS shall send a "UE TEST LOOP MODE C MBMS PACKET COUNTER REQUEST" message and wait for the UE to respond with a "UE TEST LOOP MODE C MBMS PACKET COUNTER RESPONSE" reporting the received RLC SDU counter value. Store this counter in the variable $M_{ok}$.

4. SS shall compute the BLER as the following:

   The BLER = $(M_{tot} – M_{ok}) / M_{tot}$

5. Repeat steps from 1 to 4 for each subtest in Table 10.1.5-1 as appropriate.

#### 10.1.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

### 10.1.5 Test requirement

For the parameters specified in Table 10.1.3-1, Table 10.1.3-2, Annex A.3.8.1, and SNR in Table 10.1.5-1,the value for the BLER in step 4 shall be below the test limit in Annex G.6.3 for all subtests.

**Table 10.1.5-1: Test requirement**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation condition | Correlation Matrix and antenna | Reference value | | MBMS UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR(dB) | |
| 1 | 10 MHz | R.37 FDD | OP.4 FDD | MBSFN channel model (Table B.2.6-1) | 1x2 low | 1 | 5 | 1-5 |
| 2 | 10 MHz | R.38 FDD | OP.4 FDD | | | | 11.9 | 1-5 |
| 3 | 10 MHz | R.39 FDD | OP.4 FDD | | | | 21.0 | 2-5 |
| | 5.0MHz | R.39-1 FDD | OP.4 FDD | | | | 21.4 | 1 |
| 4 | 1.4 MHz | R.40 FDD | OP.4 FDD | | | | 7.5 | 1-5 |

## 10.2 TDD MBMS performance (Fixed Reference Channel)

### 10.2.1 Test purpose

This test verifies the performance of TDD MBMS with a given SNR for which the average BLER remains below a given reference value.

### 10.2.2 Test applicability

This test applies to all types of E-UTRA TDD UE supporting MBMS release 9 and forward.

### 10.2.3 Minimum conformance requirements

The parameters specified in Table 10.2.3-1 are valid for all TDD tests unless otherwise stated. For the requirements defined in this section, the difference between CRS EPRE and the MBSFN RS EPRE should be set to 0 dB as the UE demodulation performance might be different when this condition is not met (e.g. in scenarios where power offsets are present, such as scenarios when reserved cells are present).

**Table 10.2.3-1: Common Test Parameters (TDD)**

| Parameter | Unit | Value |
|---|---|---|
| Number of HARQ processes | Processes | None |
| Subcarrier spacing | kHz | 15 kHz |
| Allocated subframes per Radio Frame (Note 1) | | 5 subframes |
| Number of OFDM symbols for PDCCH (Note 2) | | 2 symbols in the case of 3 PHICH symbols or 4 RS Ports; 1 or 2 symbols for other scenarios. |
| Cyclic Prefix | | Extended |
| Note1: For TDD mode, in line with TS 36.331, Uplink-Downlink Configuration 5 is proposed, up to 5 subframes (#3/4/7/8/9) are available for MBMS. Note2: 2 OFDM symbols are reserved for PDCCH in this subclause. | | |

The receive characteristic of MBMS is determined by the BLER. The requirement is valid for all RRC states for which the UE has capabilities for MBMS.

For the parameters specified in Table 10.2.3-1 and Table 10.2.3-2 and Annex A.3.8.2, the average downlink SNR shall be below the specified value for the BLER shown in Table 10.2.3-3.

**Table 10.2.3-2: Test Parameters for Testing**

| Parameter | | Unit | Test 1-4 |
|---|---|---|---|
| Downlink power allocation | $\rho_A$ | dB | 0 |
| | $\rho_B$ | dB | 0 (Note 1) |
| $N_{oc}$ at antenna port | | dBm/15kHz | -98 |
| Note 1: $P_B = 0$ | | | |

**Table 10.2.3-3: Minimum performance**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation condition | Correlation Matrix and antenna | Reference value | | MBMS UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR(dB) | |
| 1 | 10 MHz | R.37 TDD | OP.4 TDD | MBSFN channel model (Table B.2.6-1) | 1x2 low | 1 | 3.4 | 1-5 |
| 2 | 10 MHz | R.38 TDD | OP.4 TDD | | | | 11.1 | 1-5 |
| 3 | 10 MHz | R.39 TDD | OP.4 TDD | | | | 20.1 | 2-5 |
| | 5MHz | R.39-1 TDD | OP.4 TDD | | | | 20.5 | 1 |
| 4 | 1.4 MHz | R.40 TDD | OP.4 TDD | | | | 5.8 | 1-5 |

The normative reference for this requirement is TS 36.101 [2] clause 10.2.

## 10.2.4 Test description

### 10.2.4.1 Initial conditions

Test Environment: Normal, as defined in TS 36.508 [7] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 36.508 [7] clause 4.3.1.

Channel Bandwidths to be tested: As specified per test number in Table 10.2.3-3 as defined in TS 36.508 [7] clause 4.3.1.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connector (s) as shown in TS 36.508 [7] Annex A, Figure A.9.
2. The parameter settings for the cell are set up according to Table 10.2.3-1.
3. The downlink signals are initially set up according to Annex C.1 and Annex C.3.2 and uplink signals according to Annex H.1 and H.3.2.
4. Propagation conditions are set according to Annex B clause B.0.
5. SS configures 10 MBMS packets are included in one TB.
6. Ensure the UE is in State 3A-RF according to TS 36.508 [7] clause 5.2A.2. Message contents are defined in clause 10.2.4.3.

### 10.2.4.2 Test procedure

1. Initiate the variables $M_{tot}$ to 0. SNR is set according to table 10.2.5-1. Propagation conditions are set according to Annex B clause B.2.6
2. SS shall sendMBMS Packets on the MTCH radio bearer for the test time specified in Table G.6.4-1) . SS stores the count of transmitted MBMS Packets on the MTCH in the variable $M_{tot}$.
3. SS shall send a “UE TEST LOOP MODE C MBMS PACKET COUNTER REQUEST” message and wait for the UE to respond with a “UE TEST LOOP MODE C MBMS PACKET COUNTER RESPONSE” reporting the received MBMS Packet counter value. Store this counter in the variable $M_{ok}$.
4. SS shall compute the BLER as the following:

   The BLER = $(M_{tot} – M_{ok}) / M_{tot}$
5. Repeat steps from 1 to 4 for each subtest in Table 10.2.5-1 as appropriate.

### 10.2.4.3 Message contents

Message contents are according to TS 36.508 [7] clause 4.6.

## 10.2.5 Test requirement

For the parameters specified in Table 10.2.3-1, Table 10.2.3-2, Annex A.3.8. 2, and SNR in Table 10.2.5-1,the value for the BLER in step 4 shall be below the test limit in Annex G.6.3 for all subtests. shown in Table 10.2.5-1.

**Table 10.2.5-1: Test**

| Test number | Bandwidth | Reference Channel | OCNG Pattern | Propagation condition | Correlation Matrix and antenna | Reference value | | MBMS UE Category |
|---|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR(dB) | |
| 1 | 10 MHz | R.37 TDD | OP.4 TDD | MBSFN channel model (Table B.2.6-1) | 1x2 low | 1 | 4.3 | 1-5 |
| 2 | 10 MHz | R.38 TDD | OP.4 TDD | | | | 12 | 1-5 |
| 3 | 10 MHz | R.39 TDD | OP.4 TDD | | | | 21.0 | 2-5 |
| | 5MHz | R.39-1 TDD | OP.4 TDD | | | | 21.4 | 1 |
| 4 | 1.4 MHz | R.40 TDD | OP.4 TDD | | | | 6.7 | 1-5 |

# Annex A (normative): Measurement Channels

## A.1 General

A schematic overview of the encoding process for the reference measurement channels is provided in Figure A-1.

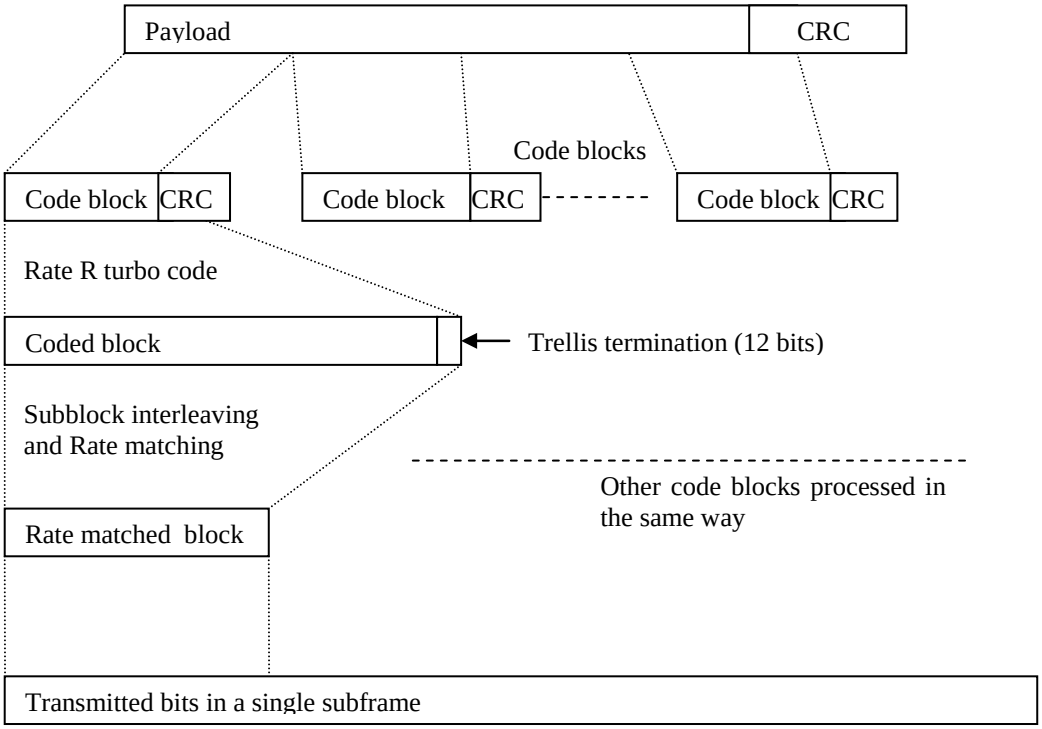



**Figure A-1: Schematic overview of the encoding process**

The throughput values defined in the measurement channels specified in Annex A, are calculated and are valid per data stream (codeword). For multi-stream (more than one codeword) transmissions, the throughput referenced in the minimum requirements is the sum of throughputs of all data streams (codewords).

The UE category entry in the definition of the reference measurement channels in Annex A is only informative and reveals the UE categories, which can support the corresponding measurement channel. Whether the measurement channel is used for testing a certain UE category or not is specified in the individual test cases.

# A.2 UL reference measurement channels

## A.2.1 General

### A.2.1.1 Applicability and common parameters

The following sections define the UL signal applicable to the Transmitter Characteristics (clause 6) and for the Receiver Characteristics (clause 7) where the UL signal is relevant.

The Reference channels in this section assume transmission of PUSCH and Demodulation Reference signal only. The following conditions apply:

- 1 HARQ transmission
- Cyclic Prefix normal
- PUSCH hopping off
- Link adaptation off
- Demodulation Reference signal as per TS 36.211 [8] subclause 5.5.2.1.2.

Where ACK/NACK is transmitted, it is assumed to be multiplexed on PUSCH as per TS 36.212 [9] subclause 5.2.2.6.

- ACK/NACK 1 bit
- ACK/NACK mapping adjacent to Demodulation Reference symbol
- ACK/NACK resources punctured into data
- Max number of resources for ACK/NACK: 4 SC-FDMA symbols per subframe
- No CQI transmitted, no RI transmitted

### A.2.1.2 Determination of payload size

The algorithm for determining the payload size $A$ is as follows; given a desired coding rate $R$ and radio block allocation $N_{RB}$

1. Calculate the number of channel bits $N_{ch}$ that can be transmitted during the first transmission of a given sub-frame.
2. Find $A$ such that the resulting coding rate is as close to $R$ as possible, that is,

   $\min\left|R-(A+24)/N_{ch}\right|$,

   subject to

   a) A is a valid TB size according to clause 7.1.7 of TS 36.213 [10] assuming an allocation of $N_{RB}$ resource blocks.

   b) Segmentation is not included in this formula, but should be considered in the TBS calculation.

   c) For RMC-s, which at the nominal target coding rate do not cover all the possible UE categories for the given modulation, reduce the target coding rate gradually (within the same modulation), until the maximal possible number of UE categories is covered.
3. If there is more than one $A$ that minimises the equation above, then the larger value is chosen per default.

### A.2.1.3 Overview of UL reference measurement channels

In Table A.2.1.3-1 are listed the UL reference measurement channels specified in annexes A.2.2 and A.2.3 of this release of TS 36.521-1. This table is informative and serves only to a better overview. The reference for the concrete reference measurement channels and corresponding implementation's parameters as to be used for testing are annexes A.2.2 and A.2.3 as appropriate.

**Table A.2.1.3-1: Overview of UL reference measurement channels**

| Duplex | Table | Name | BW | Mod | TCR | RB | RB Offset | UE Categ | Notes |
|---|---|---|---|---|---|---|---|---|---|
| ***FDD, Full RB allocation, QPSK*** | | | | | | | | | |
| *FDD* | *Table A.2.2.1.1-1* | | *1.4* | *QPSK* | *1/3* | *6* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.1-1* | | *3* | *QPSK* | *1/3* | *15* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.1-1* | | *5* | *QPSK* | *1/3* | *25* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.1-1* | | *10* | *QPSK* | *1/3* | *50* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.1-1* | | *15* | *QPSK* | *1/5* | *75* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.1-1* | | *20* | *QPSK* | *1/6* | *100* | | *≥ 1* | |
| ***FDD, Full RB allocation, 16-QAM*** | | | | | | | | | |
| *FDD* | *Table A.2.2.1.2-1* | | *1.4* | *16QAM* | *3/4* | *6* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.2-1* | | *3* | *16QAM* | *1/2* | *15* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.2-1* | | *5* | *16QAM* | *1/3* | *25* | | *≥ 1* | |
| *FDD* | *Table A.2.2.1.2-1* | | *10* | *16QAM* | *3/4* | *50* | | *≥ 2* | |
| *FDD* | *Table A.2.2.1.2-1* | | *15* | *16QAM* | *1/2* | *75* | | *≥ 2* | |
| *FDD* | *Table A.2.2.1.2-1* | | *20* | *16QAM* | *1/3* | *100* | | *≥ 2* | |
| ***FDD, Partial RB allocation, QPSK, 1.4 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.1-1* | | *1.4* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-1* | | *1.4* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-1* | | *1.4* | *QPSK* | *1/3* | *3* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-1* | | *1.4* | *QPSK* | *1/3* | *4* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-1* | | *1.4* | *QPSK* | *1/3* | *5* | | *≥ 1* | |
| ***FDD, Partial RB allocation, QPSK, 3 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *3* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *4* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *6* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-2* | | *3* | *QPSK* | *1/3* | *10* | | *≥ 1* | *Not yet used in tests* |
| ***FDD, Partial RB allocation, QPSK, 5 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.1-3* | | *5* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-3* | | *5* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-3* | | *5* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-3* | | *5* | *QPSK* | *1/3* | *6* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-3* | | *5* | *QPSK* | *1/3* | *8* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-3a* | | *5* | *QPSK* | *1/3* | *10* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-3a* | | *5* | *QPSK* | *1/3* | *15* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-3a* | | *5* | *QPSK* | *1/3* | *18* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-3a* | | *5* | *QPSK* | *1/3* | *20* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-3a* | | *5* | *QPSK* | *1/3* | *24* | | *≥ 1* | *Not yet used in tests* |
| ***FDD, Partial RB allocation, QPSK, 10 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.1-4* | | *10* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-4* | | *10* | *QPSK* | *1/3* | *2* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-4* | | *10* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.1-4* | | *10* | *QPSK* | *1/3* | *6* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.1-4* | | *10* | *QPSK* | *1/3* | *8* | | *≥ 1* | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FDD | Table A.2.2.2.1-4 | | 10 | QPSK | 1/3 | 10 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-4a | | 10 | QPSK | 1/3 | 12 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4a | | 10 | QPSK | 1/3 | 15 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4a | | 10 | QPSK | 1/3 | 16 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4a | | 10 | QPSK | 1/3 | 18 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-4a | | 10 | QPSK | 1/3 | 20 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4a | | 10 | QPSK | 1/3 | 24 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-4b | | 10 | QPSK | 1/3 | 25 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4b | | 10 | QPSK | 1/3 | 27 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-4b | | 10 | QPSK | 1/3 | 30 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4b | | 10 | QPSK | 1/3 | 36 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-4b | | 10 | QPSK | 1/3 | 40 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-4b | | 10 | QPSK | 1/3 | 48 | | ≥ 1 | |
| **FDD, Partial RB allocation, QPSK, 15 MHz** | | | | | | | | | |
| FDD | Table A.2.2.2.1-5 | | 15 | QPSK | 1/3 | 1 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5 | | 15 | QPSK | 1/3 | 2 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5 | | 15 | QPSK | 1/3 | 5 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5 | | 15 | QPSK | 1/3 | 6 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5 | | 15 | QPSK | 1/3 | 8 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5 | | 15 | QPSK | 1/3 | 9 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5a | | 15 | QPSK | 1/3 | 10 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5a | | 15 | QPSK | 1/3 | 16 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5a | | 15 | QPSK | 1/3 | 18 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5a | | 15 | QPSK | 1/3 | 20 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5a | | 15 | QPSK | 1/3 | 24 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5a | | 15 | QPSK | 1/3 | 25 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5b | | 15 | QPSK | 1/3 | 27 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5b | | 15 | QPSK | 1/3 | 36 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-5b | | 15 | QPSK | 1/3 | 40 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5b | | 15 | QPSK | 1/3 | 48 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5b | | 15 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-5b | | 15 | QPSK | 1/3 | 54 | | ≥ 1 | |
| **FDD, Partial RB allocation, QPSK, 20 MHz** | | | | | | | | | |
| FDD | Table A.2.2.2.1-6 | | 20 | QPSK | 1/3 | 1 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6 | | 20 | QPSK | 1/3 | 2 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-6 | | 20 | QPSK | 1/3 | 5 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-6 | | 20 | QPSK | 1/3 | 6 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-6 | | 20 | QPSK | 1/3 | 8 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-6 | | 20 | QPSK | 1/3 | 10 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6a | | 20 | QPSK | 1/3 | 16 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-6a | | 20 | QPSK | 1/3 | 18 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6a | | 20 | QPSK | 1/3 | 20 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6a | | 20 | QPSK | 1/3 | 24 | | ≥ 1 | Not yet used in tests |
| FDD | Table A.2.2.2.1-6a | | 20 | QPSK | 1/3 | 25 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6a | | 20 | QPSK | 1/3 | 48 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6b | | 20 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.2.2.2.1-6b | | 20 | QPSK | 1/3 | 54 | | ≥ 1 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *FDD* | *Table A.2.2.2.1-6b* | | *20* | *QPSK* | *1/5* | *75* | | *≥ 1* | |
| ***FDD, Partial RB allocation, 16-QAM, 1.4 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.2-1* | | *1.4* | *16QAM* | *3/4* | *1* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-1* | | *1.4* | *16QAM* | *3/4* | *5* | | *≥ 1* | |
| ***FDD, Partial RB allocation, 16-QAM, 3 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.2-2* | | *3* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-2* | | *3* | *16QAM* | *3/4* | *4* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-2* | | *3* | *16QAM* | *3/4* | *6* | | *≥ 1* | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ***FDD, Partial RB allocation, 16-QAM, 5 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.2-3* | | *5* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-3* | | *5* | *16QAM* | *3/4* | *8* | | *≥ 1* | |
| ***FDD, Partial RB allocation, 16-QAM, 10 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.2-4* | | *10* | *16QAM* | *3/4* | *1* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-4* | | *10* | *16QAM* | *3/4* | *12* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-4* | | *10* | *16QAM* | *1/2* | *16* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-4* | | *10* | *16QAM* | *3/4* | *30* | | *≥ 2* | |
| *FDD* | *Table A.2.2.2.2-4* | | *10* | *16QAM* | *3/4* | *36* | | *≥ 2* | |
| ***FDD, Partial RB allocation, 16-QAM, 15 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.2-5* | | *15* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-5* | | *15* | *16QAM* | *3/4* | *6* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-5* | | *15* | *16QAM* | *3/4* | *8* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-5* | | *15* | *16QAM* | *3/4* | *9* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-5* | | *15* | *16QAM* | *1/2* | *16* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-5* | | *15* | *16QAM* | *1/2* | *18* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-5a* | | *15* | *16QAM* | *1/3* | *20* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-5a* | | *15* | *16QAM* | *1/3* | *24* | | *≥ 1* | *Not yet used in tests* |
| ***FDD, Partial RB allocation, 16-QAM, 20 MHz*** | | | | | | | | | |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *3/4* | *2* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *1/2* | *16* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *1/2* | *18* | | *≥ 1* | |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *1/3* | *20* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *1/3* | *24* | | *≥ 1* | *Not yet used in tests* |
| *FDD* | *Table A.2.2.2.2-6* | | *20* | *16QAM* | *1/2* | *75* | | *≥ 2* | *Not yet used in tests* |
| ***FDD, Sustained data rate*** | | | | | | | | | |
| *FDD* | *Table A.2.2.3-1* | *R.1-1 FDD* | *10* | *QPSK* | *0.31* | *40* | | *≥ 1* | |
| *FDD* | *Table A.2.2.3-1* | *R.1-2 FDD* | *10* | *QPSK* | *0.31* | *40* | | *≥ 1* | |
| *FDD* | *Table A.2.2.3-1* | *R.1-3 FDD* | *20* | *QPSK* | *0.31* | *90* | | *≥ 2* | |
| *FDD* | *Table A.2.2.3-1* | *R.1-3A FDD* | *10* | *QPSK* | *0.31* | *40* | | *≥ 1* | |
| *FDD* | *Table A.2.2.3-1* | *R.1-4 FDD* | *20* | *QPSK* | *0.31* | *40* | | *≥ 2* | |
| ***TDD, Full RB allocation, QPSK*** | | | | | | | | | |
| *TDD* | *Table A.2.3.1.1-1* | | *1.4* | *QPSK* | *1/3* | *6* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.1-1* | | *3* | *QPSK* | *1/3* | *15* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.1-1* | | *5* | *QPSK* | *1/3* | *25* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.1-1* | | *10* | *QPSK* | *1/3* | *50* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.1-1* | | *15* | *QPSK* | *1/5* | *75* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.1-1* | | *20* | *QPSK* | *1/6* | *100* | | *≥ 1* | |
| ***TDD, Full RB allocation, 16-QAM*** | | | | | | | | | |
| *TDD* | *Table A.2.3.1.2-1* | | *1.4* | *16QAM* | *3/4* | *6* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.2-1* | | *3* | *16QAM* | *1/2* | *15* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.2-1* | | *5* | *16QAM* | *1/3* | *25* | | *≥ 1* | |
| *TDD* | *Table A.2.3.1.2-1* | | *10* | *16QAM* | *3/4* | *50* | | *≥ 2* | |
| *TDD* | *Table A.2.3.1.2-1* | | *15* | *16QAM* | *1/2* | *75* | | *≥ 2* | |
| *TDD* | *Table A.2.3.1.2-1* | | *20* | *16QAM* | *1/3* | *100* | | *≥ 2* | |
| ***TDD, Partial RB allocation, QPSK, 1.4 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.1-1* | | *1.4* | *QPSK* | *1/3* | *1* | | *≥ 1* | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *TDD* | *Table A.2.3.2.1-1* | | *1.4* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-1* | | *1.4* | *QPSK* | *1/3* | *3* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-1* | | *1.4* | *QPSK* | *1/3* | *4* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-1* | | *1.4* | *QPSK* | *1/3* | *5* | | *≥ 1* | |
| ***TDD, Partial RB allocation, QPSK, 3 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *3* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *4* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *6* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-2* | | *3* | *QPSK* | *1/3* | *10* | | *≥ 1* | *Not yet used in tests* |
| ***TDD, Partial RB allocation, QPSK, 5 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.1-3* | | *5* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-3* | | *5* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-3* | | *5* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-3* | | *5* | *QPSK* | *1/3* | *6* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-3* | | *5* | *QPSK* | *1/3* | *8* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-3a* | | *5* | *QPSK* | *1/3* | *10* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-3a* | | *5* | *QPSK* | *1/3* | *15* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-3a* | | *5* | *QPSK* | *1/3* | *18* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-3a* | | *5* | *QPSK* | *1/3* | *20* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-3a* | | *5* | *QPSK* | *1/3* | *24* | | *≥ 1* | *Not yet used in tests* |
| ***TDD, Partial RB allocation, QPSK, 10 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.1-4* | | *10* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-4* | | *10* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4* | | *10* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4* | | *10* | *QPSK* | *1/3* | *6* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-4* | | *10* | *QPSK* | *1/3* | *8* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4* | | *10* | *QPSK* | *1/3* | *10* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4a* | | *10* | *QPSK* | *1/3* | *12* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-4a* | | *10* | *QPSK* | *1/3* | *16* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4a* | | *10* | *QPSK* | *1/3* | *18* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4a* | | *10* | *QPSK* | *1/3* | *20* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4a* | | *10* | *QPSK* | *1/3* | *24* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-4a* | | *10* | *QPSK* | *1/3* | *25* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-4b* | | *10* | *QPSK* | *1/3* | *27* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4b* | | *10* | *QPSK* | *1/3* | *30* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4b* | | *10* | *QPSK* | *1/3* | *36* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-4b* | | *10* | *QPSK* | *1/3* | *40* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-4b* | | *10* | *QPSK* | *1/3* | *48* | | *≥ 1* | |
| ***TDD, Partial RB allocation, QPSK, 15 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.1-5* | | *15* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-5* | | *15* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5* | | *15* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5* | | *15* | *QPSK* | *1/3* | *6* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5* | | *15* | *QPSK* | *1/3* | *8* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-5* | | *15* | *QPSK* | *1/3* | *10* | | *≥ 1* | *Not yet used in tests* |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *TDD* | *Table A.2.3.2.1-5a* | | *15* | *QPSK* | *1/3* | *16* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-5a* | | *15* | *QPSK* | *1/3* | *18* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-5a* | | *15* | *QPSK* | *1/3* | *20* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5a* | | *15* | *QPSK* | *1/3* | *24* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5a* | | *15* | *QPSK* | *1/3* | *25* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5a* | | *15* | *QPSK* | *1/3* | *27* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5b* | | *15* | *QPSK* | *1/3* | *36* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-5b* | | *15* | *QPSK* | *1/3* | *40* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5b* | | *15* | *QPSK* | *1/3* | *48* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-5b* | | *15* | *QPSK* | *1/3* | *50* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-5b* | | *15* | *QPSK* | *1/3* | *54* | | *≥ 1* | |
| ***TDD, Partial RB allocation, QPSK, 20 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.1-6* | | *20* | *QPSK* | *1/3* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-6* | | *20* | *QPSK* | *1/3* | *2* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6* | | *20* | *QPSK* | *1/3* | *5* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6* | | *20* | *QPSK* | *1/3* | *6* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6* | | *20* | *QPSK* | *1/3* | *8* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6* | | *20* | *QPSK* | *1/5* | *10* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-6a* | | *20* | *QPSK* | *1/5* | *18* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-6a* | | *20* | *QPSK* | *1/3* | *20* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6a* | | *20* | *QPSK* | *1/3* | *24* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6a* | | *20* | *QPSK* | *1/3* | *25* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-6a* | | *20* | *QPSK* | *1/3* | *48* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.1-6a* | | *20* | *QPSK* | *1/3* | *50* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.1-6b* | | *20* | *QPSK* | *1/3* | *54* | | *≥ 1* | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *TDD* | *Table A.2.3.2.1-6b* | | *20* | *QPSK* | *1/5* | *75* | | *≥ 1* | |
| ***TDD, Partial RB allocation, 16-QAM, 1.4 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.2-1* | | *1.4* | *16QAM* | *3/4* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.2-1* | | *1.4* | *16QAM* | *3/4* | *5* | | *≥ 1* | |
| ***TDD, Partial RB allocation, 16-QAM, 3 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.2-2* | | *3* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.2-2* | | *3* | *16QAM* | *3/4* | *4* | | *≥ 1* | |
| ***TDD, Partial RB allocation, 16-QAM, 5 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.2-3* | | *5* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.2-3* | | *5* | *16QAM* | *3/4* | *8* | | *≥ 1* | |
| ***TDD, Partial RB allocation, 16-QAM, 10 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.2-4* | | *10* | *16QAM* | *3/4* | *1* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.2-4* | | *10* | *16QAM* | *3/4* | *12* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.2-4* | | *10* | *16QAM* | *1/2* | *16* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.2-4* | | *10* | *16QAM* | *1/3* | *24* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.2-4* | | *10* | *16QAM* | *3/4* | *30* | | *≥ 2* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.2-4* | | *10* | *16QAM* | *3/4* | *36* | | *≥ 2* | *Not yet used in tests* |
| ***TDD, Partial RB allocation, 16-QAM, 15 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.2-5* | | *15* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.2-5* | | *15* | *16QAM* | *1/2* | *16* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.2-5* | | *15* | *16QAM* | *3/4* | *36* | | *≥ 2* | |
| ***TDD, Partial RB allocation, 16-QAM, 20 MHz*** | | | | | | | | | |
| *TDD* | *Table A.2.3.2.2-6* | | *20* | *16QAM* | *3/4* | *1* | | *≥ 1* | *Not yet used in tests* |
| *TDD* | *Table A.2.3.2.2-6* | | *20* | *16QAM* | *1/2* | *18* | | *≥ 1* | |
| *TDD* | *Table A.2.3.2.2-6* | | *20* | *16QAM* | *3/4* | *50* | | *≥ 2* | |
| *TDD* | *Table A.2.3.2.2-6* | | *20* | *16QAM* | *1/2* | *75* | | *≥ 2* | *Not yet used in tests* |
| ***TDD, Sustained data rate*** | | | | | | | | | |
| *TDD* | *Table A.2.3.3-1* | *R.1-1 TDD* | *10* | *QPSK* | *0.43* | *40* | | *≥ 1* | |
| *TDD* | *Table A.2.3.3-1* | *R.1-2 TDD* | *10* | *QPSK* | *0.61* | *40* | | *≥ 2* | |
| *TDD* | *Table A.2.3.3-1* | *R.1-3 TDD* | *20* | *QPSK* | *0.49* | *90* | | *≥ 2* | |
| *TDD* | *Table A.2.3.3-1* | *R.1-3B TDD* | *15* | *QPSK* | *0.42* | *60* | | *≥ 2* | |
| *TDD* | *Table A.2.3.3-1* | *R.1-4 TDD* | *20* | *QPSK* | *0.49* | *90* | | *≥ 2* | |

# A.2.2 Reference measurement channels for FDD

## A.2.2.1 Full RB allocation

### A.2.2.1.1 QPSK

**Table A.2.2.1.1-1: Reference Channels for QPSK with full RB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **1.4** | **3** | **5** | **10** | **15** | **20** |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/5 | 1/6 |
| Payload size | Bits | 600 | 1544 | 2216 | 5160 | 4392 | 4584 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame (Note 1) | Bits | 1728 | 4320 | 7200 | 14400 | 21600 | 28800 |
| Total symbols per Sub-Frame | | 864 | 2160 | 3600 | 7200 | 10800 | 14400 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

### A.2.2.1.2 16-QAM

**Table A.2.2.1.2-1: Reference Channels for 16-QAM with full RB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **1.4** | **3** | **5** | **10** | **15** | **20** |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 1/2 | 1/3 | 3/4 | 1/2 | 1/3 |
| Payload size | Bits | 2600 | 4264 | 4968 | 21384 | 21384 | 19848 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 4 | 4 | 4 |
| Total number of bits per Sub-Frame | Bits | 3456 | 8640 | 14400 | 28800 | 43200 | 57600 |
| Total symbols per Sub-Frame | | 864 | 2160 | 3600 | 7200 | 10800 | 14400 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

## A.2.2.2 Partial RB allocation

For each channel bandwidth, various partial RB allocations are specified. The number of allocated RBs is chosen according to values specified in the Tx and Rx requirements. The single allocated RB case is included.

The allocated RBs are contiguous and start from one end of the channel bandwidth. A single allocated RB is at one end of the channel bandwidth.

### A.2.2.2.1 QPSK

**Table A.2.2.2.1-1: Reference Channels for 1.4MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **1.4** | **1.4** | **1.4** | **1.4** | **1.4** |
| Allocated resource blocks | | 1 | 2 | 3 | 4 | 5 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 72 | 176 | 256 | 392 | 424 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 288 | 576 | 864 | 1152 | 1440 |
| Total symbols per Sub-Frame | | 144 | 288 | 432 | 576 | 720 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | |

**Table A.2.2.2.1-2: Reference Channels for 3MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **3** | **3** | **3** | **3** | **3** | **3** | **3** |
| Allocated resource blocks | | 1 | 2 | 3 | 4 | 5 | 6 | 10 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 72 | 176 | 256 | 392 | 424 | 600 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 288 | 576 | 864 | 1152 | 1440 | 1728 | 2880 |
| Total symbols per Sub-Frame | | 144 | 288 | 432 | 576 | 720 | 864 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | | |

**Table A.2.2.2.1-3: Reference Channels for 5MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **5** | **5** | **5** | **5** | **5** |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 72 | 176 | 424 | 600 | 808 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 288 | 576 | 1440 | 1728 | 2304 |
| Total symbols per Sub-Frame | | 144 | 288 | 720 | 864 | 1152 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | |

**Table A.2.2.2.1-3a: Reference Channels for 5MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **5** | **5** | **5** | **5** | **5** |
| Allocated resource blocks | | 10 | 15 | 18 | 20 | 24 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 872 | 1320 | 1864 | 1736 | 2472 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 2880 | 4320 | 5184 | 5760 | 6912 |
| Total symbols per Sub-Frame | | 1440 | 2160 | 2592 | 2880 | 3456 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | |

**Table A.2.2.2.1-4: Reference Channels for 10MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **10** | **10** | **10** | **10** | **10** | **10** |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 | 10 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 72 | 176 | 424 | 600 | 808 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 288 | 576 | 1440 | 1728 | 2304 | 2880 |
| Total symbols per Sub-Frame | | 144 | 288 | 720 | 864 | 1152 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-4a: Reference Channels for 10MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **10** | **10** | **10** | **10** | **10** | **10** |
| Allocated resource blocks | | 12 | 15 | 16 | 18 | 20 | 24 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 1224 | 1320 | 1384 | 1864 | 1736 | 2472 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 3456 | 4320 | 4608 | 5184 | 5760 | 6912 |
| Total symbols per Sub-Frame | | 1728 | 2160 | 2304 | 2592 | 2880 | 3456 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-4b: Reference Channels for 10MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **10** | **10** | **10** | **10** | **10** | **10** |
| Allocated resource blocks | | 25 | 27 | 30 | 36 | 40 | 48 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 2216 | 2792 | 2664 | 3752 | 4136 | 4264 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 7200 | 7776 | 8640 | 10368 | 11520 | 13824 |
| Total symbols per Sub-Frame | | 3600 | 3888 | 4320 | 5184 | 5760 | 6912 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-5: Reference Channels for 15MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **15** | **15** | **15** | **15** | **15** | **15** |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 | 9 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 72 | 176 | 424 | 600 | 808 | 776 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 288 | 576 | 1440 | 1728 | 2304 | 2592 |
| Total symbols per Sub-Frame | | 144 | 288 | 720 | 864 | 1152 | 1296 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-5a: Reference Channels for 15MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **15** | **15** | **15** | **15** | **15** | **15** |
| Allocated resource blocks | | 10 | 16 | 18 | 20 | 24 | 25 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 872 | 1384 | 1864 | 1736 | 2472 | 2216 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 2880 | 4608 | 5184 | 5760 | 6912 | 7200 |
| Total symbols per Sub-Frame | | 1440 | 2304 | 2592 | 2880 | 3456 | 3600 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-5b: Reference Channels for 15MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **15** | **15** | **15** | **15** | **15** | **15** |
| Allocated resource blocks | | 27 | 36 | 40 | 48 | 50 | 54 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 2792 | 3752 | 4136 | 4264 | 5160 | 4776 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 7776 | 10368 | 11520 | 13824 | 14400 | 15552 |
| Total symbols per Sub-Frame | | 3888 | 5184 | 5760 | 6912 | 7200 | 7776 |
| UE Category | | ≥ 1 | | | | | |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-6: Reference Channels for 20MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **20** | **20** | **20** | **20** | **20** | **20** |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 | 10 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 72 | 176 | 424 | 600 | 808 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 288 | 576 | 1440 | 1728 | 2304 | 2880 |
| Total symbols per Sub-Frame | | 144 | 288 | 720 | 864 | 1152 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-6a: Reference Channels for 20MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **20** | **20** | **20** | **20** | **20** | **20** |
| Allocated resource blocks | | 16 | 18 | 20 | 24 | 25 | 48 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | Bits | 1384 | 1864 | 1736 | 2472 | 2216 | 4264 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 4608 | 5184 | 5760 | 6912 | 7200 | 13824 |
| Total symbols per Sub-Frame | | 2304 | 2592 | 2880 | 3456 | 3600 | 6912 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.1-6b: Reference Channels for 20MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value |
|---|---|---|---|---|
| Channel bandwidth | **MHz** | **20** | **20** | **20** |
| Allocated resource blocks | | 50 | 54 | 75 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/5 |
| Payload size | Bits | 5160 | 4776 | 4392 |
| Transport block CRC | Bits | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 14400 | 15552 | 21600 |
| Total symbols per Sub-Frame | | 7200 | 7776 | 10800 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | |

## A.2.2.2.2 16-QAM

**Table A.2.2.2.2-1: Reference Channels for 1.4MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | **MHz** | **1.4** | **1.4** |
| Allocated resource blocks | | 1 | 5 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 |
| Payload size | Bits | 408 | 2152 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 576 | 2880 |
| Total symbols per Sub-Frame | | 144 | 720 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | |

**Table A.2.2.2.2-2: Reference Channels for 3MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value |
|---|---|---|---|---|
| Channel bandwidth | **MHz** | **3** | **3** | **3** |
| Allocated resource blocks | | 1 | 4 | 6 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 | 3/4 |
| Payload size | Bits | 408 | 1736 | 2600 |
| Transport block CRC | Bits | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 576 | 2304 | 3456 |
| Total symbols per Sub-Frame | | 144 | 576 | 864 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | |

**Table A.2.2.2.2-3: Reference Channels for 5MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | **MHz** | **5** | **5** |
| Allocated resource blocks | | 1 | 8 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 |
| Payload size | Bits | 408 | 3496 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 576 | 4608 |
| Total symbols per Sub-Frame | | 144 | 1152 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | |

**Table A.2.2.2.2-4: Reference Channels for 10MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **10** | **10** | **10** | **10** | **10** |
| Allocated resource blocks | | 1 | 12 | 16 | 30 | 36 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 | 1/2 | 3/4 | 3/4 |
| Payload size | Bits | 408 | 5160 | 4584 | 12960 | 15264 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 3 | 3 |
| Total number of bits per Sub-Frame | Bits | 576 | 6912 | 9216 | 17280 | 20736 |
| Total symbols per Sub-Frame | | 144 | 1728 | 2304 | 4320 | 5184 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | |

**Table A.2.2.2.2-5: Reference Channels for 15MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **15** | **15** | **15** | **15** | **15** | **15** |
| Allocated resource blocks | | 1 | 6 | 8 | 9 | 16 | 18 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 | 3/4 | 3/4 | 1/2 | 1/2 |
| Payload size | Bits | 408 | 2600 | 3496 | 3864 | 4584 | 5160 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 576 | 3456 | 4608 | 5184 | 9216 | 10368 |
| Total symbols per Sub-Frame | | 144 | 864 | 1152 | 1296 | 2304 | 2592 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.2.2.2.2-5a: Reference Channels for 15MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | **MHz** | **15** | **15** |
| Allocated resource blocks | | 20 | 24 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | 16QAM | 16QAM |
| Target Coding rate | | 1/3 | 1/3 |
| Payload size | Bits | 4008 | 4776 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | 11520 | 13824 |
| Total symbols per Sub-Frame | | 2880 | 3456 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | |

**Table A.2.2.2.2-6: Reference Channels for 20MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|---|
| Channel bandwidth | **MHz** | **20** | **20** | **20** | **20** | **20** | **20** | **20** |
| Allocated resource blocks | | 1 | 2 | 16 | 18 | 20 | 24 | 75 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 | 1/2 | 1/2 | 1/3 | 1/3 | 1/2 |
| Payload size | Bits | 408 | 840 | 4584 | 5160 | 4008 | 4776 | 21384 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 | 4 |
| Total number of bits per Sub-Frame | Bits | 576 | 1152 | 9216 | 10368 | 11520 | 13824 | 43200 |
| Total symbols per Sub-Frame | | 144 | 288 | 2304 | 2592 | 2880 | 3456 | 10800 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | | |

### A.2.2.3 Reference measurement channels for sustained downlink data rate provided by lower layers

**Table A.2.2.3-1: Uplink Reference Channels for sustained data-rate test (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference Channel | | R.1-1 FDD | R.1-2 FDD | R.1-3 FDD | R.1-3A FDD | R.1-4 FDD | FFS |
| Channel Bandwidth | MHz | 10 | 10 | 20 | 10 | 20 | |
| Allocated Resource Blocks | | 40 (Note 2) | 40 (Note 2) | 90 (Note 3) | 40(Note 2) | 90 (Note 3) | |
| Allocated Sub-Frames per Radio-Frame | | 10 | 10 | 10 | 10 | 10 | |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | |
| Coding Rate | | 0.31 | 0.31 | 0.31 | 0.31 | 0.31 | |
| Information Bit Payload per Sub-Frame | Bits | 3496 | 3496 | 7992 | 3496 | 7992 | |
| Number of Code Blocks per Sub-Frame (Note 1) | | 1 | 1 | 2 | 1 | 2 | |
| Modulation Symbols per Sub-Frame | | 5760 | 5760 | 12960 | 5760 | 12960 | |
| Binary Channel Bits per Sub-Frame | | 11520 | 11520 | 25920 | 11520 | 25920 | |
| Max Throughput over 1 Radio-Frame | Mbps | 3.496 | 3.496 | 7.992 | 3.496 | 7.992 | |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 2 | |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: RB-s 5-44 allocated with PUSCH.<br>Note 3: RB-s 5-94 allocated with PUSCH. | | | | | | | |

## A.2.3 Reference measurement channels for TDD

For TDD the measurement channel is based on DL/UL configuration ratio of 2DL:2UL.

### A.2.3.1 Full RB allocation

#### A.2.3.1.1 QPSK

**Table A.2.3.1.1-1: Reference Channels for QPSK with full RB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/5 | 1/6 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 600 | 1544 | 2216 | 5160 | 4392 | 4584 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 1728 | 4320 | 7200 | 14400 | 21600 | 28800 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 864 | 2160 | 3600 | 7200 | 10800 | 14400 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

### A.2.3.1.2 16-QAM

**Table A.2.3.1.2-1: Reference Channels for 16-QAM with full RB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 1/2 | 1/3 | 3/4 | 1/2 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 2600 | 4264 | 4968 | 21384 | 21384 | 19848 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks - C | | 1 | 1 | 1 | 4 | 4 | 4 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 3456 | 8640 | 14400 | 28800 | 43200 | 57600 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 864 | 2160 | 3600 | 7200 | 10800 | 14400 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

## A.2.3.2 Partial RB allocation

For each channel bandwidth, various partial RB allocations are specified. The number of allocated RBs is chosen according to values specified in the Tx and Rx requirements. The single allocated RB case is included.

The allocated RBs are contiguous and start from one end of the channel bandwidth. A single allocated RB is at one end of the channel bandwidth.

### A.2.3.2.1 QPSK

**Table A.2.3.2.1-1: Reference Channels for 1.4MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 1.4 | 1.4 | 1.4 | 1.4 |
| Allocated resource blocks | | 1 | 2 | 3 | 4 | 5 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 72 | 176 | 256 | 392 | 424 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 288 | 576 | 864 | 1152 | 1440 |
| Total symbols per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 288 | 432 | 576 | 720 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | |

**Table A.2.3.2.1-2: Reference Channels for 3MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Allocated resource blocks | | 1 | 2 | 3 | 4 | 5 | 6 | 10 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 72 | 176 | 256 | 392 | 424 | 600 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 288 | 576 | 864 | 1152 | 1440 | 1728 | 2880 |
| Total symbols per Sub-Frame | | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 288 | 432 | 576 | 720 | 864 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | | |

**Table A.2.3.2.1-3: Reference Channels for 5MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 5 | 5 | 5 | 5 | 5 |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 72 | 176 | 424 | 600 | 808 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | | | | | |
| For Sub-Frame 2,3,7,8 | | 288 | 576 | 1440 | 1728 | 2304 |
| Total symbols per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 288 | 720 | 864 | 1152 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | |

**Table A.2.3.2.1-3a: Reference Channels for 5MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 5 | 5 | 5 | 5 | 5 |
| Allocated resource blocks | | 10 | 15 | 18 | 20 | 24 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 872 | 1320 | 1864 | 1736 | 2472 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | Bits | | | | | |
| For Sub-Frame 2,3,7,8 | | 2880 | 4320 | 5184 | 5760 | 6912 |
| Total symbols per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | | 1440 | 2160 | 2592 | 2880 | 3456 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | |

**Table A.2.3.2.1-4: Reference Channels for 10MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 | 10 |
| Uplink-Downlink Configuration | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 72 | 176 | 424 | 600 | 808 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 288 | 576 | 1440 | 1728 | 2304 | 2880 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 288 | 720 | 864 | 1152 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.1-4a: Reference Channels for 10MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Allocated resource blocks | | 12 | 16 | 18 | 20 | 24 | 25 |
| Uplink-Downlink Configuration | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 1224 | 1384 | 1864 | 1736 | 2472 | 2216 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 3456 | 4608 | 5184 | 5760 | 6912 | 7200 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 1728 | 2304 | 2592 | 2880 | 3456 | 3600 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.1-4b: Reference Channels for 10MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | 10 |
| Allocated resource blocks | | 27 | 30 | 36 | 40 | 48 |
| Uplink-Downlink Configuration | | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 2792 | 2664 | 3752 | 4136 | 4264 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 7776 | 8640 | 10368 | 11520 | 13824 |
| Total symbols per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | | 3888 | 4320 | 5184 | 5760 | 6912 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | |

**Table A.2.3.2.1-5: Reference Channels for 15MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 15 | 15 | 15 | 15 | 15 | 15 |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 | 10 |
| Uplink-Downlink Configuration | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 72 | 176 | 424 | 600 | 808 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 288 | 576 | 1440 | 1728 | 2304 | 2880 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 288 | 720 | 864 | 1152 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.1-5a: Reference Channels for 15MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 15 | 15 | 15 | 15 | 15 | 15 |
| Allocated resource blocks | | 16 | 18 | 20 | 24 | 25 | 27 |
| Uplink-Downlink Configuration | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 1384 | 1864 | 1736 | 2472 | 2216 | 2792 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 4608 | 5184 | 5760 | 6912 | 7200 | 7776 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 2304 | 2592 | 2880 | 3456 | 3600 | 3888 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.1-5b: Reference Channels for 15MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 15 | 15 | 15 | 15 | 15 |
| Allocated resource blocks | | 36 | 40 | 48 | 50 | 54 |
| Uplink-Downlink Configuration | | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 3752 | 4136 | 4264 | 5160 | 4776 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 10368 | 11520 | 13824 | 14400 | 15552 |
| Total symbols per Sub-Frame | | | | | | |
| For Sub-Frame 2,3,7,8 | | 5184 | 5760 | 6912 | 7200 | 7776 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | |

**Table A.2.3.2.1-6: Reference Channels for 20MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 20 | 20 | 20 | 20 | 20 | 20 |
| Allocated resource blocks | | 1 | 2 | 5 | 6 | 8 | 10 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 72 | 176 | 424 | 600 | 808 | 872 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 288 | 576 | 1440 | 1728 | 2304 | 2880 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 288 | 720 | 864 | 1152 | 1440 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.1-6a: Reference Channels for 20MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 20 | 20 | 20 | 20 | 20 | 20 |
| Allocated resource blocks | | 18 | 20 | 24 | 25 | 48 | 50 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 1864 | 1736 | 2472 | 2216 | 4264 | 5160 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 5184 | 5760 | 6912 | 7200 | 13824 | 14400 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 2592 | 2880 | 3456 | 3600 | 6912 | 7200 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.1-6b: Reference Channels for 20MHz QPSK with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | MHz | 20 | 20 |
| Allocated resource blocks | | 54 | 75 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | QPSK | QPSK |
| Target Coding rate | | 1/3 | 1/5 |
| Payload size | | | |
| For Sub-Frame 2,3,7,8 | Bits | 4776 | 4392 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | Bits | 15552 | 21600 |
| Total symbols per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | | 7776 | 10800 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | |

### A.2.3.2.2 16-QAM

**Table A.2.3.2.2-1: Reference Channels for 1.4MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 1.4 |
| Allocated resource blocks | | 1 | 5 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 |
| Payload size | | | |
| For Sub-Frame 2,3,7,8 | Bits | 408 | 2152 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | Bits | 576 | 2880 |
| Total symbols per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 720 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | |

**Table A.2.3.2.2-2: Reference Channels for 3MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | MHz | 3 | 3 |
| Allocated resource blocks | | 1 | 4 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 |
| Payload size | | | |
| For Sub-Frame 2,3,7,8 | Bits | 408 | 1736 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | Bits | 576 | 2304 |
| Total symbols per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 576 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | |

**Table A.2.3.2.2-3: Reference Channels for 5MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value |
|---|---|---|---|
| Channel bandwidth | MHz | 5 | 5 |
| Allocated resource blocks | | 1 | 8 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 |
| Modulation | | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 |
| Payload size | | | |
| For Sub-Frame 2,3,7,8 | Bits | 408 | 3496 |
| Transport block CRC | Bits | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 |
| Total number of bits per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | Bits | 576 | 4608 |
| Total symbols per Sub-Frame | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 1152 |
| UE Category | | ≥ 1 | ≥ 1 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | |

**Table A.2.3.2.2-4: Reference Channels for 10MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value | Value | Value |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Allocated resource blocks | | 1 | 12 | 16 | 24 | 30 | 36 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 3/4 | 1/2 | 1/3 | 3/4 | 3/4 |
| Payload size | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 408 | 5160 | 4584 | 4776 | 12960 | 15264 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 1 | 1 | 3 | 3 |
| Total number of bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 576 | 6912 | 9216 | 13824 | 17280 | 20736 |
| Total symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 1728 | 2304 | 3456 | 4320 | 5184 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | | | |

**Table A.2.3.2.2-5: Reference Channels for 15MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value |
|---|---|---|---|---|
| Channel bandwidth | MHz | 15 | 15 | 15 |
| Allocated resource blocks | | 1 | 16 | 36 |
| Uplink-Downlink Configuration(Note 2) | | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 1/2 | 3/4 |
| Payload size | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 408 | 4584 | 15264 |
| Transport block CRC | Bits | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 3 |
| Total number of bits per Sub-Frame | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 576 | 9216 | 20736 |
| Total symbols per Sub-Frame | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 2304 | 5184 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | |

**Table A.2.3.2.2-6: Reference Channels for 20MHz 16-QAM with partial RB allocation**

| Parameter | Unit | Value | Value | Value | Value |
|---|---|---|---|---|---|
| Channel bandwidth | MHz | 20 | 20 | 20 | 20 |
| Allocated resource blocks | | 1 | 18 | 50 | 75 |
| Uplink-Downlink Configuration (Note 2) | | 1 | 1 | 1 | 1 |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding rate | | 3/4 | 1/2 | 3/4 | 1/2 |
| Payload size | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 408 | 5160 | 21384 | 21384 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 |
| Number of code blocks per Sub-Frame (Note 1) | | 1 | 1 | 4 | 4 |
| Total number of bits per Sub-Frame | | | | | |
| For Sub-Frame 2,3,7,8 | Bits | 576 | 10368 | 28800 | 43200 |
| Total symbols per Sub-Frame | | | | | |
| For Sub-Frame 2,3,7,8 | | 144 | 2592 | 7200 | 10800 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | | |

### A.2.3.3 Reference measurement channels for sustained downlink data rate provided by lower layers

**Table A.2.3.3-1: Uplink Reference Channels for sustained data-rate test (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference Channel | | R.1-1 TDD | R.1-2 TDD | R.1-3 TDD | R.1-3B TDD | R.1-4 TDD | FFS |
| Channel Bandwidth | MHz | 10 | 10 | 20 | 15 | 20 | |
| Uplink-Downlink Configuration (Note 2) | | 5 | 5 | 5 | 5 | 5 | |
| Allocated Resource Blocks | | 40 (Note 3) | 40 (Note 3) | 90 (Note 5) | 60 (Note 4) | 90 (Note 5) | |
| Allocated Sub-Frames per Radio-Frame | | 1 | 1 | 1 | 1 | 1 | |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | 12 | 12 | |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | |
| Coding Rate | | | | | | | |
| For Sub-Frame 2 | | 0.43 | 0.61 | 0.49 | 0.42 | 0.49 | |
| Information Bit Payload per Sub-Frame | Bits | | | | | | |
| For Sub-Frame 2 | | 4968 | 6968 | 12576 | 7224 | 12576 | |
| Number of Code Blocks per Sub-Frame (Note 1) | | | | | | | |
| For Sub-Frame 2 | | 1 | 2 | 3 | 2 | 3 | |
| Modulation Symbols per Sub-Frame | | | | | | | |
| For Sub-Frame 2 | | 5760 | 5760 | 12960 | 8640 | 10240 | |
| Binary Channel Bits per Sub-Frame | | | | | | | |
| For Sub-Frame 2 | | 11520 | 11520 | 25920 | 17280 | 25920 | |
| Max Throughput over 1 Radio-Frame | Mbps | 0.4968 | 0.6968 | 1.2576 | 0.7224 | 1.2576 | |
| UE Category | | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 | ≥ 2 | |
| Note 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)<br>Note 2: As per Table 4.2-2 in TS 36.211 [8]<br>Note 3: RB-s 5-44 allocated with PUSCH.<br>Note 4: RB-s 7-66 allocated with PUSCH.<br>Note 5: RB-s 5-94 allocated with PUSCH. | | | | | | | |

# A.3 DL reference measurement channels

## A.3.1 General

The number of available channel bits varies across the sub-frames due to PBCH and PSS/SSS overhead. The payload size per sub-frame is varied in order to keep the code rate constant throughout a frame.

No user data is scheduled on subframes #5 in order to facilitate the transmission of system information blocks (SIB).

The algorithm for determining the payload size $A$ is as follows; given a desired coding rate $R$ and radio block allocation $N_{\mathrm{RB}}$

1. Calculate the number of channel bits $N_{\mathrm{ch}}$ that can be transmitted during the first transmission of a given sub-frame.

2. Find $A$ such that the resulting coding rate is as close to $R$ as possible, that is,

$$\min\left|R-(A+24)/N_{ch}\right|,$$

subject to

a) A is a valid TB size (according to TS 36.213 [10] clause 7.1.7) assuming an allocation of $N_{\mathrm{RB}}$ resource blocks

b) Segmentation is not included in this formula, but should be considered in the TBS calculation

3. If there is more than one *A* that minimizes the equation above, then the larger value is chosen per default.

4. For TDD, the measurement channel is based on DL/UL configuration ratio of 2DL+DwPTS (12 OFDM symbol): 2UL.

### A.3.1.1 Overview of DL reference measurement channels

In Table A.3.1.1-1 are listed the DL reference measurement channels specified in annexes A.3.2 to A.3.9 of this release of TS 36.521-1. This table is informative and serves only to a better overview. The reference for the concrete reference measurement channels and corresponding implementation's parameters as to be used for testing are annexes A.3.2 to A.3.9 as appropriate.

**Table A.3.1.1-1: Overview of DL reference measurement channels**

| Duplex | Table | Name | BW | Mod | TCR | RB | RB Offset | UE Categ | Notes |
|---|---|---|---|---|---|---|---|---|---|
| **FDD, Receiver requirements** | | | | | | | | | |
| FDD | Table A.3.2-1 | | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| FDD | Table A.3.2-1 | | 3 | QPSK | 1/3 | 15 | | ≥ 1 | |
| FDD | Table A.3.2-1 | | 5 | QPSK | 1/3 | 25 | | ≥ 1 | |
| FDD | Table A.3.2-1 | | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.3.2-1 | | 15 | QPSK | 1/3 | 75 | | ≥ 1 | |
| FDD | Table A.3.2-1 | | 20 | QPSK | 1/3 | 100 | | ≥ 1 | |
| **TDD, Receiver requirements** | | | | | | | | | |
| TDD | Table A.3.2-2 | | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| TDD | Table A.3.2-2 | | 3 | QPSK | 1/3 | 15 | | ≥ 1 | |
| TDD | Table A.3.2-2 | | 5 | QPSK | 1/3 | 25 | | ≥ 1 | |
| TDD | Table A.3.2-2 | | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.2-2 | | 15 | QPSK | 1/3 | 75 | | ≥ 1 | |
| TDD | Table A.3.2-2 | | 20 | QPSK | 1/3 | 100 | | ≥ 1 | |
| **FDD, Receiver requirements, Maximum input level for UE Categories 3-5** | | | | | | | | | |
| FDD | Table A.3.2-3 | | 1.4 | 64QAM | 3/4 | 6 | | - | |
| FDD | Table A.3.2-3 | | 3 | 64QAM | 3/4 | 15 | | - | |
| FDD | Table A.3.2-3 | | 5 | 64QAM | 3/4 | 25 | | - | |
| FDD | Table A.3.2-3 | | 10 | 64QAM | 3/4 | 50 | | - | |
| FDD | Table A.3.2-3 | | 15 | 64QAM | 3/4 | 75 | | - | |
| FDD | Table A.3.2-3 | | 20 | 64QAM | 3/4 | 100 | | - | |
| **FDD, Receiver requirements, Maximum input level for UE Categories 1** | | | | | | | | | |
| FDD | Table A.3.2-3a | | 1.4 | 64QAM | 3/4 | 6 | | - | |
| FDD | Table A.3.2-3a | | 3 | 64QAM | 3/4 | 15 | | - | |
| FDD | Table A.3.2-3a | | 5 | 64QAM | 3/4 | 18 | | - | |
| FDD | Table A.3.2-3a | | 10 | 64QAM | 3/4 | 17 | | - | |
| FDD | Table A.3.2-3a | | 15 | 64QAM | 3/4 | 17 | | - | |
| FDD | Table A.3.2-3a | | 20 | 64QAM | 3/4 | 17 | | - | |
| **FDD, Receiver requirements, Maximum input level for UE Categories 2** | | | | | | | | | |
| FDD | Table A.3.2-3b | | 1.4 | 64QAM | 3/4 | 6 | | - | |
| FDD | Table A.3.2-3b | | 3 | 64QAM | 3/4 | 15 | | - | |
| FDD | Table A.3.2-3b | | 5 | 64QAM | 3/4 | 25 | | - | |
| FDD | Table A.3.2-3b | | 10 | 64QAM | 3/4 | 50 | | - | |
| FDD | Table A.3.2-3b | | 15 | 64QAM | 3/4 | 75 | | - | |
| FDD | Table A.3.2-3b | | 20 | 64QAM | 3/4 | 83 | | - | |
| **TDD, Receiver requirements, Maximum input level for UE Categories 3-5** | | | | | | | | | |
| TDD | Table A.3.2-4 | | 1.4 | 64QAM | 3/4 | 6 | | - | |
| TDD | Table A.3.2-4 | | 3 | 64QAM | 3/4 | 15 | | - | |
| TDD | Table A.3.2-4 | | 5 | 64QAM | 3/4 | 25 | | - | |
| TDD | Table A.3.2-4 | | 10 | 64QAM | 3/4 | 50 | | - | |
| TDD | Table A.3.2-4 | | 15 | 64QAM | 3/4 | 75 | | - | |
| TDD | Table A.3.2-4 | | 20 | 64QAM | 3/4 | 100 | | - | |
| **TDD, Receiver requirements, Maximum input level for UE Categories 1** | | | | | | | | | |
| TDD | Table A.3.2-4a | | 1.4 | 64QAM | 3/4 | 6 | | - | |
| TDD | Table A.3.2-4a | | 3 | 64QAM | 3/4 | 15 | | - | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| TDD | Table A.3.2-4a | | 5 | 64QAM | 3/4 | 18 | | - | |
| TDD | Table A.3.2-4a | | 10 | 64QAM | 3/4 | 17 | | - | |
| TDD | Table A.3.2-4a | | 15 | 64QAM | 3/4 | 17 | | - | |
| TDD | Table A.3.2-4a | | 20 | 64QAM | 3/4 | 17 | | - | |
| **TDD, Receiver requirements, Maximum input level for UE Categories 2** | | | | | | | | | |
| TDD | Table A.3.2-4b | | 1.4 | 64QAM | 3/4 | 6 | | - | |
| TDD | Table A.3.2-4b | | 3 | 64QAM | 3/4 | 15 | | - | |
| TDD | Table A.3.2-4b | | 5 | 64QAM | 3/4 | 25 | | - | |
| TDD | Table A.3.2-4b | | 10 | 64QAM | 3/4 | 50 | | - | |
| TDD | Table A.3.2-4b | | 15 | 64QAM | 3/4 | 75 | | - | |
| TDD | Table A.3.2-4b | | 20 | 64QAM | 3/4 | 83 | | - | |
| **FDD, Transmitter requirements** | | | | | | | | | |
| FDD | Table A.3.2A-1 | | 1.4 | QPSK | 1/8-1/3 | 3 | | ≥ 1 | |
| FDD | Table A.3.2A-1 | | 3 | QPSK | 1/3 | 4 | | ≥ 1 | |
| FDD | Table A.3.2A-1 | | 5 | QPSK | 1/3 | 8 | | ≥ 1 | |
| FDD | Table A.3.2A-1 | | 10 | QPSK | 1/3 | 16 | | ≥ 1 | |
| FDD | Table A.3.2A-1 | | 15 | QPSK | 1/3 | 25 | | ≥ 1 | |
| FDD | Table A.3.2A-1 | | 20 | QPSK | 1/3 | 30 | | ≥ 1 | |
| **TDD, Transmitter requirements** | | | | | | | | | |
| TDD | Table A.3.2A-2 | | 1.4 | QPSK | 1/8-1/3 | 3 | | ≥ 1 | |
| TDD | Table A.3.2A-2 | | 3 | QPSK | 1/3 | 4 | | ≥ 1 | |
| TDD | Table A.3.2A-2 | | 5 | QPSK | 1/3 | 8 | | ≥ 1 | |
| TDD | Table A.3.2A-2 | | 10 | QPSK | 1/3 | 16 | | ≥ 1 | |
| TDD | Table A.3.2A-2 | | 15 | QPSK | 1/3 | 25 | | ≥ 1 | |
| TDD | Table A.3.2A-2 | | 20 | QPSK | 1/3 | 30 | | ≥ 1 | |
| **FDD, PDSCH Performance, Single-antenna transmission (CRS)** | | | | | | | | | |
| FDD | Table A.3.3.1-1 | R.4 FDD | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| FDD | Table A.3.3.1-1 | R.42 FDD | 20 | QPSK | 1/3 | 100 | | ≥ 1 | |
| FDD | Table A.3.3.1-1 | R.2 FDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.3.3.1-2 | R.3-1 FDD | 5 | 16QAM | 1/2 | 25 | | ≥ 1 | |
| FDD | Table A.3.3.1-2 | R.3 FDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| FDD | Table A.3.3.1-3 | R.5 FDD | 3 | 64QAM | 3/4 | 15 | | ≥ 1 | |
| FDD | Table A.3.3.1-3 | R.6 FDD | 5 | 64QAM | 3/4 | 25 | | ≥ 2 | |
| FDD | Table A.3.3.1-3 | R.7 FDD | 10 | 64QAM | 3/4 | 50 | | ≥ 2 | |
| FDD | Table A.3.3.1-3 | R.8 FDD | 15 | 64QAM | 3/4 | 75 | | ≥ 2 | |
| FDD | Table A.3.3.1-3 | R.9 FDD | 20 | 64QAM | 3/4 | 100 | | ≥ 3 | |
| FDD | Table A.3.3.1-3a | R.6-1 FDD | 5 | 64QAM | 3/4 | 18 | | ≥ 1 | |
| FDD | Table A.3.3.1-3a | R.7-1 FDD | 10 | 64QAM | 3/4 | 17 | | ≥ 1 | |
| FDD | Table A.3.3.1-3a | R.8-1 FDD | 15 | 64QAM | 3/4 | 17 | | ≥ 1 | |
| FDD | Table A.3.3.1-3a | R.9-1 FDD | 20 | 64QAM | 3/4 | 17 | | ≥ 1 | |
| FDD | Table A.3.3.1-3a | R.9-2 FDD | 20 | 64QAM | 3/4 | 83 | | ≥ 2 | |
| **FDD, PDSCH Performance, Single-antenna transmission (CRS), Single PRB (Channel edge)** | | | | | | | | | |
| FDD | Table A.3.3.1-4 | R.0 FDD | 3 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| FDD | Table A.3.3.1-4 | R.1 FDD | 10 / 20 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| **FDD, PDSCH Performance, Single-antenna transmission (CRS), Single PRB (MBSFN Configuration)** | | | | | | | | | |
| FDD | Table A.3.3.1-5 | R.29 FDD | 10 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| **FDD, PDSCH Performance, Multi-antenna transmission (CRS), Two antenna ports** | | | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FDD | Table A.3.3.2.1-1 | R.10 FDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.3.3.2.1-1 | R.11 FDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| FDD | Table A.3.3.2.1-1 | R.11-2 FDD | 5 | 16QAM | 1/2 | 25 | | ≥ 1 | |
| FDD | Table A.3.3.2.1-1 | R.11-3 FDD | 10 | 16QAM | 1/2 | 40 | | ≥ 1 | |
| FDD | Table A.3.3.2.1-1 | R.30 FDD | 20 | 16QAM | 1/2 | 100 | | ≥ 2 | |
| FDD | Table A.3.3.2.1-1 | R.35 FDD | 10 | 64QAM | 1/2 | 50 | | ≥ 2 | |
| **FDD, PDSCH Performance, Multi-antenna transmission (CRS), Four antenna ports** | | | | | | | | | |
| FDD | Table A.3.3.2.2-1 | R.12 FDD | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| FDD | Table A.3.3.2.2-1 | R.13 FDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.3.3.2.2-1 | R.14 FDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| FDD | Table A.3.3.2.2-1 | R.14-1 FDD | 10 | 16QAM | 1/2 | 6 | | ≥ 1 | |
| FDD | Table A.3.3.2.2-1 | R.14-2 FDD | 10 | 16QAM | 1/2 | 3 | | ≥ 1 | |
| FDD | Table A.3.3.2.2-1 | R.36 FDD | 10 | 64QAM | 1/2 | 50 | | ≥ 2 | |
| **TDD, PDSCH Performance, Single-antenna transmission (CRS)** | | | | | | | | | |
| TDD | Table A.3.4.1-1 | R.4 TDD | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| TDD | Table A.3.4.1-1 | R.2 TDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.4.1-2 | R.3-1 TDD | 5 | 16QAM | 1/2 | 25 | | ≥ 1 | |
| TDD | Table A.3.4.1-2 | R.3 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.1-3 | R.5 TDD | 3 | 64QAM | 3/4 | 15 | | ≥ 1 | |
| TDD | Table A.3.4.1-3 | R.6 TDD | 5 | 64QAM | 3/4 | 25 | | ≥ 2 | |
| TDD | Table A.3.4.1-3 | R.7 TDD | 10 | 64QAM | 3/4 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.1-3 | R.8 TDD | 15 | 64QAM | 3/4 | 75 | | ≥ 2 | |
| TDD | Table A.3.4.1-3 | R.9 TDD | 20 | 64QAM | 3/4 | 100 | | ≥ 3 | |
| TDD | Table A.3.4.1-3a | R.6-1 TDD | 5 | 64QAM | 3/4 | 18 | | ≥ 1 | |
| TDD | Table A.3.4.1-3a | R.7-1 TDD | 10 | 64QAM | 3/4 | 17 | | ≥ 1 | |
| TDD | Table A.3.4.1-3a | R.8-1 TDD | 15 | 64QAM | 3/4 | 17 | | ≥ 1 | |
| TDD | Table A.3.4.1-3a | R.9-1 TDD | 20 | 64QAM | 3/4 | 17 | | ≥ 1 | |
| TDD | Table A.3.4.1-3a | R.9-2 TDD | 20 | 64QAM | 3/4 | 83 | | ≥ 2 | |
| **TDD, PDSCH Performance, Single-antenna transmission (CRS), Single PRB (Channel edge)** | | | | | | | | | |
| TDD | Table A.3.4.1-4 | R.0 TDD | 3 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| TDD | Table A.3.4.1-4 | R.1 TDD | 10 / 20 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| **TDD, PDSCH Performance, Single-antenna transmission (CRS), Single PRB (MBSFN Configuration)** | | | | | | | | | |
| TDD | Table A.3.4.1-5 | R.29 TDD | 10 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| **TDD, PDSCH Performance, Multi-antenna transmission (CRS), Two antenna ports** | | | | | | | | | |
| TDD | Table A.3.4.2.1-1 | R.10 TDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.4.2.1-1 | R.11 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.2.1-1 | R.11-1 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.2.1-1 | R.11-2 TDD | 5 | 16QAM | 1/2 | 25 | | ≥ 1 | |
| TDD | Table A.3.4.2.1-1 | R.11-3 TDD | 10 | 16QAM | 1/2 | 40 | | ≥ 1 | |
| TDD | Table A.3.4.2.1-1 | R.30 TDD | 20 | 16QAM | 1/2 | 100 | | ≥ 2 | |
| TDD | Table A.3.4.2.1-1 | R.35 TDD | 10 | 64QAM | 1/2 | 50 | | ≥ 2 | |
| **TDD, PDSCH Performance, Multi-antenna transmission (CRS), Four antenna ports** | | | | | | | | | |
| TDD | Table A.3.4.2.2-1 | R.12 TDD | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| TDD | Table A.3.4.2.2-1 | R.13 TDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.4.2.2-1 | R.14 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.2.2-1 | R.14-1 TDD | 10 | 16QAM | 1/2 | 6 | | ≥ 1 | |
| TDD | Table A.3.4.2.2-1 | R.14-2 TDD | 10 | 16QAM | 1/2 | 3 | | ≥ 1 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| TDD | Table A.3.4.2.2-1 | R.36 TDD | 10 | 64QAM | 1/2 | 50 | | ≥ 2 | |
| **TDD, PDSCH Performance, Single antenna port (DRS)** | | | | | | | | | |
| TDD | Table A.3.4.3.1-1 | R.25 TDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.4.3.1-1 | R.26 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.3.1-1 | R.26-1 TDD | 5 | 16QAM | 1/2 | 25 | | ≥ 1 | |
| TDD | Table A.3.4.3.1-1 | R.27 TDD | 10 | 64QAM | 3/4 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.3.1-1 | R.27-1 TDD | 10 | 64QAM | 3/4 | 18 | | ≥ 1 | |
| TDD | Table A.3.4.3.1-1 | R.28 TDD | 10 | 16QAM | 1/2 | 1 | | ≥ 1 | |
| **TDD, PDSCH Performance, Two antenna ports (DRS)** | | | | | | | | | |
| TDD | Table A.3.4.3.2-1 | R.31 TDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.4.3.2-1 | R.32 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.3.2-1 | R.32-1 TDD | 5 | 16QAM | 1/2 | [25] | | ≥ 1 | |
| TDD | Table A.3.4.3.2-1 | R.33 TDD | 10 | 64QAM | 3/4 | 50 | | ≥ 2 | |
| TDD | Table A.3.4.3.2-1 | R.33-1 TDD | 10 | 64QAM | 3/4 | [18] | | ≥ 1 | |
| TDD | Table A.3.4.3.2-1 | R.34 TDD | 10 | 64QAM | 1/2 | 50 | | ≥ 2 | |
| **FDD, PDCCH / PCFICH Performance** | | | | | | | | | |
| FDD | Table A.3.5.1-1 | R.15 FDD | 10 | PDCCH | | | | | |
| FDD | Table A.3.5.1-1 | R.16 FDD | 1.4 | PDCCH | | | | | |
| FDD | Table A.3.5.1-1 | R.17 FDD | 10 | PDCCH | | | | | |
| FDD | Table A.3.5.1-1A | R.16_1 FDD | 10 | PDCCH | | | | | |
| FDD | Table A.3.5.1-1A | R.17_1 FDD | 5 | PDCCH | | | | | |
| FDD | Table A.3.5.1-2 | | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.3.5.1-2 | | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| FDD | Table A.3.5.1-2 | | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| **TDD, PDCCH / PCFICH Performance** | | | | | | | | | |
| TDD | Table A.3.5.2-1 | R.15 TDD | 10 | PDCCH | | | | | |
| TDD | Table A.3.5.2-1 | R.16 TDD | 1.4 | PDCCH | | | | | |
| TDD | Table A.3.5.2-1 | R.17 TDD | 10 | PDCCH | | | | | |
| TDD | Table A.3.5.2-1A | R.16_1 TDD | 10 | PDCCH | | | | | |
| TDD | Table A.3.5.2-1A | R.17_1 TDD | 5 | PDCCH | | | | | |
| TDD | Table A.3.5.2-2 | | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.5.2-2 | | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| TDD | Table A.3.5.2-2 | | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| **FDD / TDD, PHICH Performance** | | | | | | | | | |
| FDD / TDD | Table A.3.6-1 | R.18 | 10 | PHICH | | | | | |
| FDD / TDD | Table A.3.6-1 | R.19 | 1.4 | PHICH | | | | | |
| FDD / TDD | Table A.3.6-1 | R.20 | 10 | PHICH | | | | | |
| FDD / TDD | Table A.3.6-1 | R.24 | 10 | PHICH | | | | | |
| FDD / TDD | Table A.3.6-1A | R.19_1 | 10 | PHICH | | | | | |
| FDD / TDD | Table A.3.6-1A | R.20_1 | 5 | PHICH | | | | | |
| **FDD, PMCH Performance** | | | | | | | | | |
| FDD | Table A.3.8.1-1 | R.40 FDD | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| FDD | Table A.3.8.1-1 | R.37 FDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| FDD | Table A.3.8.1-2 | R.38 FDD | 10 | 16QAM | 1/2 | 50 | | ≥ 1 | |
| FDD | Table A.3.8.1-3 | R.39-1 FDD | 5 | 64QAM | 2/3 | 25 | | ≥ 1 | |
| FDD | Table A.3.8.1-3 | R.39 FDD | 10 | 64QAM | 2/3 | 50 | | ≥ 2 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **TDD, PMCH Performance** | | | | | | | | | |
| TDD | Table A.3.8.2-1 | R.40 TDD | 1.4 | QPSK | 1/3 | 6 | | ≥ 1 | |
| TDD | Table A.3.8.2-1 | R.37 TDD | 10 | QPSK | 1/3 | 50 | | ≥ 1 | |
| TDD | Table A.3.8.2-2 | R.38 TDD | 10 | 16QAM | 1/2 | 50 | | ≥ 1 | |
| TDD | Table A.3.8.2-3 | R.39-1 TDD | 5 | 64QAM | 2/3 | 25 | | ≥ 1 | |
| TDD | Table A.3.8.2-3 | R.39 TDD | 10 | 64QAM | 2/3 | 50 | | ≥ 2 | |
| **FDD, Sustained data rate** | | | | | | | | | |
| FDD | Table A.3.9.1-1 | R.31-1 FDD | 10 | 64QAM | 0.40 | | | ≥ 1 | |
| FDD | Table A.3.9.1-1 | R.31-2 FDD | 10 | 64QAM | 0.59-0.64 | | | ≥ 2 | |
| FDD | Table A.3.9.1-1 | R.31-3 FDD | 20 | 64QAM | 0.59-0.62 | | | ≥ 2 | |
| FDD | Table A.3.9.1-1 | R.31-3A FDD | 10 | 64QAM | 0.85-0.90 | | | ≥ 2 | |
| FDD | Table A.3.9.1-1 | R.31-4 FDD | 20 | 64QAM | 0.87-0.90 | | | ≥ 3 | |
| **TDD, Sustained data rate** | | | | | | | | | |
| TDD | Table A.3.9.2-1 | R.31-1 TDD | 10 | 64QAM | 0.40 | | | ≥ 1 | |
| TDD | Table A.3.9.2-1 | R.31-2 TDD | 10 | 64QAM | 0.59-0.64 | | | ≥ 2 | |
| TDD | Table A.3.9.2-1 | R.31-3 TDD | 20 | 64QAM | 0.59-0.62 | | | ≥ 2 | |
| TDD | Table A.3.9.2-1 | R.31-3B TDD | 15 | 64QAM | 0.87-0.90 | | | ≥ 2 | |
| TDD | Table A.3.9.2-1 | R.31-4 TDD | 20 | 64QAM | 0.87-0.90 | | | ≥ 3 | |

## A.3.2 Reference measurement channel for receiver characteristics

Tables A.3.2-1 and A.3.2-2 are applicable for measurements on the Receiver Characteristics (clause 7) with the exception of sub-clause 7.4 (Maximum input level).

Tables A.3.2-3, A.3.2-3a, A.3.2-3b, A3.2-4, A3.2-4a and A.3.2-4b are applicable for sub-clause 7.4 (Maximum input level).

Tables A.3.2-1 and A.3.2-2 also apply for the modulated interferer used in Clauses 7.5, 7.6 and 7.8 with test specific bandwidths.

**Table A.3.2-1: Fixed Reference Channel for Receiver Requirements (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding Rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 408 | 1320 | 2216 | 4392 | 6712 | 8760 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 152 | 872 | 1800 | 4392 | 6712 | 8760 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 1 | 1 | 1 | 1 | 2 | 2 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 1 | 1 | 1 | 1 | 2 | 2 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 1368 | 3780 | 6300 | 13800 | 20700 | 27600 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 528 | 2940 | 5460 | 12960 | 19860 | 26760 |
| Max. Throughput averaged over 1 frame | kbps | 341.6 | 1143.2 | 1952.8 | 3952.8 | 6040.8 | 7884 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz

Note 2: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]

Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2-2: Fixed Reference Channel for Receiver Requirements (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel Bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Uplink-Downlink Configuration (Note 6) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | 4 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Number of HARQ Processes | Processes | 7 | 7 | 7 | 7 | 7 | 7 |
| Maximum number of HARQ transmission | | 1 | 1 | 1 | 1 | 1 | 1 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target coding rate | | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Information Bit Payload per Sub-Frame | Bits | | | | | | |
| For Sub-Frame 4, 9 | | 408 | 1320 | 2216 | 4392 | 6712 | 8760 |
| For Sub-Frame 1, 6 | | n/a | 968 | 1544 | 3240 | 4968 | 6712 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 208 | 1064 | 1800 | 4392 | 6712 | 8760 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frame 4, 9 | | 1 | 1 | 1 | 1 | 2 | 2 |
| For Sub-Frame 1, 6 | | n/a | 1 | 1 | 1 | 1 | 2 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 1 | 1 | 1 | 2 | 2 |
| Binary Channel Bits Per Sub-Frame | Bits | | | | | | |
| For Sub-Frame 4, 9 | | 1368 | 3780 | 6300 | 13800 | 20700 | 27600 |
| For Sub-Frame 1, 6 | | n/a | 3276 | 5556 | 11256 | 16956 | 22656 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 672 | 3084 | 5604 | 13104 | 20004 | 26904 |
| Max. Throughput averaged over 1 frame | kbps | 102.4 | 564 | 932 | 1965.6 | 3007.2 | 3970.4 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |

Note 1: For normal subframes(0,4,5,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
Note 2: For 1.4MHz, no data shall be scheduled on special subframes (1&6) to avoid problems with insufficient PDCCH performance
Note 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: As per Table 4.2-2 in TS 36.211 [8]

**Table A.3.2-3: Fixed Reference Channel for Maximum input level for UE Categories 3-5 (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 2984 | 8504 | 14112 | 30576 | 46888 | 61664 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 6456 | 12576 | 28336 | 45352 | 61664 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | 1 | 2 | 3 | 5 | 8 | 11 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | n/a | 2 | 3 | 5 | 8 | 11 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 4104 | 11340 | 18900 | 41400 | 62100 | 82800 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 8820 | 16380 | 38880 | 59580 | 80280 |
| Max. Throughput averaged over 1 frame | kbps | 2387.2 | 7448.8 | 12547 | 27294 | 42046 | 55498 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2-3a: Fixed Reference Channel for Maximum input level for UE Category 1 (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 18 | 17 | 17 | 17 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 2984 | 8504 | 10296 | 10296 | 10296 | 10296 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 6456 | 8248 | 10296 | 10296 | 10296 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | 1 | 2 | 2 | 2 | 2 | 2 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | n/a | 2 | 2 | 2 | 2 | 2 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 4104 | 11340 | 13608 | 14076 | 14076 | 14076 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 8820 | 11088 | 14076 | 14076 | 14076 |
| Max. Throughput averaged over 1 frame | kbps | 2387.2 | 7448.8 | 9079.6 | 9266.4 | 9266.4 | 9266.4 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2-3b: Fixed Reference Channel for Maximum input level for UE Category 2 (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 83 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 2984 | 8504 | 14112 | 30576 | 46888 | 51024 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 6456 | 12576 | 28336 | 45352 | 51024 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | 1 | 2 | 3 | 5 | 8 | 9 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | n/a | 2 | 3 | 5 | 8 | 9 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 4104 | 11340 | 18900 | 41400 | 62100 | 68724 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 8820 | 16380 | 38880 | 59580 | 66204 |
| Max. Throughput averaged over 1 frame | kbps | 2387.2 | 7448.8 | 12547 | 27294 | 42046 | 45922 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz

Note 2: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]

Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2-4: Fixed Reference Channel for Maximum input level for UE Categories 3-5 (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Uplink-Downlink Configuration (Note 6) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame | | 4 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Number of HARQ Processes | Processes | 7 | 7 | 7 | 7 | 7 | 7 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 2984 | 8504 | 14112 | 30576 | 46888 | 61664 |
| For Sub-Frames 1,6 | Bits | n/a | 6968 | 11448 | 23688 | 35160 | 46888 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 6968 | 12576 | 30576 | 45352 | 61664 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 2 | 3 | 5 | 8 | 11 |
| For Sub-Frames 1,6 | | n/a | 2 | 2 | 4 | 6 | 8 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | n/a | 2 | 3 | 5 | 8 | 11 |
| Binary Channel Bits per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 4104 | 11340 | 18900 | 41400 | 62100 | 82800 |
| For Sub-Frames 1,6 | | n/a | 9828 | 16668 | 33768 | 50868 | 67968 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 9252 | 16812 | 39312 | 60012 | 80712 |
| Max. Throughput averaged over 1 frame | kbps | 596.8 | 3791.2 | 6369.6 | 13910 | 20945 | 27877 |

Note 1: For normal subframes(0,4,5,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
Note 2: For 1.4MHz, no data shall be scheduled on special subframes(1&6) to avoid problems with insufficient PDCCH performance
Note 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: As per Table 4.2-2 in TS 36.211 [8]

**Table A.3.2-4a: Fixed Reference Channel for Maximum input level for UE Category 1 (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 18 | 17 | 17 | 17 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Uplink-Downlink Configuration (Note 5) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame | | 4 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Number of HARQ Processes | Processes | 7 | 7 | 7 | 7 | 7 | 7 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 2984 | 8504 | 10296 | 10296 | 10296 | 10296 |
| For Sub-Frames 1,6 | Bits | n/a | 6968 | 8248 | 7480 | 7480 | 7480 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 6968 | 8248 | 10296 | 10296 | 10296 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 2 | 2 | 2 | 2 | 2 |
| For Sub-Frames 1,6 | | n/a | 2 | 2 | 2 | 2 | 2 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | n/a | 2 | 2 | 2 | 2 | 2 |
| Binary Channel Bits per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 4104 | 11340 | 13608 | 14076 | 14076 | 14076 |
| For Sub-Frames 1,6 | | n/a | 9828 | 11880 | 11628 | 11628 | 11628 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 9252 | 11520 | 14076 | 14076 | 14076 |
| Max. Throughput averaged over 1 frame | kbps | 596.8 | 3791.2 | 4533.6 | 4584.8 | 4584.8 | 4584.8 |

Note 1: For normal subframes(0,4,5,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.

Note 2: For 1.4MHz, no data shall be scheduled on special subframes(1&6) to avoid problems with insufficient PDCCH performance

Note 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]

Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

Note 5: As per Table 4.2-2 in TS 36.211 [8]

**Table A.3.2-4b: Fixed Reference Channel for Maximum input level for UE Category 2 (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 83 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Uplink-Downlink Configuration (Note 5) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame | | 4 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Number of HARQ Processes | Processes | 7 | 7 | 7 | 7 | 7 | 7 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 2984 | 8504 | 14112 | 30576 | 46888 | 51024 |
| For Sub-Frames 1,6 | Bits | n/a | 6968 | 11448 | 23688 | 35160 | 39232 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 6968 | 12576 | 30576 | 45352 | 51024 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 2 | 3 | 5 | 8 | 9 |
| For Sub-Frames 1,6 | | n/a | 2 | 3 | 5 | 7 | 7 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | n/a | 2 | 3 | 5 | 8 | 9 |
| Binary Channel Bits per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 4104 | 11340 | 18900 | 41400 | 62100 | 68724 |
| For Sub-Frames 1,6 | | n/a | 9828 | 16668 | 33768 | 50868 | 56340 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | n/a | 9252 | 16380 | 39312 | 60012 | 66636 |
| Max. Throughput averaged over 1 frame | kbps | 596.8 | 3791.2 | 6369.6 | 13910 | 20945 | 23154 |

Note 1: For normal subframes(0,4,5,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
Note 2: For 1.4MHz, no data shall be scheduled on special subframes(1&6) to avoid problems with insufficient PDCCH performance
Note 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
Note 5: As per Table 4.2-2 in TS 36.211 [8]

# A.3.2A Downlink Reference measurement channel for TX characteristics

Tables A.3.2A-1 and A.3.2A-2 describes the reference measurement channels to be used on the downlink during Transmitter Characteristics (clause 6) for FDD and TDD respectively. The number of allocated resource blocks have been defined (partial allocation) to allow the transmission of PBCH, PSS/SSS and system information mapped on PDSCH.

**Table A.3.2A-1: Fixed DL PDSCH Dedicated Reference Channel for TX Requirements (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 3 | 4 | 8 | 16 | 25 | 30 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target Coding Rate | | (Note 4) | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 | Bits | 88 | 328 | 680 | 1384 | 2216 | 2664 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Sub-Frame | | 1 | 1 | 1 | 1 | 1 | 1 |
| Code block CRC size | Bits | 0 | 0 | 0 | 0 | 0 | 0 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1, 2, 3, 4, 6, 7, 8, 9 | Bits | 684 | 1008 | 2016 | 4416 | 6900 | 8280 |
| For Sub-Frames 5 | | 540 | 1008 | 2016 | 4416 | 6900 | 8280 |
| For Sub-Frames 0 | | 264 | 1008 | 2016 | 4416 | 6900 | 8280 |
| Max. Throughput averaged over 1 frame | kbps | 88 | 328 | 680 | 1384 | 2216 | 2664 |
| UE-Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz

Note 2: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]

Note 3: The PDSCH shall be assigned to the UE under test with a set of allocated localized virtual resource blocks starting from one end of the channel.

Note 4: To ensure constant transport block size in 1.4MHz, the code rate for subframes varies approx. within {1/8-1/3}

**Table A.3.2A-2: Fixed DL PDSCH Dedicated Reference Channel for TX Requirements (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel Bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 3 | 4 | 8 | 16 | 25 | 30 |
| Uplink-Downlink Configuration (Note 6) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | 4 | 4 | 4 | 4 | 4 | 4 |
| Number of HARQ Processes | Processes | 7 | 7 | 7 | 7 | 7 | 7 |
| Maximum number of HARQ transmission | | 1 | 1 | 1 | 1 | 1 | 1 |
| Modulation | | QPSK | QPSK | QPSK | QPSK | QPSK | QPSK |
| Target coding rate | | (Note 5) | 1/3 | 1/3 | 1/3 | 1/3 | 1/3 |
| Information Bit Payload per Sub-Frame | Bits | | | | | | |
| For Sub-Frame 1, 6 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0, 4, 5, 9 | | 88 | 328 | 680 | 1384 | 2216 | 2664 |
| Transport block CRC | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks | | 1 | 1 | 1 | 1 | 1 | 1 |
| Code block CRC size | | 0 | 0 | 0 | 0 | 0 | 0 |
| Binary Channel Bits Per Sub-Frame | Bits | | | | | | |
| For Sub-Frame 1, 6 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 4, 9 | | 684 | 1008 | 2016 | 4416 | 6900 | 8280 |
| For Sub-Frame 0 | | 336 | 1008 | 2016 | 4416 | 6900 | 8280 |
| For Sub-Frame 5 | | 612 | 1008 | 2016 | 4416 | 6900 | 8280 |
| Max. Throughput averaged over one frame | kbps | 35.2 | 131.2 | 272 | 553.6 | 886.4 | 1065.6 |
| UE-Category | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 |

Note 1: For normal subframes (0,4,5,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
Note 2: For simplicity, no data shall be scheduled on special subframes (1&6).
Note 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 4: The PDSCH shall be assigned to the UE under test with a set of allocated localized virtual resource blocks starting from one end of the channel.
Note 5: To ensure constant transport block size in 1.4MHz, the code rate for subframes varies approx. within {1/8-1/3}.
Note 6: As per Table 4.2-2 in TS 36.211 [8]

# A.3.3 Reference measurement channel for PDSCH performance requirements (FDD)

## A.3.3.1 Single-antenna transmission (Common Reference Symbols)

**Table A.3.3.1-1: Fixed Reference Channel QPSK R=1/3**

| Parameter | Unit | | Value | | | | | |
|---|---|---|---|---|---|---|---|---|
| Reference channel | | R.4 FDD | | | R.42 FDD | R.2 FDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 20 | 10 | 15 | 20 |
| Allocated resource blocks (Note4) | | 6 | | | 100 | 50 | | |
| Allocated subframes per Radio Frame | | 10 | | | 10 | 10 | | |
| Modulation | | QPSK | | | QPSK | QPSK | | |
| Target Coding Rate | | 1/3 | | | 1/3 | 1/3 | | |
| Information Bit Payload (Note4) | | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 408 | | | 8760 | 4392 | | |
| For Sub-Frame 5 | Bits | n/a | | | n/a | n/a | | |
| For Sub-Frame 0 | Bits | 152 | | | 8760 | 4392 | | |
| Number of Code Blocks per Sub-Frame (see Note 3 and 4) | | 1 | | | | 1 | | |
| Binary Channel Bits (Note 4) | | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 1368 | | | 27600 | 13800 | | |
| For Sub-Frame 5 | Bits | n/a | | | n/a | n/a | | |
| For Sub-Frame 0 | Bits | 528 | | | 26760 | 12960 | | |
| Max. Throughput averaged over 1 frame (Note4) | Mbps | 0.342 | | | 7.884 | 3.953 | | |
| UE Category | | ≥ 1 | | | >=1 | ≥ 1 | | |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
Note 4: Given per component carrier per codeword

**Table A.3.3.1-2: Fixed Reference Channel 16QAM R=1/2**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | | R.3-1 FDD | R.3 FDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | 25 | 50 | | |
| Allocated subframes per Radio Frame | | | | 10 | 10 | | |
| Modulation | | | | 16QAM | 16QAM | | |
| Target Coding Rate | | | | 1/2 | 1/2 | | |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | | 6456 | 14112 | | |
| For Sub-Frame 5 | Bits | | | n/a | n/a | | |
| For Sub-Frame 0 | Bits | | | 5736 | 12960 | | |
| Number of Code Blocks per Sub-Frame (see Note 3) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | | | 2 | 3 | | |
| For Sub-Frame 5 | | | | n/a | n/a | | |
| For Sub-Frame 0 | | | | 1 | 3 | | |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | | 12600 | 27600 | | |
| For Sub-Frame 5 | Bits | | | n/a | n/a | | |
| For Sub-Frame 0 | Bits | | | 10920 | 25920 | | |
| Max. Throughput averaged over 1 frame | Mbps | | | 5.738 | 12.586 | | |
| UE Category | | | | ≥ 1 | ≥ 2 | | |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.3.1-3: Fixed Reference Channel 64QAM R=3/4**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | R.5 FDD | R.6 FDD | R.7 FDD | R.8 FDD | R.9 FDD |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | 15 | 25 | 50 | 75 | 100 |
| Allocated subframes per Radio Frame | | | 10 | 10 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 8504 | 14112 | 30576 | 46888 | 61664 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 6456 | 12576 | 28336 | 45352 | 61664 |
| Number of Code Blocks per Sub-Frame (see Note 3) | | | 2 | 3 | 5 | 8 | 11 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 11340 | 18900 | 41400 | 62100 | 82800 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 8820 | 16380 | 38880 | 59580 | 80280 |
| Max. Throughput averaged over 1 frame | Mbps | | 7.449 | 12.547 | 27.294 | 42.046 | 55.498 |
| UE Category | | | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 | ≥ 3 |
| Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz<br>Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.3.3.1-3a: Fixed Reference Channel 64QAM R=3/4**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | R.6-1 FDD | R.7-1 FDD | R.8-1 FDD | R.9-1 FDD | R.9-2 FDD |
| Channel bandwidth | MHz | | 5 | 10 | 15 | 20 | 20 |
| Allocated resource blocks (Note 3) | | | 18 | 17 | 17 | 17 | 83 |
| Allocated subframes per Radio Frame | | | 10 | 10 | 10 | 10 | 10 |
| Modulation | | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 10296 | 10296 | 10296 | 10296 | 51024 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 8248 | 10296 | 10296 | 10296 | 51024 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | | 2 | 2 | 2 | 2 | 9 |
| For Sub-Frame 5 | | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | | 2 | 2 | 2 | 2 | 9 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 13608 | 14076 | 14076 | 14076 | 68724 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 11088 | 14076 | 14076 | 14076 | 66204 |
| Max. Throughput averaged over 1 frame | Mbps | | | 9.266 | 9.266 | 9.266 | 45.922 |
| UE Category | | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 |
| Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz<br>Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]<br>Note 3: Localized allocation started from RB #0 is applied.<br>Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.3.3.1-4: Fixed Reference Channel Single PRB (Channel Edge)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | R.0 FDD | | R.1 FDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10/20 | 15 | 20 |
| Allocated resource blocks | | | 1 | | 1 | | |
| Allocated subframes per Radio Frame | | | 10 | | 10 | | |
| Modulation | | | 16QAM | | 16QAM | | |
| Target Coding Rate | | | 1/2 | | 1/2 | | |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 224 | | 256 | | |
| For Sub-Frame 5 | Bits | | n/a | | n/a | | |
| For Sub-Frame 0 | Bits | | 224 | | 256 | | |
| Number of Code Blocks per Sub-Frame (see Note 3) | | | 1 | | 1 | | |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 504 | | 552 | | |
| For Sub-Frame 5 | Bits | | n/a | | n/a | | |
| For Sub-Frame 0 | Bits | | 504 | | 552 | | |
| Max. Throughput averaged over 1 frame | Mbps | | 0.202 | | 0.230 | | |
| UE Category | | | ≥ 1 | | ≥ 1 | | |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.3.1-5: Fixed Reference Channel Single PRB (MBSFN Configuration)**

| Parameter | Unit | Value |
|---|---|---|
| Reference channel | | R.29 FDD (MBSFN) |
| Channel bandwidth | MHz | 10 |
| Allocated resource blocks | | 1 |
| MBSFN Configuration | | TBD |
| Allocated subframes per Radio Frame | | 4 |
| Modulation | | 16QAM |
| Target Coding Rate | | 1/2 |
| Information Bit Payload | | |
| For Sub-Frames 4,9 | Bits | 256 |
| For Sub-Frame 5 | Bits | n/a |
| For Sub-Frame 0 | Bits | 256 |
| For Sub-Frame 1,2,3,6,7,8 | Bits | 0 (MBSFN) |
| Number of Code Blocks per Sub-Frame (see Note 3) | | 1 |
| For Sub-Frames 4,9 | | 1 |
| For Sub-Frame 5 | | n/a |
| For Sub-Frame 0 | | 1 |
| For Sub-Frame 1,2,3,6,7,8 | | 0 (MBSFN) |
| Binary Channel Bits Per Sub-Frame | | |
| For Sub-Frames 4,9 | Bits | 552 |
| For Sub-Frame 5 | Bits | n/a |
| For Sub-Frame 0 | Bits | 552 |
| For Sub-Frame 1,2,3,6,7,8 | Bits | 0 (MBSFN) |
| Max. Throughput averaged over 1 frame | kbps | 76.8 |
| UE Category | | ≥ 1 |

Note 1: 2 symbols allocated to PDCCH
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit

## A.3.3.2 Multi-antenna transmission (Common Reference Symbols)

### A.3.3.2.1 Two antenna ports

**Table A.3.3.2.1-1: Fixed Reference Channel two antenna ports**

| Parameter | Unit | Value | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Reference channel | | | R.10 FDD | R.11 FDD | R.11-2 FDD | R.11-3 FDD | R.30 FDD | R.35 FDD |
| Channel bandwidth | MHz | | 10 | 10 | 5 | 10 | 20 | 10 |
| Allocated resource blocks | | | 50 | 50 | 25 | 40 | 100 | 50 |
| Allocated subframes per Radio Frame | | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | | QPSK | 16QAM | 16QAM | 16QAM | 16QAM | 64QAM |
| Target Coding Rate | | | 1/3 | 1/2 | 1/2 | 1/2 | 1/2 | 1/2 |
| Information Bit Payload | | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 4392 | 12960 | 5736 | 10296 | 25456 | 19848 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 4392 | 12960 | 4968 | 10296 | 25456 | 18336 |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 1 | 3 | 1 | 2 | 5 | 4 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 1 | 3 | 1 | 2 | 5 | 3 |
| Binary Channel Bits Per Sub-Frame | | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | | 13200 | 26400 | 12000 | 21120 | 52800 | 39600 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 12384 | 24768 | 10368 | 19488 | 51168 | 37152 |
| Max. Throughput averaged over 1 frame | Mbps | | 3.953 | 11.664 | 5.086 | 9.266 | 22.910 | 17.712 |
| UE Category | | | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 8
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

### A.3.3.2.2 Four antenna ports

**Table A.3.3.2.2-1: Fixed Reference Channel four antenna ports**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.12 FDD | R.13 FDD | R.14 FDD | R.14-1 FDD | R.14-2 FDD | R.36 FDD |
| Channel bandwidth | MHz | 1.4 | 10 | 10 | 10 | 10 | 10 |
| Allocated resource blocks | | 6 | 50 | 50 | 6 | 3 | 50 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 | 10 |
| Modulation | | QPSK | QPSK | 16QAM | 16QAM | 16QAM | 64QAM |
| Target Coding Rate | | 1/3 | 1/3 | 1/2 | 1/2 | 1/2 | 1/2 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 408 | 4392 | 12960 | [1544] | [744] | 18336 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 152 | 3624 | 11448 | n/a | n/a | 18336 |
| Number of Code Blocks per Sub-Frame (see Note 3) | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | 1 | 1 | 3 | 1 | 1 | 3 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 1 | 2 | n/a | n/a | 3 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 1248 | 12800 | 25600 | [3072] | [1536] | 38400 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 480 | 12032 | 24064 | n/a | n/a | 36096 |
| Max. Throughput averaged over 1 frame | Mbps | 0.342 | 3.876 | 11.513 | [1.235] | [0.595] | 16.502 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 1 | ≥ 2 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 8
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

## A.3.4 Reference measurement channel for PDSCH performance requirements (TDD)

### A.3.4.1 Single-antenna transmission (Common Reference Symbols)

**Table A.3.4.1-1: Fixed Reference Channel QPSK R=1/3**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.4 TDD | | | R.2 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | | | 50 | | |
| Uplink-Downlink Configuration (Note 4) | | 1 | | | 1 | | |
| Allocated subframes per Radio Frame (D+S) | | 4+2 | | | 4+2 | | |
| Modulation | | QPSK | | | QPSK | | |
| Target Coding Rate | | 1/3 | | | 1/3 | | |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | 408 | | | 4392 | | |
| For Sub-Frames 1,6 | Bits | n/a | | | 3240 | | |
| For Sub-Frame 5 | Bits | n/a | | | n/a | | |
| For Sub-Frame 0 | Bits | 208 | | | 4392 | | |
| Number of Code Blocks per Sub-Frame (Note 5) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | | | 1 | | |
| For Sub-Frames 1,6 | | n/a | | | 1 | | |
| For Sub-Frame 5 | | n/a | | | n/a | | |
| For Sub-Frame 0 | | 1 | | | 1 | | |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 1368 | | | 13800 | | |
| For Sub-Frames 1,6 | Bits | n/a | | | 11256 | | |
| For Sub-Frame 5 | Bits | n/a | | | n/a | | |
| For Sub-Frame 0 | Bits | 672 | | | 13104 | | |
| Max. Throughput averaged over 1 frame | Mbps | 0.102 | | | 1.966 | | |
| UE Category | | ≥ 1 | | | ≥ 1 | | |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: For BW=1.4 MHz, the information bit payloads of special subframes are set to zero (no scheduling) to avoid problems with insufficient PDCCH performance at the test point.
Note 3: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 8
Note 4: As per Table 4.2-2 in TS 36.211 [8]
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.4.1-2: Fixed Reference Channel 16QAM R=1/2**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | | R.3-1 TDD | R.3 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | 25 | 50 | | |
| Uplink-Downlink Configuration (Note 3) | | | | 1 | 1 | | |
| Allocated subframes per Radio Frame (D+S) | | | | 4+2 | 4+2 | | |
| Modulation | | | | 16QAM | 16QAM | | |
| Target Coding Rate | | | | 1/2 | 1/2 | | |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | | | 6456 | 14112 | | |
| For Sub-Frames 1,6 | Bits | | | 5160 | 11448 | | |
| For Sub-Frame 5 | Bits | | | n/a | n/a | | |
| For Sub-Frame 0 | Bits | | | 5736 | 12960 | | |
| Number of Code Blocks per Sub-Frame (see Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | | | 2 | 3 | | |
| For Sub-Frames 1,6 | | | | 1 | 2 | | |
| For Sub-Frame 5 | | | | n/a | n/a | | |
| For Sub-Frame 0 | | | | 1 | 3 | | |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | | | 12600 | 27600 | | |
| For Sub-Frames 1,6 | Bits | | | 11112 | 22512 | | |
| For Sub-Frame 5 | Bits | | | n/a | n/a | | |
| For Sub-Frame 0 | Bits | | | 11208 | 26208 | | |
| Max. Throughput averaged over 1 frame | Mbps | | | 2.897 | 6.408 | | |
| UE Category | | | | ≥ 1 | ≥ 2 | | |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: As per Table 4.2-2 in TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.4.1-3: Fixed Reference Channel 64QAM R=3/4**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | R.5 TDD | R.6 TDD | R.7 TDD | R.8 TDD | R.9 TDD |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | 15 | 25 | 50 | 75 | 100 |
| Uplink-Downlink Configuration (Note 3) | | | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | | 8504 | 14112 | 30576 | 46888 | 61664 |
| For Sub-Frames 1,6 | Bits | | 6968 | 11448 | 23688 | 35160 | 46888 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 6968 | 12576 | 30576 | 45352 | 61664 |
| Number of Code Blocks per Sub-Frame (see Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | | 2 | 3 | 5 | 8 | 11 |
| For Sub-Frames 1,6 | | | 2 | 2 | 4 | 6 | 8 |
| For Sub-Frame 5 | | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | | 2 | 3 | 5 | 8 | 11 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | | 11340 | 18900 | 41400 | 62100 | 82800 |
| For Sub-Frames 1,6 | Bits | | 9828 | 16668 | 33768 | 50868 | 67968 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 9252 | 16812 | 39312 | 60012 | 80712 |
| Max. Throughput averaged over 1 frame | Mbps | | 3.791 | 6.370 | 13.910 | 20.945 | 27.877 |
| UE Category | | | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 | ≥ 3 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: As per Table 4.2-2 TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.4.1-3a: Fixed Reference Channel 64QAM R=3/4**

| Parameter | Unit | | Value | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | R.6-1 TDD | R.7-1 TDD | R.8-1 TDD | R.9-1 TDD | R.9-2 TDD |
| Channel bandwidth | MHz | | 5 | 10 | 15 | 20 | 20 |
| Allocated resource blocks (Note 3) | | | 18 | 17 | 17 | 17 | 83 |
| Uplink-Downlink Configuration (Note 4) | | | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | | 10296 | 10296 | 10296 | 10296 | 51024 |
| For Sub-Frames 1,6 | Bits | | 8248 | 7480 | 7480 | 7480 | 39232 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 8248 | 10296 | 10296 | 10296 | 51024 |
| Number of Code Blocks per Sub-Frame (Note 5) | | | | | | | |
| For Sub-Frames 4,9 | | | 2 | 2 | 2 | 2 | 9 |
| For Sub-Frames 1,6 | | | 2 | 2 | 2 | 2 | 7 |
| For Sub-Frame 5 | | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | | 2 | 2 | 2 | 2 | 9 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | | 13608 | 14076 | 14076 | 14076 | 68724 |
| For Sub-Frames 1,6 | Bits | | 11880 | 11628 | 11628 | 11628 | 56340 |
| For Sub-Frame 5 | Bits | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | | 11520 | 14076 | 14076 | 14076 | 66636 |
| Max. Throughput averaged over 1 frame | Mbps | | 4.534 | 4.585 | 4.585 | 4.585 | 23.154 |
| UE Category | | | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 1 | ≥ 2 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [4]
Note 3: Localized allocation started from RB #0 is applied.
Note 4: As per Table 4.2-2 TS 36.211 [4]
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.4.1-4: Fixed Reference Channel Single PRB**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | R.0 TDD | | R.1 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10/20 | 15 | 20 |
| Allocated resource blocks | | | 1 | | 1 | | |
| Uplink-Downlink Configuration (Note 3) | | | 1 | | 1 | | |
| Allocated subframes per Radio Frame (D+S) | | | 4+2 | | 4+2 | | |
| Modulation | | | 16QAM | | 16QAM | | |
| Target Coding Rate | | | 1/2 | | 1/2 | | |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | | 224 | | 256 | | |
| For Sub-Frames 1,6 | Bits | | 208 | | 208 | | |
| For Sub-Frame 5 | Bits | | n/a | | n/a | | |
| For Sub-Frame 0 | Bits | | 224 | | 256 | | |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | | 1 | | 1 | | |
| For Sub-Frames 1,6 | | | 1 | | 1 | | |
| For Sub-Frame 5 | | | n/a | | n/a | | |
| For Sub-Frame 0 | | | 1 | | 1 | | |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | | 504 | | 552 | | |
| For Sub-Frames 1,6 | Bits | | 456 | | 456 | | |
| For Sub-Frame 5 | Bits | | n/a | | n/a | | |
| For Sub-Frame 0 | Bits | | 504 | | 552 | | |
| Max. Throughput averaged over 1 frame | Mbps | | 0.109 | | 0.118 | | |
| UE Category | | | ≥ 1 | | ≥ 1 | | |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: As per Table 4.2-2 in TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.4.1-5: Fixed Reference Channel Single PRB (MBSFN Configuration)**

| Parameter | Unit | Value |
|---|---|---|
| Reference channel | | R.29 TDD (MBSFN) |
| Channel bandwidth | MHz | 10 |
| Allocated resource blocks | | 1 |
| MBSFN Configuration | | [TBD] |
| Uplink-Downlink Configuration (Note 3) | | 1 |
| Allocated subframes per Radio Frame (D+S) | | 2+2 |
| Modulation | | 16QAM |
| Target Coding Rate | | 1/2 |
| Information Bit Payload | | |
| For Sub-Frames 4,9 | Bits | 0 (MBSFN) |
| For Sub-Frames 1,6 | Bits | 208 |
| For Sub-Frame 5 | Bits | n/a |
| For Sub-Frame 0 | Bits | 256 |
| Number of Code Blocks per Sub-Frame (Note 4) | | |
| For Sub-Frames 4,9 | Bits | 0 (MBSFN) |
| For Sub-Frames 1,6 | Bits | 1 |
| For Sub-Frame 5 | Bits | n/a |
| Binary Channel Bits Per Sub-Frame | | |
| For Sub-Frames 4,9 | Bits | 0 (MBSFN) |
| For Sub-Frames 1,6 | Bits | 456 |
| For Sub-Frame 5 | Bits | n/a |
| For Sub-Frame 0 | Bits | 552 |
| Max. Throughput averaged over 1 frame | kbps | 67.2 |
| UE Category | | ≥ 1 |

Note 1: 2 symbols allocated to PDCCH
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: as per Table 4.2-2 in TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

## A.3.4.2 Multi-antenna transmission (Common Reference Symbols)

### A.3.4.2.1 Two antenna ports

**Table A.3.4.2.1-1: Fixed Reference Channel two antenna ports**

| Parameter | Unit | Value | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Reference channel | | R.10 TDD | R.11 TDD | R.11-1 TDD | R.11-2 TDD | R.11-3 TDD | R.30 TDD | R.35 TDD |
| Channel bandwidth | MHz | 10 | 10 | 10 | 5 | 10 | 20 | 10 |
| Allocated resource blocks | | 50 | 50 | 50 | 25 | 40 | 100 | 50 |
| Uplink-Downlink Configuration (Note 3) | | **1** | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | QPSK | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 64QAM |
| Target Coding Rate | | 1/3 | 1/2 | 1/2 | 1/2 | 1/2 | 1/2 | 1/2 |
| Information Bit Payload | | | | | | | | |
| For Sub-Frames 4,9 | Bits | 4392 | 12960 | 12960 | 5736 | 10296 | 25456 | 19848 |
| For Sub-Frames 1,6 | | 3240 | 9528 | 9528 | 5160 | 9144 | 22920 | 15840 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | | n/a | n/a |
| For Sub-Frame 0 | Bits | 4392 | 12960 | n/a | 4968 | 10296 | 25456 | n/a |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 3 | 3 | 1 | 2 | 5 | 4 |
| For Sub-Frames 1,6 | | 1 | 2 | 2 | 1 | 2 | 4 | 3 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 3 | n/a | 1 | 2 | 5 | n/a |
| Binary Channel Bits Per Sub-Frame | | | | | | | | |
| For Sub-Frames 4,9 | Bits | 13200 | 26400 | 26400 | 12000 | 21120 | 52800 | 39600 |
| For Sub-Frames 1,6 | | 10656 | 21312 | 21312 | 10512 | 16992 | 42912 | 31968 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 12528 | 25056 | n/a | 10656 | 19776 | 51456 | n/a |
| Max. Throughput averaged over 1 frame | Mbps | 1.966 | 5.794 | 4.498 | 2.676 | 4.918 | 12.221 | 7.138 |
| UE Category | | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 2 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 3: As per Table 4.2-2 in TS 36.211 [8]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

### A.3.4.2.2 Four antenna ports

**Table A.3.4.2.2-1: Fixed Reference Channel four antenna ports**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.12 TDD | R.13 TDD | R.14 TDD | R.14-1 TDD | R.14-2 TDD | R.36 TDD |
| Channel bandwidth | MHz | 1.4 | 10 | 10 | 10 | 10 | 10 |
| Allocated resource blocks | | 6 | 50 | 50 | 6 | 3 | 50 |
| Uplink-Downlink Configuration (Note 4) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | QPSK | QPSK | 16QAM | 16QAM | 16QAM | 64QAM |
| Target Coding Rate | | 1/3 | 1/3 | 1/2 | 1/2 | 1/2 | 1/2 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | 408 | 4392 | 12960 | [1544] | [744] | 18336 |
| For Sub-Frames 1,6 | Bits | n/a | 3240 | 9528 | n/a | n/a | 15840 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 208 | 4392 | n/a | n/a | n/a | n/a |
| Number of Code Blocks per Sub-Frame (Note 5) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 1 | 3 | 1 | 1 | 3 |
| For Sub-Frames 1,6 | | n/a | 1 | 2 | n/a | n/a | 3 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 1 | n/a | n/a | n/a | n/a |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 1248 | 12800 | 25600 | [3072] | [1536] | 38400 |
| For Sub-Frames 1,6 | | n/a | 10256 | 20512 | n/a | n/a | 30768 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 624 | 12176 | n/a | n/a | n/a | n/a |
| Max. Throughput averaged over 1 frame | Mbps | 0.102 | 1.966 | 4.498 | [0.309] | [0.149] | 6.835 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 1 | ≥ 2 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: For BW=1.4 MHz, the information bit payloads of special subframes are set to zero (no scheduling) to avoid problems with insufficient PDCCH performance at the test point.
Note 3: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8]
Note 4: As per Table 4.2-2 in TS 36.211 [8]
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

## A.3.4.3 Reference Measurement Channels for UE-Specific Reference Symbols

### A.3.4.3.1 Single antenna port (Cell Specific)

The reference measurement channels in Table A.3.4.3.1-1 apply for verifying demodulation performance for UE-specific reference symbols with one cell-specific antenna port.

**Table A.3.4.3.1-1: Fixed Reference Channel for DRS**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.25 TDD | R.26 TDD | R.26-1 TDD | R.27 TDD | R.27-1 TDD | R.28 TDD |
| Channel bandwidth | MHz | 10 | 10 | 5 | 10 | 10 | 10 |
| Allocated resource blocks | | 50 [4] | 50 [4] | 25 [4] | 50 [4] | 18 [6] | 1 |
| Uplink-Downlink Configuration (Note 3) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | QPSK | 16QAM | 16QAM | 64QAM | 64QAM | 16QAM |
| Target Coding Rate | | 1/3 | 1/2 | 1/2 | 3/4 | 3/4 | 1/2 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | 4392 | 12960 | 5736 | 28336 | 10296 | 224 |
| For Sub-Frames 1,6 | Bits | 3240 | 9528 | 4584 | 22920 | 8248 | 176 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 2984 | 9528 | 3880 | 22152 | 10296 | 224 |
| Number of Code Blocks per Sub-Frame (see Note 5) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 3 | 1 | 5 | 2 | 1 |
| For Sub-Frames 1,6 | | 1 | 2 | 1 | 4 | 2 | 1 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 2 | 1 | 4 | 2 | 1 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 12600 | 25200 | 11400 | 37800 | 13608 | 504 |
| For Sub-Frames 1,6 | Bits | 10356 | 20712 | 10212 | 31068 | 11340 | 420 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 10332 | 20664 | 7752 | 30996 | 13608 | 504 |
| Max. Throughput averaged over 1 frame | Mbps | 1.825 | 5.450 | 2.452 | 12.466 | 4.738 | 0.102 |
| UE Category | | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 1 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8].
Note 3: as per Table 4.2-2 in TS 36.211 [8].
Note 4: For R.25, R.26 and R.27, 50 resource blocks are allocated in sub-frames 1–9 and 41 resource blocks (RB0–RB20 and RB30–RB49) are allocated in sub-frame 0. For R.26-1, 25 resource blocks are allocated in sub-frames 1–9 and 17 resource blocks (RB0–RB7 and RB16–RB24) are allocated in sub-frame 0.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Localized allocation started from RB #0 is applied.

### A.3.4.3.2 Two antenna ports (Cell Specific)

The reference measurement channels in Table A.3.4.3.2-1 apply for verifying demodulation performance for CDM-multiplexed UE specific reference symbols with two cell-specific antenna ports.

**Table A.3.4.3.2-1: Fixed Reference Channel for CDM-multiplexed DM RS**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.31 TDD | R.32 TDD | R.32-1 TDD | R.33 TDD | R.33-1 TDD | R.34 TDD |
| Channel bandwidth | MHz | 10 | 10 | 5 | 10 | 10 | 10 |
| Allocated resource blocks | | 50[4] | 50[4] | [25[4]] | 50[4] | [18][b] | 50[4] |
| Uplink-Downlink Configuration (Note 3) | | 1 | 1 | 1 | 1 | 1 | 1 |
| Allocated subframes per Radio Frame (D+S) | | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 | 4+2 |
| Modulation | | QPSK | 16QAM | 16QAM | 64QAM | 64QAM | 64QAM |
| Target Coding Rate | | 1/3 | 1/2 | 1/2 | 3/4 | 3/4 | 1/2 |
| Information Bit Payload | | | | | | | |
| For Sub-Frames 4,9 | Bits | 3624 | 11448 | [5736] | 27376 | [9528] | 18336 |
| For Sub-Frames 1,6 | | 2664 | 7736 | [3112] | 16992 | [7480] | 11832 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 2984 | 9528 | [3496] | 22152 | [9528] | 14688 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | | |
| For Sub-Frames 4,9 | | 1 | 2 | 1 | 5 | 2 | 3 |
| For Sub-Frames 1,6 | | 1 | 2 | 1 | 3 | 2 | 2 |
| For Sub-Frame 5 | | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 2 | 1 | 4 | 2 | 3 |
| Binary Channel Bits Per Sub-Frame | | | | | | | |
| For Sub-Frames 4,9 | Bits | 12000 | 24000 | [10800] | 36000 | [12960] | 36000 |
| For Sub-Frames 1,6 | | 7872 | 15744 | [6528] | 23616 | [10368] | 23616 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 9840 | 19680 | [7344] | 29520 | [12960] | 29520 |
| Max. Throughput averaged over 1 frame | Mbps | 1.556 | 4.79 | [2.119] | 11.089 | [4.354] | 7.502 |
| UE Category | | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 2 | ≥ 1 | ≥ 2 |

Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For subframe 1&6, only 2 OFDM symbols are allocated to PDCCH.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [8].
Note 3: as per Table 4.2-2 in TS 36.211 [8].
Note 4: For R.31, R.32, R.33 and R.34, 50 resource blocks are allocated in sub-frames 4,9 and 41 resource blocks (RB0–RB20 and RB30–RB49) are allocated in sub-frame 0 and the DwPTS portion of sub-frames 1,6 . For R.32-1, 25 resource blocks are allocated in sub-frames 4, 9 and 17 resource blocks (RB0–RB7 and RB16–RB24) are allocated in sub-frame 0 and the DwPTS portion of sub-frames 1, 6.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Localized allocation started from RB#0 is applied.

# A.3.5 Reference measurement channels for PDCCH/PCFICH performance requirements

## A.3.5.1 FDD

**Table A.3.5.1-1: Reference Channel FDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| Reference channel | | R.15 FDD | R.16 FDD | R.17 FDD |
| Number of transmitter antennas | | 1 | 2 | 4 |
| Channel bandwidth | MHz | 10 | 1.4 | 10 |
| Number of OFDM symbols for PDCCH | symbols | 2 | 2 | 2 |
| Aggregation level | CCE | 8 | 2 | 4 |
| DCI Format | | Format 1 | Format 2 | Format 2 |
| Cell ID | | 0 | 0 | 0 |
| Payload (without CRC) | Bits | 31 | 31 | 46 |
| | | | | |

**Table A.3.5.1-1A: Reference Channel FDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | R.16_1 FDD | R.17_1 FDD |
| Number of transmitter antennas | | 2 | 4 |
| Channel bandwidth | MHz | 10 | 5 |
| Number of OFDM symbols for PDCCH | symbols | 2 | 2 |
| Aggregation level | CCE | 4 | 2 |
| DCI Format | | Format 2 | Format 2 |
| Cell ID | | 0 | 0 |
| Payload (without CRC) | Bits | 43 | 42 |
| | | | |

**Table A.3.5.1-2: Additional PDSCH Reference Channel FDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| Number of transmitter antennas | | 1 | 2 | 4 |
| Channel bandwidth | MHz | 10 | 1.4 | 10 |
| Allocated Resource Blocks | | 50 | 6 | 50 |
| Modulation | | QPSK | QPSK | QPSK |
| Target Coding Rate | | 1/3 | 1/3 | 1/3 |
| Information Bit Payload | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 4392 | 504 | 4392 |
| For Sub-Frame 5 | | n/a | n/a | n/a |
| For Sub-Frame0 | Bits | 4392 | 256 | 3624 |
| Number of Code Blocks per Sub-Frame | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | | 1 | 1 | 1 |
| For Sub-Frame 5 | | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 1 | 1 |
| Binary Channel Bits Per Sub-Frame | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 13800 | 1584 | 12800 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a |
| For Sub-Frame 0 | | 12960 | 768 | 12032 |
| Max. Throughput averaged over 1 frame | Mbps | 3.953 | 0.429 | 3.876 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: 2 symbols allocated to PDCCH for all BW. | | | | |

## A.3.5.2 TDD

**Table A.3.5.2-1: Reference Channel TDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| Reference channel | | R.15 TDD | R.16 TDD | R.17 TDD |
| Number if transmitter antennas | | 1 | 2 | 4 |
| Channel bandwidth | MHz | 10 | 1.4 | 10 |
| Number of OFDM symbols for PDCCH | symbols | 2 | 2 | 2 |
| Aggregation level | CCE | 8 | 2 | 4 |
| DCI Format | | Format 1 | Format 2 | Format 2 |
| Cell ID | | 0 | 0 | 0 |
| Payload (without CRC) | Bits | 34 | 34 | 49 |
| | | | | |

**Table A.3.5.2-1A: Reference Channel TDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | R.16_1 TDD | R.17_1 TDD |
| Number if transmitter antennas | | 2 | 4 |
| Channel bandwidth | MHz | 10 | 5 |
| Number of OFDM symbols for PDCCH | symbols | 2 | 2 |
| Aggregation level | CCE | 4 | 2 |
| DCI Format | | Format 2 | Format 2 |
| Cell ID | | 0 | 0 |
| Payload (without CRC) | Bits | 46 | 45 |

**Table A.3.5.2-2: Additional PDSCH Reference Channel TDD**

| Parameter | Unit | Value | | |
|---|---|---|---|---|
| Number of transmitter antennas | | 1 | 2 | 4 |
| Channel bandwidth | MHz | 10 | 1.4 | 10 |
| Uplink-Downlink Configuration (Note 2) | | 0 | 0 | 0 |
| Allocated Resource Blocks | | 50 | 6 | 50 |
| Modulation | | QPSK | QPSK | QPSK |
| Target Coding Rate | | 1/3 | 1/3 | 1/3 |
| Information Bit Payload | | | | |
| For Sub-Frame 1,6 | Bits | 3240 | 328 | 3240 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 4392 | 256 | 4392 |
| Number of Code Blocks per Sub-Frame | | | | |
| For Sub-Frame 1,6 | | 1 | 1 | 1 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a |
| For Sub-Frame 0 | | 1 | 1 | 1 |
| Binary Channel Bits Per Sub-Frame | | | | |
| For Sub-Frame 1,6 | Bits | 11256 | 1152 | 10256 |
| For Sub-Frame 5 | Bits | n/a | n/a | n/a |
| For Sub-Frame 0 | Bits | 13104 | 936 | 12176 |
| Max. Throughput averaged over 1 frame | Mbps | 1.087 | 0.091 | 1.164 |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 |
| Note 1: 2 symbols allocated to PDCCH for all BW.<br>Note 2: As per Table 4.2-2 in TS 36.211 [8] | | | | |

## A.3.6 Reference measurement channels for PHICH performance requirements

**Table A.3.6-1: Reference Channel FDD/TDD**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.18 | R.19 | R.20 | R.24 |
| Number of transmitter antennas | | 1 | 2 | 4 | 1 |
| Channel bandwidth | MHz | 10 | 1.4 | 10 | 10 |
| User roles (Note 1) | | [W I1 I2] | [W I1 I2] | [W I1 I2] | [W I1] |
| Resource allocation (Note 2) | | [(0,0) (0,1) (0,4)] | [(0,0) (0,1) (0,4)] | [(0,0) (0,1) (0,4)] | [(0,0) (0,1)] |
| Power offsets (Note 3) | dB | [-4 0 -3] | [-4 0 -3] | [-4 0 -3] | [+3 0] |
| Payload (Note 4) | | [A R R] | [A R R] | [A R R] | [A R] |
| Note 1: W=wanted user, I1=interfering user 1, I2=interfering user 2.<br>Note 2: The resource allocation per user is given as (N_group_PHICH, N_seq_PHICH).<br>Note 3: The power offsets (per user) represent the difference of the power of BPSK modulated symbol per PHICH relative to the first interfering user.<br>Note 4: A=fixed ACK, R=random ACK/NACK. | | | | | |

**Table A.3.6-1A: Reference Channel FDD/TDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | R.19_1 | R.20_1 |
| Number of transmitter antennas | | 2 | 4 |
| Channel bandwidth | MHz | 10 | 5 |
| User roles (Note 1) | | [W I1 I2] | [W I1 I2] |
| Resource allocation (Note 2) | | [(0,0) (0,1) (0,4)] | [(0,0) (0,1) (0,4)] |
| Power offsets (Note 3) | dB | [-4 0 -3] | [-4 0 -3] |
| Payload (Note 4) | | [A R R] | [A R R] |
| Note 1: W=wanted user, I1=interfering user 1, I2=interfering user 2.<br>Note 2: The resource allocation per user is given as (N_group_PHICH, N_seq_PHICH).<br>Note 3: The power offsets (per user) represent the difference of the power of BPSK modulated symbol per PHICH relative to the first interfering user.<br>Note 4: A=fixed ACK, R=random ACK/NACK. | | | |

# A.3.7 [FFS]

# A.3.8 Reference measurement channels for MBMS performance requirements

## A.3.8.1 FDD

**Table A.3.8.1-1: Fixed Reference Channel QPSK R=1/3**

| Parameter | PMCH | | | | | | |
|---|---|---|---|---|---|---|---|
| | Unit | Value | | | | | |
| Reference channel | | R.40 FDD | | | R.37 FDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | | | 50 | | |
| Allocated subframes per Radio Frame (Note 1) | | 6 | | | 6 | | |
| Modulation | | QPSK | | | QPSK | | |
| Target Coding Rate | | 1/3 | | | 1/3 | | |
| Information Bit Payload (Note 2) | | | | | | | |
| For Sub-Frames 1,2,3,6,7,8 | Bits | 408 | | | 3624 | | |
| For Sub-Frames 0,4,5,9 | Bits | n/a | | | n/a | | |
| Number of Code Blocks per Subframe (Note 3) | | 1 | | | 1 | | |
| Binary Channel Bits Per Subframe | | | | | | | |
| For Sub-Frames 1,2,3,6,7,8 | Bits | 1224 | | | 10200 | | |
| For Sub-Frames 0,4,5,9 | Bits | n/a | | | n/a | | |
| MBMS UE Category | | ≥ 1 | | | ≥ 1 | | |
| Note 1: For FDD mode, up to 6 subframes (#1/2/3/6/7/8) are available for MBMS, in line with TS 36.331.<br>Note 2: 2 OFDM symbols are reserved for PDCCH; and reference signal allocated as per TS 36.211.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit). | | | | | | | |

**Table A.3.8.1-2: Fixed Reference Channel 16QAM R=1/2**

| Parameter | PMCH | | | | | | |
|---|---|---|---|---|---|---|---|
| | Unit | Value | | | | | |
| Reference channel | | | | | R.38 FDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | | 50 | | |
| Allocated subframes per Radio Frame (Note 1) | | | | | 6 | | |
| Modulation | | | | | 16QAM | | |
| Target Coding Rate | | | | | 1/2 | | |
| Information Bit Payload (Note 2) | | | | | | | |
| For Sub-Frames 1,2,3,6,7,8 | Bits | | | | 9912 | | |
| For Sub-Frames 0,4,5,9 | Bits | | | | n/a | | |
| Number of Code Blocks per Subframe (Note 3) | | | | | 2 | | |
| Binary Channel Bits Per Subframe | | | | | | | |
| For Sub-Frames 1,2,3,6,7,8 | Bits | | | | 20400 | | |
| For Sub-Frames 0,4,5,9 | Bits | | | | n/a | | |
| MBMS UE Category | | | | | ≥ 1 | | |
| Note 1: For FDD mode, up to 6 subframes (#1/2/3/6/7/8) are available for MBMS, in line with TS 36.331.<br>Note 2: 2 OFDM symbols are reserved for PDCCH; and reference signal allocated as per TS 36.211.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit). | | | | | | | |

**Table A.3.8.1-3: Fixed Reference Channel 64QAM R=2/3**

| Parameter | PMCH | | | | | | |
|---|---|---|---|---|---|---|---|
| | Unit | Value | | | | | |
| Reference channel | | | | 39-1 FDD | R.39 FDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | 25 | 50 | | |
| Allocated subframes per Radio Frame(Note1) | | | | 6 | 6 | | |
| Modulation | | | | 64QAM | 64QAM | | |
| Target Coding Rate | | | | 2/3 | 2/3 | | |
| Information Bit Payload (Note 2) | | | | | | | |
| For Sub-Frames 1,2,3,6,7,8 | Bits | | | 9912 | 19848 | | |
| For Sub-Frames 0,4,5,9 | Bits | | | n/a | n/a | | |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | 2 | 4 | | |
| Binary Channel Bits Per Subframe | | | | | | | |
| For Sub-Frames 1,2,3,6,7,8 | Bits | | | 15300 | 30600 | | |
| For Sub-Frames 0,4,5,9 | Bits | | | n/a | n/a | | |
| MBMS UE Category | | | | ≥ 1 | ≥ 2 | | |
| Note 1: For FDD mode, up to 6 subframes (#1/2/3/6/7/8) are available for MBMS, in line with TS 36.331.<br>Note 2: 2 OFDM symbols are reserved for PDCCH; and reference signal allocated as per TS 36.211.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit). | | | | | | | |

## A.3.8.2 TDD

**Table A.3.8.2-1: Fixed Reference Channel QPSK R=1/3**

| Parameter | PMCH |
|---|---|

| | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.40 TDD | | | R.37 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | | | 50 | | |
| Uplink-Downlink Configuration(Note 1) | | 5 | | | 5 | | |
| Allocated subframes per Radio Frame | | 5 | | | 5 | | |
| Modulation | | QPSK | | | QPSK | | |
| Target Coding Rate | | 1/3 | | | 1/3 | | |
| Information Bit Payload (Note 2) | | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | 408 | | | 3624 | | |
| For Sub-Frames 0,1,2,5,6 | Bits | n/a | | | n/a | | |
| Number of Code Blocks per Subframe (Note 3) | | 1 | | | 1 | | |
| Binary Channel Bits Per Subframe | | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | 1224 | | | 10200 | | |
| For Sub-Frames 0,1,2,5,6 | Bits | n/a | | | n/a | | |
| MBMS UE Category | | ≥ 1 | | | ≥ 1 | | |
| Note 1: For TDD mode, in line with TS 36.331, Uplink-Downlink Configuration 5 is proposed, up to 5 subframes (#3/4/7/8/9) are available for MBMS.<br>Note 2: 2 OFDM symbols are reserved for PDCCH; reference signal allocated as per TS 36.211.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit). | | | | | | | |

**Table A.3.8.2-2: Fixed Reference Channel 16QAM R=1/2**

| Parameter | PMCH | | | | | | |
|---|---|---|---|---|---|---|---|
| | Unit | Value | | | | | |
| Reference channel | | | | | R.38 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | | 50 | | |
| Uplink-Downlink Configuration(Note 1) | | | | | 5 | | |
| Allocated subframes per Radio Frame | | | | | 5 | | |
| Modulation | | | | | 16QAM | | |
| Target Coding Rate | | | | | 1/2 | | |
| Information Bit Payload (Note 2) | | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | | | | 9912 | | |
| For Sub-Frames 0,1,2,5,6 | Bits | | | | n/a | | |
| Number of Code Blocks per Subframe (Note 3) | | | | | 2 | | |
| Binary Channel Bits Per Subframe | | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | | | | 20400 | | |
| For Sub-Frames 0,1,2,5,6 | Bits | | | | n/a | | |
| MBMS UE Category | | | | | ≥ 1 | | |
| Note 1: For TDD mode, in line with TS 36.331, Uplink-Downlink Configuration 5 is proposed, up to 5 subframes (#3/4/7/8/9) are available for MBMS.<br>Note 2: 2 OFDM symbols are reserved for PDCCH; reference signal allocated as per TS 36.211.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit). | | | | | | | |

**Table A.3.8.2-3: Fixed Reference Channel 64QAM R=2/3**

| Parameter | PMCH |
|---|---|

| | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | | | R.39-1 TDD | R.39 TDD | | |
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | 25 | 50 | | |
| Uplink-Downlink Configuration(Note 1) | | | | 5 | 5 | | |
| Allocated subframes per Radio Frame | | | | 5 | 5 | | |
| Modulation | | | | 64QAM | 64QAM | | |
| Target Coding Rate | | | | 2/3 | 2/3 | | |
| Information Bit Payload (Note 2) | | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | | | 9912 | 19848 | | |
| For Sub-Frames 0,1,2,5,6 | Bits | | | n/a | n/a | | |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | 2 | 4 | | |
| Binary Channel Bits Per Subframe | | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | | | 15300 | 30600 | | |
| For Sub-Frames 0,1,2,5,6 | Bits | | | n/a | n/a | | |
| MBMS UE Category | | | | ≥ 1 | ≥ 2 | | |
| Note 1: For TDD mode, in line with TS 36.331, Uplink-Downlink Configuration 5 is proposed, up to 5 subframes (#3/4/7/8/9) are available for MBMS.<br>Note 2: 2 OFDM symbols are reserved for PDCCH; reference signal allocated as per TS 36.211.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit). | | | | | | | |

# A.3.9 Reference measurement channels for sustained downlink data rate provided by lower layers

## A.3.9.1 FDD

**Table A.3.9.1-1: Fixed Reference Channel for sustained data-rate test (FDD)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.31-1 FDD | R.31-2 FDD | R.31-3 FDD | R.31-3A FDD | R.31-4 FDD |
| Channel bandwidth | MHz | 10 | 10 | 20 | 10 | 20 |
| Allocated resource blocks | | Note 5 | Note 6 | Note 7 | Note 6 | Note 7 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Coding Rate | | | | | | |
| For Sub-Frame 1,2,3,4,6,7,8,9, | | 0.40 | 0.59 | 0.59 | 0.85 | 0.88 |
| For Sub-Frame 5 | | 0.40 | 0.64 | 0.62 | 0.89 | 0.87 |
| For Sub-Frame 0 | | 0.40 | 0.63 | 0.61 | 0.90 | 0.90 |
| Information Bit Payload | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 10296 | 25456 | 51024 | 36696 | 75376 |
| For Sub-Frame 5 | Bits | 10296 | 25456 | 51024 | 35160 | 71112 |
| For Sub-Frame 0 | Bits | 10296 | 25456 | 51024 | 36696 | 75376 |
| Number of Code Blocks per Sub-Frame (Note 3) | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 2 | 5 | 9 | 6 | 13 |
| For Sub-Frame 5 | | 2 | 5 | 9 | 6 | 12 |
| For Sub-Frame 0 | Bits | 2 | 5 | 9 | 6 | 13 |
| Binary Channel Bits Per Sub-Frame | | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 26100 | 43200 | 86400 | 43200 | 86400 |
| For Sub-Frame 5 | | 26100 | 39744 | 82080 | 39744 | 82080 |
| For Sub-Frame 0 | Bits | 26100 | 40752 | 83952 | 40752 | 83952 |
| Number of layers | | 1 | 2 | 2 | 2 | 2 |
| Max. Throughput averaged over 1 frame | Mbps | 10.296 | 25.456 | 51.024 | 36.542 | 74.950 |
| UE Category | | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 | ≥ 3 |

Note 1: 1 symbol allocated to PDCCH for all tests
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [4]
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
Note 4: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths
Note 5: Resource blocks $n_{PRB}$ = 6..14,30..49 are allocated for the user data in all sub-frames
Note 6: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9
Note 7: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9

### A.3.9.2 TDD

**Table A.3.9.2-1: Fixed Reference Channel for sustained data-rate test (TDD)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.31-1 TDD | R.31-2 TDD | R.31-3 TDD | R.31-3B TDD | R.31-4 TDD |
| Channel bandwidth | MHz | 10 | 10 | 20 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 | Note 8 |
| Uplink-Downlink Configuration (Note 3) | | 5 | 5 | 5 | 5 | 5 |
| Allocated subframes per Radio Frame (D+S) | | 8+1 | 8+1 | 8+1 | 8+1 | 8+1 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| Coding Rate | | | | | | |
| For Sub-Frames 3,4,7,8,9 | | 0.40 | 0.59 | 0.59 | 0.87 | 0.88 |
| For Sub-Frame 1 | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 5 | | 0.40 | 0.64 | 0.62 | 0.88 | 0.87 |
| For Sub-Frame 6 | | 0.40 | 0.60 | 0.60 | 0.88 | 0.88 |
| For Sub-Frame 0 | | 0.40 | 0.62 | 0.61 | 0.90 | 0.90 |
| Information Bit Payload | | | | | | |
| For Sub-Frames 3,4,6,7,8,9 | Bits | 10296 | 25456 | 51024 | [51024] | 75376 |
| For Sub-Frame 1 | Bits | 0 | 0 | 0 | 0 | 0 |
| For Sub-Frame 5 | Bits | 10296 | 25456 | 51024 | [51024] | 71112 |
| For Sub-Frame 0 | Bits | 10296 | 25456 | 51024 | [51024] | 75376 |
| Number of Code Blocks per Sub-Frame (Note 4) | | | | | | |
| For Sub-Frames 3,4,6,7,8,9 | | 2 | 5 | 9 | 9 | 13 |
| For Sub-Frame 1 | | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 5 | | 2 | 5 | 9 | 9 | 12 |
| For Sub-Frame 0 | | 2 | 5 | 9 | 9 | 13 |
| Binary Channel Bits Per Sub-Frame | | | | | | |
| For Sub-Frames 3,4,7,8,9 | Bits | 26100 | 43200 | 86400 | [58752] | 86400 |
| For Sub-Frame 1 | Bits | n/a | n/a | n/a | n/a | n/a |
| For Sub-Frame 5 | Bits | 26100 | 40176 | 82512 | [58320] | 82512 |
| For Sub-Frame 6 | Bits | 26100 | 42768 | 85968 | [58320] | 85968 |
| For Sub-Frame 0 | Bits | 26100 | 41184 | 84384 | [56736] | 84384 |
| Number of layers | | 1 | 2 | 2 | 2 | 2 |
| Max. Throughput averaged over 1 frame | Mbps | 8.237 | 20.365 | 40.819 | 40.819 | 59.874 |
| UE Category | | ≥ 1 | ≥ 2 | ≥ 2 | ≥ 2 | ≥ 3 |

Note 1: 1 symbol allocated to PDCCH for all tests
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [4]
Note 3: As per Table 4.2-2 in TS 36.211 [4]
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths
Note 6: Resource blocks $n_{PRB}$ = 6..14,30..49 are allocated for the user data in all subframes
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,6,7,8,9
Note 8: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,6,7,8,9
Note 9: Resource blocks $n_{PRB}$ = 4..71 are allocated for the user data in all sub-frames

## A.4 CQI reference measurement channels

This section defines the DL signal applicable to the reporting of channel quality information (Clause 9.2 and 9.3).

In Table A.4-0 are listed the UL/DL reference measurement channels specified in annex A.4 of this release of TS 36.521-1. This table is informative and serves only to a better overview. The reference for the concrete reference measurement channels and corresponding implementation's parameters as to be used for testing are the other tables of this annex as appropriate.

**Table A.4-0: Overview of CSI reference measurement channels**

| Duplex | Table | Name | BW | Mod | TCR | RB | RB Offset | UE Categ | Notes |
|---|---|---|---|---|---|---|---|---|---|
| **CSI Performance, PDSCH, Full allocation** | | | | | | | | | |
| FDD | Table A.4-1 | | 10 | CQI | CQI | 50 | | | |
| TDD | Table A.4-2 | | 10 | CQI | CQI | 50 | | | |
| **CSI Performance, PDSCH, Partial allocation (6 RB-s)** | | | | | | | | | |
| FDD | Table A.4-4 | | 10 | CQI | CQI | 6 | | | |
| TDD | Table A.4-5 | | 10 | CQI | CQI | 6 | | | |
| **CSI Performance, PDSCH, Partial allocation (15 RB-s)** | | | | | | | | | |
| FDD | Table A.4-7 | | 10 | CQI | CQI | 15 | | | |
| TDD | Table A.4-8 | | 10 | CQI | CQI | 15 | | | |
| **CSI Performance, PUSCH for PUCCH reporting mode** | | | | | | | | | |
| FDD | Table A.4-10 | | 10 | QPSK | 1/3 | 6 | | ≥ 1 | |
| TDD | Table A.4-11 | | 10 | QPSK | 1/3 | 6 | | ≥ 1 | |
| **CSI Performance, PDSCH, Partial allocation (3 RB-s)** | | | | | | | | | |
| FDD | Table A.4-12 | | 10 | CQI | CQI | 3 | | | |
| TDD | Table A.4-13 | | 10 | CQI | CQI | 3 | | | |

The reference channels in Table A.4-1, A.4-2, A.4-4 and A.4-5 comply with the CQI definition specified in Sec. 7.2.3 of TS 36.213 [10]. Table A.4-3 and A.4-6 specify the transport format corresponding to each CQI for single antenna transmission. Table A.4-3a specifies the transport format corresponding to each CQI for dual antenna transmission.

**Table A.4-1: Reference channel for CQI requirements (FDD) full PRB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 8 | 8 | 8 | 8 | 8 | 8 |
| Modulation | | | | | Table A.4-3 / Table A.4-3a | | |
| Target coding rate | | | | | Table A.4-3 / Table A.4-3a | | |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: Only subframes 1,2,3,4,6,7,8, and 9 are allocated to avoid PBCH and synchronization signal overhead | | | | | | | |

**Table A.4-2: Reference channel for CQI requirements (TDD) full PRB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 15 | 25 | 50 | 75 | 100 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 4 | 4 | 4 | 4 | 4 | 4 |
| Modulation | | | | | Table A.4-3 / Table A.4-3a | | |
| Target coding rate | | | | | Table A.4-3 / Table A.4-3a | | |
| Number of HARQ Processes | Processes | 10 | 10 | 10 | 10 | 10 | 10 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: UL-DL configuration 2 is used and only subframes 3, 4, 8, and 9 are allocated to avoid PBCH and synchronization signal overhead | | | | | | | |

**Table A.4-3: Transport format corresponding to each CQI index for 50 PRB allocation single antenna transmission**

| CQI index | Modulation | Target code rate | Imcs | Information Bit Payload (Subframes 1,2,3,4,6,7,8,9) | Binary Channel Bits Per Sub-Frame (Subframes 1,2,3,4,6,7,8,9) | Actual Code rate |
|---|---|---|---|---|---|---|
| 0 | out of range | out of range | DTX | - | - | - |
| 1 | QPSK | 0.0762 | 0 | 1384 | 12600 | 0.1117 |
| 2 | QPSK | 0.1172 | 0 | 1384 | 12600 | 0.1117 |
| 3 | QPSK | 0.1885 | 2 | 2216 | 12600 | 0.1778 |
| 4 | QPSK | 0.3008 | 4 | 3624 | 12600 | 0.2895 |
| 5 | QPSK | 0.4385 | 6 | 5160 | 12600 | 0.4114 |
| 6 | QPSK | 0.5879 | 8 | 6968 | 12600 | 0.5549 |
| 7 | 16QAM | 0.3691 | 11 | 8760 | 25200 | 0.3486 |
| 8 | 16QAM | 0.4785 | 13 | 11448 | 25200 | 0.4552 |
| 9 | 16QAM | 0.6016 | 16 | 15264 | 25200 | 0.6067 |
| 10 | 64QAM | 0.4551 | 18 | 16416 | 37800 | 0.4349 |
| 11 | 64QAM | 0.5537 | 21 | 21384 | 37800 | 0.5663 |
| 12 | 64QAM | 0.6504 | 23 | 25456 | 37800 | 0.6741 |
| 13 | 64QAM | 0.7539 | 25 | 28336 | 37800 | 0.7503 |
| 14 | 64QAM | 0.8525 | 27 | 31704 | 37800 | 0.8394 |
| 15 | 64QAM | 0.9258 | 27 | 31704 | 37800 | 0.8394 |
| Note1: Sub-frame#0 and #5 are not used for the corresponding requirement. | | | | | | |

**Table A.4-3a: Transport format corresponding to each CQI index for 50 PRB allocation dual antenna transmission**

| CQI index | Modulation | Target code rate | Imcs | Information Bit Payload (Subframes 1,2,3,4,6,7,8,9) | Binary Channel Bits Per Sub-Frame (Subframes 1,2,3,4,6,7,8,9) | Actual Code rate |
|---|---|---|---|---|---|---|
| 0 | out of range | out of range | DTX | - | - | - |
| 1 | QPSK | 0.0762 | 0 | 1384 | 12000 | 0.1173 |
| 2 | QPSK | 0.1172 | 0 | 1384 | 12000 | 0.1173 |
| 3 | QPSK | 0.1885 | 2 | 2216 | 12000 | 0.1867 |
| 4 | QPSK | 0.3008 | 4 | 3624 | 12000 | 0.3040 |
| 5 | QPSK | 0.4385 | 6 | 5160 | 12000 | 0.4320 |
| 6 | QPSK | 0.5879 | 8 | 6968 | 12000 | 0.5827 |
| 7 | 16QAM | 0.3691 | 11 | 8760 | 24000 | 0.3660 |
| 8 | 16QAM | 0.4785 | 13 | 11448 | 24000 | 0.4780 |
| 9 | 16QAM | 0.6016 | 15 | 14112 | 24000 | 0.5890 |
| 10 | 64QAM | 0.4551 | 18 | 16416 | 36000 | 0.4567 |
| 11 | 64QAM | 0.5537 | 20 | 19848 | 36000 | 0.5520 |
| 12 | 64QAM | 0.6504 | 22 | 22920 | 36000 | 0.6373 |
| 13 | 64QAM | 0.7539 | 24 | 27376 | 36000 | 0.7611 |
| 14 | 64QAM | 0.8525 | 26 | 30576 | 36000 | 0.8500 |
| 15 | 64QAM | 0.9258 | 27 | 31704 | 36000 | 0.8813 |
| Note1: Sub-frame#0 and #5 are not used for the corresponding requirement. The next subframe (i.e. sub-frame#1 or #6) shall be used for the retransmission. | | | | | | |

**Table A.4-4: Reference channel for CQI requirements (FDD) 6 PRB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 6 | 6 | 6 | 6 | 6 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 8 | 8 | 8 | 8 | 8 | 8 |
| Modulation | | | | | Table A.4-6 | | |
| Target coding rate | | | | | Table A.4-6 | | |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: Only subframes 1,2,3,4,6,7,8, and 9 are allocated to avoid PBCH and synchronization signal overhead | | | | | | | |

**Table A.4-5: Reference channel for CQI requirements (TDD) 6 PRB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 6 | 6 | 6 | 6 | 6 | 6 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 4 | 4 | 4 | 4 | 4 | 4 |
| Modulation | | | | | Table A.4-6 | | |
| Target coding rate | | | | | Table A.4-6 | | |
| Number of HARQ Processes | Processes | 10 | 10 | 10 | 10 | 10 | 10 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: UL-DL configuration 2 is used and only subframes 3, 4, 8, and 9 are allocated to avoid PBCH and synchronization signal overhead | | | | | | | |

**Table A.4-6: Transport format corresponding to each CQI index for 6 PRB allocation**

| CQI index | Modulation | Target code rate | Imcs | Information Bit Payload (Subframes 1,2,3,4,6,7,8,9) | Binary Channel Bits Per Sub-Frame (Subframes 1,2,3,4,6,7,8,9) | Actual Code rate |
|---|---|---|---|---|---|---|
| 0 | out of range | out of range | DTX | - | - | - |
| 1 | QPSK | 0.0762 | 0 | 152 | 1512 | 0.1005 |
| 2 | QPSK | 0.1172 | 0 | 152 | 1512 | 0.1005 |
| 3 | QPSK | 0.1885 | 2 | 256 | 1512 | 0.1693 |
| 4 | QPSK | 0.3008 | 4 | 408 | 1512 | 0.2698 |
| 5 | QPSK | 0.4385 | 6 | 600 | 1512 | 0.3968 |
| 6 | QPSK | 0.5879 | 8 | 808 | 1512 | 0.5344 |
| 7 | 16QAM | 0.3691 | 11 | 1032 | 3024 | 0.3413 |
| 8 | 16QAM | 0.4785 | 13 | 1352 | 3024 | 0.4471 |
| 9 | 16QAM | 0.6016 | 16 | 1800 | 3024 | 0.5952 |
| 10 | 64QAM | 0.4551 | 19 | 2152 | 4536 | 0.4744 |
| 11 | 64QAM | 0.5537 | 21 | 2600 | 4536 | 0.5732 |
| 12 | 64QAM | 0.6504 | 23 | 2984 | 4536 | 0.6578 |
| 13 | 64QAM | 0.7539 | 25 | 3496 | 4536 | 0.7707 |
| 14 | 64QAM | 0.8525 | 27 | 3752 | 4536 | 0.8272 |
| 15 | 64QAM | 0.9258 | 27 | 3752 | 4536 | 0.8272 |
| Note1: Sub-frame#0 and #5 are not used for the corresponding requirement. | | | | | | |

**Table A.4-7: Reference channel for CQI requirements (FDD) partial PRB allocation**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | 15 (Note 3) | | |
| Subcarriers per resource block | | | | 12 | | |
| Allocated subframes per Radio Frame | | | | 8 | | |
| Modulation | | Table A.4-9 | | | | |
| Target coding rate | | Table A.4-9 | | | | |
| Number of HARQ processes | | | | 8 | | |
| Maximum number of HARQ transmissions | | | | 1 | | |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: Only subframes 1,2,3,4,6,7,8, and 9 are allocated to avoid PBCH and synchronization signal overhead<br>Note 3: Centered within the Transmission Bandwidth Configuration (Figure 5.6-1) | | | | | | |

**Table A.4-8: Reference channel for CQI requirements (TDD) partial PRB allocation**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | | | 15 (Note 3) | | |
| Subcarriers per resource block | | | | 12 | | |
| Allocated subframes per Radio Frame | | | | 4 | | |
| Modulation | | Table A.4-9 | | | | |
| Target coding rate | | Table A.4-9 | | | | |
| Number of HARQ processes | | | | 10 | | |
| Maximum number of HARQ transmissions | | | | 1 | | |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: UL-DL configuration 2 is used and only subframes 3, 4, 8, and 9 are allocated to avoid PBCH and synchronization signal overhead<br>Note 3: Centered within the Transmission Bandwidth Configuration (Figure 5.6-1) | | | | | | |

**Table A.4-9: Transport format corresponding to each CQI index for 15 PRB allocation**

| CQI index | Modulation | Target code rate | Imcs | Information Bit Payload (Subframes 1,2,3,4,6,7,8,9) | Binary Channel Bits Per Sub-Frame (Subframes 1,2,3,4,6,7,8,9) | Actual Code rate |
|---|---|---|---|---|---|---|
| 0 | out of range | out of range | DTX | - | - | - |
| 1 | QPSK | 0.0762 | 0 | 392 | 3780 | 0.1037 |
| 2 | QPSK | 0.1172 | 0 | 392 | 3780 | 0.1037 |
| 3 | QPSK | 0.1885 | 2 | 648 | 3780 | 0.1714 |
| 4 | QPSK | 0.3008 | 4 | 1064 | 3780 | 0.2815 |
| 5 | QPSK | 0.4385 | 6 | 1544 | 3780 | 0.4085 |
| 6 | QPSK | 0.5879 | 8 | 2088 | 3780 | 0.5524 |
| 7 | 16QAM | 0.3691 | 11 | 2664 | 7560 | 0.3524 |
| 8 | 16QAM | 0.4785 | 13 | 3368 | 7560 | 0.4455 |
| 9 | 16QAM | 0.6016 | 16 | 4584 | 7560 | 0.6063 |
| 10 | 64QAM | 0.4551 | 19 | 4968 | 11340 | 0.4381 |
| 11 | 64QAM | 0.5537 | 21 | 6456 | 11340 | 0.5693 |
| 12 | 64QAM | 0.6504 | 23 | 7480 | 11340 | 0.6596 |
| 13 | 64QAM | 0.7539 | 25 | 8504 | 11340 | 0.7499 |
| 14 | 64QAM | 0.8525 | 27 | 9528 | 11340 | 0.8402 |
| 15 | 64QAM | 0.9258 | 27 | 9528 | 11340 | 0.8402 |
| Note1: Sub-frame#0 and #5 are not used for the corresponding requirement. | | | | | | |

**Table A.4-10: Uplink reference channels for transmitting CSI reports on PUSCH, when being in a PUCCH reporting mode (FDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 10 | 10 | 10 | | | |
| Allocated resource blocks | | 6 | 6 | 6 | | | |
| DFT-OFDM Symbols per Sub-Frame | | 12 | 12 | 12 | | | |
| Modulation | | QPSK | QPSK | QPSK | | | |
| Target Coding rate | | 1/3 | 1/3 | 1/3 | | | |
| Allocated Sub-Frames (Note 1) | | 1, 3, 5, 7 | 3, 8 | 0, 1, 2, 3, 4, 5, 6, 7, 8,9 | | | |
| Payload size | Bits | 600 | 600 | 600 | | | |
| Transport block CRC | Bits | 24 | 24 | 24 | | | |
| Number of code blocks per Sub-Frame (Note 2) | | 1 | 1 | 1 | | | |
| Total number of bits per Sub-Frame | Bits | 1728 | 1728 | 1728 | | | |
| Total symbols per Sub-Frame | | 864 | 864 | 864 | | | |
| UE Category | | ≥ 1 | ≥ 1 | ≥ 1 | | | |
| Note 1: The remaining subframes are not allocated with data. All the allocation details specified in the reference channel are valid only for the allocated subframes.<br>Note 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.4-11: Uplink reference channels for transmitting CSI reports on PUSCH, when being in a PUCCH reporting mode (TDD)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 10 | | | | | |
| Allocated resource blocks | | 6 | | | | | |
| Uplink-Downlink Configuration (Note 1) | | 2 | | | | | |
| DFT-OFDM Symbols per Sub-Frame | | 12 | | | | | |
| Modulation | | QPSK | | | | | |
| Target Coding rate | | 1/3 | | | | | |
| Allocated Sub-Frames (Note 2) | | 2, 7 | | | | | |
| Payload size | Bits | 600 | | | | | |
| Transport block CRC | Bits | 24 | | | | | |
| Number of code blocks per Sub-Frame (Note 3) | | 1 | | | | | |
| Total number of bits per Sub-Frame | Bits | 1728 | | | | | |
| Total symbols per Sub-Frame | | 864 | | | | | |
| UE Category | | ≥ 1 | | | | | |
| Note 1: As per Table 4.2-2 in TS 36.211 [4]<br>Note 2: The remaining subframes are not allocated with data. All the allocation details specified in the reference channel are valid only for the allocated subframes.<br>Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) | | | | | | | |

**Table A.4-12: Reference channel for CQI requirements (FDD) 3 PRB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 3 | 3 | 3 | 3 | 3 | 3 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 8 | 8 | 8 | 8 | 8 | 8 |
| Modulation | | | | | Table A.4-12 | | |
| Target coding rate | | | | | Table A.4-12 | | |
| Number of HARQ Processes | Processes | 8 | 8 | 8 | 8 | 8 | 8 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: Only subframes 1,2,3,4,6,7,8, and 9 are allocated to avoid PBCH and synchronization signal overhead | | | | | | | |

**Table A.4-13: Reference channel for CQI requirements (TDD) 3 PRB allocation**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Channel bandwidth | MHz | 1.4 | 3 | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | 3 | 3 | 3 | 3 | 3 | 3 |
| Subcarriers per resource block | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated subframes per Radio Frame | | 4 | 4 | 4 | 4 | 4 | 4 |
| Modulation | | | | | Table A.4-12 | | |
| Target coding rate | | | | | Table A.4-12 | | |
| Number of HARQ Processes | Processes | 10 | 10 | 10 | 10 | 10 | 10 |
| Maximum number of HARQ transmissions | | 1 | 1 | 1 | 1 | 1 | 1 |
| Note 1: 3 symbols allocated to PDCCH.<br>Note 2: UL-DL configuration 2 is used and only subframes 3, 4, 8, and 9 are allocated to avoid PBCH and synchronization signal overhead. | | | | | | | |

**Table A.4-14: Transport format corresponding to each CQI index for 3 PRB allocation**

| CQI index | Modulation | Target code rate | Imcs | Information Bit Payload (Subframes 1,2,3,4,6,7,8,9) | Binary Channel Bits Per Sub-Frame (Subframes 1,2,3,4,6,7,8,9) | Actual Code rate |
|---|---|---|---|---|---|---|
| 0 | out of range | out of range | DTX | - | - | - |
| 1 | QPSK | 0.0762 | 0 | 56 | 756 | 0.0741 |
| 2 | QPSK | 0.1172 | 1 | 88 | 756 | 0.1164 |
| 3 | QPSK | 0.1885 | 2 | 144 | 756 | 0.1905 |
| 4 | QPSK | 0.3008 | 5 | 224 | 756 | 0.2963 |
| 5 | QPSK | 0.4385 | 7 | 328 | 756 | 0.4339 |
| 6 | QPSK | 0.5879 | 9 | 456 | 756 | 0.6032 |
| 7 | 16QAM | 0.3691 | 12 | 584 | 1512 | 0.3862 |
| 8 | 16QAM | 0.4785 | 13 | 744 | 1512 | 0.4921 |
| 9 | 16QAM | 0.6016 | 16 | 904 | 1512 | 0.5979 |
| 10 | 64QAM | 0.4551 | 19 | 1064 | 2268 | 0.4691 |
| 11 | 64QAM | 0.5537 | 21 | 1288 | 2268 | 0.5679 |
| 12 | 64QAM | 0.6504 | 23 | 1480 | 2268 | 0.6526 |
| 13 | 64QAM | 0.7539 | 25 | 1736 | 2268 | 0.7654 |
| 14 | 64QAM | 0.8525 | 27 | 1864 | 2268 | 0.8219 |
| 15 | 64QAM | 0.9258 | 27 | 1864 | 2268 | 0.8219 |
| Note1: Sub-frame#0 and #5 are not used for the corresponding requirement. | | | | | | |

# A.5 OFDMA Channel Noise Generator (OCNG)

## A.5.1 OCNG Patterns for FDD

The following OCNG patterns are used for modelling allocations to virtual UEs (which are not under test). The OCNG pattern for each sub frame specifies the allocations that shall be filled with OCNG, and furthermore, the relative power level of each such allocation.

In each test case the OCNG is expressed by parameters OCNG_RA and OCNG_RB which together with a relative power level ( $\gamma$ ) specifies the PDSCH EPRE-to-RS EPRE ratios in OFDM symbols with and without reference symbols, respectively. The relative power, which is used for modelling boosting per virtual UE allocation, is expressed by:

$$\gamma_i = PDSCH_i_RA / OCNG_RA = PDSCH_i_RB / OCNG_RB,$$

where $\gamma_i$ denotes the relative power level of the *i:th* virtual UE. The parameter settings of OCNG_RA, OCNG_RB, and the set of relative power levels $\gamma$ are chosen such that when also taking allocations to the UE under test into account, as given by a PDSCH reference channel, a transmitted power spectral density that is constant on an OFDM symbol basis is targeted.

Moreover the OCNG pattern is accompanied by a PCFICH/PDCCH/PHICH reference channel which specifies the control region. For any aggregation and PHICH allocation, the PDCCH and any unused PHICH groups are padded with resource element groups with a power level given respectively by PDCCH_RA/RB and PHICH_RA/RB as specified in the test case such that a total power spectral density in the control region that is constant on an OFDM symbol basis is targeted.

### A.5.1.1 OCNG FDD pattern 1: One sided dynamic OCNG FDD pattern

This OCNG Pattern fills with OCNG all empty PRB-s (PRB-s with no allocation of data or system information) of the DL sub-frames, when the unallocated area is continuous in frequency domain (one sided).

**Table A.5.1.1-1: OP.1 FDD: One sided dynamic OCNG FDD Pattern**

| **Relative power level $\gamma_{PRB}$ [dB]** | | | **PDSCH Data** |
|---|---|---|---|
| **Subframe** | | | |
| 0 | 5 | 1 – 4, 6 – 9 | |
| **Allocation** | | | |
| First unallocated PRB – Last unallocated PRB | First unallocated PRB – Last unallocated PRB | First unallocated PRB – Last unallocated PRB | |
| 0 | 0 | 0 | Note 1 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS according to transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## A.5.1.2 OCNG FDD pattern 2: Two sided dynamic OCNG FDD pattern

This OCNG Pattern fills with OCNG all empty PRB-s (PRB-s with no allocation of data or system information) of the DL sub-frames, when the unallocated area is discontinuous in frequency domain (divided in two parts by the allocated area – two sided), starts with PRB 0 and ends with PRB $N_{RB}-1$.

**Table A.5.1.2-1: OP.2 FDD: Two sided dynamic OCNG FDD Pattern**

| Relative power level $\gamma_{PRB}$ [dB] | | | PDSCH Data |
|---|---|---|---|
| Subframe | | | |
| 0 | 5 | 1 – 4, 6 – 9 | |
| Allocation | | | |
| 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ($N_{RB}-1$) | 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ($N_{RB}-1$) | 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ($N_{RB}-1$) | |
| 0 | 0 | 0 | Note 1 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS according to transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## A.5.1.3 OCNG FDD pattern 3: 49 RB OCNG allocation with MBSFN in 10 MHz

**Table A.5.1.3-1: OP.3 FDD: OCNG FDD Pattern 3**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data | PMCH Data |
|---|---|---|---|---|---|---|
| | Subframe | | | | | |
| | 0 | 5 | 4, 9 | 1 – 3, 6 – 8 | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 – 49 | 0 | 0 (Allocation: all empty PRB-s) | 0 | N/A | Note 1 | N/A |
| 0 – 49 | N/A | N/A | N/A | 0 | N/A | Note 2 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH subframes shall contain cell-specific Reference Signals only in the first symbol of the first time slot. The parameter $\gamma_{PRB}$ is used to scale the power of PMCH.

Note 3: If two or more transmit antennas are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas according to transmission mode 2. The transmit power shall be equally split between all the transmit antennas used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

## A.5.2 OCNG Patterns for TDD

The following OCNG patterns are used for modelling allocations to virtual UEs (which are not under test). The OCNG pattern for each sub frame specifies the allocations that shall be filled with OCNG, and furthermore, the relative power level of each such allocation.

In each test case the OCNG is expressed by parameters OCNG_RA and OCNG_RB which together with a relative power level ( $\gamma$ ) specifies the PDSCH EPRE-to-RS EPRE ratios in OFDM symbols with and without reference symbols, respectively. The relative power, which is used for modelling boosting per virtual UE allocation, is expressed by:

$$\gamma_i = PDSCH_i_RA / OCNG_RA = PDSCH_i_RB / OCNG_RB,$$

where $\gamma_i$ denotes the relative power level of the *i:th* virtual UE. The parameter settings of OCNG_RA, OCNG_RB, and the set of relative power levels $\gamma$ are chosen such that when also taking allocations to the UE under test into account, as given by a PDSCH reference channel, a transmitted power spectral density that is constant on an OFDM symbol basis is targeted.

Moreover the OCNG pattern is accompanied by a PCFICH/PDCCH/PHICH reference channel which specifies the control region. For any aggregation and PHICH allocation, the PDCCH and any unused PHICH groups are padded with resource element groups with a power level given respectively by PDCCH_RA/RB and PHICH_RA/RB as specified in the test case such that a total power spectral density in the control region that is constant on an OFDM symbol basis is targeted.

### A.5.2.1 OCNG TDD pattern 1: One sided dynamic OCNG TDD pattern

This OCNG Pattern fills with OCNG all empty PRB-s (PRB-s with no allocation of data or system information) of the subframes available for DL transmission (depending on TDD UL/DL configuration), when the unallocated area is continuous in frequency domain (one sided).

**Table A.5.2.1-1: OP.1 TDD: One sided dynamic OCNG TDD Pattern**

| **Relative power level $\gamma_{PRB}$ [dB]** | | | | **PDSCH Data** |
|---|---|---|---|---|
| **Subframe (only if available for DL)** | | | | |
| 0 | 5 | 3, 4, 7, 8, 9 and 6 (as normal subframe) Note 2 | 1 and 6 (as special subframe) Note 2 | |
| **Allocation** | | | | |
| First unallocated PRB – Last unallocated PRB | First unallocated PRB – Last unallocated PRB | First unallocated PRB – Last unallocated PRB | First unallocated PRB – Last unallocated PRB | |
| 0 | 0 | 0 | 0 | **Note 1** |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211.

Note 3: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS according to transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## A.5.2.2 OCNG TDD pattern 2: Two sided dynamic OCNG TDD pattern

This OCNG Pattern fills with OCNG all empty PRB-s (PRB-s with no allocation of data or system information) of the subframes available for DL transmission (depending on TDD UL/DL configuration), when the unallocated area is discontinuous in frequency domain (divided in two parts by the allocated area – two sided), starts with PRB 0 and ends with PRB $N_{RB}-1$.

**Table A.5.2.2-1: OP.2 TDD: Two sided dynamic OCNG TDD Pattern**

| Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data |
|---|---|---|---|---|
| Subframe (only if available for DL) | | | | |
| 0 | 5 | 3, 4, 6, 7, 8, 9 (6 as normal subframe) Note 2 | 1,6 (6 as special subframe) Note 2 | |
| Allocation | | | | |
| 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ( $N_{RB}-1$) | 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ( $N_{RB}-1$) | 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ( $N_{RB}-1$) | 0 – (First allocated PRB-1) and (Last allocated PRB+1) – ( $N_{RB}-1$) | |
| [0] | [0] | [0] | [0] | Note 1 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211.

Note 3: If two or more transmit antennas with CRS are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas with CRS according to transmission mode 2. The parameter $\gamma_{PRB}$ applies to each antenna port separately, so the transmit power is equal between all the transmit antennas with CRS used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

## A.5.2.3 OCNG TDD pattern 3: 49 RB OCNG allocation with MBSFN in 10 MHz

**Table A.5.2. 3-1: OP.3 TDD: OCNG TDD Pattern 3 for 5ms downlink-to-uplink switch-point periodicity**

| Allocation $n_{PRB}$ | Relative power level $\gamma_{PRB}$ [dB] | | | | PDSCH Data | PMCH Data |
|---|---|---|---|---|---|---|
| | Subframe | | | | | |
| | 0 | 5 | 4, 9 Note 2 | 1, 6 | | |
| 1 – 49 | 0 | 0 (Allocation: all empty PRB-s) | N/A | 0 | Note 1 | N/A |
| 0 – 49 | N/A | N/A | 0 | N/A | N/A | Note 3 |

Note 1: These physical resource blocks are assigned to an arbitrary number of virtual UEs with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. The parameter $\gamma_{PRB}$ is used to scale the power of PDSCH.

Note 2: Subframes available for DL transmission depends on the Uplink-Downlink configuration in Table 4.2-2 in 3GPP TS 36.211.

Note 3: Each physical resource block (PRB) is assigned to MBSFN transmission. The data in each PRB shall be uncorrelated with data in other PRBs over the period of any measurement. The MBSFN data shall be QPSK modulated. PMCH symbols shall not contain cell-specific Reference Signals

Note 4: If two or more transmit antennas are used in the test, the OCNG shall be transmitted to the virtual users by all the transmit antennas according to transmission mode 2. The transmit power shall be equally split between all the transmit antennas used in the test. The antenna transmission modes are specified in section 7.1 in 3GPP TS 36.213.

N/A: Not Applicable

# Annex B (normative): Propagation Conditions

The propagation conditions and channel models for various environments are specified. For each environment a propagation model is used to evaluate the propagation pathless due to the distance. Channel models are formed by combining delay profiles with a Doppler spectrum, with the addition of correlation properties in the case of a multi-antenna scenario.

## B.0 No interference

The downlink connection between the System Simulator and the UE is without Additive White Gaussian Noise, and has no fading or multipath effects.

## B.1 Static propagation condition

The downlink connection between the System Simulator and the UE is an Additive White Gaussian Noise (AWGN) environment with no fading or multipath effects.

For 2 port transmission the channel matrix is defined in the frequency domain by

$$\mathbf{H} = \begin{pmatrix} 1 & j \\ 1 & -j \end{pmatrix}$$

### B.1.1 Definition of Additive White Gaussian Noise (AWGN) Interferer

Note that the AWGN interferer can be used in static propagation conditions, or in conjunction with multi-path fading.

The acceptable uncertainties of the AWGN interferer are defined in Annex F.

## B.2 Multi-path fading Propagation Conditions

The multipath propagation conditions consist of several parts:

- A delay profile in the form of a "tapped delay-line", characterized by a number of taps at fixed positions on a sampling grid. The profile can be further characterized by the r.m.s. delay spread and the maximum delay spanned by the taps.
- A combination of channel model parameters that include the Delay profile and the Doppler spectrum, that is characterized by a classical spectrum shape and a maximum Doppler frequency
- A set of correlation matrices defining the correlation between the UE and eNodeB antennas in case of multi-antenna systems.

### B.2.1 Delay profiles

The delay profiles are selected to be representative of low, medium and high delay spread environments. The resulting model parameters are defined in Table B.2.1-1 and the tapped delay line models are defined in Tables B.2.1-2, B.2.1-3 and B.2.1-4.

**Table B.2.1-1: Delay profiles for E-UTRA channel models**

| Model | Number of channel taps | Delay spread (r.m.s.) | Maximum excess tap delay (span) |
|---|---|---|---|
| Extended Pedestrian A (EPA) | 7 | 45 ns | 410 ns |
| Extended Vehicular A model (EVA) | 9 | 357 ns | 2510 ns |
| Extended Typical Urban model (ETU) | 9 | 991 ns | 5000 ns |

**Table B.2.1-2: Extended Pedestrian A model (EPA)**

| Excess tap delay [ns] | Relative power [dB] |
|---|---|
| 0 | 0.0 |
| 30 | -1.0 |
| 70 | -2.0 |
| 90 | -3.0 |
| 110 | -8.0 |
| 190 | -17.2 |
| 410 | -20.8 |

**Table B.2.1-3: Extended Vehicular A model (EVA)**

| Excess tap delay [ns] | Relative power [dB] |
|---|---|
| 0 | 0.0 |
| 30 | -1.5 |
| 150 | -1.4 |
| 310 | -3.6 |
| 370 | -0.6 |
| 710 | -9.1 |
| 1090 | -7.0 |
| 1730 | -12.0 |
| 2510 | -16.9 |

**Table B.2.1-4: Extended Typical Urban model (ETU)**

| Excess tap delay [ns] | Relative power [dB] |
|---|---|
| 0 | -1.0 |
| 50 | -1.0 |
| 120 | -1.0 |
| 200 | 0.0 |
| 230 | 0.0 |
| 500 | 0.0 |
| 1600 | -3.0 |
| 2300 | -5.0 |
| 5000 | -7.0 |

## B.2.2 Combinations of channel model parameters

Table B.2.2-1 shows propagation conditions that are used for the performance measurements in multi-path fading environment for low, medium and high Doppler frequencies.

**Table B.2.2-1: Channel model parameters**

| Model | Maximum Doppler frequency |
|---|---|
| EPA 5Hz | 5 Hz |
| EVA 5Hz | 5 Hz |
| EVA 70Hz | 70 Hz |
| ETU 70Hz | 70 Hz |
| ETU 300Hz | 300 Hz |

## B.2.3 MIMO Channel Correlation Matrices

### B.2.3.1 Definition of MIMO Correlation Matrices

Table B.2.3.1-1 defines the correlation matrix for the eNodeB

**Table B.2.3.1-1: eNodeB correlation matrix**

| | One antenna | Two antennas | Four antennas |
|---|---|---|---|
| eNode B Correlation | $R_{eNB} = 1$ | $R_{eNB} = \begin{pmatrix} 1 & \alpha \\ \alpha^* & \mathbf{1} \end{pmatrix}$ | $R_{eNB} = \begin{pmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{pmatrix}$ |

Table B.2.3.1-2 defines the correlation matrix for the UE:

**Table B.2.3.1-2: UE correlation matrix**

| | One antenna | Two antennas | Four antennas |
|---|---|---|---|
| UE Correlation | $R_{UE} = 1$ | $R_{UE} = \begin{pmatrix} 1 & \beta \\ \beta^* & \mathbf{1} \end{pmatrix}$ | $R_{UE} = \begin{pmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^* & \beta^{4/9*} & \beta^{1/9*} & 1 \end{pmatrix}$ |

Table B.2.3.1-3 defines the channel spatial correlation matrix $R_{spat}$. The parameters, α and β in Table B.2.3.1-3 defines the spatial correlation between the antennas at the eNodeB and UE.

**Table B.2.3.1-3: $R_{spat}$ correlation matrices**

| | |
|---|---|
| 1x2 case | $R_{spat} = R_{UE} = \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix}$ |
| 2x2 case | $R_{spat} = R_{eNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha \\ \alpha^* & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix} = \begin{bmatrix} 1 & \beta & \alpha & \alpha\beta \\ \beta^* & 1 & \alpha\beta^* & \alpha \\ \alpha^* & \alpha^*\beta & 1 & \beta \\ \alpha^*\beta^* & \alpha^* & \beta^* & 1 \end{bmatrix}$ |
| 4x2 case | $R_{spat} = R_{eNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix}$ |
| 4x4 case | $R_{spat} = R_{eNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{bmatrix} \otimes \begin{pmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^* & \beta^{4/9*} & \beta^{1/9*} & 1 \end{pmatrix}$ |

For cases with more antennas at either eNodeB or UE or both, the channel spatial correlation matrix can still be expressed as the Kronecker product of $R_{eNB}$ and $R_{UE}$ according to $R_{spat} = R_{eNB} \otimes R_{UE}$.

## B.2.3.2 MIMO Correlation Matrices at High, Medium and Low Level

The $\alpha$ and $\beta$ for different correlation types are given in Table B.2.3.2-1.

**Table B.2.3.2-1**

| Low correlation | | Medium Correlation | | High Correlation | |
|---|---|---|---|---|---|
| α | β | α | β | α | β |
| 0 | 0 | 0.3 | 0.9 | 0.9 | 0.9 |

The correlation matrices for high, medium and low correlation are defined in Table B.2.3.2-2, B.2.3.2-3 and B.2.3.2-4,as below.

The values in the Table B.2.3.2-2 table have been adjusted for the 4x2 and 4x4 high correlation cases to insure the correlation matrix is positive semi-definite after round-off to 4 digit precision. This is done using the equation:

$$\mathbf{R}_{high} = [\mathbf{R}_{spatial} + aI_n]/(1+a)$$

Where the value "a" is a scaling factor such that the smallest value is used to obtain a positive semi-definite result. For the 4x2 high correlation case, a=0.00010. For the 4x4 high correlation case, a=0.00012.

The same method is used to adjust the 4x4 medium correlation matrix in Table B.2.3.2-3 to insure the correlation matrix is positive semi-definite after round-off to 4 digit precision with a = 0.00012.

**Table B.2.3.2-2: MIMO correlation matrices for high correlation**

| | |
|---|---|
| 1x2 case | $R_{high} = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$ |
| 2x2 case | $R_{high} = \begin{pmatrix} 1 & 0.9 & 0.9 & 0.81 \\ 0.9 & 1 & 0.81 & 0.9 \\ 0.9 & 0.81 & 1 & 0.9 \\ 0.81 & 0.9 & 0.9 & 1 \end{pmatrix}$ |
| 4x2 case | $R_{high} = \begin{bmatrix} 1.0000 & 0.8999 & 0.9883 & 0.8894 & 0.9542 & 0.8587 & 0.8999 & 0.8099 \\ 0.8999 & 1.0000 & 0.8894 & 0.9883 & 0.8587 & 0.9542 & 0.8099 & 0.8999 \\ 0.9883 & 0.8894 & 1.0000 & 0.8999 & 0.9883 & 0.8894 & 0.9542 & 0.8587 \\ 0.8894 & 0.9883 & 0.8999 & 1.0000 & 0.8894 & 0.9883 & 0.8587 & 0.9542 \\ 0.9542 & 0.8587 & 0.9883 & 0.8894 & 1.0000 & 0.8999 & 0.9883 & 0.8894 \\ 0.8587 & 0.9542 & 0.8894 & 0.9883 & 0.8999 & 1.0000 & 0.8894 & 0.9883 \\ 0.8999 & 0.8099 & 0.9542 & 0.8587 & 0.9883 & 0.8894 & 1.0000 & 0.8999 \\ 0.8099 & 0.8999 & 0.8587 & 0.9542 & 0.8894 & 0.9883 & 0.8999 & 1.0000 \end{bmatrix}$ |
| 4x4 case | $R_{high} = \begin{bmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.8999 & 0.8894 & 0.8587 & 0.8099 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.8894 & 0.8999 & 0.8894 & 0.8587 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.8587 & 0.8894 & 0.8999 & 0.8894 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8099 & 0.8587 & 0.8894 & 0.8999 \\ 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 0.9541 & 0.9430 & 0.9105 & 0.8587 \\ 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9430 & 0.9541 & 0.9430 & 0.9105 \\ 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9105 & 0.9430 & 0.9541 & 0.9430 \\ 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8587 & 0.9105 & 0.9430 & 0.9541 \\ 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 \\ 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 \\ 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 \\ 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 \\ 0.8999 & 0.8894 & 0.8587 & 0.8099 & 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.8894 & 0.8999 & 0.8894 & 0.8587 & 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.8587 & 0.8894 & 0.8999 & 0.8894 & 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.8099 & 0.8587 & 0.8894 & 0.8999 & 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{bmatrix}$ |

**Table B.2.3.2-3: MIMO correlation matrices for medium correlation**

| | |
|---|---|
| 1x2 case | N/A |
| 2x2 case | $R_{medium} = \begin{pmatrix} 1 & 0.9 & 0.3 & 0.27 \\ 0.9 & 1 & 0.27 & 0.3 \\ 0.3 & 0.27 & 1 & 0.9 \\ 0.27 & 0.3 & 0.9 & 1 \end{pmatrix}$ |
| 4x2 case | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9000 & 0.8748 & 0.7873 & 0.5856 & 0.5271 & 0.3000 & 0.2700 \\ 0.9000 & 1.0000 & 0.7873 & 0.8748 & 0.5271 & 0.5856 & 0.2700 & 0.3000 \\ 0.8748 & 0.7873 & 1.0000 & 0.9000 & 0.8748 & 0.7873 & 0.5856 & 0.5271 \\ 0.7873 & 0.8748 & 0.9000 & 1.0000 & 0.7873 & 0.8748 & 0.5271 & 0.5856 \\ 0.5856 & 0.5271 & 0.8748 & 0.7873 & 1.0000 & 0.9000 & 0.8748 & 0.7873 \\ 0.5271 & 0.5856 & 0.7873 & 0.8748 & 0.9000 & 1.0000 & 0.7873 & 0.8748 \\ 0.3000 & 0.2700 & 0.5856 & 0.5271 & 0.8748 & 0.7873 & 1.0000 & 0.9000 \\ 0.2700 & 0.3000 & 0.5271 & 0.5856 & 0.7873 & 0.8748 & 0.9000 & 1.0000 \end{pmatrix}$ |
| 4x4 case | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.3000 & 0.2965 & 0.2862 & 0.2700 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.2965 & 0.3000 & 0.2965 & 0.2862 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.2862 & 0.2965 & 0.3000 & 0.2965 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.2700 & 0.2862 & 0.2965 & 0.3000 \\ 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 0.5855 & 0.5787 & 0.5588 & 0.5270 \\ 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.5787 & 0.5855 & 0.5787 & 0.5588 \\ 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.5588 & 0.5787 & 0.5855 & 0.5787 \\ 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.5270 & 0.5588 & 0.5787 & 0.5855 \\ 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 \\ 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 \\ 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 \\ 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 \\ 0.3000 & 0.2965 & 0.2862 & 0.2700 & 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.2965 & 0.3000 & 0.2965 & 0.2862 & 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.2862 & 0.2965 & 0.3000 & 0.2965 & 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.2700 & 0.2862 & 0.2965 & 0.3000 & 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{pmatrix}$ |

**Table B.2.3.2-4: MIMO correlation matrices for low correlation**

| | |
|---|---|
| 1x2 case | $R_{low} = \mathbf{I}_2$ |
| 2x2 case | $R_{low} = \mathbf{I}_4$ |
| 4x2 case | $R_{low} = \mathbf{I}_8$ |
| 4x4 case | $R_{low} = \mathbf{I}_{16}$ |

In Table B.2.3.2-4, $\mathbf{I}_d$ is the $d \times d$ identity matrix.

## B.2.4 Propagation conditions for CQI tests

[For Channel Quality Indication (CQI) tests, the following additional multi-path profile is used:

$$h(t,\tau) = \delta(\tau) + a\exp(-i2\pi f_D t)\delta(\tau - \tau_d)$$

in continuous time $(t,\tau)$ representation, with $\tau_d$ the delay, $a$ a constant and $f_D$ the Doppler frequency.]

## B.2.5 FFS

## B.2.6 MBSFN Propagation Channel Profile

Table B.2.6-1 shows propagation conditions that are used for the MBSFN performance requirements in multi-path fading environment in an extended delay spread environment.

**Table B.2.6-1: Propagation Conditions for Multi-Path Fading Environments for MBSFN Performance Requirements in an extended delay spread environment**

| Extended Delay Spread | |
|---|---|
| Maximum Doppler frequency [5Hz] | |
| Relative Delay [ns] | Relative Mean Power [dB] |
| 0 | 0 |
| 30 | -1.5 |
| 150 | -1.4 |
| 310 | -3.6 |
| 370 | -0.6 |
| 1090 | -7.0 |
| 12490 | -10 |
| 12520 | -11.5 |
| 12640 | -11.4 |
| 12800 | -13.6 |
| 12860 | -10.6 |
| 13580 | -17.0 |
| 27490 | -20 |
| 27520 | -21.5 |
| 27640 | -21.4 |
| 27800 | -23.6 |
| 27860 | -20.6 |
| 28580 | -27.0 |

# B.3 High speed train scenario

The high speed train condition for the test of the baseband performance is a non fading propagation channel with one tap. Doppler shift is given by

$$f_s(t) = f_d \cos\theta(t) \quad \text{(B.3.1)}$$

where $f_s(t)$ is the Doppler shift and $f_d$ is the maximum Doppler frequency. The cosine of angle $\theta(t)$ is given by

$$\cos\theta(t) = \frac{D_s/2 - vt}{\sqrt{D_{\min}{}^2 + (D_s/2 - vt)^2}},\ 0 \le t \le D_s/v \quad \text{(B.3.2)}$$

$$\cos\theta(t) = \frac{-1.5D_s + vt}{\sqrt{D_{\min}{}^2 + (-1.5D_s + vt)^2}}, \quad D_s/v < t \le 2D_s/v \tag{B.3.3}$$

$$\cos\theta(t) = \cos\theta(t \mod (2D_s/v)), \quad t > 2D_s/v \tag{B.3.4}$$

where $D_s/2$ is the initial distance of the train form eNodeB, and $D_{\min}$ is eNodeB Railway track distance, both in meters; $v$ is the velocity of the train in m/s, $t$ is time in seconds.

Doppler shift and cosine angle are given by equation B.3.1 and B.3.2-B.3.4 respectively, where the required input parameters listed in table B.3-1 and the resulting Doppler shift is shown in Figure B.3-1 are applied for all frequency bands.

**Table B.3-1: High speed train scenario**

| Parameter | Value |
|---|---|
| $D_s$ | 300 m |
| $D_{\min}$ | 2 m |
| $v$ | 300 km/h |
| $f_d$ | 750 Hz |

NOTE 1: Parameters for HST conditions in table B.3-1 including $f_d$ and Doppler shift trajectories presented on figure B.3-1 were derived for Band7.

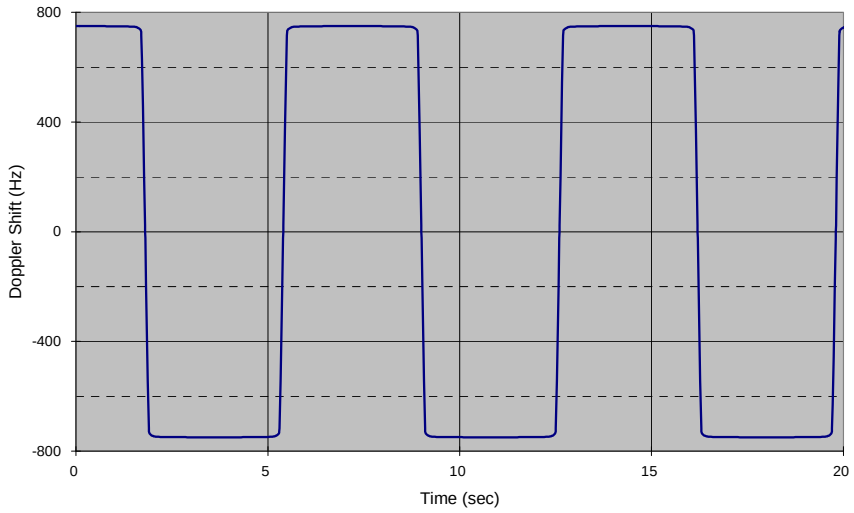



**Figure B.3-1: Doppler shift trajectory**

# B.4 Beamforming Model

## B.4.1 Single-layer beamforming (Antenna port 5, 7 or 8)

Single-layer transmission on antenna port 5 or on antenna port 7 or 8 without a simultaneous transmission on the other antenna port, is defined by using a precoder vector $W(i)$ of size $2\times1$ randomly selected with the number of layers $\upsilon = 1$ from Table 6.3.4.2.3-1 in TS 36.211 [8] as beamforming weights. This precoder takes as an input the

signal $y^{(p)}(i)$, $i = 0,1,...,M_{\text{symb}}^{\text{ap}} - 1$, for antenna port $p \in \{5, 7, 8\}$, with $M_{\text{symb}}^{\text{ap}}$ the number of modulation symbols including the user-specific reference symbols (DRS), and generates a block of signals $y_{bf}(i) = \left[y_{bf}(i) \quad \tilde{y}_{bf}(i)\right]^T$ the elements of which are to be mapped onto the same physical RE but transmitted on different antenna elements:

$$\begin{bmatrix} y_{bf}(i) \\ \tilde{y}_{bf}(i) \end{bmatrix} = W(i) y^{(p)}(i)$$

Single-layer transmission on antenna port 7 or 8 with a simultaneous transmission on the other antenna port, is defined by using a pair of precoder vectors $W_1(i)$ and $W_2(i)$ each of size $2 \times 1$, which are not identical and randomly selected with the number of layers $\upsilon = 1$ from Table 6.3.4.2.3-1 in TS 36.211 [8], as beamforming weights, and normalizing the transmit power as follows:

$$\begin{bmatrix} y_{bf}(i) \\ \tilde{y}_{bf}(i) \end{bmatrix} = \frac{1}{\sqrt{2}} \left( W_1(i) y^{(7)}(i) + W_2(i) y^{(8)}(i) \right)$$

The precoder update granularity is according to Table 8.3.2-1.

## B.4.2 Dual-layer beamforming (antenna ports 7 and 8)

Dual-layer transmission on antenna ports 7 and 8 is defined by using a precoder matrix $W(i)$ of size $2 \times 2$ randomly selected with the number of layers $\upsilon = 2$ from Table 6.3.4.2.3-1 in TS 36.211 [8] as beamforming weights. This precoder takes as an input a block of signals for antenna ports 7 and 8, $y(i) = \left[y^{(7)}(i) \quad y^{(8)}(i)\right]^T$, $i = 0,1,...,M_{\text{symb}}^{\text{ap}} - 1$, with $M_{\text{symb}}^{\text{ap}}$ being the number of modulation symbols per antenna port including the user-specific reference symbols, and generates a block of signals $y_{bf}(i) = \left[y_{bf}(i) \quad \tilde{y}_{bf}(i)\right]^T$ the elements of which are to be mapped onto the same physical RE but transmitted on different antenna elements:

$$\begin{bmatrix} y_{bf}(i) \\ \tilde{y}_{bf}(i) \end{bmatrix} = W(i) \begin{bmatrix} y^{(7)}(i) \\ y^{(8)}(i) \end{bmatrix},$$

The precoder update granularity is according to Table 8.3.2-1.

# Annex C (normative): Downlink Physical Channels

This annex specifies the downlink physical channels that are needed for setting a connection and channels that are needed during a connection.

## C.0 Downlink signal levels

The downlink power settings in Table C.0-1 are used unless otherwise specified in a test case.

If the UE has two Rx antennas, the downlink signal is applied to each one. Both UE Rx antennas shall be connected.

If the UE has one Rx antenna, the downlink signal is applied to it.

**Table C.0-1: Default Downlink power levels**

| | Unit | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1.4 MHz | 3 MHz | 5 MHz | 10 MHz | 15 MHz | 20 MHz |
| Number of RBs | | 6 | 15 | 25 | 50 | 75 | 100 |
| Channel BW Power | dBm | -66 | -62 | -60 | -57 | -55 | -54 |
| RS EPRE | dBm/15kHz | -85 | -85 | -85 | -85 | -85 | -85 |
| Note 1: The channel bandwidth powers and RB allocations are informative, based on -85dBm/15kHz RS_EPRE, then scaled according to the number of RBs and rounded to the nearest integer dBm value. Full RE allocation with no boost or deboost is assumed in this calculation, but allocation may vary during setup.<br>Note 2: The power level is specified at each UE Rx antenna. | | | | | | | |

The default signal level uncertainty is +/-3dB at each test port, for any level specified. If the uncertainty value is critical for the test purpose, a tighter uncertainty is specified for the related test case in Annex F.

## C.1 General

Table C.1-1 describes the mapping of downlink physical channels and signals to physical resources for FDD.

Table C.1-2 describes the mapping of downlink physical channels and signals to physical resources for TDD.

**Table C.1-1: Mapping of downlink physical channels and signals to physical resources for FDD**

| Physical channel | Time Domain Location | Frequency Domain Location | Note |
|---|---|---|---|
| RS | Symbols 0, 4 of each subframe for antenna port 0 & 1<br>Symbol 1 of each subframe for antenna port 2 & 3 | Downlink system bandwidth dependent. | Mapping rule is specified in TS36.211 6.10.1.2<br>- CELL_ID = 0 |
| PBCH | Symbols 0 to 3 of slot 1 of subframe 0 of each radio frame | Occupies 72 subcarriers centered on the DC subcarrier | Mapping rule is specified in TS36.211 Section 6.6.4 (Note 2) |
| PSS | Symbol 6 of slot 0 and 10 of each radio frame | Occupies 62 subcarriers centered on the DC subcarrier. Additional 10 subcarriers (5 on each side) adjacent to the centered 62 subcarriers are reserved. | Mapping rule is specified in TS36.211 Section 6.11.1.2 |
| SSS | Symbol 5 of slots 0 and 10 of each radio frame | Occupies 62 subcarriers centered on the DC subcarrier. Additional 10 subcarriers (5 on each side) adjacent to the centered 62 subcarriers are reserved. | Mapping rule is specified in TS36.211 Section 6.11.2.2 |
| PCFICH | Symbol 0 of each subframe | Downlink system bandwidth dependent. Maps into 4 REGs uniformly spread in the frequency domain over the whole system bandwidth. | Mapping rule is specified in TS36.211 Section 6.7.4 (Note 1)<br>- CELL_ID = 0 |
| PHICH | Symbol 0 of each subframe | Downlink system bandwidth dependent. Each PHICH group maps into 3 REGs in the frequency domain on the REGs not assigned to PCFICH over the whole system bandwidth, | Mapping rule is specified in TS36.211 Section 6.9.3 (Note 1)<br>- CELL_ID = 0<br>- Ng = 1<br>- Normal PHICH duration<br>-Number of PHICH groups = 1(BW=1.4MHz)/2(BW=3MHz)/ 4(BW=5MHz)/7(BW=10MHz)/ 10(BW=15MHz)/13(BW=20MHz) |
| PDCCH | Symbols 0, 1, 2, 3 of each subframe for 1.4 MHz<br><br>Symbols 0, 1, 2, of each subframe for 3 and 5 MHz<br><br>Symbols 0, 1 of each subframe for 10, 15 and 20 MHz | The remaining REGs not allocated to both PCFICH and PHICH are used for PDCCH | Mapping rule is specified in TS36.211 Section 6.8.5 (Note 1) |
| PDSCH | All remaining OFDM symbols of each subframe not allocated to PDCCH | For Subframe 0,<br>REs not allocated to RS, PSS, SSS and PBCH, is allocated to PDSCH<br><br>For Subframe 5,<br>REs not allocated to RS, PSS and SSS, is allocated to PDSCH<br><br>For other subframes,<br>REs not allocated to RS, is allocated to PDSCH | Note that there are reserved REs that are not used for transmission of any physical channels (Note 3 ) & (Note 4) which need to be taken into account when allocating REs to PDSCH |

Note 1: In case a single cell-specific RS is configured, cell-specific RS shall be assume to be present on antenna ports 0 and 1 for the purpose of mapping a symbol-quadruplet to a REG (resource-element group). (See TS 36.211 Section 6.2.4).

Note 2: PBCH is mapped into RE assuming RS from 4 antennas are used at the eNB transmitter, irrespective of the actual number of Tx antenna. Resource elements assumed to be reserved for RS but not used for transmission of RS shall not be used for transmission of any physical channel. (See TS 36.211 Section 6.6.4).

Note 3: In slot 0 and slot 10 of each radio frame, there are reserved REs for PSS and SSS that are not used for transmission of any physical channels. (See TS 36.211 Section 6.11.1.2 & 6.11.2.2).

Note 4: REs used for RS transmission on any of the antenna ports in a slot shall not be used for any transmission on any other antenna port in the same slot and set to zero. (See TS 36.211 Section 6.10.1.2).

**Table C.1-2: Mapping of downlink physical channels and signals to physical resources for TDD**

| Physical channel | Time Domain Location | Frequency Domain Location | Note |
|---|---|---|---|
| RS | Symbols 0, 4 of each subframe for antenna port 0 & 1<br>Symbol 1 of each subframe for antenna port 2 & 3 | Downlink system bandwidth dependent. | Mapping rule is specified in TS36.211[8] 6.10.1.2<br>- CELL_ID = 0 |
| PBCH | Symbols 0 to 3 of slot 1 of subframe 0 of each radio frame | Occupies 72 subcarriers centered on the DC subcarrier | Mapping rule is specified in TS36.211[8] Section 6.6.4 (Note 3) |
| PSS | Symbol 2 of slot 2 and 12 of each radio frame | Occupies 62 subcarriers centered on the DC subcarrier. Additional 10 subcarriers (5 on each side) adjacent to the centered 62 subcarriers are reserved. | Mapping rule is specified in TS36.211[8] Section 6.11.1.2 |
| SSS | Symbol 6 of slots 1 and 11 of each radio frame | Occupies 62 subcarriers centered on the DC subcarrier. Additional 10 subcarriers (5 on each side) adjacent to the centered 62 subcarriers are reserved. | Mapping rule is specified in TS36.211[8] Section 6.11.2.2 |
| PCFICH | Symbol 0 of each subframe and special subframe | Downlink system bandwidth dependent. Maps into 4 REGs uniformly spread in the frequency domain over the whole system bandwidth. | Mapping rule is specified in TS36.211[8] Section 6.7.4 (Note 2)<br>- CELL_ID = 0 |
| PHICH | Symbol 0 of each subframe and special subframe | Downlink system bandwidth dependent. Each PHICH group maps into 3 REGs in the frequency domain on the REGs not assigned to PCFICH over the whole system bandwidth, | Mapping rule is specified in TS36.211[8] Section 6.9.3 (Note 2)<br>- CELL_ID = 0<br>- Ng = 1<br>- Normal PHICH duration<br>-Number of PHICH groups = 1(BW=1.4MHz)/2(BW=3MHz)/ 4(BW=5MHz)/7(BW=10MHz)/ 10(BW=15MHz)/13(BW=20MHz) |
| PDCCH | For normal subframes(0,4,5,9)<br>Symbols 0, 1, 2, 3 of each subframe for 1.4 MHz<br>Symbols 0, 1, 2, of each subframe for 3 and 5 MHz<br>Symbols 0, 1 of each subframe for 10, 15 and 20 MHz<br><br>For special subframe (1&6)<br>Symbols 0, 1 of each subframe for all BWs | The remaining REGs not allocated to both PCFICH and PHICH are used for PDCCH | Mapping rule is specified in TS36.211[8] Section 6.8.5 (Note 2) |
| PDSCH | ,All remaining OFDM symbols of each subframe not allocated to PDCCH with the following exception:<br>For 1.4MHz,no data shall be scheduled on special subframes (1&6) to avoid problems with insufficient PDCCH performance | For Subframe 0,<br>REs not allocated to RS, SSS and PBCH, is allocated to PDSCH<br><br>For Subframe 5,<br>REs not allocated to RS and SSS, is allocated to PDSCH<br><br>For Subframe 1 and 6,<br>REs not allocated to RS, PSS, GP and UpPTS is allocated to PDSCH<br><br>For other downlink subframes,<br>REs not allocated to RS is allocated to PDSCH | Note that there are reserved REs that are not used for transmission of any physical channels (Note 4 ) & (Note 5) which need to be taken into account when allocating REs to PDSCH |
| Note 1: The mapping is based on the default TDD configuration for subframe assignment and special subframe patterns (see 36.508 [7]subclause 4.6.3)<br>Note 2: In case a single cell-specific RS is configured, cell-specific RS shall be assume to be present on antenna ports 0 and 1 for the purpose of mapping a symbol-quadruplet to a REG (resource-element group). (See TS | | | |

36.211[8] Section 6.2.4).
Note 3: PBCH is mapped into RE assuming RS from 4 antennas are used at the eNB transmitter, irrespective of the actual number of Tx antenna. Resource elements assumed to be reserved for RS but not used for transmission of RS shall not be used for transmission of any physical channel. (See TS 36.211[8] Section 6.6.4).
Note 4: In slot 1,2,11 and 12 of each radio frame, there are reserved REs for PSS and SSS that are not used for transmission of any physical channels. (See TS 36.211[8] Section 6.11.1.2 & 6.11.2.2).
Note 5: REs used for RS transmission on any of the antenna ports in a slot shall not be used for any transmission on any other antenna port in the same slot and set to zero. (See TS 36.211[8] Section 6.10.1.2).

# C.2 Set-up

Table C.2-1 describes the downlink Physical Channels that are required for connection set up.

**Table C.2-1: Downlink Physical Channels required for connection set-up**

| Physical Channel | EPRE Ratio | Note |
|---|---|---|
| **PBCH** | PBCH_RA = 0 dB | |
| | PBCH_RB = 0 dB | |
| **PSS** | PSS_RA = 0 dB | |
| **SSS** | SSS_RA = 0 dB | |
| **PCFICH** | PCFICH_RB = 0 dB | |
| **PDCCH** | PDCCH_RA = 0 dB | |
| | PDCCH_RB = 0 dB | |
| **PDSCH** | PDSCH_RA = 0 dB | |
| | PDSCH_RB = 0 dB | |
| **PHICH** | PHICH_RA = 0 dB | Note 2 |
| | PHICH_RB = 0 dB | |
| Note 1: No boosting is applied.<br>Note 2: PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | |

Table C.2-2 describes the configuration of PDSCH and PDCCH before measurement for FDD and Table C.2-3 for TDD.

**Table C.2-2: PDSCH and PDCCH configuration for FDD**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| Allocated resource blocks | | 6 | |
| MCS Index | | - | TB Size with transmitting message in 1TTI |
| Number of HARQ processes | Processes | 8 | |
| Maximum number of HARQ transmission | | 5 | |
| Aggregation level | CCE | 2 | Note 4 |
| DCI Format for PDSCH | | Format 1A | |
| DCI Format for PUSCH | | Format 0 | |
| Note 1: 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz.<br>Note 2: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8].<br>Note 3: The PDSCH shall be occupied 6 resource blocks centered on the DC subcarrier.<br>Note 4: For PDCCH using SI-RNTI, Aggregation level:<br>a)Tables C.3.0-3, C.3.1-3, and C.3.2-3 for RF tests<br>b)Table A.2.1-1 of 36.521-3 for RRM tests. | | | |

**Table C.2-3: PDSCH and PDCCH configuration for TDD**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| Allocated resource blocks | | 6 | |
| MCS Index | | 0 | TB Size with transmitting message in 1TTI |
| Number of HARQ processes (Note 1) | Processes | 7 | |
| Maximum number of HARQ transmission | | 4 | |
| Aggregation level | CCE | 2 | Note 5 |
| DCI Format for PDSCH | | Format 1A | |
| DCI Format for PUSCH | | Format 0 | |

Note 1: Number of HARQ processes shall be determined by UL/DL configuration, for configuration other than 1,the process number shall be set per TS 36.213 [10] Table 8-1.
Note 2: For normal downlink subframes, 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW. 3 symbols allocated to PDCCH for 5 MHz and 3 MHz. 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
Note 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [8].
Note 4: The PDSCH shall be occupied 6 resource blocks centered on the DC subcarrier.
Note 5: For PDCCH using SI-RNTI, Aggregation level:
a)Tables C.3.0-3, C.3.1-3, and C.3.2-3 for RF tests
b)Table A.2.2-1 of 36.521-3 for RRM tests

# C.3 Connection

The following clauses describes the downlink Physical Channels that are transmitted during a connection i.e., when measurements are done.

## C.3.0 Measurement of Transmitter Characteristics

Table C.3.0-1 is applicable for measurements on the Transmitter Characteristics (clause 6).

**Table C.3.0-1: Downlink Physical Channels transmitted during a connection (FDD and TDD)**

| Physical Channel | EPRE Ratio | Note |
|---|---|---|
| PBCH | PBCH_RA = 0 dB | |
| | PBCH_RB = 0 dB | |
| PSS | PSS_RA = 0 dB | |
| SSS | SSS_RA = 0 dB | |
| PCFICH | PCFICH_RB = 0 dB | |
| PDCCH | PDCCH_RA = 0 dB | |
| | PDCCH_RB = 0 dB | |
| PDSCH | PDSCH_RA = 0 dB | |
| | PDSCH_RB = 0 dB | |
| PHICH | PHICH_RB = 0 dB | Note 1 |

Note 1: PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group.

NOTE 1: No boosting is applied.

**Table C.3.0-2: Power allocation for OFDM symbols and reference signals**

| Parameter | Unit | Value | Note |
|---|---|---|---|
| Transmitted power spectral density $I_{or}$ | dBm/15 kHz | Test specific | 1. $I_{or}$ shall be kept constant throughout all OFDM symbols |
| Cell-specific reference signal power ratio $E_{RS} / I_{or}$ | | 0 dB | |

**Table C.3.0-3: PDCCH Aggregation Level (in CCE-s)**

| Bandwidth | DCI for DL (SI-RNTI) | DCI for DL (C-RNTI) | DCI for UL (C-RNTI) | Notes |
|---|---|---|---|---|
| 1.4 MHz | 4 | 1 | 1 | Note 1 |
| 3 MHz | 4 | 4 | 4 | Note 1 |
| 5 MHz | 4 | 4 | 4 | Note 1 |
| 10 MHz | 8 | 8 | 8 | Note 1 |
| 15 MHz | 8 | 8 | 8 | Note 1 |
| 20 MHz | 8 | 8 | 8 | Note 1 |
| Note 1: No DL data allocated on TDD special subframes | | | | |

# C.3.1 Measurement of Receiver Characteristics

Table C.3.1-1 is applicable for measurements on the Receiver Characteristics (clause 7).

**Table C.3.1-1: Downlink Physical Channels transmitted during a connection (FDD and TDD)**

| Physical Channel | EPRE Ratio | Note |
|---|---|---|
| PBCH | PBCH_RA = 0 dB | |
| | PBCH_RB = 0 dB | |
| PSS | PSS_RA = 0 dB | |
| SSS | SSS_RA = 0 dB | |
| PCFICH | PCFICH_RB = 0 dB | |
| PDCCH | PDCCH_RA = 0 dB | |
| | PDCCH_RB = 0 dB | |
| PDSCH | PDSCH_RA = 0 dB | |
| | PDSCH_RB = 0 dB | |
| PHICH | PHICH_RB = 0 dB | Note 1 |
| OCNG | OCNG_RA = 0 dB | |
| | OCNG_RB = 0 dB | |
| Note 1: PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | |

NOTE 1: No boosting is applied.

**Table C.3.1-2: Power allocation for OFDM symbols and reference signals**

| Parameter | Unit | Value | Note |
|---|---|---|---|
| Transmitted power spectral density $I_{or}$ | dBm/15 kHz | Test specific | 1. $I_{or}$ shall be kept constant throughout all OFDM symbols |
| Cell-specific reference signal power ratio $E_{RS} / I_{or}$ | | 0 dB | |

**Table C.3.1-3: PDCCH Aggregation Level (in CCE-s)**

| Bandwidth | DCI for DL (SI-RNTI) | DCI for DL (C-RNTI) | DCI for UL (C-RNTI) | Notes |
|---|---|---|---|---|
| 1.4 MHz | 4 | 4 | 2 | Note 1, 2 |
| 3 MHz | 4 | 4 | 2 | Note 2 |
| 5 MHz | 8 | 8 | 4 | Note 2 |
| 10 MHz | 8 | 8 | 8 | Note 2 |
| 15 MHz | 8 | 8 | 8 | Note 2 |
| 20 MHz | 8 | 8 | 8 | Note 2 |
| Note 1: No DL data allocated on TDD special subframes<br>Note 2: No DL data allocated on subframe 5 | | | | |

## C.3.2 Measurement of Performance requirements

Table C.3.2-1 is applicable for measurements in which uniform RS-to-EPRE boosting for all downlink physical channels.

**Table C.3.2-1: Downlink Physical Channels transmitted during a connection (FDD and TDD)**

| Physical Channel | EPRE Ratio | Note |
|---|---|---|
| PBCH | PBCH_RA = $\rho_A$ | |
| | PBCH_RB = $\rho_B$ | |
| PSS | PSS_RA = $\rho_A$ | |
| SSS | SSS_RA = $\rho_A$ | |
| PCFICH | PCFICH_RB = $\rho_B$ | |
| PDCCH | PDCCH_RA = $\rho_A$ | |
| | PDCCH_RB = $\rho_B$ | |
| PDSCH | PDSCH_RA = $\rho_A$ | |
| | PDSCH_RB = $\rho_B$ | |
| PHICH | PHICH_RB = $\rho_B$ | Note 1 |
| OCNG | OCNG_RA = $\rho_A$ | |
| | OCNG_RB = $\rho_B$ | |
| Note 1: PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | |

NOTE 1: $\rho_A = \rho_B = 0$ dB means no RS boosting.

NOTE 2: $\rho_A$ denotes the ratio of PDSCH EPRE to cell-specific RS EPRE among PDSCH REs in all the OFDM symbols not containing cell-specific RS. $\rho_B$ denotes the ratio of PDSCH EPRE to cell-specific RS EPRE among PDSCH REs in all the OFDM symbols containing cell-specific RS.

**Table C.3.2-2: Power allocation for OFDM symbols and reference signals**

| Parameter | Unit | Value | Note |
|---|---|---|---|
| Total transmitted power spectral density $I_{or}$ | dBm/15 kHz | Test specific | 1. $I_{or}$ shall be kept constant throughout all OFDM symbols |
| Cell-specific reference signal power ratio $E_{RS} / I_{or}$ | | Test specific | 1. Applies for antenna port *p* |

**Table C.3.2-3: PDCCH Aggregation Level (in CCE-s) for PDSCH demodulation and PMI performance tests**

| Bandwidth | DCI for DL (SI-RNTI) | DCI for DL (C-RNTI) | DCI for UL (C-RNTI) | Notes |
|---|---|---|---|---|
| 1.4 MHz | 4 | 4 | 2 | Note 1, 2 |
| 3 MHz | 4 | 4 | 2 | Note 2 |
| 5 MHz | 8 | 8 | 4 | Note 2 |
| 10 MHz | 8 | 8 | 8 | Note 2 |
| 15 MHz | 8 | 8 | 8 | Note 2 |
| 20 MHz | 8 | 8 | 8 | Note 2 |
| Note 1: No DL data allocated on TDD special subframes<br>Note 2: No DL data allocated on subframe 5 | | | | |

**Table C.3.2-4: PDCCH Aggregation Level for CQI and RI performance tests (in CCE-s)**

| Bandwidth | DCI for DL (SI-RNTI) | DCI for DL (C-RNTI) | DCI for UL (C-RNTI) | Notes |
|---|---|---|---|---|
| 1.4 MHz | | | | |
| 3 MHz | | | | |
| 5 MHz | | | | |
| 10 MHz | 8 | 8 | 8 | Note 1,2 |
| 15 MHz | | | | |
| 20 MHz | | | | |
| Note 1: 3 symbols allocated to PDCCH<br>Note 2: No DL data allocated on subframes 0 and 5 for FDD and 0, 1, 5 and 6 for TDD | | | | |

# Annex D (normative): Characteristics of the Interfering Signal

## D.1 General

Some RF performance requirements for the E-UTRA UE receiver are defined with interfering signals present in addition to the wanted signal. When the wanted channel band width is wider than or equal to 5MHz, a modulated 5MHz full band width E-UTRA down link signal, and in some cases an additional CW signal, are used. For wanted channel band widths below 5MHz, the band width of the modulated interferer should be equal to the channel band width of the wanted signal.

## D.2 Interference signals

Table D.2-1 describes the modulated interferer for different channel band width options.

**Table D.2-1: Description of modulated E-UTRA interferer**

| | Channel bandwidth | | | | | |
|---|---|---|---|---|---|---|
| | **1.4 MHz** | **3 MHz** | **5 MHz** | **10 MHz** | **15 MHz** | **20 MHz** |
| RB | 6 | 15 | 25 | 25 | 25 | 25 |
| $BW_{Interferer}$ | 1.4 MHz | 3 MHz | 5 MHz | 5 MHz | 5 MHz | 5 MHz |

# Annex E (normative): Global In-Channel TX-Test

Note: Clauses E.2.2 to E.5.9.3 are descriptions, which assume no power ramping adjacent to the measurement period. *Power ramping adjacent to the measurement period requires exclusion periods, decribed in clause E.7*

## E.1 General

The global in-channel TX test enables the measurement of all relevant parameters that describe the in-channel quality of the output signal of the TX under test in a single measurement process.

The parameters describing the in-channel quality of a transmitter, however, are not necessarily independent. The algorithm chosen for description inside this annex places particular emphasis on the exclusion of all interdependencies among the parameters.

## E.2 Signals and results

### E.2.1 Basic principle

The process is based on the comparison of the actual **output signal of the TX under test**, received by an ideal receiver, with a **reference signal**, that is generated by the measuring equipment and represents an ideal error free received signal. All signals are represented as equivalent (generally complex) baseband signals.

The description below uses numbers as examples. These numbers are taken from frame structure 1 with normal CP length and 20 MHz bandwidth. The application of the text below, however, is not restricted to this frame structure and bandwidth.

### E.2.2 Output signal of the TX under test

The output signal of the TX under test is acquired by the measuring equipment and stored for further processsing. It is sampled at a sampling rate of 30.72 Msps. In the time domain it comprises at least 10 uplink subframes. The measurement period is derived by concatenating the correct number of individual uplink slots until the correct measurement period is reached. The output signal is named z(v). Each slot is modelled as a signal with the following parameters: demodulated data content, carrier frequency, amplitude and phase for each subcarrier, timing, carrier leakage.

NOTE 1: TDD

For frame structure type 2, subframes with special fields (UpPTS) do not undergo any evaluation. Since the uplink subframes are not continuous, the 20 slots should be extracted from more than 1 continuous radio frame:

Figure E.2.2-1 is an example for uplink-downlink configuration 1 (DSUUDDSUUD) as specified in TS 36.211 [8] Table 4.2-2,assuming all uplink subframes are active.

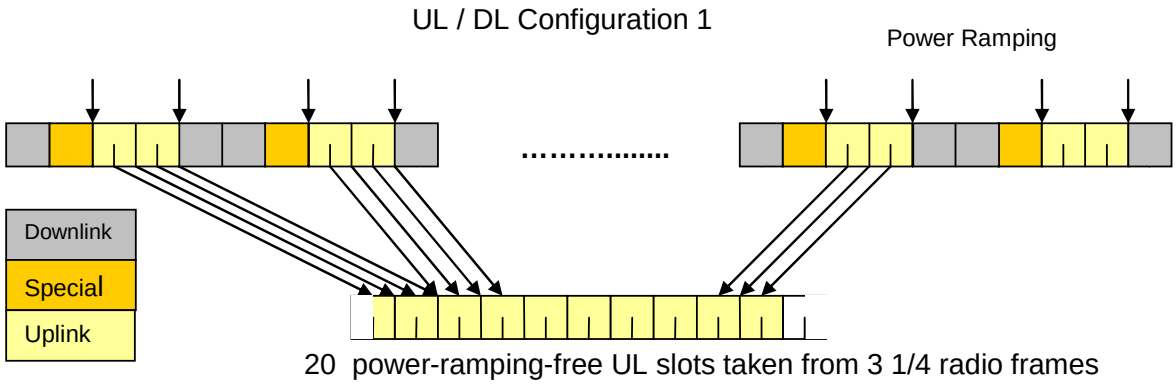



**Figure E.2.2-1: Example of uplink – downlink configuration 1**

## E.2.3 Reference signal

Two types of reference signal are defined:

The reference signal $i_1(v)$ is constructed by the measuring equipment according to the relevant TX specifications, using the following parameters: demodulated data content, nominal carrier frequency, nominal amplitude and phase for each subcarrier, nominal timing, no carrier leakage. It is represented as a sequence of samples at a sampling rate of 30.72 Msps in the time domain.

The reference signal $i_2(v)$ is constructed by the measuring equipment according to the relevant TX specifications, using the following parameters: restricted data content: nominal reference symbols, (all modulation symbols for user data symbols are set to 0V) , nominal carrier frequency, nominal amplitude and phase for each applicable subcarrier, nominal timing, no carrier leakage. It is represented as a sequence of samples at a sampling rate of 30.72 Msps in the time domain.

NOTE: The PUCCH is off during the time under test.

## E.2.4 Measurement results

The measurement results, achieved by the global in channel TX test are the following:

- Carrier Frequency error
- EVM (Error Vector Magnitude)
- Carrier leakage
- Unwanted emissions, falling into non allocated resource blocks.
- EVM equalizer sSpectrum flatness

## E.2.5 Measurement points

The unwanted emission falling into non-allocated RB(s) is calculated directly after the FFT as described below. In contrast to this, the EVM for the allocated RB(s) is calculated after the IDFT. The samples after the TX-RX chain equalizer are used to calculate EVM equalizer spectrum flatness. Carrier frequency error and carrier leakage is calculated in the block “RF correction”.

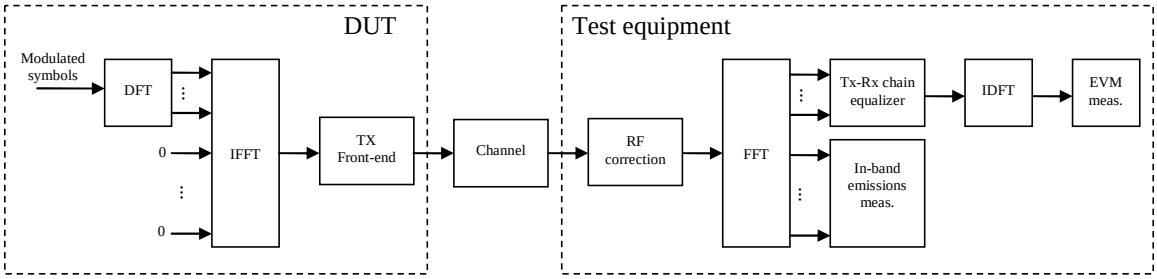



**Figure E.2.5-1: EVM measurement points**

# E.3 Signal processing

## E.3.1 Pre FFT minimization process

Before applying the pre-FFT minimization process, z(ν) and i(ν) are portioned into 20 pieces, comprising one slot each. Each slot is processed separately. Sample timing, Carrier frequency and baseband-I/Q offset (corresponding carrier leakage in RF) in z(ν) are jointly varied in order to minimise the difference between z(ν) and i(ν). Best fit (minimum difference) is achieved when the RMS difference value between z(ν) and i(ν) is an absolute minimum.

The carrier frequency variation and the IQ variation are the measurement results: Carrier Frequency Error and Carrier leakage.

From the acquired samples 20 carrier frequencies and 20 carrier leakages can be derived.

NOTE 1: The minimisation process, to derive carrier leakage and RF error can be supported by Post FFT operations. However the minimisation process defined in the pre FFT domain comprises all acquired samples (i.e. it does not exclude the samples in between the FFT widths and it does not exclude the bandwidth outside the transmission bandwidth configuration

NOTE 2: The algorithm would allow to derive Carrier Frequency error and Sample Frequency error of the TX under test separately. However there are no requirements for Sample Frequeny error. Hence the algorithm models the RF and the sample frequency commonly (not independently). It returns one error and does not distinuish between both.

After this process the samples z(ν) are called $z^0(\nu)$.

## E.3.2 Timing of the FFT window

The FFT window length is 2048 samples per OFDM symbol. 7 FFTs (14336 samples) cover less than the acquired number of samples (15360 samples) The position in time for FFT must be determined.

In an ideal signal, the FFT may start at any instant within the cyclic prefix without causing an error. The TX filter, however, reduces the window. The EVM requirements shall be met within a window W<CP. There are three different instants for FFT:

Centre of the reduced window, called $\Delta\tilde{c}$, $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2.

The timing of the measured signal is determined in the pre FFT domain as follows, using $z^0(\nu)$ and $i_2(\nu)$ :

1. The measured signal is delay spread by the TX filter. Hence the distinct boarders between the OFDM symbols and between Data and CP are also spread and the timing is not obvious.

2. In the Reference Signal $i_2(\nu)$ the timing is known.

3. Correlation between (1.) and (2.) will result in a correlation peak. The meaning of the correlation peak is approx. the "impulse response" of the TX filter. The meaning of "impulse response" assumes that the autocorrelation of

the reference signal $i_2(v)$ is a Dirac peak and that the correlation between the reference signal $i_2(v)$ and the data in the measured signal is 0. The correlation peak, (the highest, or in case of more than one, the earliest) indicates the timing in the measured signal.

From the acquired samples 20 timings can be derived.

For all calculations, except EVM, the number of samples in $z^0(v)$ is reduced to 7 blocks of samples, comprising 2048 samples (FFT width) and starting with $\Delta\tilde{c}$ in each OFDM symbol including the demodulation reference signal.

For the EVM calculation the output signal under test is reduced to 14 blocks of samples, comprising 2048 samples (FFT width) and starting with $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2 in each OFDM symbol including the demodulation reference signal.

The number of samples, used for FFT is reduced compared to $z^0(v)$. This subset of samples is called z'(v).

The timing of the centre $\Delta\tilde{c}$ with respect to the different CP length in a slot is as follows: (Frame structure 1, normal CP length)

$\Delta\tilde{c}$ is on $T_f$=72 within the CP of length 144 (in OFDM symbol 1 to 6)

$\Delta\tilde{c}$ is on $T_f$=88 (=160-72) within the CP of length 160 (in OFDM symbol 0)

## E.3.3 Post FFT equalisation

Perform 7 FFTs on z'(v), one for each OFDM symbol in a slot using the timing $\Delta\tilde{c}$, including the demodulation reference symbol. The result is an array of samples, 7 in the time axis t times 2048 in the frequency axis f. The samples represent the DFT coded data symbols (in OFDM-symbol 0,1,2,4,5and 6 in each slot) and demodulation reference symbols ( OFDM symbol 3 in each slot) in the allocated RBs and inband emissions in the non allocated RBs within the transmission BW.

Only the allocated resource blocks in the frequency domain are used for equalisation.

The nominal demodulation reference symbols and nominal DFT coded data symbols are used to equalize the measured data symbols. (Location for equalization see Figure E.2.5-1)

NOTE: The nomenclature inside this note is local and not valid outside.

The nominal DFT coded data symbols are created by a demodulation process. The location to gain the demodulated data symbols is "EVM" in Figure E.2.5-1. A demodulation process as follows is recommended:

1. Equalize the measured DFT coded data symbols using the reference symbols for equalisation. Result: Equalized DFT coded data symbols
2. iDFT transform the equalized DFT coded data symbols: Result: Equalized data symbols
3. Decide for the nearest constellation point: Result: Nominal data symbols
4. DFT transform the nominal data symbols: Result: Nominal DFT coded data symbols

At this stage we have an array of Measured DFT coded data-Symbols and reference-Symbols (MS(f,t))

versus an array of Nominal DFT coded data-Symbols and reference Symbols (NS(f,t))

(complex, the arrays comprise 6 DFT coded data symbols and 1 demodulation reference symbol in the time axis and the number of allocated subcarriers in the frequency axis.)

MS(f,t) and NS(f,t) are processed with a least square (LS) estimator, to derive one equalizer coefficient per time slot and per allocated subcarrier. EC(f)

$$EC(f) = \frac{\sum_{t=0}^{6} NS(f,t)^* NS(f,t)}{\sum_{t=0}^{6} NS(f,t)^* MS(f,t)}$$

With * denoting complex conjugation.

EC(f) are used to equalize the DFT-coded data symbols. The measured DFT-coded data and the references symbols are equalized by:

$$Z'(f,t) = MS(f,t) \cdot EC(f)$$

With · denoting multiplication.

Z'(f,t), restricted to the data symbol (excluding t=3) is used to calculate EVM, as described in E.4.1.

EC(f) is used in E.4.4 to calculate EVM equalizer spectral flatness.

NOTE: although an exclusion period for EVM may be applicable in E.7, the post FFT minimisation process is done over 7 symbols (6 DFT-coded data symbols and 1 reference symbol).

The samples of the non allocated resource blocks within the transmission bandwidth configuration in the post FFT domain are called Y(f,t) (f covering the non allocated subcarriers within the transmission bandwidth configuration, t covering the OFDM symbols during 1 slot).

# E.4 Derivation of the results

## E.4.1 EVM

For EVM create two sets of Z'(f,t)., according to the timing " $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2" using the equalizer coefficients from E.3.3.

Perform the iDFTs on Z'(f,t). The IDFT-decoding preserves the meaning of t but transforms the variable f (representing the allocated sub carriers) into an another variable g, covering the same count and representing the demodulated symbols. The samples in the post IDFT domain are called iZ'(g, t). The equivalent ideal samples are called iI(g,t). Those samples of Z'(f,t), carrying the reference symbols (=symbol 3) are not iDFT processed.

The EVM is the difference between the ideal waveform and the measured and equalized waveform for the allocated RB(s)

$$EVM = \sqrt{\frac{\sum_{t \in T} \sum_{g \in G} \left| iZ'(g,t) - iI(g,t) \right|^2}{|G| \cdot |T| \cdot P_0}},$$

where

t covers the count of demodulated symbols with the considered modulation scheme being active within the measurement period, (i.e. symbol 0,1,2,4,5and 6 in each slot, →|T|=6 )

g covers the count of demodulated symbols with the considered modulation scheme being active within the allocated bandwidth. (|G|=12* $L_{CRBs}$ (with $L_{CRBs}$ : number of allocated resource blocks)).

$iZ'(g,t)$ are the samples of the signal evaluated for the EVM.

$iI(g,t)$ is the ideal signal reconstructed by the measurement equipment, and

$P_0$ is the average power of the ideal signal. For normalized modulation symbols $P_0$ is equal to 1.

From the acquired samples 40 EVM value can be derived, 20 values for the timing $\Delta\tilde{c}$ –W/2 and 20 values for the timing $\Delta\tilde{c}$ +W/2

## E.4.2 Averaged EVM

EVM is averaged over all basic EVM measurements.

The averaging comprises 20 UL slots (for frame structure 2: excluding special fields(UpPTS))

$$\overline{EVM} = \sqrt{\frac{1}{20}\sum_{i=1}^{20} EVM_i^2}$$

The averaging is done separately for timing $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2 leading to $\overline{\mathrm{EVM}}_{\mathrm{l}}$ and $\overline{\mathrm{EVM}}_{\mathrm{h}}$

$EVM_{\mathrm{final}} = \max(\overline{\mathrm{EVM}}_{l}, \overline{\mathrm{EVM}}_{h})$ is compared against the test requirements.

## E.4.3 In-band emissions measurement

The in-band emissions are a measure of the interference falling into the non-allocated resources blocks.

Explanatory Note:

The inband emission measurement is only meaningful with allocated RBs next to non allocated RB. The allocated RBs are necessary but not under test. The non allocated RBs are under test. The RB allocation for this test is as follows: The allocated RBs are at one end of the channel BW, leaving the other end unallocated. The number of allocated RBs is smaller than half of the number of RBs, available in the channel BW. This means that the vicinity of the carrier in the centre is unallocated.

There are 3 types of inband emissions:

1. General
2. IQ image
3. Carrier leakage

*Carrier leakage* are inband emissions next to the carrier.

*IQ image* are inband emissions symmetrically (with respect to the carrier) on the other side of the allocated RBs.

*General* are applied to all unallocated RBs.

For each evaluated RB, the minimum requirement is calculated as the higher of $P_{RB}$ - 30 dB and the power sum of all limit values (General, IQ Image or Carrier leakage) that apply.

In specific the following combinations:

- Power (General)
- Power (General + Carrier leakage)
- Power (General + IQ Image)

1 and 2 is expressed in terms of power in one non allocated RB under test, normalized to the average power of an allocated RB (unit dB).

3 is expressed in terms of power in one non allocated RB, normalized to the power of all allocated RBs. (unit dBc).

This is the reason for two formulas $Emissions_{\ relative}$

Create one set of Y(t,f) per slot according to the timing "$\Delta\tilde{c}$"

For the non-allocated RBs below the in-band emissions are calculated as follows

$$Emissions_{absolute}(\Delta_{RB}) = \begin{cases} \frac{1}{|T_s|}\sum_{t\in T_s} \sum_{\max(f_{\min},(c_l+12\cdot\Delta_{RB}*\Delta f))}^{c_l+(12\cdot\Delta_{RB}+11)*\Delta f} |Y(t,f)|^2, \Delta_{RB} < 0 \\ \frac{1}{|T_s|}\sum_{t\in T_s} \sum_{c_h+(12\cdot\Delta_{RB}-11)*\Delta f}^{\min(f_{\max},(c_h+12\cdot\Delta_{RB}*\Delta f))} |Y(t,f)|^2, \Delta_{RB} > 0 \end{cases},$$

where

the upper formula represents the in band emissions below the allocated frequency block and the lower one the in band emissions above the allocated frequency block.

$T_s$ is a set of $|T_s|$ SC-FDMA symbols with the considered modulation scheme being active within the measurement period,

$\Delta_{RB}$ is the starting frequency offset between the allocated RB and the measured non-allocated RB (e.g. $\Delta_{RB} = 1$ for the first upper or $\Delta_{RB} = -1$ for the first lower adjacent RB),

$f_{\min}$ and $f_{\max}$ are the lower and upper edge of the UL transmission BW configuration,

$c_l$ and $c_h$ are the lower and upper edge of the allocated BW,

$\Delta f$ is 15kHz,and

$Y(t, f)$ is the frequency domain signal evaluated for in-band emissions as defined in the subsection E.3.3

The allocated RB power per RB and the total allocated RB power are given by:

$$P_{RB} = \frac{1}{|T_s| \cdot L_{CRBs}} \sum_{t\in T_s} \sum_{c_l}^{c_l+(12\cdot L_{CRBs}-1)*\Delta f} |\mathrm{MS}(t,f)|^2 \ [\mathrm{dBm/180\,kHz}]$$

$$P_{All-RBs} = \frac{1}{|T_s|} \sum_{t\in T_s} \sum_{c_l}^{c_l+(12\cdot L_{CRBs}-1)*\Delta f} |\mathrm{MS}(t,f)|^2 \ [\mathrm{dBm}]$$

The relative in-band emissions, applicable for General and IQ image, are given by:

$$Emissions_{relative}(\Delta_{RB}) = 10\cdot\log_{10}\left(\frac{Emissions_{absolute}(\Delta_{RB})}{\frac{1}{|T_s|\cdot L_{CRBs}}\sum_{t\in T_s}\sum_{c_l}^{c_l+(12\cdot L_{CRBs}-1)*\Delta f}|\mathrm{MS}(t,f)|^2}\right)[\mathrm{dB}]$$

$$= Emissions_{absolute}(\Delta_{RB})[\mathrm{dBm/180\,kHz}] - P_{RB}\ [\mathrm{dBm/180\,kHz}]$$

where

$L_{CRBs}$ is the number of allocated resource blocks,

and

$\mathrm{MS}(t, f)$ is the frequency domain samples for the allocated bandwidth, as defined in the subsection E.3.3.

The relative in-band emissions, applicable for carrier leakage, is given by:

$$Emissions_{relative} = 10 \cdot \log_{10}\left(\frac{Emissions_{absolute}(RBnextDC)}{\frac{1}{|T_s|}\sum_{t \in T_S}\sum_{c_l}^{c_l+(12\cdot L_{CRBs}-1)*\Delta f}|\mathrm{MS}(t,f)|^2}\right)[\mathrm{dBc}]$$

$$= Emissions_{absolute}(RBnextDC)[\mathrm{dBm/180kHz}] - P_{All-RBs}[\mathrm{dBm}]$$

where RBnextDC means: Resource Block next to the carrier.

This is one RB, namely the central one in case of an odd number of RBs in the channel BW.

This is one pair of RBs, namely the immediately adjacent RBs to the carrier in case of an even number of RBs in the channel BW.

Although an exclusion period may be applicable in the time domain, when evaluating EVM (clause E.7), the inband emissions measurement interval is defined over one complete slot in the time domain.

From the acquired samples 20 functions for general in band emissions and IQ image inband emissions can be derived. 20 values or 20 pairs of carrier leakage inband emissions can be derived. They are compared against different limits.

## E.4.4 EVM equalizer spectrum flatness

For EVM equalizer spectrum flatness use EC(f) as defined in E.3.3. Note, EC(f) represents equalizer coefficient $f \in F$ , f is the allocated subcarriers within the transmission bandwidth ((|F|=12* $L_{CRBs}$ )

From the acquired samples 20 functions EC(f) can be derived.

EC(f) is broken down to 2 functions:

$$\mathrm{EC}_1(\mathrm{f}),\ f \in Range\ \ 1$$

$$\mathrm{EC}_2(\mathrm{f}),\ f \in Range\ \ 2$$

Where Range 1 and Range 2 are as defined in Table 6.5.2.4.5-1 for normal condition and Table 6.5.2.4.5-2 for extreme condition

The following peak to peak ripple is calculated:

$\mathrm{RP}_1 = 20 * \log(\max(|\mathrm{EC}_1(\mathrm{f})|) / \min(|\mathrm{EC}_1(\mathrm{f})|))$ ,which denote the maximum ripple in Range 1

$\mathrm{RP}_2 = 20 * \log(\max(|\mathrm{EC}_2(\mathrm{f})|) / \min(|\mathrm{EC}_2(\mathrm{f})|))$ ,which denote the maximum ripple in Range 2

$\mathrm{RP}_{12} = 20 * \log(\max(|\mathrm{EC}_1(\mathrm{f})|) / \min(|\mathrm{EC}_2(\mathrm{f})|))$ ,which denote the maximum ripple between the upper side of Range 1 and lower side of Range 2

$\mathrm{RP}_{21} = 20 * \log(\max(|\mathrm{EC}_2(\mathrm{f})|) / \min(|\mathrm{EC}_1(\mathrm{f})|))$ ,which denote the maximum ripple between the upper side of Range 2 and lower side of Range 1

## E.4.5 Frequency error and Carrier leakage

See E.3.1.

# E.4.6 EVM of Demodulation reference symbols ($EVM_{DMRS}$)

For the purpose of EVM $_{DMRS}$, the steps E.2.2 to E.4.2 are repeated 6 times, constituting 6 EVM $_{DMRS}$ sub-periods. The only purpose of the repetition is to cover the longer gross measurement period of EVM $_{DMRS}$ (120 time slots) and to derive the FFT window timing per sub-period.

The bigger of the EVM results in one 20 TS period corresponding to the timing $\Delta\tilde{c}$ –W/2 or $\Delta\tilde{c}$ +W/2 is compared against the limit. (Clause E.4.2) This timing is re-used for EVM $_{DMRS}$ in the equivalent EVM $_{DMRS}$ sub-period.

For EVM the demodulation reference symbols are exlcuded, while the data symbols are used. For $EVM_{DMRS}$ the data symbols are excluded, while the demodulation references symbols are used.This is illustrated in figure E.4.6-1

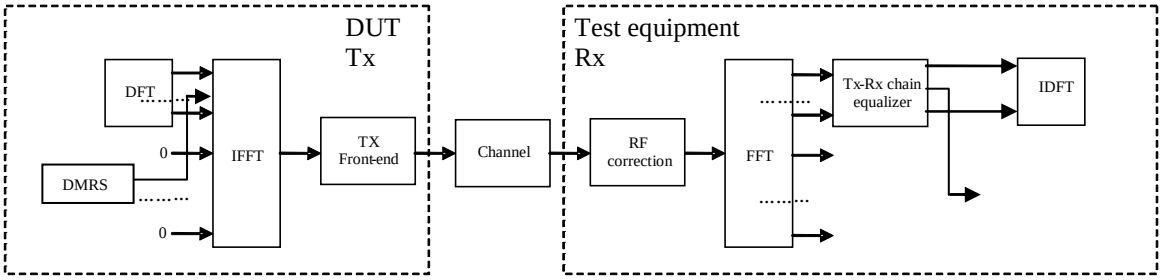



**Figure E.4.6-1: $EVM_{DMRS}$ measurement points**

Re-use the folowing formula from E.3.3:

$$Z'(f,t) = MS(f,t) \cdot EC(f)$$

To calculate $EVM_{DMRS}$ , the data symbol ( t=0,1,2,4,5,6) in Z'(f,t) are excluded and only the reference symbol (t=3) is used.

The EVM $_{DMRS}$ is the difference between the ideal waveform and the measured and equalized waveform for the allocated RB(s)

$$EVM_{DMRS} = \sqrt{\frac{\sum_{t \in T} \sum_{f \in F} \left| Z'(f,t) - I(f,t) \right|^2}{|T| \cdot P_0 \cdot |F|}},$$

where

t covers the count of demodulation reference symbols (i.e. only symbol 3 in each slot, so count =1)

f covers the count of demodulation reference symbols within the allocated bandwidth. (|F|=12* $L_{CRBs}$ (with $L_{CRBs}$ : number of allocated resource blocks)).

$Z'(f,t)$ are the samples of the signal evaluated for the EVM $_{DMRS}$

$I(f,t)$ is the ideal signal reconstructed by the measurement equipment, and

$P_0$ is the average power of the ideal signal. For normalized modulation symbols $P_0$ is equal to 1.

20 such results are generated per measurement sub-period.

## E.4.6.1 1st average for EVM $_{DMRS}$

EVM $_{DMRS}$ is averaged over all basic EVM $_{DMRS}$ measurements in one sub-period

The averaging comprises 20 UL slots (for frame structure 2: excluding special fields(UpPTS))

$$1stEVM_{DMRS} = \sqrt{\frac{1}{20}\sum_{i=1}^{20} EVM_{DMRS_i}^2}$$

The timing is taken from the EVM for the data. 6 of those results are achieved from the samples. In general the timing is not the same for each result.

### E.4.6.2 Final average for EVM DMRS

$$finalEVM_{DMRS} = \sqrt{\frac{1}{6}\sum_{i=1}^{6} 1stEVM_{DMRS_i}^2}$$

## E.5 EVM and inband emissions for PUCCH

For the purpose of worst case testing, the PUCCH shall be located on the edges of the Transmission Bandwidth Configuration (6,15,25,50,75,100 RBs).

The EVM for PUCCH ($EVM_{PUCCH}$) is averaged over 20 slots. At least 20 TSs shall be transmitted by the UE without power change. SRS multiplexing shall be avoided during this period. The following transition periods are applicable: One OFDM symbol on each side of the slot border (instant of band edge alternation).

The description below is generic in the sense that all 6 PUCCH formats are covered. Although the number of OFDM symbols in one slot is 6 or 7 (depending on the cyclic prefix length), the text below uses 7 without excluding 6.

### E.5.1 Basic principle

The basis principle is the same as described in E.2.1

### E.5.2 Output signal of the TX under test

The output signal of the TX under test is processed same as described in E.2.2

### E.5.3 Reference signal

The reference signal is defined same as in E.2.3. Same as in E.2.3, $i_1(v)$ is the ideal reference for $EVM_{PUCCH}$ and $i_2(v)$ is used to estimate the FFT window timing.

Note PUSCH is off during the PUCCH measurement period.

### E.5.4 Measurement results

The measurement results are:

- $EVM_{PUCCH}$
- Inband emissions with the sub-results: General in-band emission, IQ image (according to: 36.101. Annex F.4, Clause starting with: “At this stage the ….”)

### E.5.5 Measurement points

The measurement points are illustrated in the figure below:

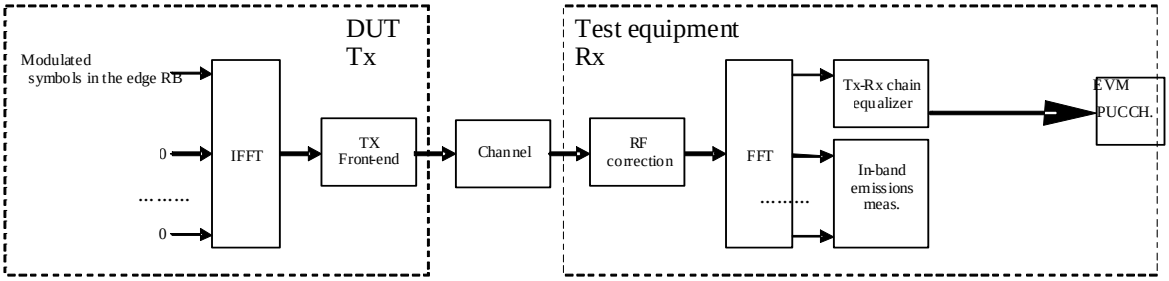



**Figure E.5.5-1**

## E.5.6 Pre FFT minimization process

The pre FFT minimisation process is the same as describes in clause E.3.1.

NOTE: although an exclusion period for $EVM_{PUCCH}$ is applicable in E.5.9.1, the pre FFT minimisation process is done over the complete slot.

RF error, and carrier leakage are necessary for best fit of the measured signal towards the ideal signal in the pre FFT domain. However they are not used to compare them against the limits.

## E.5.7 Timing of the FFT window

Timing of the FFT window is estimated with the same method as described in E.3.2.

## E.5.8 Post FFT equalisation

The post FFT equalisation is described separately without reference to E.3.3:

Perform 7 FFTs on z'(ν), one for each OFDM symbol in a slot using the timing $\Delta\tilde{c}$, including the demodulation reference symbol. The result is an array of samples, 7 in the time axis t times 2048 in the frequency axis f. The samples represent the OFDM symbols (data and reference symbols) in the allocated RBs and inband emissions in the non allocated RBs within the transmission BW.

Only the allocated resource blocks in the frequency domain are used for equalisation.

The nominal reference symbols and **nominal** OFDM data symbols are used to equalize the measured data symbols.

Note: (The nomenclature inside this note is local and not valid outside)

The nominal OFDM data symbols are created by a demodulation process. A demodulation process as follows is recommended:

1. Equalize the measured OFDM data symbols using the reference symbols for equalisation. Result: Equalized OFDM data symbols

2. Decide for the nearest constellation point, however not independent for each subcarrier in the RB. 12 constellation points are decided dependent, using the applicable CAZAC sequence. Result: Nominal OFDM data symbols

At this stage we have an array of Measured data-Symbols and reference-Symbols (MS(f,t))

versus an array of Nominal data-Symbols and reference Symbols (NS(f,t))

The arrays comprise in sum 7 data and reference symols, depending on the PUCCH format, in the time axis and the number of allocated sub-carriers in the frequency axis.

MS(f,t) and NS(f,t) are processed with a least square (LS) estimator, to derive one equalizer coefficient per time slot and per allocated subcarrier. EC(f)

$$EC(f) = \frac{\sum_{t=0}^{6} NS(f,t)^{*} NS(f,t)}{\sum_{t=0}^{6} MS(f,t)^{*} NS(f,t)}$$

With * denoting complex conjugation.

EC(f) are used to equalize the OFDM data together with the demodulation reference symbols by:

$$Z'(f,t) = MS(f,t) \cdot EC(f)$$

With · denoting multiplication.

Z'(f,t) is used to calculate $EVM_{PUCCH}$, as described in E.5.9 1

NOTE: although an exclusion period for $EVM_{PUCCH}$ is applicable in E.5.9.1, the post FFT minimisation process is done over 7 OFDM symbols.

The samples of the non allocated resource blocks within the transmission bandwidth configuration in the post FFT domain are called Y(f,t) (f covering the non allocated subcarriers within the transmission bandwidth configuration, t covering the OFDM symbols during 1 slot).

# E.5.9 Derivation of the results

## E.5.9.1 $EVM_{PUCCH}$

For $EVM_{PUCCH}$ create two sets of Z'(f,t)., according to the timing " $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2" using the equalizer coefficients from E.5.8

The $EVM_{PUCCH}$ is the difference between the ideal waveform and the measured and equalized waveform for the allocated RB(s)

$$EVM_{PUCCH} = \sqrt{\frac{\sum_{t \in T} \sum_{f \in F} \left| Z'(f,t) - I(f,t) \right|^2}{|T| \cdot P_0 \cdot |F|}},$$

where

the OFDM symbols next to slot boarders (instant of band edge alternation) are excluded:

t covers less than the count of demodulated symbols in the slot (|T|= 5)

f covers the count of subcarriers within the allocated bandwidth. (|F|=12)

$Z'(f,t)$ are the samples of the signal evaluated for the $EVM_{PUCCH}$

$I(f,t)$ is the ideal signal reconstructed by the measurement equipment, and

$P_0$ is the average power of the ideal signal. For normalized modulation symbols $P_0$ is equal to 1.

From the acquired samples 40 $EVM_{PUCCH}$ value can be derived, 20 values for the timing $\Delta\tilde{c}$ –W/2 and 20 values for the timing $\Delta\tilde{c}$ +W/2

## E.5.9.2 Averaged $EVM_{PUCCH}$

$EVM_{PUCCH}$ is averaged over all basic $EVM_{PUCCH}$ measurements

The averaging comprises 20 UL slots (for frame structure 2: excluding special fields(UpPTS))

$$\overline{EVM}_{PUCCH} = \sqrt{\frac{1}{20}\sum_{i=1}^{20} EVM_{PUCCH\,i}^{2}}$$

The averaging is done separately for timing $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2 leading to $\overline{EVM}_{PUCCH,low}$ and $\overline{EVM}_{PUCCH,high}$

$EVM_{PUCCH,final} = \max(\overline{EVM}_{PUCCH,low}, \overline{EVM}_{PUCCH,high})$ is compared against the test requirements.

## E.5.9.3 In-band emissions measurement

The in-band emissions are a measure of the interference falling into the non-allocated resources blocks

Create one set of Y(t,f) per slot according to the timing "$\Delta\tilde{c}$"

For the non-allocated RBs the in-band emissions are calculated as follows

$$Emissions_{absolute}(\Delta_{RB}) = \begin{cases} \frac{1}{|T_s|}\sum_{t\in T_s} \sum_{\max(f_{\min},(c_l+12\cdot\Delta_{RB}*\Delta f))}^{c_l+(12\cdot\Delta_{RB}+11)*\Delta f} |Y(t,f)|^2, \Delta_{RB} < 0 \\ \frac{1}{|T_s|}\sum_{t\in T_s} \sum_{c_h+(12\cdot\Delta_{RB}-11)*\Delta f}^{\min(f_{\max},(c_h+12\cdot\Delta_{RB}*\Delta f))} |Y(t,f)|^2, \Delta_{RB} > 0 \end{cases},$$

where

the upper formula represents the inband emissions below the allocated frequency block and the lower one the inband emissions above the allocated frequency block.

$T_s$ is a set of $|T_s|$ OFDM symbols in the measurement period,

$\Delta_{RB}$ is the starting frequency offset between the allocated RB and the measured non-allocated RB (e.g. $\Delta_{RB} = 1$ for the first upper or $\Delta_{RB} = -1$ for the first lower adjacent RB),

$f_{\min}$ and $f_{\max}$ are the lower and upper edge of the UL system BW,

$c_l$ and $c_h$ are the lower and upper edge of the allocated BW,

$\Delta f$ is 15kHz,and

$Y(t,f)$ is the frequency domain signal evaluated for in-band emissions as defined in the subsection E.5.8

The relative in-band emissions are, given by

$$Emissions_{relative}(\Delta_{RB}) = 10*\log_{10}\frac{Emissions_{absolute}(\Delta_{RB})}{\frac{1}{|T_s|\cdot L_{CRBs}}\sum_{t\in T_s}\sum_{c_l}^{c_l+(12\cdot L_{CRBs}-1)*\Delta f}|\mathrm{MS}(t,f)|^2}[dB]$$

where

$L_{CRBs}$ is the number of allocated RBs, which is always 1 in case of PUCCH

and $\mathrm{MS}(t,f)$ is the frequency domain samples for the allocated bandwidth, as defined in the subsection E.5.8

Although an exclusion period for EVM is applicable in E.5.9.1, the inband emissions measurement interval is defined over one complete slot in the time domain.

From the acquired samples 20 functions for inband emissions can be derived.

Since the PUCCH allocation is always on the upper or lower band-edge, the opposite to the allocated one represents the IQ image, and the remaining inner RBs represent the general inband emissions. They are compared against different limits.

# E.6 EVM for PRACH

The description below is generic in the sense that all 5 PRACH formats are covered. The numbers, used in the text below are taken from PRACH format#0 without excluding the other formats. The sampling rate for the PUSCH, 30.72 Msps in the time domain, is re-used for the PRACH. The carrier spacing of the PUSCH is 12 (format 0 to 3) and 2 (format 4) times of the PRACH. This results in an oversampling factor of 12 (format 0 to 3) and 2 (format 4), when acquiring the time samples for the PRACH. The pre-FFT algorithms (clauses E.6.6 and E.6.7) use all time samples, although oversampled. For the FFT the time samples are decimated by the factor of 12 (format 0 to 3) and 2 (format 4), resulting in the same FFT size as for the other transmit modulation tests (2048). Decimation requires a decision, which samples are used and which ones are rejected. The algorithm in E.6.6, Timing of the FFT window, can also be used the decide about the used samples.

## E.6.1 Basic principle

The basis principle is the same as described in E.2.1

## E.6.2 Output signal of the TX under test

The output signal of the TX under test is processed same as described in E.2.2

The measurement period is different:

- 2 PRACH preambles are recorded for format 0and 1,
- 1 PRACH preamble is recorded for format 2 and 3, each containing 1 CP and 2 preamble sequences
- 10 RPRACH preambles are recorded for format 4.

## E.6.3 Reference signal

The test description in 6.5.2.1.4.1A is based on non contention based access:

- PRACH configuration index (responsible for Preamble format, System frame number and subframe number)
- Preamble ID
- Preamble power

signalled to the UE, defines the reference signal unambiguously, such that no demodulation process is necessary to gain the reference signal.

The reference signal i(v) is constructed by the measuring equipment according to the relevant TX specifications, using the following parameters: the applicable Zadoff Chu sequence, nominal carrier frequency, nominal amplitude and phase for each subcarrier, nominal timing, no carrier leakage. It is represented as a sequence of samples at a sampling rate of 30.72 Msps in the time domain.

## E.6.4 Measurement results

The measurement result is:

- $EVM_{PRACH}$

## E.6.5 Measurement points

The measurement points are illustrated in the figure below:

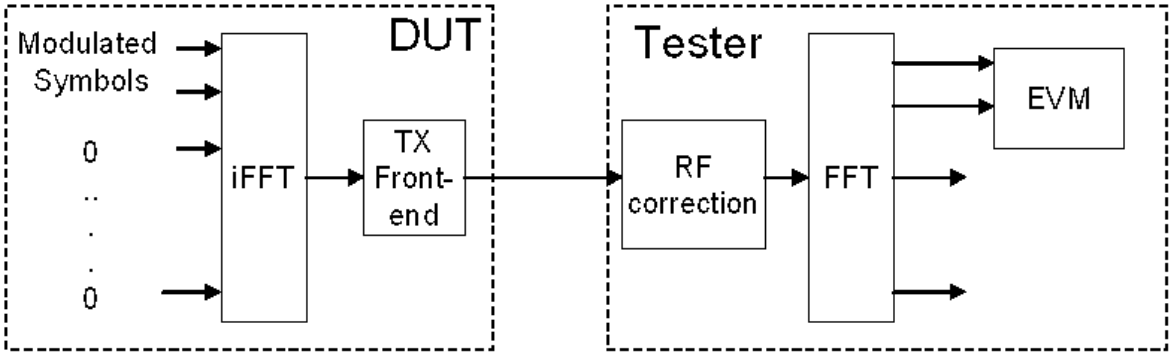



**Figure E.6.5-1**

## E.6.6 Pre FFT minimization process

The pre-FFT minimization process is applied to each PRACH preamble separately. The time period for the pre- FFT minimisation process includes the complete CP and Zadoff-Chu sequence (in other words, the power transition period is per definition outside of this time period) Sample timing, Carrier frequency and I/Q offset in z(v) are jointly varied in order to minimise the difference between z(v) and i(v). Best fit (minimum difference) is achieved when the RMS difference value between z(v) and i(v) is an absolute minimum.

After this process the samples z(v) are called $z^0(v)$.

RF error, and carrier leakage are necessary for best fit of the measured signal towards the ideal signal in the pre FFT domain. However they are not used to compare them against the limits.

## E.6.7 Timing of the FFT window

The FFT window length is 24576 samples for preamble format 0, however in the measurement period is at least 27744 samples are taken. The position in time for FFT must be determined.

In an ideal signal, the FFT may start at any instant within the cyclic prefix without causing an error. The TX filter, however, reduces the window. The EVM requirements shall be met within a window W<CP.

The reference instant for the FFT start is the centre of the reduced window, called $\Delta\tilde{c}$ ,

EVM is measured at the following two instants: $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2.

The timing of the measured signal $z^0(v)$ with respect to the ideal signal i(v) is determined in the pre FFT domain as follows:

Correlation between $z^0(v)$ and i(v) will result in a correlation peak. The meaning of the correlation peak is approx. the "impulse response" of the TX filter. The correlation peak, (the highest, or in case of more than one, the earliest) indicates the timing in the measured signal with respect to the ideal signal.

W is different for different preamble formats and shown in TableE.6.7-1.

**Table E.6.7-1EVM window length for PRACH**

| Preamble format | Cyclic prefix length[1] $N_{cp}$ | Nominal FFT size[2] | EVM window length *W* in FFT samples | Ratio of *W* to CP[3] |
|---|---|---|---|---|
| 0 | 3168 | 24576 | 3072 | 96.7% |
| 1 | 21024 | 24576 | 20928 | 99.5% |
| 2 | 6240 | 49152 | 6144 | 98.5% |
| 3 | 21024 | 49152 | 20928 | 99.5% |
| 4 | 448 | 4096 | 432 | 96.4% |
| Note 1: The unit is number of samples, sampling rate of 30.72MHz is assumed<br>Note 2: Decimation of time samples by 12(format 0 to 3) and factor 2 (format 4) is assumed, leading to a uniform FFT size of 2048 for all formats.<br>Note 3: These percentages are informative | | | | |

The number of samples, used for FFT is reduced compared to $z^0(\nu)$. This subset of samples is called $z''(\nu)$.

The sample frequency 30.72 MHz is oversampled with respect to the PRACH-subcarrier spacing of 1.25kHz (format 0 to 3) and 7.5kHz (format 4). EVM is based on 2048 samples per PRACH preamble and requires decimation of the time samples by the factor of 12 (format 0 to 3) and factor 2 (format 4). The final number of samples per PRACH preamble, used for FFT is reduced compared to $z''(\nu)$ by the factor of 12 (format 0 to 3) and factor 2 (format 4). This subset of samples is called $z'(\nu)$.

# E.6.8 Post FFT equalisation

Equalisation is not applicable for the PRACH.

# E.6.9 Derivation of the results

## E.6.9.1 $EVM_{PRACH}$

Perform FFT on $z'(\nu)$ and $i(\nu)$ using the FFT timing $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2.

For format 2 and 3 the first and the repeated preamble sequence are FFT-converted separately. using the standard FFT length 0f 2048

The $EVM_{PRACH}$ is the difference between the ideal waveform and the measured and equalized waveform for the allocated RB(s).

$$EVM_{PRACH} = \sqrt{\frac{\sum_{f \in F} \left| Z'(f) - I(f) \right|^2}{N_{ZC} \cdot P_0}},$$

where

f covers the count of demodulated symbols within the allocated bandwidth.

$Z'(f)$ are the samples of the signal evaluated for the $EVM_{PRACH}$

$I(f)$ is the ideal signal reconstructed by the measurement equipment, and

$P_0$ is the average power of the ideal signal. For normalized modulation symbols $P_0$ is equal to 1.

$N_{ZC}$ is random access preamble sequence length.

From the acquired samples 4 $EVM_{PRACH}$ value can be derived, 2 values for the timing $\Delta\tilde{c}$ –W/2 and 2 values for the timing $\Delta\tilde{c}$ +W/2 (4 and 2 applies for format 0,1,2,3. 20 and 10 applies for format 4).

### E.6.9.2 Averaged $EVM_{PRACH}$

$EVM_{PRACH}$ is averaged over all basic $EVM_{PRACH}$ measurements

$$\overline{EVM}_{PRACH} = \sqrt{\frac{1}{2}\sum_{i=1}^{2} EVM_{PRACH\,i}^{2}}$$

(i= 2 applies for format 0,1,2,3. i= 10 applies for format 4)

The averaging is done separately for timing $\Delta\tilde{c}$ –W/2 and $\Delta\tilde{c}$ +W/2 leading to $\overline{EVM}_{PRACH,low}$ and $\overline{EVM}_{PRACH,high}$

$EVM_{PRACH,final} = \max(\overline{EVM}_{PRACH,low}, \overline{EVM}_{PRACH,high})$ is compared against the test requirements.

# E.7 EVM with exclusion period

## E.7.1 General

EVM with exclusion periods is defined in clause 6.5.2.1.1, third paragraph. For PUCCH entire symbols are excluded, if applicable. For PUSCH fractions of symbols are excluded, if applicable. The exclusion period for PUSCH is defined at the air interface, leading to exclusion periods in the EVM domain. The necessary mapping is described in this clause.

## E.7.2 The model

The exclusion period in the time domain has corresponding periods in the quasi time domains (Table E.7.2). The mapping of corresponding periods needs only scaling and cyclic shifting.

The algorithm below uses a sampling frequency 30.72 MHz and FFT-width 2048 for all bandwidths. Bandwidth-adapted sampling frequencies and FFT-widths are not excluded. Only normal cyclic prefix is mentioned in the model without excluding the extended CP.

**Table. E.7.2: Model for mapping exclusion period in the time domain**

| | TX | | | | Channel | EVM meter | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Operation | | DFT | | iFFT | | | FFT | | iDFT | |
| Meaning | Modulation symbols | | Precoded symbols | | BB samples | BB samples | | Precoded symbols | | demodulated symbols |
| No of samples | allocated Sub Carriers | | allocated subcarriers + unallocated subcarriers = 2048 | | allocated subcarriers + unallocated subcarriers + CP samples | 2048, position depending on EVM window | | allocated subcarriers | | allocated subcarriers |
| Domain | Quasi time domain | | Frequency domain | | Time domain | Time domain | | Frequency domain | | Quasi time domain |
| text below | | 1 | 2 | 3 | 4 | 7 | 7 | 8 | 9 | 11 |

1. A sequence of complex valued modulation symbols are Transform-Precoded (DFT) according to 36.211 clause 5.3.3. The size of this transformation is the number of allocated subcarriers.

2. The outcome of (1) is supplemented by 0 for the non allocated subcarriers. In sum 2048 subcarriers.

3. The baseband time signal (without CP) is then calculated by a iFFT according to 36.211 clause 5.6

4. (3) is then supplemented by a cyclic prefix (144 or 160 samples) leading to 2192 or 2208 samples. (144 CP samples = 144 tail samples from the data field)

5. (4) is transmitted over the channel and sampled by the EVM meter.

6. In case of an exclusion period those samples of (5) are marked, where the exclusion applies. The exclusion period is an unbroken leading or lagging exclusion period next to a subframe or timeslot boarder.

7.Depending on early or late EVM-window a subset of 2048 samples (out of 2192 or 2208 samples) are the input for the subsequent FFT . These samples may or may not comprise marked samples. The result are 2048 frequency domain samples.

8. The non allocated subcarriers are removed from the 2048 samples.

9. (8) is then iDFT transformed. The result are demodulated complex valued symbols in the same domain as (1)

10. Step 7, 8 and 9 are modified by an equalizer algorithm.
    For the purpose of this clause, the equalizer partly re-does step 4 (CP insertion):
    The equalizer algorithm cuts that subset of CP samples, covered by the FFT, from the head and copies it to the tail of the data field.

11. The result of (10) is: complex valued symbols in the same sequence as in (1) They are compared with (1) symbol by symbol for EVM. Due to exclusion in the time domain (6) we have marked corresponding symbols, which are disregarded for EVM.

12. From step 1 to 4 the number of samples is expanded. A subset of expanded samples is marked as excluded. Form step 6 to step 9 the number of samples is compressed, leading to a non integer number of samples, marked as excluded. The number of marked samples in this domain is rounded up at the expense of the EVM samples

## E.7.3 Illustration

The figures below illustrate the cyclic shift due to the equalizer and scaling.

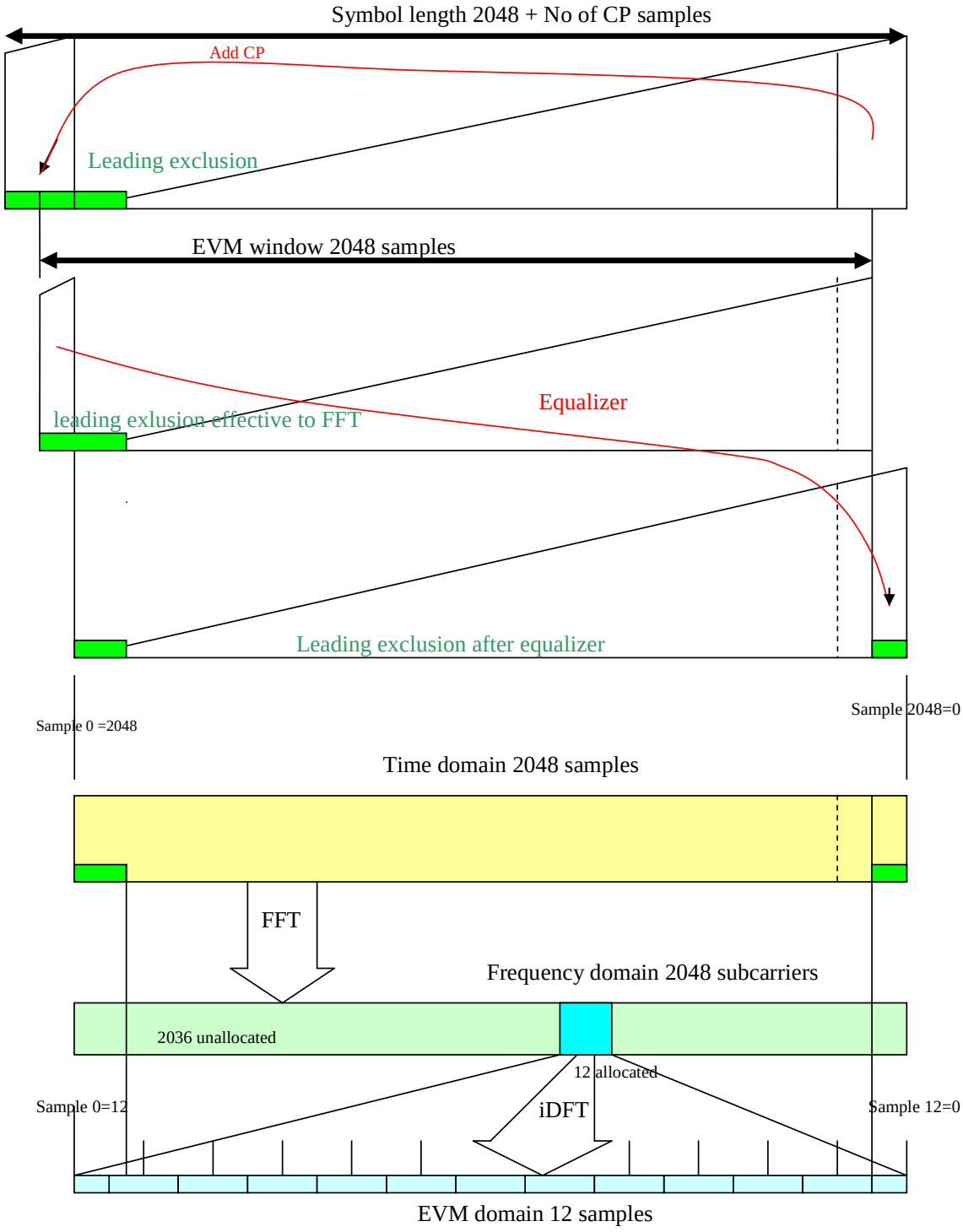



**Figure E.7.3-1: leading exclusion period (when number of RBs=1)**

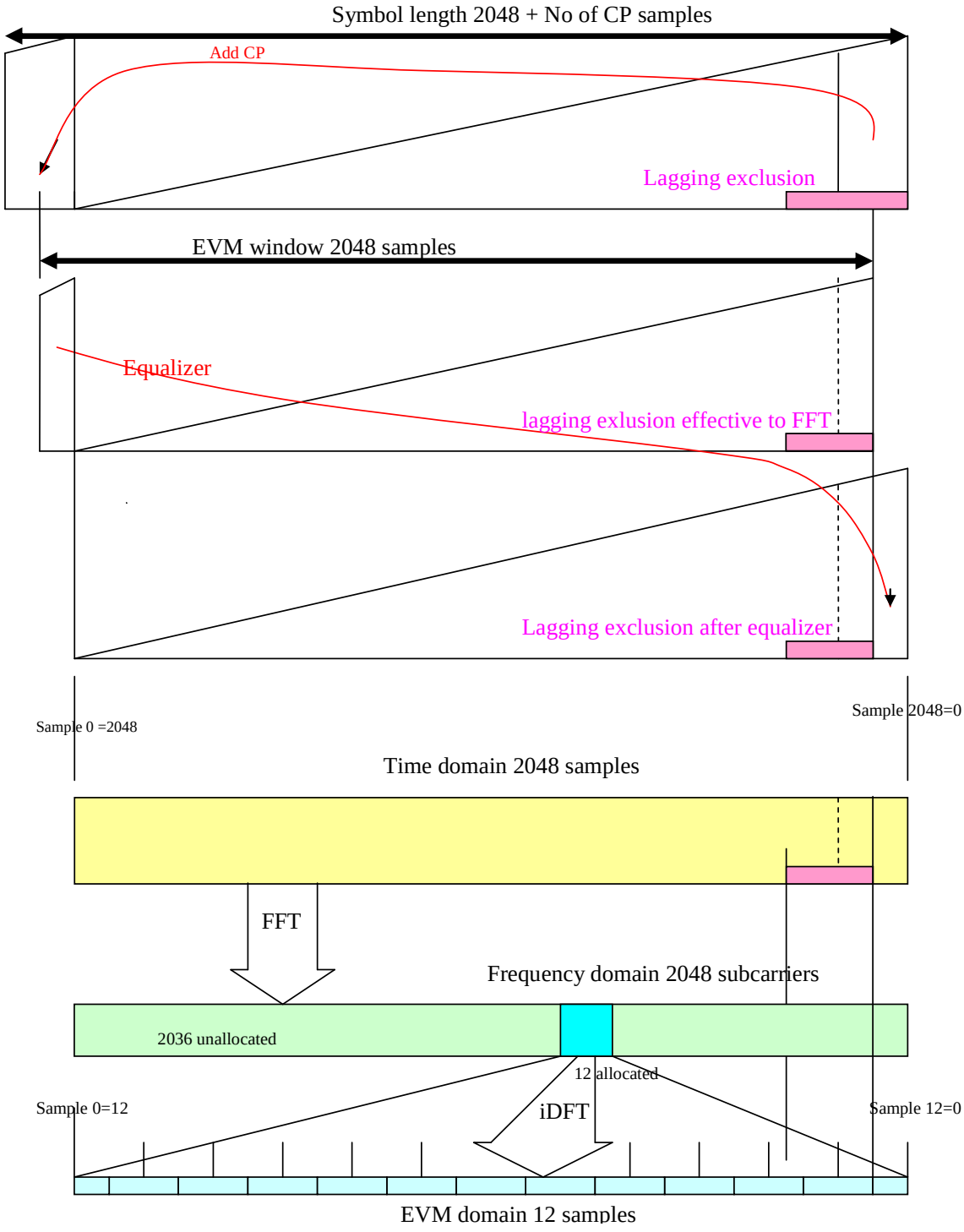



**Figure E.7.3-2: lagging exclusion period (when number of RBs=1)**

Legend to figure E.7.3-1

The figure contains 6 related subfigures.

The upper 3 triangles illustrate the cyclic shift due to EVM windowing and equalizer.

The lower 3 boxes illustrate the mapping from the time domain, where the exclusion period is defined, into the EVM domain, where EVM samples are actually excluded.

Cyclic shift

The leading exclusion period covers the entire CP and a part of the data field. The EVM window masks a part of the exclusion period. For the purpose of this annex, the equalizer re-arranges the time domain samples into the original order, splitting the exclusion period into two parts.

Mapping

The split exclusion period (after cyclic shift) is copied from above. The frequency domain is derived by Fast Fourier Transformation from the time domain and represent the frequency domain with 2048 subcarriers, 12 of them are allocated, the rest is unallocated. The 12 allocated subcarriers are iDFT transformed into the EVM domain comprising 12 samples. Note that all 3 domains are displayed cyclically: the leftmost sample is identical to the rightmost sample.

The two transformations map the time domain into the EVM domain, carrying out a compression of samples 12/2048. In spite of the compression, there is a correspondence of ranges in the time domain and in the EVM domain. One sample in the EVM domain comprises a range, which is influenced from (and only from) the equivalent samples in the time domain vertically above. Note that this correspondence holds irrespective of the position of the 12 allocated samples in the frequency domain.

**Example leading exclusion (figure E.7.3-1)**

| Sample No in the EVM domain | Influence(exclusion) from the time domain | EVM exclusion |
|---|---|---|
| 12=0 | full | excluded |
| 1 | partly | excluded |
| 2 to 11 | none | counted |

**Example lagging exclusion example (figure E.7.3-2)**

| Sample No in the EVM domain | Influence(exclusion) from the time domain | EVM exclusion |
|---|---|---|
| 12=0 | none | counted |
| 1 to 9 | none | counted |
| 10 | partly | excluded |
| 11 | full | excluded |

## E.7.4 Formula

The exclusion period is defined in μs at the air interface.

Convert the μs's into No of samples in the time domain.

No of exclusion samples (before EVM windowing and equalizer) is calculated from:

No of exclusion sample = ceil (30.72 * Exclusion μs)

then, EVM windowing and equalizer is applied by 2048 samples based cyclic shift process.

(the upper 3 triangles illustrate in fig. E.7.3-1 and -2.)

Determine the indices **k**, to be excluded in the time domain, according to fig. E.7.3-1 and -2

(after application of EVM windowing and equaliser, original sample order,

**k** = subset from the set (0 to 2047) )

The indices **l** in the EVM domain, to be excluded, are:

$\mathbf{l}$ = [ round ($\mathbf{k}$*12* $L_{CRBs}$ / 2048)]mod(12*$L_{CRBs}$)

with $L_{CRBs}$ number of allocated resource blocks

# Annex F (normative): Measurement uncertainties and Test Tolerances

## F.1 Acceptable uncertainty of Test System (normative)

The maximum acceptable uncertainty of the Test System is specified below for each test, where appropriate. The Test System shall enable the stimulus signals in the test case to be adjusted to within the specified range, and the equipment under test to be measured with an uncertainty not exceeding the specified values. All ranges and uncertainties are absolute values, and are valid for a confidence level of 95 %, unless otherwise stated.

A confidence level of 95 % is the measurement uncertainty tolerance interval for a specific measurement that contains 95 % of the performance of a population of test equipment.

For RF tests it should be noted that the uncertainties in clause F.1 apply to the Test System operating into a nominal 50 ohm load and do not include system effects due to mismatch between the DUT and the Test System.

The downlink signal uncertainties apply at each receiver antenna connector.

### F.1.1 Measurement of test environments

The measurement accuracy of the UE test environments defined in TS 36.508 subclause 4.1, Test environments shall be.

- Pressure ±5 kPa.
- Temperature ±2 degrees.
- Relative Humidity ±5 %.
- DC Voltage ±1,0 %.
- AC Voltage ±1,5 %.
- Vibration 10 %.
- Vibration frequency 0,1 Hz.

The above values shall apply unless the test environment is otherwise controlled and the specification for the control of the test environment specifies the uncertainty for the parameter.

## F.1.2 Measurement of transmitter

**Table F.1.2-1: Maximum Test System Uncertainty for transmitter tests**

| Subclause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 6.2.2 UE Maximum Output Power | ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.2.3 Maximum Power Reduction | ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.2.3A Maximum Power Reduction (MPR) for CA | TBD | |
| 6.2.4 UE Maximum Output Power with additional requirements | ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.2.5 Configured UE transmitted Output Power | ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.3.2 Minimum Output Power | ±1.0 dB, f ≤ 3.0GHz<br>±1.3 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.3.3 Transmission ON/OFF Power | Transmission OFF Power: ±1.5 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.3.4.1 General ON/OFF time mask | Transmission ON/OFF Power: ±1.5 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.3.4.2 PRACH and SRS time mask | Transmission ON/OFF Power: ±1.5 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.3.5.1 Power Control Absolute power tolerance | ±1.0 dB, f ≤ 3.0GHz<br>±1.4 dB, 3.0GHz < f ≤ 4.2GHz | Overall system uncertainty comprises two quantities:<br>1. Downlink signal level uncert<br>2. Uplink level measurement uncertainty<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared.<br>Test System uncertainty = [SQRT (DL level uncert $^2$ + UL measurement uncert $^2$)]<br>f ≤ 3.0GHz<br>DL signal level uncert ± 0.7dB<br>UL meas't uncert ± 0.7dB<br><br>3.0GHz < f ≤ 4.2GHz<br>DL signal level uncert ± 1.0dB<br>UL meas't uncert ± 1.0dB |
| 6.3.5.2 Power Control Relative power tolerance | ±0.7 dB | |
| 6.3.5.3 Aggregate power control tolerance | ±0.7 dB | |
| 6.5.1 Frequency Error | ±15 Hz<br>DL Signal level: ±0.7 dB, f ≤ 3.0GHz<br>DL Signal level: ±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.5.2.1 Error Vector Magnitude | PUSCH : ± 2.5%<br>PUCCH: ± 2.5%<br>PRACH: ± 2.5% | |
| 6.5.2.1A PUSCH-EVM with exclusion period | PUSCH : ± 2.5% | |

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| 6.5.2.2 Carrier leakage | ±0.8dB | |
| 6.5.2.3 In-band emissions for non allocated RB | ±0.8dB | |
| 6.5.2.4 EVM equalizer Spectrum flatness | ±1.4dB | |
| 6.6.1 Occupied bandwidth | 1.4MHz, 3MHz: 30kHz<br>5MHz, 10MHz: 100kHz<br>15MHz, 20MHz: 300kHz | |
| 6.6.2.1 Spectrum Emission Mask | ±1.5 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.6.2.2 Additional Spectrum Emission Mask | ±1.5 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.6.2.3 Adjacent Channel Leakage power Ratio | ±0.8 dB | |
| 6.6.2.3A Adjacent Channel Leakage power Ratio for CA | TBD | |
| 6.6.2.4 Additional ACLR requirements | ±0.8 dB | |
| 6.6.3.1 Transmitter Spurious emissions | 9kHz < f ≦ 4 GHz: ± 2.0 dB<br>4 GHz < f ≦ 19 GHz: ± 4.0 dB | |
| 6.6.3.2 Spurious emission band UE co-existence | ± 2.0 dB for results > -60 dBm, f ≤ 3.0GHz<br>±2.5 dB, 3.0GHz < f ≤ 4.2GHz<br>± 3.0 dB for results ≤ -60 dBm, f ≤ 3.0GHz<br>±3.6 dB, 3.0GHz < f ≤ 4.2GHz | |
| 6.6.3.2_1 Spurious emission band UE co-existence (Release 9 and forward) | ± 2.0 dB for results > -60 dBm<br>± 3.0 dB for results ≤ -60 dBm | |
| 6.6.3.3 Additional spurious emissions | 9kHz < f ≦ 4 GHz: ± 2.0 dB<br><br>NS-07<br>769 ≤ f ≤ 775 MHz: ± 1.5 dB | |

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|---|---|---|
| 6.7 Transmit intermodulation | ± 2.6 dB, f ≤ 3.0GHz<br>±3.6 dB, 3.0GHz < f ≤ 4.2GHz | Overall system uncertainty comprises four quantities:<br>1. Wanted signal setting error<br>2. CW Interferer level error<br>3. Wanted signal meas. error<br>4. Intermodulation product measurement error<br><br>The relative level of the wanted signal and the CW interferer has 2 x effect on the intermodulation product.<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared to provide the combined effect.<br><br>Test System uncertainty = SQRT [(2 x SQRT (Wanted setting_error$^2$ + CW_level_error$^2$ ) $^2$ + Wanted_level_meas error$^2$ + Intermodulation product measurement error$^2$]<br><br>f ≤ 3.0GHz<br>Wanted signal setting ± 0.7dB<br>CW Interferer level ± 1.0dB<br>Wanted signal meas ± 0.7dB<br>Intermodulation product measurement error ± 0.7dB<br><br>3.0GHz < f ≤ 4.2GHz<br>Wanted signal setting ± 1.0dB<br>CW Interferer level ± 1.3dB<br>Wanted signal meas ± 1.0dB<br>Intermodulation product measurement error ± 1.0dB |

## F.1.3 Measurement of receiver

**Table F.1.3-1: Maximum Test System Uncertainty for receiver tests**

| Subclause | Maximum Test System Uncertainty[1] | Derivation of Test System Uncertainty |
|---|---|---|
| 7.3.1 Reference sensitivity power level; Minimum requirements (QPSK) | Downlink power ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 7.4 Maximum input level | Downlink power ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz<br>Uplink power measurement ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | |
| 7.5 Adjacent Channel Selectivity (ACS) | ACS value ±1.1 dB, f ≤ 3.0GHz<br>±1.5 dB, 3.0GHz < f ≤ 4.2GHz<br>Uplink power measurement ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | Overall ACS uncertainty comprises three quantities:<br>1. Wanted signal level error<br>2. Interferer signal level error<br>3. Additional impact of interferer ACLR<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared to provide the ratio error of the two signals. The interferer ACLR effect is systematic, and is added aritmetically.<br>Test System uncertainty = [SQRT (wanted_level_error$^2$ + interferer_level_error$^2$)] + ACLR effect.<br>f ≤ 3.0GHz<br>Wanted signal level ± 0.7dB<br>Interferer signal level ± 0.7dB<br>3.0GHz < f ≤ 4.2GHz<br>Wanted signal level ± 1.0dB<br>Interferer signal level ± 1.0dB<br>f ≤ 4.2GHz<br>Impact of interferer ACLR 0.1dB |

| | | |
|---|---|---|
| 7.6.1 In-band blocking | Blocking ±1.4 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz<br>Uplink power measurement ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | Overall blockinguncertainty can have these contributions:<br><br>1. Wanted signal level error<br>2. Interferer signal level error<br>3. Interferer ACLR<br>4. Interferer broadband noise<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared to provide the ratio error of the two signals. The Interferer ACLR or Broadband noise effect is systematic, and is added aritmetically.<br><br>Test System uncertainty = [SQRT (wanted_level_error$^2$ + interferer_level_error$^2$)] + ACLR effect + Broadband noise effect.<br><br><u>In-band blocking, using modulated interferer:</u><br>f ≤ 3.0GHz<br>Wanted signal level ± 0.7dB<br>Interferer signal level:<br>± 0.7dB<br>3.0GHz < f ≤ 4.2GHz<br>Wanted signal level ± 1.0dB<br>Interferer signal level ± 1.0dB<br><br>f ≤ 4.2GHz<br>Interferer ACLR 0.4dB<br>Broadband noise not applicable |
| 7.6.2 Out of-band blocking | Wanted signal f ≤ 3.0GHz<br>Blocking, 1MHz < $f_{interferer}$ ≦ 3 GHz: ±1.3 dB<br>Blocking, 3 GHz < $f_{interferer}$ ≦ 12.75 GHz: ±3.2 dB<br>Uplink power measurement ±0.7 dB<br><br>Wanted signal 3.0GHz < f ≤ 4.2GHz<br>Blocking, 1MHz < $f_{interferer}$ ≦ 3 GHz: ±1.5 dB<br>Blocking, 3 GHz < $f_{interferer}$ ≦ 12.75 GHz: ±3.3 dB<br>Uplink power measurement ±1.0 dB | <u>Out of band blocking, using CW interferer:</u><br>f ≤ 3.0GHz<br>Wanted signal level ± 0.7dB<br>3.0GHz < f ≤ 4.2GHz<br>Wanted signal level ± 1.0dB<br>Interferer signal level:<br>± 1.0dB up to 3GHz<br>± 3.0dB up to 12.75GHz<br>Interferer ACLR not applicable<br>Impact of interferer<br>Broadband noise 0.1dB<br><br>Figures are combined to give Test System uncertainty, using formula given for 7.6.1 |
| 7.6.3 Narrow band blocking | Blocking ±1.3 dB, f ≤ 3.0GHz<br>±1.8 dB, 3.0GHz < f ≤ 4.2GHz<br>Uplink power measurement ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | <u>Narrow band blocking, using CW interferer:</u><br>Wanted signal level ± 0.7dB<br>Interferer signal level:<br>± 1.0dB<br>Interferer ACLR not applicable<br>Impact of interferer<br>Broadband noise 0.1dB<br><br>Figures are combined to give Test System uncertainty, using formula given for 7.6.1 |
| 7.7 Spurious response | Same as 7.6.2 | Same as 7.6.2. |

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|---|---|---|
| 7.8.1 Wide band intermodulation | Intermodulation ±1.4 dB, f ≤ 3.0GHz<br>±2.6 dB, 3.0GHz < f ≤ 4.2GHz<br>Uplink power measurement ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz | Overall intermodulationuncertainty comprises three quantities:<br>1. Wanted signal level error<br>2. CW Interferer level error<br>3. Modulated Interferer level error<br><br>Effect of interferer ACLR has not been included as modulated interferer has larger frequency offset<br><br>The effect of the closer CW signal has twice the effect.<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared to provide the combined effect of the three signals.<br><br>Test System uncertainty = SQRT [(2 x CW_level_error)$^2$ +(mod interferer_level_error)$^2$ +(wanted signal_level_error)$^2$]<br><br>f ≤ 3.0GHz<br>Wanted signal level ± 0.7dB<br>CW Interferer level ± 0.5dB<br>Mod Interferer level ± 0.7dB<br>3.0GHz < f ≤ 4.2GHz<br>Wanted signal level ± 1.0dB<br>CW Interferer level ± 0.8dB<br>Mod Interferer level ± 1.0dB |
| 7.9 Spurious emissions | 30MHz ≤ f ≤ 4.0GHz: ± 2.0 dB<br>4 GHz < f ≤ 19 GHz: ± 4.0 dB | |
| Note 1: Unless otherwise noted, only the Test System stimulus error is considered here. The effect of errors in the throughput measurements due to finite test duration is not considered. | | |

# F.1.4 Measurement of performance requirements

**Table F.1.4-1: Maximum Test System Uncertainty for Performance Requirements**

| Subclause | Maximum Test System Uncertainty[1] | Derivation of Test System Uncertainty |
|---|---|---|
| 8.2.1.1.1 Multiple PRBs<br>- Propagation Condition EVA5<br>- Propagation Condition ETU70<br>- Propagation Condition ETU300 | ± 0.8 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.1.1.1 Multiple PRBs<br>- Propagation Condition HST | ± 0.6 dB | Overall system uncertainty for HST condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |

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|---|---|---|
| 8.2.1.1.1 Single PRB<br>- Propagation Condition ETU70 | ± 0.8 dB | Overall system uncertainty for fading condition comprises three quantities:<br>1. Average Signal-to-noise ratio uncertainty<br>2. Signal-to noise ratio variation for single PRB<br>3. Fading profile power uncertainty<br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty$^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx |
| 8.2.1.1.2 Single PRB | ± 0.8 dB | Same as 8.2.1.1.1 Single PRB |
| 8.2.1.2.1<br>- Propagation Condition EVA5 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + Fading profile power uncertainty$^2$ + (0.25 x AWGN flatness and signal flatness)$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.1.2.1<br>- Propagation Condition HST | ± 0.6 dB | Overall system uncertainty for HST condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty$^2$ + (0.25 x AWGN flatness and signal flatness)$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.1.2.2 | ± 0.9 dB | Same as 8.2.1.2.1 Propagation Condition EVA5 |

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| 8.2.1.3.1 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for MIMO<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.1.3.2 | ± 0.9 dB | Same as 8.2.1.3.1 |
| 8.2.1.4.1 | ± 0.9 dB | Same as 8.2.1.3.1 |
| 8.2.1.4.2 | ± 0.9 dB | Same as 8.2.1.3.1 |
| 8.2.2.1.1 Multiple PRBs<br>- Propagation Condition EVA5<br>- Propagation Condition ETU70<br>- Propagation Condition ETU300 | ± 0.8 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br>AWGN flatness and signal flatness ±2.0 dB |

| | | |
|---|---|---|
| 8.2.2.1.1 Multiple PRBs<br>- Propagation Condition HST | ± 0.6 dB | Overall system uncertainty for HST condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.2.1.1 Single PRB<br>- Propagation Condition ETU70 | ± 0.8 dB | Overall system uncertainty for fading condition comprises three quantities:<br>1. Average Signal-to-noise ratio uncertainty<br>2. Signal-to noise ratio variation for single PRB<br>3. Fading profile power uncertainty<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>Test System uncertainty = SQRT (Average signal-to-noise ratio uncertainty $^2$ + Signal-to-noise ratio variation$^2$ + Fading profile power uncertainty $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Signal-to-noise ratio variation ±0.5 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx |
| 8.2.2.1.2 Single PRB | ± 0.8 dB | Same as 8.2.2.1.1 Single PRB |
| 8.2.2.2.1<br>- Propagation Condition EVA5 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB |

| | | |
|---|---|---|
| 8.2.2.2.1<br>- Propagation Condition HST | ± 0.6 dB | Overall system uncertainty for HST condition comprises two quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.2.2.2 | ± 0.9 dB | Same as 8.2.2.2.1 Propagation Condition EVA5 |
| 8.2.2.3.1 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for MIMO<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.2.2.3.2 | ± 0.9 dB | Same as 8.2.2.3.1 |
| 8.2.2.4.1 | ± 0.9 dB | Same as 8.2.2.3.1 |
| 8.2.2.4.2 | ± 0.9 dB | Same as 8.2.2.3.1 |

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|---|---|---|
| 8.3.2.1.1 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB |
| 8.3.2.1.1_1 | ± 0.9 dB | Same as 8.3.2.1.1 |
| 8.3.2.1.2 | ± 0.9 dB | Same as 8.2.2.3.1 |
| 8.3.2.1.3 | ± 0.9 dB | Same as 8.2.2.3.1 |
| 8.3.2.2.1 | ± 0.9 dB | Same as 8.2.2.3.1 |
| 8.4.1.1 | ± 0.8 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.2 dB |

| | | |
|---|---|---|
| 8.4.1.2.1 | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |
| 8.4.1.2.2 | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for MIMO<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |

| | | |
|---|---|---|
| 8.4.2.1 | ± 0.8 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.2 dB |
| 8.4.2.2.1 | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |

| | | |
|---|---|---|
| 8.4.2.2.2 | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for MIMO<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |
| 8.5.1.1 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |

| | | |
|---|---|---|
| 8.5.1.2.1 | ± 1.1 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.6 dB |
| 8.5.1.2.2 | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for MIMO<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |

| | | |
|---|---|---|
| 8.5.2.1 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for single Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |
| 8.5.2.2.1 | ± 1.1 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.6 dB |

| | | |
|---|---|---|
| 8.5.2.2.2 | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. Result variation due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + variation due to finite test time $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for MIMO<br>AWGN flatness and signal flatness ±2.0 dB<br>Result variation due to finite test time ±0.4 dB |
| 8.7.1 FDD sustained data rate performance | Downlink absolute power uncertainty, averaged over $BW_{Config}$ ±1.0 dB<br><br>Downlink EVM ≤ 3% | 3% EVM is equivalent to a Test system downlink SNR of 30.5dB. The noise from the Test system is then sufficiently below that required for the UE to demodulate the signal with the required % success rate. Under these conditions the UE throughput is limited by the Reference measurement channel and the UE capability, and not by the Test system EVM. |
| 8.7.2 TDD sustained data rate performance | Same as 8.7.1 | Same as 8.7.1 |
| 10.1 | ± 0.9 dB | Overall system uncertainty for fading conditions comprises three quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br><br>Items 1, 2 and 3 are assumed to be uncorrelated so can be root sum squared:<br><br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br><br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for Tx Diversity<br>AWGN flatness and signal flatness ±2.0 dB |
| 10.2 | ± 0.9 dB | Same as 10.1 |
| [Other tests FFS] | | |
| In addition, the following Test System uncertainties and related constraints apply: | | |
| AWGN Bandwidth | | ≥ 1.08MHz, 2.7MHz, 4.5MHz, 9MHz, 13.5MHz, 18MHz;<br>$N_{RB}$ x 180kHz according to $BW_{Config}$ |

| | |
|---|---|
| AWGN absolute power uncertainty, averaged over $BW_{Config}$ | ±3 dB |
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{Config}$ | ±2 dB |
| AWGN peak to average ratio | ≥10 dB @0.001% |
| Signal-to noise ratio uncertainty, averaged over downlink transmission Bandwidth | ±0.3 dB |
| Signal-to noise ratio variation for any resource block, relative to average over downlink transmission Bandwidth | ±0.5 dB |
| Fading profile power uncertainty | Test-specific |
| Fading profile delay uncertainty, relative to frame timing | ±5 ns (excludes absolute errors related to baseband timing) |
| Note 1: Only the overall stimulus error is considered here. The effect of errors in the throughput measurements due to finite test duration is not considered.<br>Note 2: The AWGN parameters apply to all test cases except 8.7.1 and 8.7.2. The fading parameters apply to test cases using fading | |

## F.1.5 Measurement of Channel State Information reporting

**Table F.1.5-1: Maximum Test System Uncertainty for Channel State Information reporting**

| Subclause | Maximum Test System Uncertainty[1] | Derivation of Test System Uncertainty |
|---|---|---|
| 9.2.1.1 FDD CQI Reporting under AWGN conditions – PUCCH 1-0 | ± 0.3 dB | Signal-to-noise ratio uncertainty ±0.3 dB<br><br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br><br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.2.1.2 TDD CQI Reporting under AWGN conditions – PUCCH 1-0 | ± 0.3 dB | Same as 9.2.1.1 |
| 9.2.2.1 FDD CQI Reporting under AWGN conditions – PUCCH 1-1 | ± 0.3 dB | Signal-to-noise ratio uncertainty ±0.3 dB<br><br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br><br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.2.2.2 TDD CQI Reporting under AWGN conditions – PUCCH 1-1 | ± 0.3 dB | Same as 9.2.1.1 |
| 9.3.1.1.1 FDD CQI Reporting under fading conditions – PUSCH 3-0 | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br><br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br><br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br><br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.3.1.1.2 TDD CQI Reporting under fading conditions – PUSCH 3-0 | ± 0.6 dB | Same as 9.3.1.1.1 |
| 9.3.2.1.1 FDD CQI Reporting under fading conditions – PUCCH 1-0 | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br><br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br><br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br><br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.3.2.1.2 TDD CQI Reporting under fading conditions – PUCCH 1-0 | ± 0.6 dB | Same as 9.3.2.1.1 |

| | | |
|---|---|---|
| 9.3.3.1.1 FDD CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0 | ± 1.2 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Ior/Iot ratio uncertainty ±1.0 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Ior/Iot ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*Ior absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.3.3.1.2 TDD CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0 | ± 1.2 dB | Same as 9.3.3.1.1 |
| 9.3.4.1.1 FDD CQI Reporting under fading conditions – PUSCH 2-0 | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.3.4.1.2 TDD CQI Reporting under fading conditions – PUSCH 2-0 | ± 0.6 dB | Same as 9.3.4.1.1 |
| 9.3.4.2.1 FDD CQI Reporting under fading conditions – PUCCH 2-0 | ± 0.6 dB | Same as 9.3.4.1.1 |
| 9.3.4.2.2 TDD CQI Reporting under fading conditions – PUCCH 2-0 | ± 0.6 dB | Same as 9.3.4.1.1 |
| 9.4.1.1.1 FDD PMI Reporting – PUSCH 3-1 (Single PMI) | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.4.1.1.2 TDD PMI Reporting – PUSCH 3-1 (Single PMI) | ± 0.6 dB | Same as 9.4.1.1.1 |

| | | |
|---|---|---|
| 9.4.1.2.1 FDD PMI Reporting – PUCCH 2-1 (Single PMI) | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.4.1.2.2 TDD PMI Reporting – PUCCH 2-1 (Single PMI) | ± 0.6 dB | Same as 9.4.1.2.1 |
| 9.4.2.1.1 FDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.4.2.1.1_1 FDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.4.2.1.2 TDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | ± 0.6 dB | Same as 9.4.2.1.1 |
| 9.4.2.1.2_1 TDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | ± 0.6 dB | Same as 9.4.2.1.1 |
| 9.4.2.2.1 FDD PMI Reporting – PUSCH 2-2 (Multiple PMI) | ± 0.6 dB | Same as 9.4.2.1.1 |
| 9.4.2.2.2 TDD PMI Reporting – PUSCH 2-2 (Multiple PMI) | ± 0.6 dB | Same as 9.4.2.1.1 |

| | | |
|---|---|---|
| 9.5.1.1 FDD RI Reporting– PUCCH 1-1 | ± 0.6 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty ±0.5 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect*<br>*AWGN absolute power uncertainty ±3.0 dB not expected to have any significant effect* |
| 9.5.1.2 TDD RI Reporting– PUSCH 3-1 | ± 0.6 dB | Same as 9.5.1.1 |
| In addition, the following Test System uncertainties and related constraints apply: | | |
| AWGN Bandwidth | | ≥ 1.08MHz, 2.7MHz, 4.5MHz, 9MHz, 13.5MHz, 18MHz;<br>$N_{RB}$ x 180kHz according to $BW_{Config}$ |
| AWGN absolute power uncertainty, averaged over $BW_{Config}$ | | ±3 dB |
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{Config}$ | | ±2 dB |
| AWGN peak to average ratio | | ≥10 dB @0.001% |
| Signal-to noise ratio uncertainty, averaged over downlink transmission Bandwidth | | ±0.3 dB |
| Signal-to noise ratio variation for any resource block, relative to average over downlink transmission Bandwidth | | ±0.5 dB |
| Fading profile power uncertainty | | Test-specific |
| Fading profile delay uncertainty, relative to frame timing | | ±5 ns (excludes absolute errors related to baseband timing) |
| Note 1: Only the overall stimulus error is considered here. The effect of errors in the throughput measurements due to finite test duration is not considered.<br>Note 2: The AWGN parameters apply to all test cases except 9.3.3.1.1 and 9.3.3.1.2. The fading parameters apply to test cases using fading | | |

# F.2 Interpretation of measurement results (normative)

The measurement results returned by the Test System are compared – without any modification – against the Test Requirements as defined by the shared risk principle.

The Shared Risk principle is defined in ETR 273-1-2 clause 6.5.

The actual measurement uncertainty of the Test System for the measurement of each parameter shall be included in the test report.

The recorded value for the Test System uncertainty shall be, for each measurement, equal to or lower than the appropriate figure in clause F.1 of the present document.

If the Test System for a test is known to have a measurement uncertainty greater than that specified in clause F.1, it is still permitted to use this apparatus provided that an adjustment is made value as follows:

Any additional uncertainty in the Test System over and above that specified in clause F.1 shall be used to tighten the Test Requirement, making the test harder to pass. For some tests, for example receiver tests, this may require modification of stimulus signals. This procedure will ensure that a Test System not compliant with clause F.1does not increase the chance of passing a device under test where that device would otherwise have failed the test if a Test System compliant with clause F.1 had been used.

# F.3 Test Tolerance and Derivation of Test Requirements (informative)

The Test Requirements in the present document have been calculated by relaxing the Minimum Requirements of the core specification using the Test Tolerances defined in this clause. When the Test Tolerance is zero, the Test Requirement will be the same as the Minimum Requirement. When the Test Tolerance is non-zero, the Test Requirements will differ from the Minimum Requirements, and the formula used for the relaxation is given in this clause.

The Test Tolerances are derived from Test System uncertainties, regulatory requirements and criticality to system performance. As a result, the Test Tolerances may sometimes be set to zero.

The test tolerances should not be modified for any reason e.g. to take account of commonly known test system errors (such as mismatch, cable loss, etc.).

The downlink Test Tolerances apply at each receiver antenna connector.

## F.3.1 Measurement of test environments

The UE test environments are set to the values defined in TS 36.508 subclause 4.1, without any relaxation. The applied Test Tolerance is therfore zero.

## F.3.2 Measurement of transmitter

**Table F.3.2-1: Derivation of Test Requirements (Transmitter tests)**

| Test | Minimum Requirement in TS 36.101 | Test Tolerance (TT) | Test Requirement in TS 36.521-1 |
|---|---|---|---|
| 6.2.2 UE Maximum Output Power | f ≤ 3.0GHz | | Formula:<br>Upper limit + TT, Lower limit - TT |
| | Power class 1: [FFS] | 0.7 dB | Power class 1: [FFS] |
| | Power class 2: [FFS] | 0.7 dB | Power class 2: [FFS] |
| | Power class 3: 23dBm ±2 dB | 0.7 dB | Power class 3: 23dBm ±2.7 dB |
| | Power class 4: [FFS] | 0.7 dB | Power class 4: [FFS] |
| | 3.0GHz < f ≤ 4.2GHz | | |
| | Power class 3: 23dBm +2/-3 dB | 1.0 dB | Power class 3: 23dBm +3.0/-4.0 dB |
| 6.2.3 Maximum Power Reduction | Power class 3 | | Formula:<br>Upper limit + TT,<br>Lower limit – MPR – TT<br>Power class 3: |
| | f ≤ 3.0GHz | | |
| | QPSK: MPR ≤ 1dB | 0.7 dB | QPSK: 23dBm +2.7 / - 3.7dB |
| | 16QAM: MPR ≤ 1dB | 0.7 dB | 16QAM: 23dBm +2.7 / - 3.7dB |
| | 16QAM: MPR ≤ 2dB | 0.7 dB | 16QAM: 23dBm +2.7 / - 4.7dB |
| | 1 | | |
| | 3.0GHz < f ≤ 4.2GHz | | |
| | QPSK: MPR ≤ 1dB | 1.0 dB | QPSK: 23dBm +3.0 / - 5.0dB |
| | 16QAM: MPR ≤ 1dB | 1.0 dB | 16QAM: 23dBm +3.0 / - 5.0dB |
| | 16QAM: MPR ≤ 2dB | 1.0 dB | 16QAM: 23dBm +3.0 / - 6.0dB |
| 6.2.3A Maximum Power Reduction (MPR) for CA | TBD | TBD | TBD |

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|---|---|---|---|
| 6.2.4 UE Maximum Output Power with additional requirements | For the UE maximum output power modified by MPR and A-MPR, the power limits specified in TS 36.101 [2] clause 6.2.5 apply.<br><br>For transmission configurations (Figure 5.4.2-1) confined within FUL_low and FUL_low + 4 MHz or FUL_high – 4 MHz and FUL_high, the power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB.<br><br>Power class 3:<br><br>QPSK: MPR ≤ 1dB<br><br>16QAM: Depending on the number RB allocated:<br>16QAM: MPR ≤ 1dB<br>16QAM: MPR ≤ 2dB<br><br>For network signalled value NS_03 , NS_04 (5MHz only), NS_05, to NS_06: A-MPR ≤ 1dB<br><br>For network signalled value NS-04; Depending on the RB_start and RB allocation (10MHz, 15MHz and 20MHz):<br><br>For 10MHz<br>Region A with RB_start=0 – 12: A-MPR ≤ 3dB.<br><br>Region B with RB_start=13 – 36 : A-MPR ≤ 2dB.<br><br>Region C with RB_start=37 – 49 : A-MPR ≤ 3dB.<br><br>For 15MHz<br><br>Region A with RB_start=0 – 18: A-MPR ≤ 3dB.<br><br>Region B with RB_start=19 – 55 : A-MPR ≤ 2dB.<br><br>Region C with RB_start=56 – 74 : A-MPR ≤ 3dB.<br><br>For 20MHz<br>Region A with RB_start=0 – 24: A-MPR ≤ 3dB.<br><br>Region B with RB_start=25 – 74 : A-MPR ≤ 2dB.<br><br>Region C with RB_start=75 – 99 : A-MPR ≤ 3dB.<br><br>***ETSI*** | 0.7 dB | Formula:<br>Upper limit + TT,<br>A: Lower limit – TT,<br>B: (UE Maximum Output Power from 6.2.2) - T($P_{CMAX}$) – MPR – TT,<br>C: (UE Maximum Output Power from 6.2.2) - T($P_{CMAX}$) – A-MPR – TT,<br>D: (UE Maximum Output Power from 6.2.2) - T($P_{CMAX}$) – A-MPR – MPR – TT<br><br>Power class 3:<br><br>Test Requirement Configuration ID versus Formula Above<br><br>Network signalled value NS_03:<br><br>[A]:2, 5, 10, 15, 20, 25<br>[B]:1, 3, 7<br>[C]:9, 14, 19, 24<br>[D]:4, 6, 8, 11, 12, 13, 16, 17, 18, 21, 22, 23, 26, 27<br><br>Network signalled value NS_04 (5, 10, 15, 20MHz):<br><br>[A] 3<br>[B] 10, 11, 19, 20, 28, 29<br>[C] 2, 6, 7, 14, 15, 16, 23, 24, 25, 32<br>[D] 1, 4, 5, 8, 9, 12, 13, 17, 18, 21, 22, 26, 27, 30, 31<br><br>Network signalled value NS_05:<br><br>[A]:1, 3, 4, 7, 8, 11, 12<br>[B]:2, 5, 9, 13<br>[C]:None<br>[D]:6, 10, 14<br><br>Network signalled value NS_06:<br><br>[A]:2, 5, 8, 11, 14, 17<br>[B]:1, 3, 4, 6, 7, 9, 10, 12, 13, 15, 16, 18<br>[C]:None<br>[D]:None<br><br>Network signalled value NS_07:<br><br>[A]:3, 8, 12<br>[B]:7, 9<br>[C]:1, 2, 5, 13, 15<br>[D]:4, 6, 10, 11, 14, 16<br><br>Network signalled value NS_08:<br><br>[A]:1, 2, 4, 5, 11, 12<br>[B]:3, 6, 13<br>[C]:None<br>[D]:7, 8, 9, 10, 14, 15, 16, 17<br>Network signalled value NS_11:<br>[A]:5c, 6b<br>[B]:8b |

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| 6.2.5 Configured UE transmitted Output Power | <br>f ≤ 3.0GHz<br>13 ≤ PCMAX < 18 ± 5.0<br>8 ≤ PCMAX < 13 ± 6.0<br>-40 ≤ PCMAX < 8 ± 7.0<br><br>3.0GHz < f ≤ 4.2GHz<br>13 ≤ PCMAX < 18 ± 5.0<br>8 ≤ PCMAX < 13 ± 6.0<br>-40 ≤ PCMAX < 8 ± 7.0 | <br><br>0.7 dB<br>0.7 dB<br>0.7 dB<br><br><br>1.0 dB<br>1.0 dB<br>1.0 dB | Formula:<br>Upper limit + TT, Lower limit – TT<br><br>13 ≤ PCMAX < 18 ± 5.7<br>8 ≤ PCMAX < 13 ± 6.7<br>-40 ≤ PCMAX < 8 ± 7.7<br><br><br>13 ≤ PCMAX < 18 ± 6.0<br>8 ≤ PCMAX < 13 ± 7.0<br>-40 ≤ PCMAX < 8 ± 8.0 |
| 6.3.2 Minimum Output Power | <br>f ≤ 3.0GHz<br>-40 dBm<br><br>3.0GHz < f ≤ 4.2GHz<br>-40 dBm | <br><br>1.0 dB<br><br><br>1.3 dB | Formula:<br>Minimum Requirement + TT<br>UE min. ouput power =–39 dBm<br><br><br>UE min. ouput power =–38.7 dBm |
| 6.3.3 Transmission ON/OFF Power | <br>f ≤ 3.0GHz<br>≤ -50 dBm<br><br>3.0GHz < f ≤ 4.2GHz<br>≤ -50 dBm | <br><br>1.5 dB<br><br><br>1.8 dB | Formula:<br>Minimum Requirement + TT<br>UE OFF Power ≤ -48.5 dBm<br><br><br>UE OFF Power ≤ -48.2 dBm |
| 6.3.4.1 General ON/OFF time mask | <br><br><br><br>f ≤ 3.0GHz<br>OFF Power ≤ -50 dBm<br>ON Power BW dependent<br><br>3.0GHz < f ≤ 4.2GHz<br>OFF Power ≤ -50 dBm<br>ON Power BW dependent<br><br>Transmission ON Power value depends on the test parameters. In the particular test case parameters the ON power measurement has minimum requirements of ±6.0 dB | <br><br><br><br><br>1.5 dB<br>1.5 dB<br><br><br>1.8 dB<br>1.8 dB | Formulae:<br>OFF Power Minimum Req't + TT<br>ON Power Upper limit + TT, Lower limit – TT<br><br>UE OFF Power ≤ -48.5 dBm<br>UE ON Power: Test value ± 7.5 dB<br><br><br>UE OFF Power ≤ -48.2 dBm<br>UE ON Power: Test value ± 7.8 dB |
| 6.3.4.2 PRACH and SRS time mask | <br><br><br><br>f ≤ 3.0GHz<br>OFF Power ≤ -50 dBm<br>ON Power BW dependent<br><br>3.0GHz < f ≤ 4.2GHz<br>OFF Power ≤ -50 dBm<br>ON Power BW dependent<br><br>Transmission ON Power value depends on the test parameters. In the particular test case parameters the ON power measurement has minimum requirements of ±6.0 dB | <br><br><br><br><br>1.5 dB<br>1.5 dB<br><br><br>1.8 dB<br>1.8 dB | Formulae:<br>OFF Power Minimum Req't + TT<br>ON Power Upper limit + TT, Lower limit – TT<br><br>UE OFF Power ≤ -48.5 dBm<br>UE ON Power: Test value ± 7.5 dB<br><br><br>UE OFF Power ≤ -48.2 dBm<br>UE ON Power: Test value ± 7.8 dB |
| 6.3.5.1 Power Control Absolute power tolerance | <br><br>f ≤ 3.0GHz<br>Normal conditions ± 9.0 dB<br>Extreme conditions ± 12.0 dB<br><br>3.0GHz < f ≤ 4.2GHz<br>Normal conditions ± 9.0 dB<br>Extreme conditions ± 12.0 dB | <br><br><br>1.0 dB<br>1.0 dB<br><br><br>1.4 dB<br>1.4 dB | Formula:<br>Upper limit + TT, Lower limit – TT<br><br>Normal conditions ± 10.0 dB<br>Extreme conditions ± 13.0 dB<br><br><br>Normal conditions ± 10.4 dB<br>Extreme conditions ± 13.4 dB |

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| 6.3.5.2 Power Control Relative power tolerance | TS 36.101 [2] clause 6.3.5.1<br><br>All combinations of PUSCH and PUCCH transitions:<br><br>ΔP < 2; ±2.5 dB<br>2 ≤ ΔP < 3; ±3.0 dB<br>3 ≤ ΔP < 4; ±3.5 dB<br>4 ≤ ΔP ≤ 10; ±4.0 dB<br>10 ≤ ΔP < 15; ±5.0 dB<br>15 ≤ ΔP; ±6.0 dB | 0.7 dB | Formula:<br>Upper limit + TT, Lower limit – TT<br>All combinations of PUSCH and PUCCH transitions:<br><br>ΔP < 2; ±3.2 dB<br>2 ≤ ΔP < 3; ±3.7 dB<br>3 ≤ ΔP < 4; ±4.2 dB<br>4 ≤ ΔP < 10; ±4.7 dB<br>10 ≤ ΔP < 15; ±5.7 dB<br>15 ≤ ΔP; ±6.7 dB |
| 6.3.5.3 Aggregate power control tolerance | Aggregate power control tolerance within 21 ms:<br><br>PUCCH = ±2.5 dB<br>PUSCH = ±3.5 dB | 0.7 dB | Formula:<br>Upper limit + TT, Lower limit - TT<br>PUCCH = ±3.2 dB<br>PUSCH = ±4.2 dB |
| 6.5.1 Frequency Error | <br><br><br><br>Modulated carrier, f ≤ 4.2GHz<br>Within ±0.1 ppm compared to the received carrier frequency<br><br>f ≤ 3.0GHz<br>DL power: Refsens<br><br>3.0GHz < f ≤ 4.2GHz<br>DL power: Refsens | <br><br><br><br>15 Hz<br><br><br>0.7 dB<br><br><br>1.0 dB | Formulae:<br>Modulated carrier frequency: Upper limit + TT, Lower limit – TT<br>DL power: Refsens + TT<br><br>Modulated carrier frequency error = ±(0.1 ppm + 15 Hz)<br><br><br>Refsens +0.7dB<br><br><br>Refsens +1.0dB |
| 6.5.2.1 Error Vector Magnitude | EVM limit:<br>BPSK :17.5 %<br>QPSK: 17.5 %<br>16QAM: 12.5 % | 0% | Formula:<br>Minimum Requirement + TT |
| 6.5.2.1A PUSCH-EVM with exclusion period | EVM limit:<br>QPSK: 17.5 %<br>16QAM: 12.5 % | 0% | Formula:<br>Minimum Requirement + TT |
| 6.5.2.2 Carrier leakage | For Output power >0 dBm<br>-25dBc<br><br>For -30 dBm ≤ Output power ≤0 dBm<br>-20dBc<br><br>For -40 dBm ≤ Output power < -30 dBm<br>-10dBc | 0.8dB | Formula:<br>Minimum Requirement + TT |

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| 6.5.2.3 In-band emissions for non allocated RB | For general emissions:<br>$\max\{ -25-10\cdot\log_{10}(N_{RB}/L_{CRE}$<br>$20\cdot\log_{10} EVM -3-5\cdot(\lvert\Delta_{RB}\rvert-1)$<br>$-57\ dBm/180\ kHz - P_{RB}\}$<br>For IQ image:<br>-25dB<br><br>For Carrier leakage:<br><br>Output power >0 dBm<br>-25dBc<br><br>-30 dBm ≤ Output power ≤0 dBm<br>-20dBc<br><br>-40 dBm ≤ Output power < -30 dBm<br>-10dBc<br>For each evaluated RB, the test requirement is calculated as the higher of $P_{RB}-30$ dB and the power sum of all limit values (General, IQ Image or Carrier leakage) | 0.8dB | Formula:<br>Minimum Requirement + TT |
| 6.5.2.4 EVM equalizer Spectrum flatness | Normal conditions.<br><br>If (F-FUL_low ≥ [3MHz])&(FUL_high-F≥ [3MHz])<br>4 dB<br>else<br>8 dB<br><br>maximum coefficient in Range 1 - the minimum coefficient in Range 2<br>5 dB<br>the maximum coefficient in Range 2 - the minimum coefficient in Range<br>7 dB<br>Extreme conditions:<br><br>If (F-FUL_low ≥ [5MHz])&(FUL_high-F≥ [5MHz])<br>4 dB<br>else<br>12 dB<br><br>maximum coefficient in Range 1 - the minimum coefficient in Range 2<br>6 dB<br>the maximum coefficient in Range 2 - the minimum coefficient in Range<br>10 dB | 1.4dB | Formula:<br>Minimum Requirement + TT |

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| 6.6.1 Occupied bandwidth | For 1.4 MHz channel bandwidth:<br>Occupied channel bandwidth = 1.4 MHz<br><br>For 3.0 MHz channel bandwidth:<br>Occupied channel bandwidth = 3.0 MHz<br><br>For 5 MHz channel bandwidth:<br>Occupied channel bandwidth = 5 MHz<br><br>For 10 MHz channel bandwidth:<br>Occupied channel bandwidth = 10 MHz<br><br>For 15 MHz channel bandwidth:<br>Occupied channel bandwidth = 15 MHz<br><br>For 20 MHz channel bandwidth:<br>Occupied channel bandwidth = 20 MHz | 0kHz | Formula:<br>Minimum Requirement + TT |
| 6.6.2.1 Spectrum Emission Mask | For 1.4 MHz BW:<br>-10 dBm / 30kHz<br>-25dBm to -10dBm / 1MHz<br>For 3 MHz BW:<br>-13 dBm / 30kHz<br>-25dBm to -10dBm / 1MHz<br>For 5 MHz BW:<br>-15dBm / 30kHz<br>-25dBm to -10dBm / 1MHz<br>For 10 MHz BW:<br>-18dBm / 30kHz<br>-25dBm to -10dBm / 1MHz<br>For 15 MHz BW:<br>-20dBm / 30kHz<br>-25dBm to -10dBm / 1MHz<br>For 20 MHz BW:<br>-21dBm / 30kHz<br>-25dBm to -10dBm / 1MHz | All cases:<br><br>f ≤ 3.0GHz<br>1.5dB<br><br>3.0GHz < f ≤ 4.2GHz<br>1.8dB | Formula:<br>Minimum Requirement + TT<br><br>Note: The Test Tolerance would be 0dB for $\Delta f_{OOB} \geq 2$ x Channel Bandwidth, but taking into account the filter position, the Test requirements specified all have $\Delta f_{OOB} < 2$ x Channel Bandwidth |

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| 6.6.2.2 Additional Spectrum Emission Mask | For 1.4 MHz BW:<br>NS_03, NS_04<br>-10 dBm / 30 kHz<br>-25 dBm to -13 dBm / 1MHz<br><br>NS_06 or NS_07<br>-13 dBm / 30 kHz<br>-13 dBm / 100 kHz<br>-25 dBm to -13 dBm / 1MHz<br><br>For 3 MHz BW:<br>NS_03, NS_04<br>-13 dBm / 30 kHz<br>-25 dBm to -13 dBm / 1 MHz<br><br>NS_06 or NS_07<br>-13 dBm / 30 kHz<br>-13 dBm / 100kHz<br>-25 dBm to -13 dBm / 1 MHz<br><br>For 5 MHz BW:<br>NS_03, NS_04<br>-15 dBm / 30 kHz<br>-25 dBm to -13 dBm / 1 MHz<br><br>NS_06 or NS_07<br>-15 dBm / 30 kHz<br>-13 dBm / 100 kHz<br>-25 dBm to -13 dBm / 1 MHz<br><br>For 10 MHz BW:<br>NS_03, NS_04,<br>-18 dBm / 30 kHz<br>-25 dBm to - 13dBm / 1 MHz<br><br>NS_06 or NS_07<br>-18 dBm / 30 kHz<br>-13 dBm / 100 kHz<br>-25 dBm to - 13dBm / 1 MHz<br><br>For 15 MHz BW:<br>NS_03, NS_04<br>-20 dBm / 30kHz<br>-25 dBm to -13 dBm / 1 MHz<br><br>For 20 MHz BW:<br>NS_03, NS_04<br>-21 dBm / 30 kHz<br>-25 dBm to -13 dBm / 1 MHz | All cases:<br><br>f ≤ 3.0GHz<br>1.5dB<br><br>3.0GHz < f ≤ 4.2GHz<br>1.8dB | Formula:<br>Minimum Requirement + TT<br><br>Note: The Test Tolerance would be 0dB for $\Delta f_{OOB}$ ≥ 2 x Channel Bandwidth, but taking into account the filter position, the Test requirements specified all have $\Delta f_{OOB}$ < 2 x Channel Bandwidth |
| 6.6.2.3 Adjacent Channel Leakage power Ratio | If the adjacent channel power is greater than –50 dBm then the ACLR shall be higher than the values specified below.<br><br>E-UTRA ACLR:<br>30 dB<br><br>UTRA ACLR:<br>33 dB for UTRA ACLR 1<br>36 dB for UTRA ACLR 2 | 0 dB<br><br><br><br><br>0.8 dB<br><br><br>0.8 dB<br>0.8 dB | Formula:<br>ACLR Minimum Requirement + TT<br><br>Formula:<br>ACLR Minimum Requirement - TT<br>E-UTRA ACLR:<br>29.2 dB<br><br>UTRA ACLR:<br>32.2 dB for UTRA ACLR 1<br>35.2 dB for UTRA ACLR 2 |
| 6.6.2.3A Adjacent Channel Leakage power Ratio for CA | TBD | TBD | TBD |

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| 6.6.2.4 Additional ACLR requirements | If the adjacent channel power is greater than –50 dBm then the ACLR shall be higher than the values specified below.<br><br>E-UTRA ACLR:<br>43 dB for UTRA ACLR 2 | 0 dB<br><br>0.8 dB | Formula:<br>ACLR Minimum Requirement + TT<br><br>Formula:<br>ACLR Minimum Requirement – TT<br><br>E-UTRA ACLR:<br>42.2 dB for UTRA ACLR 2 |
| 6.6.3.1 Transmitter Spurious emissions | 9 kHz ≤ f < 150 kHz:<br>-36dBm / 1kHz<br>150 kHz ≤ f < 30 MHz:<br>-36dBm / 10kHz<br>30 MHz ≤ f < 1 GHz:<br>-36dBm / 100kHz<br>1 GHz ≤ f < 12.75 GHz:<br>-30dBm / 1MHz<br>12.75 GHz ≤ f < 19 GHz:<br>-30dBm / 1MHz | 0 dB | Formula:<br>Minimum Requirement + TT |
| 6.6.3.2 Spurious emission band UE co-existence | -35 dBm / 6.25kHz<br><br>-36 dBm / 100kHz<br><br>-41 dBm / 300kHz<br><br>-37 dBm / 1MHz<br>-40 dBm / 1MHz<br>-50 dBm / 1MHz<br><br>Frequencies as detailed in core requirement | 0 dB | Formula:<br>Minimum Requirement + TT |
| 6.6.3.2_1 Spurious emission band UE co-existence (Release 9 and forward) | -35 dBm / 6.25kHz<br><br>-36 dBm / 100kHz<br><br>-41 dBm / 300kHz<br><br>-37 dBm / 1MHz<br>-40 dBm / 1MHz<br>-50 dBm / 1MHz<br><br>Frequencies as detailed in core requirement | 0 dB | Formula:<br>Minimum Requirement + TT |
| 6.6.3.3 Additional spurious emissions | NS_05<br>1884.5MHz ≤ f ≤ 1919.6MHz:<br>-41dBm / 300kHz<br>1884.5MHz ≤ f ≤ 1915.7MHz:<br>-41dBm / 300kHz<br>NS_07<br>769MHz ≤ f ≤ 775MHz<br>-57dBm / 6.25kHz<br>NS_08<br>860MHz ≤ f ≤ 895MHz<br>-40dBm / 1MHz<br>NS_09<br>1475.9MHz ≤ f ≤ 1510.9MHz<br>-35dBm / 1MHz | <br><br>0 dB<br><br>0 dB<br><br><br>1.5dB<br><br>0 dB<br><br><br>0 dB | Formula:<br>Minimum Requirement + TT<br>-41dBm / 300kHz<br><br>-41dBm / 300kHz<br><br><br>-55.5 dBm / 6.25kHz<br><br><br>-40dBm / 1MHz<br><br><br>-35dBm / 1MHz |
| 6.7 Transmit intermodulation | Intermodulation Product<br>5MHz -29 dBc<br>10MHz -35 dBc<br>CW Interferer level = -40 dBc | 0 dB | Formula: CW interferer Minimum Requirement– TT<br><br>Intermod Products limits remain unchanged.<br><br>CW interferer level = -40 dBc |

NOTE: Section 6.6.3.3 in the table shall be reviewed after June 2012 because of PHS band operation change

## F.3.3 Measurement of receiver

**Table F.3.3-1: Derivation of Test Requirements (Receiver tests)**

| Test | Minimum Requirement in TS 36.101 | Test Tolerance (TT) | Test Requirement in TS 36.521-1 |
|---|---|---|---|
| 7.3.1 Reference sensitivity power level; Minimum requirements (QPSK) | Reference sensitivity power level:<br><br>For 1.4MHz<br>-102.2dBm<br>-103.2dBm<br>-105.2dBm<br>-106.2dBm<br><br>For 3MHz<br>-99.2dBm<br>-100.2dBm<br>-102.2dBm<br><br>For 5MHz<br>-97dBm<br>-98dBm<br>-99dBm<br>-100dBm<br>-96.5dBm Band 9 with Multi band<br><br>For 10MHz<br>-94dBm<br>-95dBm<br>-96dBm<br>-97dBm<br>-93.5dBm Band 9 with Multi band<br><br>For 15MHz<br>-92.2dBm<br>-93.2dBm<br>-94.2dBm<br>-95.2dBm<br>-91.7dBm Band 9 with Multi band<br><br>For 20MHz<br>-91dBm<br>-92dBm<br>-93dBm<br>-94dBm<br>-90.5dBm Band 9 with Multi band<br><br>T-put limit = 95% of maximum for the Ref Meas channel | All cases:<br><br>f ≤ 3.0GHz<br>0.7dB<br><br>3.0GHz < f ≤ 4.2GHz<br>1.0 dB | Formula: Reference sensitivity power level + TT<br><br>T-put limit unchanged |
| 7.4 Maximum input level | Signal level -25dBm<br><br>T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | f ≤ 3.0GHz0.7 dB<br><br>3.0GHz < f ≤ 4.2GHz<br>1.0 dB<br><br>f ≤ 3.0GHz<br>0dB, -3.4dB | Formula: Maximum input level - TT<br><br>Signal level:<br>f ≤ 3.0GHz: -25.7 dBm<br>3.0GHz < f ≤ 4.2GHz: -26.0 dBm<br><br>T-put limit unchanged<br><br>Uplink power measurement window comprises four quantities:<br>1. UE power step size 1dB<br>2. UE Power step tolerance ±1dB |

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| | | 3.0GHz < f ≤ 4.2GHz<br>0dB, -4.0dB | 3. Test system power measurement at top of window:<br>f ≤ 3.0GHz ±0.7 dB<br>3.0GHz < f ≤ 4.2GHz ±1.0 dB.<br>4. Test system power measurement at bottom of window:<br>f ≤ 3.0GHz ±0.7 dB<br>3.0GHz < f ≤ 4.2GHz ±1.0 dB.<br><br>Items 1 to 4 are added arithmetically:<br>Overall UL power window size:<br>f ≤ 3.0GHz:<br>(1dB+1dB+0.7dB+0.7dB) =3.4dB<br>3.0GHz < f ≤ 4.2GHz:<br>(1dB+1dB+1dB+1dB) =4dB<br><br>Top of window is aligned to UL power requirement, hence +0dB, -3.4dB or +0dB, -4.0dB according to frequency |

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| 7.5 Adjacent Channel Selectivity (ACS) | Case 1:<br>Wanted signal power, all BWs:<br>(REFSENS + 14 dB)<br><br>Interferer signal power<br>For 1.4 MHz, 3 MHz, 5 MHz, 10 MHz BW:<br>(REFSENS + 45.5 dB)<br>For 15 MHz BW:<br>(REFSENS + 42.5 dB)<br>For 20 MHz BW:<br>(REFSENS + 39.5 dB)<br><br>Case 2:<br>Wanted signal power<br>For 1.4 MHz, 3 MHz, 5 MHz, 10 MHz BW: -56.5 dBm<br>For 15 MHz BW: -53.5 dBm<br>For 20 MHz BW: -50.5 dBm<br><br>Interferer signal power, all BWs:<br>-25 dBm<br><br>T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | 0 dB<br><br>f ≤ 3.0GHz<br>0dB, -3.4dB<br><br>3.0GHz < f ≤ 4.2GHz<br>0dB, -4.0dB | Formula:<br>Wanted signal power + TT<br><br>Interferer signal power unchanged<br><br>T-put limit unchanged<br><br>Uplink power measurement window same as 7.4 |
| 7.6.1 In-band blocking | Wanted signal power:<br>(REFSENS + BW dependent value)<br><br>Interferer signal power:<br>-56dBm or -44dBm<br><br>T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | 0 dB<br><br>f ≤ 3.0GHz<br>0dB, -3.4dB<br><br>3.0GHz < f ≤ 4.2GHz<br>0dB, -4.0dB | Formula:<br>Wanted signal power + TT<br><br>Interferer signal power unchanged<br><br>T-put limit unchanged<br><br>Uplink power measurement window same as 7.4 |
| 7.6.2 Out of-band blocking | Wanted signal power:<br>(REFSENS + BW dependent value)<br><br>Interferer signal power:<br>-44dBm, -30dBm or -15dBm<br><br>T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | 0 dB<br><br>f ≤ 3.0GHz<br>0dB, -3.4dB<br><br>3.0GHz < f ≤ 4.2GHz<br>0dB, -4.0dB | Formula:<br>Wanted signal power + TT<br><br>Interferer signal power unchanged<br><br>T-put limit unchanged<br><br>Uplink power measurement window same as 7.4 |
| 7.6.3 Narrow band blocking | Wanted signal power,:<br>(REFSENS + BW dependent value)<br>Interferer signal power:<br>-55dBm | 0 dB | Formula:<br>Wanted signal power + TT<br><br>Interferer signal power unchanged |

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| | T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | <br><br><br>f ≤ 3.0GHz<br>0dB, -3.4dB<br><br>3.0GHz < f ≤ 4.2GHz<br>0dB, -4.0dB | T-put limit unchanged<br><br>Uplink power measurement window same as 7.4 |
| 7.7 Spurious response | Wanted signal power:<br>(REFSENS + BW dependent value)<br><br>Interferer signal power:<br>-44dBm<br><br>T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | 0 dB<br><br><br><br><br><br>f ≤ 3.0GHz<br>0dB, -3.4dB<br><br>3.0GHz < f ≤ 4.2GHz<br>0dB, -4.0dB | Formula:<br>Wanted signal power + TT<br><br>Interferer signal power unchanged<br><br>T-put limit unchanged<br><br>Uplink power measurement window same as 7.4 |
| 7.8.1 Wide band intermodulation | Wanted signal power:<br>For 1.4 MHz BW:<br>(REFSENS + 12 dB)<br>For 3 MHz BW:<br>(REFSENS + 8 dB)<br>For 5 MHz and 10MHz BW:<br>(REFSENS + 6 dB)<br>For 15 MHz BW:<br>(REFSENS + 7 dB)<br>For 20 MHz BW:<br>(REFSENS + 9 dB)<br><br>CW Interferer power, aall BWs:<br>-46 dBm<br><br>Modulated Interferer power:, aall BWs:<br>-46 dBm<br><br>T-put limit = 95% of maximum for the Ref Meas channel<br><br>Uplink power | 0 dB<br><br><br><br><br>

## F.3.4 Measurement of performance requirements

**Table F.3.4-1: Derivation of Test Requirements (performance tests)**

| Test | Minimum Requirement in TS 36.133 | Test Tolerance (TT) | Test Requirement in TS 36.521-1 |
|---|---|---|---|
| 8.2.1.1.1 Multiple PRBs<br>- Prop'n Condition EVA5<br>- Prop'n Condition ETU70<br>- Prop'n Condition ETU300 | SNRs as specified | 0.8dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.1.1 Multiple PRBs<br>- Prop'n Condition HST | SNR as specified | 0.6dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.1.1 Single PRB<br>- Prop'n Condition ETU70 | SNRs as specified | 0.8dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.1.2 Single PRB | SNR as specified | 0.8dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.2.1<br>- Prop'n Condition EVA5 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.2.1<br>- Prop'n Condition HST | SNR as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.2.2 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.3.1 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.3.2 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.4.1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.1.4.2 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.1.1 Multiple PRBs<br>- Prop'n Condition EVA5<br>- Prop'n Condition ETU70<br>- Prop'n Condition ETU300 | SNRs as specified | 0.8dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.1.1 Multiple PRBs<br>- Prop'n Condition HST | SNR as specified | 0.6dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.1.1 Single PRB<br>- Prop'n Condition ETU70 | SNRs as specified | 0.8dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.1.2 Single PRB | SNR as specified | 0.8dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.2.1<br>- Prop'n Condition EVA5 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.2.1<br>- Prop'n Condition HST | SNR as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.2.2 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |

| | | | |
|---|---|---|---|
| 8.2.2.3.1 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.3.2 | SNR as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.4.1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.2.2.4.2 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.3.2.1.1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.3.2.1.1_1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.3.2.1.2 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.3.2.1.3 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.3.2.2.1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.4.1.1 | SNR as specified | 0.8 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.4.1.2.1 | SNR as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.4.1.2.2 | SNR as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.4.2.1 | SNR as specified | 0.8 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.4.2.2.1 | SNR as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.4.2.2.2 | SNR as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.5.1.1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.5.1.2.1 | SNR as specified | 1.1 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.5.1.2.2 | SNR as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.5.2.1 | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.5.2.2.1 | SNR as specified | 1.1 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 8.5.2.2.2 | SNR as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |

| | | | |
|---|---|---|---|
| 8.7.1 FDD sustained data rate performance | Downlink power -85dBm/15kHz | 0 dB | Formula: Downlink power + TT<br>T-put limit unchanged |
| 8.7.2 TDD sustained data rate performance | Same as 8.7.1 | Same as 8.7.1 | Same as 8.7.1 |
| 10.1 | SNR as specified | 0.9dB | Formula: SNR + TT<br>T-put limit unchanged |
| 10.2 | SNR as specified | 0.9dB | Formula: SNR + TT<br>T-put limit unchanged |
| [Other tests FFS] | | | |

# F.3.5 Measurement of Channel State Information reporting

**Table F.3.5-1: Derivation of Test Requirements (Channel State Information reporting tests)**

| Test | Minimum Requirement in TS 36.101 | Test Tolerance (TT) | Test Requirement in TS 36.521-1 |
|---|---|---|---|
| 9.2.1.1 FDD CQI Reporting under AWGN conditions – PUCCH 1-0 | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR and limits unchanged |
| 9.2.1.2 TDD CQI Reporting under AWGN conditions – PUCCH 1-0 | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR and limits unchanged |
| 9.2.2.1 FDD CQI Reporting under AWGN conditions – PUCCH 1-1 | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR and limits unchanged |
| 9.2.2.2 TDD CQI Reporting under AWGN conditions – PUCCH 1-1 | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR and limits unchanged |
| 9.3.1.1.1 FDD CQI Reporting under fading conditions – PUSCH 3-0 | SNRs as specified<br>$\alpha$ 2%<br>$\beta$ 55%<br>$\gamma$ 1.10<br>BLER 0.05 | SNR 0dB<br>$\alpha$ 0%<br>$\beta$ 0%<br>$\gamma$ 0.01<br>BLER 0 | SNR unchanged<br>$\alpha$ limit unchanged<br>$\beta$ limit unchanged<br>$\gamma$ 1.09 as per Table G.5.4-1<br>BLER limit unchanged |
| 9.3.1.1.2 TDD CQI Reporting under fading conditions – PUSCH 3-0 | Same as 9.3.1.1.1 | Same as 9.3.1.1.1 | Same as 9.3.1.1.1 |
| 9.3.2.1.1 FDD CQI Reporting under fading conditions – PUCCH 1-0 | SNRs as specified<br>$\alpha$ 20%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0dB<br>$\alpha$ 0%<br>$\gamma$ 0.01<br>BLER 0 | SNR and $\alpha$, BLER limits unchanged<br>$\alpha$ limit unchanged<br>$\gamma$ 1.04 as per Table G.5.4-1<br>BLER limit unchanged |
| 9.3.2.1.2 TDD CQI Reporting under fading conditions – PUCCH 1-0 | Same as 9.3.2.1.1 | Same as 9.3.2.1.1 | Same as 9.3.2.1.1 |
| 9.3.3.1.1 FDD CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0 | Ior and Iot as specified<br>$\alpha$ 60%<br>$\gamma$ 1.60 | Ior, Iot 0dB<br>$\alpha$ 20%<br>$\gamma$ 0.10 | Ior and Iot unchanged<br>$\alpha$ 40%, Formula: Min Req't – Test Tol<br>$\gamma$ 1.50, Formula: Min Req't – Test Tol<br>The effect of AWGN flatness and signal flatness on the $\alpha$ requirement was derived by simulation.<br>AWGN flatness / signal flatness and the statistical effect of a finite test time both affect the T-put result. The Test Tolerance comprises two quantities:<br>1. Effect of AWGN flatness and signal flatness, derived by simulation<br>2. Statistical effect as per Table G.5.4-1<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared<br>T-put requirement Test Tol = SQRT (AWGN flatness and signal flatness effect $^2$ + Statistical effect $^2$)<br>AWGN flatness and signal flatness effect 0.1, Statistical effect 0.01, giving overall effect 0.10 |

| | | | |
|---|---|---|---|
| 9.3.3.1.2 TDD CQI Reporting under fading conditions and frequency-selective interference – PUSCH 3-0 | Same as 9.3.3.1.1 | Same as 9.3.3.1.1 | Same as 9.3.3.1.1 |
| 9.3.4.1.1 FDD CQI Reporting under fading conditions – PUSCH 2-0 | SNRs as specified<br>Limits as in the Test Procedure<br>$\gamma$ 1.2 | <br><br>$\gamma$ 0.01 | SNR unchanged<br><br>$\gamma$ 1.19 as per Table G.5.4-1 |
| 9.3.4.1.2 TDD CQI Reporting under fading conditions – PUSCH 2-0 | Same as 9.3.4.1.1 | Same as 9.3.4.1.1 | Same as 9.3.4.1.1 |
| 9.3.4.2.1 FDD CQI Reporting under fading conditions – PUCCH 2-0 | SNRs as specified<br>Limits as in the Test Procedure<br>$\gamma$ 1.15 | <br><br>$\gamma$ 0.01 | SNR unchanged<br><br>$\gamma$ 1.14 as per Table G.5.4-1 |
| 9.3.4.2.2 TDD CQI Reporting under fading conditions – PUCCH 2-0 | Same as 9.3.4.2.1 | Same as 9.3.4.2.1 | Same as 9.3.4.2.1 |
| 9.4.1.1.1 FDD PMI Reporting – PUSCH 3-1 (Single PMI) | $\gamma$ 1.10 | $\gamma$ 0.01 | $\gamma$ 1.09 as per Table G.5.4-1 |
| 9.4.1.1.2 TDD PMI Reporting – PUSCH 3-1 (Single PMI) | Same as 9.4.1.1.1 | Same as 9.4.1.1.1 | Same as 9.4.1.1.1 |
| 9.4.1.2.1 FDD PMI Reporting – PUCCH 2-1 (Single PMI) | $N_{oc}$ as specified in test prodecedure<br><br>$\gamma$ 1.2 | <br><br>$\gamma$ 0.01 | $N_{oc}$ unchanged<br>Ior/ $N_{oc}$=$SNR_{rnd}$ is a result of an approach according to G.5.2 and is reused unchanged as setting in procedure step 3.<br>$\gamma$ 1.19 as per Table G.5.4-1 |
| 9.4.1.2.2 TDD PMI Reporting – PUCCH 2-1 (Single PMI) | Same as 9.4.1.2.1 | Same as 9.4.1.2.1 | Same as 9.4.1.2.1 |
| 9.4.2.1.1 FDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | $N_{oc}$ as specified in test prodecedure<br><br>$\gamma$ 1.20 | $\gamma$ 0.01 | $N_{oc}$ unchanged<br>Ior/ $N_{oc}$=$SNR_{rnd}$ is a result of an approach according to G.5.2 and is reused unchanged as setting in procedure step 3.<br>$\gamma$ 1.19 as per Table G.5.4-1 |
| 9.4.2.1.1_1 FDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | Same as 9.4.2.1.1 | Same as 9.4.2.1.1 | Same as 9.4.2.1.1 |
| 9.4.2.1.2 TDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | Same as 9.4.2.1.1 | Same as 9.4.2.1.1 | Same as 9.4.2.1.1 |
| 9.4.2.1.2_1 TDD PMI Reporting – PUSCH 1-2 (Multiple PMI) | Same as in 9.4.2.1.1 | Same as in 9.4.2.1.1 | Same as in 9.4.2.1.1 |
| 9.4.2.2.1 FDD PMI Reporting – PUSCH 2-2 (Multiple PMI) | Same as in 9.4.2.1.1 | Same as in 9.4.2.1.1 | Same as in 9.4.2.1.1 |
| 9.4.2.2.2 TDD PMI Reporting – PUSCH 2-2 (Multiple PMI) | $N_{oc}$ as specified in test prodecedure<br><br>$\gamma$ 1.15 | <br><br>$\gamma$ 0.01 | $N_{oc}$ unchanged<br>Ior/ $N_{oc}$=$SNR_{rnd}$ is a result of an approach according to G.5.2 and is reused unchanged as setting in procedure step 3.<br>$\gamma$ 1.14 as per Table G.5.4-1 |
| 9.5.1.1 FDD RI Reporting– PUCCH 1-1 | SNRs as specified<br>Test 1: $\gamma_2$ 1.00<br>Test 2: $\gamma_1$ 1.05<br>Test 3: $\gamma_2$ 1.10 | SNR 0dB<br>$\gamma_2$ 0.01<br>$\gamma_1$ 0.01<br>$\gamma_2$ 0.01 | SNR unchanged<br>$\gamma_2$ 0.99 as per Table G.5.4-1<br>$\gamma_1$ 1.04 as per Table G.5.4-1<br>$\gamma_2$ 1.09 as per Table G.5.4-1 |

| | | | |
|---|---|---|---|
| 9.5.1.2 TDD RI Reporting– PUSCH 3-1 | Same as 9.5.1.1 | Same as 9.5.1.1 | Same as 9.5.1.1 |

# Annex G (normative): Statistical Testing

## G.1 General

FFS.

## G.2 Statistical testing of receiver characteristics

### G.2.1 General

The test of receiver characteristics is two fold.

1. A signal or a combination of signals is offered to the RX port(s) of the receiver.
2. The ability of the receiver to demodulate /decode this signal is verified by measuring the throughput.

In (2) is the statistical aspect of the test and is treated here.

The minimum requirement for all receiver tests is >95% of the maximum throughput.

All receiver tests are performed in static propagation conditions. No fading conditions are applied.

### G.2.2 Mapping throughput to error ratio

a) The measured information bit throughput R is defined as the sum (in kilobits) of the information bit payloads successfully received during the test interval, divided by the duration of the test interval (in seconds).

b) In measurement practice the UE indicates successfully received information bit payload by signalling an ACK to the SS.
If payload is received, but damaged and cannot be decoded, the UE signals a NACK.

c) Only the ACK and NACK signals, not the data bits received, are accessible to the SS.
The number of bits is known in the SS from knowledge of what payload was sent.

d) For the reference measurement channel, applied for testing, the number of bits is different in different subframes, however in a radio frame it is fixed during one test.

e) The time in the measurement interval is composed of successfully received subframes (ACK), unsuccessfully received subframes (NACK) and no reception at all (DTX-subframes).

f) DTX-subframes may occur regularly according the appliccable reference measurment channel (regDTX).
In real live networks this is the time when other UEs are served. In TDD these are the UL and special subframes. regDTX vary from test to test but are fixed within the test.

g) Additional DTX-subframes occur statistically when the UE is not responding ACK or NACK where it should. (statDTX)
This may happen when the UE was not expecting data or decided that the data were not intended for it.

The pass / fail decision is done by observing the:

- number of NACKs
- number of ACKs and
- number of statDTXs (regDTX is implicitly known to the SS)

The ratio (NACK + statDTX)/(NACK+ statDTX + ACK) is the Error Ratio (ER). Taking into account the time consumed by the ACK, NACK, and DTX-TTIs (regular and statistical), ER can be mapped unambiguously to throughput for any single reference measurement channel test.

## G.2.3 Design of the test

The test is defined by the following design principles (see clause G.x, Theory….):

1. The early decision concept is applied.
2. A second limit is introduced: Bad DUT factor M>1
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

   To decide the test fail

   Cusomer Risk is applied based on the specified DUT quality

The test is defined by the following parameters:

1. Limit ER = 0.05 (Throughput limit = 95%)
2. Bad DUT factor M=1.5 (selectivity)
3. Confidence level CL = 95% (for specified DUT and Bad DUT-quality)

## G.2.4 Numerical definition of the pass fail limits

**Table G.2.4-1: pass fail limits**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 67 | NA | 39 | 763 | 500 | 78 | 1366 | 1148 | 117 | 1951 | 1828 |
| 1 | 95 | NA | 40 | 778 | 516 | 79 | 1381 | 1166 | 118 | 1965 | 1845 |
| 2 | 119 | 2 | 41 | 794 | 532 | 80 | 1396 | 1183 | 119 | 1980 | 1863 |
| 3 | 141 | 7 | 42 | 810 | 548 | 81 | 1412 | 1200 | 120 | 1995 | 1881 |
| 4 | 162 | 14 | 43 | 826 | 564 | 82 | 1427 | 1217 | 121 | 2010 | 1899 |
| 5 | 183 | 22 | 44 | 842 | 580 | 83 | 1442 | 1234 | 122 | 2025 | 1916 |
| 6 | 202 | 32 | 45 | 858 | 596 | 84 | 1457 | 1252 | 123 | 2039 | 1934 |
| 7 | 222 | 42 | 46 | 873 | 612 | 85 | 1472 | 1269 | 124 | 2054 | 1952 |
| 8 | 241 | 53 | 47 | 889 | 629 | 86 | 1487 | 1286 | 125 | 2069 | 1969 |
| 9 | 259 | 64 | 48 | 905 | 645 | 87 | 1502 | 1303 | 126 | 2084 | 1987 |
| 10 | 278 | 76 | 49 | 920 | 661 | 88 | 1517 | 1321 | 127 | 2099 | 2005 |
| 11 | 296 | 88 | 50 | 936 | 678 | 89 | 1532 | 1338 | 128 | 2113 | 2023 |
| 12 | 314 | 100 | 51 | 952 | 694 | 90 | 1547 | 1355 | 129 | 2128 | 2040 |
| 13 | 332 | 113 | 52 | 967 | 711 | 91 | 1562 | 1373 | 130 | 2143 | 2058 |
| 14 | 349 | 126 | 53 | 983 | 727 | 92 | 1577 | 1390 | 131 | 2158 | 2076 |
| 15 | 367 | 140 | 54 | 998 | 744 | 93 | 1592 | 1407 | 132 | 2172 | 2094 |
| 16 | 384 | 153 | 55 | 1014 | 760 | 94 | 1607 | 1425 | 133 | 2187 | 2111 |
| 17 | 401 | 167 | 56 | 1029 | 777 | 95 | 1623 | 1442 | 134 | 2202 | 2129 |
| 18 | 418 | 181 | 57 | 1045 | 793 | 96 | 1637 | 1459 | 135 | 2217 | 2147 |
| 19 | 435 | 195 | 58 | 1060 | 810 | 97 | 1652 | 1477 | 136 | 2231 | 2165 |
| 20 | 452 | 209 | 59 | 1076 | 827 | 98 | 1667 | 1494 | 137 | 2246 | 2183 |
| 21 | 469 | 224 | 60 | 1091 | 844 | 99 | 1682 | 1512 | 138 | 2261 | 2201 |
| 22 | 486 | 238 | 61 | 1106 | 860 | 100 | 1697 | 1529 | 139 | 2275 | 2218 |
| 23 | 503 | 253 | 62 | 1122 | 877 | 101 | 1712 | 1547 | 140 | 2290 | 2236 |
| 24 | 519 | 268 | 63 | 1137 | 894 | 102 | 1727 | 1564 | 141 | 2305 | 2254 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 25 | 536 | 283 | 64 | 1153 | 911 | 103 | 1742 | 1582 | 142 | 2320 | 2272 |
| 26 | 552 | 298 | 65 | 1168 | 928 | 104 | 1757 | 1599 | 143 | 2334 | 2290 |
| 27 | 569 | 313 | 66 | 1183 | 944 | 105 | 1772 | 1617 | 144 | 2349 | 2308 |
| 28 | 585 | 328 | 67 | 1199 | 961 | 106 | 1787 | 1634 | 145 | 2364 | 2326 |
| 29 | 602 | 343 | 68 | 1214 | 978 | 107 | 1802 | 1652 | 146 | 2378 | 2344 |
| 30 | 618 | 359 | 69 | 1229 | 995 | 108 | 1817 | 1669 | 147 | 2393 | 2361 |
| 31 | 634 | 374 | 70 | 1244 | 1012 | 109 | 1832 | 1687 | 148 | 2408 | 2379 |
| 32 | 650 | 389 | 71 | 1260 | 1029 | 110 | 1847 | 1704 | 149 | 2422 | 2397 |
| 33 | 667 | 405 | 72 | 1275 | 1046 | 111 | 1861 | 1722 | 150 | 2437 | 2415 |
| 34 | 683 | 421 | 73 | 1290 | 1063 | 112 | 1876 | 1740 | 151 | 2452 | 2433 |
| 35 | 699 | 436 | 74 | 1305 | 1080 | 113 | 1891 | 1757 | 152 | 2466 | 2451 |
| 36 | 715 | 452 | 75 | 1321 | 1097 | 114 | 1906 | 1775 | 153*) | NA | 2469 |
| 37 | 731 | 468 | 76 | 1336 | 1114 | 115 | 1921 | 1793 | | | |
| 38 | 747 | 484 | 77 | 1351 | 1131 | 116 | 1936 | 1810 | *) note 2 in G.2.5 | | |

NOTE 1: The first column is the number of errors (ne = number of NACK + statDTX)

NOTE 2: The second column is the number of samples for the pass limit ($ns_p$ , ns=Number of Samples= number of `NACK + statDTX + ACK`)

NOTE 3: The third column is the number of samples for the fail limit ($ns_f$)

## G.2.5 Pass fail decision rules

The pass fail decision rules apply for a single test, comprising one component in the test vector. The over all Pass /Fail conditions are defined in clause G.2.1.5.

Having observed 0 errors, pass the test at 67+ samples, otherwise continue

Having observed 1 error, pass the test at 95+ otherwise continue

Having observed 2 errors, pass the test at 119+ samples, fail the test at 2- samples, otherwise continue

Etc. etc.

Having observed 151 errors, pass the test at 2452+ samples, fail the test at 2433- samples, otherwise continue

Having observed 152 errors, pass the test at 2466+ samples, fail the test at 2451- samples.

Where x+ means: x or more, x- means x or less

NOTE 1: an ideal DUT passes after 67 samples. The maximum test time is 2466 samples.

NOTE 2: It is allowed to deviate from the early decision concept by postponing the decision (pass/fail or continue). Postponing the decision to or beyond the end of table Table G.2.4-1 requires a pass fail decision against the test limit: pass the DUT for ER<0.0618, otherwise fail.

## G.2.6 Test conditions for receiver tests

**Table G.2.6-1: Test conditions for receiver tests**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | Over all Pass/Fail condition |
|---|---|---|---|
| 7.3 Reference sensitivity level | Yes: the inherent receiver noise is assumed to be AWGN | tbd | To pass 7.3 each component in the test vector must pass |
| 7.4 Maximum input level | Unclear: in case, clipping causes errors, errors are data dependent. Statistical independence is assumed. | tbd | To pass 7.4 each component in the test vector must pass |
| 7.5 Adjacent Channel Selectivity (ACS) | Unclear: errors are data dependent on the interferers data. Statistical independence is assumed. | tbd | To pass 7.5 each component in the test vector must pass |
| 7.6.1 In-band blocking | Unclear: errors are data dependent on the interferers data. Statistical independence is assumed. | tbd | To pass 7.6.1 each component in the test vector must pass |
| 7.6.2 Out of-band blocking | yes: it is assumed that the CW interferer causes errors, which are independent and time invariant. | tbd | To pass 7.6.2, all except [tbd] components in the test vector must pass |
| 7.6.3 Narrow band blocking | yes: it is assumed that the CW interferer causes errors, which are independent and time invariant. | tbd | To pass 7.6.3 each component in the test vector must pass |
| 7.7 Spurious response | yes: it is assumed that the CW interferer causes errors, which are independent and time invariant. | tbd | To pass 7.7 each component in the test vector must pass |
| 7.8.1 Wide band Intermodulation | Unclear: errors are dependent on the data content of the interferer. Statistical independence is assumed. | tbd | To pass 7.8.1 each component in the test vector must pass |

# G.3 Statistical testing of Performance Requirements with throughput

## G.3.1 General

The test of receiver performance characteristics is two fold.

1. A signal or a combination of signals is offered to the RX port(s) of the receiver.
2. The ability of the receiver to demodulate /decode this signal is verified by measuring the throughput.

In (2) is the statistical aspect of the test and is treated here.

The minimum requirement for all receiver performance tests is either 70% or 30% of the maximum throughput.

All receiver performance tests are performed in fading conditions. In addition to the statistical considerations, this requires the definition of a minimum test time.

## G.3.2 Mapping throughput to error ratio

G.2.2 applies

## G.3.3 Design of the test

The test is defined by the following design principles (see clause G.x, Theory….):

1. The standard concept is applied. (not the early decision concept)
2. A second limit is introduced: The second limit is different, whether 30% or 70% throughput is tested.
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

   To decide the test fail:

   Cusomer Risk is applied based on the specified DUT quality

The test is defined by the following parameters:

1a) Limit Error Ratio = 0.3 (in case 70% Throughput is tested) or

1b) Limit Througput = 0.3 (in case 30% Throughput is tested)

2a) Bad DUT factor M=1.378 (selectivity)

2b) Bad DUT factor m=0.692 (selectivity)

justification see: TS 34.121 Clause F.6.3.3

3) Confidence level CL = 95% (for specified DUT and Bad DUT-quality)

## G.3.4 Pass Fail limit

Testing Throughput = 30%, then the test limit is

Number of successes (ACK) / number of samples $\geq$ 59 / 233

Testing Throughput = 70% then the test limit is

Number of fails (NACK and statDTX) / number of samples $\leq$ 66 / 184

We have to distinguish 3 cases:

a) The duration for the number of samples (233 or 184) is greater than the minimum test time:

   Then the number of samples (233 or 184) is predefined and the decision is done according to the number of events (59 successes or 66 fails)

b) Since subframe 0 and 5 contain less bits than the remaining subframes, it is allowed to predefine a number of samples contained in an integer number of frames In this case test-limit-ratio applies.

c) The minimum test time is greater than the duration for the number of samples:

   The minimum testtime is predefined and the decision is done comparing the measured ratio at that instant against the test-limit-ratio.

NOTE: The test time for most of the tests is governed by the Minimum Test Time

## G.3.5 Minimum Test time

If a pass fail decision in G.3.4 can be achieved earlier than the minimum test time, then the test shall not be decided, but continued until the minimum test time is elapsed.

The tables below contain the minimum number of subframes for FDD and TDD.

By simulations the minimum number of active subframes (carrying DL payload) was derived (MNAS),

then adding incative subframes to the active ones (e.g. subframe 5 contains no DL payload. For TDD additional subframes contain no DL payload)

then rounding up to full thousand and

then adding a bias of 1000 (BMNSF).

Simulation method to derive minimum test time:

With a level, corresponding a throughput at the test limit (here 30% or 70% of the max. throughput) the preliminary throughput versus time converges towards the final throughput. The allowance of ± 0.2 dB around the above mentioned level is predefined by RAN5 to find the minimum test time. The allowance of ±0.2 dB maps through the function "final throughput versus level" into a throughput corridor. The minimum test time is achieved when the preliminary throughput escapes the corridor the last time. The two functions "final throughput versus level" and "preliminary throughput versus time" are simulation results, which are done individual for each demodulation scenario. HST-scenarios and scenarios with MNAS ≥ 50000 are derived differently.

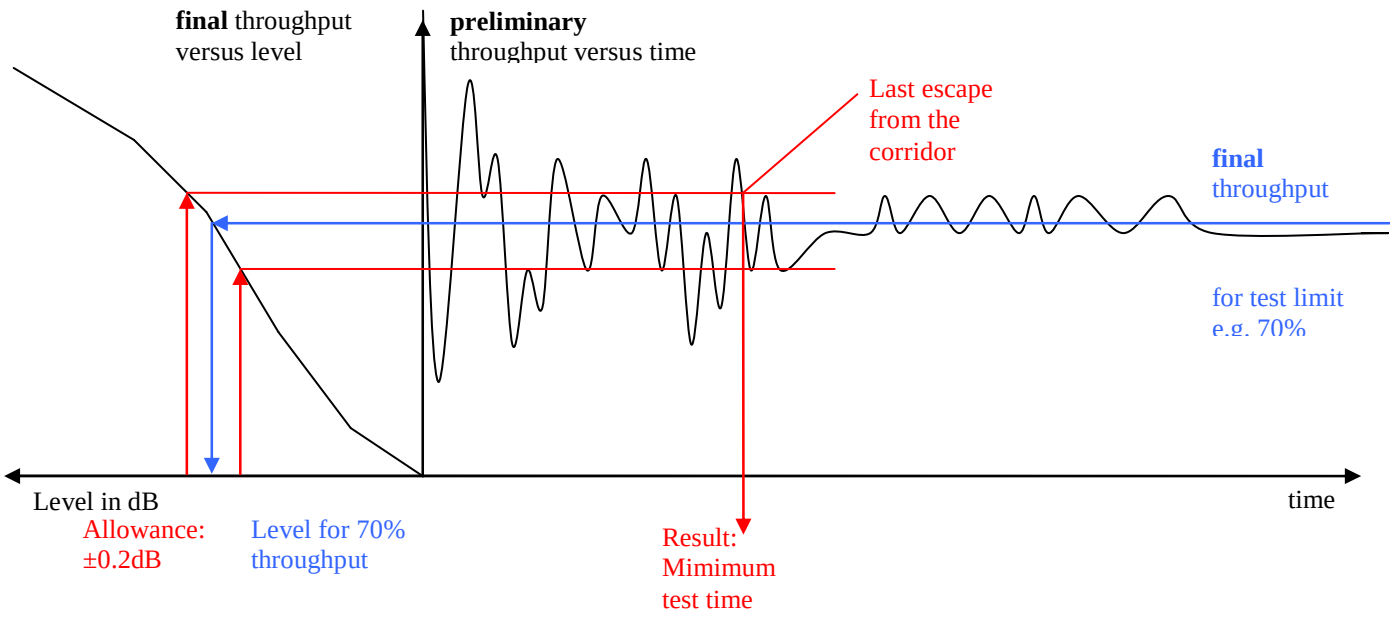



**Fig. G.3.5-1: Simulation method to derive minimum test time**

**Table G.3.5-1: Minimum Test time for PDSCH Single Antenna Port Performance**

| Test No | Demod. scenario | Demodulation scenario plain text:<br><br>RMC (Bandwidth, allocated RBs, modulation, coding)<br>Antenna (configuration, correlation)<br>Propagation condition, Doppler<br>[additional parameters, if applicable]<br><br>(info only) | Minimum Number of Active Subframes (MNAS) to reach the corridor<br><br>(Simulation, info only) | Minimum Number of Subframes (MNS) to reach the corridor<br><br>(MNS = active and inactive subframes)<br><br>(Calculation, info only) | | Biased Minimum Number of SubFrames (BMNSF)<br><br>BMNSF= $1000 * \lceil \frac{MNS}{1000} \rceil$ +1000<br><br>(mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [1.1] | R.2<br>(10 MHz, full, QPSK, 1/3)<br>(1x2 Low)<br>EVA,5 | 38 764 | 43 072 | 77 528 | 45 000 | 79 000 |
| 2 | [1.2] | R.2<br>(10 MHz, full, QPSK, 1/3)<br>(1x2 Low)<br>ETU,70 | 2 764 | 3 072 | 5 528 | 5 000 | 7 000 |
| 3 | [1.3] | R.2<br>(10 MHz, full, QPSK, 1/3)<br>(1x2 Low)<br>ETU,300 | 1 424 | 1 583 | 2 848 | 3 000 | 4 000 |
| 4 | [1.4] | R.2<br>(10 MHz , full, QPSK, 1/3)<br>(1x2 Low)<br>HST | 28 800 | NA | NA | 28 800 | 57 600 |
| 5 | [2.1] | R.4<br>(1.4 MHz, full, QPSK, 1/3)<br>(1x2 Low)<br>EVA,5 | 44 354 | 49 283 | 147 847 | 51 000 | 149 000 |
| 6 | [1.5] | R.3<br>(10 MHz, full, 16QAM, ½)<br>(1x2 Low)<br>EVA,5 | 39 020 | 43 356 | 78 040 | 45 000 | 80 000 |
| 7 | [1.6] | R.3<br>(10 MHz, full, 16QAM, ½)<br>(1x2 Low)<br>ETU,70 | 1 366 | 1 518 | 2 732 | 3 000 | 4 000 |
| 8 | [1.7] | R.3<br>(10 MHz, full, 16QAM, ½)<br>(1x2 High)<br>ETU,300 | 3 189 | 3 544 | 6 378 | 5 000 | 8 000 |
| 9 | [2.2] | R.5<br>(3 MHz, full, 64QAM, ¾)<br>(1x2 Low)<br>EVA,5 | 50 000 | 55 556 | 100 000 | 57 000 | 101 000 |
| 10 | [2.3] | R.6<br>(5 MHz, full, 64QAM, 3/4)<br>(1x2 Low)<br>EVA,5 | 48 847 | 54 275 | 97 694 | 56 000 | 99 000 |
| 11 | [1.8] | R.7<br>(10 MHz, full, 64QAM, ¾)<br>(1x2 Low)<br>EVA,5 | 46 524 | 51 694 | 93 048 | 53 000 | 95 000 |
| 12 | [1.9] | R.7<br>(10 MHz, full, 64QAM, ¾)<br>(1x2 Low)<br>ETU,70 | 4 722 | 5 247 | 9 444 | 7 000 | 11 000 |
| 13 | [1.10] | R.7<br>(10 MHz, full, 64 QAM, 3/4) (1x2High)<br>EVA,5 | 100 000 | 111 112 | 200 000 | 113 000 | 201 000 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 14 | [2.4] | R.8<br>(15 MHz, full, 64QAM, ¾)<br>(1x2 Low)<br>EVA,5 | 48 434 | 53 816 | 96 868 | 55 000 | 98 000 |
| 15 | [2.5] | R.9<br>(20 MHz, full, 64QAM,3/4)<br>(1x2 Low)<br>EVA,5 | 100 000 | 111 112 | 200 000 | 113 000 | 201 000 |
| 16 | [3.1] | R.0<br>(3 MHz, 1PRB,16QAM,½)<br>(1x2 Low)<br>ETU,70 | 5 710 | 6 345 | 11 420 | 8 000 | 13 000 |
| 17 | [3.2] | R.1<br>(10MHz,1PRB,16QAM,½)<br>(1x2 Low)<br>ETU,70 | 9 234 | 10 260 | 18 468 | 12 000 | 20 000 |
| 18 | [3.3] | R.1<br>(20MHz,1PRB,16QAM,½)<br>(1x2 Low)<br>ETU,70 | 13 373 | 14 859 | 26 746 | 16 000 | 28 000 |

**Table G.3.5-2: Minimum Test time for PDSCH Single Antenna Port Performance with 1 PRB**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [3.4] | R.29<br>(10MHz,1PRB,16QAM,½)<br>(1x2 Low)<br>ETU,70<br>[MBFSN] | 5 246 | 17 487 | 17 487 | 19 000 | 19 000 |

**Table G.3.5-3: Minimum Test time for PDSCH Transmit diversity 2x2**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [7.1] | R11<br>(10MHz, full, 16QAM ½)<br>(2x2 Med)<br>EVA,5<br>[SFBC, Space Frequency Block Code] | 50 000 | 55 556 | 100 000 | 57 000 | 101 000 |
| 2 | [7.2] | R.10<br>(10MHz, Full, QPSK, 1/3)<br>(2x2 low)<br>HST<br>[SFBC] | 28 800 | NA | NA | 28 800 | 57 600 |

**Table G.3.5-4: Minimum Test time for PDSCH Transmit diversity 4x2**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [7.3] | R.12<br>(1.4MHz, full, QPSK 1/3)<br>(4x2 med)<br>EPA,5<br>[SFBC-FSTD, SFBC-Frequency Shifted Transmit Diversity] | 150 000 | 166 667 | 300 000 | 168 000 | 301 000 |

**Table G.3.5-5: Minimum Test time for PDSCH Open Loop Spacial Multiplexing 2x2**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [6.1] | R.11<br>(10MHz, Full, 16QAM, ½)<br>(2x2 Low)<br>EVA,70<br>[LD-CDD, Large Delay-Cyclic Delay Diversity] | 7 600 | 8 445 | 19 000 | 10 000 | 20 000 |

**Table G.3.5-6: Minimum Test time for PDSCH Open Loop Spacial Multiplexing 4x2**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [6.2] | R.14<br>(10MHz, full, 16 QAM, ½)<br>(4x2 low)<br>EVA,70<br>[LD-CDD] | 4 860 | 5 400 | 12 150 | 7 000 | 14 000 |

**Table G.3.5-7: Minimum Test time for PDSCH Closed LoopSingle/Multilayer Spacial Multiplexing 2x2**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [4.1] | R.10<br>(10MHz,6PRB,QPSK,1/3)<br>(2x2 Low)<br>EVA,5<br>[SCW, Single CodeWord] | 49 140 | 54 600 | 98 280 | 56 000 | 100 000 |
| 2 | [4.2] | R.10<br>(10MHz, Full, QPSK, 1/3)<br>(2x2 High)<br>EPA,5<br>[SCW] | 50 000 | 55 556 | 100 000 | 57 000 | 101 000 |
| 3 | [5.1] | R.11<br>(10MHz,full, 16QAM ½)<br>(2x2Low)<br>EVA,5<br>[MCW, Multiple Code Word] | 34 266 | 38 074 | 85 665 | 40 000 | 87 000 |
| 4 | [5.2] | R.11<br>(10MHz, full, 16QAM ½)<br>(2x2Low)<br>ETU,70<br>[MCW] | 2 736 | 3 040 | 6 840 | 5 000 | 8 000 |

**Table G.3.5-8: Minimum Test time for PDSCH Closed LoopSingle/Multilayer Spacial Multiplexing 4x2**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [4.3] | R.13<br>(10 MHz,6PRB,QPSK1/3)<br>(4x2 Low)<br>EVA,5<br>[SCW] | [26 528] | 29 476 | 53 056 | 31 000 | 55 000 |
| 2 | [5.3] | R.14<br>(10MHz,6PRB,16QAM ½)<br>(4x2low)<br>EVA5<br>[MCW] | 26 066 | 28 963 | 65 165 | 30 000 | 67 000 |

**Table G.3.5-9: Minimum Test time for PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 8 and forward)**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [11.1] | R.25<br>(10 MHz, full, QPSK 1/3)<br>(1x2 Low)<br>EPA,5 | 38 879 | 43 199 | 77 758 | 45 000 | 79 000 |
| 2 | [11.2] | R.26<br>(10MHz, full, 16QAM ½)<br>(1x2 Low)<br>EPA5 | 47 781 | 53 090 | 95 562 | 55 000 | 97 000 |
| 3 | [11.3] | R.27<br>(10MHz, full, 64QAM 3/4)<br>(1x2 Low)<br>EPA,5 | 48 685 | 54 095 | 97 370 | 56 000 | 99 000 |
| 4 | [11.4] | R.28<br>(10MHz, 1PRB, 16QAM ½)<br>(1x2 Low)<br>EPA,5 | 100 000 | 111 112 | 200 000 | 113 000 | 201 000 |

**Table G.3.5-10: Minimum Test time for PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 9 and forward)**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|
| | | | | FDD | TDD |
| 1<br>NA | | | | | |
| 2 | [ . ] | R.26<br>(5MHz, full, 16QAM ½)<br>(2x2 Low)<br>EPA5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for antenna configuration, MNSF is reused from Table G.3.5-9, Test 2 | 55 000 | 97 000 |
| 3 | [ . ] | R.27<br>(10MHz, part, 64QAM 3/4)<br>(2x2 Low)<br>EPA,5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for antenna configuration, MNSF is reused from Table G.3.5-9, Test 3 | 56 000 | 99 000 |
| 4<br>NA | | | | | |

**Table G.3.5-11: Minimum Test time for PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 without simultaneous transmission**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|
| | | | | FDD | TDD |
| 1 | [ . ] | R.31<br>(10 MHz, full, QPSK 1/3)<br>(2x2 Low)<br>EVA,5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 1 | 45 000 | 79 000 |
| 2 | [ . ] | R.32<br>(10MHz, full, 16QAM ½)<br>(2x2 Medium)<br>EPA5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 2 | 55 000 | 97 000 |
| | | R.32-1<br>(5MHz, full, 16QAM ½)<br>(2x2 Medium)<br>EPA5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, channel BW, antenna configuration, MNSF is reused from Table G.3.5-9, Test 2 | | |
| 3 | [ . ] | R.33<br>(10MHz, full, 64QAM 3/4)<br>(2x2 Low)<br>EPA,5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 3 | 56 000 | 99 000 |
| | | R.33-1<br>(10MHz, part, 64QAM 3/4)<br>(2x2 Low)<br>EPA5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, allocation, antenna configuration, MNSF is reused from Table G.3.5-9, Test 3 | | |

**Table G.3.5-12: Minimum Test time for PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 with a simultaneous transmission**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|
| | | | | FDD | TDD |
| 4 | [ . ] | R.32<br>(10MHz, full, 16QAM ½)<br>(2x2 Medium)<br>EPA,5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 2 | 55 000 | 97 000 |
| 5 | | R.33<br>(10MHz, full, 64QAM 1/2)<br>(2x2 Low)<br>EPA,5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 3 | 56 000 | 99 000 |

**Table G.3.5-13: Minimum Test time for PDSCH Dual-layer Spatial Multiplexing**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|
| | | | | FDD | TDD |
| 1 | [ . ] | R.31<br>(10 MHz, full, QPSK 1/3)<br>(2x2 Low)<br>EVA,5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 1 | [45 000] | [79 000] |
| 2 | [ . ] | R.32<br>(10MHz, full, 16QAM ½)<br>(2x2 Medium)<br>EPA5 | Note:<br>MNAS is not simulated. Because of same demodulation scenario except for RMC, antenna configuration, MNSF is reused from Table G.3.5-9, Test 2 | [55 000] | [97 000] |

## G.3.6 Test conditions for receiver performance tests

**Table G.3.6: Test conditions for receiver performance tests**

**Table G.3.6-1: Single Antenna Port Performance (Cell-specific Reference Symbols) for test case 8.2.1.1 and 8.2.2.1 demodulation of PDSCH**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.2.1.1 FDD PDSCH Single Antenna Port Performance (Cell-Specific Reference Symbols) | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.2.1.1 and 8.2.2.1each component in the test vector must pass<br><br>For UEs, supporting multiple E_UTRA-bands (number of bands =B), the number of repetitions must be multiplied by B.<br><br>If a test is defined over a BW>(BW of the E_UTRA band), the test is not applicable and reduces the number of repetitions.<br><br>If a test is defined over a BW, which is not supported in the E_UTRAN band, the test is not applicable and reduces the number of repetitions. |
| | | QPSK | 5 | 5 | 5 | |
| | | 16QAM | 0 | 3 | 3 | |
| 8.2.1.2 TDD PDSCH Single Antenna Port Performance (Cell-Specific Reference Symbols) | subframes are independent | 64 QAM | 1 | 6 | 7 | |
| | | 1PRB | 4 | 4 | 4 | |
| | | Σ | 10 | 18 | 19 | |

**Table G.3.6-2: Transmit Diversity Performance (Cell-specific Reference Symbols) for test case 8.2.1.2 and 8.2.2.2 demodulation of PDSCH**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.2.1.2<br><br>FDD PDSCH Transmit Diversity Performance (Cell-Specific Reference Symbols) | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.2.1.2 and 8.2.2.2 each component in the test vector must pass<br><br>For UEs, supporting multiple E_UTRA-bands (number of bands =B), the number of repetitions must be multiplied by B.<br><br>If a test is defined over a BW, which is not supported in the E_UTRAN band, the test is not applicable and reduces the number of repetitions. |
| | | QPSK | 2 | 2 | 2 | |
| 8.2.2.2<br><br>TDD PDSCH Transmit Diversity Performance (Cell-Specific Reference Symbols) | subframes are independent | 16QAM | 0 | 1 | 1 | |
| | | Σ | 2 | 3 | 3 | |

**Table G.3.6-3: Open Loop Spatial Multiplexing Performance (Cell-specific Reference Symbols) for test case 8.2.1.3 and 8.2.2.3 demodulation of PDSCH**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.2.1.3<br><br>FDD PDSCH Open Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols) | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.2.1.3 and 8.2.2.3 each component in the test vector must pass |
| | | 16QAM | 0 | 2 | 2 | |
| 8.2.2.3<br><br>TDD PDSCH Open Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols) | subframes are independent | Σ | 0 | 2 | 2 | |

**Table G.3.6-4: Closed Loop Spatial Multiplexing Performance (Cell-specific Reference Symbols) for test case 8.2.1.4 and 8.2.2.4 demodulation of PDSCH**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.2.1.4<br><br>FDD PDSCH Closed Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols) | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.2.1.4 and 8.2.2.4 each component in the test vector must pass |
| | | Single layer QPSK | 3 | 3 | 3 | |
| 8.2.2.4<br><br>TDD PDSCH Closed Loop Spatial Multiplexing Performance (Cell-Specific Reference Symbols) | subframes are independent | Multi layer 16QAM | 0 | 3 | 3 | |
| | | Σ | 3 | 6 | 6 | |

**Table G.3.6-5: TDD PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 8 and forward)**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.3.2.1.1<br><br>TDD PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 8 and forward) | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.3.2.1 each component in the test vector must pass |
| | | QPSK | 1 | 1 | 1 | |
| | | 16QAM | 1 | 2 | 2 | |
| | | 64 QAM | 0 | 1 | 1 | |
| | | Σ | 2 | 4 | 4 | |

**Table G.3.6-6: TDD PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 9 and forward)**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.3.2.1.1_1<br><br>TDD PDSCH Single-layer Spatial Multiplexing on antenna port 5 (Release 9 and forward) | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.3.2.1.1_1 each component in the test vector must pass |
| | | 16QAM | 1 | 0 | 0 | |
| | | 64 QAM | 1 | 0 | 0 | |
| | | Σ | 2 | 0 | 0 | |
| | | | | | | |

**Table G.3.6-7: TDD PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 without a simultaneous transmission**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.3.2.1.2 TDD PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 without a simultaneous transmission | subframes are independent | CAT | 1 | 2-5 | | To pass 8.3.2.1.2 each component in the test vector must pass |
| | | QPSK | 1 | 1 | | |
| | | 16QAM | 1 | 1 | | |
| | | 64 QAM | 1 | 1 | | |
| | | Σ | 3 | 3 | | |

**Table G.3.6-8: TDD PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 with a simultaneous transmission**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.3.2.1.3<br><br>TDD PDSCH Single-layer Spatial Multiplexing on antenna port 7 or 8 with a simultaneous transmission | subframes are independent | CAT | 1 | 2-5 | | To pass 8.3.2.1.3 each component in the test vector must pass |
| | | 16QAM | 0 | 1 | | |
| | | 64 QAM | 0 | 1 | | |
| | | Σ | 0 | 2 | | |
| | | | | | | |

**Table G.3.6-9: TDD PDSCH Dual-layer Spatial Multiplexing**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | | | | Over all Pass/Fail condition |
|---|---|---|---|---|---|---|
| 8.3.2.2.1<br><br>TDD PDSCH Dual-layer Spatial Multiplexing | subframes are independent | CAT | 1 | 2 | 3-5 | To pass 8.3.2.2.1 each component in the test vector must pass |
| | | QPSK | 1 | 1 | 1 | |
| | | 16QAM | 1 | 2 | 2 | |
| | | 64 QAM | 0 | 1 | 1 | |
| | | Σ | 2 | 4 | 4 | |

# G.4 Statistical testing of Performance Requirements with probability of misdetection

## G.4.1 General

The test of receiver performance characteristics is two fold.

1. A signal or a combination of signals is offered to the RX port(s) of the receiver.

2. The ability of the receiver to demodulate /decode this signal is verified by analyzing the reaction of the UE to this signal.

In (2) is the statistical aspect of the test and is treated here.

The minimum requirement for those receiver performance tests are 1% or 0.1% misdetection probability

All receiver performance tests are performed in fading conditions. In addition to the statistical considerations, this requires the definition of a minimum test time.

NOTE: all demodulation performance tests (state from version 9.5.0) require a minimum test time, which exceeds the maximum test time in tables G.4.4. Under this circumstances only the test limit at the end of tables G.4.4.-1 resp. G.4.4.-2 is applicable.

## G.4.2 Mapping the UE reaction to error ratio

The UE can not indicate the detection or misdetection of the physical channel under test directly. Indirect methods are described in the procedure of the applicable test.

## G.4.3 Design of the test

G.2.3 applies, exception:

Limit ER = 0.01 and ER = 0.001

## G.4.4 Numerical definition of the pass fail limits

**Table G.4.4-1 pass fail limits for ER = 0.01**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 344 | NA | 40 | 3929 | 2553 | 80 | 7033 | 5874 | 120 | 10036 | 9354 |
| 1 | 485 | NA | 41 | 4009 | 2632 | 81 | 7109 | 5960 | 121 | 10110 | 9442 |
| 2 | 607 | 10 | 42 | 4089 | 2712 | 82 | 7185 | 6046 | 122 | 10184 | 9530 |
| 3 | 719 | 33 | 43 | 4168 | 2792 | 83 | 7261 | 6131 | 123 | 10259 | 9619 |
| 4 | 826 | 66 | 44 | 4247 | 2873 | 84 | 7336 | 6217 | 124 | 10333 | 9707 |
| 5 | 929 | 107 | 45 | 4327 | 2953 | 85 | 7412 | 6303 | 125 | 10407 | 9796 |
| 6 | 1029 | 152 | 46 | 4406 | 3034 | 86 | 7488 | 6389 | 126 | 10481 | 9884 |
| 7 | 1127 | 202 | 47 | 4484 | 3115 | 87 | 7564 | 6475 | 127 | 10555 | 9972 |
| 8 | 1223 | 255 | 48 | 4563 | 3196 | 88 | 7639 | 6561 | 128 | 10629 | 10061 |
| 9 | 1317 | 311 | 49 | 4642 | 3278 | 89 | 7715 | 6648 | 129 | 10703 | 10150 |
| 10 | 1409 | 370 | 50 | 4720 | 3359 | 90 | 7790 | 6734 | 130 | 10777 | 10238 |
| 11 | 1501 | 430 | 51 | 4799 | 3441 | 91 | 7866 | 6820 | 131 | 10851 | 10327 |
| 12 | 1592 | 492 | 52 | 4877 | 3523 | 92 | 7941 | 6907 | 132 | 10925 | 10416 |
| 13 | 1681 | 555 | 53 | 4955 | 3605 | 93 | 8017 | 6993 | 133 | 10999 | 10504 |
| 14 | 1770 | 620 | 54 | 5033 | 3688 | 94 | 8092 | 7080 | 134 | 11073 | 10593 |
| 15 | 1858 | 686 | 55 | 5111 | 3770 | 95 | 8167 | 7167 | 135 | 11147 | 10682 |
| 16 | 1946 | 754 | 56 | 5189 | 3853 | 96 | 8242 | 7253 | 136 | 11221 | 10771 |
| 17 | 2032 | 822 | 57 | 5267 | 3935 | 97 | 8317 | 7340 | 137 | 11295 | 10860 |
| 18 | 2119 | 891 | 58 | 5344 | 4018 | 98 | 8393 | 7427 | 138 | 11369 | 10949 |
| 19 | 2204 | 961 | 59 | 5422 | 4101 | 99 | 8468 | 7514 | 139 | 11442 | 11038 |
| 20 | 2290 | 1032 | 60 | 5499 | 4185 | 100 | 8543 | 7601 | 140 | 11516 | 11127 |
| 21 | 2374 | 1103 | 61 | 5577 | 4268 | 101 | 8618 | 7688 | 141 | 11590 | 11216 |
| 22 | 2459 | 1175 | 62 | 5654 | 4352 | 102 | 8693 | 7775 | 142 | 11664 | 11305 |
| 23 | 2543 | 1248 | 63 | 5731 | 4435 | 103 | 8768 | 7863 | 143 | 11737 | 11394 |
| 24 | 2627 | 1321 | 64 | 5809 | 4519 | 104 | 8843 | 7950 | 144 | 11811 | 11483 |
| 25 | 2710 | 1395 | 65 | 5886 | 4603 | 105 | 8917 | 8037 | 145 | 11885 | 11573 |
| 26 | 2793 | 1470 | 66 | 5963 | 4687 | 106 | 8992 | 8125 | 146 | 11958 | 11662 |
| 27 | 2876 | 1544 | 67 | 6039 | 4771 | 107 | 9067 | 8212 | 147 | 12032 | 11751 |
| 28 | 2958 | 1620 | 68 | 6116 | 4855 | 108 | 9142 | 8300 | 148 | 12105 | 11840 |
| 29 | 3040 | 1696 | 69 | 6193 | 4940 | 109 | 9216 | 8387 | 149 | 12179 | 11930 |
| 30 | 3122 | 1772 | 70 | 6270 | 5024 | 110 | 9291 | 8475 | 150 | 12252 | 12019 |
| 31 | 3204 | 1848 | 71 | 6346 | 5109 | 111 | 9366 | 8562 | 151 | 12326 | 12109 |
| 32 | 3285 | 1925 | 72 | 6423 | 5193 | 112 | 9440 | 8650 | 152 | 12399 | 12198 |
| 33 | 3366 | 2003 | 73 | 6499 | 5278 | 113 | 9515 | 8738 | 153 | 12473 | 12288 |
| 34 | 3447 | 2080 | 74 | 6576 | 5363 | 114 | 9589 | 8826 | 154 | 12546 | 12377 |
| 35 | 3528 | 2158 | 75 | 6652 | 5448 | 115 | 9664 | 8914 | 155 | 12620 | 12467 |
| 36 | 3609 | 2237 | 76 | 6728 | 5533 | 116 | 9738 | 9002 | 156 | 12693 | 12556 |
| 37 | 3689 | 2315 | 77 | 6805 | 5618 | 117 | 9813 | 9090 | 157 | 12767 | 12646 |
| 38 | 3769 | 2394 | 78 | 6881 | 5704 | 118 | 9887 | 9178 | 158 | 12840 | 12736 |
| 39 | 3850 | 2473 | 79 | 6957 | 5789 | 119 | 9962 | 9266 | 159 | 12913 | 12826 |
| | | | | | | | | | 160 | NA | 12915 |
| | | | | | | | | | Test limit = 1.2352E-2 | | |

**Table G.4.4-2 pass fail limits for ER = 0.001**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 3463 | NA | 41 | 40174 | 26265 | 82 | 71961 | 60368 | 123 | 102723 | 96075 |
| 1 | 4874 | 4 | 42 | 40971 | 27063 | 83 | 72720 | 61225 | 124 | 103465 | 96958 |
| 2 | 6096 | 99 | 43 | 41766 | 27863 | 84 | 73479 | 62083 | 125 | 104208 | 97842 |
| 3 | 7226 | 329 | 44 | 42559 | 28666 | 85 | 74237 | 62941 | 126 | 104949 | 98726 |
| 4 | 8298 | 658 | 45 | 43352 | 29471 | 86 | 74995 | 63801 | 127 | 105691 | 99610 |
| 5 | 9330 | 1059 | 46 | 44142 | 30279 | 87 | 75752 | 64661 | 128 | 106432 | 100495 |
| 6 | 10332 | 1513 | 47 | 44932 | 31088 | 88 | 76509 | 65522 | 129 | 107173 | 101380 |
| 7 | 11310 | 2009 | 48 | 45720 | 31899 | 89 | 77265 | 66383 | 130 | 107914 | 102266 |
| 8 | 12269 | 2539 | 49 | 46507 | 32713 | 90 | 78020 | 67246 | 131 | 108655 | 103152 |
| 9 | 13212 | 3096 | 50 | 47293 | 33528 | 91 | 78776 | 68109 | 132 | 109395 | 104039 |
| 10 | 14141 | 3677 | 51 | 48078 | 34345 | 92 | 79530 | 68973 | 133 | 110135 | 104926 |
| 11 | 15058 | 4278 | 52 | 48861 | 35164 | 93 | 80285 | 69838 | 134 | 110875 | 105813 |
| 12 | 15965 | 4896 | 53 | 49644 | 35984 | 94 | 81038 | 70704 | 135 | 111614 | 106701 |
| 13 | 16863 | 5530 | 54 | 50425 | 36807 | 95 | 81792 | 71570 | 136 | 112353 | 107589 |
| 14 | 17753 | 6177 | 55 | 51205 | 37631 | 96 | 82544 | 72437 | 137 | 113092 | 108478 |
| 15 | 18635 | 6836 | 56 | 51985 | 38456 | 97 | 83297 | 73305 | 138 | 113830 | 109367 |
| 16 | 19511 | 7507 | 57 | 52763 | 39283 | 98 | 84049 | 74173 | 139 | 114569 | 110257 |
| 17 | 20380 | 8188 | 58 | 53541 | 40112 | 99 | 84800 | 75042 | 140 | 115307 | 111146 |
| 18 | 21244 | 8878 | 59 | 54317 | 40942 | 100 | 85551 | 75911 | 141 | 116045 | 112037 |
| 19 | 22103 | 9576 | 60 | 55092 | 41773 | 101 | 86302 | 76782 | 142 | 116782 | 112927 |
| 20 | 22957 | 10282 | 61 | 55867 | 42606 | 102 | 87052 | 77653 | 143 | 117520 | 113818 |
| 21 | 23806 | 10995 | 62 | 56641 | 43440 | 103 | 87802 | 78524 | 144 | 118257 | 114710 |
| 22 | 24652 | 11715 | 63 | 57414 | 44276 | 104 | 88552 | 79396 | 145 | 118994 | 115602 |
| 23 | 25493 | 12441 | 64 | 58186 | 45113 | 105 | 89301 | 80269 | 146 | 119730 | 116494 |
| 24 | 26331 | 13173 | 65 | 58957 | 45951 | 106 | 90050 | 81143 | 147 | 120466 | 117386 |
| 25 | 27166 | 13911 | 66 | 59728 | 46790 | 107 | 90798 | 82017 | 148 | 121203 | 118279 |
| 26 | 27997 | 14654 | 67 | 60497 | 47631 | 108 | 91546 | 82891 | 149 | 121939 | 119173 |
| 27 | 28826 | 15401 | 68 | 61266 | 48472 | 109 | 92293 | 83766 | 150 | 122674 | 120066 |
| 28 | 29651 | 16154 | 69 | 62035 | 49315 | 110 | 93041 | 84642 | 151 | 123410 | 120960 |
| 29 | 30474 | 16910 | 70 | 62802 | 50159 | 111 | 93787 | 85518 | 152 | 124145 | 121855 |
| 30 | 31294 | 17671 | 71 | 63569 | 51004 | 112 | 94534 | 86395 | 153 | 124880 | 122749 |
| 31 | 32111 | 18436 | 72 | 64335 | 51851 | 113 | 95280 | 87273 | 154 | 125615 | 123644 |
| 32 | 32927 | 19204 | 73 | 65100 | 52698 | 114 | 96026 | 88151 | 155 | 126349 | 124540 |
| 33 | 33740 | 19976 | 74 | 65865 | 53546 | 115 | 96771 | 89029 | 156 | 127083 | 125435 |
| 34 | 34551 | 20752 | 75 | 66629 | 54396 | 116 | 97516 | 89908 | 157 | 127818 | 126332 |
| 35 | 35360 | 21531 | 76 | 67393 | 55246 | 117 | 98261 | 90788 | 158 | 128551 | 127228 |
| 36 | 36166 | 22312 | 77 | 68156 | 56097 | 118 | 99005 | 91668 | 159 | 129285 | 128125 |
| 37 | 36971 | 23097 | 78 | 68918 | 56950 | 119 | 99750 | 92548 | 160 | 130019 | 129022 |
| 38 | 37775 | 23885 | 79 | 69679 | 57803 | 120 | 100493 | 93429 | 161 | 130752 | 129919 |
| 39 | 38576 | 24676 | 80 | 70440 | 58657 | 121 | 101237 | 94311 | 162 | NA | 130817 |
| 40 | 39376 | 25469 | 81 | 71201 | 59512 | 122 | 101980 | 95193 | Test limit = 1.2345E-3 | | |

NOTE 1: The first column is the number of errors (ne = number of misdetections)

NOTE 2: The second column is the number of samples for the pass limit ($ns_p$ , ns=Number of Samples= number misdetections + number of detections )

NOTE 3: The third column is the number of samples for the fail limit ($ns_f$)

NOTE 4: The test limit at the end of the table is applicable, when the minimum test time in clause 3.5 governs the test. Pass the test for ER $\leq$ Test limit, otherwise fail.

## G.4.5 Pass fail decision rules

G.2.5 applies

NOTE: For ER=0.01 an ideal DUT passes after 344 samples. The maximum test time is 12913 samples. .For ER=0.001 an ideal DUT passes after 3463 samples. The maximum test time is 130752 samples.

# G.4.6 Minimum Test time

**Table G.4.6-1: Minimum Test time for Demodulation of PCFICH/PDCCH**

| Test No | Demod. Scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [8.1] | R.15 (10 MHz, 8CCE, full, QPSK 1/3) (1x2 Low) ETU70 | 200 000 | 222 222 | 400 000 | 224 000 | 401 000 |
| 1 | [8.2] | R.16 (1.4MHz, 2CCE, full, QPSK 1/3) (2x2 Low) EPA5 | 200 000 | 222 222 | 400 000 | 224 000 | 401 000 |
| 1 Rel 9 … | [ ] | R.16-1 (10MHz, 4CCE, full, QPSK 1/3) (2x2 Low) EVA70 | 200 000 | 222 222 | 400 000 | 224 000 | 401 000 |
| 1 | [8.3] | R.17 (10MHz, 4CCE, full, QPSK 1/3) (4x2 Medium) EVA5 | 200 000 | 222 222 | 400 000 | 224 000 | 401 000 |
| 1 Rel 9 … | [ ] | R.17-1 (5MHz, 2CCE, full, QPSK 1/3) (4x2 Medium) EPA5 | 200 000 | 222 222 | 400 000 | 224 000 | 401 000 |
| Note: Simulation method to derive MNAS is based on finite test time and its effect on test system uncertainty specified in clause F.1.4. | | | | | | | |

**Table G.4.6-2: Minimum Test time for Demodulation of PHICH**

| Test No | Demod. scenario | Demodulation scenario (info only) | MNAS (Simulation) | MNS (Calculation) | | MNSF (Min No Sub Frames, mandatory) | |
|---|---|---|---|---|---|---|---|
| | | | | FDD | TDD | FDD | TDD |
| 1 | [9.1] | R.18<br>(10 MHz, full, QPSK 1/3)<br>(1x2 Low)<br>ETU70 | 200 000 | 200 000 | 500 000 | 200 000 | 500 000 |
| 2 | [9.4] | R.24<br>(10MHz, full, 16QAM ½)<br>(1x2 Low)<br>ETU70 | 200 000 | 200 000 | 500 000 | 200 000 | 500 000 |
| 1 | [9.2] | R.19<br>(1.4MHz, full, 64QAM 3/4)<br>(2x2 Low)<br>EPA5 | 200 000 | 200 000 | 500 000 | 200 000 | 500 000 |
| 1 Rel. 9… | [ ] | R.19-1<br>(10MHz, full, 64QAM 3/4)<br>(2x2 Low)<br>EVA70 | 200 000 | 200 000 | 500 000 | 200 000 | 500 000 |
| 1 | [9.3] | R.20<br>(10MHz, 1PRB, 16QAM ½)<br>(4x2 Medium)<br>EVA5 | 200 000 | 200 000 | 500 000 | 200 000 | 500 000 |
| 1 Rel. 9… | [ ] | R.20-1<br>(5MHz, 1PRB, 16QAM ½)<br>(4x2 Medium)<br>EPA5 | 200 000<br>R.20 has 5MHz BW Scenario 9.3. has EVA5 | 200 000 | 500 000 | 200 000 | 500 000 |
| Note: Simulation method to derive MNAS is based on finite test time and its effect on test system uncertainty specified in clause F.1.4. | | | | | | | |

# G.4.7 Test conditions for receiver performance tests

**Table G.4.7: Test conditions for receiver performance tests**

| Test | Statistical independence | Number of components in the test vector, as specified in the test requirements and initial conditions of the applicable test | Over all Pass/Fail condition Restrictions and extensions see Table G.3.6-1 |
|---|---|---|---|
| 8.4.1.1 FDD PCFICH/PDCCH Single-antenna Port Performance | A misdetection is an independent event | 1 | NA |
| 8.4.1.2 FDD PCFICH/PDCCH Transmit Diversity Performance | A misdetection is an independent event | 2 | To pass 8.4.1.2 each component in the test vector must pass |
| 8.4.2.1 TDD PCFICH/PDCCH Single-antenna Port Performance | A misdetection is an independent event | 1 | NA |
| 8.4.2.2 TDD PCFICH/PDCCH Transmit Diversity Performance | A misdetection is an independent event | 2 | To pass 8.4.2.2 each component in the test vector must pass |
| 8.5.1.1 FDD PHICH Single-antenna Port Performance | A misdetection is an independent event | 2 | To pass 8.5.1.1 each component in the test vector must pass |
| 8.5.1.2 FDD PHICH Transmit Diversity Performance | A misdetection is an independent event | 2 | To pass 8.5.1.2 each component in the test vector must pass |
| 8.5.2.1 TDD PHICH Single-antenna Port Performance | A misdetection is an independent event | 2 | To pass 8.5.2.1 each component in the test vector must pass |
| 8.5.2.2 TDD PHICH Transmit Diversity Performance | A misdetection is an independent event | 2 | To pass 8.5.2.2 each component in the test vector must pass |

# G.5 Measuring throughput ratio

## G.5.1 General

Annex G.5 is applicable for clauses 9.3, 9.4 and 9.5. Common to those clauses is, that a throughput ratio γ is measured. These clauses are tested exclusively with “slow” multipath fading profiles. Hence the test time is governed by test time due to fading, and number of samples due to statistical significance is not applicable.

The test requirements in clause 9.4 are a ratio of 2 throughput tests according to $\gamma = \frac{t_{ue}}{t_{rnd}}$. The denominator must be established by an approach, resulting in the denominator throughput $t_{rnd}$ and the reference $SNR_{rnd}$, the latter is reused to measure the nominator throughput.

The test requirements in clauses 9.3 and 9.5 are a ratio of 2 throughput tests according to $\gamma = \frac{t_{subband}}{t_{median}}$, $\gamma = \frac{t_{reported}}{t_{fix}}$ etc.

Nominator and denominator are ordinary throughput tests

$t_{ue}$, $t_{rnd}$ , $t_{reported}$, $t_{fix}$ , $t_{subband}$, $t_{median}$, $t_{wideband}$ are throughputs, derived under different conditions and are defined in clause s 9.3, 9.4 and 9.5.

$SNR_{rnd}$ is the signal noise ratio, derived together with $t_{rnd}$ and is defined in clause 9.4.

## G.5.2 Establishing $t_{rnd}$

Adjust SNR such that the measured throughput is $58\% \leq t_{rnd} \leq 62\%$.

The resulting SNR is declared $SNR_{rnd}$

To achieve statistical significance the final throughput measurement must be done with MNS samples, given table G.5.4-1

The approach, leading to $t_{rnd}$ and $SNR_{rnd}$ is not specified.

## G.5.3 Measuring T-put

To achieve statistical significance the final throughput measurement must be done with MNS samples, given in table.G.5.4 -1. Number of samples due to statistical significance is not applicable.

For measuring $t_{subband}$, $t_{wideband}$, $t_{reported}$ and $t_{fix}$, the SS collects ACK, NACK and statDTX from the UE and records the time, elapsed from the beginning of the test. The payload size, received by the UE and acknowledged towards the SS, may vary within a test versus time (e.g. due to subband changes upon a UE report) Throughput is calculated in the SS by summing up the payload, associated to each ACK, from the start of the test and dividing the accumulated payload in kilobits by the time in seconds, elapsed from the beginning of the test. This is similar but not same as in G.2.2. (Main difference in bullet d, where the payload size is constant).

For measuring $t_{median}$, $t_{ue}$, and $t_{rnd}$, the SS collects ACK, NACK and statDTX from the UE and records the time, elapsed from the beginning of the test. The payload size, received by the UE and acknowledged towards the SS, is constant. Throughput can be calculated in the SS by multiplying the payload size with the number of ACKs and dividing the accumulated payload in kilobits by the time in seconds, elapsed from the beginning of the test, being associated to the following ratio: ACK/ (ACK+NACK + DTX).

## G.5.4 Number of samples for throughput ratios

TT for γ and MNS are based on theoretical estimations.

**Table G.5.4-1: Test time for testing throughput ratios**

| Test | Demodulation scenario: RMC (Bandwidth, allocated RBs, modulation, coding) [Antenna configuration, correlation] Propagation condition, Doppler | Γ | Minimum Number of Subframes (MNS) (MNS = active and inactive Subframes, more details in Annex G.3.5 and table G.3.5-1) | | Γ including TT | BLER |
|---|---|---|---|---|---|---|
| | | | FDD | TDD | | |
| 9.3.1.1.1<br>9.3.1.1.2 | (10 MHz, 6, variable modulation and coding) [1x2, full] Special propagation according to clause B.4.2, 5Hz | 1.1 | 100000<br>For denominator- and nominator-measurement each | 170000<br>For denominator- and nominator-measurement each | γ =1.09 | BLER=0.05, no TT<br>No of samples: subset of ACKs and NACKs in the MNS for throughput. |
| 9.3.2.1.1<br>9.3.2.1.2 | (10 MHz, full, variable modulation and coding) [1x2, high] EPA5 | 1.05 | 100000<br>For denominator- and nominator-measurement each | 170000<br>For denominator- and nominator-measurement each | γ =1.04 | BLER=0.02, no TT<br>No of samples for FDD: subset of ACKs and NACKs in the MNS for throughput.<br>No of samples for TDD: subset of *filtered* ACKs and NACKs in the MNS for throughput. |
| 9.3.3.1.1<br>9.3.3.1.2 | 10 MHz, full(however unequal SNR), variable modulation and coding) [1x2, full] Special: propagation according to clause B. 2.4, 5Hz | 1.6 | 100000<br>For denominator- and nominator-measurement each | 170000<br>For denominator- and nominator-measurement each | 1.50 | |
| 9.3.4.1.1<br>9.3.4.1.2 | Same as 9.3.3 | 1.2 | 100000 | 170000 | 1.19 | |
| 9.3.4.2.1<br>9.3.4.2.1 | Same as 9.3.3 | 1.15 | 100000 | 170000 | 1.14 | |
| 9.4.1.1.1<br>9.4.1.1.2 | R. 10 (10 MHz, full, QPSK, 1/3) (2x2 Low) EVA,5 | 1.1 | 100000 | 170000 | γ =1.09 | |
| 9.4.1.2.1<br>9.4.1.2.2 | R.14 (10 MHz, full, | 1.2 | 100000 | 170000 | 1.19 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | QPSK, 1/3) (4x2 Low) EVA,5 | | | | | |
| 9.4.2.1.1 9.4.2.1.2 | R.30 (20 MHz, full, 16QAM, 1/2) (2x2 Low) EPA5 | 1.2 | 100000 | 170000 | γ =1.19 | |
| 9.4.2.1.1 _1 9.4.2.1.2 _1 | R.11-3 (10 MHz, full, 16QAM, 1/2) (2x2 Low) EPA5 | 1.2 | 100000 | 170000 | γ =1.19 | |
| 9.4.2.2.1 | R.14-2 FDD (10MHz, 3, 16 QAM, ½, 4x2low, EVA5) | 1.2 | 100000 | 170000 | 1.19 | |
| 9.4.2.2.2 | R.14-2 TDD (10MHz, 3, 16QAM, ½, 4x2low, EVA5) | 1.15 | 100000 | 170000 | 1.14 | |
| 9.5.1.1 9.5.1.2 | (10MHz, full, variable modulation and coding) (2x2, low or high according to test) (2x2, EPA5) | Test 2$\gamma_1$=1 .05 Test 1$\gamma_2$=1 Test 3$\gamma_2$=1 .1 | 100000 | 170000 | Test2$\gamma_1$= 1.04 Test1$\gamma_2$= 0.99 Test3$\gamma_2$= 1.09 | |

# G.6 Statistical testing of MBMS Performance

## G.6.1 General

The system simulator sends MBMS packets to the UE under test. The number of packets, the SS sends, is predefined by the test time in G.6.4. The UE under tests demodulates the MBMS packets and counts the successfully received number of MBMS packets into the UE internal MBMS packet counter. The SS reads out the counter and issues a pass fail decision.

## G.6.2 Mapping of MBMS Packet ratio to BLER

The minimum requirements are designed in terms of BLER = 1%. 10 MBMS packets are included in one Transport block, constant throughout the test. The ratio of correct received MBMS packets ($M_{ok}$) to transmitted packets ($M_{tot}$) is assumed to be equal to the ratio of correct received transport blocks to transmitted transport blocks. This assumption ignores the unlikely case that payload and CRC for a transport block are consistent but wrong.

## G.6.3 Design of the test

The minimum requirement for all MBMS tests is BLER = 1%. All MBMS tests are performed under a fading scenario: MBSFN channel model (Table B.2.6-1). It is obvious, that this fading scenario requires a minimum test time, greater than the time, required for statistical significance. In addition the test design with the MBMS packet counter in the UE suggest a fixed test time. Without simulation the test time is set to [200 000] active subframes (Transport blocks), leading to [333 333] subframes (333sec) for FDD and [400 000] subframes (400sec) for TDD. In order to avoid the fail of a good DUT due to statistical uncertainty the minimum requirement of BLER = 1% leads to a

Test limit = 1.2352 %. (refer Table G.4.4-1)

This means a DUT actually on the limit (1%) is measured and passed with a confidence level of greater than 95%.

## G.6.4 Test time for MBMS performance tests

**Table G.6.4-1: Minimum Test time for MBMS**

| Test No | Demodulation scenario (info only) | Min No of Sub Frames | |
|---|---|---|---|
| | | FDD (6 of 10 subframes are active) | TDD (5 of 10 subframes are active) |
| 1 | R.37 (10 MHz, full, QPSK 1/3) (1x2 Low) MBSFN channel model | [333 333] | [400 000] |
| 2 | R.38 (10MHz, full, 16QAM 1/2) (1x2 Low) MBSFN channel model | [333 333] | [400 000] |
| 3 | R.39 (10 MHz, full, 64QAM 2/3) (1x2 Low) MBSFN channel model | [333 333] | [400 000 |
| 3 | R.39-1 (5 MHz, full, 64QAM 2/3) (1x2 Low) MBSFN channel model | [333 333] | [400 000] |
| 4 | R.40 (1.4 MHz, full, QPSK 1/3) (1x2 Medium) MBSFN channel model | [333 333] | [400 000] |

# G.X Theory to derive the numbers in Table G.2.1.3-1 (Informative)

Editor's note: this section of the Annex G is for information only and it described the background theory and information to derive the entries in the table G.2.1.3-1.

## G.X.1 Error Ratio (ER)

The Error Ratio (ER) is defined as the ratio of number of errors (ne) to all results, number of samples (ns).

(1-ER is the success ratio).

## G.X.2 Test Design

A statistical test is characterised by:

Test-time, Selectivity and Confidence level.

## G.X.3 Confidence level

The outcome of a statistical test is a decision. This decision may be correct or in-correct. The Confidence Level CL describes the probability that the decision is a correct one. The complepement is the wrong decision probability (risk) D = 1-CL

## G.X.4 Introduction: Supplier Risk versus Customer Risk

There are two targets of decision:

1. (a) A measurement on the pass-limit shows, that the DUT has the specified quality or is better with probability CL (CL e.g.95%) This shall lead to a “pass decision”

   The pass-limit is on the good side of the specified DUT-quality. A more stringent CL (CL e.g.99%) shifts the pass-limit farer into the the good direction. Given the quality of the DUTs is distributed, a greater CL passes less and better DUTs.

   A measurement on the bad side of the pass-limit is simply “not pass” (undecided or artificial fail).

   (aa) Complementary:

   A measurement on the fail-limit shows, that the DUT is worse than the specified quality with probability CL.

   The fail-limit is on the bad side of the specified DUT-quality. A more stringent CL shifts the fail-limit farer into the the bad direction. Given the quality of the DUTs is distributed, a greater CL fails less and worse DUTs.

   A measurement on the good side of the fail-limit is simply “not fail”.

2. (b) A DUT, known to have the specified quality, shall be measured and decided pass with probability CL. This leads to the test limit.

   For CL e.g. 95%, the test limit is on the bad side of the specified DUT-quality. CL e.g.99% shifts the pass-limit farer into the the bad direction. Given the DUT-quality is distributed, a greater CL passes more and worse DUTs.

   (bb) A DUT, known to be an ($\varepsilon \rightarrow 0$) beyond the specified quality, shall be measured and decided fail with probability CL.

   For CL e.g.95%, the test limit is on the good side of the specified DUT-quality.

NOTE 1: the different sense for CL in (a), (aa) versus (b), (bb)

NOTE 2: for constant CL in all 4 bullets (a) is equivalent to (bb) and (aa) is equivalent to (b)

## G.X.5 Supplier Risk versus Customer Risk

The table below summarizes the different targets of decision.

**Table G.X.5-1 Equivalent statements**

| | **Equivalent statements, using different cause-to-effect-directions, and assuming CL = constant >1/2** | |
|---|---|---|
| cause-to-effect-directions | Known measurement result → estimation of the DUT's quality | Known DUT's quality → estimation of the measurement's outcome |
| Supplier Risk | A measurement on the pass-limit shows, that the DUT has the specified quality or is better (a) | A DUT, known to have an (ε→0) beyond the specified DUT-quality, shall be measured and decided fail (bb) |
| Customer Risk | A measurement on the fail-limit shall shows, that the DUT is worse than the specified quality (aa) | A DUT, known to have the specified quality, shall be measured and decided pass (b) |

The shaded area shown the direct interpretation of Supplier Risk and Customer Risk.

The same statements can be based on other DUT-quality-definitions.

## G.X.6 Introduction: Standard test versus early decision concept

In standard statistical tests, a certain number of results (ns) is predefined in advance to the test. After ns results the number of bad results (ne) is counted and the error ratio (ER) is calculated by ne/ns.

Applying statistical theory, a decision limit can be designed, against which the calculated ER is compared to derive the decision. Such a limit is one decision point and is characterised by:

- D: the wrong decision probability (a predefined parameter)
- ns: the number of results (a fixed predefined parameter)
- ne: the number of bad results (the limit based on just ns)

In the formula for the limit, D and ns can be understood as variable parameter and variable. However the standard test execution requires fixed ns and D. The property of such a test is: It discriminate between two states only, depending on the test design:

- pass (with CL) / undecided (undecided in the sense: finally undecided)
- fail (with CL) / undecided (undecided in the sense: finally undecided)
- pass(with CL) / fail (with CL) (however against two limits).

In contrast to the standard statistical tests, the early decision concept predefines a set of (ne,ns) co-ordinates, representing the limit-curve for decision. After each result a preliminary ER is calculated and compared against the limit-curve. After each result one may make the decision or not (undecided for later decision) The parameters and variables in the limit-curve for the early decision concept have a similar but not equal meaning:

- D: the wrong decision probability (a predefined parameter)
- ns: the number of results (a variable parameter)
- ne: the number of bad results (the limit. It varies together with ns)

To avoid a "final undecided" in the standard test, a second limit must be introduced and the single decision co-ordinate (ne,ns) needs a high ne, leading to a fixed (high) test time. In the early decision concept, having the same selectivity and the same confidence level an "undecided" need not to be avoided, as it can be decided later. A perfect DUT will hit the decision coordinate (ne,ns) with ne=0. This test time is short.

# G.X.7 Standard test versus early decision concept

For Supplier Risk:

The wrong decision probability D in the standard test is the probability, to decide a DUT in-correct in the single decision point. In the early decision concept there is a probability of in-correct decisions d at each point of the limit-curve. The sum of all those wrong decision probabilities accumulate to D. Hence d<D

For Customer Risk:

The correct decision probability CL in the standard test is the probability, to decide a DUT correct in the single decision point. In the early decision concept there is a probability of correct decisions cl at each point of the limit-curve. The sum of all those correct decision probabilities accumulate to CL. Hence cl<CL or d>D

# G.X.8 Selectivity

There is no statistical test which can discriminate between a limit DUT and a DUT which is an ($\varepsilon \rightarrow 0$) apart from the limit in finite time and high confidence level CL. Either the test discriminates against one limit with the results pass (with CL)/undecided or fail (with CL)/undecided, or the test ends in a result pass (with CL)/fail (with CL) but this requires a second limit.

For CL>1/2, a (measurement-result = specified-DUT-quality), generates undecided in test "supplier risk against pass limit" (a, from above) and also in the test "customer risk against the fail limit " (aa)

For CL>1/2, a DUT, known to be on the limit, will be decided pass for the test "customer risk against pass limit" (b) and also "supplier risk against fail limit" (bb).

This overlap or undecided area is not a fault or a contradiction, however it can be avoided by introducing a Bad or a Good DUT quality according to:

- Bad DUT quality: specified DUT-quality * M (M>1)
- Good DUT quality: specified DUT-qualityt * m (m<1)

Using e.g M>1 and CL=95% the test for different DUT qualities yield different pass probabilities:

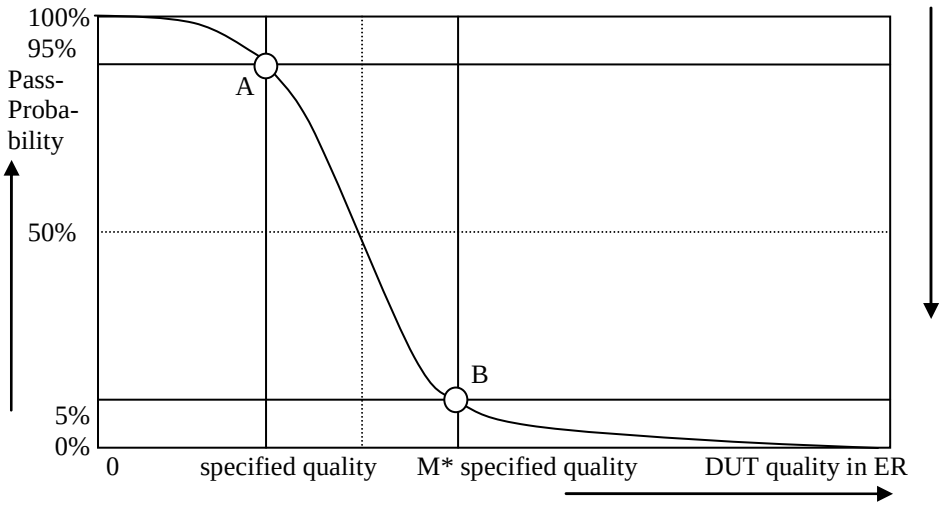



**Figure G.X.8-1: Pass probability versus DUT quality**

# G.X.9 Design of the test

The receiver characteristic test are defined by the following design principles:

1. The early decision concept is applied.
2. A second limit is introduced: Bad DUT factor M>1
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

   To decide the test fail

   Cusomer Risk is applied based on the specified DUT quality

The receiver characteristic test are defined by the following parameters:

1. Limit ER = 0.05
2. Bad DUT factor M=1.5 (selectivity)
3. Confidence level CL = 95% (for specified DUT and Bad DUT-quality)

This has the following consequences:

1. A measurement on the fail limit is connected with 2 equivalent statements:

| | |
|---|---|
| A measurement on the fail-limit shows, that the DUT is worse than the specified DUT-quality | A DUT, known have the specified quality, shall be measured and decided pass |

2. A measurement on the pass limit is connected with the complementary statements:

| | |
|---|---|
| A measurement on the pass limit shows, that the DUT is better than the Bad DUT-quality. | A DUT, known to have the Bad DUT quality, shall be measured and decided fail |

   The left comumn is used to decide the measurement.

   The right column is used to verify the design of the test by simulation.

   The simulation is based on the two fulcrums A and B only in Figure G.x.8-1

3. Test time

   The minimum and maximum test time is fixed.

   The average test time is a function of the DUT's quality.

   The individual test time is not predictable.

4. The number of decision co-ordinates (ne,ns) in the early decision concept is responsible for the selectivity of the test and the maximum test time. Having fixed the number of decision co-ordinates there is still freedom to select the individual decision co-ordinates in many combinations, all leading to the same confidence level.

# G.X.10 Simulation to derive the pass fail limits in Table G.2.1.3-1

There is freedom to design the decision co-ordinates (ne,ns).

The binomial distribution and its inverse is used to design the pass and fail limits. Note that this method is not unique and that other methods exist.

$$fail(ne, d_f) := \frac{ne}{(ne + qnbinom(d_f, ne, ER))}$$

$$pass(ne, cl_p, M) := \frac{ne}{(ne + qnbinom(cl_p, ne, ER \cdot M))}$$

Where

- fail(..) is the error ratio for the fail limit
- pass(..) is the error ratio for the pass limit
- ER is the specified error ratio 0.05
- ne is the number of bad results. This is the variable in both equations
- M is the Bad DUT factor M=1.5
- $d_f$ is the wrong decision probability of a single (ne,ns) co-ordinate for the fail limit.
  It is found by simulation to be $d_f = 0.004$
- $cl_p$ is the confidence level of a single (ne,ns) co-ordinate for the pass limit.
  It is found by simulation to be $cl_p = 0.9975$
- qnbinom(..): The inverse cumulative function of the negative binomial distribution

The simulation works as follows:

- A large population of limit DUTs with true ER = 0.05 is decided against the pass and fail limits.
- $cl_p$ and $d_f$ are tuned such that CL (95%) of the population passes and D (5%) of the population fails.
- A population of Bad DUTs with true ER = M*0.05 is decided against the same pass and fail limits.
- $cl_p$ and $d_f$ are tuned such that CL (95%) of the population fails and D (5%) of the population passes.
- This procedure and the relationship to the measurement is justified in clause G.x.9. The number of DUTs decrease during the simulation, as the decided DUTs leave the population. That number decreases with an approximately exponential characteristics. After 169 bad results all DUTs of the population are decided.

NOTE: The exponential decrease of the population is an optimal design goal for the decision co-ordinates (ne,ns), which can be achieved with other formulas or methods as well.

# Annex H (normative): Uplink Physical Channels

## H.0 Uplink Signal Levels

Editor's note: The configuration of SRS is FFS

Unless otherwise specified in the test case, the uplink power settings result from the default configuration of the UE described in 3GPP TS 36.508 [7].

## H.1 General

This annex specifies the uplink physical channels that are needed for setting a connection and channels that are needed during a connection. Table H.1-1 describes the mapping of uplink physical channels and signals to physical resources for FDD. Table H.1-2 describes the mapping of uplink physical channels and signals to physical resources for TDD.

**Table H.1-1: Mapping of uplink physical channels and signals to physical resources for FDD**

| Physical channel | Time Domain Location | Frequency Domain Location | Note |
|---|---|---|---|
| PRACH | Allowed for the parameter *prach-Configuration Index* provided by higher layers | [Allowed for the parameter *prach-FrequencyOffset* provided by higher layers] | Mapping rule is specified in TS36.211 Section 5.7.1 |
| DMRS | For PUCCH:<br>Symbols 2 to 4 of each slot (PUCCH format: 1, 1a, 1b)<br><br>Symbol 1 and 5 of each slot (PUCCH format: 2, 2a, 2b)<br><br>For PUSCH:<br>Symbol 3 of each slot | Uplink system bandwidth dependent. | Mapping rule of DMRS for PUCCH is specified in TS36.211 5.5.2.2.2<br><br>Mapping rule of DMRS for PUSCH is specified in TS36.211 5.5.2.1.2 |
| PUCCH | Slot 0 and 1 of each subframe | [Each 12 subcarriers of both ends of the bandwidth] | Mapping rule is specified in TS36.211 Section 5.4.3 |
| PUSCH | All remaining SC-FDMA symbols of each subframe not allocated to DMRS | RBs allocated according to Reference Measurement channel in Annex A.2 | Mapping rule is specified in TS36.211 Section 5.4.2 |

**Table H.1-2: Mapping of uplink physical channels and signals to physical resources for TDD**

| Physical channel | Time Domain Location | Frequency Domain Location | Note |
|---|---|---|---|
| PRACH | Allowed for the parameters $(t_{RA}^0, t_{RA}^1, t_{RA}^2)$ in *prach-Configuration Index* provided by higher layers | For format 0-3, the frequency location allowed is by *prach-FrequencyOffset* and $(f_{RA})$ in *prach-Configration Index* provided by higher layers. Preamble format 4 is mapped only on UpPTS, where the frequency location allowed is only by $(f_{RA})$ in *prach-Configration Index* provided by higher layers. | Mapping rule is specified in TS36.211 Section 5.7.1 |
| DMRS | For PUCCH:<br>Symbols 2 to 4 of each slot (PUCCH format: 1, 1a, 1b)<br><br>Symbol 1 and 5 of each slot (PUCCH format: 2, 2a, 2b)<br><br>For PUSCH:<br>Symbol 3 of each slot | Uplink system bandwidth dependent. | Mapping rule of DMRS for PUCCH is specified in TS36.211 5.5.2.2.2<br><br>Mapping rule of DMRS for PUSCH is specified in TS36.211 5.5.2.1.2 |
| PUCCH | Slot 0 and 1 of each subframe | [Each 12 subcarriers of both ends of the bandwidth] | Mapping rule is specified in TS36.211 Section 5.4.3 |
| PUSCH | All remaining SC-FDMA symbols of each subframe not allocated to DMRS | RBs allocated according to Reference Measurement channel in Annex A.2 | Mapping rule is specified in TS36.211 Section 5.4.2 |

NOTE: PUSCH, PUCCH, DMRS are not present in UpPTS for TDD.

# H.2 Set-up

Table H.2-1 describes the uplink physical channels that are required for connection set up.

**Table H.2-1: Uplink Physical Channels required for connection set-up**

| Physical Channel |
|---|
| PRACH |
| DMRS |
| PUCCH |
| PUSCH |

# H.3 Connection

The following clauses describes the uplink physical channels that are transmitted during a connection i.e., when measurements are done.

**Table H.3-1: Uplink Physical Channels required during a connection**

[Table contents FFS]

## H.3.0 Measurement of Transmitter Characteristics

[FFS]

## H.3.1 Measurement of Receiver Characteristics

[FFS]

## H.3.2 Measurement of Performance Requirements

[FFS]

# Annex I (informative): Handling requirements and tests for different releases

This annex gives guidance on how minimum requirements in different releases of 3GPP TS 36.101 [2] are handled in the specification 3GPP TS 36.521-1.

## I.1 Mapping of minimum requirements to conformance tests

Basic principles of deriving and specifying conformance tests in 3GPP TS 36.521-1 from minimum requirements in 3GPP TS 36.101 [2]:

a) Reuse as far as possible the already specified tests even for later releases (if applicable)

b) Avoid repeating and redundant test standardization

c) Minimize the maintenance work

The application of these principles results in the following conformance test drafting strategy:

d) Minimum requirements specified (i.e. the corresponding subclause is present) only in one release of 3GPP TS 36.101, are mapped to a single conformance test in 3GPP TS 36.521-1. The test is also applicable only to the implied release.

e) Minimum requirements specified (i.e. the same corresponding subclause is present) in multiple releases of 3GPP TS 36.101, are mapped to a:

   i) Single conformance test in 3GPP TS 36.521-1, if the minimum requirements are identical in all implied releases of 3GPP TS 36.101. The test is also applicable to all those releases.

   ii) Set of conformance tests in 3GPP TS 36.521-1 (“original test” and “additional tests”), if the minimum requirements are not identical in all implied releases of 3GPP TS 36.101. Each test in the set is applicable to the appropriate release(s).

Bullets d) and e) are described graphically in Figure I.1-1.

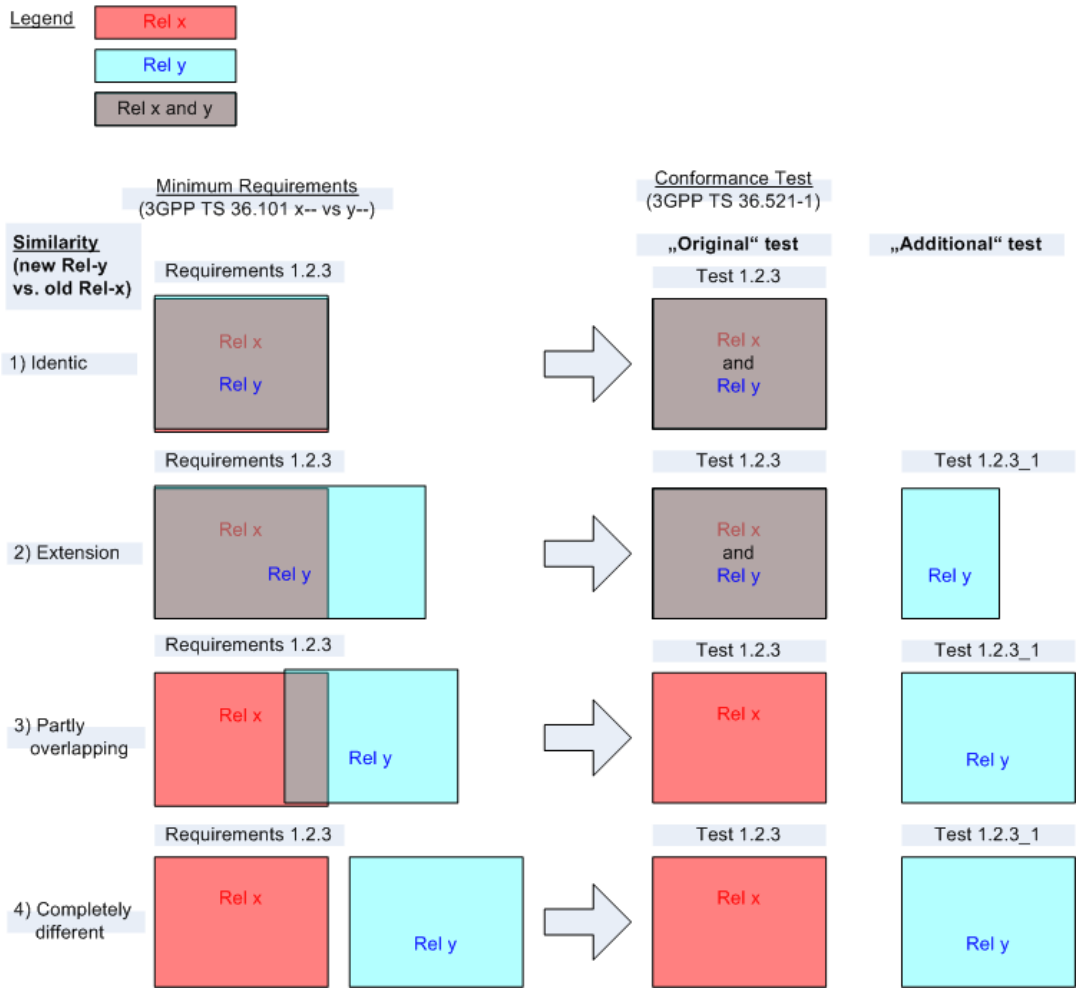



**Figure I.1-1: Mapping of minimum requirements to conformance tests, when minimum requirements are specified in multiple releases**

f) The difference between minimum requirements specified in the same subclause of different releases of 3GPP TS 36.101 as per e(ii) can consist in:

i) Different requirement values / additional test points
→ separate “additional” test is specified in 3GPP TS 36.521-1.

ii) New tested features (specific for new technology releases)
→ separate “additional” test is specified in 3GPP TS 36.521-1 for each new feature.

iii) Different combination of the above
→ bullets ii) and i) apply consecutively, i.e. further “additional” tests as per i) are specified within present “additional” tests as per ii).

# I.2 Structure of subclauses

The general effort for the structure of 3GPP TS 36.521-1 is:

a) Keep the structure, numbering and titling of conformance tests in 3GPP TS 36.521-1 as similar as possible with the corresponding minimum requirements in 3GPP TS 36.101 [2].

This means:

b) In general to each subclause with minimum requirement in 3GPP TS 36.101, should correspond a:

i) Subclause in 3GPP TS 36.521-1 with number and title similar with the implied reference subclause in 3GPP TS 36.101, containing the mapped conformance tests acc. to Annex I.1 bullets d and e(i).

ii) Set of subclauses in 3GPP TS 36.521-1 with root number and title similar with the implied reference subclause in 3GPP TS 36.101, containing the mapped set of conformance tests ("original tests" and "additional tests") acc. to Annex I.1 bullet e(ii).

c) Tests' numbers and titles for conformance tests set as per bullet b(ii):

i) The "original test" bears the root number and title.

ii) For the "additional tests":
- The root number inherited from RAN4 is followed by an RAN5 relevant numbering extension preceded by an underscore ("_").
- The root title inherited from RAN4 is followed by an RAN5 relevant title extension put in brackets.

d) Tests' numbers and titles for different type of "additional" tests:

i) For "additional" tests acc. to Annex I.1 f(i), the test number extension consists in Arabic numerals ("1, 2, 3, .."), while the title extension indicates the release applicability of the test.

ii) For "additional" tests acc. to Annex I.1 f(ii), the test number extension consists in Latin letters ("A, B, C, …") , while the title extension indicates the new feature tested in abbreviated form. The letters chosen for the number extension should possibly be used for the same new feature throughout the specification and possibly comply with the allocation used by RAN4 as described in clause 4.1.

iii) For combined "additional" tests acc. to Annex I.1 f(iii), the test number extension consists in Latin letters ("A, B, C, …") as per ii) followed without any separation character by Arabic numerals as per i), while the title extension indicates the new feature tested and the release applicability of the test.

e) Examples:

i) Minimum requirements in 3GPP TS 36.101 R4-Nr. R4-Title

ii) Conformance tests in 3GPP TS 36.521-1 (RAN5 extensions in **bold**)

| Single / Original test | Nr. Title<br>(Nr / Title similar to R4-NR / R4-Title) |
|---|---|
| Add. test acc. to Annex I.2 d(i) | Nr **_1**. Title **(Rel 9 and forward)**<br>Nr **_2**. Title **(Rel 10 and forward)** |
| Add. test acc. to Annex I.2 d(ii) | Nr **_A**. Title **(CA)**<br>Nr **_B**. Title **(UL-MIMO)** |
| Add. test acc. to Annex I.2 d(iii) | Nr **_A**. Title **(CA)**<br>Nr **_A1**. Title **(CA Rel 11 and forward)**<br>Nr **_B**. Title **(UL-MIMO)** |

# I.3 Description of the additional tests

In case when the additional conformance tests are procedurally very similar to the original tests, in order to reduce the standardization and maintenance work (as per Annex I.1), the content of subclauses of the additional tests may be reduced to a minimum by referencing to the analogue subclauses of the original tests (provided they are similar), and specifying the exceptions (requirement-tables, test parameter tables etc). This method is to apply very carefully, in order to make sure that possible differences in the core functionality of different E-UTRAN releases remain considered.

# Annex J (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment** | **Old** | **New** |
| 2007-08 | RAN5 #36 | R5-072185 | | | Skeleton proposed for RAN5#36Athens | | 0.0.1 |
| 2007-08 | RAN5 #36 | R5-072419 | | | Update the skeleton base on R4-071234_TR36.803.0.4.0.doc | 0.0.1 | 0.0.2 |
| 2007-08 | RAN5 #36 | R5-072424 | | | Update with editorial changes | 0.0.2 | 0.0.3 |
| 2007-11 | RAN5 #37 | R5-073043 | | | Update document with some info as following:<br>Section 5: Frequency band information<br>Section 6.2: Maximum output power<br>Section 6.5: Output RF spectrum emissions<br>Section 6.5.1: Occupied bandwidth<br>Section 6.5.2: Out of band emission<br>Section 6.5.3: Spurious emissions | 0.0.3 | 0.0.4 |
| 2007-11 | RAN5 #37 | R5-073360 | | | Editorial change to split MOP and UE Power classes | 0.0.4 | 0.0.5 |
| 2008-03 | RAN5 #38 | R5-080069 | | | Editorial changes to sync up with 36.101 v1.0.0 as much as feasible for the moment:<br>Update definitions, symbols and abbreviations<br>Update frequency bands, channel bandwidth, channel numbers information.<br>Restructure document to move “frequency error” sub-section inside Transmit signal quality.<br>Add “additional spectrum Emission Mask” sub-test (mask A,B,C) section to address the regulatory requirements that are not met with the general mask (OOB and spurious emission).<br>Add “Additional ACLR requirements” to address additional requirements that the network might indicate to the UE via signalling for a specific deployment scenario (in terms of additional requirements for UTRA/ACLR2<br>Restructure “Spurious Emission” to indicate we need to have 3 test cases to address: “E-UTRA Spurious Emission” requirements , “Spurious Emission band UE co-existence” requirements, and “Additional spurious emissions” requirements<br>Separate wide band and narrow band intermodulation in the intermodulation characteristics | 0.0.5 | 0.0.6 |
| 2008-03 | RAN5 #38 | R5-080408 | | | LTE Reference Sensitivity test Text proposal | | 0.0.7 |
| 2008-03 | RAN5 #38 | R5-080409 | | | LTE Maximum Rx input level test Text proposal | | 0.0.7 |
| 2008-03 | RAN5 #38 | R5-080410 | | | LTE Adjacent Channel Selectivity test Text proposal | | 0.0.7 |
| 2008-03 | RAN5 #38 | R5-080064 | | | LTE RF Receiver tests, General section Text proposal | | 0.0.7 |
| 2008-03 | RAN5 #38 | R5-080412 | | | LTE RF: transmission modulation initial EVM test proposal | | 0.0.7 |
| 2008-03 | RAN5 Workshop-UE LTE Test (9-11 April) | R5w0800027 | | | Modify styles and formats of tables and others according to drafting rules.<br>Add some definitions and abbreviations<br>Modified section 6.2 structure to be aligned with 36.101 v8.1.0<br>Modify tables of requirements to remove 1.6 MHz and 3.2MHz channel bandwidth according to new requirements 36.101 v8.1.0 | | 0.0.9 |
| 2008-03 | RAN5 Workshop-UE LTE Test (9-11 April) | R5w0800028 | | | Following TPs have been included:<br>R5w080013r1<br>R5w080014r1<br>R5w080008r2<br>R5w080009r2<br>R5w080040r1<br>R5w080015r1<br>R5w080016r1<br>R5w080017r1<br>R5w080018r2 | 0.0.9 | 0.1.0 |
| 2008-05 | RAN5#39 | R5-081046 | | | 36-521-1 alignment of measurement state for test cases | 0.1.0 | 0.1.1 |
| 2008-05 | RAN5#39 | R5-081042 | | | Following approved TPs have been included:<br>R5-081040 36.521-1 after April LTE-RF workshop<br>R5-081415 36-521-1 alignment of measurement state for test cases – also the measurement state for each test cases has been updated according to R5-081404<br>R5-081416 Cover for LTE E-UTRAN RRC_IDLE State Mobility text proposal<br>R5-081417 Cover for LTE E-UTRAN RRC_CONNECTED State Mobility text proposal<br>R5-081404 LTE Rx Intermodulation test case text proposal | 0.1.1 | 0.2.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | R5-081409 Annex structure for Measurement uncertainty & Test Tools<br>R5-081405 Text Proposal for TS36.521-1 TC7.6 Blocking Characteristics<br>R5-081406 Text Proposal for TS36.521-1 TC7.7 Spurious Response<br>R5-081403 Text Proposal for TS36.521-1 TC7.9 Spurious Emissions<br>R5-081410 Uncertainties and Test Tools for subset of UE tests<br>R5-081331 Clarification of diversity characteristics section for multiple UE antennas<br>R5-081335 36-521-1 update of nominal and additional channel bandwidths | | |
| 2008-06 | RAN5 #39bis | R5-082029 | | | Following approved TPs have been included:<br>R5-082129: Restructure of TS 36.521-1 and RRM proposal (Split of RRM from 36.521-1 v0.2.0 in its own specification 36.521-3.)<br>R5-082166: Text Proposal for Annex C Downlink Physical Channels<br>R5-082130: Text Proposal for Chan bandwidths in TS 36.521-1<br>R5-082155: Text Proposal for LTE Tx Minimum Output Power<br>R5-082027: Text Proposal for Occupied bandwidth in TS 36.521-1<br>R5-082171: Text Proposal for LTE Adjacent Channel Leakage power Ratio<br>R5-082134: Text Proposal for LTE Tx Spurious Emissions<br>R5-082135: Text Proposal for LTE UE Maximum Output Power<br>R5-082136: Text Proposal for LTE Spectrum Emission Mask<br>R5-082138: UE Spurious Emissions Measurement uncertainty & Test Tolerances<br>R5-082169: LTE Spectrum Emission Mask test uncertainties and TTs<br>R5-082151: LTE UE Max Power and ACLR tests uncertainties and TTs<br>R5-082152: Text proposal for LTE Transmit OFF Power<br>R5-082153: LTE UE Max Rx Input and ACS test cases update<br>R5-082082: LTE Rx Intermodulation test case uncertainties and TTs<br>R5-082093: Text Proposal for TS36.521-1 TC7.6 Blocking Characteristics<br>R5-082154: Text Proposal for TS36.521-1 TC7.7 Spurious Response<br>R5-082167: OBW Measurement uncertainty & Test Tolerances<br>R5-082158: Cover for LTE Performance Requirement text proposal<br>R5-082159: Text Proposal for LTE Demodulation of PCFICH/PDCCH and PHICH<br>R5-082156: Text proposal for LTE Tx Minimum Output Power Uncertainty<br>R5-082157: Text proposal for LTE Tx Minimum Output Power Tolerance<br>R5-082164: Statistical testing of receiver characteristics<br>R5-082170: Cover for LTE Propagation Conditions Text Proposal<br>Editorial changes to align tables and figures numbering with R5-082025 | 0.2.0 | 0.3.0 |
| 2008-08 | RAN5 #40 | R5-083163 | | | Following approved TPs have been included:<br>R5-083804: LTE Demodulation Performance text proposal<br>R5-083159: LTE-RF Occupied bandwidth test case / measurement uncertainty and TT text proposal<br>R5-083160: Transmission OFF power: TP, measurement uncertainty and test tolerances proposal<br>R5-083805: Frequency Error test case / measurement uncertainty and TT test proposal<br>R5-083162: Propagation conditions correction text proposal<br>R5-083220:Text Proposal for LTE Tx Minimum Output Power<br>R5-083806: TP of section 8 for E-UTRAN TDD in 36.521-1<br>R5-083344: Test Tolerance and System uncertainty for OBW test<br>R5-083848:Test Tolerance and System uncertainty for Reference sensitivity test<br>R5-083840: Test Tolerances for Spectrum Emission Mask<br>R5-083808: Reference Measurement Channel for LTE UE | 0.3.0 | 1.0.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Receiver tests<br>R5-083350: Test Tolerance and System uncertainty for Blocking and Spurious response<br>R5-083366: Text Proposal for LTE Reporting of CQI/PMI<br>R5-083810: LTE PBCH Demodulation Performance Requirements<br>R5-083482: LTE-RF TP for Test Case 7.6 Blocking Characteristics<br>R5-083809: LTE-RF TP for Test Case 7.7 Spurious Response<br>R5-083484: LTE-RF TP for Test Case 7.9 Spurious Emissions<br>R5-083811: Annex E Global In-Channel TX-Test<br>R5-083163: TS 36.521-1 after RAN5#40 | | |
| 2008-10 | RAN5 #40Bis | R5-084072 | | | Following approved TPs have been included:<br>R5-084072 TS 36.521-1 after RAN5#40Bis<br>R5-084300 LTE-RF TP for Definitions Symbols and Abbreviations<br>R5-084304 LTE-RF-TP for general section<br>R5-084036 Test Tolerances for additional SEM<br>R5-084303 LTE-RF TP for Channel bandwidths and frequency range<br>R5-084305 LTE-RF TP for new Absolute Power Tolerance test case<br>R5-084067 LTE-RF TP for Transmission OFF test case<br>R5-084318 LTE-RF TP for Transmission Modulation test cases<br>R5-084069 LTE-RF Investigation of E-UTRA-TDD Frequency Error test case applicability<br>R5-084319 LTE-RF TP for Frequency Error test case<br>R5-084309 Text Proposal for LTE Tx Spurious Emissions<br>R5-084111 Text Proposal for LTE Adjacent Channel Leakage power Ratio<br>R5-084320 Text Proposal for LTE Additional Spectrum Emission Mask<br>R5-084310 Test Tolerances for additional spurious emission<br>R5-084311 Text Proposal for Occupied bandwidth<br>R5-084321 Text Proposal for LTE Spectrum Emission Mask<br>R5-084060 Modification to section 7.2 Diversity characteristics<br>R5-084312 References in 36.521-1 tests initial conditions<br>R5-084148 Update of Reference Measurement Channel for LTE UE Rx tests<br>R5-084167 LTE-RF TP for TC7.9 Spurious Emissions<br>R5-084075 LTE DL Reference Measurement Channel for PDSCH (FDD) text proposal<br>R5-084077 LTE Measurement of Performance Requirements text proposal<br>R5-084313 LTE Demodulation of PDSCH Test Requirements text proposal<br>R5-084147 Specification of DL propagation conditions for LTE UE tests<br>R5-084315 Text Proposal for LTE Demodulation of PCFICH/PDCCH<br>R5-084323 Text Proposal for Annex E Global In-Channel | 1.0.0 | 1.1.0 |
| 2008-12 | RAN#42 | RP-080863 | | | Approval of version 2.0.0 at RAN#42, then put to version 8.0.0. | 2.0.0 | 8.0.0 |
| 2008-01 | | | | | Editorial corrections. | 8.0.0 | 8.0.1 |
| 2009-03 | RAN#43 | R5-086011 | 0001 | - | TP for In-band emissions | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086012 | 0002 | - | TP for Spectrum flatness | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086013 | 0003 | - | TP for IQ-component | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086064 | 0004 | - | LTE-RF: UE max output power | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086093 | 0005 | - | Clarification of measurement period in minimum output power test procedure | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086094 | 0006 | - | Clarification of measurement period in transmit OFF power test procedure | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086120 | 0007 | - | Update of Max.input level test | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086125 | 0008 | - | Addition of UL Reference Measurement Channels in Annex A2 | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086160 | 0009 | - | correction for Maximum Power Reduction (MPR) | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086167 | 0010 | - | LTE-RF: TDD applicability and CR for Blocking Characteristics and Spurious Response | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086168 | 0011 | - | LTE-RF: TDD applicability and CR for Spurious Emissions | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086239 | 0012 | - | Update of Symbols | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086401 | 0013 | - | LTE-RF: TX-RX channel freq separation | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086405 | 0014 | - | Update of 6.7 Transmit intermodulation test | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086406 | 0015 | - | Update of initial conditions for Tx and Rx test cases | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086408 | 0016 | - | Update of Adjacent Channel Leakage power Ratio | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086409 | 0017 | - | Removal of [ ] from Clause 7 Receiver Characteristics | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086413 | 0018 | - | Updates to Demodulation of PCFICH/PDCCH test case | 8.0.1 | 8.1.0 |

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|---|---|---|---|---|---|---|---|
| 2009-03 | RAN#43 | R5-086414 | 0019 | - | Text proposal for Reporting of Channel State Information | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086415 | 0020 | - | Correction of RS_EPRE powers for default DL signal levels | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086416 | 0021 | - | Update of DL Reference Measurement Channels in Annex A3 | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086417 | 0022 | - | Update to Annex E | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086425 | 0023 | - | Update of General text in clause 6 | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086426 | 0024 | - | Clarification of measurement bandwidth in spectrum emission mask test | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086428 | 0025 | - | Demodulation of TDD PHICH test requirements text proposal | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-086429 | 0026 | - | Demodulation of TDD PCFICH/PDCCH test requirements text proposal | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-090306 | 0027 | - | New Annex H for Uplink Physical Channels | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-090308 | 0028 | - | Text proposal for Reporting of Channel State Information | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-090403 | 0029 | - | CR to 36.521-1: Update of Spurious Emissions test cases | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-090404 | 0030 | - | CR to 36.521-1: Update of ACLR test case | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-090443 | 0031 | - | LTE-RF: Correction to 36.521-1 Frequency error test case | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-090488 | 0032 | - | LTE TDD applicability for Transmit intermodulation test case | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091002 | 0033 | - | LTE Demodulation of PDSCH Test Requirements text proposal | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091004 | 0034 | - | LTE-RF: CR for UE max power test case | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091007 | 0035 | - | LTE-RF: TDD Applicability and CR for Spectrum Emission Mask and Additional Spectrum Emission Mask | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091008 | 0036 | - | LTE-RF Investigation of E-UTRA-TDD for Occupied bandwidth test case applicability | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091009 | 0037 | - | LTE-RF: Investigation of E-UTRA-TDD for Adjacent Channel Leakage power Ratio test case applicability | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091011 | 0038 | - | LTE-RF: TDD applicability and CR for Maximum Input Level | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091012 | 0039 | - | LTE-RF: TDD applicability and CR for Adjacent Channel Selectivity (ACS) | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091017 | 0040 | - | Removal of Rx Narrowband Intermod 7.8.2 | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091019 | 0041 | - | Relocation of 36.521-1 Annex C DL mapping | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091020 | 0042 | - | Removal of "Out-of-synchronization handling of output power" heading | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091023 | 0043 | - | Test requirements of TDD PDSCH demodulation performance with user-specific reference symbols | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091024 | 0044 | - | CR to 36.521-1: Update of Annex F.3.2 Measurement of transmitter | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091025 | 0045 | - | CR to 36.521-1: Update of SEM and Additional SEM test cases | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091077 | 0046 | - | CR to 36.521-1: Addition of test combinations for test cases with MPR application | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091082 | 0047 | - | Spurious emission requirements on PHS band including the future plan in Japan | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091101 | 0048 | - | LTE-RF: CR for MPR test case | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091106 | 0049 | - | Update of Reference sensitivity test in 7.3 | 8.0.1 | 8.1.0 |
| 2009-03 | RAN#43 | R5-091111 | 0050 | 1 | Update of initial conditions for Rx tests | 8.0.1 | 8.1.0 |
| 2009-05 | RAN#44 | R5-092144 | 0051 | - | LTE-RF: Resubmission of R5-086424 UE output power dynamics 36.521-1 v8.1.0 (re-submit no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092146 | 0052 | - | LTE-RF: CR for UE configured UE transmitted output power test case (re-submit no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092147 | 0053 | - | LTE-RF: CR for UE minimum output power test case (re-submit no change) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092149 | 0054 | - | LTE-RF: CR for Power Control Absolute power tolerance test case (re-submit no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092150 | 0055 | - | LTE-RF: CR for Power Control Relative power tolerance test case (re-submit no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092151 | 0056 | - | LTE-RF: New test case for Aggregate power control tolerance (re-submit no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092263 | 0057 | - | Text proposal for Reporting of Channel State Information | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092264 | 0058 | - | Propagation conditions for CQI tests | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092265 | 0059 | - | Correction to Demodulation of PDCCH/PCFICH test cases | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092273 | 0060 | - | Mapping of downlink physical channels for TDD | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092277 | 0061 | - | Annex A RMC updates | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092369 | 0062 | - | Update of A.3.4.3 for RMC with UE-specific RS | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092372 | 0063 | - | Maintenance on Initial configurations for Perf TCs | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092436 | 0064 | - | CR to 36.521-1: Update of ACLR test case | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092442 | 0065 | - | CR to 36.521-1: Update of Spurious Emissions test case | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092467 | 0066 | - | LTE-RF: Transmit OFF Power update | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092473 | 0067 | - | LTE_RF - Update on TC 7.7 Spurious Response (re-submit with no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092474 | 0068 | - | LTE_RF - Update on TC 7.9 Spurious Emissions (re-submit with no changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092527 | 0069 | - | Update of TDD PDSCH test cases | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092602 | 0070 | - | LTE-RF: CR for Maximum Power Reduction test case (re- | 8.1.0 | 8.2.0 |

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|---|---|---|---|---|---|---|---|
| | | | | | submit no changes) | | |
| 2009-05 | RAN#44 | R5-092603 | 0071 | - | TP for Demodulation of TDD PDCCH/PCFICH | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092605 | 0072 | - | Mapping of uplink physical channels for FDD | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092606 | 0073 | - | Update of Annex C | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092607 | 0074 | - | CR to 36.521-1: Update of test parameters for Demodulation of PDSCH (FDD) tests | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092614 | 0075 | - | Update of SEM test case | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092642 | 0076 | - | Update of transmit quality test cases | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092643 | 0077 | - | Text proposal for TDD part of CQI Reporting under Fading conditions | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092644 | 0078 | - | Text proposal for TDD part of CQI Reporting under AWGN conditions | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092645 | 0079 | - | LTE-RF: Update of Additional Spectrum Emission mask Test case with TDD Uplink Test configuration | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092649 | 0080 | - | LTE-RF: CR for TDD DL RMC to be used in TX test cases | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092653 | 0081 | - | LTE-RF: CR for Additional Maximum Power Reduction test case | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092661 | 0082 | - | RMC update for PDCCH/PCFICH performance requirement | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | RP-090444 | 1161 | - | Test frequencies for Additional Spurious Emission test case | 8.6.0 | 8.7.0 |
| 2009-05 | RAN#44 | R5-092366 | 0084 | - | Update of 7.3.1 | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092440 | 0085 | - | LTE-RF: CR for UE max output power test case | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092472 | 0086 | - | LTE_RF - Update on TC 7.6 Blocking Characteristics (re-submit with changes) | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092636 | 0087 | - | CR to 36.521-1 Addition of frequencies for band 18 and band 19 | 8.1.0 | 8.2.0 |
| 2009-05 | RAN#44 | R5-092652 | 0088 | 2 | Improved stability of TC 7.8.5 Power Control in the DL fro F-DPCH to HSUPA TC 5.2D and 5.13.2B | 8.1.0 | 8.2.0 |
| - | - | - | - | - | Editorial corrections | 8.2.0 | 8.2.1 |
| 2009-09 | RAN#45 | R5-094032 | 0089 | - | Correction CR to 36.521-1: Update of Requirements for Demodulation of PDSCH (FDD) tests | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094034 | 0090 | - | Correction CR to 36.521-1: Update of General Requirements for Demodulation tests | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094214 | 0091 | - | Update of In-band emissions | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094215 | 0092 | - | TDD Initial downlink channel setting | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094216 | 0093 | - | Correction to Annex B | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094248 | 0094 | - | CR to 36.521-1: Update to ACLR test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094250 | 0095 | - | CR to 36.521-1: Update to UE max output power test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094281 | 0096 | - | Mapping of uplink physical channels for TDD | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094282 | 0097 | - | LTE-RF: CR for notes in TDD DL RMC to be used in TX test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094283 | 0098 | - | LTE-RF: message update to keep Tx power constant for some Rx test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094313 | 0099 | - | LTE-RF: CR to test case for Aggregate power control tolerance | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094317 | 0100 | - | LTE-RF: CR for UE minimum output power test case for TDD | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094318 | 0101 | - | LTE-RF: CR for Power Control Relative power tolerance test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094319 | 0102 | - | In band emission for non-allocated RB | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094320 | 0103 | - | LTE RF: correction for subclause 6.6.2.2.5 (A-SEM) supported band list | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094362 | 0104 | - | Correction of RMCs (36.521 Annex A) | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094363 | 0105 | - | Usage of the Global In-Channels TX-Test across different Signal Quality tests. | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094365 | 0106 | - | LTE TX: 1to2 RX antenna | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094367 | 0107 | - | Correction to 6.6.2.2 Additional Spectrum Emission Mask | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094370 | 0108 | - | Correction to 6.6.2.3 ACLR | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094371 | 0109 | - | Correction to 6.7 TX Intermodulation | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094374 | 0110 | - | Correction to 7.6.1 In-Band Blocking | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094375 | 0111 | - | UE category (36.521 clause 8) | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094378 | 0112 | - | Completion of Global in-Channel TX-Test (36.521 Annex E) | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094379 | 0113 | - | Completion of Global in-Channel TX-Test with PRACH (36.521 Annex E) | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094380 | 0114 | - | Completion of Statistical testing (36.521 Annex G) | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094385 | 0115 | - | Correction to Annex D.2 Interference signals | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094439 | 0116 | - | Update for ACS | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094661 | 0117 | - | LTE RF - Core update on TC7.6.2 Out-of-band Blocking | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094663 | 0118 | - | LTE RF - Symbols Update on UL transmission configurations | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094665 | 0119 | - | LTE RF - Clarification for Test Configurations in General Section | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094668 | 0120 | - | LTE RF - Applicability of 6.2.3 MPR | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094671 | 0121 | - | LTE RF - Verification of UE Output Power in Out of Band Emission tests | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094684 | 0122 | - | CR to 36.521-1: Update to UE max output power test case | 8.2.1 | 8.3.0 |

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| 2009-09 | RAN#45 | R5-094686 | 0123 | - | LTE-RF CR to 36.521-1: Update the E-UTRA channel numbers | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094687 | 0124 | - | LTE-RF: CR for UE maximum power reduction(MPR) test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094699 | 0125 | - | Update to SEM and spurious emissions TC | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094706 | 0126 | - | Resubmission-Update to the Requirements for frequency-selective fading test | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094717 | 0127 | - | Update of SEM | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094718 | 0128 | - | Update of initial conditions with Annex references | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094721 | 0129 | - | Update of 6.7 Tx Inter Mod | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094725 | 0130 | - | Correction to E-UTRA channel numbers for Band 2 | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094726 | 0131 | - | Correction to Tx spurious emissions | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094757 | 0132 | - | Update of TDD PHICH test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094874 | 0133 | - | Correction to Demodulation of PDCCH/PCFICH test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094902 | 0134 | - | Addition of 15 MHz and 20 MHz bandwidths and corresponding sensitivity requirements into band 38 | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094903 | 0135 | - | Correction CR to 36.521-1: Update of Transmitter tests network signalled parameter value | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094905 | 0136 | - | Update of TDD PDSCH test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094908 | 0137 | - | LTE-RF: CR for Power Control Absolute power tolerance test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094909 | 0138 | - | Update to Output Power dynamics test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094913 | 0139 | - | Clarification for downlink signal setting in RX tests | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094914 | 0140 | - | UL RB allocation for receiver tests | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094915 | 0141 | - | Update of TDD PCFICH/PDCCH test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094921 | 0142 | - | Correction to CQI performance test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094922 | 0143 | - | Test description for CQI test cases under AWGN conditions | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094923 | 0144 | - | Resubmission - Requirements for PMI reporting ( Single and Multiple PMI) | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094966 | 0145 | - | CR to 36.521-1: Addition of A-MPR for band 19 | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094976 | 0146 | - | Without loop back: 6.2.2 UE maximum output power | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094977 | 0147 | - | Without loop back: 6.3.2 Minimum output power | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094979 | 0148 | - | LTE-RF: CR for UE configured UE transmitted output power test case | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094980 | 0149 | - | CR to 36.521-1: Definition of Maximum Power state in TX/RX test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094982 | 0150 | 1 | Correction of Tx general description | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094986 | 0151 | - | Update of 6.6.1OBW | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094989 | 0152 | - | Correction to 1PRB tests in Demodulation of PDSCH | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094995 | 0153 | - | Correction CR to 36.521-1: Update of Requirements for Additional Maximum Power Reduction (A-MPR) test | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094996 | 0154 | - | Correction to Demodulation of PHICH test cases | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-094997 | 0155 | - | EVM TC update | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-095300 | 0156 | - | LTE-RF: test description update | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-095301 | 0157 | - | Correction CR to 36.521-1: Addition of measurement uncertainty and test tolerances for A-MPR | 8.2.1 | 8.3.0 |
| 2009-09 | RAN#45 | R5-095304 | 0158 | - | Sorting out Demodulation of PDSCH for FDD | 8.2.1 | 8.3.0 |
| 2009-09 | - | - | - | - | TOC update and Annexes' titles formattings | 8.3.0 | 8.3.1 |
| 2009-12 | RAN#46 | R5-095515 | 0159 | - | Correction CR to 36.521-1: Additional Spectrum Emission Mask test need to be updated to include the network signalled value "NS_07ö message contents exceptions | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095589 | 0160 | - | Update for test period description in the general section | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095657 | 0161 | - | LTE-RF: CR for Power Control Absolute power tolerance test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095661 | 0162 | - | LTE-RF: CR for UE minimum output power test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095735 | 0163 | - | Corrections to Annex A.4 | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095766 | 0164 | - | LTE-RF: CR for In band emission for non-allocated RB | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095790 | 0165 | - | Completion of Statistical testing (36.521 Annex G) | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-095791 | 0166 | - | Corrections to Annex E | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096058 | 0167 | - | Removal of [ ] from 7.6.1, 7.8.1, and 7.5 of Annex F3.3 | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096096 | 0168 | - | Update on 8.2.1 | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096105 | 0169 | - | LTE RF: Symbols Update on Configured UE Transmitted Power | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096204 | 0170 | - | LTE-RF: CR to Transmission signal quality | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096208 | 0171 | - | LTE-RF: CR for Power Control Relative power tolerance test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096210 | 0172 | - | LTE-RF: CR to ON/OFF Time mask test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096211 | 0173 | - | Measurement period for TX-Tests | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096213 | 0174 | - | CR to 36.521-1: Update to Spurious Emissions test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096214 | 0175 | - | CR to 36.521-1: Update to ACLR test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096219 | 0176 | - | LTE-RF: CR for UE configured UE transmitted output power test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096222 | 0177 | - | Test description for CQI test cases under AWGN conditions | 8.3.1 | 8.4.0 |

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|---|---|---|---|---|---|---|---|
| 2009-12 | RAN#46 | R5-096223 | 0178 | - | LTE RF: Blocking Characteristics update | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096224 | 0179 | - | LTE RF: Spurious Response Update | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096228 | 0180 | - | LTE-RF: CR for MPR test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096229 | 0204 | 2 | CR to 36.521-1: Update to A-MPR test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096230 | 0181 | - | LTE RF: Applicability of 6.2.4 A-MPR | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096231 | 0182 | - | Correction to Demodulation of PHICH test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096235 | 0183 | - | Introduction of CQI reporting test with frequency-selective interference | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096239 | 0184 | - | Update to the test procedure and message contents of TDD PMI reporting test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096240 | 0205 | - | CR to 36.521-1: Update to Derivation of Test Requirements for A-MPR | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096241 | 0185 | - | Measurement uncertainties and Test Tolerances for transmit quality test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096242 | 0186 | - | Update for 36.521-1 Annex A | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096289 | 0187 | - | CR on 36.521-1, 'Introduction of clause 8.2.1.1 test case uncertainties and Test Tolerances' | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096306 | 0188 | - | Update to the test procedure of SEM test cases of 36.521-1 | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096311 | 0189 | - | Update of 6.6.1 OBW | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096312 | 0190 | - | Correction to SEM | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096313 | 0191 | - | Update of 6.7 Transmit intermodulation | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096315 | 0192 | - | CR to 36.521-1: Update to UE max output power test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096316 | 0193 | - | CR to 36.521-1: Update to Additional Spurious Emissions test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096317 | 0194 | - | CR to TDD PHICH demodulation test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096318 | 0195 | - | Correction to FDD PMI reporting test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096320 | 0196 | - | Tx power range and core update for Receiver tests | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096322 | 0197 | - | Update on 7.4, 7.5, and 7.8.1 | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096323 | 0198 | - | Introduction of RI reporting test | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096333 | 0199 | - | Update to 6.5 Transmit signal quality test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096334 | 0200 | - | LTE-RF: CR for Aggregate power control tolerance test case | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096335 | 0201 | - | Correction CR to 36.521-1: Update for Demodulation of PDSCH (FDD) tests to correct CR merges results from RAN5#44 | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096336 | 0206 | 1 | Update TDD PDSCH test cases | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096338 | 0202 | - | Number of used HARQ processes in DL Performance tests | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096342 | 0207 | 2 | Minimum test time for performance tests | 8.3.1 | 8.4.0 |
| 2009-12 | RAN#46 | R5-096718 | 0203 | - | LTE RF: A-SEM update and A-MPR verification | 8.3.1 | 8.4.0 |
| 2010-03 | RAN#47 | R5-100353 | 0208 | - | LTE-RF CR to 36.521-1:TIME MASK test case updated | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100354 | 0209 | - | LTE-RF: CR for A-MPR notation in NS_07 | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100403 | 0210 | - | LTE-RF: CR for Tx Intermodulation test case | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100404 | 0211 | - | LTE-RF: CR for OBW measurement period alignment | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100408 | 0212 | - | Reporting mode, Reporting Interval and Editorial corrections for demodulation | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100456 | 0213 | - | Misc update on MAC padding in Rx and performance sections | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100566 | 0214 | - | Missing Test limits in 36.521-1 Annex G | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100567 | 0215 | - | Wrong references from 36.521-1 clauses 8.4 and 8.5 into Annex G | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100569 | 0216 | - | Typos in 36.521-1, Annex E | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100571 | 0217 | - | Minimum test time for performance tests | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100572 | 0218 | - | Correction to 6.6.3.3 Additional spurious emissions | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100790 | 0219 | - | DL-RMC-s for transmitter tests: Corrections | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100800 | 0220 | - | Update of Test environment for RF test | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100803 | 0221 | - | Spectrum emission mask: Correction to uplink configuration | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100807 | 0222 | - | Performance tests: Scheduling of retransmissions | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100810 | 0223 | - | UL-RMC-s: Corrections and completion | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100814 | 0224 | - | Corrections to Cl 5.4.2.1 of TS 36.521-1 | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100815 | 0225 | - | LTE-RF: CR for UE configured UE transmitted output power test case | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100816 | 0226 | - | LTE-RF: CR for Power Control Relative power tolerance test case | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100822 | 0227 | - | CR to 36.521-1: Update to Maximum output power | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100823 | 0228 | - | CR to 36.521-1: Update to ACLR test case | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100825 | 0229 | - | CR to 36.521-1: Update to Additional Tx spurious emissions test case | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100826 | 0230 | - | RMC-s and OCNG patterns: Update according 36.101 8.8.0 | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100827 | 0231 | - | Receiver and performance tests: Update use of OCNG according 36.101 8.8.0 | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100828 | 0232 | - | Update of PDSCH Demodulation Tests | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100831 | 0233 | - | Introduction of clause 8.2.1.2, 8.2.1.3, 8.2.1.4 test case uncertainties and Test Tolerances | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100832 | 0234 | - | Clarifications on DRS performance test case | 8.4.0 | 8.5.0 |

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|---|---|---|---|---|---|---|---|
| 2010-03 | RAN#47 | R5-100833 | 0235 | - | Misc update on MAC padding in PDCCH, CSI test | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100834 | 0236 | - | Updates to the TDD portion of CQI reporting test cases under AWGN | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100838 | 0237 | - | Editorial Correction to 8.2.1.3 | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100839 | 0238 | - | Update on Annex C for 36.521-1 | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100840 | 0239 | - | Update on MAC padding in TDD PMI test case 9.4 of 36.521-1. | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100841 | 0240 | - | Correction to CQI test cases under AWGN conditions | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100842 | 0241 | - | Correction to CQI test cases under fading conditions | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100843 | 0242 | - | Correction to PMI reporting test cases | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100845 | 0243 | - | CSI: Corrections to tests titles and RI clause structure | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100848 | 0244 | - | CR to 36.521-1: Update LTE RF test cases with test requirements for extended LTE1500MHz | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100886 | 0245 | - | Transmitter characteristics: UE Categories and other corrections | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100887 | 0246 | - | CR to 36.521-1: Update to Tx spurious emissions and Spurious emission band UE co-existence | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100888 | 0247 | - | Clarification on notes in Max Power | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100889 | 0248 | - | Maximum input level: Corrections w.r.t. UE categories | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100891 | 0249 | - | Correction to PDCCH demodulation test cases | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100892 | 0250 | - | Correction to PHICH demodulation test cases | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100907 | 0251 | - | Update of RI reporting test case | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | R5-100909 | 0252 | - | Correction to set UL power in Rx TCs | 8.4.0 | 8.5.0 |
| 2010-03 | RAN#47 | - | - | - | Moved to v9.0.0 with no change | 8.5.0 | 9.0.0 |
| 2010-06 | RAN#48 | R5-103102 | 0253 | - | CR to 36.521-1: Update of EARFCN for band 21 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103103 | 0254 | - | CR to 36.521-1: Update of A-MPR test case with band 21 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103104 | 0255 | - | CR to 36.521-1: Update of Additional Spurious test case with band 21 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103106 | 0256 | - | CR to 36.521-1: Update to ACLR test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103108 | 0257 | - | CR to 36.521-1: Update of Reference sensitivity level test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103226 | 0258 | - | CR to 36.521-1: Update of UE RF requirements for LTE, Band 20 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103263 | 0259 | - | LTE-RF:Updates of PDCCH demodulation test cases (FDD and TDD) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103265 | 0260 | - | LTE-RF:CR for TDD ACK/NACK feedback mode in CQI BLER test cases | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103288 | 0261 | - | PDCCH Aggregation level for RF tests | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103291 | 0262 | - | Update and correction to UE maximum output power requirements | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103293 | 0263 | - | Editorial correction in In-band blocking test | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103296 | 0264 | - | Correction to additional spectrum emission mask test configuration | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103300 | 0265 | - | Corrections to Uplink RMC-s | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103450 | 0266 | - | LTE-RF: editorial CR for TC 7.6.2 and 7.7 | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103471 | 0267 | - | Minimum test time for performance tests | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103476 | 0268 | - | EVM with exclusion period (annex) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103521 | 0269 | - | CR on 36.521-1 for updating the "Reporting of Channel State Information" | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103525 | 0270 | - | CR on 36.521-1 for corrections in UE RF requirements | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103598 | 0271 | - | Correction to notes in Max Power | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103602 | 0272 | - | Clarification of measurement conditions for Rx spurious emission | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103726 | 0273 | - | CR to 36.521-1: Update of Spurious emission band UE co-existence test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103727 | 0274 | - | LTE-RF: CR for Prach time mask test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103728 | 0275 | - | LTE-RF: CR for General ON/OFF time mask test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103729 | 0276 | - | LTE-RF: Update to spectrum flatness test case and relevant annexes | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103730 | 0277 | - | LTE-RF:CR for test case of In-band emissions | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103731 | 0278 | - | EVM with exclusion period (test) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103732 | 0279 | - | CR to 36.521-1 on Correction to Demodulation Requirements for PDSCH | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103733 | 0280 | - | CR to 36.521-1: Update PDCCH DCI Formats for Open Loop and Closed Loop Spatial Multiplexing Test Cases | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103751 | 0281 | - | Misc update in CSI tests | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103752 | 0282 | - | Correction of the statistical part in PMI and RI tests | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103753 | 0283 | - | LTE-RF:CR to downlink RMCs for TX characteristics | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103754 | 0284 | - | LTE-RF: Update of annex C | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103756 | 0285 | - | Measuring throughput ratios (Annex G) | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103763 | 0286 | - | LTE-RF: CR for Minimum output power test case | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103764 | 0287 | - | Performance, CSI reporting and uncertainties for UEs with multiple Rx antennas | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103771 | 0288 | - | Introduction of clause 8.4.1 and 8.5.1 test case uncertainties | 9.0.0 | 9.1.0 |

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| | | | | | and Test Tolerances | | |
| 2010-06 | RAN#48 | R5-103778 | 0291 | - | Uplink power for receiver tests | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103780 | 0292 | 1 | Addition of the exceptional message for In-band emissions | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103781 | 0289 | - | Correction to 6.5.2.1 EVM | 9.0.0 | 9.1.0 |
| 2010-06 | RAN#48 | R5-103782 | 0290 | - | Correction to CQI reporting | 9.0.0 | 9.1.0 |
| 2010-09 | RAN#49 | R5-104090 | 0294 | - | Corrections to Spectrum emission mask test regarding UE category | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104091 | 0295 | - | Missing note in Additional spurious emission test with NS_07 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104095 | 0296 | - | PDCCH Aggregation level for CSI tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104096 | 0297 | - | Default initial and connection Uplink power for RF tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104212 | 0298 | - | Limits on Uplink power for Receiver tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104244 | 0299 | - | Correction to Demodulation Requirements | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104461 | 0300 | - | CR to 36.521-1: Editorial Corrections for Closed Loop Spatial Multiplexing Test Cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104478 | 0301 | - | Correction to Test requirements in 6.5.2.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104482 | 0302 | - | Correction to 8.2.1.1 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104520 | 0303 | - | 36521-1 General update of sections 00 to 08: missing Introduction references formatting | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104583 | 0304 | - | No necessity to apply - consecutive time slots for EVM | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104584 | 0305 | - | Correction to E.4.4 EVM equalizer spectrum flatness | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104630 | 0306 | - | Correction of table reference in In-band emissions test | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104808 | 0307 | - | CR to 36.521-1: Update to Additional Spectrum Emission Mask test case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104809 | 0308 | - | CR to 36.521-1: Update to Spurious emission band UE co-existence test case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104810 | 0309 | - | LTE-RF: CR for Max Output Power | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104811 | 0310 | - | LTE-RF: CR for Freq Error | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104812 | 0311 | - | Introduction of exclusion period for PUCCH-EVM test in clause 6.5.3 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104813 | 0312 | - | Correction to Demodulation UE-Specific Reference Symbols | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104814 | 0313 | - | Uncertainties and Test Tolerances for CSI Test cases 9.2.1.1 and 9.2.1.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104815 | 0314 | - | Uncertainties and Test Tolerances for CSI Test cases 9.2.2.1 and 9.2.2.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104816 | 0315 | - | UE applicability for CSI test cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104817 | 0316 | - | Update of CQI reporting TCs under fading conditions | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104818 | 0317 | - | Update of Reporting of Precoding Matrix Indicator TCs | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104819 | 0318 | - | Correction of the statistical part 9.3.1.1.1 (CQI Reporting under fading conditions) | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104820 | 0319 | - | Correction of the statistical part 9.3.3.1.1 (CQI Reporting under fading conditions) | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104821 | 0320 | - | Correction of the statistical part 9.3.2.1.1 (CQI Reporting under fading conditions) | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104822 | 0321 | - | Update and new RMC-s for CQI tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104823 | 0322 | - | Correction of EVM calculation in annex | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104824 | 0323 | - | Introduction of exclusion period for PUCCH-EVM test in Annex E | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104844 | 0324 | - | Pcmax changes to Configured UE Transmitted Output Power | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104845 | 0325 | - | Clarification on the frequency range with net work signal in 6.6.3.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104846 | 0326 | - | Update of editor's notes | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104847 | 0327 | - | Removal of Extreme Conditions in 6.2.3 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104850 | 0328 | - | Corrections to Test procedure loop in CSI tests | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104851 | 0329 | - | Introduction of TDD CQI Reporting under fading conditions and frequency-selective interference test case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104852 | 0330 | - | Introduction of TDD RI Reporting test case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104853 | 0331 | - | Update of CQI reporting TCs under AWGN conditions | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104854 | 0332 | - | Update of FDD RI Reporting TC | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104857 | 0333 | - | CR to 36.521-1 LTE UE Tx_RX test cases band 20 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104861 | 0334 | - | Corrections to Test requirements for MPR test | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104863 | 0335 | - | Clarification on notes in Max Power | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104872 | 0336 | - | Correction to 6.3.5.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104873 | 0337 | - | Numbering and alignment of TDD PHICH demod test cases | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104874 | 0338 | - | Correction to test numbering for exceptional messages in 8.2.x.x | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104875 | 0339 | - | Correction to 9.2.2 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104876 | 0340 | - | Correction to the test procedures of 9.3.x | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104877 | 0341 | - | Correction to 9.4.x | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104878 | 0342 | - | The new reference of connection diagram for 9.3.3 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104879 | 0343 | - | Correction to 6.3.4.1 and 6.3.5.1 | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-104888 | 0344 | - | Update of Annex C.2 for AG level | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-105055 | 0345 | - | Introduction of a new RF test case (8.7) to verify downlink | 9.1.0 | 9.2.0 |

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| | | | | | sustained data rate performance | | |
| 2010-09 | RAN#49 | R5-105061 | 0347 | - | CR to 36.521-1: Modification to Additional Maximum Power Reduction Test Case | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-105062 | 0348 | - | Modification to Additional Spectrum Emission Mask | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-105063 | 0349 | - | Modification to Additional Spurious Emissions | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-105064 | 0350 | - | Modification to Maximum Power Reduction | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | R5-105065 | 0351 | - | Modification to Adjacent Channel Leakage Power Ratio | 9.1.0 | 9.2.0 |
| 2010-09 | RAN#49 | RP-100987 | 0352 | - | Correction of status for RF performance test case | 9.1.0 | 9.2.0 |
| 2010-12 | RAN#50 | R5-106073 | 0353 | - | Corrections to receiver spurious emissions test | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106074 | 0354 | - | Update of downlink power for receiver tests | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106076 | 0355 | - | CQI: Side condition when CQI median equals min or max CQI-values | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106077 | 0356 | - | Update of the throughput-definition for multi-data stream transmission | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106078 | 0357 | - | Update of RF OCNG patterns | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106092 | 0358 | - | Correction of DCI format used in PDSCH performance test 8.2.1.4.2 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106249 | 0359 | - | CR to 36.521-1: Correction to Spurious emission band UE co-existence test case | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106250 | 0360 | - | CR to 36.521-1: Correction to Additional Tx spurious emissions test case | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106374 | 0361 | - | Correction of FDD CQI reporting test under AWGN - PUCCH 1-1 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106394 | 0362 | - | Correction of clause 9.3.1 and 9.3.3 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106399 | 0363 | - | "Correction of G.2.5 Pass fail decision rules" | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106420 | 0364 | - | Introduction of test uncertainties and tolerances for TDD PDSCH DRS test cases | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106440 | 0365 | - | Correction to unsigned numbers in Annex F.1.2 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106443 | 0366 | - | Correction to the exceptional messages in 9.4 Reporting of PMI TCs | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106491 | 0367 | - | CR to 36.521-1: Correction to Table Numbering Error in TDD PDSCH Closed Loop Single/Multi Layer Spatial Multiplexing 2x2 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106512 | 0368 | - | Transport format table clarification in CSI test cases | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106801 | 0369 | - | HARQ scheduling in TDD performance tests using special subframes | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106803 | 0370 | - | Correction to Fading Profiles in TCs 8.4 and 8.5 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106816 | 0372 | - | CR to 36.521-1: Modification to Spectrum Emissions Mask | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106817 | 0373 | - | Introduction of test uncertainties and tolerances for TDD PCFICH/PDCCH and PHICH test cases | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106818 | 0374 | - | Update of TDD PDSCH CRS Demodulation test cases | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106821 | 0375 | - | PUSCH-EVM with exclusion period | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106822 | 0376 | - | Maintenance of Band 20 for receiver tests | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106823 | 0377 | - | Completion of clause 9.3.1 and 9.3.2 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106824 | 0378 | - | Update of FDD RI Reporting TC | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106825 | 0379 | - | Correction to 9.2.2 CQI TCs | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106826 | 0380 | - | G.3.5 on PDCCH&PHICH Minimum Test Times | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106827 | 0381 | - | Completion of test time and $\frac{1}{T_F}$ -TT for clauses 9.3 to 9.5 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106828 | 0382 | - | EVM window length for PRACH | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106842 | 0383 | - | Modification of TC 8.3.2.1 - TDD PDSCH Single-layer SM Performance (UE-Specific Reference Symbols) | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106843 | 0384 | - | Power control relative power tolerance: Missing band edge relaxation | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106844 | 0385 | - | SRS time mask test procedure update | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106845 | 0386 | - | Correction of TC General ON/OFF time mask | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106846 | 0387 | - | Update of TDD CQI reporting test under frequency selective interference conditions | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106847 | 0388 | - | Update of TDD RI reporting test | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106848 | 0389 | - | Iot setting in CQI test clarification | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106850 | 0390 | - | Correction of Test Uncertainties and Test Tolerances for Reference Sensitivity-Band 4 | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106855 | 0391 | - | Correction to DL and UL RMC configurations in 6.5.1 Frequency Error | 9.2.0 | 9.3.0 |
| 2010-12 | RAN#50 | R5-106858 | 0392 | - | CR to 36.521-1: Update LTE RF test cases with test requirements for EUTRA TDD LTE band 41. | 9.2.0 | 9.3.0 |
| 2011-03 | RAN#51 | R5-110138 | 0393 | - | Spurious emission band co-existence test: Remaining old test requirement table | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110139 | 0394 | - | Transmit intermodulation test: Interferer offset for Band 20, Bandwidth 20 MHz | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110141 | 0395 | - | PMI Performance tests: Corrections to test settings | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110143 | 0396 | - | PDCCH Performance test: Correction to TDD DL RMC-s | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110144 | 0397 | - | OCNG for RF tests: Updates | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110158 | 0398 | - | CQI Performance tests: Clarification on subbands used in the | 9.3.0 | 9.4.0 |

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| | | | | | test | | |
| 2011-03 | RAN#51 | R5-110302 | 0399 | - | LTE RF: PCFICH/PDCCH Transmit Diversity Performance | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110349 | 0401 | - | Correction of OFF power measurements in 6.3.4 ON/OFF time mask | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110354 | 0402 | - | Correction of Annex E (Global In-Channel TX-Test) | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110506 | 0403 | - | Addition of 3500 MHz TDD bands into chapter 5 of 36.521-1 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110747 | 0404 | - | Updates to section 8.7 DL sustained data rate test case | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110850 | 0405 | - | Sustained data rate: Definition of UL RMC-s | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110860 | 0400 | - | Update measurement period from test procedure of 6.2.2 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110861 | 0428 | - | A-MPR Test requirement update | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110862 | 0437 | - | Measuring throughput with different payload size. | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110900 | 0406 | - | Correction to SNR reference in FDD PCFICH/PDCCH Single Antenna Port Performance Test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110901 | 0407 | - | CR to 36.521-1: Correction of Additional Maximum Power Reduction (A-MPR) test case | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110915 | 0408 | - | Uncertainties and Test Tolerances for CSI Test cases 9.3.1.1.x and 9.3.2.1.x | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110916 | 0409 | - | Uncertainties and Test Tolerances for CSI Test cases 9.3.3.1.x | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110917 | 0410 | - | Uncertainties and Test Tolerances for CSI Test cases 9.4.1.1.x and 9.4.2.1.x | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110918 | 0411 | - | Uncertainties and Test Tolerances for CSI Test cases 9.5.1.x | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110919 | 0412 | - | Editorial errors in CSI test cases 9.2.1.x Test procedure | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110933 | 0413 | - | Handling of different releases in RAN5 LTE RF specification | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110938 | 0414 | - | PRACH Time mask and EVM tests: TDD Special subframe configuration | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110939 | 0415 | - | Update on PRACH time mask for TDD | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110940 | 0416 | - | Correction of the connection diagram reference in the initial conditions for Transmitter Characteristics | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110941 | 0417 | - | Update of the reference sensitivity requirement for the 1.4MHz and 3MHz bandwidths and note correction for Band 3 and Band 9 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110942 | 0418 | - | CR Removing brackets of band 41 reference sensitivity in 36.521-1 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110949 | 0419 | - | PDCCH and PHICH performance tests: Updates and corrections | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110950 | 0420 | - | Addition of CodeBookSubsetRestriction bitmap for Multi-Layer Spatial Multiplexing | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110951 | 0421 | - | Addition of exceptional message in 8.4.1.2.2 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110952 | 0422 | - | CQI test 9.2.2.2: Update acc TS 36.101 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110953 | 0423 | - | Maximum input level test: Correction to DL-RMC | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110954 | 0424 | - | Correction of E.7, EVM with exclusion period | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110967 | 0425 | - | Updates to Additional Spurious emissions and Spurious emission band UE co-existence test | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110970 | 0426 | - | LTE RF: references to state 3A in 36.521-1 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110973 | 0429 | - | RI Performance tests: Corrections | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110975 | 0430 | - | Completion of annex G.3.5 (Minimum test time, performance tests) | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110978 | 0431 | - | Correction to Band 12 frequency range | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110979 | 0432 | - | Additional in-band blocking requirement for Band 12 | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110989 | 0433 | - | Completion of annex G.3.6 (test conditions, performance tests) | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110990 | 0434 | - | Addition of test cases of TDD PDSCH Single-layer and Dual-layer Spatial Multiplexing Performance | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110991 | 0435 | - | Correction to Times Mask and Power Control tests | 9.3.0 | 9.4.0 |
| 2011-03 | RAN#51 | R5-110992 | 0436 | - | Add requirement of QPSK with partial RB allocation into test requirement of 6.2.3 | 9.3.0 | 9.4.0 |
| 2011-04 | - | - | - | - | Added approved R5-110967 which was missing. | 9.4.0 | 9.4.1 |
| 2011-06 | RAN#52 | R5-112148 | 0438 | - | ON/OFF time mask for PRACH: PRACH configuration index | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112149 | 0439 | - | CQI tests with frequency selective scheduling mode: Random selection of Sub-Bands | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112150 | 0440 | - | DL-RMC for receiver tests: Obsolete editors note | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112211 | 0441 | - | Adding Band 24 to TS 36.521-1 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112311 | 0442 | - | Correction to the sustained data rate tests | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112312 | 0443 | - | Correction to the multi-antenna transmission tests | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112349 | 0444 | - | Update Annex G.3.6 for spatial multiplexing test cases | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112412 | 0445 | - | Correction to DL RMC for Max input for UE category 2 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112459 | 0446 | - | EVM window length for PRACH | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112460 | 0447 | - | Removal of square brackets in Annex G.5.4 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112462 | 0448 | - | Test tolerances for EVM with exclusion period | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112808 | 0449 | - | LTE RF - UEs Demodulation and CSI Band applicability | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112810 | 0450 | - | Correction on test requirement of test case 6.5.2.3 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112811 | 0451 | - | LTE RF - TC6.3.5.2 Change in Minimum Requirements | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112812 | 0452 | - | Removable of NS values for TDD | 9.4.1 | 9.5.0 |

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| 2011-06 | RAN#52 | R5-112813 | 0453 | - | Correction on initial condition for test cases of 9.3.3 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112814 | 0454 | - | Correction to 9.4 PMI test cases | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112816 | 0455 | - | Test procedure change on Tx spurious emissions test cases | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112850 | 0456 | - | Update of Initial conditions according to table 7.3.3-2 for the 5MHz and 10MHz bandwidths | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112851 | 0457 | - | Iot modelling in frequency-selective interference CQI tests | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112852 | 0458 | - | Correction to test procedure in 9.5.1 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112859 | 0459 | - | Introduction of additional PDCCH test cases for Rel-9 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112860 | 0460 | - | Introduction of additional PHICH test cases for Rel-9 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112861 | 0461 | - | Introducing UE-selected subband CQI tests for Rel-9 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112862 | 0462 | - | Introduction of new PMI test cases for UE-selected sub-band reporting for Rel-9 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112863 | 0463 | - | Addition of a new TC for FDD MBMS performance | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112864 | 0464 | - | Addition of a new TC for TDD MBMS performance | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112866 | 0465 | - | Addition of UE category 1 coverage for single-layer spatial multiplexing transmission on antenna port 7 or 8 | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112870 | 0466 | - | Removal of square brackets in annex G.3.5 (Minimum test time, performance tests) | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112871 | 0467 | - | Update minimum requirements and test requirements for spatial multiplexing test cases | 9.4.1 | 9.5.0 |
| 2011-06 | RAN#52 | R5-112873 | 0468 | - | Test procedure change on few Rx test cases | 9.4.1 | 9.5.0 |
| 2011-09 | RAN#53 | R5-113178 | 0506 | - | RF TC 8.7.2: Correction of the ACK/NACK feedback mode | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113179 | 0469 | - | RF TC-s 9.3.4: Updates to minimum requirements and corrections to test procedures | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113180 | 0470 | - | RF: New UL RMC for 10 MHz 15RB-s | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113181 | 0471 | - | RF: Use of State 3A-RF | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113233 | 0472 | - | LTE RF: EVM Annex E correction | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113238 | 0473 | - | RF TC 6.3.4.2.2: Correction to SRS configuration | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113368 | 0474 | - | Correction in test cases of 8.2.2.1.2 and 8.5.1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113369 | 0475 | - | Correction in test cases of 9.5 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113380 | 0476 | - | Update of minimum test time for PCFICH/PDCCH and PHICH tests | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113383 | 0477 | - | Correction of Table G.3.5-1 (minimum test time) | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113419 | 0478 | - | Correction for spurious emission band UE co-existence limits of Band 3 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113454 | 0479 | - | Correction to test procedure in 6.5.1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113455 | 0480 | - | Correction to Minimum requirement in 6.6.2.2 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113456 | 0481 | - | Correction to RMC for PDCCH/PCFICH performance requirements | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113458 | 0482 | - | Correction to CSI TCs | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113541 | 0483 | - | Introduction of Expanded 1900MHz Band (Band 25) into section 5 of 36.521-1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-113596 | 0506 | - | Abbreviation update and Editorial corrections in TS36.521-1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114000 | 0484 | - | Correction in 6.3.5.2 Power Control Relative power tolerance | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114001 | 0485 | - | Correction in 6.3.4.2.2 SRS time mask | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114002 | 0486 | - | Addition of PDSCH TDD performance tests for Low UE categories | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114003 | 0487 | - | Additional FDD scenarios | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114004 | 0488 | - | Addition of RMC-s for PDSCH performance tests for low UE categories | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114032 | 0489 | - | Introduction of Expanded 1900MHz Band (Band 25) into section 6 of 36.521-1 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114033 | 0490 | - | Correction on TDD MBMS performance requirements for 64QAM mode | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114034 | 0491 | - | Correction on FDD MDMS performance requirements for 64QAM mode | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114038 | 0492 | - | RF TC 6.3.5.2: Consideration of band edge relaxation in test requirements | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114039 | 0493 | - | Band 19 A-MPR refinement | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114040 | 0494 | - | Test system uncertainty and TT for 6.6.3.3 additional spurious emissions NS_07 | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114041 | 0495 | - | Correction to Additional Maximum Power Reduction | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114042 | 0496 | - | Correction to 6.3.4.2.1 PRACH time mask | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114043 | 0497 | - | Correction to Receiver Characteristics Minimum Requirements | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114044 | 0498 | - | Addition of PDSCH FDD performance tests for Low UE categories | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114045 | 0499 | - | Additional Rel-9 TDD scenarios | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114046 | 0500 | - | Update of CQI/PMI test cases | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114047 | 0501 | - | Update of RI test cases | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114048 | 0502 | - | Release dependent RMCs in PCFICH/PDCCH and PHICH tests | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114049 | 0503 | - | Addition of tests scenarios in Annex | 9.5.0 | 9.6.0 |
| 2011-09 | RAN#53 | R5-114073 | 0504 | - | Introduction of Expanded 1900MHz Band (Band 25) into | 9.5.0 | 9.6.0 |

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|---|---|---|---|---|---|---|---|
| | | | | | section 7 of 36.521-1 | | |
| 2011-09 | RAN#53 | R5-114091 | 0505 | - | Correction to test cases 10.1 and 10.2 | 9.5.0 | 9.6.0 |
| 2011-12 | RAN#54 | R5-115112 | 0507 | - | RF: Coverage of MPR and A-MPR requirements in emissions tests | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115113 | 0508 | - | RF: Update to names of some RMC-s used in different releases | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115114 | 0509 | - | RF TC 6.2.4, 6.6.2.2, 6.6.3.3: Corrections to A-MPR related tests | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115115 | 0510 | - | RF TC-s 9: Corrections to UL allocation for some CQI tests | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115116 | 0511 | - | RF TC 6.3.5.2: Missing extreme conditions in test requirements | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115117 | 0512 | - | RF TC 6.2.3: Missing test requirements for band 25 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115118 | 0513 | - | RF TC 6.5.2.1: Redundand conflicting IE-s in message contents | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115119 | 0514 | - | RF TC 6.6.3.1: Correction of reference to the connection diagram | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115324 | 0517 | - | LTE RF: CSI test case update | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115472 | 0518 | - | Corrections to the dual-layer beamforming demodulation requirements | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115475 | 0519 | - | Correction to Code Block Numbers | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115800 | 0522 | - | Delete note in PUSCH-EVM | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115812 | 0526 | - | RF: Corrections to tests with release dependent requirements | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115815 | 0527 | - | Test Frequency for Relative Power Tolerance | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115821 | 0531 | - | RF: General review of the reference measurement channels | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115822 | 0532 | - | Corretion for codebook subset restriction insingle-layer closed loop spatial multiplexing test | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115830 | 0533 | - | Uncertainties and Test Tolerances for 3000MHz to 4200MHz, Tx Test cases | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115831 | 0534 | - | Uncertainties and Test Tolerances for 3000MHz to 4200MHz, Rx Test cases | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115832 | 0535 | - | Support for band 22, 42 and 43 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115837 | 0536 | - | Update LTE RF test cases with test requirements for FDD LTE Band 23 in 36.521-1 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115875 | 0537 | - | Correction on FDD and TDD MBMS conformance requirements | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115877 | 0538 | - | Introduction of test system uncertainties and TT to new TCs in clauses 9.3.4 to 9.4.2.2.2 | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115879 | 0539 | - | Uncertainties and Test Tolerances for Sustained data rate test cases | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115891 | 0540 | - | Requirement change in UE spurious emissions for Band 7 and 38 co-existence (Rel-8 only) | 9.6.0 | 9.7.0 |
| 2011-12 | RAN#54 | R5-115162 | 0515 | - | Definitions, symbols and abbreviations for CA RF | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115565 | 0520 | - | In-band blocking for CA (New) | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115568 | 0521 | - | Spurious Response for CA (New) | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115801 | 0523 | - | Out-of-Band blocking for CA (new) | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115802 | 0524 | - | Narrow band blocking for CA (new) | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115803 | 0525 | - | Addition of new RMCs for CA feature related Chapter8 test cases | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115816 | 0528 | - | General Section for CA RF | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115817 | 0529 | - | Frequency Band and Channel Assignments for CA RF | 9.7.0 | 10.0.0 |
| 2011-12 | RAN#54 | R5-115818 | 0530 | - | UE Maximum Output Power for intra-band contiguous CA (new) | 9.7.0 | 10.0.0 |
| 2012-03 | RAN#55 | R5-120080 | 0587 | - | Removal of technical content in 36.521-1 v9.7.0 and substitution with pointer to the next Release | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120134 | 0541 | - | RF: General clarification on the count of ACK / NACKs in throughput calculation in CQI tests | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120137 | 0542 | - | RF: Corrections to message contents in some MIMO demodulation performance tests | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120138 | 0543 | - | RF: Updates to handling requirements and tests for different releases | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120140 | 0544 | - | RF: Coverage of MPR and A-MPR requirements in Tx emissions tests | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120243 | 0545 | - | Tx Test cases Uplink power limit window for 3000MHz to 4200MHz | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120244 | 0546 | - | Rx Test cases Uplink power limit window for 3000MHz to 4200MHz | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120252 | 0547 | - | Global in-channel TX-test, delete note | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120288 | 0548 | - | LTE RF - Update of abbreviations and addition of RBstart | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120314 | 0549 | - | Clarification for the starting point of RB in 6.3.5.2 Relative Power test | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120338 | 0550 | - | Correction to MCS value in Table A.4-3 | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120343 | 0551 | - | Introduction of new maximum input level test case for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120351 | 0552 | - | CA RF - Changes in aggregated CC configurations | 10.0.0 | 10.1.0 |

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|---|---|---|---|---|---|---|---|
| 2012-03 | RAN#55 | R5-120355 | 0553 | - | CA RF - updates to 7.7A Spurious response for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120370 | 0554 | - | Test Tolerance for 6.6.3.3 additional spurious emissions NS_07 | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120426 | 0555 | - | Correction to Frequency Range for Spurious Emission Requirements | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120521 | 0556 | - | Introduction to Maximum Power Reduction for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120524 | 0557 | - | Introduction to Maximum Power Reduction for CA in Annex | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120526 | 0558 | - | Introduction to Adjacent Channel Leakage power Ratio (ACLR) for CA in Annex | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120801 | 0559 | - | Correction of TC 6.5.2.1 Error Vector Magnitude (EVM) for Test requirement | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120802 | 0560 | - | Correction of TC 8.2.1.3.1 FDD PDSCH Open Loop Spatial Multiplexing 2x2 for 8.2.1.3.1.4.1 Initial conditions | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120803 | 0561 | - | Correction of TC8.2.1.3.2 FDD PDSCH Open Loop Spatial Multiplexing 4x2 for 8.2.1.3.2.4.1 Initial conditions | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120804 | 0562 | - | Correction of TC 8.2.2.3.1 TDD PDSCH Open Loop Spatial Multiplexing 2x2 for 8.2.2.3.1.4.1 Initial conditions | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120805 | 0563 | - | Correction of TC 8.2.2.3.2 TDD PDSCH Open Loop Spatial Multiplexing 4x2 for 8.2.2.3.2.4.1 Initial conditions | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120816 | 0564 | - | Addition of frequency band and channel assignments for UL MIMO | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120825 | 0565 | - | Introduction of Chapter8 Single antenna port and Open loop spatial Multiplexing test cases for CA capable UE's. | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120828 | 0566 | - | LTE RF - UE Co-ex test point clarification for bands 7 and 38 | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120833 | 0567 | - | Addition of receiver image section 7.10 | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120835 | 0568 | - | Correction to Reporting of Rank Indicator (RI) FDD Test | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120837 | 0569 | - | Introduction to Adjacent Channel Leakage power Ratio (ACLR) for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120839 | 0570 | - | Addition of a new TC 6.2.2B for UE Maximum Output Power for UL-MIMO | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120842 | 0571 | - | Harmonic exceptions in LTE UE to UE co-existence tests | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120843 | 0572 | - | Clarification for upper and lower interferers in 7.6.3 Narrow Band Blocking | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120844 | 0573 | - | RF: Correction of frequency range for out of band blocking test | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120845 | 0574 | - | Correction and completion to clause 10 MBMS performance | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120874 | 0575 | - | RF: New RMC-s and updates to the RMC-s overview tables | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120875 | 0576 | - | RF: Test frequencies for UE co-existence emissions under Note 13 | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120877 | 0577 | - | UE Transmit OFF power for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120878 | 0578 | - | Frequency error for Intraband CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120879 | 0579 | - | Occupied bandwidth for intra-band contiguous CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120880 | 0580 | - | Transmitter Spurious emissions for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120881 | 0581 | - | Reference sensitivity level for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120882 | 0582 | - | CA RF - Addition of test description to 6.2.2A MOP for intra-band CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120884 | 0583 | - | CA RF - updates to 7.6.2A OOB blocking for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120885 | 0584 | - | CA RF - updates to 7.6.3A Narrow-band blocking for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120899 | 0585 | - | CA RF - updates to 7.6.1A In-band Blocking for CA | 10.0.0 | 10.1.0 |
| 2012-03 | RAN#55 | R5-120900 | 0586 | - | Correction to UE Spurious Emissions | 10.0.0 | 10.1.0 |

# History

| Document history | | |
|---|---|---|
| V10.0.0 | March 2012 | Publication |
| V10.1.0 | March 2012 | Publication |
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